Theory and Methods in Causal Inference

Lecture Notes

Jae Kwang Kim

Department of Statistics

Iowa State University

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Contents

Preface

These lecture notes accompany the graduate course **STAT 5900B: Theory and Methods in Causal Inference** at the Department of Statistics, Iowa State University.

The course targets first-year PhD students in statistics. It assumes familiarity with probability theory, mathematical statistics, and linear models at the level of a graduate sequence, but no prior exposure to causal inference.

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Organization

The notes are organized in three parts, with a fourth part on policy learning deferred to a later release.

Part I — Foundations (Chapters 1–3) builds the graphical and conceptual foundation of causal inference. Chapter 1 introduces the three languages of causality — structural equation models (SEMs), directed acyclic graphs (DAGs), and potential outcomes — and explains how they relate. Chapter 2 develops DAGs and d-separation as tools for reading off conditional independence from causal structure. Chapter 3 introduces the do-calculus and the main identification criteria (back-door, front-door, do-calculus rules) that translate causal queries into statistical ones.

Part II — Identification (Chapters 4–9) bridges the graphical framework and observed data. Chapter 4 connects the potential outcomes framework to the do-calculus and establishes the consistency, exchangeability, and positivity assumptions. Chapter 5 covers randomization and the back-door adjustment formula. Chapter 6 develops propensity score methods — weighting, matching, and the balancing property. Chapter 7 develops instrumental variables: the Wald estimand, the local average treatment effect (LATE) for compliers, and the three IV assumptions. Chapter 8 covers mediation analysis and the front-door criterion, distinguishing the controlled direct effect from the natural direct and indirect effects. Chapter 9 addresses sensitivity analysis and partial identification: the three canonical sensitivity models (Rosenbaum Γ, E-value, marginal sensitivity model), Manski's no-assumption bounds, and the connection between sensitivity analysis and modern estimation methods.

Part III — Estimation (Chapters 10–13) covers semiparametric efficiency theory and its modern applications. Chapter 10 introduces the estimation framework — estimating equations, asymptotic linearity, influence functions, Z-estimation with nuisance parameters, and the semiparametric efficiency bound — that underlies the rest of Part III. Chapter 11 develops the augmented inverse probability weighted (AIPW) estimator from three complementary perspectives — bias correction, optimal augmentation, and the efficient influence function — and establishes double robustness and semiparametric efficiency. Chapter 12 addresses flexible nuisance estimation with orthogonal scores and cross-fitting, culminating in the double machine learning (DML) estimator and its rate conditions. Chapter 13 covers estimation under instrumental variables: the Wald estimator, two-stage least squares, GMM and overidentification tests, weak instruments, generalized empirical likelihood, and the control function approach.

Part IV — Policy Learning and Sequential Decision Making (Chapters 14–16, deferred) will cover policy evaluation, optimal policy learning, and dynamic treatment regimes.

Appendices

Appendix A provides graphical intuition for conditional independence and d-separation. **Appendix B** introduces Single World Intervention Graphs (SWIGs). **Appendix C** develops the formal foundations of pathwise differentiability and efficient influence functions for readers who want the semiparametric geometry underlying Chapter 10.

Notation

Throughout these notes:

- $\mathrm{do}(T = t)$ denotes the do-operator (intervention).
- $Y(t)$ denotes the potential outcome under $T = t$.
- $\mathbb{E}[\,\cdot\,]$ denotes expectation; $\perp\!\!\!\perp$ denotes conditional independence.
- $\mathcal{G}$ denotes a DAG; $\mathcal{G}_{\overline{T}}$ denotes the mutilated graph after intervening on T.
- $\pi(x) = P(T = 1 \mid X = x)$ denotes the propensity score.
- $\mu_t(x) = \mathbb{E}(Y \mid T = t, X = x)$ denotes the outcome regression.

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Part I

Foundations

Chapter 1

Introduction

Learning Objectives

By the end of this chapter, students should be able to:

1. Articulate the distinction between an interventional distribution $P(y \mid \mathrm{do}(T{=}t))$ and an observational conditional distribution $P(y \mid T{=}t)$, and explain why they differ, in general, under confounding or endogeneity.
2. Describe the same causal question in all three languages — SEM, DAG graph surgery, and potential outcomes — and begin to translate between them.
3. Derive the two Gaussian laws explicitly in the linear confounded model, and identify $\alpha\gamma\sigma_U^2/\sigma_T^2$ as the difference between the observational and causal mean slopes.
4. State the two-step paradigm (identification then estimation) and locate each step in the causal inference pipeline.

This chapter is an orientation, not a complete treatment. It introduces four ideas — the interventional/observational distinction, the causal trinity, the Gaussian confounded model as a running example, and the identification-versus-estimation paradigm — that will be developed rigorously in the chapters that follow. After one reading, students should have a map of the territory; the details will become clear as the course proceeds. The remarks on NPSEM-IE and on scope, together with Section 1.2.6, are optional on a first reading.

1.1 Motivation: Why Causal Inference Is Hard

1.1.1 The Central Question

Causal inference is concerned with a deceptively simple question: what would happen to Y if we were to *set* $T = t$? This is fundamentally different from the question that standard statistical procedures answer: what is the distribution of Y among units *observed* to have $T = t$?

Definition: Interventional vs. Observational Distribution

- The **interventional distribution** $P(y \mid \mathrm{do}(T{=}t))$ is the distribution of Y in a hypothetical world where T has been *externally set* to t, regardless of the usual data-generating mechanism for T.
- The **observational conditional distribution** $P(y \mid T{=}t)$ is the distribution of Y among the subpopulation of units for whom $T = t$ was *observed* to hold.

These two distributions coincide when the treatment assignment is *unconfounded* with the outcome — in the simplest case, when T shares no common causes with Y. Randomization is the canonical way to enforce unconfoundedness, but it is not the only way: any data-generating mechanism in which T is unconfounded with Y will produce the same coincidence. Under confounding, the two distributions generally differ. In the linear model of Section 1.3, this discrepancy appears as *endogeneity bias* in the OLS regression coefficient.

Example: Drug Trial with Unmeasured Severity

Does a new drug ($T = 1$) reduce blood pressure (Y)? Suppose sicker patients are more likely to take the drug, so that unmeasured severity U is a common cause of T and Y. Then:

- *Observational study*: $\mathbb{E}[Y \mid T{=}1] > \mathbb{E}[Y \mid T{=}0]$. Drug users have higher blood pressure on average. A naive analyst concludes the drug is harmful.
- *Randomized trial*: $\mathbb{E}[Y \mid \mathrm{do}(T{=}1)] < \mathbb{E}[Y \mid \mathrm{do}(T{=}0)]$. Forcing everyone to take the drug reduces blood pressure. The drug is beneficial.

Same population. Different questions. Opposite answers. The entire discrepancy is caused by the back-door path $T \leftarrow U \rightarrow Y$, which the observational quantity $\mathbb{E}[Y \mid T{=}1]$ cannot remove.

The Identification Problem

The observed-data law contains $P(y \mid T{=}t)$, which can be estimated from data. The identification problem is whether that observational law also determines $P(y \mid \mathrm{do}(T{=}t))$. The goal of *identification theory* is to find conditions under which it does.

1.2 The Causal Trinity: Three Languages for One Idea

The same causal question — what happens when we set $T = t$? — can be expressed in three closely related languages: the *structural equation model* (SEM), the *directed acyclic graph* (DAG) with graph surgery, and the *potential outcomes* framework. Collectively, we call these the **causal trinity**. The three frameworks are often intertranslatable, though exact equivalence requires additional assumptions about the causal model; we return to this point in the NPSEM-IE remark below and in Section 1.2.6. Each language illuminates a different facet of the same underlying idea, and fluency in all three is essential for modern causal reasoning.

1.2.1 Language 1: Structural Equation Models

A structural equation model (SEM) represents the causal data-generating mechanism as a system of equations, one for each *modeled variable*, together with a joint distribution for the *background inputs*.[1] The variables on the left-hand sides of the structural equations are generated within the modeled system. The background inputs have no structural equations in the model, and their joint distribution is taken as given; under the NPSEM-IE assumption adopted below, that joint distribution factorizes into independent marginals. The background inputs are the root sources of randomness; the modeled variables inherit their variability, directly or indirectly, from them.

The simplest interesting SEM has three observed variables: a covariate X, a treatment T, and an outcome Y, each generated by its own equation:

$$X = f_X(\varepsilon_X), \qquad T = f_T(X, \delta), \qquad Y = f_Y(X, T, \varepsilon), \tag{1.1}$$

where ε_X, δ, and ε are mutually independent background disturbances. In this system, X, T, and Y are generated within the model, while the three disturbances are background inputs. As a concrete interpretation, consider a job-training program: background characteristics such as education and motivation (X) influence both enrollment in the program (T) and subsequent employment outcomes (Y), while T may also affect Y directly (Wang et al. 2026). On the surface, Equation 1.1 looks like a system of nonparametric regression models. What makes it *structural* is that each equation is read as an autonomous mechanism: it describes how its left-hand variable is *generated*, and it remains valid when other equations in the system are replaced by interventions.

Equation 1.1 poses no essential causal difficulty: the common cause X of T and Y is observed, so it can be adjusted for (Chapters 3–5 develop this in detail). The hard case — and the running setup for the rest of this chapter — arises when the common cause is *unobserved*. Writing U for such a variable, and

[1] SCM texts often call the background inputs *exogenous variables* and the variables generated by structural equations *endogenous variables*. To avoid conflict with econometric usage, these notes reserve the terms *exogeneity* and *endogeneity* primarily for equation-specific restrictions.

suppressing its structural equation because we do not model how it arises, we obtain the three-variable confounded SEM over (U, T, Y):

$$T = f_T(U, \delta), \qquad Y = f_Y(T, U, \varepsilon), \tag{1.2}$$

where U, δ, and ε are *mutually independent* background variables, with joint distribution $P_{U,\delta,\varepsilon} = P_U \otimes P_\delta \otimes P_\varepsilon$. The variable U is an unobserved background cause of both T and Y (a confounder), while δ and ε are equation-specific disturbances. Note the contrast with Equation 1.1: a variable whose generating equation we leave unspecified is treated as a background input. The marginal distributions of the background variables — including $p(u)$, to which we refer by this notation later — are suppressed in Equation 1.2.

Remark: Error Independence in the NPSEM-IE

The assumption that the background inputs in Equation 1.2 — U, δ, and ε — are *jointly independent* is a substantive restriction, not merely a technical convenience. Because U is unobserved and enters both f_T and f_Y, error independence does not remove the dependence between T and $Y(t)$ induced by U; it restricts only the additional channel of dependence running through the equation-specific errors (δ, ε). Richardson and Robins (2014) call SEMs with this structure *Nonparametric Structural Equation Models with Independent Errors* (NPSEM-IEs).

A weaker single-world model is Robins's FFRCISTG — the Finest Fully Randomized Causally Interpretable Structured Tree Graph (Robins 1986; Richardson and Robins 2014). For the version in which all modeled variables are subject to intervention, an FFRCISTG admits a structural-equation representation with possibly *dependent* equation-specific errors; the NPSEM-IE is a strict submodel obtained by imposing independence of those errors. The added independence is not innocuous: as shown in Chapter 4 and Appendix B, it introduces *cross-world* restrictions that are not empirically testable even under randomized treatment assignment. See Wang et al. (2026) for a careful side-by-side comparison.

These notes adopt the NPSEM-IE as a convenient *sufficient* model: error independence ensures the DAG's Markov condition holds, making the Markov factorization of the joint density immediate, and the counterfactual construction integrates cleanly with the graphical identification theory developed in later chapters. The identification results we derive — the back-door criterion, the do-calculus, the IV assumptions — generally require less than the full collection of NPSEM-IE cross-world restrictions; the error-independence assumption is a modeling choice, not an identification necessity.

Observational regime. Throughout this chapter, U is treated as unobserved. Because U enters both f_T and f_Y, treatment status carries information about U: conditioning on T changes the distribution of U, so the conditional distribution $P(y \mid T{=}t)$ generally differs from the interventional distribution $P(y \mid \mathrm{do}(T{=}t))$. This is *confounding*. If U is omitted from an outcome regression, its contribution enters the regression disturbance, which is then correlated with T — even though the structural disturbance ε itself is independent of T under the NPSEM-IE; this is *endogeneity* in the usual econometric sense. Common-cause confounding thus manifests as endogeneity in the specified outcome equation.

Remark: Equation-Specific Exogeneity

In these notes, exogeneity is defined relative to a particular equation. For an outcome equation with structural disturbance η, conditional mean exogeneity may be expressed as $\mathbb{E}(\eta \mid T, X) = 0$; for a particular linear estimator, a weaker moment restriction such as $\mathbb{E}[T\eta] = 0$ may suffice. Full statistical independence between T and η is stronger than either requirement. Exogeneity does not require the treatment to be independent of every background variable: in the confounded model of Section 1.3, either $\alpha = 0$ or $\gamma = 0$ makes T exogenous for the outcome equation, although only the first makes T independent of U.

Interventional regime. The do-operator replaces the equation for T with a constant t, yielding the *mutilated system*:

$$U \sim p(u), \qquad T = t \text{ (fixed)}, \qquad Y = f_Y(t, U, \varepsilon).$$

Now T and U are independent by construction: the back-door path through U has been severed. The interventional density $P(y \mid \mathrm{do}(T{=}t))$ is therefore the distribution of $f_Y(t, U, \varepsilon)$ when U is drawn from its marginal $p(u)$, not from the conditional $p(u \mid T{=}t)$. We define the *potential outcome* under treatment t as the random variable $Y(t) := f_Y(t, U, \varepsilon)$; by construction, $Y(t)$ has distribution $P(y \mid \mathrm{do}(T{=}t))$.

The surgery just performed relies on a substantive assumption that deserves a name. Only the equation for T was replaced; the function f_Y, the distribution of (U, ε), and every parameter they contain were carried over from the observational world unchanged. This invariance is *autonomy* (Haavelmo 1943): each structural equation describes a mechanism that continues to operate — with the same functional form and the same parameters — regardless of how its inputs come about, whether from the natural data-generating process or from an external intervention. It is precisely this across-world invariance that gives the mutilated system its predictive content — without autonomy, quantities estimated in the observational regime would say nothing about the interventional one. The three frameworks connect interventional quantities to observed outcomes in related but not identical ways: autonomy and modularity specify how mechanisms change under intervention, whereas consistency links observed outcomes to the potential outcomes corresponding to realized treatments. Section 1.2.4 compares these roles directly.

1.2.2 Language 2: Directed Acyclic Graphs

A DAG makes the same surgery graphically visible.

> Definition: Graph Surgery
>
> In the observational DAG, arrows into T encode the mechanisms that determine T (including back-door paths through unobserved variables). Under $\mathrm{do}(T{=}t)$, all arrows into T are *deleted*, producing the mutilated graph $\mathcal{G}_{\overline{T}}$. This removes spurious back-door associations into T while preserving the directed causal pathways from T to its descendants. The effect of T on Y then propagates along whatever directed $T \to \cdots \to Y$ paths remain.

Graph surgery has an exact distributional counterpart. Under the Markov factorization — immediate for an NPSEM-IE — the observational joint distribution factors into one conditional per node, $P(v_1, \dots, v_p) = \prod_j P(v_j \mid \mathrm{Pa}(v_j))$. *Modularity* states that the intervention $\mathrm{do}(T{=}t)$ changes this factorization in exactly one place: the factor for T is removed, and every remaining factor is kept as it is, now evaluated at the fixed value $T{=}t$. Arrow deletion at the level of the graph is factor deletion at the level of the distribution. For the confounded triangle of Equation 1.2, the observational joint is $p(u)\,P(t \mid u)\,P(y \mid t, u)$; removing the treatment factor $P(t \mid u)$ and setting $T{=}t$ in what remains gives

$$P(u, y \mid \mathrm{do}(T{=}t)) = p(u)\,P(y \mid t, u), \qquad \text{hence} \qquad P(y \mid \mathrm{do}(T{=}t)) = \int P(y \mid t, u)\,p(u)\,du.$$

Two features of this calculation deserve emphasis. First, the right-hand side contains only pieces of the observational model — the marginal $p(u)$ and the conditional $P(y \mid t, u)$ — so the interventional distribution is *determined* by the observational joint of (U, T, Y) together with the graph; no new distributions need to be postulated. Second, determined is not the same as estimable: the retained factors involve the unobserved U, so the display cannot be evaluated from data on (T, Y) alone. Whether the same interventional quantity can be re-expressed using observed variables only is the identification problem of Chapter 3; when an *observed* back-door set is available, the display above — with that set in place of U — becomes the adjustment formula. Modularity is thus the DAG's counterpart of the autonomy property of Section 1.2.1; its general form, the *truncated factorization*, is developed in Chapter 3.

1.2.3 Language 3: Potential Outcomes

> Definition: Potential Outcome [@rubin1974estimating]
>
> The potential-outcomes framework takes the collection $\{Y(t) : t \in \mathcal{T}\}$ as *primitive*: $Y(t)$ is the outcome that would have been observed had T been set to t, possibly contrary to fact. Within the SEM Equation 1.2, the same potential outcome is *constructed* as $Y(t) = f_Y(t, U, \varepsilon)$, the solution for Y in the mutilated system.

Definition: SUTVA

The *stable unit treatment value assumption* (SUTVA) has two components: (i) *no interference* — unit i's potential outcome depends only on unit i's own treatment, not on the treatments of other units; and (ii) *no hidden versions of treatment* — each treatment level t corresponds to a single, well-defined intervention. SUTVA makes the scalar notation $Y_i(t)$ adequate; the separate *consistency* assumption $Y_i = Y_i(T_i)$ then connects the observed outcome to the potential outcome at the treatment actually received. Both are developed more fully in Chapter 4.

The potential-outcomes framework has two practical advantages that explain its popularity in the statistics community. First, it is particularly useful for defining causal parameters precisely. The average treatment effect is $\mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$, and more refined estimands are equally direct; for example, the average treatment effect on the treated is $\mathbb{E}[Y(1) - Y(0) \mid T{=}1]$. The ATT conditions on the treatment actually received, so its second term, $\mathbb{E}[Y(0) \mid T{=}1]$, is a genuinely counterfactual quantity: it cannot be written as a do-expression, because the do-operator describes interventional distributions only. Addressing such estimands graphically requires augmenting the DAG with counterfactual variables, as in the single-world intervention graphs of Appendix B. Second, statistical inference is more convenient in the potential-outcomes framework: once an estimand is written as a functional of $(Y(1), Y(0), T, X)$, the machinery of estimation — estimators, standard errors, and the semiparametric efficiency theory of Part III — applies directly. These estimands are developed systematically in Chapter 4.

1.2.4 Two Questions Every Framework Must Answer

The three languages just introduced differ in notation and emphasis, but each must answer the same two questions. First, how is an intervention *represented*, and how is the interventional world connected to the observed data? Autonomy and modularity answer this by specifying how the system changes under intervention: the treatment mechanism is replaced while all remaining mechanisms are preserved. Consistency plays a different role: it links the observed outcome to the potential outcome corresponding to the treatment actually received. Second, what *effect-specific condition* licenses adjustment for observed covariates? The table below, adapted from Table 1 of Wang et al. (2026), summarizes both answers side by side. Here T denotes the treatment, Y the outcome, and X observed pretreatment covariates.

Framework	**Representation of intervention and link to observed outcomes**	**Effect-specific condition for adjustment given X**
Potential outcomes	Define $Y(t)$; *consistency* links potential and observed outcomes through $Y = Y(T)$, assumed alongside SUTVA	*Weak ignorability*: $Y(t) \perp\!\!\!\perp T \mid X$ for each treatment level t
SEM	*Autonomy* (Haavelmo 1943; Pearl 2009): do($T{=}t$) replaces the structural equation for T with the constant t, while every other equation and the distribution of the background disturbances remain unchanged	Conditional on X, the treatment mechanism carries no information about the structural inputs determining $Y(t)$
Causal DAG	*Modularity* (graph surgery): do($T{=}t$) deletes all arrows into T, and the interventional distribution is given by the truncated factorization (Chapter 3)	X blocks every back-door path from T to Y and contains no descendant of T (Chapter 3)

In all three representations, an additional support condition — *positivity* — is required for nonparametric adjustment. For discrete treatment, $P(T{=}t \mid X{=}x) > 0$ for each relevant pair (t, x); for continuous treatment, t must lie in the conditional support of $T \mid X{=}x$, with positive conditional density in the relevant neighborhood (Chapter 4).

A note on terminology: the word *modularity* is used in the literature for both the SEM and the DAG rows of the table. In the SEM literature it is a synonym for autonomy — the mechanism generating T can be replaced without altering the other equations — while in the graphical literature it refers to the arrow-deletion semantics of the do-operator. The two usages express the same idea in different languages, but readers should be prepared to encounter both.

Remark: Scope of the Assumptions

The assumptions used in the three languages differ in scope. Weak ignorability and the back-door condition are *effect-specific*: they concern a particular treatment–outcome pair and an adjustment set. The NPSEM-IE is *system-wide*: it jointly constructs counterfactual variables for interventions throughout the model and imposes restrictions across intervention worlds. *Causal sufficiency* is *graph-wide*: it rules out latent common causes among the variables represented in the graph. None of these system-wide properties by itself implies that every causal effect is identified from the observed-data distribution; identification remains a separate question, taken up in Section 1.4. See Wang et al. (2026) for an extended discussion.

1.2.5 Roadmap: How the Trinity Organizes These Notes

The three Parts of the course each draw on the three languages introduced above, with different languages moving into the foreground as the subject matter demands. Understanding this organization in advance will help you see why each language is introduced when it is.

Part I — Foundations: the graphical language (Chapters 1–3). Part I builds the graphical foundation. The DAG and the do-operator are the primary language here because they make the interventional/observational distinction *syntactically explicit*: confounding is visible as an open back-door path, and identifying assumptions are visible as graph criteria (back-door, front-door, do-calculus). A mistake that is easy to commit silently in potential-outcomes notation becomes a type error in do-notation, and is therefore caught before it propagates. The SEM appears as the generative backbone wherever closed-form calculations are needed, particularly in the Gaussian confounded running example.

Part II — Identification: designs and research strategies (Chapters 4–9). Part II introduces the potential outcomes framework (Chapter 4) and then studies the specific research designs through which observational and experimental data support identification: randomization and back-door adjustment (Chapter 5), propensity score methods (Chapter 6), instrumental variables (Chapter 7), mediation and front-door identification (Chapter 8), and sensitivity analysis when identifying assumptions are only partially credible (Chapter 9). This is where the potential-outcomes and graphical languages are most explicitly translated into each other: ignorability ($Y(t) \perp\!\!\!\perp T \mid X$) is the potential-outcomes counterpart of the back-door criterion, and each design is stated both graphically and in potential-outcomes form before the resulting identification formula is derived.

Part III — Estimation: semiparametric and machine-learning methods (Chapters 10–13). Once identification has been established in the language of either DAGs or potential outcomes, Part III turns to estimation. The semiparametric efficiency framework — estimating equations, influence functions, doubly robust estimators, and double machine learning — is largely language-neutral: it works with whatever moment condition the identification step produces. Chapter 13 applies these tools to estimation under instrumental variables. Fluency in all three languages is expected by this point, as Part III draws freely on all of them.

1.2.6 Historical Roots of the Causal Trinity (Optional)

The three languages did not emerge from a single research program. They grew independently in different disciplines over more than a century, driven by different scientific questions and different communities. Understanding this history explains why the three schools sometimes use incompatible terminology for the same idea, and why unifying them is intellectually non-trivial.

Structural Equation Models: genetics and econometrics. The oldest strand begins with the biologist Sewall Wright (1921), who introduced *path analysis* to decompose correlations among hereditary traits into direct and indirect causal contributions — arguably the first use of a directed graph for causal reasoning. The framework was transplanted into economics by Haavelmo (1943), who argued that economic relationships should be modeled as autonomous structural equations robust to policy interventions — an early statement of what Pearl would later call "graph surgery". The Cowles Commission (Koopmans, Klein, and others) formalized simultaneous equation systems throughout the 1940s–50s, and this SEM tradition became the standard language of causal inference in econometrics. Lucas's (1976) critique — that reduced-form equations break down when policy changes the data-generating process — is essentially a statement of the interventional/observational distinction, decades before that phrase existed. SEMs were independently adopted by sociologists and psychologists (LISREL, factor analysis), making this the most interdisciplinary of the three schools.

Potential Outcomes: experimental statistics and epidemiology. Neyman (1923) introduced the notation $Y_i(t)$ in the context of randomized agricultural experiments, asking what a plot's yield *would have been* under an unobserved treatment. Fisher (1935) placed randomization at the foundation of causal inference via the randomization distribution. The crucial extension to *observational* studies was made by Rubin (1974), who formalized the assignment mechanism; the stable unit treatment value assumption was subsequently labeled SUTVA in Rubin (1980). Holland's (1986) *JASA* paper "Statistics and Causal Inference" brought the framework to mainstream statistics under the dictum "no causation without manipulation". Robins (1986) extended potential outcomes to longitudinal treatments and time-varying confounding via g-computation. Imbens and Angrist (1994) connected the framework to instrumental variables and defined the LATE, bringing potential outcomes into empirical economics. Today the framework is dominant in statistics, epidemiology, and much of economics.

DAGs and do-calculus: computer science and philosophy. Pearl developed Bayesian networks through the 1980s as a representation for probabilistic reasoning in artificial intelligence. The pivotal step came in Pearl (1995), where the *do-operator* was introduced: a formal syntax for distinguishing interventional from observational distributions within a graph-based algebra. The 2000 monograph *Causality* synthesized DAGs, the do-calculus, identification theory, and the connections to potential outcomes into a unified framework. In parallel, Spirtes, Glymour, and Scheines (1993) — working from *philosophy* at Carnegie Mellon — developed constraint-based algorithms (PC, FCI) for learning causal structure from data. The DAG framework was rapidly adopted by epidemiology (Hernán and Robins) and is now the primary language of causal inference in computer science and AI.

Convergence. The three frameworks are closely related and often intertranslatable, but exact equivalence requires additional assumptions about the causal model; see Wang et al. (2026) for a recent side-by-side comparison. In these notes we work mainly in an NPSEM-IE setting, where translation between the languages is especially clean. Under weaker single-world models — for example, Robins's FFRCISTG (Robins 1986) — some nested cross-world counterfactuals represented by an NPSEM-IE are not encoded without additional cross-world structure, so causal interpretations or identification results involving those quantities require assumptions beyond the single-world model. The historical differences between the three schools are primarily in notation, native strengths, and disciplinary culture, with substantive differences appearing mainly in how much cross-world structure each framework commits to.

1.3 The Gaussian Linear Confounded Model

Section 1.2 introduced three complementary ways to represent an intervention. We now make the distinction between observation and intervention quantitatively explicit. In a confounded structural model, observing $T{=}t$ selects units according to a treatment-dependent distribution of the latent common cause U, whereas intervening to set $T{=}t$ leaves U distributed according to its population law. In the linear model below, the coefficient of t in the observational conditional mean decomposes into the causal slope β plus an omitted-variable confounding term — *association* = *causation* + *confounding*, in the slogan of Wang et al. (2026) — while the full observational and interventional distributions additionally differ in variance. General conditions under which interventional distributions can be identified from observational data are deferred to Chapters 3–5.

1.3.1 Setup

The Gaussian linear confounded model is our primary running example throughout this course. It is simple enough to permit closed-form calculations yet rich enough to illustrate every key concept. We use it here not to develop any particular identification strategy, but to make the gap between observational and interventional reasoning mathematically explicit; various identification strategies for this model appear in later chapters.

> Example: Gaussian Linear Confounded Model
>
> The structural equations are
>
> $$Y = \beta T + \gamma U + \varepsilon, \qquad T = \alpha U + \delta, \tag{1.3}$$
>
> where U is an unobserved confounder and the three primitive variables are mutually independent:
>
> $$U \sim \mathcal{N}(0, \sigma_U^2), \qquad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2), \qquad \delta \sim \mathcal{N}(0, \sigma_\delta^2).$$
>
> Here β is the causal effect of T on Y, γ is the direct effect of the confounder U on Y, and α is the direct effect of U on T. The joint independence of (U, ε, δ) is the NPSEM-IE condition.

The endogeneity of T is immediate. From the analyst's perspective, the composite error in the Y equation is $\gamma U + \varepsilon$, so:

$$\mathrm{Cov}(T,\ \gamma U + \varepsilon) = \gamma\,\mathrm{Cov}(\alpha U + \delta,\ U) = \gamma\alpha\sigma_U^2 \neq 0 \quad (\text{when } \alpha \neq 0 \text{ and } \gamma \neq 0).$$

Consequently, OLS regressing Y on T does not estimate β.

1.3.2 Two Explicit Densities

We now derive both densities in closed form. Write $\sigma_T^2 = \alpha^2\sigma_U^2 + \sigma_\delta^2$ for the marginal variance of T.

Interventional density. Set $T = t$ externally. The back-door path through U is severed, so U is independent of T under the intervention and is drawn from its marginal $\mathcal{N}(0, \sigma_U^2)$. Then $Y \mid \mathrm{do}(T{=}t) = \beta t + \gamma U + \varepsilon$ with $U \perp\!\!\!\perp \varepsilon$. Hence

$$Y \mid \mathrm{do}(T{=}t) \sim \mathcal{N}\big(\beta t,\ \ \gamma^2\sigma_U^2 + \sigma_\varepsilon^2\big)\,. \tag{1.4}$$

Observational density. Observe $T = t$. Since Y and T are both linear combinations of the independent normals (U, ε, δ), the pair (Y, T) is jointly normal. Their covariance is

$$\mathrm{Cov}(Y, T) = \mathrm{Cov}(\beta T + \gamma U + \varepsilon,\ T) = \beta\sigma_T^2 + \gamma\alpha\sigma_U^2.$$

Applying the conditional normal formula:

$$Y \mid T{=}t \sim \mathcal{N}\left(\left(\beta + \frac{\gamma\alpha\sigma_U^2}{\sigma_T^2}\right)t,\ \ \frac{\gamma^2\sigma_U^2\sigma_\delta^2}{\sigma_T^2} + \sigma_\varepsilon^2\right). \tag{1.5}$$

1.3.3 The Endogeneity Gap

Equations Equation 1.4 and Equation 1.5 differ in both mean and variance:

Table 1.1: Equal when $\alpha = 0$ or $\gamma = 0$ (no confounding).

	Interventional	**Observational**
Mean	βt	$\left(\beta + \frac{\gamma\alpha\sigma_U^2}{\sigma_T^2}\right)t$
Variance	$\gamma^2\sigma_U^2 + \sigma_\varepsilon^2$	$\frac{\gamma^2\sigma_U^2\sigma_\delta^2}{\sigma_T^2} + \sigma_\varepsilon^2$
Distribution of U	$p(u)$	$p(u \mid T{=}t)$
Identified by	RCT; the slope β also by IV (Chapter 7)	OLS

In this linear model, the coefficient of t in the observational mean Equation 1.5 is also the probability limit of the OLS slope from regressing Y on T, since both equal $\mathrm{Cov}(Y, T)/\mathrm{Var}(T)$. The comparison of the two means is therefore exactly the decomposition promised by the slogan in this section's opening:

$$\underbrace{\beta + \frac{\gamma\alpha\sigma_U^2}{\sigma_T^2}}_{\text{association: observational slope}} = \underbrace{\beta}_{\text{causation}} + \underbrace{\gamma \cdot \frac{\alpha\sigma_U^2}{\sigma_T^2}}_{\text{confounding: omitted-variable bias}}.$$

The confounding term is the classical *omitted-variable bias* (developed in Section 1.3.4 below) — the endogeneity bias of the OLS coefficient anticipated in the definition above — and the example shows exactly why: the omitted U affects the outcome with coefficient γ, and its regression on the included regressor has slope $\mathbb{E}[U \mid T{=}t] = (\alpha\sigma_U^2/\sigma_T^2)\,t$ by the conditional normal formula. The bias is the product of the two — the effect of the omitted variable on Y times the regression of the omitted variable on T — and it vanishes when either factor does ($\gamma = 0$ or $\alpha = 0$). It takes the sign of $\gamma\alpha$ (positive here). Numerically, with $\alpha = \gamma = 1$ and $\sigma_U^2 = \sigma_\delta^2 = 1$:

$$\text{bias} = \frac{1 \times 1 \times 1}{1^2 \times 1 + 1} = \frac{1}{2} = 0.5.$$

1.3.4 Omitted-Variable Bias

The decomposition just derived is an instance of a completely general regression identity that involves no causal assumptions — only second moments. Consider three population least-squares regressions of centered random variables, in notation mirroring the running model: the *full model*, regressing Y on (T, X) with coefficients (β, γ); the *reduced model*, regressing Y on T alone with coefficient $\tilde\beta$; and the *auxiliary model*, regressing the omitted variable X on T with coefficient $\tilde\alpha = \operatorname{Cov}(X,T)/\operatorname{Var}(T)$. (Tilded coefficients come from regressions on T alone.)

Theorem 1.1 (Omitted-Variable Bias)**.** *Let (T, X, Y) be centered random variables with finite second moments, $\operatorname{Var}(T) > 0$, and (T, X) not perfectly collinear. Then*

$$\tilde\beta = \beta + \gamma\,\tilde\alpha, \qquad \textit{i.e.,} \qquad \underbrace{\tilde\beta - \beta}_{\textit{omitted-variable bias}} = \gamma\,\tilde\alpha.$$

The bias in the reduced-model coefficient is the product of the omitted variable's coefficient in the full model and the coefficient from regressing the omitted variable on the included regressor.

Proof. By the normal equations of the full regression, $Y = \beta T + \gamma X + e$ with $\mathbb{E}[eT] = \mathbb{E}[eX] = 0$. Hence

$$\tilde\beta = \frac{\operatorname{Cov}(Y,T)}{\operatorname{Var}(T)} = \beta + \gamma\,\frac{\operatorname{Cov}(X,T)}{\operatorname{Var}(T)} = \beta + \gamma\,\tilde\alpha. \qquad \square$$

The identity can equivalently be routed through the *treatment model*, the regression of T on X with coefficient $\alpha = \operatorname{Cov}(T,X)/\operatorname{Var}(X)$: since $\operatorname{Cov}(X,T) = \alpha\operatorname{Var}(X)$, the auxiliary coefficient is $\tilde\alpha = \alpha\operatorname{Var}(X)/\operatorname{Var}(T)$ and the bias is $\gamma\,\alpha\operatorname{Var}(X)/\operatorname{Var}(T)$.

Geometric interpretation. Population least squares is orthogonal projection in the Hilbert space of centered, square-integrable random variables with inner product $\langle A, B\rangle = \mathbb{E}[AB]$. The full model is the projection of Y onto the plane $\operatorname{span}\{T, X\}$: the normal equations say exactly that $Y = \beta T + \gamma X + e$ with $e \perp \operatorname{span}\{T, X\}$. The reduced coefficient is the projection onto the line $\operatorname{span}\{T\}$, $\tilde\beta\,T = \Pi_T Y$. Because the line lies inside the plane, we may project in two stages:

$$\Pi_T Y = \Pi_T(\beta T + \gamma X + e) = \beta T + \gamma\,\Pi_T X + 0 = (\beta + \gamma\tilde\alpha)\,T,$$

since $\Pi_T e = 0$ (e is orthogonal to the whole plane, hence to the line) and $\Pi_T X = \tilde\alpha\,T$ — which is precisely the auxiliary regression. Theorem 1.1 is thus the tower property of nested projections, and the omitted-variable bias is the *shadow* that the omitted component γX casts on the line spanned by the included regressor. The bias vanishes precisely when $\gamma X \perp T$: either there is no omitted component ($\gamma = 0$) or it is orthogonal to the included regressor ($\tilde\alpha = 0$, equivalently $\alpha = 0$). A complementary fact, the Frisch–Waugh–Lovell theorem, reads off the *full*-model coefficient from the same picture: β is the coefficient of Y on the component of T orthogonal to X. The same projection geometry, in infinite-dimensional form, underlies the semiparametric efficiency theory of Part III and Appendix C.

Confirmation in the Gaussian model. Take $X = U$ in the running example Equation 1.3. Because ε is independent of (T, U), the structural coefficients β and γ are also the full-model regression coefficients. The auxiliary coefficient is $\tilde{\alpha} = \mathrm{Cov}(U, T)/\mathrm{Var}(T) = \alpha\sigma_U^2/\sigma_T^2$ — the slope of $\mathbb{E}[U \mid T{=}t]$ computed in the preceding subsection — where α enters because the treatment equation $T = \alpha U + \delta$ is itself the population regression of T on U. Theorem 1.1 then gives

$$\tilde{\beta} - \beta = \gamma\,\tilde{\alpha} = \gamma\,\alpha\,\frac{\sigma_U^2}{\sigma_T^2},$$

exactly the confounding term of the decomposition display above. One caution: the theorem is assumption-free regression algebra, so it attaches no causal meaning to β by itself. In this example the full-model coefficient *is* the causal slope because the structural model is linear and U closes the only back-door path; in general, whether a full-model coefficient has a causal interpretation is precisely the identification question of Section 1.4.

1.3.5 Lab: Simulating the Two Densities

The analytical results of this section were illustrated numerically by simulation using $n = 10{,}000$ draws from the structural equations with parameters $\beta = 2$, $\alpha = 1$, $\gamma = 1$, $\sigma_U = \sigma_\varepsilon = \sigma_\delta = 1$. The R code is distributed separately as `chapter1_lab.R`.

Experiment 1: OLS bias. OLS was applied to the observational pairs (Y_i, T_i). By the omitted-variable bias formula, the OLS probability limit is $\beta + \gamma\alpha\sigma_U^2/\sigma_T^2 = 2 + 0.5 = 2.5$. The results are consistent with this:

	Simulated	**Theory**	**Formula**
OLS slope	2.5147	2.5000	$\beta + \gamma\alpha\sigma_U^2/\sigma_T^2$
True β	—	2.0000	β
Bias	0.5147	0.5000	$\gamma\alpha\sigma_U^2/\sigma_T^2$

OLS overestimates the causal effect by 25%.

Experiment 2: Two-sample comparison at $t_0 = 1$. Since (Y, T) is jointly normal, the exact observational conditional distribution $Y \mid T{=}1$ is available in closed form from Equation 1.5: it is $\mathcal{N}(2.5,\ 1.5)$. The observational sample was therefore drawn directly from this distribution; no bandwidth or neighborhood selection was required. The interventional sample was generated by setting $T = 1$ externally and drawing fresh $U_i^* \sim \mathcal{N}(0, 1)$ and $\varepsilon_i^* \sim \mathcal{N}(0, 1)$ independently, giving $Y_i^* = \beta + \gamma U_i^* + \varepsilon_i^* \sim \mathcal{N}(2,\ 2)$.

	Mean (simulated)	**Mean (theory)**	**SD (simulated)**	**SD (theory)**
$Y \mid T{=}1$	2.497	2.500	1.229	$\sqrt{1.5} = 1.225$
$Y \mid \mathrm{do}(T{=}1)$	2.001	2.000	1.428	$\sqrt{2} = 1.414$

The variance reduction in the observational distribution reflects the ratio $\sigma^2_{\text{obs}}/\sigma^2_{\text{int}} = 1.5/2 = 0.75$: conditioning on $T{=}1$ pins down the value of $\alpha U + \delta$, which through the correlation with γU restricts the spread of Y.

Experiment 3: Density overlay for varying α. The figure below shows the interventional density $\mathcal{N}(\beta t_0,\ \gamma^2\sigma_U^2 + \sigma_\varepsilon^2)$ (solid) and the observational density Equation 1.5 (broken line styles), both evaluated at $t_0 = 1$, for $\alpha \in \{0, 0.3, 0.7, 1.0\}$. The interventional density does not depend on α and remains fixed; as α increases over the plotted range, the observational curve shifts rightward (growing bias) and narrows (reduced variance). The bias is not globally monotone in α, however: differentiating $b(\alpha) = \gamma\alpha\sigma_U^2/(\alpha^2\sigma_U^2 + \sigma_\delta^2)$ shows that it peaks at $\alpha = \sigma_\delta/\sigma_U$ — exactly $\alpha = 1$ under the figure's parameters — and vanishes as $\alpha \to \infty$, because increasing α inflates $\mathrm{Var}(T)$ faster than $\mathrm{Cov}(T, U)$.

1.4 The Two-Step Paradigm

Causal inference is a two-step discipline. The two steps are logically distinct and require different tools.

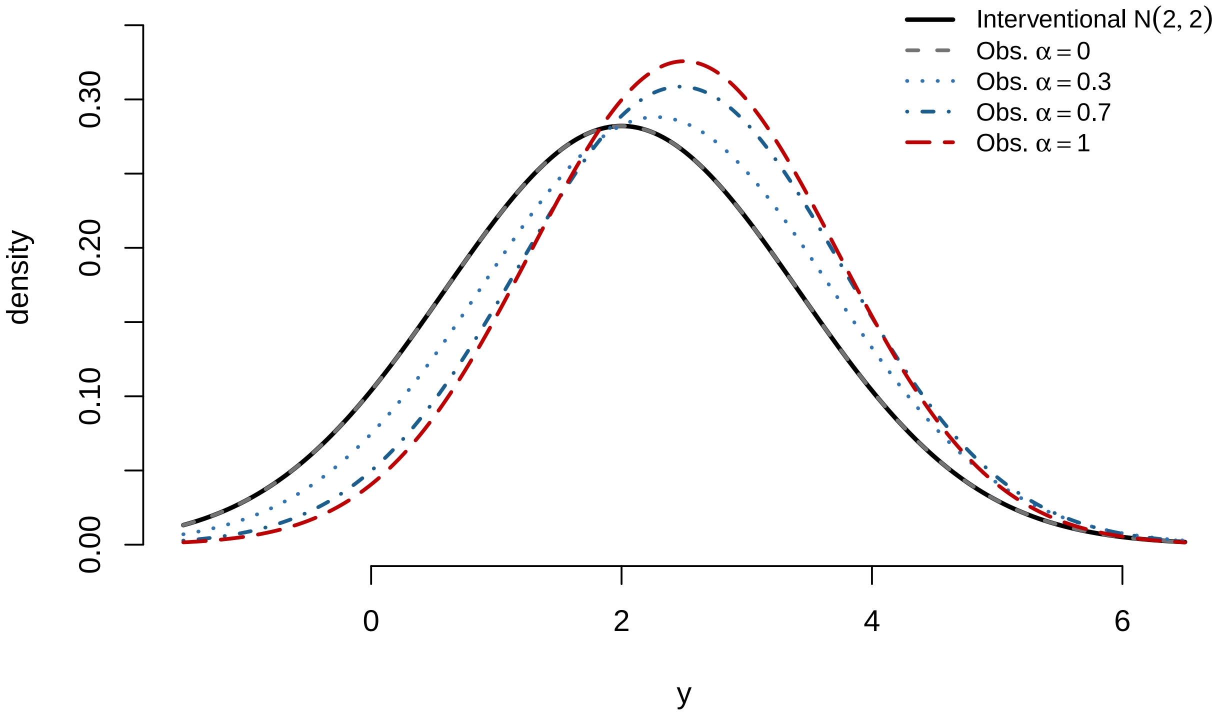

Figure 1.1: Interventional density $N(2,2)$ (solid black) versus observational density for $\alpha \in \{0, 0.3, 0.7, 1.0\}$ (broken line styles), evaluated at $t_0 = 1$. Parameters: $\beta = 2$, $\gamma = 1$, $\sigma_U = \sigma_\varepsilon = \sigma_\delta = 1$. At $\alpha = 0$ there is no confounding and the two densities coincide.

1.4.1 Step 1: Identification

Definition: Identification

A causal quantity $P(y \mid \mathrm{do}(T{=}t))$ is *identified* from the observational distribution P if there exists a functional Ψ, valid across the assumed model class, such that

$$P(y \mid \mathrm{do}(T{=}t)) = \Psi(P),$$

and $\Psi(P)$ depends only on the observable joint distribution. Equivalently, relative to a class $\mathcal{M}$ of causal models, a causal quantity Q is identified if any two models in $\mathcal{M}$ that induce the same observed-data distribution also agree on Q:

$$P^{\mathrm{obs}}_{M_1} = P^{\mathrm{obs}}_{M_2} \implies Q(M_1) = Q(M_2) \qquad \text{for all } M_1, M_2 \in \mathcal{M}.$$

The second formulation makes explicit what *failure* of identification means: two causal models that are indistinguishable from observational data yet disagree on the causal quantity. Chapter 3 exhibits exactly such a construction.

Example: Identification and Its Failure in the Gaussian Model

Return to the Gaussian confounded model Equation 1.3 and suppose only (T, Y) is observed. Consider two parameter settings:

- M_1 (all causal): $\beta = 1$, $\alpha = \gamma = 0$, $\sigma_\delta^2 = 1$, $\sigma_\varepsilon^2 = 2$;
- M_2 (all confounding): $\beta = 0$, $\alpha = 1$, $\gamma = 2$, $\sigma_U^2 = \sigma_\delta^2 = \frac{1}{2}$, $\sigma_\varepsilon^2 = 1$.

Both settings induce the *same* observed law: (T, Y) is bivariate normal with mean zero, $\mathrm{Var}(T) = 1$, $\mathrm{Var}(Y) = 3$, and $\mathrm{Cov}(T, Y) = 1$. In particular, the observed slope $\mathrm{Cov}(T, Y)/\mathrm{Var}(T) = 1$ in both — but it is all causation under M_1 and all confounding under M_2: the two terms of the slogan cannot be separated from the observed law alone. The interventional distributions differ accordingly: $Y \mid \mathrm{do}(T{=}t) \sim \mathcal{N}(t, 2)$ under M_1, but $\mathcal{N}(0, 3)$ under M_2. No functional Ψ of the law of

(T, Y) can return both answers, so $P(y \mid \mathrm{do}(T{=}t))$ is *not identified* from the distribution of (T, Y) alone. If instead U is also observed, the two models induce different laws for (U, T, Y), and the functional $\Psi(P) = \int P(y \mid t, u)\, p(u)\, du$ — the modularity calculation of Section 1.2.2 — identifies the interventional distribution. Observing the confounder restores identification.

Identification is a population-level question, posed relative to a causal model class, the observed-data law, and the support of that law: given the assumed graph and assumptions, can the interventional density be expressed as a functional of the observed-data distribution? It does not depend on sample size.

The main tools are the back-door criterion (introduced in Chapter 3, applied in Chapters 4–6), the front-door criterion (Chapters 3 and 8), the three rules of the do-calculus (Chapter 3), and IV assumptions (Chapter 7).

1.4.2 Step 2: Estimation

Once $\Psi(P)$ has been identified, the statistical problem is to estimate the functional $\Psi(P)$ from n observations $O_1, \dots, O_n$ as efficiently as possible. The main tools are outcome regression and standardization (the g-formula, Chapter 5), efficient influence functions (EIF, Part III), inverse probability weighting (IPW, Chapter 6 and Part III), doubly robust estimators (Part III), and double machine learning (DML, Part III).

1.4.3 A Preview: Identification and Estimation in the Flu-Vaccination Example

The following example illustrates the two steps with a binary outcome and a fully observed confounder, so no advanced machinery is required.

Scenario. A public health agency observes 1,000 individuals over one flu season. Each person either received a flu vaccine ($T = 1$) or did not ($T = 0$), and either became infected ($Y = 1$) or did not ($Y = 0$). Age group X (elderly: $X{=}1$; young: $X{=}0$) is recorded for all individuals. Elderly people are both more likely to be vaccinated (due to public health campaigns) and more susceptible to infection (due to weaker immune systems), so X confounds the relationship between T and Y.

Observed data. The 1,000 individuals break down as follows (with within-cell infection rates in parentheses):

	Vaccinated ($T = 1$): n	**Infected**	**Unvaccinated ($T = 0$): n**	**Infected**	**Total**
Elderly ($X{=}1$)	360	108 (30%)	40	20 (50%)	400
Young ($X{=}0$)	60	3 (5%)	540	54 (10%)	600
Total	420	111 (26.4%)	580	74 (12.8%)	1,000

Simpson's paradox at a glance. The infection-rate comparisons between vaccinated and unvaccinated — within each stratum and in aggregate — tell contradictory stories:

Stratum	**Vaccinated**	**Unvaccinated**	**Difference**	**Conclusion**
Elderly ($X{=}1$)	30%	50%	−20 pp	vaccine helps
Young ($X{=}0$)	5%	10%	−5 pp	vaccine helps
Aggregate (raw)	**26.4%**	**12.8%**	**+13.6 pp**	**vaccine harms??**

Within every stratum the vaccine reduces infection. Yet in the aggregate the vaccinated group has a *higher* infection rate. The reversal occurs because the vaccinated group is disproportionately elderly (86% of vaccinated vs. 7% of unvaccinated): a high-risk group is systematically over-represented among the treated, making the raw comparison misleading. The back-door adjustment formula, developed in Chapter 3, corrects this by standardizing to the *population* age distribution rather than the treatment-conditional one.

We use the back-door formula here as a preview: Chapters 3–5 derive its graphical and potential-outcomes justifications and develop the corresponding estimators systematically.

Step 1: Identification. The only back-door path is $T \leftarrow X \rightarrow Y$. The set $\{X\}$ satisfies the back-door criterion: it blocks this path (fork at X) and contains no descendant of T. We also assume positivity, $P(T{=}t \mid X{=}x) > 0$ for all four (t, x) pairs — an assumption about the population with which the table is consistent, since every treatment-by-stratum cell is occupied. By the back-door formula:

$$P(Y{=}1 \mid \mathrm{do}(T{=}t)) = \sum_{x \in \{0,1\}} P(Y{=}1 \mid T{=}t, X{=}x)\, P(X{=}x). \tag{1.6}$$

The causal quantity on the left has been expressed entirely as a functional of the observed-data distribution on the right.

Step 2: Estimation. Plugging sample proportions into Equation 1.6, with $\hat{P}(X{=}1) = 0.4$ and $\hat{P}(X{=}0) = 0.6$:

$$\hat{P}(Y{=}1 \mid \mathrm{do}(T{=}1)) = 0.30 \times 0.4 + 0.05 \times 0.6 = 0.15,$$

$$\hat{P}(Y{=}1 \mid \mathrm{do}(T{=}0)) = 0.50 \times 0.4 + 0.10 \times 0.6 = 0.26.$$

The estimated causal risk difference is $\hat{\Delta}_{\text{causal}} = 0.15 - 0.26 = -0.11$: under the assumed causal DAG, consistency, and positivity, the standardized estimate indicates an 11-percentage-point reduction in infection risk. This is a plug-in point estimate; quantifying its uncertainty is deferred to later chapters.

Why the naive comparison fails. The raw difference $0.264 - 0.128 = +0.136$ is precisely the aggregate row of the Simpson table above: it mixes strata in treatment-dependent proportions, importing the high elderly infection rate into the vaccinated group. Under the assumed causal model, the standardized estimate -0.11 is the correct target because it re-weights each stratum by $P(X{=}x)$, the population share, not by $P(X{=}x \mid T{=}t)$.

> The Two Steps, Concretely
>
> **Step 1 (identification)** established — from the assumed causal graph and its intervention semantics — that the back-door formula Equation 1.6 expresses the causal quantity as a functional of observable probabilities. **Step 2 (estimation)** replaced the population probabilities in that formula with sample proportions.
> These two steps are logically separate: Step 1 is a mathematical argument about the causal model that does not depend on sample size; Step 2 is a statistical problem conditional on the identification formula obtained in Step 1. A researcher who skips Step 1 and reports the raw difference conflates the two and draws the wrong conclusion.

1.5 Summary

1. **Interventional vs. observational distribution.** Causal inference answers questions about interventions $P(y \mid \mathrm{do}(T{=}t))$, not about observations $P(y \mid T{=}t)$. In the Gaussian confounded model, the two densities coincide when $\alpha\gamma = 0$. Only $\alpha = 0$ corresponds to exogeneity in the sense $T \perp\!\!\!\perp U$; $\gamma = 0$ instead means that U no longer affects Y. In either case, T is exogenous *for the outcome equation* and OLS recovers β. When both channels are active ($\alpha \neq 0$ and $\gamma \neq 0$), the two distributions differ; the gap in mean slopes is the endogeneity bias of the OLS coefficient.

2. **The causal trinity.** The same causal question can be expressed in three languages — SEM (equation surgery), DAG (graph surgery), and potential outcomes ($Y(t)$). In the SEM framework, $Y(t)$ is constructed as the solution for Y in the mutilated system, so $Y(t)$ has distribution $P(y \mid \mathrm{do}(T{=}t))$ by construction. Fluency in all three languages is essential for modern causal reasoning.

3. **Endogeneity bias in the Gaussian confounded model.** In the Gaussian linear confounded model, the coefficient of T in the observational mean $\mathbb{E}[Y \mid T{=}t]$ differs from the causal coefficient β by $\gamma\alpha\sigma_U^2/\sigma_T^2$, where $\sigma_T^2 = \alpha^2\sigma_U^2 + \sigma_\delta^2$. This is the omitted-variable bias: U drives both T (via α) and Y (via γ), and its omission from the regression biases the OLS estimand. The observational variance $\gamma^2\sigma_U^2\sigma_\delta^2/\sigma_T^2 + \sigma_\varepsilon^2$ is smaller than the interventional variance $\gamma^2\sigma_U^2 + \sigma_\varepsilon^2$.

4. **Two-step paradigm.** Causal inference is a two-step discipline: *identification* (expressing $\Psi(P)$ as a functional of observable data, a mathematical question about the causal model) followed by *estimation* (constructing $\widehat{\Psi}_n$ efficiently from n observations, a statistical question about finite samples). The two steps are analytically separate, with Step 2 carried out conditional on the identification formula obtained in Step 1.

Causal inference is the study of when, and under what assumptions, the observational distribution contains enough information to determine an interventional distribution or causal estimand. Everything in this course — the back-door criterion, do-calculus, instrumental variables, sensitivity analysis, semiparametric estimation — is an answer to that question in a particular setting.

1.6 Problems

1. Do-notation fundamentals. For each statement below, decide whether it refers to an interventional or an observational distribution, and rewrite it unambiguously using do-notation where appropriate.

(a) "The probability that a patient recovers given that they took the drug."
(b) "The probability that a patient would recover if we prescribed the drug to everyone."
(c) "Among students who attended tutoring, the average exam score was 85."
(d) "If we enrolled all students in tutoring, the average exam score would be 85."

2. Deriving the two Gaussian laws. In the Gaussian confounded model Equation 1.3, verify Equation 1.5 and Equation 1.4 by carrying out the following steps. Let $\sigma_T^2 = \alpha^2\sigma_U^2 + \sigma_\delta^2$.

(a) Show that (Y, T) is jointly normal by writing both as linear combinations of the independent normals (U, ε, δ).
(b) Compute $\mathrm{Cov}(Y, T)$ and $\mathrm{Var}(T)$.
(c) Apply the conditional normal formula to derive $\mathbb{E}[Y \mid T{=}t]$ and $\mathrm{Var}[Y \mid T{=}t]$, and hence verify Equation 1.5.
(d) Show that the OLS probability limit is $\beta + \gamma\alpha\sigma_U^2/\sigma_T^2$, and interpret this as an omitted-variable bias formula.
(e) Derive the distribution of $Y \mid \mathrm{do}(T{=}t)$ directly from the mutilated structural equation, and compare its mean and variance with those obtained in part (c).

3. The causal trinity. Consider the following verbal causal claim: "Aspirin (T) reduces the risk of heart attack (Y) because it inhibits platelet aggregation (M), but patients with pre-existing cardiovascular disease (U, unobserved) are both more likely to take aspirin and more likely to have a heart attack." Assume that all effects of aspirin on heart-attack risk operate through platelet aggregation, so there is no direct $T \to Y$ edge.

(a) Draw the causal DAG implied by this description.
(b) Write down the recursive SEM with appropriate structural functions f_T, f_M, f_Y.
(c) State the SUTVA assumption and discuss whether it is plausible in this setting.
(d) Explain why $\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0]$ does not identify the causal effect $\mathbb{E}[Y \mid \mathrm{do}(T{=}1)] - \mathbb{E}[Y \mid \mathrm{do}(T{=}0)]$ in this DAG.

4. Equation-specific exogeneity. In the Gaussian confounded model Equation 1.3, write the outcome equation as $Y = \beta T + \eta$ with structural disturbance $\eta = \gamma U + \varepsilon$. Verify that $\mathrm{Cov}(T, \eta) = \alpha\gamma\sigma_U^2$, and classify T as exogenous or endogenous for the outcome equation under each of the following cases:

(i) $\alpha \neq 0$ and $\gamma \neq 0$;
(ii) $\alpha = 0$ and $\gamma \neq 0$;
(iii) $\alpha \neq 0$ and $\gamma = 0$.

In which cases does OLS recover β? In which cases is T independent of U?

5. Omitted-variable bias. In the setting of Theorem 1.1, with full model $Y = \beta T + \gamma X + e$, reduced-model coefficient $\tilde{\beta}$, and auxiliary coefficient $\tilde{\alpha} = \mathrm{Cov}(X, T)/\mathrm{Var}(T)$:

(a) Show that the bias can equivalently be written in terms of the *treatment model* $T = \alpha X + v$, with $\alpha = \mathrm{Cov}(T, X)/\mathrm{Var}(X)$, as $\tilde{\beta} - \beta = \gamma\alpha\,\mathrm{Var}(X)/\mathrm{Var}(T)$, and explain why a variance ratio appears in this form but not in Theorem 1.1.

(b) A researcher regresses log wage (Y) on years of schooling (T), omitting ability (X). Suppose ability raises both wages ($\gamma > 0$) and schooling ($\alpha > 0$). Determine the sign of the bias in the schooling coefficient and interpret the result.

(c) Suppose instead that the omitted X is uncorrelated with T ($\alpha = 0$) but still affects Y ($\gamma \neq 0$). Show that $\tilde{\beta} = \beta$, and show that the reduced model's residual variance exceeds the full model's by $\gamma^2 \operatorname{Var}(X)$. What does omitting X cost, if not bias?

Chapter 2

DAGs and d-Separation

Learning Objectives

By the end of this chapter, students should be able to:

1. Construct a DAG from a verbal description of a causal model and identify parents, descendants, ancestors, and non-descendants of any node.
2. Classify every intermediate node on a path as a chain, fork, or collider, and apply the blocking rule for each.
3. Apply the d-separation criterion to determine whether a proposed conditional independence is implied by a DAG.
4. State the Markov factorization and use it to write the joint density of a DAG in terms of local conditional densities.
5. Translate d-separation statements into conditional-independence implications under the Markov property, distinguish these implications from faithfulness assumptions, and explain why causal interpretation requires more than the observational conditional-independence structure.
6. Recognize and avoid collider bias, including the case where a descendant of a collider is conditioned upon.

Readers who want a gentler introduction to conditional independence, the three basic path motifs, and the Markov-property interpretation may consult Appendix A before or alongside this chapter.

2.1 Directed Acyclic Graphs

DAGs allow us to translate qualitative causal assumptions into quantitative statistical restrictions. This single idea underlies everything in this chapter: once we draw a graph encoding our causal assumptions, d-separation tells us which conditional independences are implied by the graph and helps identify candidate adjustment sets for removing confounding.

Remark

In this chapter, the word *graph* does not mean a plot of data or a graph of a function. It means a collection of nodes and edges used to represent relationships among variables. Our objective is to understand how such graphs encode statistical structure — especially conditional independence, confounding, and path blocking — and how this graphical structure will later support causal identification.

2.1.1 Basic Definition

Definition: Directed Acyclic Graph

A *directed acyclic graph* (DAG) $\mathcal{G} = (\mathcal{V}, E)$ consists of a finite set of nodes $\mathcal{V}$ and a set of directed edges $E \subseteq \mathcal{V} \times \mathcal{V}$ such that there is no directed cycle: no node has a directed path of length at least one back to itself.

In a causal DAG each node represents a variable and each directed edge $A \to B$ encodes that A is a *direct* cause of B relative to the variables included in the graph: the structural mechanism generating B depends nontrivially on A when the other parents of B are held fixed. A direct edge does not preclude additional mediated influence — in the mediation example below, T affects Y both directly and through M. Strictly speaking, a drawn arrow represents genuine dependence on the parent only under a *causal minimality* convention, which we adopt informally throughout: edges are not drawn gratuitously.

The acyclicity condition is essential for the standard DAG semantics: it guarantees that the nodes admit a *topological ordering*, in which every parent precedes its children. Acyclicity alone, however, imposes no restriction on a joint distribution — by the chain rule, any density factorizes as $p(v_1, \dots, v_k) = \prod_i p(v_i \mid v_1, \dots, v_{i-1})$ along any ordering. We say that a distribution P is *Markov with respect to* $\mathcal{G}$ if, along a topological ordering, the predecessors can be replaced by the parents:

$$p(v_1, \dots, v_k) = \prod_{i=1}^{k} p(v_i \mid \mathrm{Pa}(v_i)).$$

This factorization is a property of the pair $(\mathcal{G}, P)$, not a consequence of acyclicity alone — it is precisely the substantive probabilistic restriction imposed by the graphical model, developed formally in Section 2.4. If directed feedback were present, the graph would not be a DAG, and the recursive semantics developed here would not apply without additional equilibrium or dynamic structure; cyclic systems are outside the scope of these notes.

2.1.2 Structural Relationships

Definition: Structural Relationships in a DAG

For a node $v \in \mathcal{V}$ in $\mathcal{G} = (\mathcal{V}, E)$:

- $\mathrm{Pa}(v) = \{w \in \mathcal{V} : w \to v \in E\}$ are the *parents* of v.
- $\mathrm{De}(v)$ = all nodes $w \neq v$ reachable from v by a directed path of positive length are the (strict) *descendants* of v.
- $\mathrm{An}(v)$ = all nodes $w \neq v$ with a directed path of positive length to v are the (strict) *ancestors* of v.
- $\mathrm{Nd}(v) = \mathcal{V} \setminus (\mathrm{De}(v) \cup \{v\})$ are the *non-descendants* of v.

Note that $\mathrm{Nd}(v)$ includes the parents of v; when stating the local Markov property in Section 2.4, we will exclude the parents explicitly. These sets depend on the graph; when several graphs are in play — as in Chapter 3, where graphs are modified to represent interventions — we write $\mathrm{Pa}_{\mathcal{G}}(v)$, $\mathrm{An}_{\mathcal{G}}(v)$, and so on.

Quick Terminology Summary

A *node* represents a variable, and an *edge* represents a relationship between two variables. If $A \to B$, then A is a *parent* of B and B is a *child* of A. Two nodes are *adjacent* if they are connected by an edge. A *path* is any sequence of *distinct* connected nodes, regardless of arrow direction; a *directed path* is a path whose arrows all point in the same direction. A node is an *ancestor* of another if there is a directed path from the former to the latter, and a *descendant* if the direction is reversed. A *directed cycle* is a sequence of edges $V_0 \to V_1 \to \cdots \to V_k = V_0$ with $k \geq 1$ and $V_0, \dots, V_{k-1}$ distinct; it is not a path in the above sense, since its first and last nodes coincide. A *directed acyclic graph* (DAG) is a directed graph with no directed cycles.

Example: Education, Earnings, and Family Background

Consider the causal story: education (E) affects earnings (Y); family background (B) is a common cause of both; and neighborhood (N) affects education but has no direct effect on earnings.
The Markov factorization is $p(n, b, e, y) = p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b)$. Reading off structural relationships: $\mathrm{Pa}(E) = \{N, B\}$, $\mathrm{Pa}(Y) = \{E, B\}$, $\mathrm{Pa}(N) = \mathrm{Pa}(B) = \emptyset$. The descendants of N are $\{E, Y\}$; B and N are non-descendants of each other.

Example 2.1 (The IV DAG)**.** The canonical instrumental variable model involves four nodes: Z (instrument), T (treatment), Y (outcome), and U (unobserved confounder).

$\mathrm{Pa}(T) = \{Z, U\}$, $\mathrm{Pa}(Y) = \{T, U\}$, $\mathrm{Pa}(Z) = \mathrm{Pa}(U) = \emptyset$. The descendants of Z are $\{T, Y\}$; U and Z are non-descendants of each other.

Example: The Mediation DAG

In a mediation model, treatment T affects outcome Y both directly and through an intermediate variable M (the mediator). An unobserved confounder U creates a back-door path $T \leftarrow U \rightarrow Y$, making T endogenous. The $T \rightarrow M$ relation is unconfounded: the only back-door path from T to M, namely $T \leftarrow U \rightarrow Y \leftarrow M$, is blocked by the collider at Y. The $M \rightarrow Y$ relation, by contrast, has two open back-door paths, $M \leftarrow T \rightarrow Y$ and $M \leftarrow T \leftarrow U \rightarrow Y$, both of which are blocked by conditioning on T. (This example previews the notions of back-door paths, colliders, and blocking, defined formally in Section 2.2 and Section 2.3; read it for the picture now, and return to verify each claim after studying d-separation.)
$\mathrm{Pa}(T) = \{U\}$, $\mathrm{Pa}(M) = \{T\}$, $\mathrm{Pa}(Y) = \{M, T, U\}$, $\mathrm{Pa}(U) = \emptyset$. The total effect of T on Y travels along two paths: the direct path $T \rightarrow Y$ and the indirect path $T \rightarrow M \rightarrow Y$. This graph is a standard mediation DAG, not a front-door DAG. Because there is a direct path $T \rightarrow Y$ bypassing M, the front-door criterion does not apply here. A front-door graph would require that every directed path from T to Y pass through M.

2.2 The Three Motifs

A DAG encodes direct causal relationships through its edges, but variables can also be statistically related through longer routes that mix directed and reversed edges. The concept of a *path* formalizes these routes, and every intermediate node on a path plays exactly one of three structural roles: a link in a *chain*, the apex of a *fork*, or a *collider*. This section introduces the three motifs through concrete examples and then develops the blocking rules that determine whether each motif transmits or blocks statistical dependence, depending on which variables are conditioned upon. Section 2.3 then assembles these single-path rules into the d-separation criterion. The key insight is that conditioning can both *close* channels that were open (by blocking a chain or fork) and *open* channels that were closed (by activating a collider).

2.2.1 Paths and the Three Motifs

Definition: Path

A *path* between nodes X and Y in $\mathcal{G}$ is any sequence of distinct nodes $X = V_0, V_1, \ldots, V_k = Y$ such that each consecutive pair is connected by an edge in either direction.

Every intermediate node V_m on a path is classified by the orientation of the two edges meeting it. The node V_m is a **collider** on the path if both arrows point *into* it: $V_{m-1} \rightarrow V_m \leftarrow V_{m+1}$. Otherwise V_m is a **non-collider**. The non-collider cases have familiar names: in a **chain** ($V_{m-1} \rightarrow V_m \rightarrow V_{m+1}$, or its mirror image $V_{m-1} \leftarrow V_m \leftarrow V_{m+1}$), V_m is a causal intermediary through which influence is transmitted; in a **fork** ($V_{m-1} \leftarrow V_m \rightarrow V_{m+1}$), V_m is a common cause of its two neighbors on the path. Chain and fork are useful motif names, but the formal d-separation rule of Section 2.3 distinguishes only colliders from non-colliders. The collider is the crucial asymmetric case whose behavior under conditioning is the opposite of the other two.

The three motifs correspond to three familiar causal patterns, one per configuration. Each story below should be read as an *idealized three-node model*: the stated conclusions hold for the displayed graph, and need not hold in the corresponding real-world setting, where omitted common causes, additional paths, or measurement error may intervene.

Chain: Smoking → Tar → Cancer. Let A = smoking, M = tar deposits in the lungs, B = lung cancer. The path $A \to M \to B$ is a chain: smoking causes tar accumulation, which in turn causes cancer, so M is a causal intermediary. Marginally, the chain transmits association: heavy smokers have elevated cancer rates.

Fork: Poverty → Poor Diet; Poverty → Lack of Exercise. Let M = poverty, A = poor diet, B = lack of exercise. The path $A \leftarrow M \to B$ is a fork: poverty is a common cause of both outcomes. Unconditionally, the fork also transmits association: diet quality and exercise are correlated — not because one causes the other, but because both are driven by poverty.

Collider: Accident → Hospitalization ← Cancer. Let A = traffic accident, B = cancer, M = hospitalization. Both accidents and cancer independently cause hospitalization, so M is a collider: a common effect of its two neighbors. The collider is the odd one out already at the marginal level: in the general population, having a traffic accident tells you nothing about your cancer risk, $A \perp\!\!\!\perp B$. The path is *closed by default*.

What happens to each motif when the middle node is conditioned upon is the subject of the next subsection.

2.2.2 Blocking Rules

Conditioning on a node affects the paths it lies on differently depending on its structural role:

Structure	**Pattern**	**Effect of conditioning**	**Key intuition**
Chain	$A \to M \to B$	Blocks the path	Causal transmission
Fork	$A \leftarrow M \to B$	Blocks the path	Common cause
Collider	$A \to M \leftarrow B$	**Opens** the path (conditioning on M or a descendant of M)	Common effect

The critical asymmetry — colliders are *closed by default* and *opened* by conditioning, while chains and forks are *open by default* and *closed* by conditioning — is what makes collider bias so easy to overlook in practice. Returning to the three stories of Section 2.2.1 anchors each rule to a concrete setting.

Chain. If we condition on tar level (e.g., restrict to patients with the same measured tar burden), smoking no longer predicts cancer within that stratum: once the tar level is fixed, the causal channel is fully accounted for. Conditioning on the intermediary M *blocks* the path.

Fork. Conditioning on poverty status removes the spurious association: among people at the same level of poverty, the diet–exercise correlation disappears. Conditioning on the common cause M *blocks* the path.

Collider. Among hospitalized patients — conditioning on M — accident and cancer typically become negatively associated. If we learn that a hospitalized patient was *not* in a traffic accident, cancer or another serious illness becomes more likely as an explanation for the admission. Conditioning on the common effect M *opens* the path. (Opening the path licenses dependence; the negative sign reflects the monotone selection mechanism — either cause suffices to produce hospitalization — not the arrow pattern alone.) This is Berkson's bias (Berkson 1946).

Three-Node Motifs at a Glance

The three basic path structures behave differently under conditioning. In a *chain* $A \to M \to B$, the middle node M transmits association, so conditioning on M blocks the path. In a *fork* $A \leftarrow M \to B$, the middle node M is a common cause, so conditioning on M removes the spurious association. In a *collider* $A \to M \leftarrow B$, the path is blocked by default, but conditioning on M — or on any descendant of M — opens the path and induces association. These three motifs are the local building blocks of d-separation in arbitrary DAGs.

The blocking rules need not be taken on faith. The next example derives the chain rule directly from the Markov factorization stated in Section 2.1; the remark that follows explains how the remaining rules are obtained by the same kind of computation.

Example 2.2 (A Direct Probability Proof for the Chain)**.** Consider the chain $X_1 \to X_2 \to X_3$. By the Markov factorization,

$$p(x_1, x_2, x_3) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2).$$

Conditioning on $X_2 = x_2$, for any x_2 with $p(x_2) > 0$:

$$p(x_1, x_3 \mid x_2) = \frac{p(x_1, x_2, x_3)}{p(x_2)} = \frac{p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2)}{p(x_2)}.$$

Rearranging,

$$p(x_1, x_3 \mid x_2) = \frac{p(x_1)\, p(x_2 \mid x_1)}{p(x_2)}\, p(x_3 \mid x_2) = p(x_1 \mid x_2)\, p(x_3 \mid x_2).$$

Hence $X_1 \perp\!\!\!\perp X_3 \mid X_2$. This derivation shows concretely how a graphical blocking statement becomes an ordinary conditional-independence identity in the distribution. The corresponding fork case $X_1 \leftarrow X_2 \to X_3$ can be verified similarly and is left as an exercise.

Remark: Where the Blocking Rules Come From

Each rule is a small theorem about any distribution that is Markov with respect to the corresponding three-node graph. The chain rule is exactly the example above, and the fork case is identical after rewriting the factorization (Problem 8). For the collider, the factorization $p(a, b, m) = p(a)\, p(b)\, p(m \mid a, b)$ integrates over m to $p(a, b) = p(a)\, p(b)$ — closed by default — and conditioning on M generically destroys the product (Problem 5). The descendant clause is the same computation one step removed: for a child D of M,

$$p(a, b \mid d) \;\propto\; p(a)\, p(b) \int p(m \mid a, b)\, p(d \mid m)\, \mathrm{d}m,$$

which generically does not factorize in (a, b): observing D is observing a noisy proxy of the collider. Two qualifications: an open path *licenses* dependence but does not force it (the generic-parameter reading of Section 2.3), and the three-node computations are only base cases — Theorem 2.1 extends the correspondence to every path in every DAG. In short, the blocking rules are definitions *engineered* so that graphical blocking matches conditional independence under the Markov property, and *verified* here by elementary computation.

2.3 d-Separation

The blocking rules of Section 2.2 settle the status of a *single* path. In any realistic DAG, however, two variables are typically joined by several paths at once, and association can flow along any one of them: a statistical relationship is ruled out by the graph only when *every* route is shut. *d-Separation* — the “d” stands for *directional*, because the rules depend on the orientation of the arrows along each path — is the criterion that performs exactly this aggregation. Given two nodes and a conditioning set $\mathbf{S}$, it examines each path in turn and declares the nodes *separated* precisely when $\mathbf{S}$ blocks all of the paths. It is a purely graphical computation: it can be carried out from the arrows alone, without reference to any probability distribution.

Why does this bookkeeping deserve a section of its own? Because d-separation is the bridge over which the graph speaks about data. The soundness theorem below (Theorem 2.1) shows that, for any distribution that is Markov with respect to the DAG, every d-separation statement translates into a genuine conditional independence of that distribution. The criterion thereby converts qualitative causal assumptions — arrows — into concrete probabilistic restrictions, and it is the engine behind the identification criteria of later chapters: the back-door criterion of Chapter 3 is, at bottom, a d-separation requirement in a modified graph.

2.3.1 The d-Separation Criterion

Before stating the formal criterion, it is useful to summarize the intuition: a path stays active unless we block a non-collider on it, or unless it contains a collider that has not been conditioned on (and has no conditioned-on descendant). The definition below simply turns this intuition into a rule that applies to any path in any DAG.

Definition 2.1 (d-Blocking and d-Separation)**.** A path π is *d-blocked* by a set $\mathbf{S}$ if:

1. π contains a non-collider M (a chain or fork node) with $M \in \mathbf{S}$; or
2. π contains a collider $A \to C \leftarrow B$ with $C \notin \mathbf{S}$ and no descendant of C is in $\mathbf{S}$.

Nodes X and Y are *d-separated* by $\mathbf{S}$, written $(X \perp\!\!\!\perp Y \mid \mathbf{S})_{\mathcal{G}}$, if every path between X and Y in $\mathcal{G}$ is d-blocked by $\mathbf{S}$. If some path is not d-blocked, X and Y are *d-connected* given $\mathbf{S}$, written $(X \not\!\perp\!\!\!\perp Y \mid \mathbf{S})_{\mathcal{G}}$. The subscript $\mathcal{G}$ always marks a statement about the graph; unsubscripted statements such as $X \perp\!\!\!\perp Y \mid \mathbf{S}$ refer to the distribution P.

Example 2.3 (Applying the Criterion to the Three Toy Settings)**.** We revisit the three motif stories of Section 2.2.1 and verify each blocking claim of Section 2.2.2 directly using the definition.

(i) Chain — Smoking → Tar → Cancer. Let A = smoking, M = tar, B = cancer. The only path between A and B is the chain $A \to M \to B$.

- $\mathbf{S} = \emptyset$: the path contains no conditioned-on non-collider, so clause (1) does not apply. The collider clause (2) is also irrelevant (there is no collider). The path is *not* d-blocked. $\Rightarrow (A \not\!\perp\!\!\!\perp B)_{\mathcal{G}}$ (generically, smokers have higher cancer rates).
- $\mathbf{S} = \{M\}$: the chain node $M \in \mathbf{S}$, so clause (1) applies and the path is d-blocked. $\Rightarrow (A \perp\!\!\!\perp B \mid M)_{\mathcal{G}}$ (at fixed tar level, smoking no longer predicts cancer).

(ii) Fork — Poverty → Poor Diet; Poverty → Lack of Exercise. Let M = poverty, A = poor diet, B = lack of exercise. The only path is the fork $A \leftarrow M \to B$.

- $\mathbf{S} = \emptyset$: no non-collider on the path is in $\mathbf{S}$, so the path is not d-blocked. $\Rightarrow (A \not\!\perp\!\!\!\perp B)_{\mathcal{G}}$ (generically, poor diet and lack of exercise are correlated through shared poverty).
- $\mathbf{S} = \{M\}$: the fork node $M \in \mathbf{S}$, so clause (1) applies and the path is d-blocked. $\Rightarrow (A \perp\!\!\!\perp B \mid M)_{\mathcal{G}}$ (at fixed poverty status, the diet–exercise association disappears).

(iii) Collider — Accident → Hospitalization ← Cancer. Let A = accident, M = hospitalization, B = cancer. The only path is the collider $A \to M \leftarrow B$.

- $\mathbf{S} = \emptyset$: the path contains collider M with $M \notin \mathbf{S}$ and no descendant of M in $\mathbf{S}$, so clause (2) applies and the path is d-blocked. $\Rightarrow (A \perp\!\!\!\perp B)_{\mathcal{G}}$ (accidents and cancer are independent in the general population).
- $\mathbf{S} = \{M\}$: now $M \in \mathbf{S}$, so clause (2) no longer blocks the path. No other clause applies, so the path is *open*. $\Rightarrow (A \not\!\perp\!\!\!\perp B \mid M)_{\mathcal{G}}$ (generically, among hospitalized patients, ruling out an accident raises the probability of cancer — Berkson's bias).

The following theorem is what gives d-separation its statistical force: it guarantees that every d-separation statement in the graph corresponds to a genuine conditional independence in the joint distribution.

Theorem 2.1 (Soundness of d-Separation (Pearl 2009, Theorem 1.2.4))**.** *If* $(X \perp\!\!\!\perp Y \mid \mathbf{S})_{\mathcal{G}}$*, then* $X \perp\!\!\!\perp Y \mid \mathbf{S}$ *in every distribution that is Markov with respect to* $\mathcal{G}$*.*

Remark: Soundness, Not Converse

Theorem 2.1 states that d-separation implies conditional independence for every distribution Markov with respect to $\mathcal{G}$. The converse need not hold without additional assumptions such as *faithfulness*: a conditional independence may occur in the data because of special parameter values even when the corresponding nodes are not d-separated in the graph.
For the same reason, the dependence conclusions drawn in our examples — statements of the form "the path is open, hence $A \not\!\perp\!\!\!\perp B$" — are not consequences of the Markov property alone. An open path licenses dependence but does not force it; the dependence holds generically, i.e., except for special parameter cancellations, and is guaranteed under faithfulness. We adopt this generic reading throughout the examples in this chapter.

Example: Unfaithfulness through Parameter Cancellation

The caveat above is not merely hypothetical. Consider the linear Gaussian model

$$M = aX + \varepsilon_M, \qquad Y = bM + cX + \varepsilon_Y,$$

with X, ε_M, ε_Y mutually independent, corresponding to the DAG with edges $X \to M$, $M \to Y$, and $X \to Y$. Here X and Y are d-connected: both $X \to Y$ and $X \to M \to Y$ are open paths. Yet

$$\mathrm{Cov}(X, Y) = b\,\mathrm{Cov}(X, M) + c\,\mathrm{Var}(X) = (ab + c)\,\mathrm{Var}(X),$$

which is zero whenever $c = -ab$: the associations transmitted along the two open paths cancel exactly, and $X \perp\!\!\!\perp Y$ despite the d-connection (in the jointly Gaussian case, zero covariance implies independence). Such parameter configurations are precisely what the faithfulness assumption rules out.

Within this finite-dimensional linear-Gaussian parameterization, the cancellation set $c = -ab$ has Lebesgue measure zero, which is why the generic reading of open paths is usually harmless; the claim is about this parameter space, not about arbitrary nonparametric causal models. Near-cancellations can still occur in practice — for instance when direct and mediated effects have opposite signs — making the observed dependence empirically weak.

2.3.2 Practical Ways to Check d-Separation

The formal definition of d-separation can appear abstract when applied to larger graphs. Two complementary algorithmic viewpoints are useful in practice, both equivalent to the definition. On a first reading, path-by-path application of the definition — systematized by the tracing rules below — is the essential skill; the moral-graph method may be treated as optional second-pass material.

1. Active-Path Tracing. Imagine releasing a ball from X along each path and asking whether it can reach Y, given that nodes in $\mathbf{S}$ are observed (shaded). On any given path, the ball follows four local rules:

- At a non-collider (chain or fork node) *not* in $\mathbf{S}$: ball passes through.
- At a non-collider in $\mathbf{S}$: ball stops.
- At a collider not in $\mathbf{S}$ (and no descendant in $\mathbf{S}$): ball stops.
- At a collider in $\mathbf{S}$ (or with a descendant in $\mathbf{S}$): ball passes through.

If the ball can reach Y from X along some path, then X and Y are d-connected given $\mathbf{S}$: the graph does not imply $X \perp\!\!\!\perp Y \mid \mathbf{S}$. If every path is stopped, then $(X \perp\!\!\!\perp Y \mid \mathbf{S})_{\mathcal{G}}$. These rules restate the d-blocking criterion path by path; a related, direction-sensitive search algorithm — *Bayes-Ball*, which visits each node at most a bounded number of times rather than enumerating paths — is developed by Shachter (1998).

2. Moral Graph Transformation. There is also a graph-transformation approach, which connects DAG models to undirected graphical models (Markov random fields).

1. Restrict the DAG to the *ancestral set* of $\{X, Y\} \cup \mathbf{S}$: keep the nodes X, Y, and $\mathbf{S}$ themselves together with all of their ancestors, and discard everything else.
2. *Moralize* the graph: for every collider $A \to C \leftarrow B$ in the ancestral subgraph, add an undirected edge $A - B$ (“marry the parents”).
3. Remove all edge directions to obtain an undirected graph.
4. Delete all nodes in $\mathbf{S}$ (and their incident edges).

X and Y are d-separated by $\mathbf{S}$ in $\mathcal{G}$ *if and only if* they are disconnected in the resulting undirected graph. The moralization step is what makes the collider rule visible in the undirected representation: without adding the A–B edge, the path through C would appear disconnected even when C is conditioned upon, which would give the wrong answer. A full treatment is in Lauritzen (1996, Ch. 3).

These two viewpoints — active-path tracing and moral-graph separation — are equivalent algorithmic ways of evaluating the d-separation criterion. Students are encouraged to use whichever they find most natural, and to cross-check with the other when in doubt.

Example 2.4 (Checking d-Separation in a Four-Node DAG). Consider the DAG with edges $Z \to T$, $U \to T$, $T \to Y$, $U \to Y$.

Query 1. Is $(Z \perp\!\!\!\perp U)_{\mathcal{G}}$, i.e. $\mathbf{S} = \emptyset$?

Path tracing. The only path from Z to U is $Z \to T \leftarrow U$. Node T is a collider with $T \notin \mathbf{S}$ and no descendant of T in $\mathbf{S}$, so the ball is stopped at T. No ball can reach U, hence $(Z \perp\!\!\!\perp U)_{\mathcal{G}}$.

Moral graph. (1) Ancestral set of $\{Z, U\} \cup \mathbf{S} = \{Z, U\}$: neither Z nor U has any parents, so the ancestral set is $\{Z, U\}$ itself; nodes T and Y are discarded. (2) Moralize the subgraph $\{Z, U\}$: it has no edges and no colliders, so nothing is added. (3) Undirected graph: isolated nodes Z and U. (4) Delete $\mathbf{S} = \emptyset$: nothing changes. Z and U are disconnected $\Rightarrow (Z \perp\!\!\!\perp U)_{\mathcal{G}}$. Note that the key step is restricting to the ancestral set: this removes T from the graph before moralization, so the collider T never has a chance to create a moral edge between Z and U.

Query 2. Is $(Z \perp\!\!\!\perp U \mid T)_{\mathcal{G}}$, i.e. $\mathbf{S} = \{T\}$?

Path tracing. Same path $Z \to T \leftarrow U$. Now $T \in \mathbf{S}$: the collider is observed, so the ball passes through T. The path is open, hence $(Z \not\!\perp\!\!\!\perp U \mid T)_{\mathcal{G}}$.

Moral graph. (1) Ancestral set of $\{Z, U\} \cup \mathbf{S} = \{Z, U, T\}$: the parents of T are Z and U, both already in the set; node Y is discarded. (2) Moralize: the subgraph contains the collider $Z \to T \leftarrow U$, so add the moral edge $Z - U$. (3) Undirected graph: edges $Z - T$, $U - T$, $Z - U$. (4) Delete $\mathbf{S} = \{T\}$ and its incident edges. Remaining graph: single edge $Z - U$. Z and U are connected $\Rightarrow (Z \not\!\perp\!\!\!\perp U \mid T)_{\mathcal{G}}$. The moralization step is decisive: it adds the edge $Z - U$ *before* T is deleted, ensuring the dependence induced by conditioning on the collider is visible in the undirected graph.

Comparing the two queries: Z and U are marginally independent (the instrument is exogenous) but become dependent once we condition on T. This is collider bias — of which Berkson's bias is the classic instance — treated in full in Section 2.5.

Example 2.5 (Practice DAG — Eight-Node Graph). Consider the DAG below with nodes W, X_1, T, M_1, M_2, Y, C, and D.

For each query below, state whether the independence holds and identify every relevant path and its blocking status.

(1) Is $X_1 \perp\!\!\!\perp Y$? *No.* There are four paths from X_1 to Y. Two are open: $X_1 \leftarrow W \to Y$ (fork at W, unblocked) and $X_1 \leftarrow W \to T \to M_1 \to Y$ (fork at W, then chain; also open). Two pass through the collider at C: $X_1 \to C \leftarrow M_2 \leftarrow M_1 \to Y$ and $X_1 \to C \leftarrow M_2 \leftarrow M_1 \leftarrow T \leftarrow W \to Y$. With $\mathbf{S} = \emptyset$, neither C nor any descendant of C is in $\mathbf{S}$, so both collider paths are blocked. One open path suffices: $(X_1 \not\!\perp\!\!\!\perp Y)_{\mathcal{G}}$.

(2) Is $X_1 \perp\!\!\!\perp Y \mid W$? *Yes.* Both open paths above pass through the fork W; conditioning on W blocks them. The two remaining paths each contain the collider at C with $C \notin \{W\}$ (and no descendant of C in $\{W\}$), so both remain blocked. All paths blocked $\Rightarrow (X_1 \perp\!\!\!\perp Y \mid W)_{\mathcal{G}}$.

(3) Is $T \perp\!\!\!\perp M_2 \mid M_1$? *Yes.* There are four paths between T and M_2. The direct path $T \to M_1 \to M_2$ is a chain with $M_1 \in \{M_1\}$ — blocked. The path $T \leftarrow W \to X_1 \to C \leftarrow M_2$ has a collider at C with $C \notin \{M_1\}$ — blocked. The path $T \leftarrow W \to Y \leftarrow M_1 \to M_2$ has a collider at Y ($W \to Y \leftarrow M_1$) with $Y \notin \{M_1\}$ — blocked (and also blocked at the fork node $M_1 \in \{M_1\}$). Finally, the path $T \to M_1 \to Y \leftarrow W \to X_1 \to C \leftarrow M_2$ is blocked at the chain node $M_1 \in \{M_1\}$ (and also at the colliders Y and C). All paths blocked $\Rightarrow (T \perp\!\!\!\perp M_2 \mid M_1)_{\mathcal{G}}$.

(4) Does conditioning on C open a collider path between X_1 and M_2? *Yes.* C is a collider on the path $X_1 \to C \leftarrow M_2$, which is blocked marginally and is opened by conditioning on C. Note, however, that X_1 and M_2 are *not* marginally d-separated: the path $X_1 \leftarrow W \to T \to M_1 \to M_2$ is open even without any conditioning. A cleaner comparison conditions on W throughout. First, $(X_1 \perp\!\!\!\perp M_2 \mid W)_{\mathcal{G}}$: conditioning on W blocks the fork path above, and the two collider paths remain blocked. Second, $(X_1 \not\!\perp\!\!\!\perp M_2 \mid \{W, C\})_{\mathcal{G}}$: adding C to the conditioning set opens $X_1 \to C \leftarrow M_2$. Conditioning on the collider thus introduces a new, non-causal source of association — it does not create the only connection between X_1 and M_2, but it destroys an independence that held given W alone.

(5) Does conditioning on D open the path $X_1 \to C \leftarrow M_2$? *Yes.* D is a descendant of C. By clause (2) of the d-separation definition, a collider path is unblocked whenever the collider *or any of its* *descendants* is in the conditioning set. Since $D \in \{D\}$, the collider at C is activated, opening the path $X_1 \to C \leftarrow M_2$ even though C itself is not conditioned on.

2.4 The Markov Property and Factorization

Up to this point, we have used DAGs qualitatively, to decide which paths are open or blocked. We now connect the graph to probability algebra: the same parent structure that governs d-separation also determines how the joint distribution factorizes. Appendix A provides a gentler preview of the same ideas.

Without structural assumptions, any joint distribution can always be written by repeated conditioning, but that generic representation is often too high-dimensional to reveal much structure. A DAG becomes statistically meaningful because, together with the Markov property, it replaces the generic factorization by a sparse one involving only the parents of each node. This is the sense in which a graphical model is not merely a picture: it imposes probabilistic structure on the joint distribution. Whether that structure is empirically testable depends on which variables are observed; implications involving only observed variables yield conditional independences that can be checked against data, while those involving latent variables generally cannot.

Definition: Markov Factorization

Let $\mathcal{G}$ be a DAG with nodes $V_1, \dots, V_k$. A distribution P *factorizes according to* $\mathcal{G}$ (equivalently, is *Markov with respect to* $\mathcal{G}$) if its joint density satisfies

$$p(v_1, \dots, v_k) \;=\; \prod_{i=1}^{k} p(v_i \mid \mathrm{Pa}(v_i))\,. \tag{2.1}$$

The product does not depend on how the nodes are ordered.

Under Markov compatibility — the assumption that P satisfies Equation 2.1 for the graph at hand — the parent sets determine the sparse factorization of the joint law; the factorization, not the picture alone, is the bridge between causal structure and probability.

Definition: Local Markov Property

A distribution P satisfies the *local Markov property* with respect to $\mathcal{G}$ if, for every node $V_i \in \mathcal{V}$,

$$V_i \perp\!\!\!\perp \big[\mathrm{Nd}(V_i) \setminus \mathrm{Pa}(V_i)\big] \,\Big|\, \mathrm{Pa}(V_i).$$

That is, once we condition on the direct causes of a node, that node is independent of all variables that are neither its descendants nor its parents.

Remark: Three Equivalent Formulations

A distribution P satisfies the *global Markov property* with respect to $\mathcal{G}$ if every d-separation in $\mathcal{G}$ corresponds to a conditional independence in P. Theorem 2.1 establishes one arrow of the picture: the factorization implies the global property. The remaining arrows are elementary: the local property is the special case of the global one obtained by separating each node from its non-descendants given its parents, and the local property yields the factorization by applying the chain rule along a topological ordering. For DAG models with densities, under the regularity conditions recorded in Appendix A, the three formulations are therefore equivalent (Lauritzen 1996, Theorem 3.27): the Markov factorization Equation 2.1, the local Markov property, and the global Markov property each imply the other two. Any one of the three can therefore serve as the definition of "P is Markov with respect to $\mathcal{G}$"; the factorization is what is most often used directly in identification arguments.

What does the equivalence buy in the smallest possible case? For the three-node chain, the global property contains exactly one substantive statement, $X_1 \perp\!\!\!\perp X_3 \mid X_2$ — and the factorization delivers it in the two lines already carried out above. We record the result for reference: it is the miniature version of Theorem 2.1 that the reader has proved by hand.

Proposition 2.1 (Conditional Independence in the Three-Node Chain)**.** *Let $X_1 \to X_2 \to X_3$ be a chain with Markov factorization $p(x_1, x_2, x_3) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2)$. Then $X_1 \perp\!\!\!\perp X_3 \mid X_2$. The analogous result for the fork $X_1 \leftarrow X_2 \to X_3$ is left as an exercise.*

Remark: Edges as Scientific Claims

In a causal DAG, an arrow is typically drawn only when a direct dependence-generating relation is believed to be present. For that reason, adjacent nodes are usually expected to be statistically dependent unless special parameter cancellations occur. We do not formalize minimality or faithfulness here, but this intuition explains why edges are not included gratuitously: every arrow in the graph represents a substantive scientific claim. For ordinary causal interpretation we therefore work with a minimal graph, in which displayed arrows are intended as active direct causal inputs. For certain *nonidentification* arguments, however, Chapter 3 adopts the weaker allowed-parent convention, under which a structural function may ignore some displayed parents.

Example: Markov Factorization in the Education Example

For the DAG with edges $N \to E$, $B \to E$, $B \to Y$, $E \to Y$:

$$p(n, b, e, y) = p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b).$$

A regression of Y on E alone is generally confounded by B, which lies on the open back-door path $E \leftarrow B \to Y$. Graphically, conditioning on B blocks this path. After adding the intervention semantics and positivity condition developed in Chapter 3, this graphical fact yields the adjustment formula

$$P\{y \mid \mathrm{do}(E{=}e)\} \;=\; \int P(y \mid e, b)\, \mathrm{d}P_B(b).$$

Example: Markov Factorization in the IV DAG

For the IV DAG, including the latent U:

$$p(z, t, y, u) = p(z)\, p(u)\, p(t \mid z, u)\, p(y \mid t, u).$$

Since U is unobserved, the observable factorization is obtained by marginalizing:

$$p(z, t, y) = \int p(z)\, p(u)\, p(t \mid z, u)\, p(y \mid t, u)\, \mathrm{d}u.$$

In general, this observed law does not factorize as $p(z)\, p(t \mid z)\, p(y \mid t)$: the latent common cause U makes T endogenous in the outcome equation, and the integral over u does not disappear. Exactly which graphical mechanism produces this failure — a collider opened by conditioning on T — is the subject of the next section.

2.5 Collider Bias and the IV DAG

Throughout the d-separation analyses of Section 2.3, the collider was the asymmetric case: closed by default, and opened — often unintentionally — by conditioning. This section studies the resulting phenomenon — *collider bias* — systematically: first by defining collider bias and illustrating it through Berkson-type selection, and then through a full d-separation analysis of the instrumental-variables DAG, where a collider is simultaneously the central danger for the naive analyst and, in later chapters, part of the identification strategy itself.

2.5.1 Berkson's Bias

Definition: Collider Bias

Collider bias is the distortion of a target association or causal-effect estimate caused by conditioning on a common effect (a collider), or on a descendant of that collider, thereby opening a non-causal path. In a path $A \to C \leftarrow B$, the path is blocked by default, but conditioning on C — or on any descendant of C — can make A and B statistically dependent along it. If A and B are marginally independent, this creates an association where none existed; if they are already associated through other paths, collider conditioning distorts the existing association.

Collider Bias vs. Confounding

Unlike confounding, which is *removed* by conditioning on a common cause, collider bias is *created* by conditioning on a common effect. This is why conditioning on more variables is not always safer in causal analysis: adding a collider or a descendant of a collider to the adjustment set introduces bias rather than reducing it.

Example: Collider Bias in the IV DAG

In the IV DAG, T is a collider on the path $Z \to T \leftarrow U$. Marginally, $(Z \perp\!\!\!\perp U)_{\mathcal{G}}$ (the instrument is exogenous). However,

$$(Z \perp\!\!\!\perp U)_{\mathcal{G}} \qquad \text{but} \qquad (Z \not\perp\!\!\!\perp U \mid T)_{\mathcal{G}}$$

(established graphically in Example 2.4). Conditioning on T — for example, analyzing within a stratum $T = t$ — makes the instrument Z appear associated with U, invalidating the IV argument within that stratum.

Example: Collider Bias through Selection — Talent and Wealth

A similar phenomenon arises whenever selection depends on two otherwise unrelated variables. Suppose both talent (A) and wealth (B) increase the probability of admission to an elite school, so admission (S) is a collider: $A \to S \leftarrow B$.

In the general population, talent and wealth may be independent: the path $A \to S \leftarrow B$ is blocked by the collider S, so $A \perp\!\!\!\perp B$. But among admitted students — conditioning on S — the two typically become negatively associated: because both variables contribute to the same selection event, lower talent makes higher wealth more likely as the explanation for admission, and vice versa. (As with Berkson's bias, the negative sign follows from the monotone selection mechanism, not from the arrow pattern alone.) Conditioning on admission therefore induces association between variables that were not associated in the population. This is again collider bias: admission is a common effect of talent and wealth, and restricting to admitted students is precisely conditioning on that common effect.

2.5.2 Full d-Separation Analysis of the IV DAG

We work through the IV DAG with edges $Z \to T$, $T \to Y$, $U \to T$, $U \to Y$. This is the same graph as in Example 2.4, but now U is treated as *unobserved*. That assumption is precisely what earns the name "IV DAG": because U cannot be conditioned on, the back-door path $T \leftarrow U \to Y$ cannot be blocked by adjustment, and the instrument Z motivates an alternative identification strategy. The graph alone does not nonparametrically identify $P(y \mid \mathrm{do}(T{=}t))$; Chapter 7 introduces the additional assumptions — such as linearity or monotonicity — under which particular IV estimands are identified.

Before listing the paths, it is worth classifying them by *type*, since students often conflate causal and non-causal sources of dependence.

Endpoints	Path	Type	Why
$Z \leftrightarrow Y$	$Z \to T \to Y$	*Directed causal*	Directed from Z to Y; carries the IV signal

Endpoints	**Path**	**Type**	**Why**
$Z \leftrightarrow Y$	$Z \to T \leftarrow U \to Y$	*Non-causal, collider-blocked*	Blocked by collider at T unless T (or a descendant) is conditioned on
$Z \leftrightarrow U$	$Z \to T \leftarrow U$	*Non-causal, collider-blocked*	Collider at T; dormant unless T conditioned on
$Z \leftrightarrow U$	$Z \to T \to Y \leftarrow U$	*Non-causal, collider-blocked*	Collider at Y; dormant unless Y (or a descendant) conditioned on

The key distinction: a *causal path* from Z to Y is one on which every arrow points from Z toward Y; every other path is *non-causal*. A non-causal path does not represent a mechanism by which Z produces Y, but it can transmit statistical association. Non-causal paths have no single default blocking status: a back-door fork such as $T \leftarrow U \to Y$ is open without any conditioning, whereas the three non-causal paths in the present IV DAG each contain a collider (T on the first two, Y on the third) and are blocked unless the corresponding collider, or one of its descendants, is conditioned upon. With this in mind, the four questions become much easier to diagnose.

Question 1. Is $Z \perp\!\!\!\perp Y$? There are two paths between Z and Y. The directed path $Z \to T \to Y$ is open, since it is a chain and we are not conditioning on T. The path $Z \to T \leftarrow U \to Y$ contains a collider at T; since we are not conditioning on T or any descendant of T, that path is blocked. Because the causal path remains open, $(Z \not\!\perp\!\!\!\perp Y)_{\mathcal{G}}$.

Question 2. Is $Z \perp\!\!\!\perp Y \mid T$? The causal path $Z \to T \to Y$ is blocked because conditioning on T blocks the chain. The path $Z \to T \leftarrow U \to Y$ contains a collider at T, and because we are conditioning on T, that collider is opened. $\Rightarrow (Z \not\!\perp\!\!\!\perp Y \mid T)_{\mathcal{G}}$.

Question 3. Is $Z \perp\!\!\!\perp Y \mid \{T, U\}$? The causal path is blocked at T. The path $Z \to T \leftarrow U \to Y$ is opened at the collider T, but the same path then passes through U, and because we are also conditioning on U, the path is blocked there. Hence both paths are blocked, so $(Z \perp\!\!\!\perp Y \mid \{T, U\})_{\mathcal{G}}$.

Question 4. Is $Z \perp\!\!\!\perp U$? There are two paths between Z and U: $Z \to T \leftarrow U$ and $Z \to T \to Y \leftarrow U$. The first contains a collider at T; the second contains a collider at Y. Since we are not conditioning on T, on Y, or on any descendant of either, both colliders remain closed, so both paths are blocked. Hence $(Z \perp\!\!\!\perp U)_{\mathcal{G}}$.

Taken together, the four questions give a graphical account of the structure behind the three IV assumptions, together with one critical warning.

Relevance. Question 1 shows that the graph places Z on an open causal path to Y through T. Relevance, however, concerns the first stage: it requires that changing Z actually shifts the distribution of T — the edge $Z \to T$ together with a nonzero first-stage effect. A drawn arrow is compatible with an arbitrarily weak (or, absent a minimality convention, even null) effect, so relevance is a substantive condition beyond the displayed topology.

Exogeneity. Question 4 expresses exogeneity: every path from Z to the confounder U is a dormant collider path, so the graph implies $Z \perp\!\!\!\perp U$.

Exclusion. Question 3 is the d-separation footprint of the exclusion restriction. The restriction itself is structural: every directed path from Z to Y must pass through T. In this four-node graph, that is equivalent to Z being absent from the mechanism generating Y; in a richer graph, absence from Y's own structural equation would rule out only a direct $Z \to Y$ edge, and the path formulation is the general one. The conditional independence $(Z \perp\!\!\!\perp Y \mid \{T, U\})_{\mathcal{G}}$ is an *implication* of that structure, not its definition. Note also that verifying the corresponding probabilistic independence directly would require observing U, which is unavailable by assumption. The IV strategy instead uses exclusion and exogeneity *jointly*: in a centered linear outcome model without additional covariates, the two assumptions yield the orthogonality restriction $\mathbb{E}[Z\varepsilon] = 0$, where ε is the structural disturbance in the outcome equation — exclusion keeps Z out of that equation, and exogeneity makes Z orthogonal to the latent determinants collected in ε. Chapter 7 introduces observed covariates and states the covariate-adjusted version, $\mathbb{E}[\varepsilon Z \mid X] = 0$, precisely.

Collider warning. Question 2 is the warning: conditioning on T alone — as a naive analyst might do — blocks the directed path $Z \to T \to Y$ and opens the non-causal path $Z \to T \leftarrow U \to Y$. The remaining association between Z and Y within levels of T is therefore collider-induced and non-causal; it contains no open causal component from Z to Y.

It is important to recognize that the IV DAG encodes the substantive assumptions behind instrument validity graphically — relevance, exogeneity, and exclusion — but does not make them testable in any strong sense. Some observed-data implications of the IV DAG may be checked for compatibility with data, but the core IV assumptions are not generally fully testable from observational data alone.

2.6 Worked Example: The Education–Earnings DAG

This section is the template for how we will use DAGs throughout the course. Starting from a single causal graph, we move step by step through the full pipeline: first specify the DAG, then read off the factorization, then use d-separation to identify the implied conditional independences, and finally interpret the result in causal terms. Later chapters will follow the same pattern, but with richer identification questions.

Practice: d-separation workflow. Before working through the example below, return to Example 2.5 and rework each of the five queries from scratch, without looking at the answers. For each query, follow the same three steps: (1) list every path between the two nodes of interest; (2) classify every intermediate node on each path as a chain, fork, or collider; and (3) determine which paths are blocked or open after conditioning on the given set. This is exactly the procedure that d-separation always requires, in graphs of any size.

The causal story. Education (E) affects earnings (Y); family background (B) is a common cause of both education and earnings; neighborhood (N) affects education but has no direct effect on earnings. All four displayed variables are observed. In addition, we assume *causal sufficiency* relative to these variables: no common cause of any pair among N, B, E, and Y has been omitted. This is a substantive simplifying assumption, not a consequence of observing the displayed variables — in reality, neighborhood and family background are often associated through broader socioeconomic factors, which would constitute exactly such an omitted common cause; we exclude it to keep the d-separation analysis tractable.

Step 1 — Markov factorization. The parents are $\mathrm{Pa}(N) = \mathrm{Pa}(B) = \emptyset$, $\mathrm{Pa}(E) = \{N, B\}$, and $\mathrm{Pa}(Y) = \{E, B\}$. The Markov factorization is therefore

$$p(n, b, e, y) \;=\; p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b).$$

Every factor on the right-hand side involves only observed variables, so in principle each term is estimable from data. This is the starting point for identification: the full joint distribution is expressed in terms of estimable quantities.

Step 2 — d-Separation. There are two paths between N and Y: Path 1 is $N \to E \to Y$ (a chain through E); Path 2 is $N \to E \leftarrow B \to Y$ (a collider at E, followed by the fork leg $B \to Y$). We examine four queries.

Query (i): Is $(N \perp\!\!\!\perp Y)_{\mathcal{G}}$? Path 1 is a chain with nothing conditioned on, so it is open. Path 2 has a collider at E, and since E is not conditioned on, that path is blocked. Because Path 1 remains open, $(N \not\!\perp\!\!\!\perp Y)_{\mathcal{G}}$.

Query (ii): Is $(N \perp\!\!\!\perp Y \mid E)_{\mathcal{G}}$? Path 1 is a chain, and conditioning on E blocks it. Path 2 has a collider at E, and conditioning on E opens it. The activated path $N \to E \leftarrow B \to Y$ continues through B, which is not conditioned on, so the path is open. Hence $(N \not\!\perp\!\!\!\perp Y \mid E)_{\mathcal{G}}$. This is precisely the collider bias of Section 2.5: conditioning on E opens the path $N \to E \leftarrow B \to Y$, thereby inducing a spurious association between N and Y through B.

Query (iii): Is $(N \perp\!\!\!\perp Y \mid \{E, B\})_{\mathcal{G}}$? Path 1 is blocked by conditioning on E. Path 2 is opened at the collider E, but then blocked at B because B is also conditioned on. Both paths are blocked: $(N \perp\!\!\!\perp Y \mid E, B)_{\mathcal{G}}$.

Query (iv): Is $(N \perp\!\!\!\perp B)_{\mathcal{G}}$? The only path is $N \to E \leftarrow B$, which has a collider at E. Since E is not conditioned on, the path is blocked. Therefore $(N \perp\!\!\!\perp B)_{\mathcal{G}}$.

Step 3 — Conditional Independence. We now move from the graphical analysis to the probabilistic implications, being careful about the direction of the inference. By Theorem 2.1, the graph *entails* the two independence statements

$$N \perp\!\!\!\perp B, \qquad N \perp\!\!\!\perp Y \mid E, B.$$

The two remaining graphical conclusions are d-*connections*: N and Y are d-connected marginally, and also given E. These mean only that the graph does not imply the corresponding independences; they become dependence predictions,

$$N \not\perp\!\!\!\perp Y, \qquad N \not\perp\!\!\!\perp Y \mid E,$$

under the additional faithfulness (or generic-parameter) reading discussed in the soundness remark of Section 2.3.1.

Because all four variables are observed, all four empirical relationships can in principle be investigated. Under the Markov assumption alone, however, only violations of the two entailed conditional independences contradict the graphical model — and agreement with those restrictions does not by itself verify the causal graph. The d-connection given E is especially important as a warning to practitioners. If the target is the total causal effect of neighborhood N on earnings Y, conditioning on education E blocks the mediated causal path $N \to E \to Y$ and opens the collider path $N \to E \leftarrow B \to Y$: the adjustment distorts the very quantity being estimated rather than removing bias.

Step 4 — Identification. We wish to identify $P(y \mid \mathrm{do}(E{=}e))$, the distribution of earnings under an intervention that sets education to e. The back-door criterion (Chapter 3) requires an adjustment set $\mathbf{S}$ that (i) blocks every back-door path from E to Y and (ii) contains no descendant of E.

The only back-door path is $E \leftarrow B \to Y$, a fork at B. Consider two candidate adjustment sets:

- $\mathbf{S} = \{B\}$: this blocks the path $E \leftarrow B \to Y$, and B is not a descendant of E. Hence $\{B\}$ is valid.
- $\mathbf{S} = \{N\}$: this does not block the path $E \leftarrow B \to Y$, so $\{N\}$ is invalid.

This illustrates why every back-door path must be blocked: although N is upstream of E, conditioning on N does nothing to close the confounding fork $E \leftarrow B \to Y$.

As a preview of the back-door adjustment formula derived rigorously in Chapter 3, with $\mathbf{S} = \{B\}$ the interventional distribution takes the form

$$P(y \mid \mathrm{do}(E{=}e)) \;=\; \sum_b P(y \mid e, b)\, P(b).$$

Every term on the right-hand side involves only the observed distribution, so the causal effect is identified entirely from observational data — the payoff of the graphical analysis. Beyond the path criterion, the formula also relies on the causal interpretation of the DAG with its intervention semantics, and on a *positivity* condition: for discrete E, $P(E{=}e \mid B{=}b) > 0$ for every relevant b; for continuous E, the intervention level e must lie in the conditional support of E given $B = b$. (For continuous B, the sum over b is likewise replaced by an integral with respect to the distribution of B.) The path analysis identifies B as the appropriate adjustment variable; Chapter 3 shows, using intervention graphs, why this graphical condition yields the standardization formula above.

2.7 The Big Picture

The concepts developed in this chapter form the graphical foundation for the identification and estimation theory developed later in the course. The central logic runs from a qualitative causal graph to an identification formula for an interventional quantity:

Each arrow represents a distinct step in causal reasoning. First, we encode substantive assumptions in a DAG. Second, we use d-separation to read off the conditional independence structure implied by that graph. Third, under the Markov property, we translate those graphical statements into probabilistic restrictions. Fourth, adding the intervention semantics developed in Chapter 3 — the causal reading of arrows, modularity, and d-separation in suitably modified graphs — we use those restrictions to derive identification formulas for interventional quantities such as $P(y \mid \mathrm{do}(T{=}t))$. Observational conditional independences alone do not yield causal identification; the intervention step is what carries the causal content. Section 2.6 illustrates the first three steps in full and previews the fourth through the back-door formula.

In more abstract form, the logic of the course runs

$$\begin{aligned}\text{substantive causal assumptions} &\implies \text{causal DAG, with intervention semantics (Ch. 3)}\\ &\implies \text{Markov factorization and d-separation relations}\\ &\implies \text{conditional independences under the Markov property}\\ &\implies \text{identification formulas in later chapters.}\end{aligned}$$

This chapter establishes the graph syntax, the Markov semantics, and the d-separation machinery in this chain; the intervention step is the subject of Chapter 3. Chapters 3 and beyond use the same graphical machinery to derive specific identification results, including back-door adjustment, front-door adjustment, and do-calculus formulas.

2.8 Summary

1. **DAGs as causal structure.** A DAG is a directed acyclic graph whose nodes represent variables and whose directed edges represent direct causal relationships. Once a DAG and a treatment–outcome target are specified, the graph shows which paths are causal, which are noncausal, and which variables may block or open those paths.
2. **Three-node motifs.** Every path is built from three local structures: chains, forks, and colliders. Chains and forks are open by default and are blocked by conditioning on the middle node. Colliders are blocked by default and are opened by conditioning on the collider or on one of its descendants.
3. **d-Separation and conditional independence.** The d-separation criterion is a graphical rule: X and Y are d-separated by a set $\mathbf{S}$ exactly when every path between them is blocked by $\mathbf{S}$. Under the Markov property, a d-separation statement implies the corresponding conditional independence in the distribution; a d-connection implies dependence only under the additional faithfulness assumption. Some such implications may be assessed empirically, but agreement with the data does not by itself verify the graph, and assumptions involving unobserved variables are generally not testable.
4. **Markov factorization.** The local Markov property yields the factorization $p(v_1, \dots, v_k) = \prod_{i=1}^{k} p(v_i \mid \mathrm{Pa}(v_i))$, which expresses the joint distribution in terms of local conditional distributions. This factorization is the bridge from graphical structure to probability calculus and underlies later identification arguments.
5. **Collider bias.** Conditioning on a collider — or on a descendant of a collider — can induce a new association between otherwise independent variables or distort an existing one (Berkson 1946). This is the opposite of confounding adjustment: conditioning on a common cause removes spurious association, whereas conditioning on a common effect creates it.
6. **A practical workflow.** To analyze a DAG, proceed in four steps: identify the graph structure, determine which paths are open or blocked, translate d-separation statements into conditional independences under the Markov property, and then interpret those independences in light of the causal question. This workflow will be reused throughout the rest of the course.

2.9 Problems

1. Warm-up: a single collider. Consider the DAG $X \to Y \leftarrow Z$, where X and Z have no other connections.

(a) Identify the structural role of Y on the path $X \to Y \leftarrow Z$.
(b) Are X and Z d-separated marginally? Apply the d-separation criterion to the only path between X and Z, and state which blocking rule applies.
(c) Are X and Z d-separated given Y? Explain what happens to the path when Y is conditioned on, state what the faithfulness assumption would add to the graphical conclusion, and describe in one sentence the real-world phenomenon this illustrates.

2. d-Separation practice. Consider the DAG: $A \to B \to D$, $A \to C \to D$, $B \to E$, $C \to E$.

(a) List all paths between A and E. (*Hint*: there are four paths in total; two pass through D.)
(b) For each path, identify the role (chain, fork, collider) of each intermediate node.

(c) Does $\{B, C\}$ d-separate A and E?
(d) Does $\{D\}$ d-separate B and C? What type of node is D on the path $B \to D \leftarrow C$?

3. Berkson's bias. Suppose X and Y are independent standard normal variables, and let $S = \mathbf{1}[X+Y > 0]$ (selected into a sample).

(a) Verify analytically that $\mathrm{Cov}(X, Y \mid S{=}1) < 0$. (*Hint*: rotate coordinates. $R = (X+Y)/\sqrt{2}$ and $Q = (X-Y)/\sqrt{2}$ are independent standard normals, and selection is simply $R > 0$.)
(b) Draw the DAG for (X, Y, S) and identify S as a collider.
(c) Explain in one sentence why restricting the analysis to the subsample with $S = 1$ biases estimates of any association between X and Y, and name the type of bias this illustrates.

4. Markov factorization and collider activation. Consider the DAG with edges $A \to E$, $A \to W$, $F \to E$, $E \to W$, where A = ability, F = family income, E = education, W = wages.

(a) Write down the Markov factorization $p(a, f, e, w)$.
(b) Is $(F \perp\!\!\!\perp W)_{\mathcal{G}}$? List all paths between F and W and determine which are open.
(c) Is $(F \perp\!\!\!\perp W \mid E)_{\mathcal{G}}$? Identify the role of E on each path and explain whether conditioning on E opens or closes each one.
(d) A researcher regresses W on E and F, omitting A. Is the coefficient on E a causal effect of education on wages? Explain using the graph.

5. Soundness of d-separation for the collider. Consider the collider $A \to M \leftarrow B$ with Markov factorization $p(a, m, b) = p(a)\, p(b)\, p(m \mid a, b)$.

(a) By marginalizing over M, show that $A \perp\!\!\!\perp B$ in the joint distribution (i.e., $p(a, b) = p(a)\, p(b)$). This verifies Theorem 2.1 for the case $\mathbf{S} = \emptyset$, where Example 2.3 established d-separation graphically.
(b) Show that conditioning on M *need not* preserve the marginal independence: write out $p(a, b \mid m)$ and explain why it generally does not factorize into $p(a \mid m)\, p(b \mid m)$. Under what special parameterizations could conditional independence nevertheless occur? What does this imply, generically, about A and B among hospitalized patients in the accident–hospitalization–cancer example?
(c) Explain in one sentence why parts (a) and (b) together are consistent with Theorem 2.1. (*Hint*: Theorem 2.1 is a one-directional statement.)

6. Terminology check. Consider the DAG with edges $U \to X$, $U \to Z$, $X \to W$, $Z \to W$, $W \to Y$.

(a) Identify the parents, children, ancestors, descendants, and non-descendants of node W.
(b) List all pairs of adjacent nodes (connected by a single edge).
(c) Which pairs of nodes are connected by a directed path? List every such pair and all corresponding directed paths (some pairs are connected by more than one).
(d) Write the Markov factorization $p(u, x, z, w, y)$ implied by this DAG.

7. Markov factorization and local Markov property. Consider the DAG with edges $X_1 \to X_2$, $X_1 \to X_3$, $X_2 \to X_4$, $X_3 \to X_4$.

(a) Write the joint density $p(x_1, x_2, x_3, x_4)$ implied by the Markov factorization.
(b) State the local Markov property for each of the four nodes. For each node, identify the conditioning set $\mathrm{Pa}(X_i)$ and the independence set $\mathrm{Nd}(X_i) \setminus \mathrm{Pa}(X_i)$.
(c) Is $(X_2 \perp\!\!\!\perp X_3)_{\mathcal{G}}$? Is $(X_2 \perp\!\!\!\perp X_3 \mid X_1)_{\mathcal{G}}$? Justify each answer by listing all paths between X_2 and X_3 and checking whether each is blocked.

8. Toy proof: conditional independence in the fork. Consider the fork $X_1 \leftarrow X_2 \to X_3$ with Markov factorization $p(x_1, x_2, x_3) = p(x_2)\, p(x_1 \mid x_2)\, p(x_3 \mid x_2)$.

(a) Show directly, by conditioning on $X_2 = x_2$, that $X_1 \perp\!\!\!\perp X_3 \mid X_2$. Follow the same three steps used in Example 2.2 for the chain.
(b) Does the DAG imply $X_1 \perp\!\!\!\perp X_3$ marginally? Justify your answer graphically (using d-separation) and algebraically: show that $p(x_1, x_3) = \int p(x_2)\, p(x_1 \mid x_2)\, p(x_3 \mid x_2)\, dx_2$ generally does not factorize into $p(x_1)\, p(x_3)$, although special parameterizations can make it do so.
(c) Explain in one sentence what the fork represents substantively and why conditioning on the common cause X_2 removes the association between X_1 and X_3.

9. d-Separation in the practice DAG. Refer to the eight-node DAG in Example 2.5 (nodes W, X_1, T, M_1, M_2, Y, C, D; edges $W \to X_1$, $W \to T$, $W \to Y$, $T \to M_1$, $M_1 \to Y$, $M_1 \to M_2$, $X_1 \to C$, $M_2 \to C$,

$C \to D$). For each query below, state whether it is true or false and justify your answer by listing all relevant paths and determining whether each is blocked or open. This problem is intended as retrieval practice: do not refer to the worked solutions until you have written your own answer.

(a) $X_1 \perp\!\!\!\perp Y$
(b) $X_1 \perp\!\!\!\perp Y \mid W$
(c) $T \perp\!\!\!\perp M_2 \mid M_1$
(d) Are X_1 and M_2 d-separated given W? Given $\{W, C\}$? Identify the specific path that conditioning on C opens, and explain why the association it transmits is non-causal.
(e) Does conditioning on D (a descendant of C) open the path $X_1 \to C \leftarrow M_2$? State which clause of the d-separation definition applies.

Chapter 3

Graphical Identification and the Do-Calculus

Learning Objectives

By the end of this chapter, students should be able to:

1. Distinguish conditioning from intervention and construct the modified graphs $\mathcal{G}_{\overline{X}}$ and $\mathcal{G}_{\underline{X}}$.
2. Determine whether an observed set satisfies the back-door criterion and write the corresponding standardization formula, including its positivity requirement.
3. State the qualitative roles of the three do-calculus rules and use them in a basic derivation of the back-door formula.
4. Recognize the prototypical front-door structure and explain the two-stage logic of the front-door identifying functional.
5. Explain nonidentification through observationally equivalent causal models and state the practical meaning of completeness of the do-calculus.

How to read this chapter. Section 3.1–Section 3.3 are the core material: intervention graphs, back-door identification, and the do-calculus. Section 3.4 previews the front-door strategy, which Chapter 8 develops in full, and Section 3.5 explains completeness and nonidentification, with the supporting graph machinery in Appendix A.

3.1 Intervention and Modified Graphs

In Chapter 2 we learned to read conditional independence from a DAG using d-separation. This chapter takes the next step: translating the graphical language into *identification formulas* — expressions that write the interventional distribution $P(y \mid \mathrm{do}(T{=}t))$ entirely in terms of quantities observable from data.

3.1.1 Conditioning versus Intervention

Before introducing any graph machinery, it is worth being precise about what the $\mathrm{do}(\cdot)$ operator means and why it is not the same as conditioning. The conditional distribution $P(Y \mid T{=}t)$ describes the subpopulation of units for whom T was *observed* to equal t. Because treatment assignment may be influenced by confounders, this subpopulation is not representative of the full population. The interventional distribution $P(Y \mid \mathrm{do}(T{=}t))$, by contrast, describes the population that would result if T were *externally set* to t for everyone — removing the natural process that determines T and replacing it with the forced value.

The two distributions coincide in special cases — under randomized treatment assignment, or more generally when the causal graph contains no open back-door path from T to Y (a notion made precise in Section 3.2). Note that merely *measuring* common causes does not make the unadjusted conditional $P(Y \mid T{=}t)$ equal to $P(Y \mid \mathrm{do}(T{=}t))$: measured confounders still must be adjusted for, as the remainder of this chapter develops.

The structural basis of the do-operator. In the SEM framework of Chapter 1, every variable is generated by a structural equation. For the treatment node:

$$T = f_T(\mathrm{Pa}(T), U_T),$$

where $\mathrm{Pa}(T)$ are the causal parents of T and U_T is exogenous noise. An *intervention* $\mathrm{do}(T{=}t)$ **replaces this entire equation** with the constant $T = t$. The replacement has two consequences:

1. T is no longer influenced by its parents — its dependence on $\mathrm{Pa}(T)$ and U_T is severed.
2. All other structural equations remain unchanged — the rest of the causal mechanism is unaffected. This is the *modularity* assumption of Chapter 1: intervening on T leaves the mechanisms generating every other variable intact.

This is the sense in which an intervention is a *surgical* operation on the model: it cuts out one equation and replaces it with a constant, leaving the remainder of the system intact.

Graph surgery as the graphical realization. Because $\mathrm{Pa}(T)$ no longer affects T after the intervention, every arrow pointing *into* T is removed in the post-intervention graph. The resulting graph — the *intervention graph* $\mathcal{G}_{\overline{T}}$ — represents the post-intervention world. D-separation in $\mathcal{G}_{\overline{T}}$ therefore implies conditional independence in $P(\cdot \mid \mathrm{do}(T{=}t))$, not in the original observational distribution P. This is the key link that allows graphical reasoning to answer causal questions.

Truncated factorization. Suppose the observational distribution factorizes according to $\mathcal{G}$ as

$$p(v) = \prod_{j=1}^{p} p\{v_j \mid \mathrm{Pa}(v_j)\}.$$

Under the intervention $\mathrm{do}(X{=}x)$, the factors for the intervened nodes are removed and all remaining mechanisms are preserved:

$$p(v \setminus x \mid \mathrm{do}(X{=}x)) = \prod_{V_j \notin X} p\{v_j \mid \mathrm{Pa}(v_j)\}\Big|_{X=x}.$$

Thus deleting arrows into X at the graph level corresponds to deleting the conditional factors for X at the distributional level.

3.1.2 The Graphs $\mathcal{G}_{\overline{X}}$ and $\mathcal{G}_{\underline{X}}$

Both operations modify the DAG by selectively removing edges.

Definition: The Modified Graphs $\mathcal{G}_{\overline{X}}$ and $\mathcal{G}_{\underline{X}}$

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a DAG and $X \subseteq \mathcal{V}$ a set of nodes.

- $\mathcal{G}_{\overline{X}}$ (*intervention graph*, or *mutilated graph* in Pearl's terminology) is obtained by *deleting all arrows into* X: remove every arrow $V \to X_i$ for all $X_i \in X$. This represents the intervention $\mathrm{do}(X{=}x)$, which *replaces the structural equation for* X *by the constant* x, thereby severing X's dependence on all its former parents.
- $\mathcal{G}_{\underline{X}}$ (*auxiliary graph*) is obtained by *deleting all arrows out of* X: remove every arrow $X_i \to V$ for all $X_i \in X$. This is a *technical device* used in the graphical conditions of certain do-calculus rules; it does *not* represent conditioning on X.

3.2 Back-Door Identification

The back-door criterion is the most widely used identification strategy. It gives a simple graphical condition under which conditioning on an observed set $\mathbf{S}$ is sufficient to identify the causal effect of T on Y. The formula it produces — standardization over $\mathbf{S}$ — is computable directly from the observed data distribution; Section 3.3.2 supplies the do-calculus proof.

3.2.1 The Back-Door Criterion

A *back-door path* from T to Y is a path that begins with an arrow pointing *into* T (a path beginning $T \leftarrow \cdots$). Such paths are the graphical routes along which confounding travels: when open, they can transmit association between T and Y that reflects no causal effect of T.

If an observed pretreatment set blocks every back-door path from T to Y, identification by adjustment is straightforward, subject to positivity. Measuring the relevant common causes often provides such a set, but the graphical criterion is path blocking, not an instruction to condition indiscriminately on every measured variable. The interesting case — and the one the back-door criterion is designed for — is when some confounders are *unobserved*, yet an observed set $\mathbf{S}$ can still block every spurious path.

Consider the following DAG. A drug T affects recovery Y. An unobserved variable U (socioeconomic status) is a cause of T ($U \to T$) and affects Y only indirectly, through an observed intermediate variable W on the confounding path (e.g., access to healthcare) with $U \to W$ and $W \to Y$. U does not act on Y through any other pathway.

The only back-door path is $T \leftarrow U \to W \to Y$. Although U is unobserved and cannot be adjusted for directly, W sits on this path and *is* observed. Conditioning on W blocks the path at the chain node W, and W is not a descendant of T, so $\mathbf{S} = \{W\}$ satisfies the back-door criterion. The causal effect is therefore identified even though U is never measured.

This is the key insight: the criterion does not require adjusting for the confounders themselves, only for an observed set that *intercepts* every back-door path.

Definition: Back-Door Criterion

A set of observed variables $\mathbf{S}$ with $\mathbf{S} \cap \{T, Y\} = \emptyset$ satisfies the *back-door criterion* for the causal effect of T on Y in $\mathcal{G}$ if:

1. No node in $\mathbf{S}$ is a descendant of T.
2. $\mathbf{S}$ blocks every *back-door path* from T to Y, i.e., every path that begins with an arrow pointing *into* T (a path beginning $T \leftarrow \cdots$).

Remark: Where the Name Comes From

The terminology reflects the geometry relative to T: back-door paths enter T from behind and carry *potential* non-causal association, while the directed paths $T \to \cdots \to Y$ leave through the front and represent potential causal routes through which an intervention on T may affect Y. The front-door criterion of Section 3.4 is named for the complementary idea of intercepting the causal paths at a mediator.

Remark: Equivalent Modified-Graph Check

Given the no-descendant condition, the back-door path requirement can equivalently be checked by verifying that $\mathbf{S}$ d-separates T and Y in $\mathcal{G}_{\underline{T}}$, the graph obtained by deleting arrows out of T: removing T's outgoing edges eliminates the causal paths, so only back-door paths remain. (Two scope qualifications: the criterion is sufficient but not complete — sets exist that fail it yet still support valid adjustment — and failure of every adjustment strategy does not by itself imply nonidentification, as the front-door strategy of Section 3.4 shows.)

3.2.2 The Adjustment Formula and Positivity

Throughout the identification theorems of this chapter, let $\mathcal{M}$ be a structural causal model compatible with $\mathcal{G}$, with interventions defined by modular replacement of structural equations as in Section 3.1. The theorems are statements about such models, not about arbitrary distributions that merely factorize according to $\mathcal{G}$.

Theorem 3.1 (Back-Door Adjustment Formula (Pearl 1993))**.** *If $\mathbf{S}$ satisfies the back-door criterion for the effect of T on Y in $\mathcal{G}$ and, for discrete T, the positivity condition*

$$P(T{=}t \mid \mathbf{S}{=}\mathbf{s}) \;>\; 0 \quad \textit{for } P_{\mathbf{S}}\textit{-almost every } \mathbf{s} \tag{3.1}$$

holds (for continuous T, replace Equation 3.1 by the corresponding conditional-support requirement — t must lie in the support of $T \mid \mathbf{S}{=}\mathbf{s}$ for $P_{\mathbf{S}}$-almost every $\mathbf{s}$ — together with the regularity conditions needed to define the conditional law at t; see Chapter 4), then:

$$P(y \mid \mathrm{do}(T{=}t)) \;=\; \int P(y \mid T{=}t,\; \mathbf{S}{=}\mathbf{s})\, dP(\mathbf{s}). \tag{3.2}$$

Equivalently, for any measurable function h for which the displayed expectations are well-defined and finite,

$$\mathbb{E}[h(Y) \mid \mathrm{do}(T{=}t)] \;=\; \mathbb{E}_{\mathbf{S}}\Big[\mathbb{E}[h(Y) \mid T{=}t,\; \mathbf{S}]\Big]. \tag{3.3}$$

When $\mathbf{S}$ is discrete, the integral reduces to $\sum_{\mathbf{s}} P(y \mid t, \mathbf{s})\, P(\mathbf{s})$.

Remark: Positivity Is a Data-Support Condition

Condition Equation 3.1 is the *positivity* (or *overlap*) assumption: the treatment level t must be supported within the covariate strata receiving positive population weight, since otherwise $P(y \mid t, \mathbf{s})$ is undefined on a set of positive mass. Positivity is a data-support condition, not a graphical one; both must hold for the adjustment formula to be well-defined. A population stratum in which level t is never received is a genuine identification failure; an empty cell in a finite sample is an estimation problem, not evidence of population nonpositivity. Continuous treatments require the corresponding conditional-support and regularity conditions; see Chapter 4.

The derivation uses Rules 2 and 3 of the do-calculus; the complete three-step proof is given in Section 3.3.2.

Equation Equation 3.2 is also known as the *standardization formula* or, in the epidemiological literature, the *g-formula* (Robins 1986). Standardization is the point-treatment (single-time-point) special case of Robins's g-formula, which was developed for general time-varying treatments and covariates; the multi-period version underlies the theory of dynamic treatment regimes.

The interpretation of the formula is important. Within each stratum $\mathbf{s}$, all back-door paths are blocked, so the conditional and interventional distributions coincide: $P(y \mid \mathrm{do}(t), \mathbf{s}) = P(y \mid t, \mathbf{s})$ (proved as Step 3 in Section 3.3.2). The outer integration then weights the strata by the *population marginal* distribution $P(\mathbf{S})$, rather than the treatment-conditional distribution $P(\mathbf{S} \mid T{=}t)$ that a naive comparison implicitly uses — this re-weighting, made explicit by the iterated-expectation form Equation 3.3, is what removes the confounding.

3.2.3 Invalid Adjustment Variables

Two failure modes deserve explicit warnings.

Post-Treatment Mediators

A post-treatment mediator is not, in general, a valid adjustment variable for identifying the total effect. It violates condition 1 because it is a descendant of T, and conditioning on it blocks part of the causal pathway from T to Y. Depending on the graph, conditioning on the mediator may also open, close, or leave uncontrolled noncausal paths. Consequently, the resulting adjusted functional does not in general identify the total effect. Post-treatment variables require an explicitly defined alternative estimand and additional assumptions; Chapter 8 develops the relevant direct- and indirect-effect formulations.

Pretreatment Colliders (M-Bias)

A pretreatment variable can also be invalid: if it is a *collider* on a back-door path, conditioning on it can *open* a previously blocked path rather than block one — so "include all pretreatment variables" is not a safe heuristic, and every candidate adjustment variable must be checked against the graph. The collider mechanism is developed in Chapter 2; the M-shaped graph $T \leftarrow U_1 \rightarrow C \leftarrow U_2 \rightarrow Y$, with C observed and U_1, U_2 unobserved, is the canonical example.

3.3 The Do-Calculus

General theory. Having developed back-door identification as a concrete strategy, we now introduce Pearl's three rules of do-calculus, the general transformation system of which it is one instance (Pearl 1995). Each rule uses a d-separation condition in a specified modified graph to license a transformation of an expression involving the do-operator, bringing it one step closer to a purely observational formula. Section 3.3.2 derives the back-door formula as a short sequence of these rules, and Section 3.5 presents the completeness theorem of Shpitser and Pearl (2006) and Huang and Valtorta (2006), which shows that the three rules identify *every* identifiable unconditional interventional distribution $P(y \mid \mathrm{do}(t))$ — and thereby delimits what can be identified from observational data alone.

> How to Use a Do-Calculus Rule
>
> When applying a do-calculus rule, proceed in three steps:
> 1. **Choose the relevant modified graph.** For the rule under consideration, form the graph obtained by deleting the appropriate incoming or outgoing arrows.
> 2. **Check the d-separation condition in that graph.** Set aside the probability expression for a moment and work purely graphically: list the relevant paths and determine whether they are blocked.
> 3. **Apply the corresponding algebraic transformation.** Once the graphical condition holds, use the rule to remove an observation, insert an intervention, or exchange an observation for an intervention (or vice versa).
>
> The do-calculus is therefore not a mysterious symbolic manipulation. It is a
>
> $$\text{graph modification} \longrightarrow \text{d-separation check} \longrightarrow \text{probability transformation}$$
>
> procedure.

3.3.1 The Three Transformations

In the rule statements below, the graph subscript records which incoming or outgoing arrows have been deleted before the d-separation condition is checked; the table that follows gives the exact recipe for each subscript.

Recall from Chapter 2 that P is *Markov with respect to* $\mathcal{G}$ if every d-separation in $\mathcal{G}$ implies the corresponding conditional independence under P. The do-expressions below acquire meaning through the causal semantics of Section 3.1.1: $\mathrm{do}(x)$ replaces structural equations, so the post-intervention distribution is Markov with respect to $\mathcal{G}_{\overline{X}}$.

The table below gives the recipe for each graph subscript. The subscripts should be treated as a reference system rather than memorized at the outset: first determine whether the desired operation concerns an observation, an action–observation exchange, or an action, and then use the table to select the corresponding test graph.

Subscript	**Appears in**	**Arrows deleted**
$\mathcal{G}_{\overline{X}}$	Rule 1	all arrows *into* X
$\mathcal{G}_{\overline{X}\underline{Z}}$	Rule 2	all arrows *into* X; all arrows *out of* Z
$\mathcal{G}_{\overline{X}\overline{Z}}$	Rule 3	all arrows *into* X; all arrows *into* Z

Theorem 3.2 (The Three Rules of Do-Calculus (Pearl 1995))**.** *Let $\mathcal{M}$ be a structural causal model compatible with the DAG $\mathcal{G}$ over variables $\mathcal{V}$, with interventions defined by structural-equation replacement (equivalently, the truncated factorization) as in Section 3.1.1, and let $X, Y, Z, W \subseteq \mathcal{V}$ be pairwise disjoint sets. The rules hold for every interventional distribution generated by $\mathcal{M}$. All conditional distributions in the rules below are understood at values for which the relevant regular conditional laws are defined; in the discrete case, the conditioning events are assumed to have positive probability.*

Rule 1 (Insertion/Deletion of Observations). Purpose: remove or insert an observation when it becomes irrelevant in the post-intervention graph.

$$P(y \mid \mathrm{do}(x), z, w) \;=\; P(y \mid \mathrm{do}(x), w) \quad \textit{if} \quad (Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}}}.$$

The graph $\mathcal{G}_{\overline{X}}$ is obtained by deleting all arrows into X; it represents the world after the intervention do(x)*. If Y and Z are d-separated given $\{X, W\}$ in this graph, then conditioning on Z carries no information about Y beyond what W already does.*

Rule 2 (Action/Observation Exchange). Purpose: replace an intervention by an observation, or vice versa, when the modified graph makes them equivalent.

$$P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) \;=\; P(y \mid \mathrm{do}(x), z, w) \quad \textit{if} \quad (Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}\underline{Z}}}.$$

The graph $\mathcal{G}_{\overline{X}\underline{Z}}$ is obtained by deleting arrows into X and *arrows out of Z. It is an auxiliary test graph,* not *a model of the intervention* do(z) *(which would delete arrows* into *Z). When the condition holds, intervening on Z and conditioning on Z produce the same distribution of Y; the soundness remark below explains why this deletion pattern isolates exactly the routes on which observation and intervention differ.*

Rule 3 (Insertion/Deletion of Actions). Purpose: remove or insert an intervention when it becomes irrelevant in the modified graph.

$$P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) \;=\; P(y \mid \mathrm{do}(x), w) \quad \textit{if} \quad (Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}\,\overline{Z}}},$$

stated here for the case — sufficient for every application in this course, and covering $W = \emptyset$ in particular — in which no member of Z is an ancestor of any W-node in $\mathcal{G}_{\overline{X}}$. When the condition holds, the intervention do(z) *can be deleted entirely. The general statement appears in the remark below.*

> Remark: General Form of Rule 3 (Advanced)
>
> In full generality the Rule 3 subscript is $\mathcal{G}_{\overline{X}\,\overline{Z(W)}}$, where $Z(W)$ denotes the set of Z-nodes that are *not* ancestors of any W-node in $\mathcal{G}_{\overline{X}}$. The refinement matters because conditioning on $W{=}w$ can preserve information about the natural causes of any Z-node that is ancestral to W. The Rule 3 test graph therefore retains the incoming arrows of those Z-nodes and deletes incoming arrows only for $Z(W)$; deleting all incoming arrows into Z in this case could make the graphical independence condition artificially too strong. When no Z-node is an ancestor of W — in particular whenever $W = \emptyset$ — $Z(W) = Z$ and the statement in the theorem is exact.

> Remark: What the Three Rules Cover
>
> Each rule is a *conditional independence statement in a specific intervention graph.* The three rules cover all ways of adding or removing terms from a do-expression: Rule 1 handles observations; Rule 2 exchanges actions for observations; Rule 3 inserts or deletes actions. Any sequence of such operations that converts a do-expression into a purely observational expression is an *identification proof.*

> Remark: Soundness of the Rules
>
> The three rules are *sound*: each holds for every interventional distribution generated by a structural causal model compatible with $\mathcal{G}$. The intuition for Rule 2 is instructive. Recall that the post-intervention distribution $P(\cdot \mid \mathrm{do}(x))$ is Markov with respect to $\mathcal{G}_{\overline{X}}$; the Markov factorization of $\mathcal{G}_{\overline{X}}$ is exactly P with the structural equation for X replaced by the constant x (the truncated factorization).
> Rule 2 asserts $P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) = P(y \mid \mathrm{do}(x), z, w)$ when $(Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}\underline{Z}}}$. *Why does this hold?* Observation and intervention differ only through the incoming arrows of Z: observing $Z{=}z$ leaves its natural causes active, whereas do$(Z{=}z)$ severs them. The auxiliary graph $\mathcal{G}_{\overline{X}\underline{Z}}$ deletes the *outgoing* arrows of Z, thereby removing the causal pathways that respond to the fixed value z in the same way under observation and intervention. If Y and Z are then d-separated given $\{X, W\}$, no active route through Z's natural causes remains, so the action do(z) may be exchanged for the observation z.

Rules 1 and 3 are analogous. Rule 1 uses $\mathcal{G}_{\overline{X}}$ alone to check whether Z is conditionally irrelevant for Y in the post-do(x) world. Rule 3 uses $\mathcal{G}_{\overline{X}\,\overline{Z}}$ to check whether intervening on Z has any effect on the distribution of Y in the specified context. Full proofs appear in Pearl (1995).

Rule	Statistical operation	Graphical idea
1	Add or remove an observed conditioning variable	The variable is irrelevant after the intervention
2	Exchange do(z) and observation z	Natural causes of Z no longer create a difference
3	Add or remove do(z)	Intervening on Z cannot affect Y in the modified graph

3.3.2 A Do-Calculus Derivation of Back-Door Adjustment

With the three rules in hand, we can prove the back-door adjustment formula of Section 3.2.

Proof of Theorem 3.1 via Rules 2 and 3

The derivation applies Rule 3 to the marginal of **S** and Rule 2 to the stratum-specific conditional; the substantive work is to verify that back-door condition 1 guarantees the Rule 3 separation and condition 2 the Rule 2 separation, so that the argument applies to *any* graph in which **S** satisfies the criterion. The proof manipulates $P(y \mid \mathrm{do}(t))$ in three steps.
Step 1: Introduce S by the law of total probability.

$$P(y \mid \mathrm{do}(t)) = \int P(y \mid \mathrm{do}(t),\ \mathbf{s})\, dP(\mathbf{s} \mid \mathrm{do}(t)).$$

Step 2: Simplify $P(\mathbf{s} \mid \mathrm{do}(t))$ by Rule 3. Apply Rule 3 with $X = \emptyset$, $Z = T$, $W = \emptyset$; the required graph is $\mathcal{G}_{\overline{T}}$. Because **S** contains no descendants of T in $\mathcal{G}$ (back-door condition 1), no directed path from T reaches any node in **S** in $\mathcal{G}_{\overline{T}}$ either. And since T has no parents in $\mathcal{G}_{\overline{T}}$ and the conditioning set is empty, any *non-directed* path from T to **S** must leave T through an outgoing arrow and reverse direction at some node, creating an unconditioned collider that blocks the path. Hence $(\mathbf{S} \perp\!\!\!\perp T)_{\mathcal{G}_{\overline{T}}}$, and Rule 3 gives:

$$P(\mathbf{s} \mid \mathrm{do}(t)) = P(\mathbf{s}).$$

Step 3: Simplify $P(y \mid \mathrm{do}(t), \mathbf{s})$ by Rule 2. Apply Rule 2 with $X = \emptyset$, $Z = T$, $W = \mathbf{S}$; the required graph is $\mathcal{G}_{\underline{T}}$. In $\mathcal{G}_{\underline{T}}$, all outgoing arrows of T are deleted, so the only paths between T and Y that remain are back-door paths (those beginning with an arrow into T). Back-door condition 2 states exactly that **S** blocks every such path, i.e. $(Y \perp\!\!\!\perp T \mid \mathbf{S})_{\mathcal{G}_{\underline{T}}}$. Rule 2 therefore gives:

$$P(y \mid \mathrm{do}(t), \mathbf{s}) = P(y \mid t, \mathbf{s}).$$

Combining:

$$P(y \mid \mathrm{do}(t)) = \int P(y \mid t, \mathbf{s})\, dP(\mathbf{s}). \qquad \square$$

3.4 Front-Door Identification: A Preview

The back-door criterion requires an observed set that blocks every back-door path. When the confounder is unobserved and no observed variable intercepts the back-door path, adjustment is unavailable — yet the causal effect may still be identified by routing through an observed *mediator*. The prototypical graph is $T \to M \to Y$ with an unobserved common cause U of T and Y:

Definition: Front-Door Criterion

A set of observed variables M satisfies the *front-door criterion* for the effect of T on Y in $\mathcal{G}$ if:

1. M intercepts all directed paths from T to Y;
2. there is no unblocked back-door path from T to M; and
3. all back-door paths from M to Y are blocked by T.

The identification logic runs in two stages. The link $T \to M$ is identified *directly*: by condition 2, the empty set satisfies the back-door criterion for the effect of T on M, so $P(m \mid \mathrm{do}(t)) = P(m \mid t)$. The link $M \to Y$ is identified *by adjustment for* T: by condition 3, $\{T\}$ satisfies the back-door criterion for the effect of M on Y (in the prototypical graph it blocks $M \leftarrow T \leftarrow U \to Y$), so $P(y \mid \mathrm{do}(m)) = \int P(y \mid t', m)\, dP(t')$. Chaining the two stages — condition 1 ensuring that all of T's effect on Y passes through M — yields the *front-door formula* (Pearl 1995): under support conditions made precise in Chapter 8,

$$P(y \mid \mathrm{do}(T{=}t)) \;=\; \sum_m \underbrace{P(m \mid T{=}t)}_{\text{Stage 1}} \underbrace{\sum_{t'} P(y \mid T{=}t',\ M{=}m)\, P(t')}_{\text{Stage 2}}, \tag{3.4}$$

with integrals replacing sums for continuous variables. Stage 2 is the back-door adjustment formula for $P(y \mid \mathrm{do}(M{=}m))$ with T as the adjustment variable: it averages $P(y \mid t', m)$ over the *marginal* $P(t')$, not the conditional $P(t' \mid m)$. In this sense the front-door strategy does not remove confounding directly; it *routes around it* through an observed causal pathway, never directly comparing the treatment groups on Y.

Remark: Comparing Back-Door and Front-Door

The two criteria are not nested; neither implies the other. The back-door criterion requires an observed set blocking every back-door path from T to Y — the confounders themselves need not be observed. The front-door criterion tolerates a completely unobserved T–Y confounder, but requires an observed mediator carrying *all* of the treatment effect, with its own two conditions intact.

Here the mediator is used as an identification bridge for the *total* effect of T on Y; no natural direct or indirect effect is being defined. Those decomposition estimands require different counterfactual objects and are introduced in Chapter 8. The formula demonstrates that identification need not proceed through direct adjustment for the treatment–outcome confounders. Chapter 8 returns to the front-door strategy, derives the formula formally via the do-calculus, and examines its relationship to mediation analysis.

3.5 Identification and Nonidentification

General theory (continued). This section is included to explain the scope and limits of the do-calculus. Students need not master the ID algorithm procedurally on a first reading; the main takeaway is that identifiability is a structural property of the graph, not merely a matter of algebraic cleverness. For a first reading, treat the bow-graph example and its two-model witness as the core of this section; the ADMG and semi-Markovian terminology records the formal scope of the completeness theorem.

A natural question after seeing the three rules is: are they enough? Could there be a graph where the causal effect is identifiable in principle, but none of the rules — applied in any sequence — can derive a purely observational expression? Shpitser and Pearl (2006) and, independently, Huang and Valtorta (2006) answered this definitively.

The completeness theorem is stated for *semi-Markovian* causal models represented by *acyclic directed mixed graphs* (ADMGs), in which a bidirected edge $X \leftrightarrow Y$ compactly encodes a latent common cause of X and Y (in place of the explicit unobserved nodes, such as $U \to T$ and $U \to Y$, drawn so far); Appendix A gives the relevant graph terminology and an outline of the ID algorithm. The IV DAG and front-door graphs of this course are both special cases.

Theorem 3.3 (Completeness of the Do-Calculus (Shpitser and Pearl 2006; Huang and Valtorta 2006))**.** *Let $\mathcal{G}$ be the ADMG of a semi-Markovian causal model. An unconditional interventional distribution $P(y \mid \mathrm{do}(t))$ is identifiable from the observed joint distribution P relative to $\mathcal{G}$ if and only if it can be reduced to a purely observational expression by the three rules of do-calculus together with standard probability calculus.*

> Remark: Completeness versus Search
>
> The completeness theorem is an existence statement: failure to find a do-calculus derivation by hand does not by itself establish nonidentification. For an algorithmic decision, one applies the ID algorithm. When ID fails, the resulting *hedge* provides a graphical certificate of nonidentification.

The proof is beyond the scope of this chapter; a proof sketch — the ID algorithm, districts, and the hedge obstruction — is given in Appendix A for reference and independent reading. We ground the theorem in the simplest possible nonidentifiable graph.

Example 3.1 (The Bow Graph — The Simplest Non-Identifiable Structure)**.** The *bow graph* is the ADMG with one directed edge $T \to Y$ and one bidirected edge $T \leftrightarrow Y$ (representing an unobserved U with $U \to T$ and $U \to Y$):

The single back-door path is $T \leftarrow U \to Y$. Because U is unobserved and there are no other variables in the graph, no observed set can block this path, so the back-door criterion cannot be satisfied. The front-door criterion also fails: there is no observed mediator on any directed path from T to Y. We now show *directly* that $P(y \mid \mathrm{do}(t))$ is not determined by the observed distribution $P(T, Y)$ alone, by constructing two models that agree on $P(T, Y)$ but disagree on $P(y \mid \mathrm{do}(t))$.

Following the convention for nonidentification witnesses stated in Appendix A, a model is *compatible* with $\mathcal{G}$ if each structural equation depends on a subset of the displayed parents, so a displayed arrow may carry a null effect. Let $T, Y, U \in \{0, 1\}$ with $U \sim \text{Bernoulli}(1/2)$.

Model $\mathcal{M}_1$ (pure confounding; no causal effect of T).

$$T = U, \qquad Y = U.$$

Both T and Y are driven entirely by U; the arrow $T \to Y$ carries no causal influence. Since $T = U$ and $Y = U$, the observed joint distribution is $P(T{=}0, Y{=}0) = P(T{=}1, Y{=}1) = 1/2$ and $P(T \neq Y) = 0$.

Under either intervention, the equation $T = U$ is replaced by a constant while $Y = U$ is unchanged, so

$$P_1(Y{=}1 \mid \mathrm{do}(T{=}1)) = P_1(Y{=}1 \mid \mathrm{do}(T{=}0)) = P(U{=}1) = \tfrac{1}{2},$$

and the average treatment effect is 0.

Model $\mathcal{M}_2$ (full causal effect; U acts on Y only through T).

$$T = U, \qquad Y = T.$$

Y is determined entirely by T; U affects Y only via T. Since $T = U$ again, we have $Y = T = U$, so the observed joint distribution is again $P(T{=}0, Y{=}0) = P(T{=}1, Y{=}1) = 1/2$ and $P(T \neq Y) = 0$ — *identical* to $\mathcal{M}_1$.

Under either intervention, the equation $T = U$ is replaced by a constant, and $Y = T$ deterministically, so

$$P_2(Y{=}1 \mid \mathrm{do}(T{=}1)) = 1, \qquad P_2(Y{=}1 \mid \mathrm{do}(T{=}0)) = 0,$$

and the average treatment effect is 1.

Conclusion. The two models produce identical observed distributions $P(T, Y)$, so no observational data set, however large, can distinguish $\mathcal{M}_1$ from $\mathcal{M}_2$. Yet they assign different values to $P(Y{=}1 \mid \mathrm{do}(T{=}1))$ (1/2 versus 1) and to $P(Y{=}1 \mid \mathrm{do}(T{=}0))$ (1/2 versus 0), hence different average treatment effects (0 versus 1). Therefore $P(y \mid \mathrm{do}(t))$ is *not identified* from $P(T, Y)$ in the bow graph. The completeness theorem tells us that no identification method — the do-calculus or otherwise — can overcome this obstruction: the bow graph contains a hedge, and the ID algorithm returns FAIL.

> Remark: The Practical Force of Completeness
>
> Theorem 3.3 has a crucial practical implication: *if the ID algorithm reports nonidentification, then no estimation method — however clever — can recover a target of the form $P(y \mid \mathrm{do}(t))$ from observational data alone*, given the assumed graph structure. Nonidentification is not a limitation of

a particular technique; it is a fundamental property of the causal model.
Possible responses include imposing scientifically defensible structural restrictions that go beyond the graph (e.g., linearity or monotonicity); collecting randomized, interventional, or other auxiliary data that break the obstruction; or reporting an identified alternative, such as partial-identification bounds. Two cautions are in order. An instrumental variable helps only in combination with assumptions sufficient for the particular IV estimand (Chapters 7 and 13). And switching to a different target — say, the effect among the treated rather than the average treatment effect — helps only when that alternative target is itself identified, which must be checked, not assumed.

3.6 Summary

1. Two graph operations underpin the do-calculus: $\mathcal{G}_{\overline{X}}$ deletes arrows *into* X and models the intervention $\mathrm{do}(x)$; $\mathcal{G}_{\underline{X}}$ deletes arrows *out of* X and is used as a technical device in graphical conditions for certain do-calculus steps — it does not model conditioning on X.

2. The back-door criterion identifies $P(y \mid \mathrm{do}(t))$ when an observed set $\mathbf{S}$ blocks all back-door paths and contains no descendant of T. The general formula, valid under the stated causal model and support conditions, is $\int P(y \mid t, \mathbf{s})\, dP(\mathbf{s})$ or, in iterated-expectation form, $\mathbb{E}_{\mathbf{S}}[\mathbb{E}[h(Y) \mid T{=}t, \mathbf{S}]]$ (standardization; the point-treatment g-formula).

3. The front-door criterion identifies $P(y \mid \mathrm{do}(t))$ via an observed mediator M intercepting all causal paths from T to Y, even when the confounder U is entirely unobserved, provided the three front-door conditions hold.

4. The do-calculus has three rules, each licensed by a d-separation condition in an appropriate intervention graph. Rule 1 adds/removes observations; Rule 2 exchanges actions for observations; Rule 3 inserts or deletes interventions. The back-door formula is derivable as a short sequence of these rules (Section 3.3.2); Chapter 8 derives the front-door formula the same way.

5. **Completeness.** The do-calculus is complete for identification from observational data relative to the assumed graph (Shpitser and Pearl 2006; Huang and Valtorta 2006): an unconditional interventional distribution of the form $P(y \mid \mathrm{do}(t))$ is identifiable if and only if the do-calculus can derive a purely observational expression for it. Operationally: if the complete ID procedure fails for the target, then no functional of the observed joint distribution identifies it relative to the assumed graph, and no purely observational method can recover it without additional assumptions or new data. Failure to find a do-calculus derivation *by hand*, however, proves nothing by itself, as the completeness-versus-search remark in Section 3.5 explains.

This chapter completes the graphical foundation for point identification: causal assumptions (the graph) are converted into an identification functional $\Psi(P_{\mathrm{obs}})$. Chapters 4–9 apply the machinery to specific designs — potential outcomes and randomization (Chapters 4–5), propensity scores (Chapter 6), instrumental variables (Chapter 7), mediation and the front-door strategy (Chapter 8), and sensitivity analysis (Chapter 9) — and Part III turns to the estimation of Ψ.

3.7 Problems

1. Warm-up: graph operations. Let $\mathcal{G}$ be the DAG with edges $Z \to T$, $T \to Y$, $U \to T$, $U \to Y$ (the IV DAG, with U unobserved).

(a) Draw $\mathcal{G}_{\overline{T}}$ and identify all paths from Z to Y that remain open.
(b) Draw $\mathcal{G}_{\underline{T}}$ and identify all paths from Z to Y that remain. For each path, classify each intermediate node as a collider or a non-collider (chain or fork), and state whether the path is open or closed by default (without conditioning on any variable).
(c) Show that $(Z \perp\!\!\!\perp Y)_{\mathcal{G}_{\underline{T}}}$ holds in this four-node graph (part (b) has already classified the relevant path). Explain how this conclusion combines two graphical features: every directed path from Z to Y passes through T (the exclusion restriction), and Z has no open back-door path to Y (instrument exogeneity). Why would merely omitting the direct edge $Z \to Y$ be insufficient in a richer graph?

(d) Use Rule 1 to show that $P(y \mid \mathrm{do}(t), z) = P(y \mid \mathrm{do}(t))$ in this graph: state the required d-separation condition, verify it in $\mathcal{G}_{\overline{T}}$ by tracing paths, and interpret the result (once T is set externally, the instrument carries no further information about Y).

2. Back-door practice. Consider the DAG: $X \to T$, $X \to Y$, $T \to M$, $M \to Y$, $T \to Y$, where X is observed. Parts (a)–(f) are core; part (g) is additional practice; part (h) is retrieval from Chapter 2.

(a) List all back-door paths from T to Y.
(b) Does $\{X\}$ satisfy the back-door criterion for the effect of T on Y? Write the resulting adjustment formula.
(c) Does $\{M\}$ satisfy the back-door criterion? Explain why or why not.
(d) Does $\{X, M\}$ satisfy the back-door criterion? Identify which condition of the criterion M violates, and explain why this rules it out as an adjustment variable.
(e) Even if the formal criterion issue in (d) were somehow set aside, explain why conditioning on M blocks the indirect causal pathway $T \to M \to Y$ and therefore changes the target away from the total effect. Why does the back-door criterion deliberately rule out this operation?
(f) In a finite sample from this model you observe no treated units in the stratum $X = x_0$, where $P(X{=}x_0) > 0$. Distinguish two explanations: (i) a sample empty cell with $P(T{=}1 \mid X{=}x_0) > 0$, and (ii) population nonpositivity, $P(T{=}1 \mid X{=}x_0) = 0$. Which is an identification failure for $P(y \mid \mathrm{do}(T{=}1))$, and which is an estimation problem?
(g) Now consider a different DAG with observed C_1, C_2, unobserved U, and edges $C_1 \to T$, $C_1 \to Y$, $U \to T$, $U \to C_2$, $C_2 \to Y$, $T \to Y$. List the two back-door paths from T to Y, determine for each of $\{C_1\}$, $\{C_2\}$, and $\{C_1, C_2\}$ whether the back-door criterion is satisfied, and write the adjustment formula for any valid set. What does the validity of $\{C_1, C_2\}$, despite U being unobserved, show about the role of a variable's *position* on a back-door path?
(h) (*Retrieval.*) The education–earnings DAG of Chapter 2 has edges $N \to E$, $B \to E$, $B \to Y$, $E \to Y$ (N = neighborhood, B = background, E = education, Y = earnings), all observed. Which of $\{B\}$, $\{N\}$, and $\{N, B\}$ satisfy the back-door criterion for the effect of E on Y? Write the adjustment formula for one valid set.

3. Back-door formula: proof and uniqueness.

(a) (*Second pass.*) Let $\mathcal{M}$ be a structural causal model compatible with $\mathcal{G}$, and let $\mathbf{S}$ satisfy the back-door criterion for (T, Y) in $\mathcal{G}$ with positivity. Prove that

$$P(y \mid \mathrm{do}(t)) \;=\; \int P(y \mid t, \mathbf{s})\, dP(\mathbf{s}).$$

(*Hint*: begin with $P(y \mid \mathrm{do}(t)) = \int P(y \mid \mathrm{do}(t), \mathbf{s})\, dP\{\mathbf{s} \mid \mathrm{do}(t)\}$. Apply Rule 3 with $X = \emptyset$, $Z = T$, $W = \emptyset$: its graphical condition $(\mathbf{S} \perp\!\!\!\perp T)_{\mathcal{G}_{\overline{T}}}$ follows from back-door condition 1, and Rule 3 yields $P(\mathbf{s} \mid \mathrm{do}(t)) = P(\mathbf{s})$. Then apply Rule 2 with $X = \emptyset$, $Z = T$, $W = \mathbf{S}$: its graphical condition $(Y \perp\!\!\!\perp T \mid \mathbf{S})_{\mathcal{G}_{\underline{T}}}$ follows from back-door condition 2, and Rule 2 yields $P(y \mid \mathrm{do}(t), \mathbf{s}) = P(y \mid t, \mathbf{s})$.)

(b) Suppose $\mathbf{S}_1$ and $\mathbf{S}_2$ both satisfy the back-door criterion for (T, Y) in $\mathcal{G}$ (with positivity). Deduce from part (a) that

$$\int P(y \mid t, \mathbf{s}_1)\, dP(\mathbf{s}_1) \;=\; \int P(y \mid t, \mathbf{s}_2)\, dP(\mathbf{s}_2).$$

Interpret this result: the identification target $P(y \mid \mathrm{do}(t))$ is unique even when the adjustment set is not.

4. Nonidentification in the bow graph. Consider the bow graph of Example 3.1: $T \to Y$ together with $T \leftrightarrow Y$ (an unobserved U with $U \to T$ and $U \to Y$).

(a) Show that no observed set satisfies the back-door criterion for the effect of T on Y, and that no observed mediator satisfies the front-door criterion.
(b) Verify that the two models $\mathcal{M}_1$ and $\mathcal{M}_2$ of Example 3.1 are compatible with the graph under the stated convention. Confirm that they induce the same observed distribution $P(T, Y)$, but compute different values of *both* $P(Y{=}1 \mid \mathrm{do}(T{=}1))$ and $P(Y{=}1 \mid \mathrm{do}(T{=}0))$; hence compute the average treatment effect under each model.
(c) Explain why part (b) establishes that both the interventional distribution $P(y \mid \mathrm{do}(t))$ and the average treatment effect are not identified from $P(T, Y)$. What does the completeness theorem add beyond this particular two-model witness?

Part II

Identification

Chapter 4

Potential Outcomes and Adjustment

Learning Objectives

By the end of this chapter, students should be able to:

1. Define the potential outcome $Y(t)$, state SUTVA and consistency precisely, and explain how the consistency equation $Y = Y(T)$ links observed and potential outcomes.
2. Define the ATE and ATT, explain how they relate to each other, and re-express them as functionals of the structural equation for Y.
3. Show that the linear SEM structural coefficient β equals the ATE under homogeneous effects, and explain how the ATE and ATT can diverge under treatment-effect heterogeneity when treatment selection is associated with the individual treatment effect.
4. State the ignorability assumption, distinguish its weak (single-world) and strong (joint) forms, interpret it graphically as a back-door condition, and explain why positivity is required for nonparametric adjustment over the target population.
5. Derive the adjustment formula for the ATE under consistency, weak conditional exchangeability, and positivity, and explain how the back-door criterion provides the graphical justification for conditional exchangeability.

4.1 Motivation: A Third Language for Causality

The first three chapters developed causal inference primarily in the languages of structural equations and directed acyclic graphs. Those two languages are especially effective for expressing interventions syntactically: the structural equation model shows how an intervention replaces an assignment mechanism, and the DAG shows how the same intervention deletes incoming arrows into the treatment node. In this chapter we introduce a third language: the potential outcomes framework of Neyman (1923) and Rubin (1974).

Potential outcomes provide a direct language for defining causal estimands. Quantities such as the average treatment effect, the average treatment effect on the treated, and related causal contrasts are most naturally written in terms of counterfactual outcomes $Y(1)$, $Y(0)$, and more generally $Y(t)$. For this reason, the potential outcomes framework has become standard in statistics, biostatistics, epidemiology, and much of econometrics.

Assumptions such as ignorability and exclusion can certainly be stated in this language, but DAGs and the do-operator often make their structural content more transparent by showing which paths are blocked, which variables are confounders, and which interventions are being considered. Positivity, by contrast, is a support condition on the observed-data distribution and is not encoded by any graph. Thus the two frameworks should be viewed as complementary rather than competing: potential outcomes define the causal quantities of interest, while DAGs and the do-calculus clarify identification.

The purpose of this chapter is therefore not to replace the graphical viewpoint developed earlier, but to connect it to the potential-outcomes notation that dominates much of the applied literature. We first define potential outcomes and the standard causal estimands built from them. We then show how consistency, exchangeability, and positivity together identify the average treatment effect, and how the back-door criterion provides the graphical justification for that identification.

Section 4.2 defines the potential outcome framework rigorously. Section 4.3 develops the two principal causal estimands: ATE and ATT. Section 4.4 develops the identification chain from consistency, exchangeability, and positivity to the adjustment formula, and explains how the back-door criterion provides the graphical justification. Section 4.5 situates the potential outcomes framework relative to the do-calculus and the SEM. For readers who want a formal graphical representation of counterfactual variables, Appendix B introduces single-world intervention graphs (SWIGs).

4.2 The Neyman–Rubin Potential Outcomes Framework

4.2.1 The Potential Outcome

Definition: Potential Outcome [@neyman1923application; @rubin1974estimating]

For each unit i and each possible treatment value $t \in \mathcal{T}$, the *potential outcome* $Y_i(t)$ is the value of the outcome that unit i *would have exhibited* had its treatment been set to t, possibly contrary to fact.
The collection $\{Y_i(t) : t \in \mathcal{T}\}$ is called the *schedule of potential outcomes* for unit i. In the binary case $\mathcal{T} = \{0, 1\}$, the schedule is $(Y_i(0), Y_i(1))$.

Interpretation. $Y_i(1)$ is the outcome unit i would achieve *under treatment*; $Y_i(0)$ is the outcome under *control*. Only one of these is ever observed for any given unit. The unobserved potential outcome is called the *counterfactual*.

The Fundamental Problem of Causal Inference

(Holland 1986) For a single treatment occasion, only $Y_i(T_i)$ is observed; the pair $(Y_i(0), Y_i(1))$ is never jointly observed for the same unit, so the individual-level causal effect $Y_i(1) - Y_i(0)$ is *never directly observable*. Without much stronger structural assumptions, individual treatment effects are not identified. Causal inference therefore typically targets population summaries — such as the ATE and ATT — that can be identified from cross-unit comparisons under explicit assumptions.

The structure of the problem is visible in a small table — often called the *science table* — listing the full potential-outcome schedule next to what is actually observed:

Unit	$Y_i(0)$	$Y_i(1)$	T_i	$Y_i = Y_i(T_i)$
1	observed	?	0	$Y_1(0)$
2	?	observed	1	$Y_2(1)$
3	observed	?	0	$Y_3(0)$

Each row is missing exactly one entry, and *which* entry is missing is determined by T_i: this is the fundamental problem in tabular form. The final column is the consistency equation at work, and the impossibility of filling in the "?" entries within a row is why causal inference must rely on cross-unit comparisons — which is precisely what the exchangeability assumptions of Section 4.4 will license.

4.2.2 SUTVA, Treatment Versions, and Consistency

The potential outcome $Y_i(t)$ is only well-defined if it does not depend on the treatments assigned to *other* units, and if there is a single, unambiguous version of each treatment level. These requirements are formalized as SUTVA.

Definition: SUTVA [@rubin1980randomization]

The *stable unit treatment value assumption* has two components:
1. **No interference.** The potential outcome $Y_i(t)$ depends only on unit i's own treatment, not on the treatments of other units: $Y_i(t_1, \dots, t_n) = Y_i(t_i)$.
2. **No hidden versions.** For each labeled treatment level t, all outcome-relevant treatment versions are either explicitly encoded in t or are causally equivalent for the estimand under study, so that $Y_i(t)$ refers to a unique, well-defined intervention.

Definition: Consistency

For every unit i and every treatment level t: if $T_i = t$, then $Y_i = Y_i(t)$. Compactly, the observed outcome equals the potential outcome at the treatment actually received:

$$Y_i \;=\; Y_i(T_i). \tag{4.1}$$

For binary treatment this can be written $Y_i = T_i\,Y_i(1) + (1 - T_i)\,Y_i(0)$.

Consistency is the bridge between the potential-outcome world and the observed-data world, and it is conceptually distinct from the exchangeability and positivity assumptions of Section 4.4. It is best regarded as an assumption in its own right rather than as an automatic consequence of SUTVA, although SUTVA is what makes the scalar notation $Y_i(t)$ in Equation 4.1 adequate.

Two clarifications about SUTVA itself are worth recording. First, interference does not make potential outcomes meaningless; it makes the *scalar notation* inadequate. Under interference one writes $Y_i(t_1, \dots, t_n)$, or reduces the assignment vector through an *exposure mapping*, and causal analysis proceeds with the richer index set. Second, "no hidden versions" does not require the physical treatment received by any two treated units to be literally identical; it requires that any outcome-relevant versions be either encoded in the treatment variable or causally equivalent for the estimand at hand. When these conditions fail — vaccination that provides herd immunity, or a label $T{=}1$ that covers interventions with different effects — it is the scalar consistency equation Equation 4.1 that breaks down.

4.2.3 Connection to the Do-Operator

The notation $Y(t)$ and the expression $P(Y \mid \mathrm{do}(T{=}t))$ refer to the same causal idea viewed from two complementary angles. The potential outcome $Y(t)$ denotes the outcome that would be realized for a unit if the treatment were set to t. The do-operator, by contrast, describes an intervention at the level of the data-generating mechanism: it replaces the original treatment assignment rule by the constant value t.

When $Y(t)$ is constructed by applying the same structural intervention that defines $\mathrm{do}(T{=}t)$, equality of the resulting laws holds by construction: intervening to set $T = t$ in the underlying structural equation model produces an outcome variable with the same distribution as $Y(t)$. No interference makes the scalar unit-level notation $Y_i(t)$ adequate. The separate consistency assumption links these counterfactual variables to observed outcomes when the realized treatment equals t. Thus the potential-outcomes notation and the interventional distribution are two ways of encoding the same manipulated world.

Proposition 4.1 (Potential Outcomes and the Do-Operator)**.** *Consider the structural causal model of Chapters 1–3, and construct $Y(t)$ by replacing the structural equation for T with the constant assignment $T := t$ — the same intervention that defines $\mathrm{do}(T{=}t)$. Under no interference (SUTVA), the potential outcome has the interventional distribution:*

$$P\{Y(t) \le y\} \;=\; P\{Y \le y \mid \mathrm{do}(T{=}t)\} \qquad \text{for every } y; \tag{4.2}$$

in particular, $\mathbb{E}[Y(t)] = \mathbb{E}[Y \mid \mathrm{do}(T{=}t)]$.

Proof sketch. In the intervened SCM, whose graph is $\mathcal{G}_{\overline{T}}$, the structural equation for T is replaced by $T := t$ for every unit. The structural model then determines the value of Y for unit i from t and unit i's own background variables — which is precisely what the potential outcomes framework defines as $Y_i(t)$. SUTVA's no-interference condition ensures $Y_i(t)$ does not depend on other units' treatment values, so the marginal distribution of Y in the intervened model equals the marginal distribution of $Y(t)$. By definition of the do-operator, the former is $f(y \mid \mathrm{do}(T{=}t))$. □

This proposition should be interpreted carefully. Consistency alone does not create the link between $Y(t)$ and the do-operator; the link comes from the underlying causal model that assigns meaning to the intervention $\mathrm{do}(T{=}t)$. Consistency and no interference ensure that the *observed* outcome agrees with the appropriate potential outcome at the realized treatment, while the structural model explains how the intervention generates the counterfactual world.

This equivalence matters because it lets us move freely between two notational traditions. When defining estimands such as $\mathbb{E}[Y(1) - Y(0)]$, potential-outcome notation is often most natural. When proving identification results from a graph, the do-operator is often more convenient because it integrates directly with graph surgery and the do-calculus. In general $f(y \mid \mathrm{do}(t))$ differs from $f(y \mid T{=}t)$ when T is endogenous (Chapter 1), although equality can occur through special cancellations. Potential-outcome notation marks the same distinction — $\mathbb{E}[Y(t)]$ and $\mathbb{E}[Y \mid T{=}t]$ are different expressions — but it does not by itself supply the graphical calculus for deciding when the two coincide. In a well-specified causal model these are not competing definitions, but compatible representations of the same intervention.

4.3 Causal Estimands

4.3.1 The Average Treatment Effect and Its Relatives

Definition: ATE and ATT

For a binary treatment $T \in \{0, 1\}$:

- The *average treatment effect* (ATE) is $\tau_{\mathrm{ATE}} = \mathbb{E}[Y(1) - Y(0)]$.
- The *average treatment effect on the treated* (ATT) is $\tau_{\mathrm{ATT}} = \mathbb{E}[Y(1) - Y(0) \mid T{=}1]$.

Let $\tau = Y(1) - Y(0)$ denote the unit-level effect and let $p = P(T{=}1) \in (0, 1)$. Alongside the ATT one may define the *average treatment effect on the controls*, $\tau_{\mathrm{ATC}} = \mathbb{E}[\tau \mid T{=}0]$. The law of total expectation gives the exact decomposition

$$\tau_{\mathrm{ATE}} \;=\; p\,\tau_{\mathrm{ATT}} + (1-p)\,\tau_{\mathrm{ATC}}, \tag{4.3}$$

so the ATE and ATT coincide if and only if $\tau_{\mathrm{ATT}} = \tau_{\mathrm{ATC}}$: the mean effect among the treated must equal the mean effect among the controls. Equivalently, $\tau_{\mathrm{ATT}} - \tau_{\mathrm{ATE}} = \mathrm{Cov}(T, \tau)/p$. Treatment effects may therefore be heterogeneous across units while the ATE and ATT still coincide; what matters is whether selection into treatment is related to the effect *in the mean*. Unit-level constancy of the effect is sufficient but not necessary.

None of these quantities is directly observable. The naive estimator $\hat{\tau}_{\mathrm{naive}} = \bar{Y}_{T=1} - \bar{Y}_{T=0}$ estimates the observed treatment-group contrast $D_{\mathrm{obs}} = \mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0]$, which in general differs from the ATE. Under consistency, $D_{\mathrm{obs}} = \mathbb{E}[Y(1) \mid T{=}1] - \mathbb{E}[Y(0) \mid T{=}0]$, and adding and subtracting $\mathbb{E}[Y(0) \mid T{=}1]$ yields the exact decomposition

$$D_{\mathrm{obs}} - \tau_{\mathrm{ATE}} \;=\; \underbrace{\mathbb{E}[Y(0) \mid T{=}1] - \mathbb{E}[Y(0) \mid T{=}0]}_{\text{baseline-selection bias}} + \underbrace{(\tau_{\mathrm{ATT}} - \tau_{\mathrm{ATE}})}_{\text{effect-selection difference}}\,. \tag{4.4}$$

The first term reflects baseline differences between the two groups — units who select into treatment may have fared differently even untreated — and the second reflects selection on the size of the effect. In a randomized experiment both terms vanish. Problem 2 works through a numerical case.

Remark: When ATE and ATT Coincide

ATE and ATT coincide when $\mathbb{E}[Y(t) \mid T] = \mathbb{E}[Y(t)]$ for $t \in \{0, 1\}$, i.e. when treatment assignment is mean-independent of potential outcomes. In a randomized experiment this holds by design, and both terms of the decomposition Equation 4.4 vanish.

4.3.2 Causal Estimands as Functionals of the Structural Model

The potential outcome $Y_i(t)$ was introduced as a *hypothetical*: the value unit i would have exhibited under treatment t. The SEM framework of Chapter 1 gives this hypothetical a precise generative meaning, and thereby expresses the potential-outcome estimands as functionals of the structural equation for Y.

From structural equation to potential outcome. In the SEM, the outcome is determined by a structural equation

$$Y \;=\; g(T,\, \mathbf{X},\, U_Y), \tag{4.5}$$

where U_Y collects all sources of variation in Y not already accounted for by $(T, \mathbf{X})$. The potential outcome under $\mathrm{do}(T{=}t)$ is obtained by substituting t for T while holding everything else fixed:

$$Y_i(t) \;=\; g(t,\, \mathbf{X}_i,\, U_{Y,i}). \tag{4.6}$$

Here we assume, as throughout this chapter, that $\mathbf{X}$ is *pretreatment*: no component of $\mathbf{X}$ is a descendant of T, so setting $T = t$ leaves $\mathbf{X}_i$ unchanged. If some component of $\mathbf{X}$ were itself affected by treatment, recursive substitution would replace it by its own counterfactual value $\mathbf{X}_i(t)$ in Equation 4.6. This is precisely what graph surgery does: the mutilated graph $\mathcal{G}_{\overline{T}}$ replaces the equation for T with the constant t, leaving the equation for Y unchanged. Equation Equation 4.6 makes explicit that the potential outcome is a unit-level quantity determined by g, $\mathbf{X}_i$, and the unit's own error $U_{Y,i}$ — never by the treatments of other units, which is the no-interference component of SUTVA.

ATE and ATT as structural parameters. Substituting Equation 4.6 into the definitions gives:

$$\tau_{\mathrm{ATE}} = \mathbb{E}[g(1, \mathbf{X}, U_Y) - g(0, \mathbf{X}, U_Y)]\,, \tag{4.7}$$

$$\tau_{\mathrm{ATT}} = \mathbb{E}[g(1, \mathbf{X}, U_Y) - g(0, \mathbf{X}, U_Y) \mid T{=}1]\,. \tag{4.8}$$

Both quantities are averages of the unit-level causal effect $g(1, \mathbf{X}_i, U_{Y,i}) - g(0, \mathbf{X}_i, U_{Y,i})$ over different reference populations.

The linear SEM as a special case. In the Gaussian linear SEM of Chapter 1,

$$Y \;=\; \beta T + \gamma^\top \mathbf{X} + \varepsilon, \tag{4.9}$$

the structural equation is additive and separable in T, so $Y_i(t) = \beta t + \gamma^\top \mathbf{X}_i + \varepsilon_i$ and the unit-level effect is $Y_i(1) - Y_i(0) = \beta$ for every unit. Consequently,

$$\tau_{\mathrm{ATE}} \;=\; \tau_{\mathrm{ATT}} \;=\; \beta.$$

The structural coefficient β *is* the average treatment effect: no averaging over heterogeneity is needed because there is none. This homogeneity is a special property of the linear additive model, not a general feature.

Heterogeneous effects. In the nonparametric SEM Equation 4.5, the unit-level effect $g(1, \mathbf{X}_i, U_{Y,i}) - g(0, \mathbf{X}_i, U_{Y,i})$ varies across units. The ATE averages this over the full population; the ATT averages it over the treated subpopulation. The two differ whenever treatment selection correlates with the individual effect size — i.e., whenever units who tend to benefit more also tend to self-select into treatment.

> Remark: Equivalent in Content, Different in Emphasis
>
> The SEM representation Equation 4.6 clarifies why the potential outcomes framework and the do-calculus are *equivalent in content but different in emphasis*. The do-calculus works with the interventional distribution $f(y \mid \mathrm{do}(T{=}t))$ as a population-level object and asks when it can be recovered from observational data. The potential outcomes framework works with unit-level quantities $Y_i(t)$ and asks what population summaries (ATE, ATT) are scientifically meaningful. The SEM ties the two together: $Y_i(t) = g(t, \mathbf{X}_i, U_{Y,i})$ is the unit-level object, and integrating over the distribution of $(\mathbf{X}, U_Y)$ recovers the interventional distribution.

4.4 Ignorability, Positivity, and Adjustment

4.4.1 Exchangeability: Mean, Weak, and Strong Forms

The central identifying assumption in observational studies is that treatment assignment is as good as random after conditioning on observed covariates X.

Definition 4.1 (Strong Ignorability (Rosenbaum and Rubin 1983))**.** The treatment assignment T is *strongly ignorable* given X if:

1. **Unconfoundedness:** $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$.
2. **Overlap (positivity):** $0 < P(T{=}1 \mid X) < 1$, P_X-almost surely (see the positivity taxonomy remark in Section 4.4.4 for why the almost-sure formulation is the natural one).

Definition 4.1 states the *joint*, or *strong*, form of the exchangeability condition. For identifying the ATE, strictly weaker conditions suffice, and the hierarchy is worth recording explicitly:

- **Mean exchangeability:** $\mathbb{E}[Y(t) \mid T, X] = \mathbb{E}[Y(t) \mid X]$. Together with consistency, positivity, and the relevant moment condition, this suffices to identify the mean $\mathbb{E}[Y(t)]$.
- **Weak (single-world) exchangeability:** $Y(t) \perp\!\!\!\perp T \mid X$ for each t separately. Together with consistency and positivity, this identifies the entire marginal distribution of each $Y(t)$.
- **Strong (joint) exchangeability:** $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$. This constrains the joint counterfactual pair and is the condition stated in Definition 4.1.

Each condition in the list implies those above it, and none of the implications reverses in general. The adjustment argument below uses only weak exchangeability, one treatment level at a time; the joint form becomes relevant for functionals of the joint law of $(Y(0), Y(1))$, such as the variance of the unit-level effect. Even the joint form, however, is not generally sufficient to identify such functionals: adjustment identifies the two marginal laws, but the observed data do not determine the cross-world dependence between $Y(0)$ and $Y(1)$; the covariance term in $\mathrm{Var}\{Y(1) - Y(0)\}$ requires additional cross-world restrictions, such as rank invariance, for point identification; in the absence of such restrictions one may instead derive partial-identification bounds for the joint counterfactual functional (Appendix B).

4.4.2 Adjustment Formula under Ignorability

Proposition 4.2 (Adjustment under Conditional Exchangeability). *Fix $t \in \{0, 1\}$ and assume:*

1. ***Consistency:** $T = t$ implies $Y = Y(t)$;*
2. ***Weak conditional exchangeability:** $Y(t) \perp\!\!\!\perp T \mid X$;*
3. ***Positivity:** $P(T{=}t \mid X) > 0$, P_X-almost surely;*
4. ***Integrability:** $\mathbb{E}|Y(t)| < \infty$.*

Then $\mathbb{E}[Y(t)] = \mathbb{E}_X[\mathbb{E}(Y \mid T{=}t,\, X)]$. If the assumptions hold for both $t = 0$ and $t = 1$, the ATE is identified by the standardization formula

$$\tau_{\mathrm{ATE}} \;=\; \mathbb{E}_X\big[\mathbb{E}[Y \mid T{=}1, X] - \mathbb{E}[Y \mid T{=}0, X]\big]\,. \tag{4.10}$$

Strong ignorability (Definition 4.1) implies assumptions 2–3 for both treatment levels, so it is sufficient but not necessary; note also that consistency, which the argument uses explicitly, is not part of Definition 4.1 and must be assumed alongside it. Equation Equation 4.10 is the back-door adjustment formula of Chapter 3 written in potential-outcome notation.

Proof. For the fixed treatment level t,

$$\begin{aligned}
\mathbb{E}[Y(t)] &= \mathbb{E}_X[\mathbb{E}\{Y(t) \mid X\}] && \text{(iterated expectations)}\\
&= \mathbb{E}_X[\mathbb{E}\{Y(t) \mid T{=}t,\, X\}] && \text{(weak exchangeability)}\\
&= \mathbb{E}_X[\mathbb{E}\{Y \mid T{=}t,\, X\}] && \text{(consistency).}
\end{aligned}$$

Positivity is not an algebraic step but a support condition: it guarantees that the conditional mean $\mathbb{E}(Y \mid T{=}t, X)$ is well-defined for P_X-almost every covariate value, so the outer expectation is meaningful; integrability licenses the iterated expectations. Subtracting the $t = 0$ display from the $t = 1$ display yields Equation 4.10. □

Example: A Two-Stratum Adjustment Calculation

Suppose $X \in \{0, 1\}$ with $P(X{=}1) = 0.4$ and $P(X{=}0) = 0.6$, and the observed conditional means are

$$\mathbb{E}[Y \mid T{=}1, X{=}1] = 8, \quad \mathbb{E}[Y \mid T{=}0, X{=}1] = 6,$$
$$\mathbb{E}[Y \mid T{=}1, X{=}0] = 5, \quad \mathbb{E}[Y \mid T{=}0, X{=}0] = 4.$$

Under consistency, weak exchangeability, and positivity, Equation 4.10 gives

$$\begin{aligned}\mathbb{E}[Y(1)] &= P(X{=}1)\cdot\mathbb{E}[Y \mid T{=}1, X{=}1] + P(X{=}0)\cdot\mathbb{E}[Y \mid T{=}1, X{=}0]\\ &= 0.4\times 8 + 0.6\times 5 \;=\; 6.2,\\ \mathbb{E}[Y(0)] &= P(X{=}1)\cdot\mathbb{E}[Y \mid T{=}0, X{=}1] + P(X{=}0)\cdot\mathbb{E}[Y \mid T{=}0, X{=}0]\\ &= 0.4\times 6 + 0.6\times 4 \;=\; 4.8.\end{aligned}$$

Hence $\tau_{\text{ATE}} = \mathbb{E}[Y(1) - Y(0)] = 6.2 - 4.8 = 1.4$. The adjustment formula works by comparing treated and control outcomes within each covariate stratum and then averaging those within-stratum comparisons over the marginal distribution of X.

4.4.3 Back-Door Interpretation

For a fixed treatment level t, the weak conditional-exchangeability condition

$$Y(t) \;\perp\!\!\!\perp\; T \mid X$$

states that, after conditioning on X, treatment assignment carries no residual information about the outcome that would be observed under the intervention $T{=}t$. The joint condition $\big(Y(0), Y(1)\big) \perp\!\!\!\perp T \mid X$ of Definition 4.1 is a stronger cross-world package; the adjustment formula requires only the single-world condition for each treatment level separately. In the language of DAGs, the closely related idea is that X blocks all back-door paths from T to Y.

The graphical criterion is especially useful because it makes the source of ignorability visible. If X satisfies the back-door criterion relative to (T, Y), then, under the structural causal model semantics adopted in these notes, treatment assignment is conditionally exchangeable given X, which justifies standardization, regression adjustment, and related methods. In that case the observed conditional distribution within levels of X can be used to recover the interventional distribution, leading to the adjustment formula Equation 4.10 developed above.

Proposition 4.3 (Back-Door Criterion Implies Ignorability)**.** *Under the NPSEM/SWIG semantics adopted in these notes, if X satisfies the back-door criterion for the effect of T on Y — that is,*

1. *no node in X is a descendant of T, and*
2. *X blocks every back-door path from T to Y —*

then weak (single-world) unconfoundedness $Y(t) \perp\!\!\!\perp T \mid X$ holds for all t.

Proof sketch. The target statement $Y(t) \perp\!\!\!\perp T \mid X$ is a *single-world* counterfactual independence: it is not an ordinary d-separation statement in $\mathcal{G}$ itself, but it is represented as a d-separation statement in the single-world intervention graph $\mathcal{G}(t)$ (Appendix B). The rigorous translation can be carried out directly in the SWIG $\mathcal{G}(t)$. A stronger sufficient construction is the NPSEM-IE representation

$$Y(t) \;=\; g\big(t, \operatorname{Pa}(Y)\setminus\{T\}, U_Y\big), \qquad T \;=\; f_T\big(\operatorname{Pa}(T), U_T\big),$$

with mutually independent exogenous errors (writing, for simplicity, the case in which the non-treatment parents of Y are non-descendants of T; in general $Y(t)$ is defined by recursive substitution, with any mediating parents replaced by their own counterfactual values). Under this semantics, $Y(t)$ and T depend on disjoint exogenous errors together with their own ancestors, and the back-door criterion implies the d-separation condition on $\mathcal{G}(t)$ that makes these two sets of inputs conditionally independent given X. Functional independence of inputs then transfers to the outputs, yielding $Y(t) \perp\!\!\!\perp T \mid X$ for every t. See Appendix B for the explicit SWIG derivation, and Wang et al. (2026) for a discussion of the extra cross-world constraints that NPSEM-IE imposes beyond what is strictly required to identify the ATE. ☐

Remark: Which Direction Matters in Practice

This is the direction most important for practice: a graphical adjustment set justifies the counterfactual independence needed for standardization, regression adjustment, and propensity-score methods. The converse — whether every ignorability statement corresponds to a back-door condition — is more delicate and depends on the precise graphical-counterfactual representation used; see Appendix

B.

Remark: Independent Errors — Mild or Strong? (Advanced/Optional)

The additional independent-error assumption used in the NPSEM-IE construction — mentioned as one sufficient route in Proposition 4.3, whose conclusion can also be established directly in the SWIG — deserves careful emphasis, because its strength is easy to underestimate.
Consider the causally sufficient triangle $X \to T$, $X \to Y$, $T \to Y$ written as an NPSEM,

$$X = U_X, \qquad T = f_T(X, U_T), \qquad Y = f_Y(X, T, U_Y),$$

with $U_X \perp\!\!\!\perp U_T \perp\!\!\!\perp U_Y$ mutually independent. Two lines suffice to show that error independence *implies* the unconfoundedness half of strong ignorability (Definition 4.1). Given X, the treatment $T = f_T(X, U_T)$ is a function of U_T alone, while the entire collection $\{Y(t) = f_Y(X, t, U_Y) : t \in \{0, 1\}\}$ is a function of U_Y alone; since $U_T \perp\!\!\!\perp U_Y$ conditional on X (itself a function of U_X), independence of the inputs transfers to the outputs:

$$(Y(0), Y(1)) \perp\!\!\!\perp T \mid X.$$

In this causally sufficient model — where X contains every common cause of T and Y — an analyst who writes down a system of regression equations with mutually independent errors and reads it structurally has therefore already assumed conditional ignorability given X: the assumption commonly regarded as heroic follows from the one commonly regarded as routine (Wang et al. 2026). Independent equation-specific errors alone do not remove confounding by a shared background cause omitted from X: the model of Chapter 1, with $T = f_T(U, \delta)$, $Y = f_Y(T, U, \varepsilon)$, and (U, δ, ε) mutually independent, remains confounded by the shared U.
The implication is strict, because error independence says far more. A single error term U_Y generates the *entire array* of counterfactuals $\{Y(x, t) : \text{all } x, t\}$ simultaneously, and U_T generates the entire vector $\{T(x) : \text{all } x\}$, where $T(x) = f_T(x, U_T)$ denotes the potential treatment under X set to x. Mutual independence of the errors therefore implies the *cross-world* statement

$$X \perp\!\!\!\perp (T(x), \forall x) \perp\!\!\!\perp (Y(x, t), \forall x, t)$$

(and under the canonical representation, in which the errors are the counterfactual arrays themselves, the two statements are equivalent). This ties together counterfactuals from mutually exclusive interventions — for example, $Y(x{=}0, t{=}1)$ and $Y(x{=}1, t{=}0)$ jointly — yet no unit ever inhabits two worlds, so no experiment, not even one randomizing both X and T, can test these joint constraints. Weak, single-world ignorability, by contrast, restricts one world at a time, and it can be *guaranteed by design*: randomizing T within levels of X enforces it. Even then it is not empirically verified from the observed outcomes of a single experiment; its credibility comes from the known assignment mechanism. Indeed, there exist counterfactual models — the FFRCISTGs of Robins (1986); see Richardson and Robins (2014) — in which every single-world independence holds, so that ignorability is satisfied for each t, while cross-world dependence remains; such models are observationally and experimentally indistinguishable from an NPSEM-IE.
Why, then, does error independence *feel* mild? Because in ordinary regression practice it is a statement about the observational joint distribution only, where it is close to a bookkeeping convention. What converts it into a strong assumption is autonomy: the structural reading insists that the same equations and the same error distributions persist across interventional worlds, so that U_Y is no longer "whatever residual makes the regression fit" but the unit's latent response type, held fixed across all hypothetical treatments. The extra cross-world strength is not vacuous: it supplies one ingredient in the standard identification of the natural direct and indirect effects — quantities that the weaker FFRCISTG model leaves unidentified. The mediation formula additionally requires consistency, appropriate positivity, and the relevant treatment–mediator and mediator–outcome exchangeability conditions; these are developed in Chapter 8.

Example: Labor Training Program

Following LaLonde (1986) and Dehejia and Wahba (1999), let T = receipt of job training, Y = earnings two years later, X = (age, education, prior earnings, race, marital status), all measured before training (pretreatment).
The hidden confounder U (dashed) is drawn to represent the threat discussed below; the solid graph is the working model.
The back-door path $T \leftarrow X \rightarrow Y$ is blocked by conditioning on X, leaving only the causal path $T \rightarrow Y$ (green). Ignorability holds if this DAG is correctly specified. If there is an unobserved variable U (motivation, ability) that affects both training participation and earnings, X no longer satisfies the back-door criterion, so the graph no longer justifies ignorability; weak exchangeability then generally fails (absent special cancellations).

This is one of the central translations in causal inference: the potential-outcomes notation states the assumption in terms of counterfactual independence, while the DAG states it in terms of blocked paths. Together, consistency, weak exchangeability, and positivity identify the ATE via the adjustment formula Equation 4.10, and the back-door criterion provides the graphical justification for why that formula recovers the causal effect. Appendix B provides a formal graphical representation of this connection through single-world intervention graphs (SWIGs).

4.4.4 Overlap and Positivity

Why Overlap Is Non-Negotiable for Nonparametric Adjustment

Without overlap, some covariate strata contain only treated or only control units. In those strata, $\mathbb{E}[Y \mid T{=}0, X{=}x]$ or $\mathbb{E}[Y \mid T{=}1, X{=}x]$ is unobservable, and Equation 4.10 cannot be evaluated at those values of x. Identification of the ATE fails not because the causal structure is wrong, but because the data do not span the support needed.
The ATT can remain identified under a *weaker, one-sided* overlap condition: together with consistency and the single control-arm exchangeability condition $Y(0) \perp\!\!\!\perp T \mid X$, it suffices that $P(T{=}0 \mid X{=}x) > 0$ almost surely on the support of X in the treated subpopulation, so that the control-arm regression $\mathbb{E}[Y \mid T{=}0, X{=}x]$ is well-defined at every covariate value where a treated unit appears. Full two-sided overlap is not required: strata containing only *control* units pose no obstacle for ATT, because ATT averages only over the treated population and such strata never enter that average. Strata containing only treated units, by contrast, are precisely where the one-sided condition fails.

Remark: Three Grades of Positivity Failure

Positivity is a *data-support* condition, not a causal one: unconfoundedness is a property of the assignment mechanism, but positivity is a property of the joint distribution of (T, X). Three cases should be distinguished.
First, when positivity fails only on a set of X-values of measure zero, the adjustment formula Equation 4.10 still identifies the estimand, because the offending strata contribute zero mass to the outer expectation — this is why the condition is naturally stated almost-everywhere, $P(T{=}t \mid X) > 0$ for P_X-almost every x.
Second, a stratum of *positive* probability in which treatment level t is never received — a population zero rather than a sample zero — is a genuine identification failure for the ATE: $\mathbb{E}[Y \mid T{=}t, X{=}x]$ is undefined on a set of positive mass (though a different estimand, such as the ATT under the one-sided condition above, may remain identified). The failure is one of *nonparametric* identification by adjustment over the original target population: a parametric outcome model can extrapolate into unsupported strata, but such values are identified by the model restriction, not by the observed data.
Third, small but positive treatment probabilities, or empty cells in a finite sample, are practical *estimation* problems — sparse strata and inflated variance — and do not by themselves establish population nonpositivity. Part III returns to the estimation consequences of near-positivity violations.

Remark: Continuous Treatments (Technical; Second Reading)

For continuous T, the standardization formula is understood for P_T-almost every treatment value t such that the conditional density satisfies $f_{T|X}(t \mid x) > 0$ for P_X-almost every relevant x: regular conditional distributions are unique only almost everywhere, so a positive density at a single point does not by itself select a unique version of $\mathbb{E}[Y \mid T{=}t, X{=}x]$ at that point. Pointwise identification at a prespecified treatment value t additionally requires a regularity condition that selects a unique version of the conditional law, such as continuity in t. The same qualification applies to other identification formulas evaluated at fixed treatment or mediator values, such as the front-door formula of Chapter 8.

4.5 Where the Frameworks Agree and Diverge

4.5.1 A Systematic Comparison

Task	**Potential Outcomes**	**Do-Calculus / DAG**	**SEM**
Define causal estimands (ATE, ATT)	✓ Especially natural notation	ATE via $\mathbb{E}[Y \mid \mathrm{do}(t)]$; ATT requires counterfactual augmentation (Appendix B)	Via structural equations
Encode causal assumptions	Formal counterfactual restrictions; a causal graph may be added but is not implicit in the notation	✓ Explicit directed edges	✓ Structural equations
Read conditional independence	Requires auxiliary graph or model to read off	✓ d-separation	Read from the induced graph under dependence restrictions on the exogenous inputs
Identification from observational data	Via assignment and counterfactual assumptions plus probability algebra	✓ Back-door, front-door, do-calculus	Via functional, exclusion, rank, and background-input restrictions
Likelihood construction	Typically paired with estimating equations or semiparametric models	Identifies observed-data functionals; a likelihood requires an additional statistical model	Induced once the structural functions and error laws are sufficiently specified
Cross-world assumptions (e.g. monotonicity)	✓ Natural to state	Not expressible in single-world graphs; see Appendix B	Joint counterfactuals are defined, but restrictions such as monotonicity must be imposed separately
Standard in statistics/epidemiology	✓ Dominant	Growing rapidly	Econometrics

Our Position in This Course

All three frameworks are indispensable. We use *potential outcomes* to *define* causal estimands. We use the *do-calculus and DAGs* for *identification*. The back-door criterion is the graphical condition that justifies conditional ignorability and hence the adjustment formula through which causal effects are identified. For a formal graphical representation that hosts both frameworks simultaneously, see Appendix B.
Both notations mark the interventional/observational distinction syntactically: $\mathbb{E}[Y(t)]$ and $\mathbb{E}[Y \mid T{=}t]$ are different expressions, just as $f(y \mid \mathrm{do}(t))$ and $f(y \mid T{=}t)$ are. The do-operator is preferred

in this course because it integrates directly with graph surgery and the do-calculus — the tools by which identification is actually decided. After an interventional estimand has been identified as an observed-data functional, Part III derives estimating equations, influence functions, and efficiency bounds for that functional.

4.6 Summary

1. The potential outcome $Y(t)$ is the outcome that would be observed under the intervention $T = t$. Under the structural causal semantics adopted in these notes (with no interference), $Y(t)$ has the same distribution as the outcome under the intervention $\mathrm{do}(T{=}t)$; the separate consistency assumption links counterfactuals to data through $Y_i = Y_i(T_i)$.

2. The ATE and ATT are averages of the unit-level causal effect $Y_i(1) - Y_i(0) = g(1, \mathbf{X}_i, U_{Y,i}) - g(0, \mathbf{X}_i, U_{Y,i})$ over the full population and the treated subpopulation respectively. In the linear SEM, both equal the structural coefficient β. They diverge under heterogeneous effects when treatment selection correlates with individual effect size.

3. Consistency, weak conditional exchangeability $Y(t) \perp\!\!\!\perp T \mid X$, and positivity identify the ATE via Equation 4.10 (Proposition 4.2); strong ignorability (Definition 4.1) is a sufficient package. The back-door criterion provides a sufficient graphical condition for the conditional exchangeability assumption used in adjustment formulas.

4. Single-world intervention graphs (SWIGs) (Richardson and Robins 2014) provide a formal graphical representation of counterfactual variables, making the single-world independence $Y(t) \perp\!\!\!\perp T \mid X$, for one fixed t, a d-separation statement in a single diagram that hosts both the potential outcome and the natural treatment. The joint strong-ignorability condition involves counterfactuals from two different worlds and is not represented by any single SWIG. This material is developed in Appendix B and is not required for a first-pass understanding of adjustment.

5. The frameworks are complementary: potential outcomes define estimands; the do-calculus identifies them. The back-door criterion connects the two by providing the graphical condition under which adjustment recovers the causal effect.

Remark: How Much Extra Structure Does NPSEM-IE Impose?

A binary example, following Wang et al. (2026), makes the comparison quantitative. Take X, T, and Y all binary, so the full joint distribution of $(X,\, T(x),\, Y(t,x))$ for $t, x \in \{0,1\}$ has $2^{1+2+4} - 1 = 127$ free parameters. Three nested assumptions progressively shrink this model:

- Weak (single-world) ignorability $T \perp\!\!\!\perp Y(t) \mid X$ for each t separately: 123 parameters.
- Strong cross-world ignorability $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$: 121 parameters.
- Full NPSEM-IE (all exogenous errors mutually independent): 19 parameters.

NPSEM-IE therefore imposes $123 - 19 = 104$ additional constraints beyond what is needed for ATE identification. Most are cross-world statements such as $T(X{=}0) \perp\!\!\!\perp Y(t, X{=}1)$ that remain untestable even under randomized treatment. Working under NPSEM-IE is a substantive modeling choice, not a free consequence of using structural equations.

From Ignorability to Randomization

In observational studies, ignorability must be justified by substantive knowledge encoded in a causal graph. Because unobserved confounding can never be ruled out empirically, this justification is often debated.

Randomized experiments provide a fundamentally different solution. When treatment is assigned randomly — with T denoting the randomized assignment, which coincides with treatment received in the simple full-compliance trial — $(Y(0), Y(1)) \perp\!\!\!\perp T$ holds by design, guaranteeing ignorability *without any covariate adjustment*. Chapter 5 studies randomized experiments as the canonical design where causal effects are identifiable directly from the data-generating mechanism.

Causal inference = counterfactual questions + graphical assumptions + statistical estimation.

Identification vs. Estimation

The first four chapters have focused on *identification*: whether a causal quantity $\tau = \mathbb{E}[Y(1) - Y(0)]$ can be written as $\tau = \Phi(P(Y, T, X))$ for some functional Φ of the observed distribution. Basic estimators — regression adjustment, standardization, stratification, and inverse probability weighting — are introduced in Chapters 5–6; their systematic asymptotic, influence-function, and semiparametric analysis, together with doubly robust estimators and instrumental variables, is developed in Part III.

4.7 Problems

1. SUTVA and consistency. Suppose $n = 3$ units receive binary treatments (T_1, T_2, T_3) and unit i's outcome may depend on all three treatments: $Y_i = Y_i(T_1, T_2, T_3)$.

(a) How many potential outcomes does unit 1 have? Write them out explicitly.
(b) SUTVA imposes no interference: $Y_i(T_1, T_2, T_3) = Y_i(T_i)$. How many distinct potential outcomes remain?
(c) Give a real-world example where no-interference plausibly holds and one where it plausibly fails. In the latter case, redefine the potential outcome using *one* of: the complete assignment vector, an exposure mapping, partial interference, or cluster-level treatment, and explain what additional design or structural assumption would then be needed.

2. ATE, ATT, and selection bias. Let $(Y(0), Y(1), T) \sim P$ with $P(T{=}1) = 0.5$. Suppose $\mathbb{E}[Y(1)] = 3$, $\mathbb{E}[Y(0)] = 1$, $\mathbb{E}[Y(1) \mid T{=}1] = 4$, $\mathbb{E}[Y(0) \mid T{=}1] = 2$, $\mathbb{E}[Y(1) \mid T{=}0] = 2$, $\mathbb{E}[Y(0) \mid T{=}0] = 0$.

(a) Compute the ATE and ATT. Are they equal?
(b) Compute the naive contrast $D_{\text{obs}} = \mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0]$ (the population quantity targeted by the naive estimator). Decompose the gap between this and the ATE into a selection bias term and an ATT–ATE difference.
(c) Give a sufficient graphical condition on the DAG under which the naive contrast D_{obs} equals the ATE: the *empty set* satisfies the back-door criterion, i.e. there is no open back-door path from T to Y. State the corresponding condition in potential-outcome language (marginal ignorability), and explain why the mere existence of some nonempty adjustment set is not enough.

3. Causal estimands from the structural equation. Consider the nonparametric SEM $Y = g(T, X, U_Y)$ with binary treatment $T \in \{0, 1\}$, observed covariate X, and unobserved error U_Y with $\mathbb{E}[U_Y] = 0$; any further distributional assumptions are stated in the individual parts.

(a) Write τ_{ATE} and τ_{ATT} as expectations of $g(1, X, U_Y) - g(0, X, U_Y)$ over the appropriate distribution. Under what condition do they coincide?
(b) Specialize to the linear SEM $g(t, x, u) = \alpha + \beta t + \gamma x + u$. Show that the unit-level causal effect $Y_i(1) - Y_i(0)$ is constant across all units, and hence $\tau_{\text{ATE}} = \tau_{\text{ATT}} = \beta$.
(c) Now consider the heterogeneous SEM $g(t, x, u) = (\alpha{+}u)\, t{+}\gamma x$, where $\mathbb{E}[U_Y] = 0$, $\mathrm{Var}(U_Y) = \sigma^2 < \infty$, $\mathrm{Cov}(U_Y, T) = \rho$, $P(T{=}1) = p \in (0, 1)$, and X is independent of (T, U_Y); assume a joint law compatible with these moments (the Cauchy–Schwarz inequality gives the necessary condition $|\rho| \le \sigma\sqrt{p(1-p)}$). Compute τ_{ATE} and τ_{ATT}. Show that they differ when $\rho \neq 0$, and interpret this difference in words.
(d) In part (c), show that the population OLS slope on T in the regression of Y on $(1, T, X)$ equals $\alpha + \rho/p = \tau_{\text{ATT}}$, not $\tau_{\text{ATE}} = \alpha$. Explain why: the stratum-specific difference $\mathbb{E}[Y \mid T{=}1, X] - \mathbb{E}[Y \mid T{=}0, X]$ equals $\alpha + \rho/p$ at every X (the heterogeneity carried by U_Y is not observable through X), so adjusted OLS recovers the ATT, shifted away from $\tau_{\text{ATE}} = \alpha$ by the selection term $\mathbb{E}[U_Y \mid T{=}1] = \rho/p$. What additional structure would identify τ_{ATE}? (*Hint*: randomization of T would imply $\mathbb{E}[U_Y \mid T] = 0$. Alternatively, if a richer set of pretreatment covariates W rendered $Y(t) \perp\!\!\!\perp T \mid W$, then τ_{ATE} would be identified by standardization over W, even though $\mathbb{E}[U_Y \mid T]$ need not vanish. A valid instrument does not generally identify the ATE under treatment-effect heterogeneity; in the binary-instrument, binary-treatment setting, relevance, exogeneity, exclusion, *and monotonicity* together identify the LATE for compliers — see Chapter 7.)

4. SWIGs and ignorability. *(Requires Appendix B.)* Consider the DAG: $X \to T$, $X \to Y$, $T \to Y$, with X fully observed.

(a) Construct the SWIG $\mathcal{G}(t)$ by splitting T into its random and fixed halves. Draw the result, labeling the random half, fixed half, and the potential outcome $Y(t)$.

(b) In $\mathcal{G}(t)$, identify all paths between the random half T and $Y(t)$. Apply the SWIG d-separation rules to determine which paths are blocked and which are open before conditioning.
(c) First verify, in the original DAG, that X satisfies the back-door criterion for (T, Y). Then use d-separation in $\mathcal{G}(t)$ to verify that $(Y(t) \perp\!\!\!\perp T \mid X)_{\mathcal{G}(t)}$ holds, illustrating the direction of Proposition 4.3.
(d) Now add a hidden common cause $U \to T$, $U \to Y$. Draw the revised SWIG. Does the ignorability argument still hold? State the correct conclusion and identify what additional structure (if any) would be needed for identification.

5. Positivity and target populations. Let $X \in \{0, 1\}$ with $P(X{=}1) > 0$, and suppose $P(T{=}1 \mid X{=}1) = 1$.

(a) Is the ATE identified by nonparametric adjustment over the original target population? Explain.
(b) State conditions under which the ATT may remain identified.
(c) Contrast this population positivity failure with a finite sample that happens to contain no control units with $X = 1$ even though $P(T{=}0 \mid X{=}1) > 0$ in the population.
(d) Explain what additional assumption is being relied upon if a parametric outcome model is extrapolated into the unsupported stratum.

Chapter 5

Randomization and Back-Door Adjustment

Learning Objectives

By the end of this chapter, students should be able to:

1. Explain in do-calculus terms why, within the randomized experimental law, $p_R(y \mid T{=}t) = f(y \mid \mathrm{do}(T{=}t))$, and identify the graphical feature that produces this equality.
2. State the ignorability (unconfoundedness) assumption in both potential-outcomes and graphical language, and distinguish weak from strong ignorability.
3. Work within the finite-population randomization framework: distinguish the finite-population ATE $\bar{\tau}_n$ from the superpopulation ATE, derive the exact design-based unbiasedness and variance of the difference-in-means estimator, and explain why the Neyman variance estimator is conservative.
4. Construct the fully interacted regression-adjusted estimator in a randomized experiment, distinguish its design consistency from exact unbiasedness, and explain its asymptotic precision properties under model misspecification.
5. Derive and compute the standardization (g-formula) estimator of the ATE from a contingency table or regression output.
6. Explain the method of stratification (subclassification) and describe when and why it removes confounding bias.
7. Identify Simpson's paradox in a numerical example and resolve it using the back-door criterion.
8. Articulate why propensity score methods are needed when X is high-dimensional, motivating Chapter 6.

Reading Guide

This chapter can be read at two depths. **First reading**: the do-calculus of randomization and the *statement* of Neyman's theorem (Section 5.1); the Fisher–Neyman distinction; the two routes to ignorability and the identification results (Section 5.2); outcome regression and standardization through Section 5.3's specification discussion; stratification and the design idea of the college case study (Section 5.4); Simpson's paradox (Section 5.5); and the bridge to propensity scores. **Second reading**: the proofs of Theorem 5.1 and Theorem 5.3, the superpopulation variance derivation, the detailed numerical findings of the case study, the survey-sampling precedence remark, and the analytical dissection of the lab (Section 5.6).

5.1 Randomized Experiments

5.1.1 The Do-Calculus of Randomization

The defining feature of a randomized experiment is that the analyst assigns treatment by a *known randomized mechanism*, independent of the potential outcomes and of all pretreatment characteristics. (Under complete randomization the individual assignments are not mutually independent — their total is fixed — but the assignment vector as a whole is independent of everything substantive.) In do-calculus terms, the experimental joint law is a randomized *mixture* of treatment-specific interventional laws,

$$p_R(t, y) \;=\; p_R(t)\, f(y \mid \mathrm{do}(T{=}t)),$$

where $p_R(t)$ is the known assignment probability. The experimental law is not itself the degenerate law generated by fixing T at one value — treatment remains random — but *within* it, conditioning on the assigned treatment recovers the interventional distribution: $p_R(y \mid T{=}t) = f(y \mid \mathrm{do}(T{=}t))$. This equality, which fails in general for observational data, is what the following box establishes.

> The Fundamental Equality of Randomized Experiments
>
> In a completely randomized experiment, T is assigned independently of all pre-treatment variables. In the DAG, this means there are *no arrows into T from variables in the scientific causal system*: suppressing the exogenous randomizer, the treatment node has no substantive pretreatment parents. By Rule 2 of the do-calculus, with $X = \emptyset$, $Z = T$, $W = \emptyset$, the graphical condition $(Y \perp\!\!\!\perp T)_{\mathcal{G}_{\underline{T}}}$ holds because T is isolated in $\mathcal{G}_{\underline{T}}$ (no substantive parents in $\mathcal{G}$, and the outgoing edge $T \to Y$ is deleted). Rule 2, applied within the randomized experimental law p_R, therefore gives:
>
> $$p_R(y \mid T{=}t) \;=\; f(y \mid \mathrm{do}(T{=}t)). \tag{5.1}$$
>
> Within the randomized assignment law, conditioning on the assigned treatment $T = t$ recovers the outcome distribution under the corresponding intervention. The regime subscript R matters: for an observational law the analogous equality generally fails, which is the subject of Section 5.2.

What the equality assumes beyond the graph. A precision is worth recording about *where* the extra assumptions enter. The fundamental equality Equation 5.1, read as a statement about randomized *assignment*, holds by the graph-surgery argument alone: randomization identifies the causal effect of assignment without any compliance assumption. The further conditions below are needed to *interpret* that assignment effect as the point-treatment effect developed in this chapter.

First, SUTVA in the sense of Chapter 4 — no interference (one unit's outcome does not depend on another unit's treatment) and no hidden treatment versions — which makes the scalar notation $Y_i(t)$ adequate, together with the separate consistency assumption $Y_i = Y_i(T_i)$. These are assumptions about the *science* — the treatment and the outcome-generating mechanism — and no randomization protocol, however well executed, can guarantee them.

Second, full compliance is what equates the effect of assignment with the effect of treatment received. It is worth separating the two variables that the notation T conflates: write Z_i for the *randomized assignment* and D_i for the *treatment received*. The design randomizes Z, never D. Throughout this chapter we assume full compliance, $D_i = Z_i$, and write T_i for their common value. When compliance fails, Z remains randomized and identifies the intention-to-treat effect of *assignment*, but D is no longer randomized: the right-hand side $f(y \mid D{=}t)$ describes outcomes among those who actually received t rather than those who were assigned t, and no longer matches the interventional distribution even though the design has severed all arrows into Z. The intent-to-treat versus per-protocol distinction then becomes substantive and is the subject of Chapters 7 and 13; interference and spillover settings require separate frameworks beyond the scope of these notes.

Graphical argument. In the observational DAG, there are arrows into T from observed covariates X and unobserved confounders U, creating back-door paths from T to Y. Randomization *physically* severs these arrows: assignment is determined by an exogenous randomizer, not by X or U. Call the resulting graph the *randomized-assignment graph*: T remains a random variable, but its only parent is the randomizer. This graph shares the incoming-edge structure of the mutilated graph $\mathcal{G}_{\overline{T}}$ — no arrows into T from X or U — but the two objects should not be identified: in the randomized-assignment graph T is stochastic, whereas under $\mathrm{do}(T{=}t)$ the mutilated graph fixes T at the single value t. What matters for identification is the shared feature: there are no back-door paths to block, because none exist.

Potential outcomes statement. In the potential outcomes language, complete or Bernoulli randomization with a common assignment probability implies:

$$(Y(0),\, Y(1)) \;\perp\!\!\!\perp\; T.$$

The potential outcomes are independent of assignment *unconditionally* — no covariate adjustment is needed. This is the unconditional counterpart of strong ignorability (which conditions on covariates X). The scoping matters: in stratified, blocked, or covariate-adaptive designs, where the allocation probability depends on design variables, the natural statement is *conditional* exchangeability given those variables. Throughout this chapter, "randomized experiment" means complete randomization unless stated otherwise.

5.1.2 Estimation of the ATE in a Randomized Experiment

Lemma 5.1 (Identification under Complete Randomization)**.** *Under complete randomization,*

$$\mathbb{E}[Y(t)] \;=\; \mathbb{E}[Y \mid T{=}t], \qquad t \in \{0,1\}. \tag{5.2}$$

Proof. By the potential-outcomes/do-operator correspondence of Chapter 4, $Y(t)$ has the same distribution as Y under the interventional distribution $f(y \mid \mathrm{do}(T{=}t))$, so $\mathbb{E}[Y(t)] = \mathbb{E}[Y \mid \mathrm{do}(T{=}t)]$. In a completely randomized experiment, the treatment node has no parents in the DAG, so Equation 5.1 gives $p_R(y \mid T{=}t) = f(y \mid \mathrm{do}(T{=}t))$. Therefore, within the experimental law, $\mathbb{E}[Y(t)] = \mathbb{E}[Y \mid T{=}t]$. □

A word on what probability law the expectations refer to. Complete randomization by itself does not conjure an external superpopulation: in this lemma, $\mathbb{E}$ is taken either with respect to a randomly selected unit from the trial population (the empirical finite-population distribution supplying the law) or under the two-phase superpopulation model discussed below, in which units are sampled before being randomized. Randomization supports the exact finite-population statements of Theorem 5.1; it does not make a nonrepresentative trial sample representative of any wider population.

Under complete randomization, the ATE is estimated by the *difference-in-means* estimator:

$$\hat{\tau}_{\mathrm{DIM}} = \bar{Y}_1 - \bar{Y}_0 = \frac{1}{n_1}\sum_{i:\,T_i=1} Y_i - \frac{1}{n_0}\sum_{i:\,T_i=0} Y_i,$$

where n_1 and n_0 are the numbers of treated and control units.

Two targets, two frameworks. Before stating the exact properties of $\hat{\tau}_{\mathrm{DIM}}$, it is worth separating two estimands that the generic symbol τ_{ATE} tends to blur. The identity Equation 5.2 may be read either with respect to a uniformly selected unit from the trial population or under the two-phase superpopulation model. The theorem below instead *conditions on the realized finite population*: it treats the n enrolled units' potential outcomes as fixed numbers, targets the finite-population ATE

$$\bar{\tau}_n \;=\; \frac{1}{n}\sum_{i=1}^{n}\{Y_i(1) - Y_i(0)\},$$

and derives all randomness from the assignment mechanism alone. Randomization by itself licenses design-based inference for $\bar{\tau}_n$ — the effect among the units actually randomized (internal validity). Extending conclusions to a wider population requires an additional random sampling or transportability argument (external validity).

Theorem 5.1 (Neyman's Theorem for the Difference-in-Means Estimator)**.** *Under complete randomization, let $\mathcal{F}_n = \{(Y_i(0), Y_i(1)) : i = 1, \dots, n\}$ denote the set of potential outcomes, treated as fixed, and let the expectations and variances below be taken with respect to the randomization distribution of the assignment vector. Set $f_1 = n_1/n$ and let*

$$\bar{Y}(t) = \frac{1}{n}\sum_{i=1}^{n} Y_i(t), \qquad S_t^2 = \frac{1}{n-1}\sum_{i=1}^{n}\big(Y_i(t) - \bar{Y}(t)\big)^2,$$

$$S_{01} = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i(0) - \bar{Y}(0))(Y_i(1) - \bar{Y}(1)).$$

Then the following hold.

1. ***Unbiasedness.*** $\mathbb{E}[\hat{\tau}_{\text{DIM}} \mid \mathcal{F}_n] = \bar{Y}(1) - \bar{Y}(0) = \bar{\tau}_n$.
2. ***Design variance.***

$$\operatorname{Var}(\hat{\tau}_{\text{DIM}} \mid \mathcal{F}_n) \;=\; \frac{1}{n}\left(\frac{n_0}{n_1}\, S_1^2 + \frac{n_1}{n_0}\, S_0^2 + 2\, S_{01}\right) \;=\; \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\tau^2}{n}, \tag{5.3}$$

where $S_\tau^2 = (n-1)^{-1}\sum_{i=1}^n (\tau_i - \bar{\tau}_n)^2$ *is the finite-population variance of the unit-level effects* $\tau_i = Y_i(1) - Y_i(0)$.

3. ***Conservative variance estimator.*** *Let* $\hat{S}_1^2$ *and* $\hat{S}_0^2$ *be the within-arm sample variances. Then*

$$\hat{V} \;=\; \frac{\hat{S}_1^2}{n_1} + \frac{\hat{S}_0^2}{n_0} \tag{5.4}$$

satisfies $\mathbb{E}[\hat{V} \mid \mathcal{F}_n] - \operatorname{Var}(\hat{\tau}_{\text{DIM}} \mid \mathcal{F}_n) = S_\tau^2/n \geq 0$, *so* $\hat{V}$ *is conservative. It is unbiased in the special case of a constant unit-level effect, in which case* $S_\tau^2 = 0$.

Proof (second reading)

We work throughout conditionally on $\mathcal{F}_n$. Under complete randomization, the marginal assignment probabilities satisfy $\mathbb{E}[T_i \mid \mathcal{F}_n] = n_1/n = f_1$ for all i, and the pairwise covariances satisfy

$$\operatorname{Cov}(T_i, T_j \mid \mathcal{F}_n) \;=\; \begin{cases} f_1(1-f_1) & i = j, \\ -\dfrac{f_1(1-f_1)}{n-1} & i \neq j. \end{cases} \tag{5.5}$$

The $i \neq j$ expression follows from $\mathbb{E}[T_i T_j \mid \mathcal{F}_n] = n_1(n_1-1)/[n(n-1)]$, a consequence of sampling n_1 treated units without replacement. By SUTVA and consistency, $Y_i = T_i Y_i(1) + (1-T_i)Y_i(0)$.
(i) Unbiasedness.

$$\mathbb{E}\left[\frac{1}{n_1}\sum_{i=1}^n T_i Y_i(1) \,\middle|\, \mathcal{F}_n\right] \;=\; \frac{1}{n_1}\sum_{i=1}^n \frac{n_1}{n}\, Y_i(1) \;=\; \bar{Y}(1),$$

and similarly for the control arm. Subtracting gives (i).
(ii) Design variance. Write $Z_i(t) = Y_i(t) - \bar{Y}(t)$; since $Z_\cdot(t)$ differs from $Y_\cdot(t)$ by a constant, substituting Z for Y does not change variances or covariances. Using Equation 5.5,

$$\begin{aligned}
\operatorname{Var}\left(\frac{1}{n_1}\sum_{i=1}^n T_i Y_i(1) \,\middle|\, \mathcal{F}_n\right) &= \frac{1}{n_1^2}\sum_{i,j=1}^n \operatorname{Cov}(T_i, T_j \mid \mathcal{F}_n)\, Z_i(1) Z_j(1) \\
&= \frac{f_1(1-f_1)}{n_1^2}\left[\sum_{i=1}^n Z_i(1)^2 - \frac{1}{n-1}\sum_{i \neq j} Z_i(1) Z_j(1)\right] \\
&= \frac{f_1(1-f_1)}{n_1^2} \cdot \frac{n}{n-1}\sum_{i=1}^n Z_i(1)^2 \;=\; \frac{1}{n} \cdot \frac{n_0}{n_1}\, S_1^2,
\end{aligned}$$

where the third equality uses $\sum_i Z_i(1) = 0 \Rightarrow \sum_{i \neq j} Z_i(1) Z_j(1) = -\sum_i Z_i(1)^2$. The symmetric calculation for the control arm gives $n^{-1}(n_1/n_0) S_0^2$. For the cross-arm covariance,

$$\operatorname{Cov}\left(\tfrac{1}{n_1}\sum_i T_i Y_i(1),\; \tfrac{1}{n_0}\sum_j (1-T_j) Y_j(0) \,\middle|\, \mathcal{F}_n\right) = -\frac{f_1(1-f_1)}{n_1 n_0} \cdot \frac{n}{n-1}\sum_{i=1}^n Z_i(1) Z_i(0) \;=\; -\frac{S_{01}}{n}.$$

Combining gives the first form in Equation 5.3. For the second form, rearrange using $n_0/(n_1 n) = 1/n_1 - 1/n$ and $n_1/(n_0 n) = 1/n_0 - 1/n$:

$$\operatorname{Var}(\hat{\tau}_{\text{DIM}} \mid \mathcal{F}_n) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{1}{n}(S_1^2 + S_0^2 - 2S_{01}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\tau^2}{n},$$

using the algebraic identity $S_\tau^2 = S_1^2 + S_0^2 - 2S_{01}$.
(iii) Conservative variance estimator. Under complete randomization, the treated units form a simple random sample without replacement of size n_1, so standard finite-population theory gives $\mathbb{E}[\hat{S}_1^2 \mid \mathcal{F}_n] = S_1^2$; symmetrically for the control arm. Therefore $\mathbb{E}[\hat{V} \mid \mathcal{F}_n] = S_1^2/n_1 + S_0^2/n_0$, and subtracting the second form of Equation 5.3 yields $S_\tau^2/n \geq 0$. Equality holds iff $S_\tau^2 = 0$. $\square$

Remark: Why the Conservatism Cannot Be Removed

The correction term S_τ^2/n in Equation 5.3 depends on the unit-level effect variance. Computing S_τ^2 would require both $Y_i(0)$ and $Y_i(1)$ for every unit — precisely the information the fundamental problem of causal inference forbids us from observing. The exact randomization variance is therefore not point identified without additional restrictions on the joint potential-outcome schedule. The estimator $\widehat{V}$ is the simple, conventional conservative choice; it is not the sharpest one — for instance, bounds on the unidentified cross-term S_{01} implied by the marginal potential-outcome distributions yield tighter conservative variance procedures. Under a constant treatment effect, the correction vanishes and $\widehat{V}$ becomes exact.

Remark: Superpopulation Variance as a Two-Phase Corollary

The superpopulation variance formula that appears in most textbooks follows as a direct corollary of Theorem 5.1. Suppose the units are i.i.d. draws from a superpopulation in which $(Y_i(0), Y_i(1))$ has variances (σ_0^2, σ_1^2) with covariance σ_{01}. The sampling structure has two phases: (1) draw $\mathcal{F}_n$ i.i.d. from the superpopulation; (2) assign treatment by complete randomization given $\mathcal{F}_n$. The second phase is precisely the randomization distribution of Theorem 5.1.
By the law of total variance,

$$\mathrm{Var}(\hat{\tau}_{\mathrm{DIM}}) \;=\; \mathrm{Var}\big(\mathbb{E}[\hat{\tau}_{\mathrm{DIM}} \mid \mathcal{F}_n]\big) \;+\; \mathbb{E}\big[\mathrm{Var}(\hat{\tau}_{\mathrm{DIM}} \mid \mathcal{F}_n)\big].$$

The first term is $\mathrm{Var}(\bar{\tau}_n) = (\sigma_1^2 + \sigma_0^2 - 2\sigma_{01})/n$. The second is obtained by taking the superpopulation expectation of the design variance, using $\mathbb{E}[S_t^2] = \sigma_t^2$ and $\mathbb{E}[S_{01}] = \sigma_{01}$. A short rearrangement gives

$$\mathrm{Var}(\hat{\tau}_{\mathrm{DIM}}) \;=\; \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}. \tag{5.6}$$

The Neyman correction S_τ^2/n is exactly cancelled by the first-phase variance of $\bar{\tau}_n$ when one averages over superpopulation draws. Equation Equation 5.6 is the formula most commonly reported in textbooks; Theorem 5.1(ii) is its finite-population refinement. See Ding (2024) for a full treatment.

5.1.3 Fisher's Randomization Tests vs. Neyman's Repeated-Randomization Estimation

Both Fisher and Neyman take the randomization mechanism seriously, but they use it for different inferential purposes.

Two inferential targets. Fisher's randomization framework is built around the *sharp null hypothesis* of no treatment effect for any unit, $Y_i(1) = Y_i(0)$ for all i. Under this null, every missing potential outcome is known once the observed outcomes are given. This makes it possible to compute the exact randomization distribution of a chosen test statistic under repeated reassignments of treatment. The exactness rests on four ingredients: a fully known assignment mechanism, completely observed outcomes, a well-defined sharp null, and SUTVA's no-interference clause; remove any one and the enumeration argument breaks.

Neyman's framework instead focuses on *average causal effects*. Its design-based target is the finite-population ATE $\bar{\tau}_n$, with the superpopulation ATE arising under the two-phase sampling framework. Neyman's approach studies the repeated-sampling behavior of estimators such as the difference in sample means.

Practical distinction. Fisher's framework is naturally aligned with exact hypothesis testing under a sharp null, whereas Neyman's is naturally aligned with point estimation and uncertainty quantification. The two approaches are complementary rather than contradictory: both rely on the treatment assignment mechanism, but they answer different questions.

Remark: Randomization Licenses Inference Without Population Assumptions

A fundamental advantage of randomization is that it licenses causal inference *without any assumption about a population model*. In Fisher's design-based framework, the potential outcomes are treated as fixed numbers attached to the n units in front of the researcher. The only source of randomness is the assignment vector, drawn uniformly at random from the set of all $\binom{n}{n_1}$ possible assignments. This is a finite, enumerable set, and probability statements refer to this mechanism — not to repeated draws from a population.
The practical consequence is striking: the i.i.d. assumption is not required. Units need not be a random sample from any population; they can be a convenience sample, a census, or a small targeted group. The randomization itself — the physical act of assigning treatment by lottery — is what justifies inference. The correct contrast with observational studies is about the *treatment-assignment mechanism*, not about randomness in general. Design-based inference without a superpopulation model is available whenever a known physical randomization anchors the probability statements — probability sampling without replacement in surveys is the canonical non-causal example. What an observational causal study lacks is a known law for treatment assignment. Two further caveats keep the comparison honest: randomization of treatment does not make a convenience sample representative of any wider population, and observational data collected by probability sampling still face the identification problem even though their sampling inference is design-based.

Remark: Fisher's Sharp Null — Unidentifiable Effects, Exactly Testable Hypothesis

A seemingly paradoxical feature of the Fisher framework is that individual causal effects τ_i are *fundamentally unidentifiable* for any unit — only one of the two potential outcomes is ever observed — yet the sharp null $H_0 : Y_i(1) = Y_i(0)$ for all i is *exactly testable* via randomization.
The resolution is that the sharp null completely specifies the joint potential outcome table. Under H_0, every unit's two potential outcomes are equal, so the observed outcome equals both regardless of assignment. The full $2n$-vector is therefore known from the observed data under H_0, even though only n entries are observed. The permutation distribution of $\hat{\tau}_{\text{DIM}}$ under all $\binom{n}{n_1}$ assignments is then completely determined, and the p-value is the fraction of assignments giving a statistic as extreme as the observed one — no estimation step required.
Individual effects are untestable because any specific value of τ_i would be a statement about one unobserved potential outcome of one unit. The sharp null is testable because it is a *joint* restriction on all n units that, when true, makes the entire data vector fully observed under any assignment. This illustrates a broader principle: untestability of individual-level quantities does not imply untestability of population-level or joint hypotheses derived from them.

Remark: Further Reading on the Randomization Approach

Ding (2024) provides a comprehensive graduate-level treatment of the randomization framework, covering complete randomization, stratified experiments, regression adjustment, and beyond. The asymptotic theory underlying large-sample confidence intervals for $\hat{\tau}_{\text{DIM}}$ — specifically, a finite-population central limit theorem that requires neither i.i.d. observations nor a superpopulation model — is worked out rigorously by Li and Ding (2017). For inference about *heterogeneity* of treatment effects across units, Ding et al. (2016) develop randomization-based tests and confidence intervals under the Neyman–Rubin model.

5.2 Ignorability

5.2.1 Two Routes to the Same Estimand

Section 5.1 established that a randomized experiment achieves the fundamental equality and that the difference-in-means estimator is exactly unbiased for the finite-population ATE. The mechanism is graphical: randomization physically severs every arrow into T, so no back-door path from T to Y exists.

In observational studies, no such physical severance occurs. The treatment was not assigned by the researcher; it was selected — by patients, firms, governments, or individuals — based on their characteristics.

Those characteristics may include unmeasured variables U that also affect the outcome, opening back-door paths that observational data alone cannot close.

The key insight of this chapter is that, *when defined for the same target population*, both settings can target the same estimand — the ATE — but reach it by fundamentally different routes:

	Randomized experiment	**Observational study**
How ignorability arises	By design: the researcher severs all arrows into T	By assumption: the analyst asserts that X is a sufficient pretreatment adjustment set
Status of $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$	*Guaranteed* — holds unconditionally (X not needed)	*Assumed* — requires X to block every back-door path
Credibility	As strong as the randomization protocol	As strong as substantive knowledge of the data-generating process
Testability	Can audit the assignment mechanism	Cannot be verified from the data alone
Failure mode	Protocol violations, non-compliance	An unmeasured common cause opening a back-door path that no observed set blocks

The central pedagogical point. When statisticians speak of "controlling for confounders" or "adjusting for X," they are invoking the observational route. Statistical adjustment can *implement* ignorability once it is assumed, but it cannot *create* it. No amount of covariate adjustment — however flexible the model — can close a back-door path that no observed pretreatment variable intercepts. (When an observed variable does lie on the path, adjustment can block it even though the common cause itself is unmeasured — the graph $T \leftarrow U \rightarrow W \rightarrow Y$ of Chapter 3, with W observed, is the canonical example, and the case study of Section 5.4.4 is a real one.) This is why a well-conducted randomized experiment is considered more credible than even the most carefully adjusted observational study.

In observational studies, the three methods developed in Section 5.3–Section 5.4 all *implement* the observational route — they assume ignorability and then estimate the identified quantity. The same regression machinery reappears in Section 5.3.5 for a different purpose: *precision improvement* under randomization, where identification is supplied by the design and no ignorability assumption is needed. Chapter 6 extends the observational route with the propensity score. Chapter 7 abandons the ignorability route entirely when unobserved confounders are present.

5.2.2 Terminology

The formal definition of strong ignorability was given in Chapter 4: joint unconfoundedness $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$, together with overlap $0 < P(T{=}1 \mid X) < 1$ almost surely (Rosenbaum and Rubin 1983). The pointwise (weak) form $Y(t) \perp\!\!\!\perp T \mid X$ for each t separately is implied by the joint statement and is what is actually used to identify $\mathbb{E}[Y(t)]$ for a single t, as in Lemma 5.2 below. Functionals of the joint law of $(Y(0), Y(1))$ require additional cross-world restrictions; strong joint exchangeability constrains the assignment mechanism for the pair but is not by itself generally sufficient to identify that dependence.

Several closely related names are used in the literature: *unconfoundedness*, *selection on observables*, *no unmeasured confounders*, and *conditional exchangeability*. Conventions vary across authors — over weak versus joint exchangeability, mean versus distributional forms, and whether positivity or consistency is bundled into the package — but all express the same core idea. In a randomized experiment, the joint independence holds *unconditionally* — X is not needed, and the condition reduces to $(Y(0), Y(1)) \perp\!\!\!\perp T$, the strongest possible form.

5.2.3 Three Languages for Ignorability

Chapter 4 established that under the structural causal model semantics adopted in these notes, a set X satisfying the back-door criterion implies ignorability. The condition has parallel sufficient formulations in each of the three languages of the causal trinity.

Language	Statement of ignorability
SEM	In the point-treatment SEM $Y = f_Y(T, X, U_Y)$ with $T = f_T(X, U_T)$, let U_Y denote the *entire collection* of exogenous inputs determining $Y(t)$. A sufficient condition is $U_Y \perp\!\!\!\perp T \mid X$ — which follows, for example, from $U_T \perp\!\!\!\perp U_Y \mid X$. This presumes no shared background cause remains outside X: independence of an equation-specific noise term alone is *not* sufficient if an additional latent variable enters both mechanisms, as in the confounded SEM of Chapter 1, where $\varepsilon \perp\!\!\!\perp T$ holds yet U confounds.
DAG / do-calculus	X satisfies the back-door criterion for the effect of T on Y: X blocks all back-door paths from T to Y and contains no descendant of T (Chapter 3).
Potential outcomes	$Y(t) \perp\!\!\!\perp T \mid X$ for $t \in \{0, 1\}$ (weak ignorability; see the exchangeability hierarchy in Chapter 4).

These three rows are *corresponding sufficient formulations* under the structural causal model semantics adopted in these notes, together with consistency and no interference — not literally equivalent statements. In particular, the SEM row is sufficient in the displayed point-treatment model, but a general $Y(t)$ can depend on multiple exogenous inputs through mediators and other ancestors, and strict logical equivalence across the rows requires further bridge assumptions (Appendix B). In practice, which language is most useful depends on the task: the SEM formulation is natural when writing down a structural model and asking whether any omitted variable enters both equations; the DAG formulation is natural for reading off the criterion from a graph; the potential outcomes formulation is natural for writing down estimators and proofs.

5.2.4 What Ignorability Requires

Ignorability is a *substantive*, not a statistical, assumption. It does not require observing every common cause itself: it requires an observed *pretreatment* set X that blocks every back-door path from T to Y. An unmeasured common cause defeats adjustment only when no observed set intercepts all of the paths it creates; a variable lying downstream of the unobserved cause can block the relevant path even though the cause itself is never seen. The shorthand "no unmeasured confounders" should be read in this path-blocking sense.

Ignorability Is Untestable from Observational Data

From the observed law of (Y, T, X) alone, without additional structure, conditional exchangeability is not testable. If an unobserved variable U *directly* affects both T and Y, the two-edge back-door path $T \leftarrow U \rightarrow Y$ contains no intermediate vertex, and conditioning on observed variables cannot block it. Auxiliary variables can help — negative controls, proxies, or multiple outcomes may imply *falsifiable* restrictions on an enlarged causal model, as the case study of Section 5.4.4 illustrates — but only under their own substantive assumptions, and they do not directly verify exchangeability. Sensitivity analysis (Rosenbaum 2002) can quantify how large such an unobserved confounder would have to be to overturn a conclusion, but cannot confirm the assumption itself.

5.2.5 From Ignorability to Identification

Chapter 4 established the back-door adjustment formula from consistency, weak conditional exchangeability, positivity, and integrability. We restate the result here as a lemma, with each step of the proof traced back to the assumption it invokes. The explicit tracing makes the failure mode of each assumption transparent: a positivity failure may be visible empirically and may motivate re-targeting to a supported subpopulation, but it prevents nonparametric identification of the original target; weak ignorability and consistency can fail without an equally direct empirical signature.

Lemma 5.2 (Identification of $\mathbb{E}\{Y(t)\}$)**.** *Suppose: (i)* weak ignorability*:* $Y(t) \perp\!\!\!\perp T \mid X$*; (ii)* overlap*:* $P(T{=}t \mid X{=}x) > 0$ *for* P_X*-almost every* x*; (iii)* consistency*:* $Y = Y(T)$*; (iv)* integrability*:* $\mathbb{E}|Y(t)| < \infty$*. Then:*

$$\mathbb{E}[Y(t)] = \mathbb{E}\Big[\mathbb{E}[Y \mid T{=}t,\ X]\Big] = \int \mathbb{E}[Y \mid T{=}t,\ X{=}x]\, p(x)\, dx. \tag{5.7}$$

Proof. Apply the law of iterated expectations, then use each assumption in turn:

$$\begin{aligned}
\mathbb{E}[Y(t)] &= \mathbb{E}\Big[\mathbb{E}[Y(t) \mid X]\Big] && \text{(LIE)}\\
&= \mathbb{E}\Big[\mathbb{E}[Y(t) \mid T{=}t,\ X]\Big] && \text{(weak ignorability)}\\
&= \mathbb{E}\Big[\mathbb{E}[Y \mid T{=}t,\ X]\Big] && \text{(consistency).}
\end{aligned}$$

The step labeled "weak ignorability" uses $Y(t) \perp\!\!\!\perp T \mid X$, which gives $\mathbb{E}[Y(t) \mid X] = \mathbb{E}[Y(t) \mid T{=}t, X]$ for any value of T that has positive probability given X — and overlap guarantees this almost surely, so $\mathbb{E}[Y \mid T{=}t, X{=}x]$ is well-defined everywhere in the support of X. The step labeled "consistency" uses $Y = Y(t)$ on the event $\{T = t\}$. Integrability licenses the law of iterated expectations. □

Remark: Two Languages, One Formula

Lemma 5.2 and the back-door theorem of Chapter 3 are two proofs of the same identification formula, written in different languages.
The Chapter 3 theorem works in the graphical language. The assumption is that the adjustment set $\mathbf{S}$ satisfies the back-door criterion in the DAG. The conclusion is $P(y \mid \mathrm{do}(T{=}t)) = \int P(y \mid T{=}t, \mathbf{S}{=}\mathbf{s})\, dP(\mathbf{s})$, proved by applying Rule 3 (checked in $\mathcal{G}_{\overline{T}}$) and Rule 2 (checked in $\mathcal{G}_{\underline{T}}$).
Lemma 5.2 works in the potential-outcomes language. The assumptions are weak ignorability, overlap, and consistency. The conclusion is $\mathbb{E}[Y(t)] = \mathbb{E}[\mathbb{E}[Y \mid T{=}t, X]]$, proved via the law of iterated expectations.
Why they agree. The bridge is the structural causal semantics adopted in these notes: under the NPSEM together with SUTVA's no-interference clause, $Y(t)$ has the same distribution as Y under $\mathrm{do}(T{=}t)$, so the two targets are the same quantity. Consistency plays a different role: it connects the observed Y, within the stratum $\{T = t\}$, to the realized potential outcome. On the graphical side, when $\mathbf{S} = X$ satisfies the back-door criterion, it can be shown that $Y(t) \perp\!\!\!\perp T \mid X$ holds in the corresponding nonparametric structural equation model (see also Pearl 2009). Chapter 3 proves the formula from the graph; this chapter proves it from the potential outcomes.

The ATE and ATT follow immediately as corollaries.

Theorem 5.2 (Identification of the ATE and ATT)**.** *Under the conditions of Lemma 5.2, imposed for both treatment levels* $t \in \{0, 1\}$*:*

$$\tau_{\text{ATE}} = \int [\mathbb{E}(Y \mid T{=}1, X{=}x) - \mathbb{E}(Y \mid T{=}0, X{=}x)]\, p(x)\, dx, \tag{5.8}$$

$$\tau_{\text{ATT}} = \int [\mathbb{E}(Y \mid T{=}1, X{=}x) - \mathbb{E}(Y \mid T{=}0, X{=}x)]\, p(x \mid T{=}1)\, dx. \tag{5.9}$$

Proof. For the ATE, apply Lemma 5.2 separately to $t = 1$ and $t = 0$ and subtract.

For the ATT, the lemma cannot simply be re-weighted — it is an unconditional statement, whereas the ATT conditions on $T{=}1$ — so we argue conditionally. By the law of iterated expectations within the treated subpopulation,

$$\tau_{\text{ATT}} = \mathbb{E}_{X \mid T=1}\Big[\mathbb{E}\{Y(1) - Y(0) \mid T{=}1, X\}\Big].$$

For the treated potential outcome, consistency alone gives $\mathbb{E}\{Y(1) \mid T{=}1, X\} = \mathbb{E}(Y \mid T{=}1, X)$. For the counterfactual mean,

$$\begin{aligned}
\mathbb{E}\{Y(0) \mid T{=}1, X\} &= \mathbb{E}\{Y(0) \mid X\} && \text{by } Y(0) \perp\!\!\!\perp T \mid X\\
&= \mathbb{E}\{Y(0) \mid T{=}0, X\} && \text{by } Y(0) \perp\!\!\!\perp T \mid X \text{ again}\\
&= \mathbb{E}(Y \mid T{=}0, X) && \text{by consistency,}
\end{aligned}$$

where control-arm positivity ensures the conditional expectation in the second line is well-defined where it is needed. Substituting and integrating against $p(x \mid T{=}1)$ yields Equation 5.9. □

> Remark: The ATT Needs Less than the ATE
>
> The proof reveals that the ATT is identified under a strictly weaker package than the ATE. Because $\mathbb{E}[Y(1) \mid T{=}1] = \mathbb{E}[Y \mid T{=}1]$ by consistency alone, only the counterfactual mean $\mathbb{E}[Y(0) \mid T{=}1]$ requires adjustment. It therefore suffices to assume: (i') exchangeability for the *control* potential outcome only, $Y(0) \perp\!\!\!\perp T \mid X$; (ii') *control-arm* positivity on the treated support; and (iii)–(iv) as before. No assumption about $Y(1)$'s assignment mechanism, and no positivity for the treated arm, is needed. Matching the assumptions to the estimand can substantially weaken what must be defended.

What the proof uses and where it can fail. The table below traces each step back to the assumption it invokes.

Table 5.1: The three substantive identification assumptions and what goes wrong when each is violated. The integrability condition is omitted because it is a technical regularity condition rather than a substantive causal assumption.

Proof step	Assumption invoked	Failure mode
$\mathbb{E}[Y(t) \mid X] = \mathbb{E}[Y(t) \mid T{=}t, X]$	Weak ignorability	Unmeasured confounder U: $Y(t)$ depends on T even within X-strata
$\mathbb{E}[Y \mid T{=}t, X{=}x]$ is well-defined	Overlap	Structural zero: $P(T{=}t \mid X{=}x) = 0$ on a positive-probability set; the estimand is not identified for the original population
$\mathbb{E}[Y(t) \mid T{=}t, X] = \mathbb{E}[Y \mid T{=}t, X]$	Consistency	Interference: Y_i depends on T_j for $j \neq i$; or hidden versions of treatment

> Remark: Population Positivity versus Empirical Overlap
>
> There is still an asymmetry among the assumptions, but it must be stated carefully. *Empirical* overlap can be diagnosed: for any candidate X, one can compare the distributions of $X \mid T{=}1$ and $X \mid T{=}0$ and look for regions where one arm is absent or nearly so. *Population* positivity, by contrast, is an assumption about the data-generating distribution and is not fully testable from a finite sample — especially with continuous or high-dimensional covariates. Two distinct failure modes should not be conflated: a structural zero is an *identification* failure for the original target; a finite-sample empty cell despite positive population probability is an *estimation* problem. Restricting the analysis to a region of common support can yield an identified effect *for that restricted target population*, but it changes the estimand. Weak ignorability and consistency remain untestable from data alone. Overlap diagnostics using the propensity score are developed in Chapter 6.

Equations Equation 1.6–Equation 5.9 are the *identification results*. The next sections present methods for *estimating* the identified quantities from data.

5.2.6 Which Variables to Condition On: The Pre-Treatment Requirement

The back-door formula conditions on X. Before estimating anything, there is a prior question that is purely graphical: *which variables should be in X?* The answer is not "all available variables." Conditioning on the wrong variable can introduce bias rather than remove it.

Do Not Condition on Post-Treatment Variables when Targeting the Total Effect

A variable L is *post-treatment* if it is a descendant of T. Conditioning on L in the outcome regression can bias the estimated treatment effect, even if L would otherwise be a strong predictor of Y. There are two distinct mechanisms:

1. **Blocking a causal pathway.** If L lies on a directed path from T to Y (i.e. L is a mediator), conditioning on L blocks part of the causal effect of T and *changes the target*. Importantly, it does not by itself identify a direct effect: the conditional contrast can be distorted by mediator–outcome confounding and treatment-induced collider bias, so controlled and natural direct effects require the explicit estimands and additional assumptions of Chapter 8. What can be said in general is only that the resulting quantity is no longer the total effect.
2. **Opening a collider path.** If L has two parents — $T \to L \leftarrow U$ — then L is a *collider* on the path $T \to L \leftarrow U \to Y$. This path is blocked when L is not conditioned on. Conditioning on L *opens* the path, creating a spurious association between T and Y through U that did not exist before adjustment.

In both cases, including L in X violates condition (1) of the back-door criterion: X *must contain no descendant of* T. The back-door criterion is a *sufficient* condition, not a complete characterization of all valid adjustment sets, but excluding post-treatment variables is the correct first-pass filter for the total-effect analyses of this course.

The practical rule. Before running any regression or computing any cell means, classify every candidate covariate as pre-treatment or post-treatment using the causal graph. Only pre-treatment variables that satisfy the back-door criterion belong in X. Post-treatment variables should be left out unless the research question is specifically about a direct or mediated effect — in which case Chapter 8 provides the appropriate framework.

Remark: Choosing among Valid Adjustment Sets

The back-door criterion typically admits more than one valid set. In the education–earnings DAG of Chapter 2, both $\{B\}$ and $\{N, B\}$ satisfy the criterion for the effect of E on Y, while $\{N\}$ does not — it leaves the fork at B open. Three points govern the choice.

(a) Multiple valid sets identify the same quantity. Any set satisfying the criterion (with positivity) yields the same interventional distribution; the identification target is unique even though the adjustment sets are not.

(b) Supersets of a valid set are not automatically valid. Adding a variable can open a collider on a back-door path, so "more adjustment is safer" is not a valid heuristic. In this particular graph $\{N, B\}$ remains valid because N is not a collider on any back-door path — but this must be verified from the graph, not assumed.

(c) Smaller valid sets reduce positivity and modeling burdens, but are not uniformly more efficient. Adjusting for N in addition to B requires $P(E{=}e \mid B{=}b, N{=}n) > 0$ for all cells receiving positive weight — a stricter requirement — and sparse or empty cells in high-dimensional strata cause practical positivity failure. Statistical efficiency, however, is a separate question from graphical minimality: adding a pretreatment variable that is strongly prognostic for the outcome can *improve* precision, whereas adding one that mainly predicts treatment can inflate variance. Graphical sufficiency settles *validity*; the choice among valid sets involves positivity, dimensionality, and estimator-dependent efficiency considerations.

5.3 Regression Adjustment and Standardization

5.3.1 The Common Three-Step Logic

Both regression adjustment and standardization are implementations of the same back-door formula Equation 5.8. They differ only in *how* they estimate the conditional mean $\mu(t, x) = \mathbb{E}[Y \mid T{=}t, X{=}x]$, not in the underlying procedure. Throughout this section X denotes a set of *pre-treatment* covariates satisfying the back-door criterion.

The Three-Step Recipe: Outcome Regression / g-Formula

1. **Estimate the outcome model.** Fit a model for $\mu(t, x) = \mathbb{E}[Y \mid T{=}t, X{=}x]$ using the observed data. Any regression method may be used: OLS, logistic regression, a flexible nonparametric estimator, or cell means.
2. **Predict the treatment-specific conditional means for every unit.** For each unit i — *regardless of the treatment that unit actually received* — compute $\widehat{Y}_i(1) = \hat{\mu}(1, X_i)$ and $\widehat{Y}_i(0) = \hat{\mu}(0, X_i)$. A caution on what these predictions are: $\hat{\mu}(t, X_i)$ estimates the conditional mean $\mathbb{E}[Y(t) \mid X{=}X_i]$ — the best prediction given X_i — not the unit's own potential outcome $Y_i(t)$, which remains unobservable. The difference $\hat{\mu}(1, X_i) - \hat{\mu}(0, X_i)$ is accordingly a predicted conditional-mean contrast at X_i, not an estimate of the individual effect.
3. **Average the predicted conditional-mean contrasts.**

$$\hat{\tau}_{\text{OR}} = \frac{1}{n}\sum_{i=1}^{n}[\hat{\mu}(1, X_i) - \hat{\mu}(0, X_i)]. \tag{5.10}$$

To estimate the ATT instead, average only over the n_1 treated units. A variant replaces the treated-arm prediction by the treated unit's *observed* outcome,

$$\hat{\tau}_{\text{ATT}}^{\text{imp}} = \frac{1}{n_1}\sum_{i:\, T_i=1}\{Y_i - \hat{\mu}(0, X_i)\},$$

which requires modeling only $\mu(0, \cdot)$ and thereby mirrors the weaker ATT identification package: exchangeability and positivity are needed for the control arm only.

This estimator is called the *outcome regression* (OR) estimator or the *g-computation* estimator (Robins 1986). The g-formula name emphasizes that step 3 *averages out* the covariates X by plugging in the empirical distribution of X, exactly as the back-door formula requires.

Why the three steps correspond to the identifying formula. The back-door formula says $\tau_{\text{ATE}} = \int[\mu(1, x) - \mu(0, x)]\, p(x)\, dx$. Step 1 estimates μ; step 2 evaluates the contrast at the observed covariate value of each unit; step 3 approximates the integral by the sample average, using the empirical distribution of X as a nonparametric estimate of $p(x)$. The estimator is *plug-in*: replace every population quantity in the identification formula with its sample counterpart.

5.3.2 A Worked Example

Consider a binary covariate $X \in \{0, 1\}$ (e.g. sex) and a continuous outcome Y (e.g. earnings in thousands of dollars). An OLS model fit separately within each treatment arm gives:

	Treated ($T = 1$):		**Control** ($T = 0$):	
	n	$\bar{Y}$	n	$\bar{Y}$
$X = 0$	30	42	70	35
$X = 1$	70	58	30	50
All	100	53.2	100	39.5

The unadjusted difference in means is $53.2 - 39.5 = 13.7$. But $X{=}1$ units (with higher baseline earnings) are overrepresented in the treated arm: 70% treated vs. 30% control. X confounds the comparison.

Step 1. The estimated conditional means are $\hat{\mu}(1, 0) = 42$, $\hat{\mu}(1, 1) = 58$, $\hat{\mu}(0, 0) = 35$, $\hat{\mu}(0, 1) = 50$.

Step 2. The imputed within-stratum effects are $42 - 35 = 7$ (for $X{=}0$ units) and $58 - 50 = 8$ (for $X{=}1$ units).

Step 3. Averaging over the *marginal* distribution of X: 50% of the full sample has $X{=}0$, 50% has $X{=}1$ (equal mix in the combined sample of 200), giving

$$\hat{\tau}_{\text{ATE}} = 7 \times 0.5 + 8 \times 0.5 = 7.5.$$

After adjusting for X, the estimated treatment effect is 7.5, not 13.7. The difference arises because the unadjusted comparison conflates the treatment effect with the higher baseline earnings of $X{=}1$ units who happen to be treated more often.

5.3.3 Standardization as a Special Case

Standardization is the name traditionally used in epidemiology and biostatistics for the special case of regression adjustment in which X is categorical with finite support $\mathcal{X}$. When X is categorical, no parametric model for $\mu(t,x)$ is needed: the conditional mean is simply the sample mean of Y within each cell $(T{=}t, X{=}x)$. This is equivalent to fitting a fully saturated outcome model.

The resulting estimator is

$$\hat{\tau}_{\text{STD}} = \sum_{x\in\mathcal{X}} [\hat{\mu}(1,x) - \hat{\mu}(0,x)]\, \hat{p}(x), \tag{5.11}$$

where $\hat{\mu}(t,x) = \bar{Y}_{t,x}$ and $\hat{p}(x) = n_x/n$. Equation Equation 5.11 is *algebraically identical* to the plug-in form of Equation 5.10. Standardization is therefore regression adjustment with a saturated outcome model, and the two estimators coincide whenever X is categorical.

The worked example above *is* a standardization: the cell means were computed directly from the data without any parametric model. The formula is also known as the *g-formula* (Robins 1986) and as *direct standardization* in epidemiology.

	Regression adjustment	**Standardization**
How $\hat{\mu}(t,x)$ is estimated	Regression model: parametric (OLS, logistic) or flexible nonparametric learner	Cell means: $\bar{Y}$ within $(T{=}t, X{=}x)$
Works when X is	Continuous or moderately high-dimensional (through modeling)	Discrete and low-dimensional
Main vulnerability	Model misspecification	Sparse or empty cells: undefined cell means, instability
Steps 2–3	Identical: predict both conditional means, average the contrasts	Identical

When X is continuous, cell means are unavailable. $\hat{\mu}(t,x)$ is then estimated by a regression model and the three-step recipe applies with no other change.

5.3.4 Model Specification and What Can Go Wrong

To see exactly what consistency of the plug-in estimator requires, write $P_n f = n^{-1}\sum_{i=1}^{n} f(X_i)$ and $Pf = \mathbb{E}\{f(X)\}$, with $\hat{\mu}_t(x) = \hat{\mu}(t,x)$. Then

$$\hat{\tau}_{\text{OR}} - \tau_{\text{ATE}} = P_n(\hat{\mu}_1 - \mu_1) \;-\; P_n(\hat{\mu}_0 - \mu_0) \;+\; (P_n - P)(\mu_1 - \mu_0).$$

Suppose the observations are i.i.d., $\mathbb{E}|\mu_1(X) - \mu_0(X)| < \infty$, and the estimated regressions satisfy the *empirical-*L_1 condition

$$P_n|\hat{\mu}_t - \mu_t| = \frac{1}{n}\sum_{i=1}^{n} |\hat{\mu}(t,X_i) - \mu(t,X_i)| \;\xrightarrow{p}\; 0, \qquad t \in \{0,1\}.$$

The first two terms are then $o_p(1)$, while the law of large numbers handles the third. Consequently $\hat{\tau}_{\text{OR}} \xrightarrow{p} \tau_{\text{ATE}}$.

The empirical form of the first condition matters: because $\hat{\mu}$ is fitted and evaluated on the *same* observations, convergence in the population sense is not by itself enough — a learner can have negligible population error while behaving anomalously at its own training points. The empirical-L_1 condition follows, for example, from uniform convergence; from population-$L_1(P_X)$ convergence together with suitable Glivenko–Cantelli or stability conditions; or from independent (cross-fitted) evaluation under integrability conditions. A correctly specified finite-dimensional parametric model satisfies it under standard regularity conditions.

The condition is sufficient, *not* necessary: if $\hat{\mu}(t, \cdot)$ converges to some limit μ_t^*, the OR estimator targets $\mathbb{E}[\mu_1^*(X) - \mu_0^*(X)]$, and it is consistent whenever this integrated contrast equals τ_{ATE} even though the two functions are pointwise wrong everywhere. Section 5.3.5 and the lab of Section 5.6 are built around exactly this cancellation under randomization. In an observational study, however, no design guarantees the cancellation, and consistent estimation of the outcome regressions is the practical requirement. This is the main vulnerability.

Misspecification bias. If the true $\mu(t, x)$ is nonlinear but a linear model is used, the estimated treatment effect absorbs part of the functional form error. The bias can be in either direction. Flexible learners can reduce approximation bias, though they may introduce greater estimation variability and overfitting; in practice they are paired with sample splitting (cross-fitting), which separates fitting from evaluation. For plain outcome regression that is cross-fitting's whole role: it does not repair an inconsistent outcome model, and it does not confer first-order robustness to nuisance-estimation error. That stronger role emerges only when cross-fitting is paired with an *orthogonal* score, as in the AIPW/DML estimators of Chapter 12.

Extrapolation. Step 2 predicts $\hat{Y}_i(0)$ for treated units with covariate values that may lie outside the support of the control group. In regions with no control data, $\hat{\mu}(0, X_i)$ is extrapolated from the model rather than estimated from comparable units. The propensity score overlap condition (Chapter 6) formalizes when extrapolation is unavoidable.

5.3.5 Regression Adjustment in Randomized Experiments: Lin (2013)

In a *randomized* experiment, outcome regression can be used to reduce variance even though covariate adjustment is not needed for unbiasedness. The *regression-adjusted estimator* of Lin (2013) fits the fully interacted OLS model

$$Y_i = \alpha + \beta T_i + \gamma^\top \tilde{X}_i + \delta^\top (T_i \cdot \tilde{X}_i) + \varepsilon_i, \tag{5.12}$$

where $\tilde{X}_i = X_i - \bar{X}$ are mean-centered covariates, and takes $\hat{\beta}$ as the point estimate of the finite-population ATE $\bar{\tau}_n$. Before discussing its properties we explain, in two short steps, why $\hat{\beta}$ coincides with the three-step outcome-regression estimator fit arm-by-arm with a linear model.

Step A: Single-arm regression with a centered covariate. Consider the single-arm OLS fit minimizing $\sum_i T_i\{Y_i - \beta_0 - \beta_1^\top(X_i - \bar{X})\}^2$. The first-order condition for β_0 gives $\hat{\beta}_0 = \bar{Y}_1 - \hat{\beta}_1^\top(\bar{X}_1 - \bar{X})$. On the other hand, the three-step OR estimator with a linear within-arm model $\hat{\mu}(1, x) = \hat{a}_1 + \hat{b}_1^\top x$, averaged over all units, equals $\hat{\mu}_{1,\text{reg}} = \hat{a}_1 + \hat{b}_1^\top \bar{X}$. The within-treated slope $\hat{b}_1$ coincides with $\hat{\beta}_1$ (centering shifts the intercept but not the slope), and at the within-treated sample mean $\bar{Y}_1 = \hat{a}_1 + \hat{b}_1^\top \bar{X}_1$, so

$$\hat{a}_1 + \hat{b}_1^\top \bar{X} = \bar{Y}_1 - \hat{b}_1^\top(\bar{X}_1 - \bar{X}) = \hat{\beta}_0.$$

Hence $\hat{\beta}_0 = \hat{\mu}_{1,\text{reg}}$: *when the covariate is centered at the full-sample mean $\bar{X}$, the OLS intercept is the regression-adjusted estimator of the treated-arm mean.* The symmetric argument in the control arm gives $\hat{\mu}_{0,\text{reg}}$.

Step B: Interacted regression over both arms. Write $\widehat{Y_i(t)} = \hat{\mu}_{t,\text{reg}} + \hat{B}_1(t)^\top(X_i - \bar{X})$, where $\hat{B}_1(t)$ is the within-arm slope. The arm-specific predictions combine as

$$\hat{Y}_i = \hat{\mu}_{0,\text{reg}} + T_i(\hat{\mu}_{1,\text{reg}} - \hat{\mu}_{0,\text{reg}}) + \hat{B}_1(0)^\top(X_i - \bar{X}) + T_i(X_i - \bar{X})^\top(\hat{B}_1(1) - \hat{B}_1(0)).$$

The right-hand side is exactly the fit of the interacted regression, and reading off coefficients identifies $\hat{\beta}$ with $\hat{\mu}_{1,\text{reg}} - \hat{\mu}_{0,\text{reg}}$. Setting $\hat{\tau}_{\text{reg}} := \hat{\mu}_{1,\text{reg}} - \hat{\mu}_{0,\text{reg}}$, we have the algebraic identity

$$\hat{\beta} \;=\; \hat{\tau}_{\text{reg}}. \tag{5.13}$$

The Lin estimator is therefore not a separate method: it is the three-step outcome-regression estimator with linear within-arm models, recast as a single regression. The interacted representation makes *computation* of the estimator and of its Huber–White sandwich standard error immediate in standard software; the asymptotic validity of that variance estimator under the randomization distribution is not an algebraic consequence of the recasting but part of Lin's theoretical result.

Theorem 5.3 (Design-Consistency of the Regression-Adjusted Estimator (Lin 2013)). *Consider a sequence of finite populations with fixed covariate dimension in which $n_1/n \to \rho \in (0,1)$, the population Gram matrices converge to a nonsingular limit, and the covariates and potential outcomes satisfy moment and leverage conditions sufficient for the arm-specific OLS coefficients to be root-n consistent for their finite-population projection coefficients. Under complete randomization,*

$$\hat{\beta} - \bar{\tau}_n \xrightarrow{p} 0$$

regardless of whether the linear working models are correctly specified. The result is asymptotic: unlike $\hat{\tau}_{\mathrm{DIM}}$, the estimator $\hat{\beta}$ need not be exactly unbiased in finite samples.

Proof (second reading)

We work in the design-based framework of Theorem 5.1: the potential outcomes and covariates are treated as fixed, and the only source of randomness is the assignment vector. Let $\mathbf{x}_i = (1, X_i^\top)^\top$. By Equation 5.13, it suffices to show $\hat{\mu}_{1,\mathrm{reg}} - \bar{Y}(1) \to_p 0$; the control arm follows symmetrically.
Step 1: Linearization around an oracle coefficient. Let $\mathbf{B}_1 = (\sum_i \mathbf{x}_i \mathbf{x}_i^\top)^{-1} \sum_i \mathbf{x}_i Y_i(1)$ denote the full-population OLS coefficient one would obtain if the treated potential outcome were observed for every unit. Since $\mathbf{B}_1$ depends only on $\mathcal{F}_n$, it is fixed under the randomization distribution. Define the oracle residuals $e_i(1) = Y_i(1) - \mathbf{x}_i^\top \mathbf{B}_1$. By the normal equations, whenever $\mathbf{x}_i$ contains an intercept,

$$\sum_{i=1}^{n} e_i(1) = 0, \qquad \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i^\top \mathbf{B}_1 = \bar{Y}(1). \tag{5.14}$$

Using the normal equations for the observed-data fit, a short algebraic manipulation gives the exact decomposition

$$\hat{\mu}_{1,\mathrm{reg}} = \frac{1}{n}\sum_i \mathbf{x}_i^\top \mathbf{B}_1 + \frac{1}{n_1}\sum_i T_i (Y_i - \mathbf{x}_i^\top \mathbf{B}_1) + R_n,$$

with remainder $R_n = (n^{-1}\sum_i \mathbf{x}_i - n_1^{-1}\sum_i T_i \mathbf{x}_i)^\top (\hat{\beta}_1 - \mathbf{B}_1)$. Both factors in R_n are $O_p(n^{-1/2})$: the covariate-balance gap by the design, and $\hat{\beta}_1 - \mathbf{B}_1$ by standard within-arm OLS arguments. Hence $R_n = O_p(n^{-1})$. Using treated-unit consistency and Equation 5.14,

$$\hat{\mu}_{1,\mathrm{reg}} = \bar{Y}(1) + \frac{1}{n_1}\sum_{i=1}^{n} T_i\, e_i(1) + O_p(n^{-1}). \tag{5.15}$$

Step 2: Design-based consistency. Under complete randomization, $\Pr(T_i = 1 \mid \mathcal{F}_n) = n_1/n$, so $\mathbb{E}[n_1^{-1}\sum_i T_i e_i(1) \mid \mathcal{F}_n] = n^{-1}\sum_i e_i(1) = 0$ by Equation 5.14. The sum is a finite-population mean of the fixed quantities $\{e_i(1)\}$ under simple-random-sample assignment, so its conditional variance is $O(n^{-1})$ by standard finite-population sampling theory. By Chebyshev, $n_1^{-1}\sum T_i e_i(1) \to_p 0$. Combined with Equation 5.15, $\hat{\mu}_{1,\mathrm{reg}} - \bar{Y}(1) \to_p 0$, and hence $\hat{\beta} \to_p \bar{\tau}_n$.
The probability limit of $\hat{\beta}_1$ never appears in the conclusion: the oracle coefficient $\mathbf{B}_1$ may be replaced by any fixed, finite coefficient vector. Consistency comes from the randomization distribution of the assignment, not from correctness of the linear model. □

Under the two-phase superpopulation framework, the finite-population target itself converges, $\bar{\tau}_n \to_p \tau_{\mathrm{ATE}}$, so $\hat{\beta}$ is consistent for the superpopulation ATE as well; the design-based statement is the more primitive one.

Byproduct: a linearization of the regression estimator. Equation Equation 5.15, applied to both arms, gives

$$\hat{\tau}_{\mathrm{reg}} \;\simeq\; \bar{\tau}_n + \frac{1}{n_1}\sum_{i=1}^{n} T_i\, e_i(1) - \frac{1}{n_0}\sum_{i=1}^{n}(1 - T_i)\, e_i(0), \tag{5.16}$$

the linearization at the heart of the design-based analysis. Replacing Y_i by the smaller residual $e_i(t)$ is the mechanism behind the potential variance reduction. Marginal residual-variance reduction alone does not settle the comparison: the design variance also contains the cross-potential-outcome term, exactly as in Theorem 5.1, and it too changes under residualization. Lin's theorem establishes that, for the fully interacted specification under its regularity conditions, the *combined* asymptotic variance cannot exceed that of the unadjusted difference in means. The full calculation is in Lin (2013); see also Freedman (2008).

Remark: The Role of the Regression Model in a Randomized Experiment

For estimation of the *marginal ATE point estimate* in this completely randomized, full-compliance design, the regression model is not needed for identification; its principal role is *precision improvement*. It does not carry the burden of removing confounding.

Two properties should be kept apart. *Exact* finite-sample unbiasedness belongs to the plain difference-in-means estimator (Theorem 5.1). The regression-adjusted estimator is instead *design-consistent*: randomization alone guarantees $\hat{\beta} \to_p \bar{\tau}_n$ regardless of whether the outcome model is correctly specified — this is precisely what Theorem 5.3 states — but $\hat{\beta}$ generally carries a design bias of order n^{-1} rather than exact zero bias, and this bias vanishes asymptotically (Freedman 2008; Lin 2013). The principal statistical reason to add covariates is to reduce $\mathrm{Var}(\hat{\tau})$ by absorbing residual variation in Y unrelated to T. If the model fits well, the residual variance shrinks and the estimator becomes more precise. If the model is misspecified or the covariates are weakly predictive, the variance reduction is small or zero, but no *asymptotic* bias is introduced. (The asymptotic no-loss-of-efficiency property is specific to the fully interacted specification under Lin's regularity conditions; uninteracted adjustment can lose precision, as Freedman (2008) showed.)

This stands in sharp contrast to the observational setting, where the regression model carries a *double* burden: it must both adjust for confounding (unbiasedness) and fit the outcome surface well (efficiency). Misspecification in an observational study therefore threatens *both* properties simultaneously. In a randomized experiment — for the marginal ATE point estimator, with randomization-robust (sandwich) inference — it threatens only efficiency. A misspecified model can still misrepresent conditional effects and subgroup contrasts, and model-based (non-robust) standard errors remain invalid, as the lab of Section 5.6 illustrates.

Remark: Precedence in Survey Sampling

Regression and calibration estimators in survey sampling provide close design-based precedents for the design-consistency result of Theorem 5.3.

Isaki and Fuller (1982) formalized the notion of design consistency for finite-population estimation under general probability sampling designs and studied the regression estimator's design and model properties within that framework, establishing design consistency even when the assisting regression model is misspecified. Because a completely randomized experiment is structurally equivalent to simple random sampling without replacement from the finite population $\mathcal{F}_n$ of potential outcomes, their analysis provides, under the corresponding regularity conditions, a direct precedent for Theorem 5.3.

Deville and Särndal (1992) developed the broader framework of *calibration estimators*, which adjust the sampling weights so that weighted sample totals match known population totals for the auxiliary variables X. The generalized regression (GREG) estimator is the special case using linear distance; under regularity conditions, calibration estimators share the design-consistency property.

Lin (2013)'s contribution was to adapt the regression-estimator intuition to the finite-population Neyman–Rubin model, prove that the *fully interacted* specification cannot hurt asymptotic precision relative to the unadjusted difference in means, and show that the Huber–White sandwich variance estimator yields asymptotically valid (or conservative) confidence intervals under the randomization distribution — a presentation tailored to the experimental causal inference audience where these earlier survey sampling results were not widely known.

Remark: Lin's Result Is Specific to Randomization

Lin's consistency result is specific to randomized experiments. In an observational study there is no design-based cancellation to fall back on: consistency of $\hat{\tau}_{\mathrm{OR}}$ rests on the estimated outcome regressions, in practice on $\hat{\mu}$ converging to the true μ in the empirical-L_1 sense above. The doubly robust (AIPW) estimator of Chapter 11 achieves consistency if *either* the outcome model or the propensity score model is correctly specified, providing a robustness guarantee the plain OR estimator lacks.

5.4 Stratification

5.4.1 The Idea

Stratification (subclassification) is a nonparametric implementation of the back-door formula. Rather than modeling $\mathbb{E}[Y \mid T, X]$, the analyst divides the sample into *strata* — subgroups of units with the same or similar values of X — and estimates the treatment effect within each stratum.

Within a stratum $\mathcal{S}_k = \{i : X_i \in B_k\}$, the treated and control units have similar covariate distributions. Narrowness alone is not enough: coarsening X into bins does not automatically inherit exchangeability, and residual confounding can persist within every finite-width stratum. What justifies the method is a combination of conditions: exchangeability given the *exact* X, treatment propensity and outcome regressions that vary smoothly within strata, stratum widths that shrink with the sample size, and adequate within-stratum overlap. Under these conditions the within-stratum difference in means

$$\hat{\tau}_k = \bar{Y}_{1,k} - \bar{Y}_{0,k}$$

is an approximately unconfounded comparison: its residual coarsening bias vanishes as the strata shrink. Finite-width strata are not literally unbiased.

5.4.2 The Stratified Estimator

The overall ATE is estimated as a weighted average of within-stratum estimates:

$$\hat{\tau}_{\text{STRAT}} = \sum_{k=1}^{K} \hat{\tau}_k \cdot \frac{n_k}{n}, \tag{5.17}$$

where K is the number of strata and n_k the number of units in stratum k. Cochran (1968) showed — in his setting of a single continuous confounder under particular distributional models, not as a universal fact — that with $K = 5$ strata of equal size, roughly 90% of the bias is removed; with $K = 10$ strata, over 95%.

> Remark: Stratification, Standardization, and Survey Sampling Terminology
>
> When X is categorical, the stratified estimator Equation 5.17 and the standardization estimator Equation 5.11 are algebraically identical: both compute within-cell treatment effect estimates and aggregate them with marginal weights n_k/n. The two names reflect disciplinary tradition rather than a methodological distinction.
> When X is continuous, stratification estimates $\hat{\mu}(t, x)$ by a piecewise-constant step function, while regression adjustment fits a smooth model. Under regularity conditions and provided each stratum retains enough treated and control units, the stratified estimator approaches a nonparametric regression target as the number of strata grows and stratum widths shrink. This is a directional statement, not a free lunch: with too many strata, empty cells and within-stratum variance begin to dominate.
> In the survey sampling literature, *stratification* and *post-stratification* are distinct operations. *Stratification* is a *design-stage* decision: the population is divided into strata before sampling. *Post-stratification* is an *analysis-stage* adjustment: stratum membership is recorded after a non-stratified sample has been drawn, and the estimator is reweighted toward target stratum shares. Standardization to the *empirical* weights n_k/n is algebraically *analogous* to post-stratification rather than identical to it: traditional survey post-stratification uses externally known target-population proportions N_k/N, and those weights could likewise be substituted here when a causal effect is to be transported to a different target population.

5.4.3 Limitations

Stratification faces the *curse of dimensionality*: with p binary covariates there are 2^p cells, many of which will be empty in practice. This motivates two solutions studied in this course:

- **Regression modeling** (Section 5.3) permits interpolation across continuous or sparse cells; exact cell standardization remains nonparametric but becomes infeasible as the dimension grows.

- **Propensity score methods** (Chapter 6) collapse the multivariate X to a scalar $\pi(X) = P(T{=}1 \mid X)$ and adjust on the scalar. The Propensity Score Theorem of Chapter 6 guarantees that conditioning on the *true* score $\pi(X)$ balances the full multivariate X; an estimated score delivers only approximate balance, which must be checked with diagnostics.

5.4.4 Case Study: The Payoff to Attending a Selective College

The identification assumptions of Section 5.2, and the stratified estimator just developed, are best appreciated through a real study in which ignorability is initially not credible, and a clever design partially restores it — by *constructing the strata.* Dale and Krueger (2002) ask whether attending a more selective college causally raises later earnings. Let T denote the selectivity of the college a student attends (measured by the school's average SAT score) and Y the student's log earnings observed in 1995, roughly two decades after the 1976 entering cohort they study began college. Earlier studies had found that a 100-point increase in school-average SAT is associated with earnings 3–7 percent higher; the basic model of Dale and Krueger (2002), which adjusts for the student's own SAT score, high school rank, race, sex, athletic participation, and parental income, gives 7.6 percent in the College and Beyond survey of 30 selective institutions.

Is this a causal effect? The threat is an unmeasured confounder — a variable that influences both where a student enrolls and how much the student later earns, but that no data set records. Model the admissions decision of college j on applicant i as a threshold rule,

$$\text{admit } i \text{ to college } j \quad \Longleftrightarrow \quad Z_{ij} = \gamma_1 X_{1i} + \gamma_2 X_{2i} + e_{ij} > C_j, \tag{5.18}$$

where X_1 collects characteristics observed by both the admissions committee and the analyst (SAT score, high school rank), X_2 collects characteristics observed by the committee but *not* by the analyst (motivation, ambition, and maturity as judged from essays, interviews, and recommendation letters), e_{ij} is idiosyncratic committee taste, and C_j is the cutoff of college j. If the labor market also rewards X_2, then students at selective colleges have higher earnings capacity regardless of where they enroll, and adjustment for X_1 alone leaves the back-door path through X_2 open: ignorability given X_1 fails, and the 7.6 percent figure is biased upward.

In the potential-outcomes language of Chapter 4 the problem is stated in one line. Simplify to a binary treatment: $T = 1$ if the student enrolls in the more selective tier. The naive contrast then decomposes as

$$\underbrace{\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0]}_{\text{observed gap}} = \underbrace{\mathbb{E}[Y(1) - Y(0) \mid T{=}1]}_{\tau_{\text{ATT}}} + \underbrace{\mathbb{E}[Y(0) \mid T{=}1] - \mathbb{E}[Y(0) \mid T{=}0]}_{\text{selection bias}}, \tag{5.19}$$

obtained by adding and subtracting $\mathbb{E}[Y(0) \mid T{=}1]$ and applying consistency. The selection-bias term compares the earnings the selective-college students *would have had anyway* with the earnings of everyone else. Under the admissions model Equation 5.18 this term is *plausibly* positive, but a step of care is needed: the model governs *admission*, while T records the selectivity *attended*, and admission does not determine how students choose among the schools that admit them. Attending a selective school does require clearing its high cutoff, so attendees have high values of the latent index — and hence of X_2 — on average; if, in addition, the labor market rewards X_2 and matriculation choice does not systematically reverse this ranking, then $\mathbb{E}[Y(0) \mid T{=}1] > \mathbb{E}[Y(0) \mid T{=}0]$, and the observed gap overstates τ_{ATT}. Adjusting for X_1 shrinks the selection-bias term but cannot drive it to zero, because part of it flows through X_2.

The design. The insight of Dale and Krueger (2002) is that the admissions process itself *measures* X_2, imperfectly but repeatedly. Each accept or reject decision reveals, through Equation 5.18, whether the latent index cleared the threshold C_j. A student admitted by a college with cutoff C_1 but rejected by one with cutoff $C_2 > C_1$ has, absent noise, an index bracketed between C_1 and C_2. As accept/reject decisions accumulate over a fine range of cutoffs, the profile A of admissions outcomes pins down the index ever more tightly. Conditioning on A therefore approximates conditioning on $\gamma_1 X_1 + \gamma_2 X_2$ — a function of the unobservable — even though X_2 itself is never seen. Operationally, Dale and Krueger (2002) group students who applied to, and were accepted and rejected by, the same set of colleges ("matched applicants") and include a dummy variable for each group in the wage regression, comparing only students within a group who nonetheless enrolled at schools of different selectivity. The authors describe this as extending *selection on the observables* to "selection on the observables and unobservables."

The table below shows how the grouping works for seven hypothetical applicants. Schools whose average SAT scores fall in the same 25-point interval are treated as equally selective.

Table 5.2: Construction of matched-applicant groups. A = accepted, R = rejected; an asterisk marks the school attended. Schools with average SAT in the same 25-point interval are equally selective, so A and B share a group despite applying to slightly different schools. F is excluded (a single application provides too little screening information); G has no counterpart. Modeled on Table I of Dale and Krueger (2002).

Student	App 1	App 2	App 3	Matched group
A	1300 / R	1240 / A*	1190 / A	1
B	1305 / R	1245 / A	1185 / A*	1
C	1355 / A*	1310 / A	—	2
D	1350 / A	1305 / A*	—	2
E	1370 / A*	1315 / A	—	2
F	1200 / A*	—	—	excluded
G	1400 / R	1150 / A*	—	no match

Students A and B applied to *equivalent* — not literally identical — schools and received identical decisions, so they form group 1; within the group they enrolled at schools 55 points apart, which is exactly the within-group variation the estimator uses. Students C, D, and E form group 2; note that D chose the *less* selective option from the same admitted set — informative variation, but also precisely the kind of choice the matriculation caveat below is about. In the actual data, 6,335 of the 14,238 workers could be matched this way, forming 1,232 groups. The exclusions are not innocuous for the estimand: because only matchable applicants contribute, the design naturally targets an effect for the *matched (overlap) population* rather than the full entering cohort.

The estimator itself should look familiar, though one distinction matters: with one dummy variable per group, the wage regression compares earnings only among students who share a group label — the matched groups play exactly the role of strata, so the design principle is that of the stratified estimator Equation 5.17. The two are *not*, however, generally algebraically identical. For a binary treatment with no additional covariates, the common-slope fixed-effects regression yields

$$\hat{\beta}_{\text{FE}} = \frac{\sum_{k=1}^{K} (n_{1k} n_{0k}/n_k)\, \hat{\tau}_k}{\sum_{k=1}^{K} n_{1k} n_{0k}/n_k},$$

which weights each within-group contrast by its within-group treatment variation, whereas Equation 5.17 uses the target-population weights n_k/n. In the actual application, moreover, treatment is *continuous* school selectivity, so the regression estimates a common within-group linear slope. The matched groups supply a stratified *design*; fixed-effects regression supplies a closely related within-stratum *estimator*.

If students choose from their admitted set for reasons unrelated to X_2, then there is no $X_2 \to T$ arrow, the set (X_1, A) satisfies the back-door criterion, and ignorability $Y(t) \perp\!\!\!\perp T \mid X_1, A$ holds even though a confounder remains unmeasured. Two qualifications keep the claim honest. First, a choice driver may be safely left outside the adjustment set only if it does not *also* affect later earnings: geography or cost that influences both college choice and earnings is simply another confounder. Second, identification also requires positivity within the target admissions profiles: matched groups in which every student attended a school of the same selectivity exhibit no within-profile treatment variation. The lesson is that an unobserved common cause need not itself be measured: what identification requires is an observed pretreatment quantity that *blocks* every back-door path the unobservable opens, and a design can sometimes locate one.

Findings. Within matched-applicant groups, the large positive association between school selectivity and earnings *disappears*: most selection-adjusted specifications are near zero, and one is significantly negative. All coefficients are from Table III of Dale and Krueger (2002): the coefficient on school-average SAT (per 100 points) falls from 0.076 (s.e. 0.016) in the basic model to −0.016 (0.022) when matching on schools of similar selectivity, and to −0.106 (0.036) when matching students who applied to, and were accepted or rejected by, exactly the same schools; matching on *Barron's* selectivity categories gives 0.004 (0.016). A nested variant, the "self-revelation" model — which controls for the average SAT score of the schools the student *applied to* — gives −0.001 (0.018).

Two further results sharpen the interpretation. First, the average SAT score of the schools that *rejected* the student predicts earnings strongly (coefficient 0.072, s.e. 0.012). Under the maintained assumption that rejected-school selectivity has no causal path to later earnings (itself a substantive exclusion, not a logical fact), its predictive power is a pure signature of confounding — the application-and-admissions record is informative about X_2 — and it falsifies the unadjusted comparison without any appeal to the unobservable itself. Anticipating the language of Chapter 9, this is a *negative control* or falsification argument. (Translating its magnitude into the bias of the treatment-effect estimate requires additional calibration structure; detection and quantification are different tasks.) Second, the near-zero average masks heterogeneity: interacting selectivity with parental income shows a gain of about 8 percent (per 200 SAT points) for students from bottom-decile-income families, versus essentially nil at mean income — a reminder that "no average effect" is not "no effect for anyone."

What the Matched-Applicant Design Does Not Fix

Matching on the admissions profile addresses selection by *colleges*; it is silent about selection by *students*. If matriculation choice from the admitted set depends on X_2 — the dashed "?" arrow in the figure — a back-door path $T \leftarrow X_2 \rightarrow Y$ remains open within matched groups, and the estimate can be biased in either direction. Dale and Krueger (2002) report that, within matched groups, students with higher *observed* SAT scores were significantly more likely to enroll at the most selective school that admitted them; if unobserved ability sorts the same way, some upward bias survives, which only strengthens their null finding but would weaken a positive one. A related caveat concerns heterogeneous effects: if the payoff to selectivity varies across students and students know their own payoffs, they sort on the gain (a Roy selection model), and the regression coefficient need not identify the ATE or the ATT. Closing one back-door path is not the same as closing them all — credibility must be argued path by path.

5.5 Simpson's Paradox

Simpson's paradox — the reversal of an association when conditioning on a third variable — was introduced in Chapter 1. There, a flu-vaccination example showed that the vaccine appeared harmful in the aggregate yet beneficial within every age stratum, because elderly individuals were disproportionately vaccinated. Conditioning on age via the back-door formula recovered the correct causal effect.

This chapter's development gives that example a second layer of interpretation. The back-door formula used in Chapter 1 is precisely the standardization estimator Equation 5.11: within-stratum conditional means averaged over the *marginal* distribution of the confounder rather than its treatment-conditional distribution. The weighted average corrects the confounding; the pooled association does not, because it weights strata by $P(X \mid T)$ instead of $P(X)$. One qualifier keeps the resolution honest: standardizing over age settles the example only under the maintained causal assumptions — that age is a sufficient adjustment set, that positivity holds, and that consistency is credible. The reversal by itself is a fact about conditional proportions; it does not reveal which comparison, marginal or conditional, is the causal one. That verdict comes from the graph.

5.5.1 When Should You *Not* Condition on X?

Simpson's paradox also has a mirror image: conditioning on a variable can *create* a spurious association or make a true causal effect disappear. This occurs when X is a *mediator* or a *collider*. The back-door criterion gives the correct prescription: include X in the adjustment set only if it blocks back-door paths *without* opening collider paths.

Conditioning on a Mediator or Collider

Adjusting for a post-treatment variable X that lies on a causal pathway $T \rightarrow X \rightarrow Y$ blocks part of the causal effect of T, producing an estimate that generally differs from the total effect — with no universal direction of the discrepancy. Adjusting for a collider X with $T \rightarrow X \leftarrow Y$ opens a spurious association between T and Y that does not exist in the population. Neither mistake is detectable from the data alone; the DAG is essential.

We revisit the mediator case in detail in Chapter 8; the collider case was covered in Chapter 2.

5.6 Lab: Simulation Study of the Outcome Regression Estimator

The identification results of Section 5.2 determine the target functional *before* any outcome model is chosen. Estimating that functional by the outcome-regression plug-in then creates a bias–variance tradeoff. In an observational study, consistent estimation of the treatment-specific outcome regressions is the standard sufficient condition for plug-in consistency. Under a completely randomized design (CRD), by contrast, Theorem 5.3 gives the fully interacted linear OR estimator a separate design-consistency guarantee even when its working outcome models are misspecified; the guarantee disappears as soon as treatment is confounded.

DGP for Lab 5

Let $X \sim \text{Uniform}(0,1)$ be a single pre-treatment covariate and $\varepsilon \sim \mathcal{N}(0,1)$ independent of X. The potential outcome surfaces are:

$$Y(t) \;=\; t \;+\; 3(1+t)X^2 \;+\; \varepsilon, \qquad t \in \{0,1\}. \tag{5.20}$$

The conditional average treatment effect is $\tau(x) = 1 + 3x^2$, so $\tau_{\text{ATE}} = \mathbb{E}[1 + 3X^2] = 2$.
Two assignment mechanisms are studied:

- **CRD (complete randomization):** exactly $n/2$ units are drawn uniformly without replacement to receive treatment. The assignment vector is independent of X and of the potential outcomes.
- **Observational:** $T \mid X \sim \text{Bern}(\text{expit}(-2+5X))$, giving $\pi(0) \approx 0.12$, $\pi(0.5) \approx 0.62$, $\pi(1) \approx 0.95$. Strong confounding: high-X units are nearly always treated.

The observed outcome is $Y = Y(T)$ in both cases. Strong ignorability holds in both: in the CRD because the assignment vector is drawn independently of $(Y(0), Y(1), X)$ jointly; in the observational study because the treatment-generating mechanism is a stochastic function of X alone.

Estimators. Two outcome regression estimators of the form Equation 5.10 are compared. Both follow the three-step recipe. **Linear OR**: within each arm, fit the simple linear regression $Y \sim X$ by OLS. This model is *misspecified*: the true conditional mean $\mu(t,x) = t + 3x^2 + 3tx^2$ contains a quadratic-in-X term and a tx^2 interaction. **Local-linear OR**: within each arm, estimate $\mu(t,x_0)$ by local linear regression using a Gaussian kernel with bandwidth $h = 0.5 \cdot n^{-1/5}$. Local linear regression removes the leading-order boundary bias, which matters here because the observational propensity score places treated and control units in different regions of $[0,1]$.

Results ($n = 1{,}000$, $B = 2{,}000$ replications, seed 2024; true $\tau_{\text{ATE}} = 2.000$):

Design	**Estimator**	**Mean**	**Bias**	**SD**	**RMSE**
CRD	Linear OR	1.9964	−0.0036	0.0709	0.0710
CRD	Local-linear OR	2.0260	+0.0260	0.0685	0.0732
Observational	Linear OR	1.9674	−0.0326	0.0895	0.0952
Observational	Local-linear OR	2.0273	+0.0273	0.0951	0.0989

1. Under CRD, the linear OR estimator is consistent despite model misspecification. Conditional on each simulated finite population, Theorem 5.3 gives $\hat{\tau}_{\text{OR}} - \bar{\tau}_n \xrightarrow{p} 0$ under the randomization distribution, even though the linear outcome model is wrong. Because the simulation first draws the units i.i.d., $\bar{\tau}_n \xrightarrow{p} \tau_{\text{ATE}} = 2$; combining the two phases gives consistency. Randomization, rather than correctness of the working model, supplies the guarantee.

2. Under an observational design, the misspecified linear OR is biased. The bias grows from -0.0036 under CRD to -0.0326. The omitted X^2 and TX^2 terms cause $\hat{\mu}(t,x)$ to misrepresent the outcome surface, and because high-X units are predominantly treated (the propensity score ranges from 0.12 to 0.95), the misfit is systematically amplified in the direction of the confounding. In a CRD the same misfit is present, but the balanced assignment averages it out.

3. The local-linear OR reduces bias at the cost of higher variance. Under the observational design the local-linear OR reduces the absolute bias from 0.0326 to 0.0273, but its standard deviation is

0.0951 compared to 0.0895 for the misspecified linear model. The RMSE comparison (0.0952 vs. 0.0989) therefore favors the linear estimator, even though it is biased. This is the classic bias–variance tradeoff.

Model Misspecification and Confounding Interact

Comparing the two "Linear OR" rows: the bias is -0.0036 under CRD and -0.0326 under the observational design. The misspecification is identical in both cases — the same wrong model, the same DGP — so the difference must come entirely from the assignment mechanism.

The DGP makes the cancellation under CRD explicit. Under CRD, $T \perp\!\!\!\perp X$, so X within each arm is still Uniform$(0,1)$. The population OLS line fit to $Y(t)$ against X over $U(0,1)$ has slope $\beta_t = 3(1+t)\,\mathrm{Cov}(X^2, X)/\mathrm{Var}(X) = 3(1+t)$, giving $\hat{\mu}_{\text{lin}}(0,x) = -\frac{1}{2} + 3x$ and $\hat{\mu}_{\text{lin}}(1,x) = 6x$. The pointwise error in the estimated CATE is therefore

$$b(x) = (\tfrac{1}{2} + 3x) - (1 + 3x^2) = -\tfrac{1}{2} + 3x - 3x^2.$$

The pointwise bias vanishes only at $x = (3 \mp \sqrt{3})/6 \approx 0.21$ and 0.79. Yet its average over $X \sim U(0,1)$ is exactly zero:

$$\mathbb{E}[b(X)] = \int_0^1 (-\tfrac{1}{2} + 3x - 3x^2)\,dx = -\tfrac{1}{2} + \tfrac{3}{2} - 1 = 0.$$

The cancellation is structural, not accidental. The population OLS line is the L_2 projection of $\tau(X)$ onto the space spanned by $\{1, X\}$ under the fitting distribution. An L_2 projection onto a space containing an intercept preserves the mean, so $\mathbb{E}[\hat{\tau}_{\text{lin}}(X)] = \tau_{\text{ATE}}$ whenever the expectation is taken over the *same* distribution used for fitting. Under CRD the fitting distribution equals the evaluation distribution, so that condition holds.

Under the observational design, the fitting and evaluation distributions diverge. High-X units are almost always treated and low-X units almost never, so the model for arm $t = 1$ is fit on a sample concentrated at high X and the model for arm $t = 0$ at low X. The fitted lines are tuned to different parts of the covariate space, so when both are evaluated at the full marginal of X, the pointwise errors no longer cancel.

The narrow conclusion: for the *average* treatment effect under randomization, the misspecified linear outcome model can still give a consistent estimate. This is not a blanket endorsement of misspecified models in randomized trials. The same fitted model is biased pointwise for $\tau(x)$ at almost every x, would produce a misleading picture of treatment-effect heterogeneity, and need not give correct model-based standard errors — which is why design-based variance estimators are typically used with Lin-style adjustment.

Remark: What This Lab Does Not Show

The simulation fixes the bandwidth $h = 0.5 \cdot n^{-1/5}$ and does not perform data-driven bandwidth selection. In practice, choosing h by cross-validation changes the bias–variance tradeoff and can substantially improve the local-linear estimator's RMSE. The lab also studies only a scalar X; with a multivariate covariate, the curse of dimensionality would make local linear regression impractical, which is one motivation for the propensity score approach of Chapter 6.

5.7 Where the Propensity Score Fits

The three methods developed above — regression adjustment, standardization, and stratification — all implement the same identifying formula Equation 5.8, and all three therefore rest on the *same* causal identification conditions: consistency, conditional exchangeability, and positivity. What distinguishes them is only how they handle the conditioning on the high-dimensional covariate vector X, and what each additionally requires *at the estimation stage*:

Method	How it handles X	Method-specific estimation requirement
Outcome regression / standardization	Estimates $\mu(t, x)$: parametrically, nonparametrically, or by saturated cells	Sufficient convergence of the estimated conditional means — correct parametric specification is one route, not the only one
Stratification	Coarsens X into cells; estimates effect in each	Vanishing coarsening bias (narrow strata), adequate within-cell overlap, growing cell sizes
Propensity score (Ch. 6)	Reduces X to the scalar balancing score $\pi(X)$	Consistent score estimation, or estimator-specific balance conditions, together with overlap (a correct propensity model does not repair unmeasured confounding)

When X is low-dimensional, stratification and regression work well. When X has many components, direct stratification suffers from the curse of dimensionality. The *propensity score* provides an identification-preserving dimension reduction: Chapter 6 shows that adjusting for the scalar $\pi(X)$ suffices for identification even when X is high-dimensional. (The theorem reduces the dimension of the balancing variable, not every statistical difficulty — estimating π with high-dimensional X can itself be hard.)

Bridge to Chapter 6

Chapter 6 takes up this idea formally. It proves the balancing property ($T \perp\!\!\!\perp X \mid \pi(X)$), derives the Propensity Score Theorem, and develops two principal estimation strategies — matching on the score and inverse probability weighting (IPW), in Horvitz–Thompson and Hájek forms — together with balance and overlap diagnostics and trimming rules for near-violations of positivity. Propensity-score *subclassification* — the scalar-score analogue of stratification — is not developed separately there. Matching, likewise, need not rely on the propensity score: covariate-distance methods such as Mahalanobis-distance matching are also valid under the same identification assumptions. Chapter 6 focuses on the propensity-score-based methods because its principal theme is dimension reduction.

5.8 Summary

1. **Randomization as graph surgery.** A randomized experiment removes all arrows into T from variables in the scientific causal system, so the fundamental equality $p_R(y \mid T{=}t) = f(y \mid \mathrm{do}(T{=}t))$ holds within the experimental law and the difference-in-means estimator is exactly unbiased for the finite-population ATE $\bar{\tau}_n$, without covariate adjustment. Reaching the superpopulation ATE additionally requires a sampling or transportability argument.

2. **Ignorability and support.** Under consistency, weak conditional exchangeability, positivity for both treatment levels, and integrability, the back-door adjustment formula identifies the ATE. Strong ignorability is a sufficient package for the exchangeability and positivity parts; consistency remains a separate assumption. In an observational study, exchangeability and consistency are substantive conditions that cannot be verified from the observed law; population positivity is likewise not fully testable from a finite sample, although empirical overlap can and should be diagnosed.

3. **Three estimation strategies.** Regression adjustment (g-formula), standardization, and stratification all implement the same identifying formula via different modeling choices. In observational studies, a standard sufficient condition for regression and standardization is consistent estimation of the relevant outcome regressions, and stratification requires strata fine enough to achieve approximate balance; under a randomized experiment, regression adjustment remains design-consistent for the finite-population ATE even when used as a misspecified assisting model (Theorem 5.3).

4. **Simpson's paradox.** Pooled associations can reverse within subgroups when a confounder determines treatment selection. The correct causal estimate requires standardization over the marginal

covariate distribution, guided by the back-door criterion.

5. **The propensity score dimension reduction.** When X is high-dimensional, direct stratification or cell-by-cell standardization breaks down. The propensity score $\pi(X)$ provides a scalar balancing score that preserves the adjustment identification result. Chapter 6 develops the theory and estimation methods.

5.9 Problems

1. Randomization and the do-calculus. Let the observational DAG be $\{U \to T,\ U \to Y,\ X \to T,\ X \to Y,\ T \to Y\}$ with U unobserved.

(a) Identify all back-door paths from T to Y.
(b) Does X satisfy the back-door criterion? Does U? Explain.
(c) Now suppose treatment is randomized. Draw the modified DAG and identify all back-door paths. Show that $p_R(y \mid T{=}t) = f(y \mid \mathrm{do}(T{=}t))$ holds within the randomized experimental law using Rule 2 of the do-calculus. (*Hint*: identify the appropriate (X, Z, W) instantiation, construct $\mathcal{G}_{\underline{T}}$, and verify the d-separation condition.)
(d) Under randomization, is covariate adjustment on X necessary for unbiasedness? Is it ever beneficial? Explain.

2. Ignorability and the back-door criterion. Consider the DAG $\{X \to T,\ X \to Y,\ T \to Y,\ U \to Y\}$ with U unobserved.

(a) Does $\{X\}$ satisfy the back-door criterion? Write out the identifying formula for τ_{ATE}.
(b) Now add the edge $U \to T$. Does $\{X\}$ still satisfy the back-door criterion? What does the criterion require when both X and U affect T?
(c) Explain in words what "unconfoundedness given X" means about the role of U in the data-generating process.

3. Standardization. A study of job training (T) and earnings (Y, in thousands) produces the following cell means, with $P(X{=}0) = 0.4$ and $P(X{=}1) = 0.6$:

X	$\hat{\mu}(1, x)$	$\hat{\mu}(0, x)$	n_x/n
0	28	22	0.4
1	35	31	0.6

(a) Compute $\hat{\tau}_{\mathrm{ATE}}$ using the standardization formula.
(b) Compute $\hat{\tau}_{\mathrm{ATT}}$, given that $P(T{=}1) = 0.5$ and $P(X{=}1 \mid T{=}1) = 0.8$ — so that $P(X{=}1 \mid T{=}0) = 0.4$, consistent with the marginal $P(X{=}1) = 0.6$. The ATT-weighting density in Equation 5.9 places mass 0.2 on $X{=}0$ and 0.8 on $X{=}1$.
(c) The unadjusted difference in means is $33.6 - 25.6 = 8$. Compare to your answers in (a) and (b) and explain the discrepancy.

4. Stratified standardization. You have a binary confounder $X \in \{0, 1\}$ and form two strata. Within stratum $X{=}0$: $n_{X=0} = 600$, $\bar{Y}_1 = 10$, $\bar{Y}_0 = 8$. Within stratum $X{=}1$: $n_{X=1} = 400$, $\bar{Y}_1 = 15$, $\bar{Y}_0 = 12$.

(a) Compute the stratified ATE estimator $\hat{\tau}_{\mathrm{STRAT}}$.
(b) Suppose an unadjusted difference-in-means gives $\hat{\tau}_{\mathrm{DIM}} = 5.5$. Explain why the two estimates differ and, under the assumption that X is a sufficient adjustment variable, which one is the appropriate causal estimate.
(c) Describe one limitation that would arise if X were a continuous variable with 15 dimensions.

5. Simpson's paradox. A hospital reports that patients who receive intensive care (ICU admission, $T{=}1$) have a higher mortality rate than those who do not: $P(Y{=}1 \mid T{=}1) = 0.30$ vs. $P(Y{=}1 \mid T{=}0) = 0.10$. For concreteness, assume 1,000 patients in each treatment arm.

(a) Construct a numerical example (a $2 \times 2 \times 2$ table with disease severity X as the confounder) consistent with the pooled numbers and yet showing ICU admission *reduces* mortality within both severity strata.
(b) Compute the standardized ATE using the marginal distribution of X.

(c) Draw the DAG. Identify the back-door path that the pooled comparison fails to block.
(d) A colleague suggests that since the hospital treats patients with both mild and severe illness, the pooled statistic is the right answer. Explain, using potential outcomes notation, why this argument is incorrect.

6. Finite-population enumeration. Consider a finite population of $n = 4$ units with potential-outcome schedule

Unit i	1	2	3	4
$Y_i(0)$	0	1	1	2
$Y_i(1)$	1	3	2	4

and complete randomization with $n_1 = n_0 = 2$.

(a) Compute the finite-population ATE $\bar{\tau}_n$ and the finite-population variances S_1^2, S_0^2, S_τ^2, and the covariance S_{01}.
(b) Enumerate all $\binom{4}{2} = 6$ equally likely assignment vectors and compute $\hat{\tau}_{\text{DIM}}$ for each. Verify by direct averaging that $\mathbb{E}[\hat{\tau}_{\text{DIM}} \mid \mathcal{F}_n] = \bar{\tau}_n$ (Theorem 5.1).
(c) Compute the exact design variance from your enumeration and verify that it equals Equation 5.3.

7. Conservative Neyman variance. Continue with the schedule of the previous problem.

(a) For each of the 6 assignments, compute the Neyman variance estimator $\hat{V}$ of Equation 5.4, and compute $\mathbb{E}[\hat{V} \mid \mathcal{F}_n]$ by averaging. (*Hint*: under simple random sampling without replacement, the within-arm sample variance is unbiased for the finite-population variance.)
(b) Verify that $\mathbb{E}[\hat{V} \mid \mathcal{F}_n] - \text{Var}(\hat{\tau}_{\text{DIM}} \mid \mathcal{F}_n) = S_\tau^2/n$, as Theorem 5.1 asserts.
(c) For what modification of the schedule — keeping the same $Y_i(0)$ column — would the gap be exactly zero? Explain in terms of treatment-effect heterogeneity.

8. Lin's estimator as arm-specific standardization. Consider the fully interacted regression of Y_i on $(1,\, T_i,\, X_i - \bar{X},\, T_i(X_i - \bar{X}))$, where $\bar{X}$ is the full-sample covariate mean.

(a) Show that this regression is equivalent to fitting separate OLS regressions of Y on $(1, X)$ within each treatment arm.
(b) Let $(\hat{a}_t, \hat{b}_t)$ denote the arm-t intercept and slope. Show that the OLS coefficient on T_i equals $[\hat{a}_1 + \hat{b}_1\bar{X}] - [\hat{a}_0 + \hat{b}_0\bar{X}] = n^{-1}\sum_i\{\hat{\mu}(1, X_i) - \hat{\mu}(0, X_i)\}$, i.e. the OR estimator Equation 5.10 with linear working models.
(c) Explain why the centering at $\bar{X}$ is essential: what does the coefficient on T_i estimate if the covariate is not centered?

9. Exact unbiasedness versus design consistency. Under complete randomization, $\hat{\tau}_{\text{DIM}}$ is exactly unbiased for $\bar{\tau}_n$ in every finite sample (Theorem 5.1), while the regression-adjusted estimator $\hat{\beta}$ is only design-consistent (Theorem 5.3).

(a) Identify the step in the proof of Theorem 5.1 that delivers *exact* unbiasedness, and explain why no analogous exact argument is available for $\hat{\beta}$. (*Hint*: which quantity in the linearization of $\hat{\beta}$ is random but appears inside a nonlinear function of the assignment vector?)
(b) Locate where asymptotics enter the proof of Theorem 5.3: which remainder term is $O_p(n^{-1})$, and why is asymptotic negligibility, rather than exact unbiasedness, the relevant conclusion for it?
(c) A colleague argues that since the regression-adjusted estimator can be biased in finite samples, the unadjusted difference in means should always be preferred. Evaluate this argument, referring to the order of the bias and the variance comparison in the remark on the role of the regression model in Section 5.3.5.

Chapter 6

Propensity Score Methods

Learning Objectives

By the end of this chapter, students should be able to:

1. Define the propensity score $\pi(X) = P(T{=}1 \mid X)$ and explain why it is a *balancing score*.
2. State and prove the Balancing Property ($T \perp\!\!\!\perp X \mid \pi(X)$); prove that conditional exchangeability given X transfers to any balancing score $b(X)$, and explain separately why positivity transfers from X to $b(X)$ (the Propensity Score Theorem).
3. Select covariates for the propensity model on causal grounds, estimate the score, and evaluate the fitted design through balance and overlap diagnostics rather than treatment-prediction accuracy.
4. Derive the IPW identification formula for the ATE and explain the role of the Horvitz–Thompson representation.
5. Explain how IPW estimators can target either the ATE or the ATT depending on the weighting scheme, and how the nearest-neighbor matching estimator presented in this chapter targets the ATT.
6. Distinguish *positivity* (weak overlap) from *strong overlap*, describe the practical consequences of near-violations, and explain the estimand consequences of trimming.
7. Articulate the fundamental limitation of propensity-score methods — the unconfoundedness assumption — and explain why this motivates instrumental variables (Chapter 7).

Reading Guide

First reading: the curse-of-dimensionality motivation; the balancing property and the *statements* of Lemma 6.1 and Theorem 6.2; the whole of the design-stage material — covariate selection (Section 6.3.1), balance versus prediction, diagnostics (Section 6.3.5), and the workflow (Section 6.3.7); the matching and IPW estimator definitions with the weighted-balance identities (Lemma 6.2); positivity, trimming, and re-targeting; and the lab's headline lessons. **Second reading**: the general-balancing-score proof of Theorem 6.2; the nonstandard-inference remark for matching; the lab's estimated-score efficiency analysis and control-variate interpretation, and the level-versus-shape decomposition of the misspecification bias.

6.1 Motivation: The Curse of Dimensionality

Chapter 5 established that under consistency, conditional exchangeability, positivity, and integrability, the ATE is identified by the back-door adjustment formula, and developed three estimation strategies: regression adjustment, standardization, and stratification. All three require, in practice, that we condition on a covariate vector X. When X is low-dimensional, this is feasible. When X has many components, it is not.

The problem. Stratification requires cells $\{X = x\}$ to contain both treated and control units. With p

binary covariates, there are 2^p cells. With $p = 10$, that is 1024 cells — far more than most datasets can support. Regression adjustment avoids the cell-count problem but requires a correctly specified model for $\mathbb{E}[Y \mid T, X]$, which becomes harder to specify reliably as p grows.

The solution. Rosenbaum and Rubin (1983) showed that it is sufficient to condition on a single scalar function of X: the *propensity score* $\pi(X) = P(T{=}1 \mid X)$. The propensity score reduces the adjustment dimension without changing the identified estimand, provided unconfoundedness and positivity hold. Identification power is preserved; finite-sample efficiency, covariate balance, and robustness to model misspecification, however, depend on how well the score is estimated and how it is used in the downstream estimator.

6.2 The Propensity Score and Its Balancing Properties

The central insight of Rosenbaum and Rubin (1983) is that the p-dimensional adjustment problem of Chapter 5 reduces to a one-dimensional one. The right abstraction is the *balancing score*: any function of X that renders treatment assignment conditionally independent of the covariates. The propensity score is the canonical — and coarsest — such function.

6.2.1 Balancing Scores and the Propensity Score

Definition: Balancing Score [@rosenbaum1983central]

A function $b(X)$ is called a **balancing score** if $T \perp\!\!\!\perp X \mid b(X)$.

The full covariate vector X is trivially a balancing score: $T \perp\!\!\!\perp X \mid X$ holds because, conditional on X, the conditional distribution of X is degenerate and therefore cannot depend on treatment status. The propensity score is the *coarsest* balancing score (Lemma 6.1 below), and therefore provides the greatest dimension reduction without sacrificing identification.

Definition: Propensity Score [@rosenbaum1983central]

For a binary treatment $T \in \{0, 1\}$ and observed covariates X, the **propensity score** is $\pi(X) = P(T{=}1 \mid X)$.

In a randomized experiment, $\pi(X)$ is a known constant for all units — equal to 0.5 under balanced Bernoulli assignment, and, marginally for each unit, under complete randomization with exactly $n/2$ treated, where the assignment indicators are dependent because the treated count is fixed. In an observational study, $\pi(X)$ varies with X and must be estimated from data.

Connection to strong ignorability. Under strong ignorability $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$, the potential outcomes carry no further information about treatment assignment once X is given, so the treatment assignment mechanism satisfies

$$P(T{=}1 \mid X,\, Y(0),\, Y(1)) = P(T{=}1 \mid X) = \pi(X).$$

This equality is a *consequence* of the ignorability assumption, not part of the definition of $\pi(X)$. What unconfoundedness rules out is precisely any residual dependence of treatment on the potential outcomes after conditioning on X.

Theorem 6.1 (The Propensity Score Is a Balancing Score (Rosenbaum and Rubin 1983))**.** $T \perp\!\!\!\perp X \mid \pi(X)$.

Proof. It suffices to show $P(T{=}1 \mid X, \pi(X)) = P(T{=}1 \mid \pi(X))$. Since $\pi(X)$ is a deterministic function of X, conditioning on both X and $\pi(X)$ is the same as conditioning on X alone, so $P(T{=}1 \mid X, \pi(X)) = P(T{=}1 \mid X) = \pi(X)$. For the right-hand side, the law of iterated expectations gives

$$P(T{=}1 \mid \pi(X)) = \mathbb{E}\big[P(T{=}1 \mid X) \,\big|\, \pi(X)\big] = \mathbb{E}\big[\pi(X) \,\big|\, \pi(X)\big] = \pi(X),$$

where the last equality holds because $\pi(X)$ is measurable with respect to itself. Hence both sides equal $\pi(X)$. □

Remark: Covariate Balance and Design-Stage Analysis

Theorem 6.1 has a practical consequence that Imbens and Rubin (2015) call *design before analysis*: covariate balance can be assessed before the outcome data are consulted. The balancing property implies

$$f(X \mid T{=}1,\, \pi(X)) = f(X \mid T{=}0,\, \pi(X)) \quad \text{a.s. on } \{0 < \pi(X) < 1\},$$

i.e., at every score value where both arms occur with positive probability, the conditional distribution of X given the propensity score is the same in the two arms.
Stratifying on a discretized $\hat{\pi}(X)$ and comparing covariate distributions within strata — via standardized mean differences (Section 6.3.5), which are preferred over t-tests because they do not conflate imbalance with sample size — diagnoses whether the propensity score model achieves adequate balance, without ever looking at Y. Because the diagnostic does not involve the outcome, specifications cannot be selected on the basis of favorable effect estimates. Analysts can — and generally should — iterate the propensity model in response to observed imbalance; doing so while blinded to Y is what keeps the design stage objective.

Remark: Observed Covariates Only

Theorem 6.1 is a statement about the *observed* covariate distribution. It says that within strata of $\pi(X)$, the distribution of X is the same in the treated and control groups. It does *not* say anything about unobserved confounders U: if U affects both T and Y and is not captured by X, the balancing property fails to close the back-door path through U.

6.2.2 The Propensity Score Theorem

The balancing property has a fundamental identification consequence: unconfoundedness at X implies unconfoundedness at *any* balancing score of X, and in particular at the scalar $\pi(X)$. Before stating that result we record a structural fact about balancing scores: among all of them, the propensity score is the coarsest.

Lemma 6.1 (The Propensity Score Is the Coarsest Balancing Score (Rosenbaum and Rubin 1983))**.** *If $b(X)$ is any balancing score — that is, $T \perp\!\!\!\perp X \mid b(X)$ — then $\pi(X)$ is a function of $b(X)$: $\pi(X) = g(b(X))$ for some measurable function g.*

Proof. Since $b(X)$ is a balancing score, $P(T{=}1 \mid X,\, b(X)) = P(T{=}1 \mid b(X))$. But $b(X)$ is a function of X, so conditioning on $b(X)$ is already implicit in conditioning on X:

$$\pi(X) = P(T{=}1 \mid X) = P(T{=}1 \mid X,\, b(X)) = P(T{=}1 \mid b(X)).$$

Hence $\pi(X)$ is a measurable function of $b(X)$ alone. □

The lemma is the precise statement of the *Rosenbaum–Rubin coarsest-balancing-score property*: the propensity score is a measurable function of every balancing score, or equivalently $\sigma\{\pi(X)\} \subseteq \sigma\{b(X)\}$. Every balancing score is therefore at least as fine as $\pi(X)$, which is what makes $\pi(X)$ the coarsest balancing score and the maximally parsimonious adjustment variable. The reverse factorization need *not* hold: a general balancing score — X itself, for example — need not be a function of $\pi(X)$. (By analogy with classical sufficiency, $\pi(X)$ plays the role of a minimal sufficient reduction of X for the treatment indicator, although no parametric model is involved here.)

Theorem 6.2 (Propensity Score Theorem (Rosenbaum and Rubin 1983))**.** *Suppose joint conditional exchangeability holds: $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$. Then for any balancing score $b(X)$,*

$$(Y(0),\, Y(1)) \;\perp\!\!\!\perp\; T \,\Big|\, b(X);$$

no positivity is needed for this conditional-independence transfer. If, in addition, $0 < \pi(X) < 1$ almost surely, then $0 < P(T{=}1 \mid b(X)) < 1$ almost surely as well, so strong *ignorability given X implies strong ignorability given $b(X)$, and adjustment for $b(X)$ identifies the potential-outcome means. In particular, taking $b(X) = \pi(X)$ gives strong ignorability with the scalar propensity score in place of the full covariate vector.*

Proof (second reading)

Write $\mathbf{Y} = (Y(0), Y(1))$ and let $b(X)$ be any balancing score. By Lemma 6.1 there is a measurable function g with $\pi(X) = g(b(X))$. Since $b(X)$ is a function of X, the law of iterated expectations gives

$$\begin{aligned} P(T{=}1 \mid \mathbf{Y}, b(X)) &= \mathbb{E}\big[P(T{=}1 \mid \mathbf{Y}, X) \mid \mathbf{Y}, b(X)\big] \\ &= \mathbb{E}\big[P(T{=}1 \mid X) \mid \mathbf{Y}, b(X)\big] \quad \text{(joint ignorability)} \\ &= \mathbb{E}\big[g(b(X)) \mid \mathbf{Y}, b(X)\big] = g(b(X)). \end{aligned}$$

The same computation without $\mathbf{Y}$ in the conditioning set yields $P(T{=}1 \mid b(X)) = \mathbb{E}[\pi(X) \mid b(X)] = g(b(X))$. Hence $P(T{=}1 \mid \mathbf{Y}, b(X)) = P(T{=}1 \mid b(X))$, and because T is binary, $\mathbf{Y} \perp\!\!\!\perp T \mid b(X)$; note that positivity was not used in this argument.
If additionally $0 < \pi(X) < 1$ almost surely, positivity transfers to the balancing score: $P(T{=}1 \mid b(X)) = \pi(X) \in (0,1)$ almost surely. Taking $b(X) = \pi(X)$, with g the identity, gives the propensity score case. $\square$

Remark: Identification via the Propensity Score

Combining Theorem 6.2 with consistency and positivity yields an alternative identification formula. Define the arm-specific conditional mean functions $m_t(p) = \mathbb{E}[Y \mid T{=}t,\, \pi(X) = p]$; then $\mathbb{E}[Y(t)] = \mathbb{E}[m_t(\pi(X))]$.
A common misconception is that this means the two conditional regressions $\mathbb{E}[Y \mid \pi(X), T{=}t]$ and $\mathbb{E}[Y \mid X, T{=}t]$ are equal. They are not: they are different functions and their pointwise values generally differ. The theorem guarantees only that their marginal expectations agree:

$$\mathbb{E}[\mathbb{E}[Y \mid \pi(X),\, T{=}t]] = \mathbb{E}[\mathbb{E}[Y \mid X,\, T{=}t]] = \mathbb{E}[Y(t)].$$

Note also that the formula licenses no particular parametric specification: identification through the propensity score requires that the arm-specific functions m_t be estimated sufficiently well — nonparametrically or by a correctly specified model — and then standardized over the marginal distribution of $\pi(X)$. An arbitrary parametric regression of Y on T and $\pi(X)$ need not identify the ATE.

The Dimension Reduction

Theorem 6.2 says that instead of adjusting for the full high-dimensional vector X, it is sufficient to adjust for the scalar $\pi(X)$. This reduces the *curse of dimensionality* in covariate adjustment from a $\dim(X)$-dimensional problem to a one-dimensional one. Together with consistency and positivity, identification of the ATE follows:

$$\tau_{\text{ATE}} = \mathbb{E}\big[\mathbb{E}[Y \mid T{=}1, \pi(X)] - \mathbb{E}[Y \mid T{=}0, \pi(X)]\big].$$

6.3 Constructing the Propensity Score

The true propensity score is *unknown* in observational studies and must be estimated from the data. Its practical construction, however, requires more than choosing a binary-response algorithm. Three logically distinct decisions must be kept separate. First, *causal reasoning* determines the pretreatment adjustment set X: the set must be sufficient to support conditional exchangeability, for example by blocking every back-door path from T to Y (Chapters 3 and 5). Second, *statistical modeling* determines how the treatment probability depends on the chosen covariates. Third, *design diagnostics* determine whether the fitted score actually produces adequate balance and overlap for the chosen estimand. A good predictive model for treatment does not automatically produce a good causal design, and no statistical treatment model can repair an invalid adjustment set chosen from an incorrect causal graph.

6.3.1 Covariate Selection: What Belongs in X?

The propensity score inherits its causal validity from the adjustment set on which it is conditioned. The primary rule is therefore graphical and scientific rather than predictive: X should be a pretreatment set sufficient to block all back-door paths from T to Y. The causal assumption is $Y(t) \perp\!\!\!\perp T \mid X$; estimating $\pi(X)$ cannot make this assumption true if it was false for the chosen X.

> What the Propensity Score Cannot Do
>
> The propensity score balances the covariates included in X. It does not balance unmeasured confounders, repair adjustment for a collider, or justify conditioning on a post-treatment variable. The causal validity of X must be established — from subject-matter knowledge and the causal graph — *before* the propensity score is estimated.

Within a causally valid pretreatment set, candidate variables need not play identical roles. A useful first taxonomy classifies them by whether they predict treatment, outcome, both, or neither (Brookhart et al. 2006; VanderWeele 2019):

Variable type	**Predicts T?**	**Predicts Y?**	**Typical recommendation**
Confounder	yes	yes	include
Outcome predictor only	no or weakly	yes	usually include
Treatment predictor only	yes	no	include only with caution
Predictor of neither	no	no	usually unnecessary

Variables related to both treatment and outcome are central confounders and should ordinarily be included: they are needed to block back-door paths. Pretreatment variables that strongly predict the outcome may be useful even when they predict treatment only weakly, because they can improve precision and make residual imbalance less consequential. Variables that strongly predict treatment but are unrelated to the potential outcomes require more caution: although they improve prediction of T, they can push estimated scores toward zero or one, reduce overlap and effective sample size, and inflate the variance of weighted estimators without removing appreciable confounding bias; moreover, conditioning on such a near-instrument can *amplify* the bias arising from any residual unmeasured confounding. Pure noise variables mainly add estimation variability and may worsen finite-sample balance.

These are tendencies rather than universal rules, and they do not justify automatic variable deletion. Whether a variable is truly unrelated to the potential outcomes is a scientific judgment subject to uncertainty, and plausible pretreatment confounders should generally be retained even when their treatment-model coefficients are small or statistically insignificant.

Two exclusions, by contrast, are categorical when the target is a total effect. *Post-treatment variables* should not be conditioned on: doing so can block part of the causal effect, induce selection bias, and change the estimand. *Pretreatment colliders* can be equally damaging despite being measured before treatment: in the M-structure of panel (b) below, the variable C is pretreatment, yet conditioning on it opens the noncausal path $T \leftarrow U_1 \rightarrow C \leftarrow U_2 \rightarrow Y$ that is blocked when C is left alone (Chapter 3). "Include every available variable" is therefore not a safe default.

6.3.2 Parametric Estimation by Logistic Regression

The classical approach models

$$\log \frac{\pi(X)}{1-\pi(X)} = \beta_0 + \sum_{j=1}^{p} \beta_j X_j + \sum_{j=1}^{p} \gamma_j \, h_j(X_j) + \sum_{j<k} \delta_{jk} \, X_j X_k,$$

where the h_j are optional nonlinear transformations (polynomials, splines) and the cross-product terms capture interactions among covariates, and estimates the coefficients by maximum likelihood. Linearity on the logit scale in the raw covariates is an assumption, not a default truth: nonlinear terms and covariate-by-covariate interactions are often needed to achieve balance. (Treatment-by-covariate interactions have no place in this model — treatment is the *response* here.) Terms should be added based on causal knowledge and the balance diagnostics of Section 6.3.5, not predictive fit alone.

Variable selection by treatment-model p-values is generally inappropriate. A covariate may be an important confounder or outcome predictor even when its coefficient in the treatment model is small or statistically insignificant, and removing it can worsen causal balance without materially changing treatment-prediction accuracy. Backward selection driven by significance serves sparse prediction, which is not the objective of a propensity model.

6.3.3 Flexible and Machine-Learning Estimators

When X is high-dimensional or when the true propensity score is non-linear, machine learning methods — gradient boosted trees, random forests, regularized logistic regression — can estimate $\pi(X)$ more flexibly, reducing misspecification of the treatment model without fully prespecifying nonlinearities and interactions. Predictive accuracy alone, however, is not the relevant success criterion: hyperparameters tuned solely to classification loss (log loss, AUC) may improve prediction precisely by producing more extreme probability estimates, which destabilizes the resulting weights.

A related issue is *overfitting*: an estimator that nearly perfectly separates treated and control units in-sample produces fitted scores near one for treated units and near zero for controls, mimicking an overlap violation that need not exist in the population. Cross-fitting (Chapter 12) removes the *same-sample* component of this problem by evaluating each unit's score on a fold not used to fit it, but it does not create overlap and does not prevent extreme out-of-fold predictions: a genuinely near-deterministic treatment rule yields extreme scores on held-out data as well. Regularization, overlap diagnostics, and — when scientifically justified — trimming or re-targeting the estimand remain separate requirements. Whatever the learner, the fitted design must ultimately be evaluated through the balance and overlap it delivers; no machine-learning method eliminates the need for causal covariate selection or design diagnostics.

6.3.4 The Goal Is Balance, Not Treatment Prediction

In ordinary classification, a model is rewarded for separating the two classes. In propensity-score analysis, strong separation signals trouble: fitted scores near one for treated units and near zero for controls mean little common support, extreme inverse-probability weights, and an estimator whose value depends on a handful of observations. A high AUC may therefore indicate a difficult causal comparison rather than a successful design. Predictive criteria are useful for model fitting, but they are not sufficient criteria for choosing a propensity-score design.

The estimated score, moreover, is a design device, not the final diagnostic object. Conditioning on the *true* score balances X in distribution (Theorem 6.1), but matching or weighting approximately on an *estimated* score does not guarantee adequate finite-sample balance. After matching or weighting, balance must therefore be assessed on the original covariates and, when scientifically relevant, on nonlinear functions and interactions of them; similar distributions of $\hat{\pi}(X)$ across arms do not by themselves establish that the underlying covariates are balanced.

> **Propensity Score Estimation Is a Nuisance, Not the Goal**
>
> The propensity score is estimated in order to construct a good estimator of τ_{ATE}. Balance diagnostics assess whether the estimated score has achieved covariate balance on the *observed* covariates, but they say nothing about whether unobserved confounders are balanced. Relatedly, high treatment-prediction accuracy is not the goal: the estimated score is judged by the balance and overlap it delivers for the causal target, not by its classification performance.

6.3.5 Balance and Overlap Diagnostics

The standard scalar diagnostic for covariate balance is the *standardized mean difference* (SMD). For covariate X_j, let $\bar{X}_{1j}$ and $\bar{X}_{0j}$ denote the treated and control sample means and s_{1j}^2, s_{0j}^2 the corresponding sample variances. Then

$$\text{SMD}(X_j) = \frac{\bar{X}_{1j} - \bar{X}_{0j}}{\sqrt{(s_{1j}^2 + s_{0j}^2)/2}};$$

for a binary covariate, the numerator is the difference in proportions, with the analogous pooled scale in the denominator. Dividing by the pooled scale makes the diagnostic free of the units of X_j and — unlike a t-statistic — insensitive to the sample size, so it measures the magnitude of imbalance rather than its statistical detectability. A common informal benchmark treats $|\mathrm{SMD}| \le 0.1$ as adequate balance; this is a heuristic, not a test, and whether residual imbalance is consequential depends on how strongly the covariate predicts the outcome, on the sample size, and on the estimand. For weighted or matched analyses, the arm means are replaced by their design-weighted counterparts (with reused control units counted with their matching multiplicities); the denominator, however, should be held fixed at the pre-adjustment pooled scale, so that before-and-after values are directly comparable.

Balance of means alone can be inadequate when the outcome depends nonlinearly on the covariates. Useful supplements include variance ratios, quantile differences, empirical-distribution plots, and SMDs computed on squares, interactions, and other scientifically relevant transformations.

Balance and overlap are distinct, and both must be checked: a design can show good mean balance after weighting while relying on a few extreme weights, and raw overlap can look reasonable while nonlinear imbalance remains. Overlap diagnostics start with plots of the estimated score distributions in the two arms on a common scale. Which comparison matters depends on the estimand: for the ATT, the question is whether the control scores cover the treated-score distribution; for the ATE, both arms must cover the full target population. Weight summaries — minimum, maximum, selected quantiles, and the coefficient of variation — expose instability directly. For matching, the bookkeeping of the design plays the same role: the number of unmatched treated units, the number of distinct controls used and how often each is reused, and the distribution of match distances.

The cost of extreme weights can be summarized by the *effective sample size* of a weighted arm with weights $\{w_i\}$,

$$n_{\text{eff}} = \frac{\left(\sum_i w_i\right)^2}{\sum_i w_i^2},$$

which equals the actual sample size when the weights are constant and collapses toward 1 when a single weight dominates. For two-arm estimators it is usually most informative arm by arm. A weighted arm with 500 units but $n_{\text{eff}} = 40$ delivers roughly the precision of 40 equally weighted observations; reporting n_{eff} alongside the point estimate makes the price of near-overlap violations concrete.

6.3.6 Design Objectives Depend on the Estimand

The same propensity model need not serve every estimator equally, so the diagnostics above should be tailored to the target. For *matching*, the score must support close matches and covariate balance within the matched sample; the relevant diagnostics are match distances, residual SMDs among matched units, unmatched treated units, and control reuse. For *ATE weighting*, with weights $1/\hat{\pi}(X_i)$ for treated and $1/\{1-\hat{\pi}(X_i)\}$ for control units, both arms are re-weighted to the full target population, so weighted balance in both arms, weight tails, and arm-specific effective sample sizes are the primary concerns. For *ATT weighting*, with weights

$$w_i = T_i + (1-T_i)\,\frac{\hat{\pi}(X_i)}{1-\hat{\pi}(X_i)},$$

the treated arm is left unweighted and the central question is whether the controls can represent the treated units, so control coverage of the treated-score distribution is the key diagnostic.

> Remark: Overlap Weights
>
> The *overlap weights* $w_i = T_i\{1-\hat{\pi}(X_i)\} + (1-T_i)\,\hat{\pi}(X_i)$ emphasize units with scores near 1/2 and are bounded by construction, so trimming is not needed merely to control divergent inverse weights; covariate balance, finite-sample support in both arms, and the causal validity of X must still be assessed. The price is a change of estimand: they target the average effect in the overlap population — an instance of the re-targeting idea of Section 6.6.4 — rather than the original ATE.

The figure below displays the four weighting schemes as functions of the propensity score. The picture makes estimand targeting immediate: each scheme up-weights the units that must stand in for the population it targets, and instability appears exactly where a required weight diverges.

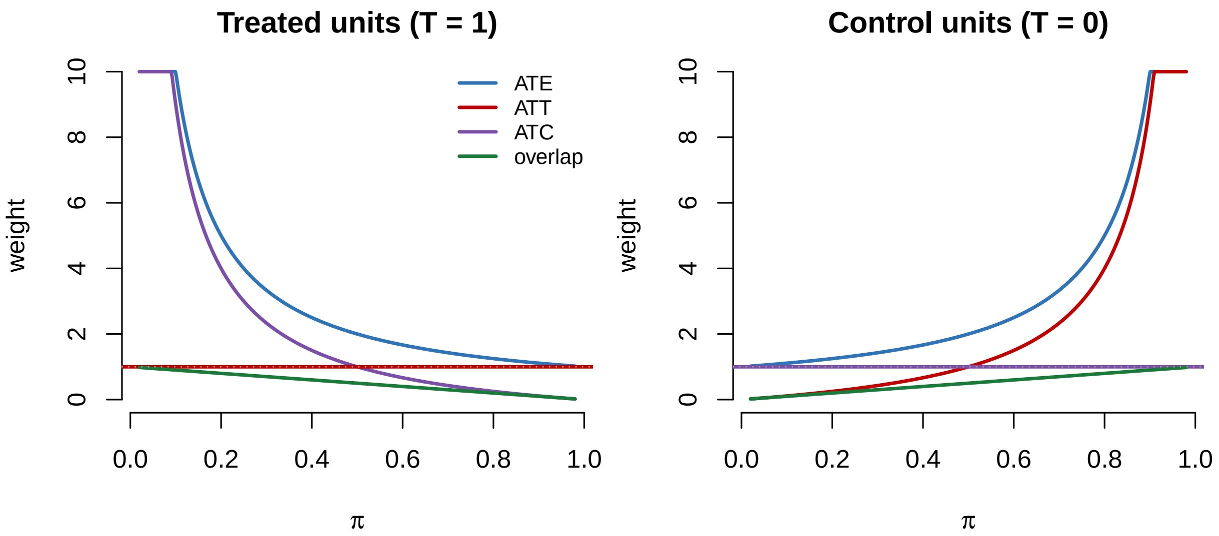

Figure 6.1: Weights attached to treated units (left) and control units (right) as functions of the propensity score, for the ATE, ATT, ATC, and overlap schemes. Each scheme up-weights the units standing in for its target population: the ATE diverges in whichever arm is scarce, the ATT leaves treated units unweighted while up-weighting high-π controls, the ATC is the mirror image, and the overlap weights are bounded. Curves truncated at weight 10.

The general principle: a propensity model cannot be evaluated independently of the estimand and the estimator that will use it.

6.3.7 A Design-Stage Workflow

Propensity-score design is typically *iterative*: specify a treatment model, construct the proposed matched or weighted sample, assess balance and overlap, revise the design when the diagnostics are inadequate, and repeat. Revisions may include nonlinear terms, interactions, an alternative link or learner, regularization, a modified matching ratio or caliper, a different weighting target, or trimming. The outcome data remain hidden throughout this cycle.

Outcome blinding, in particular, does not mean that the specification is fixed after one fit. Three levels should be distinguished. *Outcome-blind design*: treatment-effect estimates are never inspected while the propensity model is being specified. *Balance-driven revision*: altering the model in response to observed imbalance is legitimate — indeed expected. *Design lock*: once balance and overlap are judged acceptable, the matching or weighting rule is fixed, and only then are outcomes analyzed. The benefit is not that subjectivity disappears; it is that design decisions cannot be selected because they produce a preferred causal estimate.

A Propensity-Score Design Workflow

1. **Define the estimand.** Decide whether the target is the ATE, ATT, ATC, or an overlap-population effect.
2. **Specify the causal adjustment set.** Use subject-matter knowledge and the causal graph to select pretreatment variables that block all back-door paths. Exclude post-treatment variables and colliders.
3. **Specify an initial treatment model.** Include plausible nonlinear terms and interactions among covariates when needed. Do not select variables by treatment-model significance.
4. **Construct the design.** Apply the intended matching or weighting procedure.
5. **Assess covariate balance.** Examine standardized mean differences, distributional diagnostics, and balance of scientifically relevant transformations and interactions.
6. **Assess overlap and stability.** Inspect score distributions, match distances, weight tails, effective sample sizes, and the number of unmatched or trimmed units.
7. **Revise without using outcomes.** Modify the model or design until balance and overlap

are acceptable, or conclude that the original target population is not supportable.

8. **Lock the design and analyze outcomes.** After the design is fixed, estimate the causal effect and quantify uncertainty validly for the chosen procedure.

Example: Variable Selection — Higher AUC, Worse Design

Consider five candidate variables: age A and baseline severity S, each affecting both treatment and outcome; hospital preference H, strongly predictive of treatment but essentially unrelated to the potential outcomes; post-treatment adherence M; and a pretreatment collider C driven by two latent variables as in panel (b) of the figure above. Compare three propensity models:

$$\text{Model 1: } \pi(A,S), \qquad \text{Model 2: } \pi(A,S,H), \qquad \text{Model 3: } \pi(A,S,H,M,C).$$

Model 1 contains the essential adjustment set. Model 2 predicts treatment better but, if H nearly determines treatment, pushes fitted scores toward the boundaries, inflating weights and shrinking the effective sample size without removing additional confounding. Model 3 is causally invalid regardless of its fit: it conditions on the mediator M and the collider C. Qualitatively:

Model	**AUC**	**Max weight**	n_{eff}	**Causally valid?**
$\pi(A,S)$	moderate	moderate	high	yes
$\pi(A,S,H)$	high	large	low	yes, but imprecise
$\pi(A,S,H,M,C)$	very high	extreme	poor	no

The model with the highest AUC is the worst causal design.

6.4 Matching on the Propensity Score

The propensity score also motivates *matching*: for each treated unit i, find a control unit j with $\hat{\pi}(X_j) \approx \hat{\pi}(X_i)$, and use Y_j as a local estimate of $\mathbb{E}[Y(0) \mid \pi(X) = \pi(X_i)]$ (Abadie and Imbens 2006, 2016). It is important to be precise about what is being estimated: the matched control outcome Y_j is not a substitute for the individual counterfactual $Y_i(0)$, which is unobservable and contains an irreducible idiosyncratic component. Rather, Y_j is a noisy local approximation to the *conditional mean* $\mathbb{E}[Y(0) \mid \pi(X) = \pi(X_i)]$, which is identified from observed control outcomes via the chain

$$\mathbb{E}[Y(0) \mid \pi(X) = \pi(X_i)] = \mathbb{E}[Y(0) \mid \pi(X) = \pi(X_i),\, T = 0] = \mathbb{E}[Y \mid \pi(X) = \pi(X_i),\, T = 0],$$

where the first equality uses Theorem 6.2 and the second uses consistency. Averaging across treated units targets — and, under the regularity conditions discussed below, consistently estimates — the ATT:

$$\hat{\tau}_{\text{ATT}} = \frac{1}{n_1} \sum_{i:\, T_i = 1} [Y_i - Y_{\hat{\jmath}(i)}], \tag{6.1}$$

where n_1 is the number of treated units and $\hat{\jmath}(i)$ is the index of the matched control unit.

6.4.1 One-to-One Nearest Neighbor Matching

The most common matching algorithm is *nearest-neighbor* (1:1) matching (Abadie and Imbens 2006): for each treated unit i, find

$$\hat{\jmath}(i) = \arg\min_{j:\, T_j = 0} \left|\hat{\pi}(X_i) - \hat{\pi}(X_j)\right|.$$

When several controls are equally close, a convention is required — deterministic tie-breaking, averaging the outcomes of all equally close controls, or random selection; the choice affects the exact estimator and its variance, though not the identification logic. Matched control units may be selected with or without replacement. Matching with replacement typically reduces bias (each control unit is available to every treated unit, so match quality improves) but typically increases variance (a small number of control units may be used many times); neither direction is guaranteed without further conditions.

6.4.2 Caliper Matching

Caliper matching restricts matches to pairs within a maximum distance δ (the "caliper") on the propensity score scale. Treated units with no control match within the caliper are discarded. The caliper may be defined on the probability scale or — commonly in practice — on the logit scale $\operatorname{logit} \hat{\pi}(X)$, which spaces units more evenly near the boundaries of $[0, 1]$; the chosen scale must be reported, since numerical caliper values are not comparable across scales.

Because matchable treated units are retained and the rest discarded, the procedure no longer targets the original ATT: it targets an ATT restricted to the retained treated subpopulation with adequate overlap. And because the retained set depends on $\hat{\pi}$, this target is data-dependent in finite samples; it stabilizes to a fixed population estimand only as $\hat{\pi}$ converges.

Remark: Matching as Nonparametric Regression on $\pi(X)$

The key idea behind propensity score matching is that it implicitly fits a *nonparametric regression* of Y on $\pi(X)$ separately within each treatment arm. By Theorem 6.2, $(Y(0), Y(1)) \perp\!\!\!\perp T \mid \pi(X)$, so the conditional mean $\mathbb{E}[Y(t) \mid \pi(X) = p]$ is identified from observed data for each t. Nearest-neighbor matching approximates this conditional mean at each observed $\hat{\pi}(X_i)$ by averaging the outcomes of close matches, effectively performing local constant regression on the propensity score. The dimension reduction from Theorem 6.2 is therefore what makes nonparametric estimation feasible: instead of smoothing over the full covariate vector X, one smooths over the scalar $\pi(X)$.

Remark: Design Transparency

Propensity score matching provides *design transparency*: by inspecting matched pairs, the analyst can directly assess whether treated and control units are similar on observed covariates. Comparing $\bar{X}_1$ and $\bar{X}_0$ within matched pairs via standardized mean differences is a standard pre-analysis diagnostic that can be conducted without consulting the outcome Y. In practice, the propensity specification should be revised until the matched sample shows satisfactory balance on the original covariates (Section 6.3.7), not merely small distances in the estimated score.

Inference after Matching Is Nonstandard

Matching estimators with a fixed number of matches are not classical sample averages. The matching step induces dependence across units; when matching directly on a multidimensional X, the estimator carries a bias term that can dominate the $\sqrt{n}$ rate (Abadie and Imbens 2006); and the naive bootstrap fails to reproduce its sampling distribution (Abadie and Imbens 2008). Valid standard errors require purpose-built variance estimators, and when the propensity score is estimated rather than known, the estimation error in $\hat{\pi}$ must additionally be accounted for (Abadie and Imbens 2016). Consistency of the matching estimator itself also requires more than strong ignorability: sufficient accuracy (or balancing behavior) of the fitted score, smoothness of the relevant conditional mean functions, and match discrepancies that vanish asymptotically.

6.5 Inverse Probability Weighting

6.5.1 The IPW Identification Formula

The ATE admits an alternative representation as an *inverse probability weighting* (IPW) formula, in which each observed outcome is weighted by the inverse probability of the treatment actually received.

Theorem 6.3 (IPW Identification)**.** *Fix $t \in \{0, 1\}$ and assume: (i)* consistency*: $T = t$ implies $Y = Y(t)$; (ii)* weak conditional exchangeability*: $Y(t) \perp\!\!\!\perp T \mid X$; (iii)* positivity*: $P(T{=}t \mid X) > 0$ almost surely; and (iv)* integrability*: $\mathbb{E}|Y(t)| < \infty$. Then*

$$\mathbb{E}\left[\frac{T \cdot Y}{\pi(X)}\right] = \mathbb{E}[Y(1)], \qquad \mathbb{E}\left[\frac{(1-T) \cdot Y}{1-\pi(X)}\right] = \mathbb{E}[Y(0)].$$

If the conditions hold for both treatment levels,

$$\tau_{\text{ATE}} = \mathbb{E}\left[\frac{T \cdot Y}{\pi(X)} - \frac{(1-T) \cdot Y}{1-\pi(X)}\right]. \tag{6.2}$$

Proof.

$$\mathbb{E}\left[\frac{TY}{\pi(X)}\right] = \mathbb{E}\left[\frac{T\,Y(1)}{\pi(X)}\right] = \mathbb{E}\left[\frac{Y(1)}{\pi(X)}\,\mathbb{E}[T \mid X,\, Y(1)]\right] = \mathbb{E}\left[\frac{Y(1)}{\pi(X)}\,\pi(X)\right] = \mathbb{E}[Y(1)].$$

The first equality uses consistency ($Y = Y(1)$ when $T{=}1$). The second is the law of iterated expectations conditioning on $(X, Y(1))$; the factor $Y(1)/\pi(X)$ is measurable with respect to the conditioning variables and pulls out of the inner expectation. The third uses ignorability: $Y(1) \perp\!\!\!\perp T \mid X$ implies $\mathbb{E}[T \mid X, Y(1)] = \pi(X)$. Integrability, with positivity, ensures each expectation is well defined. The case $t = 0$ is symmetric. □

> Remark: Scope of the Ignorability Assumption Used
>
> Theorem 6.3 is deliberately stated arm by arm under the *marginal* independence $Y(t) \perp\!\!\!\perp T \mid X$, not the full joint independence: each per-arm identity needs only the corresponding marginal condition together with arm-specific positivity. The joint independence is invoked throughout this chapter for stylistic uniformity with the back-door theorem of Chapter 5 and because joint-distribution targets such as $\operatorname{Var}\{Y(1) - Y(0)\}$ at least require it — although even joint ignorability identifies only the two marginal potential-outcome distributions, not the cross-world dependence between $Y(0)$ and $Y(1)$, so such targets remain only partially identified without further assumptions (Chapter 4). The marginal version suffices for any estimand expressible as a contrast of marginal means.

6.5.2 The Horvitz–Thompson Estimator

The corresponding sample estimator is the *Horvitz–Thompson* (HT) IPW estimator:

$$\hat{\tau}_{\text{IPW}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{T_i Y_i}{\hat{\pi}(X_i)} - \frac{(1-T_i)Y_i}{1-\hat{\pi}(X_i)}\right]. \tag{6.3}$$

This estimator is consistent when $\hat{\pi}(X)$ converges to $\pi(X)$, given positivity and mild moment conditions — with flexible nuisance estimators evaluated on the same sample, additional conditions such as sample splitting may be required (Chapter 12) — but it can be *inefficient* due to extreme weights. In practice, a fitted score used for weighting is evaluated through weighted covariate balance, the weight distribution, and effective sample sizes.

The *normalized* (Hájek) estimator divides by the sum of weights within each group and is often more stable in practice:

$$\hat{\tau}_{\text{HJ}} = \frac{\sum_i T_i Y_i/\hat{\pi}(X_i)}{\sum_i T_i/\hat{\pi}(X_i)} - \frac{\sum_i (1-T_i)Y_i/(1-\hat{\pi}(X_i))}{\sum_i (1-T_i)/(1-\hat{\pi}(X_i))}. \tag{6.4}$$

The efficient doubly robust estimator (AIPW) of Chapter 11 combines the outcome model and the propensity score; under correct nuisance specification it can achieve higher efficiency than either pure IPW or pure outcome regression alone.

6.5.3 Why the Weights Work: Weighted-Balance Identities

The balancing theorem and the IPW formula are two faces of the same mechanism, and the connection can be made algebraic: inverse-probability weights re-create, inside each treatment arm, the covariate distribution of the target population.

Lemma 6.2 (Weighted-Balance Identities)**.** *Let $h(X)$ be any integrable function of the covariates and suppose $0 < \pi(X) < 1$ almost surely. Then*

$$\mathbb{E}\left[\frac{T\,h(X)}{\pi(X)}\right] = \mathbb{E}[h(X)] \qquad \text{and} \qquad \mathbb{E}\left[\frac{(1-T)\,h(X)}{1-\pi(X)}\right] = \mathbb{E}[h(X)].$$

Moreover, for the treated target population,

$$\frac{1}{P(T{=}1)}\,\mathbb{E}\left[(1-T)\,\frac{\pi(X)}{1-\pi(X)}\,h(X)\right] = \mathbb{E}[h(X) \mid T{=}1].$$

Proof. For the first identity, condition on X and note that $h(X)/\pi(X)$ pulls out of the inner expectation: $\mathbb{E}[Th(X)/\pi(X)] = \mathbb{E}[h(X)\,\mathbb{E}(T \mid X)/\pi(X)] = \mathbb{E}[h(X)]$. The second is symmetric. For the third,

$$\mathbb{E}\left[(1-T)\,\frac{\pi(X)}{1-\pi(X)}\,h(X)\right] = \mathbb{E}\left[\{1-\pi(X)\}\,\frac{\pi(X)}{1-\pi(X)}\,h(X)\right] = \mathbb{E}[\pi(X)\,h(X)] = \mathbb{E}[T\,h(X)],$$

and $\mathbb{E}[T\,h(X)] = P(T{=}1)\,\mathbb{E}[h(X) \mid T{=}1]$. □

Taking $h \equiv 1$ shows that the average inverse weights in each arm have expectation one — the weight-sum quantities that reappear in the lab. Taking $h(X) = X_j$ shows that the weighted treated and control samples reproduce the covariate means of the target population: the full population under the ATE weights, the treated subpopulation under the ATT odds weights applied to the control arm. This is why the weights take their particular forms. Note that no potential outcomes appear in the lemma: like the balancing theorem, it is a property of the assignment mechanism alone. Substituting the outcome Y for $h(X)$ is what requires ignorability — that substitution is exactly the content of Theorem 6.3.

6.6 Overlap and Positivity

6.6.1 Positivity and Strong Overlap

We must distinguish two conditions on the propensity score that play different roles in identification and inference.

> Definition: Positivity (Weak Overlap)
>
> The **positivity** condition — also called *weak overlap* — requires $0 < \pi(X) < 1$ almost surely. Every unit has a positive probability of receiving either treatment or control — more precisely, this holds for P_X-almost every covariate value.

Together with unconfoundedness, positivity constitutes the *strong ignorability* of Rosenbaum and Rubin (1983). Positivity is the condition needed for *identification*: without it, certain strata have no counterfactual information and the ATE is not nonparametrically identified.

> Definition: Strong Overlap
>
> The **strong overlap** condition requires that there exists $c \in (0, 1/2)$ such that $c \le \pi(X) \le 1 - c$ almost surely.

Strong overlap bounds the inverse weights uniformly away from infinity. It is a convenient *sufficient* condition for the standard $\sqrt{n}$-consistent, asymptotically normal theory of later chapters (Chapter 12), because the uniform bound provides a transparent regularity condition — but it is not necessary in every model: root-n inference can survive under weaker, estimator-specific tail and moment conditions, provided the relevant influence function retains finite variance. Positivity alone permits identification but does not guarantee stable inference: when $\pi(X)$ approaches 0 or 1 in parts of the covariate space, IPW weights blow up and influence values become extreme, even though the parameter is technically identified (Khan and Tamer 2010).

Why positivity is necessary. The identification formula requires that both $\mathbb{E}[Y \mid T{=}1, X{=}x]$ and $\mathbb{E}[Y \mid T{=}0, X{=}x]$ be defined for P_X-almost all x in the support of X. If $\pi(x) = 0$ for some x, then no treated units exist with $X = x$, so $\mathbb{E}[Y \mid T{=}1, X{=}x]$ is not estimable from the data. Similarly if $\pi(x) = 1$.

6.6.2 Practical Consequences of Near-Violations

Exact violations of overlap are rare in practice. More common are *near-violations*: some covariate strata have very few treated or control units, making the estimated propensity score close to 0 or 1.

> Near-Overlap Problems
>
> When $\hat{\pi}(X_i)$ is close to 0 or 1, the IPW weight $1/\hat{\pi}(X_i)$ or $1/(1-\hat{\pi}(X_i))$ is very large. A small number of units with extreme weights can dominate the estimator, increasing variance dramatically. This is the *practical positivity violation* problem, distinct from the theoretical violation.

Near-violations arise when a covariate is a near-perfect predictor of treatment, when the covariate distributions of treated and control groups barely overlap, or when the propensity score model is over-flexible and fits the treatment variable nearly perfectly in-sample. Severe lack of overlap is therefore not necessarily a model-fitting failure: it may reflect a genuine absence of comparable treated and control units in the target population, in which case the honest responses are those of the next two subsections — trimming or re-targeting the estimand — rather than model revision.

6.6.3 Trimming Strategies

Trimming by propensity score. Restrict the analysis to units with estimated propensity scores in a prespecified interval $[\eta, 1-\eta]$ for some small $\eta > 0$. The estimand changes to the ATE in the trimmed subpopulation:

$$\tau_{\text{trim}} = \mathbb{E}\big[Y(1) - Y(0) \mid \eta \leq \pi(X) \leq 1-\eta\big]. \tag{6.5}$$

Note that the target population is defined by the *true* score $\pi(X)$, while the practical rule is applied with the estimated score $\hat{\pi}(X)$; the two coincide asymptotically when $\hat{\pi}$ is consistent. The choice of η is a precision–target-population trade-off rather than a bias–variance trade-off in the usual sense: removing regions with extreme scores eliminates the largest weights, but it also discards observations and changes the estimand. The discrepancy between τ_{trim} and the original ATE is a change of target, not a statistical bias relative to τ_{trim}, and the net variance need not decrease monotonically in η.

Crump et al. (2009) rule. Crump et al. (2009) derive an optimal symmetric trimming threshold — distribution-dependent, minimizing the asymptotic variance of the trimmed ATE estimator under their criterion. Cutoffs near 0.1 often emerge from this calculation, so the fixed rule $\eta = 0.1$ is a common heuristic, though not a universal optimum.

6.6.4 Re-targeting the Estimand to Avoid Overlap Failures

Trimming restricts the population in order to preserve the ATE within a region of good overlap. An alternative, complementary response is to leave the population alone but change the *estimand*: target the average treatment effect on a subset of the population for which the relevant side of positivity can be guaranteed.

ATT (target the treated population). The ATT averages $Y(1) - Y(0)$ over $X \mid T{=}1$ and therefore requires only that $\mathbb{E}[Y(0) \mid X = x]$ be identified where the treated population lives. The treated-covariate law is absolutely continuous with respect to P_X with density proportional to the propensity score, $dP_{X|T=1} = \{\pi(x)/P(T{=}1)\}\, dP_X$, so treated-population mass concentrates on $\{x : \pi(x) > 0\}$; there the requirement reduces to $\pi(x) < 1$. Hence, for nonparametric identification by adjustment:

$$\tau_{\text{ATT}} \text{ is identified iff } \pi(X) < 1 \text{ a.s. on the support of } X \mid T{=}1.$$

The ATT therefore tolerates regions with $\pi(x) = 0$ (no treated units are observed there in any case); it cannot tolerate regions inside the treated support where $\pi(x) = 1$ (no comparable controls).

ATC (target the control population). Symmetrically, the ATC averages over $X \mid T{=}0$ and is nonparametrically identified by adjustment iff $\pi(X) > 0$ a.s. on the support of $X \mid T{=}0$. These equivalences are statements about nonparametric identification by adjustment, under consistency and the relevant arm-specific ignorability condition; additional outcome-model restrictions could identify the same estimands under weaker support conditions.

Practical rule. When overlap fails at covariate values where $\pi(X) \approx 0$ — treated units are scarce or absent there — re-target to the ATT. When overlap fails where $\pi(X) \approx 1$ — controls are scarce or absent — re-target to the ATC. When overlap fails on both sides, restrict to the trimmed-population estimand.

6.7 Lab: Simulation Study of IPW and Matching Estimators

This lab compares five estimators on a five-covariate data-generating process with treatment-effect heterogeneity. With five continuous confounders, direct stratification is infeasible, making propensity-score methods the natural choice. The estimators span two estimands: the ATE (targeted by the HT and Hájek IPW estimators) and the ATT (targeted by the ATT-IPW variants and by matching). Five questions drive the comparison:

1. How much variance does weight normalization save (Hájek over HT) for ATE estimation, and why does the baseline outcome mean matter?
2. Does the same variance problem afflict HT when targeting the ATT, and does Hájek normalization help as much?
3. Among estimators targeting the ATT, do IPW and matching agree on the target, and which is more efficient?
4. How large is the estimand gap between ATT and ATE, and which estimators reveal it?
5. How does replacing the true propensity score with a logistic estimate change bias and variance?

6.7.1 Part 1: Correctly Specified Propensity Score

DGP for Lab 6

Let $X_j \overset{\text{i.i.d.}}{\sim} \text{Normal}(0,1)$ for $j = 1, \dots, 5$. All five covariates enter both the treatment mechanism and the outcome, so every X_j is a genuine confounder. The true propensity score is

$$\pi(X) = \operatorname{expit}(-0.5 + 0.8X_1 + 0.5X_2 - 0.3X_3 + 0.2X_4 + 0.1X_5), \tag{6.6}$$

giving marginal treatment probability $P(T{=}1) \approx 0.40$. The potential outcomes are

$$Y(t) = (1 + 0.5X_1) \cdot t + 8 + 0.8X_1 + 0.5X_2 + 0.4X_3 + 0.3X_4 + 0.2X_5 + \varepsilon, \quad \varepsilon \sim \text{N}(0,1). \tag{6.7}$$

The CATE is $\tau(X) = 1 + 0.5X_1$. Because $\mathbb{E}[X_1] = 0$,

$$\tau_{\text{ATE}} = \mathbb{E}[1 + 0.5X_1] = 1.000 \quad \text{(exact)}.$$

Because $\pi(X)$ is increasing in X_1, treated units have higher X_1 on average, so

$$\tau_{\text{ATT}} = \mathbb{E}[1 + 0.5X_1 \mid T{=}1] \approx 1.199 \quad \text{(verified by oracle run, } n = 10^7\text{)}.$$

The gap $\tau_{\text{ATT}} - \tau_{\text{ATE}} \approx 0.199$ reflects a selection effect: units most likely to be treated also tend to have larger individual treatment effects. The intercept 8 in the outcome model represents a population baseline. Treatment is generated as $T \mid X \sim \text{Bernoulli}\{\pi(X)\}$, the error satisfies $\varepsilon \perp\!\!\!\perp (X, T)$, and units are sampled independently; strong ignorability therefore holds by construction.

With five confounders, direct cell-by-cell stratification is infeasible. The propensity score reduces this five-dimensional adjustment to a scalar. The large baseline mean ($\mathbb{E}[Y(0)] = 8$) plays a distinct role: it amplifies the consequences of extreme IPW weights, making Hájek normalization strongly recommended in this setting rather than merely helpful.

Estimators. Five estimators are compared, each applied under both the true and estimated propensity score. The estimated score is obtained by logistic regression of T on $(X_1, \dots, X_5)$, which is correctly specified since the true logit is linear in all five covariates.

Horvitz–Thompson ATE (HT-ATE), Equation 6.3. Targets the ATE. Raw inverse weights can be extreme when $\hat{\pi}(X_i)$ is near 0 or 1, and with $\mathbb{E}[Y(0)] = 8$ the variance consequences are severe.

Hájek ATE (HJ-ATE), Equation 6.4. Targets the ATE. Normalizing by the empirical weight sum within each arm caps the effective influence of any single unit.

Horvitz–Thompson ATT (HT-ATT). The ATT identification formula re-weights the *control* arm to look like the treated population while leaving the treated arm unchanged:

$$\hat{\tau}_{\mathrm{HT,ATT}} = \frac{1}{n_1} \sum_{i:\, T_i=1} Y_i \; - \; \frac{1}{n_1} \sum_{i:\, T_i=0} \frac{\hat{\pi}(X_i)}{1-\hat{\pi}(X_i)}\, Y_i.$$

Each control unit receives weight proportional to its *odds of treatment*: units whose covariates resemble the treated population receive large weight. For a control unit with $\hat{\pi}(X_i) = 0.95$ the weight is 19; combined with a baseline outcome near 8, this produces extreme variance.

Hájek ATT (HJ-ATT). The normalized counterpart replaces the raw-weight sum over the control arm with a self-normalized version:

$$\hat{\tau}_{\mathrm{HJ,ATT}} = \frac{1}{n_1} \sum_{i:\, T_i=1} Y_i \; - \; \frac{\sum_{i:\, T_i=0} \{\hat{\pi}(X_i)/(1-\hat{\pi}(X_i))\}\, Y_i}{\sum_{i:\, T_i=0} \hat{\pi}(X_i)/(1-\hat{\pi}(X_i))}.$$

The treated arm is already a simple mean and requires no normalization. This is the recommended default estimator for the ATT in practice.

One-to-one nearest-neighbor matching with replacement (NNM), Equation 6.1. Targets the ATT. Matching with replacement ensures every treated unit finds its globally closest control, preventing the finite-sample "running out of good matches" problem that arises without replacement when the propensity score distribution is wide (Abadie and Imbens 2006).

Results (n = 1,000, B = 2,000 replications, seed 42). Bias is computed relative to each estimator's own target: ATE = 1.000 for HT-ATE and HJ-ATE; ATT $\approx$ 1.199 for HT-ATT, HJ-ATT, and NNM. Monte Carlo standard errors of the reported means equal $\mathrm{SD}/\sqrt{B} \approx \mathrm{SD}/45$.

Table 6.1: Monte Carlo performance of five propensity-score estimators under known and estimated propensity scores. True ATE = 1.000 (exact); true ATT $\approx$ 1.199 (oracle, $n = 10^7$).

PS	Estimator	Estimand	Mean	Bias	SD	RMSE
Known	HT-ATE	ATE	1.008	+0.008	0.626	0.626
Known	HJ-ATE	ATE	1.003	+0.003	0.145	0.145
Known	HT-ATT	ATT	1.184	−0.016	0.725	0.725
Known	HJ-ATT	ATT	1.203	+0.003	0.131	0.131
Known	NNM	ATT	1.207	+0.008	0.130	0.130
Estimated	HT-ATE	ATE	0.998	−0.002	0.194	0.194
Estimated	HJ-ATE	ATE	1.002	+0.002	0.102	0.103
Estimated	HT-ATT	ATT	1.202	+0.002	0.251	0.251
Estimated	HJ-ATT	ATT	1.202	+0.002	0.101	0.101
Estimated	NNM	ATT	1.205	+0.006	0.121	0.121

1. Hájek normalization is strongly recommended for ATE estimation when outcomes are non-centered. With the known propensity score, HT-ATE achieves SD = 0.626, while HJ-ATE achieves SD = 0.145 — a **4.3-fold reduction** from simply normalizing the weights. The mechanism is transparent: a unit with $\pi(X_i) = 0.05$ receives raw weight 20; multiplied by an outcome near $\mathbb{E}[Y(0)] = 8$, its contribution to the HT sum is of order 160. Variability in which units are extreme across samples inflates the HT standard deviation to more than half the true effect size.

The role of the baseline can be isolated exactly. Define the average inverse weights in each arm,

$$w_T = \frac{1}{n} \sum_{i=1}^{n} \frac{T_i}{\hat{\pi}(X_i)}, \qquad w_C = \frac{1}{n} \sum_{i=1}^{n} \frac{1-T_i}{1-\hat{\pi}(X_i)},$$

each of which has expectation one under the true score (Lemma 6.2 with $h \equiv 1$) but fluctuates in finite samples. Shifting every outcome by a constant c changes the HT estimator by

$$\hat{\tau}_{\mathrm{HT}}(Y + c) = \hat{\tau}_{\mathrm{HT}}(Y) + c\,(w_T - w_C),$$

whereas the Hájek difference is exactly invariant to the shift, since each arm's self-normalized mean moves by c and the shifts cancel. With a baseline near $c = 8$, finite-sample fluctuation in $w_T - w_C$ is amplified eightfold — this, not variability in the treatment effect itself, is the dominant source of the HT variance here.

2. HT-ATT is even more unstable; Hájek normalization is equally beneficial for ATT estimation. With the known PS, HT-ATT achieves SD = 0.725, slightly *worse* than HT-ATE. The reason is structural: the ATT re-weights the control arm by the treatment odds $\pi(X)/(1-\pi(X))$. A control unit with $\pi(X) = 0.95$ receives an odds-ratio weight of 19; combined with the large baseline outcome this is as explosive as the ATE inverse-probability weight. The apparent bias of -0.016 for HT-ATT is within one Monte Carlo standard error of $0.725/\sqrt{2000} \approx 0.016$ and should be attributed to simulation noise rather than systematic inconsistency. HJ-ATT reduces SD to 0.131 — a **5.5-fold reduction** — confirming that Hájek normalization is not an ATE-specific fix but a general principle applicable whenever inverse-probability-style weights are used.

3. HJ-ATT and NNM converge to the same target with nearly identical efficiency. Under the known PS, both target the ATT ≈ 1.199 and achieve essentially the same standard deviation (0.131 vs. 0.130). This is the sharpest demonstration of the "two routes, one estimand" principle: IPW re-weights the full sample using the propensity score; matching discards some control units and retains only those nearest each treated unit on the propensity scale. Both are consistent for the ATT under strong ignorability, and at $n = 1{,}000$ they are virtually indistinguishable in efficiency. The practical implication is that the choice between IPW-ATT and matching can be made on secondary grounds — interpretability, balance diagnostics, or computational cost — rather than efficiency when the PS is known.

4. The estimand gap between ATE and ATT is large and clearly revealed by the table. The ATE estimators converge to 1.000; the ATT estimators converge to ≈ 1.199. The 0.199 gap is not a bias — it is a real difference in the estimands. High-X_1 units are more likely to be treated (Equation 6.6) and have larger treatment effects (Equation 6.7); the treated subpopulation therefore benefits more than the full population average. A researcher who applies NNM and reports its output against the ATE benchmark of 1.000 would conclude the estimator has 20% bias; in fact it is unbiased for the correct target. Declaring the estimand before looking at any results is a non-negotiable discipline.

5. Estimated PS collapses HT variance and makes HJ-ATT the most efficient ATT estimator. Under the estimated PS, HT-ATE SD falls from 0.626 to 0.194 (69% reduction); HT-ATT SD falls from 0.725 to 0.251 (65% reduction). This counterintuitive gain — the *estimated* score outperforming the *true* one — is a known phenomenon, but it is not a consequence of consistency alone: the classical result requires a correctly specified propensity model estimated by maximum likelihood (here, the logistic model is exactly correct). The likelihood score equations of the logistic fit, $\sum_i (T_i - \hat{\pi}(X_i))X_i = 0$, act as control variates: they remove the component of the IPW sampling error correlated with the score functions. IPW with the ML estimate of a correctly specified score is therefore asymptotically no less efficient than IPW with the known score, and an analogous conclusion holds for suitable nonparametric score estimators (Hirano et al. 2003).

More strikingly, HJ-ATT with estimated PS achieves SD = 0.101 — beating both NNM (SD = 0.121) and HT-ATT (SD = 0.251) by a wide margin and matching HJ-ATE (SD = 0.102) almost exactly. This is the finite-sample analog of the efficiency result of Hirano et al. (2003): a properly normalized IPW estimator for the ATT is consistent — though, as a ratio estimator, not exactly unbiased in finite samples — and can be more efficient than matching when the propensity score model is well estimated.

The Estimand Must Be Chosen Before the Estimator

In this DGP, $\tau_{\text{ATE}} = 1.000$ and $\tau_{\text{ATT}} \approx 1.199$. The gap arises because treatment status is correlated with the individual treatment effect: high-X_1 units are simultaneously more likely to be treated and more likely to benefit. This situation — treatment-effect heterogeneity correlated with propensity for treatment (Imbens and Rubin 2015) — is common in practice: individuals often select into treatment partly because they anticipate larger benefits.

IPW-ATE and HJ-ATE re-weight the full sample to represent the population; IPW-ATT and HJ-ATT re-weight only the control arm to represent the treated population; NNM directly pairs each treated unit with a similar control. All three ATT estimators target the same quantity via different mechanisms. Choosing among them should be driven by the policy question (is the ATE or ATT the right estimand?) and by practical concerns about balance and model specification, not by which produces the larger or smaller point estimate.

Remark: What Part 1 Does Not Show

Part 1 uses a correctly specified logistic propensity model, which is the best-case scenario for IPW. In practice the true PS may be nonlinear, making correct logistic specification impossible with the predictors at hand. Part 2 addresses exactly this setting. The doubly robust AIPW estimator of Chapter 11 provides a complementary fix: it is consistent if *either* the PS model or the outcome model is correctly specified, offering robustness that neither pure IPW nor pure matching can achieve alone.

6.7.2 Part 2: Robustness to PS Model Misspecification

Part 1 established that under a correctly specified PS model, IPW and NNM are both consistent for their respective estimands, with HJ-ATT and NNM delivering similar efficiency. Part 2 breaks that assumption by introducing a *nonlinear* true propensity score that the standard logistic regression cannot capture.

Modified DGP. The outcome model is unchanged from Equation 6.7. The true propensity score now contains a quadratic term in X_1, and X_2 and X_3 do not affect treatment:

$$\pi^*(X) = \text{expit}\big(-0.5 + 0.8X_1 + 0.5X_1^2 + 0.2X_4 + 0.1X_5\big). \tag{6.8}$$

The quadratic $0.5X_1^2$ means that both large positive *and* large negative values of X_1 increase the probability of treatment, creating a U-shaped propensity surface. Because $\mathbb{E}[X_1] = 0$ and the CATE is still $\tau(X) = 1 + 0.5X_1$, the ATE is unchanged at 1.000 exactly. The ATT changes because the new PS shifts which units are treated: $\tau^*_{\text{ATT}} \approx 1.152$ (oracle run, $n = 10^7$).

The *estimated* PS is still obtained by fitting a linear logistic regression of T on $(X_1, \dots, X_5)$. This model is **misspecified**: it omits the X_1^2 term and includes X_2 and X_3, which do not appear in the true logit.

Table 6.2: Part 2: Monte Carlo performance under a nonlinear true PS and a misspecified linear logistic estimate. True ATE = 1.000 (exact); true ATT $\approx$ 1.152 (oracle, $n = 10^7$, new DGP).

PS	Estimator	Estimand	Mean	Bias	SD	RMSE
True	HT-ATE	ATE	1.028	+0.028	0.824	0.825
True	HJ-ATE	ATE	1.013	+0.013	0.152	0.152
True	HT-ATT	ATT	1.186	+0.034	1.333	1.333
True	HJ-ATT	ATT	1.183	+0.031	0.212	0.215
True	NNM	ATT	1.178	+0.026	0.175	0.177
Misspecified	HT-ATE	ATE	1.478	+0.478	0.147	0.501
Misspecified	HJ-ATE	ATE	0.861	−0.139	0.087	0.164
Misspecified	HT-ATT	ATT	1.780	+0.628	0.162	0.649
Misspecified	HJ-ATT	ATT	1.306	+0.154	0.081	0.174
Misspecified	NNM	ATT	1.172	+0.020	0.138	0.139

6. PS misspecification removes the consistency guarantee for IPW; here every IPW estimator is materially biased, and the Hájek correction cannot fix this. Under the misspecified PS, HT-ATE bias is +0.478, nearly half the true effect; HJ-ATE bias is −0.139. For the ATT estimators the problem is worse: HT-ATT bias is +0.628; HJ-ATT bias is +0.154. Comparing these to the near-zero biases in the "True PS" rows makes clear that the Hájek normalization is a *variance* correction — it cannot compensate for a misspecified model that produces systematically wrong weights. IPW consistency is tied to consistency of the PS estimator; a structurally wrong model generally produces structurally wrong weights, and no normalization of wrong weights produces correct estimates. (Special calibration identities can leave particular weighted estimators consistent under particular misspecifications, but no such protection is generic, and none operates here.)

Notice also that HJ-ATE and HT-ATE have biases in *opposite directions* (−0.139 vs. +0.478). The mechanism is a multiplicative *level error* that HT inherits from misspecification but Hájek removes. Recall the average inverse weights w_T and w_C, each of which converges to 1 under a correctly specified PS. Under misspecification both can drift; in this DGP the Monte Carlo means are $w_T \approx 1.045$ and $w_C \approx 0.972$. Algebraically, $\hat{\tau}_{\text{HT}} = w_T\, \hat{\mu}_1^{\text{HJ}} - w_C\, \hat{\mu}_0^{\text{HJ}}$, so the HT bias inherits an extra contribution

$$(w_T - 1)\,\mathbb{E}[Y(1)] \;-\; (w_C - 1)\,\mathbb{E}[Y(0)] \;\approx\; 0.045 \times 9 \;+\; 0.028 \times 8 \;\approx\; +0.63,$$

which accounts for essentially the entire gap between the two biases $(+0.478 - (-0.139) = +0.617)$. With $\mathbb{E}[Y(0)] = 8$, even sub-five-percent weight-sum errors translate into bias contributions of order 0.3. What remains in HJ-ATE is a smaller *shape* error: the residual misweighting of the population that survives self-normalization. The level and shape errors are determined by different aspects of the misspecification — the first by how far the average inverse weights drift from one, the second by how the misspecified weights re-weight the population — so there is no reason for their signs to align.

7. In this simulation, matching is less sensitive to the particular PS misspecification. The NNM bias barely changes: +0.026 under the true PS versus +0.020 under the misspecified PS. Its RMSE actually *improves* slightly (0.177 → 0.139) because the misspecified logistic scores, being shrunk toward the mean of T, produce somewhat less variable matches than the true nonlinear scores do. The reason is that NNM does not require the PS model to be correct in order to produce good matches — it only requires that the estimated score roughly *orders* units so that matched pairs are approximately balanced on the true confounders. The linear logistic model, though wrong, still captures the dominant linear effects of X_1, X_4, and X_5 on treatment; the quadratic component is missed, but the ranking of units is not drastically distorted.

This insensitivity is conditional on the estimated score preserving covariate balance. Propensity-score matching can also fail badly under misspecification — if, for example, the misspecification distorts the ranking of units enough that matched pairs are imbalanced on outcome-relevant covariates. The lesson is therefore not that matching is unconditionally robust, but that its bias depends on covariate balance after matching rather than on the parametric correctness of the score itself; balance must be checked, not assumed. Theoretical analyses of when this conditional robustness holds appear in Yang et al. (2016) and Yang and Zhang (2023).

8. In this simulation, the relative advantage of matching over IPW reverses with model misspecification. In Part 1, HJ-ATT was more efficient than NNM under the estimated PS (SD = 0.101 vs. 0.121). In Part 2, NNM dominates on both bias *and* RMSE relative to HJ-ATT. The practical implication is that the efficiency advantage of IPW is contingent on the PS model being approximately correct. When there is reason to doubt the functional form of the propensity model — for instance, when treatment is driven by a continuous covariate in a nonlinear way — matching can be a more robust alternative, because it relies on local covariate similarity rather than a globally correct weighting function; whether this conditional advantage materializes depends on whether the estimated score still produces close matches on the outcome-relevant covariates.

Table 6.3: Head-to-head comparison of HJ-ATT and NNM under correct vs. misspecified PS estimation (estimated-PS column from each part).

	Part 1: correct PS model, bias	**RMSE**	**Part 2: misspecified PS model, bias**	**RMSE**
HJ-ATT	+0.002	0.101	+0.154	0.174
NNM	+0.006	0.121	+0.020	0.139

IPW Needs a Correct PS Model; Matching Needs Covariate Balance

The two lessons of this lab can be compressed into a single contrast. IPW is a *model-dependent* estimator: its consistency relies on the PS model being correctly specified. Get the model right and IPW (especially the Hájek form) is highly efficient; get it wrong and the bias from incorrect weights can be large and systematic, as the +0.154 HJ-ATT bias in Part 2 demonstrates.
Matching is a *design-dependent* estimator: its consistency relies on matched pairs being approximately balanced on the true confounders. This condition can hold even when the PS model is wrong, provided the estimated score still separates units well enough that close matches on the score imply closeness on the full covariate vector. But a misspecified score is not itself a balancing score, so this

favorable case must be demonstrated by balance diagnostics on the matched sample — including, where relevant, nonlinear transformations and interactions — not assumed. The cost is efficiency: when the PS model is correct, matching leaves variance on the table relative to IPW.
Neither dominates unconditionally. The practical recommendation is to assess the plausibility of the PS model before choosing between IPW and matching as the primary estimator — or to use both as a sensitivity check (Yang et al. 2016; Yang and Zhang 2023).

6.8 The Limits of Propensity Score Methods

6.8.1 The Untestable Assumption

The balancing property (Theorem 6.1), the coarsest-balancing-score lemma, and the weighted-balance identities (Lemma 6.2) are properties of the observed treatment-assignment mechanism: they hold for any binary treatment and require no causal assumptions. The *causal* results of this chapter — the Propensity Score Theorem, the IPW identification formula, the causal interpretation of matching — additionally rest on the unconfoundedness assumption $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$, together with consistency and the relevant positivity condition.

Unconfoundedness requires that the observed pretreatment set X be *sufficient for adjustment.* In graphical terms: X blocks every back-door path from T to Y. This does not require that every latent common cause itself be observed: an observed variable lying on a confounding path can block that path even when the common cause behind it is latent (Chapter 3). What is required is that every noncausal path be intercepted by the observed set — and this can never be verified from the data alone.

Design-Based vs. Model-Based Identification

Randomization *guarantees* ignorability: by construction, no back-door paths exist, so the interventional and observational distributions coincide. Propensity scores *assume* ignorability: the analyst hopes that all confounders have been measured and included in X, but this can never be verified from the data alone. This distinction — between identification secured by the study design and identification secured by a modeling assumption — is one of the most important dividing lines in causal inference.

Two complementary responses. When unconfoundedness is in doubt, two distinct strategies are available and they answer different questions. *Sensitivity analysis* (Chapter 9) keeps the back-door framework but quantifies how strong an unmeasured confounder would have to be to overturn the conclusion: if the answer is “implausibly strong” the analysis is robust, whereas if the answer is “modest” the conclusion is fragile. The *instrumental variable* approach (Chapter 7) replaces the unconfoundedness assumption with a different set of identifying assumptions, exploiting an external source of variation in treatment that is independent of the unobserved treatment–outcome confounders. The hidden-confounding concern is thereby relocated, not eliminated: the instrument itself must be valid — unconfounded with the outcome and excluded from affecting it directly.

6.8.2 The Road to Instrumental Variables

The limitation of back-door adjustment motivates the search for alternative identification strategies when unobserved confounders are present. One canonical alternative — the one taken up next — is the instrumental variable (IV), studied in Chapter 7. It is not the only one: front-door identification (Chapter 3) and sensitivity analysis with partial identification (Chapter 9) are others.

Strategy	**Key assumption**	**Identification mechanism**
Back-door adjustment (propensity score)	Observed pretreatment set blocks all back-door paths	Condition on X to block $T \leftarrow U \rightarrow Y$
Instrumental variables	Some confounders may be unobserved	Exploit exogenous variation $Z \rightarrow T$

Back-door adjustment removes confounding by conditioning on observed variables. Instrumental variables take a different approach: instead of blocking the confounding path $T \leftarrow U \rightarrow Y$, they isolate variation in T induced by an external source Z that is independent of U. Under additional IV assumptions, the ratio of the Z-induced change in Y to the Z-induced change in T — the Wald estimator — identifies a causal parameter. In the binary instrument and treatment setting, under the additional *monotonicity* (no-defiers) assumption, that parameter is the local average treatment effect (LATE) among compliers. This is derived formally in Chapter 7.

6.9 Summary

1. **The propensity score reduces dimension.** The propensity score $\pi(X) = P(T{=}1 \mid X)$ is a *balancing score* — $T \perp\!\!\!\perp X \mid \pi(X)$ — for any treatment-assignment mechanism, with no causal assumption required. Under unconfoundedness and positivity, the balancing property upgrades to $(Y(0), Y(1)) \perp\!\!\!\perp T \mid \pi(X)$ (Theorem 6.2), so identification requires adjustment for the scalar $\pi(X)$ alone, not the full vector X.
2. **IPW identification.** Under consistency, weak conditional exchangeability for both treatment levels, positivity, and integrability, the ATE is identified by Equation 6.2 (Theorem 6.3); joint strong ignorability supplies the exchangeability and positivity components but is stronger than necessary, and consistency remains a separate assumption. The Horvitz–Thompson IPW estimator is consistent but can be sensitive to extreme weights; the Hájek variant is often more stable in finite samples. Augmented IPW (AIPW) can achieve higher efficiency under correct nuisance specification, as established formally in Chapter 11.
3. **The matching procedure in this chapter targets the ATT.** The nearest-neighbor matching estimator presented here finds control units with similar propensity scores to each treated unit. It is designed to target the ATT and provides transparent covariate balance diagnostics, but differs fundamentally from IPW weighting in estimand and methodology. More generally, matching procedures can be designed to target the ATT, ATC, or an ATE-like estimand depending on the matching design and weighting scheme.
4. **Positivity is necessary; strong overlap is a standard sufficient condition for stable estimation.** Positivity is necessary for identification of the ATE. Strong overlap is a stronger, convenient sufficient condition under which the standard $\sqrt{n}$-consistent, asymptotically normal theory goes through. Near-violations of strong overlap create extreme IPW weights and large variance; trimming strategies address this at the cost of changing the estimand.
5. **The fundamental limitation.** Unconfoundedness is a strong, untestable assumption. When unobserved confounders are present, propensity score adjustment fails. An instrumental variable is one important alternative identification strategy (Chapter 7); others include front-door identification (Chapter 3) and sensitivity analysis with partial identification (Chapter 9).

Comparison of Identification Strategies

Design	Key assumption	Identified estimand
Randomized experiment	$(Y(0), Y(1)) \perp\!\!\!\perp T$ (by design)	ATE
Propensity-score adjustment	$(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ (sufficient adjustment set), positivity	ATE, ATT, or ATC
Instrumental variables	Relevance, exogeneity, exclusion; monotonicity for LATE	LATE (compliers)

6.10 Problems

1. Propensity score and balancing. Suppose $X = (X_1, X_2)$, where X_1 is binary and X_2 is continuous. The propensity score model is correctly specified as $\text{logit}(\pi(X)) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$.

(a) State and prove the Balancing Property $T \perp\!\!\!\perp X \mid \pi(X)$.
(b) Two units i and j have $X_i = (1, 2.3)$ and $X_j = (0, 3.5)$ but $\pi(X_i) = \pi(X_j) = 0.4$ under the *true* propensity score. Explain what equality of the true propensity score implies about the conditional distribution of X across treatment arms at this score value, and why this is a population-level balancing statement rather than an exact individual counterfactual equality: if i is treated and j is control, Y_j is a noisy estimate of a conditional mean, not of $Y_i(0)$. State the additional assumptions needed for propensity-score matching to consistently estimate the ATT.
(c) Explain why matching on $\hat{\pi}(X)$ is *not* the same as matching on X directly. Under what conditions do they give the same answer?

2. ATE, ATT, and propensity score weighting. A dataset has $n = 1000$ observations with estimated propensity scores $\hat{\pi}(X_i)$ and $n_1 = 400$ treated units.

(a) Write the Horvitz–Thompson IPW estimator of the ATE as a weighted sum of observed outcomes. What weights do treated units receive? What weights do control units receive?
(b) Derive an analogous IPW estimator for the ATT. (*Hint*: the ATT averages over the treated distribution; the denominator of the weight for control units should reflect this.)
(c) Suppose the propensity score is constant: $\pi(X) = p$ for all units. Show that the Horvitz–Thompson IPW estimator of the ATE equals

$$\frac{n_1}{np}\,\bar{Y}_1 - \frac{n_0}{n(1-p)}\,\bar{Y}_0,$$

and conclude that it reduces to the difference in means only when the realized count satisfies $n_1 = np$ exactly — as under complete randomization with fixed arm sizes — whereas under Bernoulli assignment n_1 is random and the equality fails. Show that the Hájek estimator reduces to $\bar{Y}_1 - \bar{Y}_0$ *exactly* whenever the score is constant. Under which experimental designs does $\pi(X) = 0.5$ hold for each unit: Bernoulli randomization with $P(T_i{=}1) = 0.5$, complete randomization with exactly $n/2$ treated, or both?

3. Overlap. Let $\pi(X)$ be the true propensity score and suppose overlap fails on a set $\mathcal{S}_1$ of covariate values with $P(X \in \mathcal{S}_1) > 0$: $\pi(x) = 1$ for all $x \in \mathcal{S}_1$. (For a continuous covariate, a violation at a single P_X-null point would be harmless; positivity is a statement about non-null sets.)

(a) For each of τ_{ATE} and τ_{ATT}, identify the exact formula that fails to be identified, and explain why. (*Hint*: consider whether $\mathcal{S}_1$ intersects the support of $X \mid T{=}1$.)
(b) Suppose overlap fails only on a set $\mathcal{S}$ with $P(X \in \mathcal{S}) = 0.15$. Define the *trimmed ATE* estimand. How does it differ from the ATE?
(c) A researcher applies the Crump et al. (2009) rule with $\eta = 0.1$, removing 8% of the sample. (i) List two reasons the trimmed estimator may have lower variance than the untrimmed IPW estimator. (ii) What is the cost in terms of external validity?

4. Covariate selection for the propensity model. A researcher has pretreatment covariates C, P, and Z, and a post-treatment variable M. In the assumed causal graph, C is a confounder of T and Y; P predicts the outcome but not treatment; Z strongly predicts treatment but has no causal relation to the potential outcomes; and M is a mediator on the path from T to Y.

(a) Which variables belong in the propensity model when the target is the total effect of T on Y? Justify each inclusion or exclusion using the taxonomy of Section 6.3.1.
(b) Explain how including Z may affect overlap, the distribution of the inverse-probability weights, and the effective sample size, even though it does not invalidate the adjustment set.
(c) Suppose the researcher discovers that C was in fact measured *after* treatment for a subset of units. What concerns does this raise for the propensity design?

5. Hidden confounding (conceptual preview of Chapter 9). Suppose you estimate $\hat{\tau}_{\text{ATE}} = 3.2$ using propensity-score methods, and a referee questions whether unconfoundedness holds.

(a) Define what it means for U to be a "hidden confounder" in the context of the DAG, and explain why its presence invalidates the IPW identification formula.
(b) Explain in words what it means for an estimated effect to be "robust to hidden confounding," and why such an assessment is necessarily comparative: it depends on a quantitative yardstick for how strong an unmeasured confounder would have to be to overturn the conclusion. Without doing formal calculations, sketch the narrative a researcher might construct to argue that $\hat{\tau}_{\text{ATE}} = 3.2$ would

survive any plausible hidden confounder. (Chapter 9 develops three formal yardsticks: Rosenbaum's Γ, the E-value, and the marginal sensitivity model.)

(c) How can a valid instrumental variable identify a causal parameter despite unmeasured confounding between T and Y (Chapter 7)? State the assumptions that replace treatment–outcome unconfoundedness, and explain why unmeasured confounding between the *instrument* and the outcome would still invalidate the IV design.

6. Matching vs. IPW. You have $n = 500$ observations with a binary treatment and a scalar propensity score. You compare (A) the Hájek IPW estimator targeting the ATE and (B) 1:1 nearest-neighbor matching designed to target the ATT.

(a) Explain in one sentence why estimator (A) and estimator (B) target different estimands even though both use $\hat{\pi}(X)$.

(b) In your matched sample, the standardized mean difference for covariate X_1 is 0.05 after matching and 0.45 before matching. What does this tell you about the success of matching, and what assumption does balance on observed covariates not verify?

(c) Under what condition on the treatment effect heterogeneity are the ATE and ATT equal?

Chapter 7

Instrumental Variables

Learning Objectives

By the end of this chapter, students should be able to:

1. Diagnose the endogeneity problem: explain why an unobserved confounder makes back-door adjustment fail, and describe how an instrumental variable provides an alternative identification route.
2. State the three IV assumptions — relevance, exogeneity, and exclusion — in the DAG/do-calculus, structural-equation, and potential-outcomes languages, and explain which are testable and which require institutional justification.
3. Follow how the IV assumptions identify the Wald estimand, and interpret the reduced form and first stage as its two observable components.
4. Explain what the Wald estimand identifies when treatment effects are heterogeneous — the Local Average Treatment Effect for compliers — why that estimand depends on the choice of instrument, and when it coincides with the ATE.
5. Compare IV and back-door adjustment on the assumptions each requires, the estimand each identifies, and the ways each can fail.

7.1 Why Instrumental Variables?

Chapter 6 showed how causal effects can be identified when all confounders are observed and can be blocked by conditioning on X. When some confounders are unobserved, back-door adjustment fails. This chapter develops instrumental variables (IV) as an alternative identification strategy: rather than blocking the confounding path $T \leftarrow U \rightarrow Y$, IV exploits an external variable Z whose effect on T is free of confounding by U. This chapter asks what IV identifies and under what assumptions; Chapter 13 asks how that estimand is computed and tested in practice.

7.1.1 The Endogeneity Problem

Chapter 6 established that the propensity score provides a powerful surrogate for randomization, but only under *unconfoundedness*: $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$. This assumption requires that every variable affecting both treatment and outcome is observed and included in X. In many empirical settings this is implausible:

- In labor economics, unobserved ability or motivation affects both schooling decisions and wages.
- In epidemiology, unobserved health behaviors affect both treatment uptake and outcomes.
- In survey sampling, voluntary participation is driven by factors that also affect the variables of interest.

Whenever an unobserved confounder U creates a back-door path $T \leftarrow U \rightarrow Y$, the adjustment formula fails:

$$\int f(y \mid t, x)\, p(x)\, dx \;\neq\; f(y \mid \mathrm{do}(T{=}t)).$$

The gap is the *endogeneity bias*. We need a different identification strategy.

7.1.2 The IV Idea

An *instrumental variable* Z is an observed variable that:

1. *Moves* treatment T (relevance).
2. *Does so exogenously* — Z is unrelated to the unobserved confounder U (exogeneity).
3. *Affects* Y *only through* T — Z has no direct path to Y (exclusion).

Under these three conditions, the variation in T induced by Z is free of confounding by U, so the ratio of the Z-induced variation in Y to the Z-induced variation in T becomes a meaningful target for identification. Which causal estimand it identifies — the ATE, the ATT, or the LATE for compliers — depends on a fourth structural restriction on treatment-effect heterogeneity, taken up in Section 7.3 and Section 7.7.

Instrumental variables do not eliminate the confounding path $T \leftarrow U \rightarrow Y$, nor do they make treatment as-if randomly assigned for the full population. Instead, IV identifies causal effects by isolating a component of treatment variation that is induced by Z and is therefore exogenous under the IV assumptions. In that sense, IV *avoids* confounding rather than controlling for it — a distinction that matters both for interpreting what is identified (the LATE for compliers, not the ATE) and for understanding which assumption does the heaviest lifting (exclusion, not unconfoundedness).

Example: Returns to Schooling as Motivation

A researcher wants to estimate the causal effect of years of schooling T on log wages Y. Unobserved ability U raises both schooling and wages, so $\mathrm{Cov}(T, \varepsilon) > 0$ and OLS is biased upward. No set of observed covariates X fully captures ability. An instrumental variable Z that shifts schooling for reasons unrelated to ability — such as proximity to a school or a policy change in compulsory attendance laws — provides a way to isolate the causal effect of schooling on wages.

7.2 Graphical Setup and Core Assumptions

This section is the conceptual foundation of the IV chapter. We fix the causal graph for the IV design and state the three core assumptions in the three causal languages used throughout these notes: DAGs and do-calculus, structural equations, and potential outcomes. The purpose is not merely stylistic. Each language highlights a different aspect of the design: the DAG makes the causal pathways visible, the do-calculus formalizes what those pathways imply about interventional distributions, the structural model supplies the moment conditions used in the Wald derivation, and the potential-outcomes formulation prepares the ground for the LATE framework in Section 7.7.

Before proceeding, it is important to separate two questions that are often conflated. The first is: when is an instrument valid? Validity requires the three core IV assumptions — relevance, exogeneity, and exclusion — which are the subject of this section. The second is: what does a valid instrument identify? That answer depends on a fourth structural assumption. Under linearity and homogeneous treatment effects (Section 7.3), the Wald ratio identifies a single parameter β equal to the ATE. Under heterogeneous effects with monotonicity (Section 7.7), the same Wald ratio identifies the LATE for compliers. The three IV assumptions determine whether the design is valid; the fourth assumption determines what causal quantity it recovers.

7.2.1 The IV DAG

The causal structure for a basic IV model with covariates X is:

The three IV assumptions correspond to three distinct features of this DAG:

1. **Relevance**: there is a directed path $Z \rightarrow T$ (edge labeled (1) above), so the instrument shifts treatment.
2. **Exogeneity**: conditional on observed covariates X, the instrument is free of open back-door paths to the latent determinants of the outcome — the DAG implies $Z \perp\!\!\!\perp U \mid X$ by d-separation. Intuitively, Z is as-if randomized with respect to the unobserved factors that confound T and Y.
3. **Exclusion**: every directed path from Z to Y in $\mathcal{G}$ passes through T. In the simple IV DAG above this rules out the direct edge $Z \rightarrow Y$; in richer graphs it also rules out indirect channels $Z \rightarrow M \rightarrow Y$ that bypass T through any unmodeled mediator M.

The distinction between exogeneity and exclusion is fundamental. Exogeneity says that the instrument is not confounded with the latent causes of the outcome. Exclusion says that the instrument has no causal channel to the outcome except through treatment. An instrument can satisfy one and fail the other. For example, a randomized encouragement may be exogenous by design, yet still violate exclusion if the encouragement changes outcomes through information, motivation, stigma, or administrative treatment apart from the treatment itself.

What the DAG rules out. The absence of a direct $Z \to Y$ arrow encodes the exclusion restriction. Any such arrow — no matter how small the coefficient — would violate exclusion and invalidate the Wald estimand. The DAG forces the researcher to commit to this assumption explicitly; the potential outcomes framework expresses it as the equality $Y_i(t, z) = Y_i(t, z')$.

7.2.2 The Three Assumptions in Three Languages

The same three assumptions can be expressed in three causal languages. These are parallel formulations, not literally identical statements: each language highlights a different aspect of the underlying design. The graphical formulation is most useful for causal design, because it makes confounding paths and forbidden arrows explicit. The structural formulation is most useful for deriving moment restrictions such as $\mathbb{E}[Z\varepsilon \mid X] = 0$. The potential-outcomes formulation is most useful when we later introduce compliance types and the LATE theorem. The assumptions are ordered *relevance* $\to$ *exogeneity* $\to$ *exclusion*, reflecting the natural sequence in which a researcher assesses them.

Assumption	**Potential outcomes**	**Structural / econometric**	**Do-calculus / DAG**
Relevance	$\Pr(T_i(1) \neq T_i(0)) > 0$	$\pi \neq 0$ in $T = \pi Z + \delta^\top X + \eta$	$Z \to T$ in $\mathcal{G}$ (no d-separation)
Exogeneity	$Z \perp\!\!\!\perp (Y(0), Y(1), T(0), T(1)) \mid X$	$\mathbb{E}[\varepsilon \mid Z, X] = 0$	$Z \perp\!\!\!\perp U \mid X$
Exclusion	$Y_i(t, z) = Y_i(t, z')$ for all z, z'	Z absent from structural equation for Y	$f(y \mid x, \mathrm{do}(T{=}t), z) = f(y \mid x, \mathrm{do}(T{=}t))$

Remark: Three Languages, One Identification Argument

These three formulations are aligned but not literally identical. The graphical statement encodes causal structure — which arrows are present or absent in $\mathcal{G}$. The potential-outcomes statement encodes counterfactual independence — which potential outcomes are independent of Z. The structural formulation encodes moment orthogonality — which inner product with the residual is zero. In a well-specified IV model they support the same identification argument, but students should not treat them as interchangeable symbols.

Substantively: *relevance* is about the $Z \to T$ link — Z must genuinely move T, not merely be correlated with it. *Exogeneity* is about the absence of omitted common causes linking Z to Y — Z must be as-if randomly assigned relative to the latent determinants of the outcome. *Exclusion* is about the absence of any direct causal path $Z \to Y$ that bypasses T — it rules out both a direct arrow in the DAG and any parallel channel through unmodeled variables.

Why the do-calculus column is preferred. The exclusion restriction in the do-calculus column reads

$$f(y \mid x,\, \mathrm{do}(T{=}t),\, z) \;=\; f(y \mid x,\, \mathrm{do}(T{=}t)).$$

This is a statement about the *interventional* density — the distribution of Y after we have *set* $T = t$ by do-surgery. It says that once T is fixed by intervention, Z carries no additional information about Y. The do-operator makes it impossible to confuse the exclusion restriction with the observational statement $f(y \mid x, T{=}t, z) = f(y \mid x, T{=}t)$, which is a much weaker condition.

7.2.3 Relevance

Definition: Relevance

The instrument Z is **relevant** if it has a non-zero causal effect on the treatment T within at least some stratum of covariates X. Formally, $P(T \mid \mathrm{do}(Z{=}z), X{=}x)$ varies with z for some x in the support; in the linear first-stage model $T = \pi Z + \delta^\top X + \eta$, this reduces to $\pi \neq 0$.

The graphical analogue of relevance is that Z and T are not d-separated in $\mathcal{G}$. Under faithfulness, a directed path $Z \to T$ implies an observable association between Z and T; without faithfulness, non-d-separation does not by itself guarantee a non-zero conditional covariance, and an exact cancellation between alternative paths is mathematically possible. Conversely, an observable association between Z and T can arise without a causal $Z \to T$ effect when Z and T share a common cause. Graphical non-d-separation, observable association, and a non-zero causal effect are therefore three related but not identical conditions. The conditional covariance $\mathrm{Cov}(Z, T \mid X)$ and the first-stage F-statistic are empirical diagnostics for the observable association; the causal claim that Z shifts T is design-based.

7.2.4 Exogeneity

Definition: Exogeneity (Graphical Assumption)

The instrument Z is **exogenous** given covariates X if there is no open back-door path from Z to Y through unobserved causes. In DAG terms: $Z \perp\!\!\!\perp U \mid X$ in $\mathcal{G}$ by d-separation.

Exogeneity means that the instrument carries no information about latent determinants of the outcome other than through treatment. In the structural linear model, the same requirement appears as the moment orthogonality condition $\mathbb{E}[Z\varepsilon \mid X] = 0$, because the structural residual ε is a function of U. In potential-outcomes language, Z must be independent of the counterfactual quantities that the IV argument requires — outcomes $Y(0), Y(1)$ and treatment indicators $T(0), T(1)$ — given X. These are related manifestations of the same identification requirement, but they live in different formal languages and are not literally the same statement.

Econometric implication. The moment condition $\mathbb{E}[\varepsilon Z \mid X] = 0$ is a *consequence* of the graphical assumption, not a definition of exogeneity. This distinction matters: a researcher who adopts the moment condition as a primitive has no guarantee that Z is free of back-door paths to the outcome, only that a particular linear projection is zero.

Potential-outcomes analogue. In the potential-outcomes language, exogeneity is expressed as the full independence of the instrument from all counterfactual outcomes and counterfactual treatments:

$$Z \perp\!\!\!\perp (Y(0),\, Y(1),\, T(0),\, T(1)) \mid X.$$

This is a stronger statement than the moment condition alone: it also covers the joint independence of Z with the treatment potential outcomes $T(0), T(1)$, which the LATE proof in Section 7.7 requires in addition to $\mathbb{E}[\varepsilon Z \mid X] = 0$.

Example: Quarter of Birth as an Instrument

Angrist and Krueger (1991) used quarter of birth as an instrument for schooling to estimate the return to education. In graphical terms, exogeneity requires no open back-door path from quarter of birth to wages through unobserved variables such as ability or family background; in potential-outcomes terms, it requires that birth timing is independent of what wages a student would earn under any schooling level. Both are plausible because birth timing is largely outside parental control. Bound et al. (1995) later raised concerns about weak first stages in some specifications, illustrating that even a well-motivated instrument can fail the relevance requirement in certain samples.

Key point. All three versions of exogeneity are untestable: the unobserved confounder U is, by definition, unobserved, and counterfactual potential outcomes are never jointly observed. The assumption must be justified by institutional knowledge or the design of the study. See Section 7.6 for the limited consistency check that overidentification provides.

7.2.5 Exclusion

Definition: Exclusion Restriction

The instrument Z satisfies the **exclusion restriction** if it affects the outcome Y *only* through its effect on the treatment T. In do-calculus terms:

$$f(y \mid x,\, \mathrm{do}(T{=}t),\, z) \;=\; f(y \mid x,\, \mathrm{do}(T{=}t)) \quad \text{for all } z.$$

In potential outcomes terms: $Y_i(t, z) = Y_i(t, z')$ for all $z \neq z'$.

Exclusion is a *causal* restriction, not a conditional-correlation restriction. It is emphatically *not* the observational statement $Y \perp\!\!\!\perp Z \mid T, X$, which can hold in the data even when Z has a direct structural effect on Y. What exclusion requires is that changing Z while holding treatment fixed — that is, intervening to set $T = t$ — would leave Y unchanged. In graphical terms, every directed path from Z to Y must pass through T. In the structural linear model, this means no Z term appears in the outcome equation for Y.

Why Exclusion Is the Hardest Assumption

Relevance can be tested. Exogeneity can sometimes be defended by design (e.g., randomized encouragement). But the exclusion restriction — that Z has *no direct effect* on Y whatsoever — is:

- **Untestable** in just-identified models (one instrument, one treatment).
- **Easy to violate in practice.** An instrument that "moves treatment" often also moves other inputs. For example: if Z is distance to a hospital (instrument for treatment uptake), it may also directly affect health outcomes through travel time in emergencies.
- **Consequential.** Even a small direct effect of Z on Y — if correlated with the instrument — can cause substantial bias in the Wald estimand, especially when the first stage is weak. This bias is derived explicitly in Section 7.4.

The do-calculus makes the violation visible. Adding a direct arrow $Z \to Y$ to the DAG is exactly the formal counterpart of exclusion failing: the arrow represents a structural effect of Z on Y that bypasses T. In the mutilated graph $\mathcal{G}_{\overline{T}}$ (where we have removed all arrows into T), the path $Z \to Y$ remains open. Rule 1 of the do-calculus, which would allow us to remove Z from the density of Y given $\mathrm{do}(T{=}t)$, no longer applies. The exclusion restriction is therefore visibly a statement about the structure of $\mathcal{G}_{\overline{T}}$, not of $\mathcal{G}$ itself.

Remark: Three Assumptions Are Necessary but Not Sufficient for Point Identification

The three IV assumptions are necessary but not sufficient for *point* identification of any causal estimand. Under these three assumptions alone, the observed data restrict the distribution of potential outcomes but do not pin it down to a single value; the Wald ratio is a well-defined function of observable quantities, but it does not equal any specific causal parameter without additional structure (Hernán and Robins 2006; Levis et al. 2024). A fourth *structural assumption* on the counterfactual distribution of treatment response or treatment effect heterogeneity is required.
The choice of fourth assumption determines which causal quantity the Wald estimand identifies. Two leading choices appear in this chapter: constant treatment effects (Framework 1, Section 7.3), under which the Wald estimand identifies the ATE; and monotonicity (Framework 2, Section 7.7), under which it identifies the LATE for compliers. A third important family relies on *structural mean model* (SMM) restrictions. Under an additive SMM that rules out direct effect modification of Y by Z among the treated, the Wald ratio identifies the ATT — the average effect of treatment among those who actually received it — without requiring either constant effects or monotonicity (Hernán and Robins 2006; Levis et al. 2024). Moving further to identify the population ATE from the ATT requires an additional step: that the average effect among the treated equals the average effect among the untreated.
Strikingly, across these different identification schemes the same conditional Wald formula $\mathrm{Cov}(Z, Y \mid X)/\mathrm{Cov}(Z, T \mid X)$ appears as the identifying expression in many cases. The qualification "many" is precise: when identification relies on a *multiplicative* SMM instead of an additive one, the identifying formula for the ATE takes a different form entirely and the Wald ratio is no longer consistent for

the target estimand (Hernán and Robins 2006). What changes in the additive cases is not the observed-data formula but its causal target: the same estimator can consistently estimate the ATE, the LATE, or the ATT depending on which structural assumption is maintained.

A further consequence deserves emphasis: the three core IV assumptions are all expressible within the graphical language of this course. Relevance is the directed edge $Z \to T$; exogeneity is the absence of an open back-door path from Z to Y via U; exclusion is the absence of a direct edge $Z \to Y$ in the mutilated graph $\mathcal{G}_{\overline{T}}$. The fourth assumption, by contrast, lies *outside* the graphical language in every case. Monotonicity ($T_i(1) \geq T_i(0)$ for all i) is a restriction on the joint distribution of potential treatments across two intervention values simultaneously — a cross-world statement that cannot be encoded by the presence or absence of any arrow in the DAG. Constant treatment effects similarly constrains the covariance structure of the potential outcome pair, which the DAG does not represent. This is a genuine scope limitation of the graphical approach: a DAG specifies the causal structure of a single world, but the fourth assumption always reaches across worlds.

Example: Same Observables, Different ATEs

Two distinct data-generating processes satisfy all three IV assumptions and produce identical observable statistics, yet imply different ATEs. No amount of data can distinguish them without a fourth structural assumption.

Observable statistics. Let $Z, T \in \{0, 1\}$ and let Y be a continuous outcome. Suppose

$$\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0] = 0.2 \quad \text{(first stage)},$$

$$\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0] = 0.1 \quad \text{(reduced form)},$$

giving a Wald ratio of $0.1/0.2 = 0.5$. Both DGPs share the same compliance structure — under monotonicity, $P(\text{co}) = 0.2$, $P(\text{at}) = 0.4$, $P(\text{nt}) = 0.4$ — though under DGP A the compliance types are simply a convenient labeling; what drives identification there is the constant-effect restriction.

DGP A: constant treatment effects. Every unit has the same effect $\tau_i = 0.5$. Then the reduced form is $P(\text{co}) \cdot \tau_{\text{co}} = 0.2 \times 0.5 = 0.1$ ✓, and ATE $= 0.5$.

DGP B: heterogeneous effects with monotonicity. Effects differ by type: $\tau_{\text{co}} = 0.5$, $\tau_{\text{at}} = 0$, $\tau_{\text{nt}} = 0$. The reduced form is again $0.2 \times 0.5 = 0.1$ ✓, but

$$\text{ATE} = 0.2 \times 0.5 + 0.4 \times 0 + 0.4 \times 0 = 0.1.$$

What this shows. Both DGPs produce the same first stage, the same reduced form, and therefore the same Wald ratio of 0.5. Yet the ATE is 0.5 under DGP A and 0.1 under DGP B — a fivefold difference. The *fourth* assumption is what resolves the ambiguity: under DGP A, constant effects makes the Wald ratio equal the ATE; under DGP B, monotonicity makes it equal the LATE ($= 0.5$) for compliers, a different quantity from the ATE ($= 0.1$). Committing to one fourth assumption or the other changes not the estimator but what it estimates.

7.3 Identification in the Linear Homogeneous-Effect Model

Framework 1: Constant Treatment Effect and Linear Structure

This section works entirely within a linear structural model in which the treatment effect is the *same* for every unit and every treatment contrast:

$$Y_i(t) - Y_i(t') = \beta(t - t') \quad \text{for all } i,\ t,\ t'.$$

Under this assumption, IV identifies the single parameter β, which equals the ATE, the ATT, and the LATE simultaneously. This section should therefore be read as a special-case identification result. In Section 7.7 we drop homogeneity, allow treatment effects to vary across units, and reinterpret the same Wald ratio as the LATE for compliers. The estimator does not change; what changes is the estimand.

Note on assumption strength. Constant treatment effects is the strongest form of the homogeneity family of fourth assumptions: it requires not only that unobserved confounders U do not modify the effect of T on Y, but that the effect is exactly the same for every unit within each stratum of X. Weaker homogeneity conditions — such as requiring only that U not act as an additive effect modifier on Y — can also identify the ATE under IV (Wang and Tchetgen Tchetgen 2018).

7.3.1 The Linear Structural Model

Consider the linear structural model:

$$Y = \alpha + \beta T + \gamma^\top X + \varepsilon, \tag{7.1}$$

$$T = \pi Z + \delta^\top X + \eta, \tag{7.2}$$

where ε and η are structural errors with standard deviations σ_ε and σ_η and $\mathrm{Cov}(\varepsilon, \eta) = \rho\sigma_\varepsilon\sigma_\eta \neq 0$. The non-zero covariance is the source of endogeneity: OLS applied to Equation 7.1 gives a biased estimator of β.

The parameter of interest is β, the causal effect of T on Y:

$$\beta = \mathbb{E}[Y \mid \mathrm{do}(T{=}t+1)] - \mathbb{E}[Y \mid \mathrm{do}(T{=}t)].$$

Equation Equation 7.1 is the *outcome equation* (or second stage); Equation 7.2 is the *first-stage* equation. The *reduced form* is obtained by substituting Equation 7.2 into Equation 7.1:

$$Y = \alpha + \beta\pi Z + (\beta\delta + \gamma)^\top X + (\beta\eta + \varepsilon).$$

The reduced-form coefficient on Z is $\beta\pi$: the total effect of the instrument on the outcome. Dividing by the first-stage coefficient π recovers β, provided $\pi \neq 0$.

7.3.2 The OLS Bias

To see why OLS fails, apply OLS to Equation 7.1. The probability limit of the OLS estimator is

$$\operatorname{plim} \hat{\beta}_{\mathrm{OLS}} = \beta + \frac{\mathrm{Cov}(T, \varepsilon)}{\mathrm{Var}(T)} = \beta + \frac{\rho\sigma_\varepsilon\sigma_\eta}{\pi^2\mathrm{Var}(Z) + \sigma_\eta^2}. \tag{7.3}$$

The second term is the endogeneity bias. It is zero only if $\rho = 0$ (no unobserved confounding) or if the instrument perfectly determines T (infeasible in practice). The direction of the bias depends on the sign of ρ.

The point of IV is not that OLS is always wrong, but that when T is endogenous, IV can recover a causal parameter from the exogenous variation in T that Z induces — variation that OLS mixes indiscriminately with the confounded component.

7.3.3 Derivation of the Wald Estimand

The derivation has three ingredients: the **first stage**, which measures how much the instrument moves treatment; the **reduced form**, which measures how much the instrument moves the outcome; and the **exclusion restriction**, which implies that any effect of Z on Y must operate through T. Under the homogeneous-effect linear model, these three ingredients imply that the reduced-form effect of Z on Y is exactly β times the first-stage effect of Z on T.

Assumptions Active in This Derivation

1. **Linear homogeneous-effect model:** $Y_i(t) - Y_i(t') = \beta(t - t')$ for all i, t, t', and the structural equations are linear.
2. **Relevance:** $\mathrm{Cov}(Z, T \mid X) \neq 0$.
3. **Exogeneity:** $\mathrm{Cov}(Z, \varepsilon \mid X) = 0$.
4. **Exclusion:** Z has no direct effect on Y; it does not appear in the structural equation for Y.

Assumptions 2–4 are the three core IV assumptions; Assumption 1 is the homogeneity restriction specific to this section. Under heterogeneous effects the same Wald ratio identifies a different quantity; see Section 7.7.

We derive the identification of β step by step, using the structural model as the vehicle and the do-calculus to interpret the exclusion step.

1. **Exogeneity** ($Z \perp\!\!\!\perp U \mid X$ in $\mathcal{G}$) implies, for the structural residual ε, that $\mathbb{E}[\varepsilon \mid Z, X] = \mathbb{E}[\varepsilon \mid X]$. With the normalization $\mathbb{E}[\varepsilon \mid X] = 0$ (absorbed into α), we obtain $\mathbb{E}[\varepsilon \mid Z, X] = 0$.

2. **Exclusion** (Z absent from the structural equation for Y) is what permits writing Equation 7.1 without a Z term in the first place: under exclusion, the structural residual ε depends on U but does not absorb a direct effect of Z on Y. Combined with exogeneity, $\mathbb{E}[\varepsilon \mid Z, X] = 0$ immediately yields the moment condition $\mathbb{E}[\varepsilon \cdot Z \mid X] = 0$. The graphical counterpart is in $\mathcal{G}_{\overline{T}}$: exclusion means there is no direct edge $Z \to Y$, so $Y \perp\!\!\!\perp Z \mid T, X$ holds in the mutilated graph and Rule 1 of the do-calculus removes Z from $f(y \mid x, \mathrm{do}(T{=}t), z)$.

3. Multiplying Equation 7.1 through by $(Z - \mathbb{E}[Z \mid X])$ and taking expectations, the residual term $\mathbb{E}[\varepsilon(Z - \mathbb{E}[Z \mid X])]$ vanishes by step 2, leaving

$$\mathbb{E}[(Y - \alpha - \gamma^\top X)(Z - \mathbb{E}[Z \mid X])] = \beta\, \mathbb{E}[(T - \mathbb{E}[T \mid X])(Z - \mathbb{E}[Z \mid X])].$$

   In the simpler case with no covariates X, this collapses to the **reduced-form decomposition** $\mathrm{Cov}(Y, Z) = \beta\, \mathrm{Cov}(T, Z)$, which shows that the Z–Y covariance is entirely attributable to the causal path $Z \to T \to Y$ (by exclusion), and equals β times the first-stage covariance.

4. **Relevance** ($\mathrm{Cov}(Z, T \mid X) \neq 0$) ensures the denominator is non-zero, so we can solve:

$$\beta = \frac{\mathrm{Cov}(Y,\, Z \mid X)}{\mathrm{Cov}(T,\, Z \mid X)}. \tag{7.4}$$

The Wald Estimand

In the binary instrument case ($Z \in \{0, 1\}$) without additional covariates, Equation 7.4 simplifies to

$$\beta \;=\; \frac{\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0]}{\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0]}. \tag{7.5}$$

The numerator is the *reduced form*: the total effect of Z on Y. The denominator is the *first stage*: the effect of Z on T. The exclusion restriction is what guarantees that the entire reduced form operates through T, with no parallel channel; dividing by the first stage then strips out the $Z \to T$ piece and leaves β.

The key lesson is that the Wald ratio identifies β in this section only because treatment effects are assumed to be constant across units. The observed formula Equation 7.4 is an identifying expression, not an estimand by itself. Its causal meaning depends on the structural assumptions maintained around it. In Section 7.7, the same ratio will survive, but its interpretation will change from a common causal effect to a complier-specific average effect.

Remark: Estimation of the Wald Estimand

The Wald estimand above is an identification result: it expresses the causal parameter β as a ratio of observable quantities. Estimation — that is, how to consistently estimate this ratio from a finite sample, including the correct treatment of standard errors — is taken up in Chapter 13.

Example: Returns to Schooling

In the Mincer earnings equation $\log W = \alpha + \beta S + \gamma^\top X + \varepsilon$, where S is years of schooling, the OLS estimator of β is biased upward because unobserved ability U raises both S and W ($\rho > 0$). Angrist and Krueger (1991) use quarter of birth as Z: proximity to mandatory school-leaving age at different birth quarters generates exogenous variation in completed schooling. The IV estimate of the return to schooling is approximately 0.08–0.10 per year, similar to the OLS estimate in this case, suggesting the ability bias is modest.

7.4 Why the IV Assumptions Matter

The three IV assumptions are not merely technical conditions: each one is load-bearing, and each failure mode produces a distinct, quantifiable distortion of the Wald estimand. Violating exogeneity or exclusion contaminates the reduced form, and dividing by a weak first stage magnifies that contamination. A weak instrument is therefore not merely inefficient; it makes any small violation of the identifying assumptions more consequential.

When relevance fails. If $\pi = 0$, the Wald estimand is undefined: its denominator is zero and there is no exogenous variation to exploit. When π is small but nonzero, the instrument is *weak*. The estimator's variance diverges as $\pi \to 0$ because the denominator amplifies noise, and finite-sample bias pulls the IV estimate toward the OLS estimate at a rate proportional to $1/F$, where F is the first-stage F-statistic.

When exclusion fails. If the outcome equation contains a direct effect $Y = \alpha + \beta T + \delta Z + \gamma^\top X + \varepsilon$ with $\delta \neq 0$, the reduced form picks up both the indirect path $\beta\pi$ and the direct path δ, so the Wald estimand converges to $\beta + \delta/\pi$. The bias δ/π is amplified by a weak first stage: a small direct effect combined with a weak instrument can produce large bias. This is why a weak instrument with a plausible exclusion violation is not "nearly valid" — it may be severely misleading.

When exogeneity fails. If $\mathrm{Cov}(Z, \varepsilon \mid X) \neq 0$, the IV moment condition fails and the Wald estimand converges to $\beta + \mathrm{Cov}(\varepsilon, Z)/\mathrm{Cov}(T, Z)$. This is the IV analogue of OLS omitted-variable bias: IV fails when the instrument itself is endogenous, just as OLS fails when the treatment is endogenous. Again the bias is amplified when the first stage is weak.

Table 7.1: Testability of the three IV assumptions. Although exogeneity and exclusion are not directly testable, they are not entirely beyond empirical scrutiny. The IV model imposes inequality restrictions on the observable data distribution; when these are violated, falsification tests can detect the violation (Kitagawa 2015; Hernán and Robins 2006). Importantly, such tests are not consistent against all alternatives: for most data distributions under which the assumptions fail, the tests will not reject even in large samples.

Assumption	**Directly testable?**	**Basis for assessment**
Relevance	Association testable; causal claim design-based	First-stage F-statistic and $\mathrm{Cov}(Z, T \mid X)$ test the observable association; the causal $Z \to T$ link rests on the design
Exogeneity	No	Institutional knowledge; randomization (if available); placebo regressions on pre-determined outcomes
Exclusion	No in just-identified case; partially in overidentified case	Institutional argument; overidentification test (J-test, Chapter 13) checks mutual consistency of instruments but cannot confirm all are valid

The key asymmetry is that the hardest work in defending an IV design — the institutional arguments for exogeneity and exclusion — is precisely the work that the data cannot do for the researcher.

7.5 Lab: OLS vs. IV Across Instrument Strengths

The analytic results of Section 7.3 and Section 7.4 established two facts: OLS is biased by a precise, computable amount whenever $\rho \neq 0$; and IV is consistent but its variance diverges as $\pi \to 0$. What the analytics do not immediately reveal is the *finite-sample consequence*: for weak instruments, the IV variance penalty is so severe that the biased OLS estimator can have lower mean squared error. This simulation studies estimator behavior conditional on the IV assumptions being true; it does not address the separate and fundamentally substantive question of whether a proposed instrument is valid in an applied study.

DGP for Lab 7

The simulation implements the linear structural model of Section 7.3 with no covariates. Each replication draws $n = 500$ observations from

$$Y_i = \beta T_i + \varepsilon_i, \qquad T_i = \pi Z_i + \eta_i,$$

with $Z_i \overset{\text{i.i.d.}}{\sim} \mathrm{N}(0,1)$ and the structural errors generated as

$$\eta_i = U_i, \qquad \varepsilon_i = \rho U_i + \sqrt{1-\rho^2}\,\xi_i, \qquad U_i,\, \xi_i \overset{\text{i.i.d.}}{\sim} \mathrm{N}(0,1).$$

Here U_i is the unobserved confounder shared by both equations, and ξ_i is the idiosyncratic component of ε_i. This gives $\mathrm{Var}(\varepsilon_i) = \mathrm{Var}(\eta_i) = 1$ and $\mathrm{Cov}(\varepsilon_i, \eta_i) = \rho$. The fixed parameter values are $\beta = 1$ (true causal effect) and $\rho = 0.8$ (strong positive endogeneity). The first-stage coefficient π is varied across eight values:

$$\pi \ \in \ \{0,\ 0.10,\ 0.15,\ 0.20,\ 0.30,\ 0.50,\ 1.00,\ 2.00\}.$$

The expected first-stage F-statistic is approximately $n\pi^2 = 500\pi^2$. The Wald estimand is undefined at $\pi = 0$; the Staiger–Stock rule of thumb $F \geq 10$ corresponds to $\pi \geq 0.141$ in this design.

Potential Outcomes Interpretation of the DGP

The structural DGP has an exact translation into the potential outcomes language of Chapter 4, which makes explicit *why* OLS fails and *why* IV succeeds.

Potential outcomes. The latent background factor ε_i is fixed for each unit across all counterfactual worlds. The potential outcome under treatment value t is therefore

$$Y_i(t) \ = \ \beta t + \varepsilon_i. \tag{7.6}$$

The observed outcome satisfies $Y_i = Y_i(T_i)$, which is consistency. Because ε_i does not depend on t, the individual treatment effect is constant: $Y_i(t) - Y_i(t') = \beta(t - t')$ for all i. This is exactly Framework 1: β is simultaneously the ATE, the ATT, and the LATE.

The unobserved confounder. The shared factor U_i enters both the treatment assignment and the background outcome. Because ε_i depends on U_i and $\eta_i = U_i$, units with large T_i (driven by large U_i) also tend to have large ε_i, and hence large Y_i, *even if* $\beta = 0$. Formally,

$$\mathrm{Cov}\big(Y_i(t),\, T_i\big) = \mathrm{Cov}(\varepsilon_i,\, \eta_i) = \rho \ \neq \ 0,$$

so unconfoundedness fails. The parameter ρ is precisely the *degree of violation*.

Why IV restores identification. The instrument Z_i is drawn independently of U_i and ξ_i, so $Z_i \perp\!\!\!\perp U_i$ implies $Z_i \perp\!\!\!\perp \varepsilon_i$ — exogeneity, satisfied by construction. The instrument does not appear in $Y_i(t)$ (Equation 7.6), which is the exclusion restriction: $Y_i(t, z) = Y_i(t)$ for all z. And Z_i moves T_i through π, which is relevance. All three IV assumptions hold exactly by construction.

What varies across the grid. The potential outcomes $Y_i(t)$ are *the same* for every value of π — the parameter π governs only how strongly the instrument shifts treatment. All that changes for OLS as π grows is that more of T_i's variance comes from the clean source Z_i rather than from the confounded source U_i, which partially dilutes the endogeneity bias. But because the potential outcomes themselves are unchanged, $\beta = 1$ remains the target at every row of the table.

Estimators. *OLS* regresses Y on T; from Equation 7.3 it converges to $\beta + \rho/(\pi^2 + 1) = 1 + 0.8/(\pi^2 + 1)$. The bias is always positive (since $\rho > 0$), decreasing in $|\pi|$, but never zero for finite π. *IV (Wald)* uses Z as the instrument; it is consistent for β for any $\pi \neq 0$, with finite-sample variance approximately $\sigma^2/(n\pi^2)$, diverging as $\pi \to 0$.

Results ($n = 500$, $B = 2{,}000$ replications, seed 2024; true $\beta = 1.000$):

Table 7.2: Monte Carlo performance of OLS and IV across instrument strengths. Theoretical OLS bias $= 0.8/(\pi^2 + 1)$. "—" denotes $\pi = 0$, where IV is not defined. IV mean and SD at $\pi = 0.10$ and 0.15 are dominated by extreme outliers; the IV median is ≈ 1.01 at both values.

π	F	Theory bias	OLS mean	OLS bias	OLS SD	OLS RMSE	IV mean	IV bias	IV SD	IV RMSE
0.00	1	+0.800	1.801	+0.801	0.026	0.802	—	—	—	—
0.10	6	+0.792	1.791	+0.791	0.027	0.792	0.872	−0.128	4.391	4.393
0.15	13	+0.782	1.782	+0.782	0.027	0.783	0.764	−0.236	6.487	6.491
0.20	21	+0.769	1.769	+0.769	0.027	0.770	0.940	−0.060	0.380	0.385
0.30	46	+0.734	1.735	+0.735	0.028	0.735	0.975	−0.025	0.163	0.165
0.50	127	+0.640	1.639	+0.639	0.028	0.640	0.996	−0.004	0.091	0.091
1.00	500	+0.400	1.401	+0.401	0.027	0.402	0.999	−0.001	0.045	0.045
2.00	2006	+0.160	1.160	+0.160	0.019	0.161	1.000	0.000	0.022	0.022

1. The OLS bias formula is exact. The theory-bias column matches the simulated OLS bias to within Monte Carlo error across all eight values of π. OLS is biased in the direction of ρ at every value of π, including $\pi = 0$ where there is no instrument at all. Crucially, the OLS bias *decreases* as π grows, because a stronger first stage means more of T's variance comes from the exogenous source Z. At $\pi = 2$ the bias is only 0.160, yet this still produces RMSE $= 0.161$, which is $7\times$ larger than the IV RMSE of 0.022 at the same π.

2. IV is consistent but has catastrophically heavy tails when the instrument is weak. At $\pi = 0.10$ ($F \approx 6$) and $\pi = 0.15$ ($F \approx 13$) the IV *mean* is far from the true value, suggesting bias. This is misleading: the IV *median* is ≈ 1.01 at both values, confirming that the estimator is consistent and the *typical* replication recovers $\beta = 1$. The mean is dragged off by a small fraction of replications in which the first stage happens to be near zero, making the Wald ratio explode. Indeed, the just-identified IV estimator possesses *no finite moments*, so the mean and SD entries at $\pi = 0.10$ and 0.15 have no population counterparts: they are determined by the few most extreme replications and vary substantially across seeds. The take-away is that the weak instrument problem is not inconsistency; it is uncontrolled tail behavior.

3. The RMSE crossover occurs near $F \approx 20$. OLS RMSE ranges from 0.802 down to 0.161, while IV RMSE starts infinite, is catastrophic at $\pi = 0.10$–0.15, then drops sharply to 0.385 at $\pi = 0.20$ ($F \approx 21$) and to 0.165 at $\pi = 0.30$. IV first beats OLS on RMSE at $\pi = 0.20$: this is the crossover. Below this threshold, the variance penalty of IV exceeds the bias penalty of OLS and the biased estimator is *preferable* in mean squared error terms. The Staiger–Stock rule of thumb ($F \geq 10$) is therefore slightly *too lenient*: at $F \approx 13$ the IV RMSE in this run is still an order of magnitude above OLS. A more conservative threshold of $F \geq 20$–25 is needed before IV dominates OLS in this DGP.

4. A strong instrument eliminates both problems. At $\pi = 1.00$ ($F \approx 500$), IV is virtually unbiased with RMSE $= 0.045$, while OLS still carries a bias of 0.401, giving RMSE $= 0.402$ — a **9-fold** improvement from IV. With a strong instrument, IV is simultaneously unbiased *and* more efficient than OLS in MSE terms, because OLS efficiency is illusory: its small variance is offset by a large, persistent bias.

RMSE Cannot Be the Only Criterion

The table shows that OLS has lower RMSE than IV for $\pi < 0.20$ in this DGP. This does not vindicate OLS. An estimator with RMSE $= 0.78$ because it is biased by 0.80 is useless for causal inference: the bias is systematic and does not shrink with sample size. As $n \to \infty$, IV RMSE shrinks to zero while OLS RMSE stays at 0.80. The RMSE comparison is only meaningful in *finite samples*. For a fixed n, it quantifies when a weak instrument is so unreliable that IV is not yet practically useful — but that is an argument for finding a stronger instrument, not for using OLS.

Remark: The First-Stage F-Statistic as a Diagnostic

The approximate formula $F \approx n\pi^2$ gives first-stage F-statistics of 6, 13, 21, 46, 127, 500, and 2006 for the seven non-zero π values. The Staiger–Stock threshold of $F \geq 10$ is widely used in practice and corresponds to IV RMSE within a factor of roughly two of the strong-instrument limit; the weak-instrument problem it addresses was documented forcefully by Bound et al. (1995). The simulation here suggests this threshold may understate the problem when endogeneity is strong ($\rho = 0.8$): at $F = 13$ the IV RMSE in this run is still far above any useful threshold. A researcher should report the first-stage F alongside any IV estimate, and treat values below 20–25 with particular caution.

7.6 Multiple Instruments and Overidentification

This section gives only the identification-level intuition for using more than one instrument. The computational details of 2SLS with multiple instruments, GMM weighting, and the Sargan–Hansen J-test are deferred to Chapter 13.

When there are exactly as many instruments as endogenous variables ($q = p$), the model is *just-identified*: the IV assumptions pin down the causal parameter exactly, leaving no surplus identifying variation. When $q > p$, the model is *overidentified*: the extra instruments impose additional moment restrictions beyond those needed for identification. Under the homogeneous-effect linear model, every valid instrument must imply the same structural coefficient β, so those extra restrictions are testable. Under heterogeneous treatment effects, valid instruments can legitimately identify *different* LATEs because they shift treatment for different complier populations; the overidentification test then targets equality of probability limits across instruments, and a rejection admits more interpretations than instrument invalidity. In both cases, the Sargan–Hansen J-test (Sargan 1958; Hansen 1982) formalizes the mutual-consistency check; rejection indicates that at least one of the instruments either violates exogeneity or exclusion or identifies a different LATE, but does not localize which.

The terminology distinguishes the *order condition* — a counting requirement that $q \geq p$ — from the *rank condition*, which requires the instruments to be linearly independent in the first stage. Both are necessary for identification. The order condition is met by inspection; the rank condition is testable from the first-stage coefficient matrix.

What Overidentification Does Not Do

Passing the J-test does not confirm that any instrument is valid. It only shows that the sample moment conditions are mutually compatible — a weaker conclusion than validity. In the just-identified case the test has no power at all. Multiple invalid instruments can agree with one another if they share the same violation, so a consistent set of instruments is not necessarily a valid one. Heterogeneous treatment effects or model misspecification can also affect the test's behavior. Overidentification is an opportunity for a consistency check, not a substitute for the institutional argument that makes a design credible.

Remark: GMM and Overidentification

In the overidentified case, the surplus moment conditions can also be exploited for efficient estimation: rather than discarding the extra instruments, one can combine all q moment conditions optimally. The generalized method of moments (GMM) framework provides the natural setting for this, treating the just-identified IV estimator as a special case. This connection is developed in Chapter 13.

7.7 Heterogeneous Treatment Effects and the LATE Framework

Framework 2: Heterogeneous Treatment Effects

Section 7.3 imposed a common treatment effect β for every unit. That assumption is often too strong for applied work. We now drop homogeneity and allow $\tau_i = Y_i(1) - Y_i(0)$ to vary across units. Throughout this section we work in the binary instrument, binary treatment setting ($Z, T \in \{0, 1\}$). In this more realistic setting, the Wald ratio generally no longer identifies the ATE. Instead, under an additional assumption of monotonicity, it identifies the *Local Average Treatment Effect* (LATE): the average treatment effect for the subpopulation whose treatment status is changed by the instrument. The contrast with Framework 1 should be kept explicit:

- In Framework 1, the Wald ratio identifies a single common effect $\beta = \text{ATE} = \text{ATT} = \text{LATE}$.
- In Framework 2, the same ratio identifies a subgroup-specific average effect for compliers: $\tau_{\text{LATE}} \neq \text{ATE}$ in general.

The observed-data formula remains the same; the causal target changes.

Remark: Nonidentification of the ATE Is Structural, Not Estimator-Specific

The failure is not a defect of the Wald ratio: under relevance, exogeneity, exclusion, and monotonicity alone, *no* functional of the observed joint distribution $P(Z, T, Y)$ identifies the ATE. Problem 7 constructs two models that agree on $P(Z, T, Y)$ — and hence on the Wald ratio and, as the problem shows, on the LATE — yet disagree on the ATE, in the two-model style of the completeness discussion of Chapter 3. Point identification of the ATE therefore requires something beyond the core IV assumptions: effect homogeneity as in Framework 1, an extrapolation assumption such as those discussed in Section 7.8.2, or a different target (such as the LATE itself, or partial-identification bounds, Chapter 9).

7.7.1 Compliance Types

In the binary-instrument, binary-treatment setting, each unit's response to the instrument is fully described by the pair of potential treatment decisions $(T_i(0), T_i(1))$: the treatment the unit would take under each value of Z. This pair defines the unit's *compliance type* — a latent causal classification, because $(T_i(0), T_i(1))$ is never jointly observed in data.

Definition: Compliance Types [@angrist1996identification]

For binary $Z, T \in \{0, 1\}$, define the potential treatment $T_i(z)$ as the treatment unit i would take under instrument value z. The four compliance types are:

Compliance type	$T_i(0)$	$T_i(1)$
Complier	0	1
Always-taker	1	1
Never-taker	0	0
Defier	1	0

A **complier** takes treatment if and only if the instrument is switched on. An **always-taker** takes treatment regardless of Z. A **never-taker** never takes treatment regardless of Z. A **defier** does the opposite of what the instrument suggests.

The instrument Z only shifts treatment for *compliers*: always-takers and never-takers have the same treatment status regardless of Z, so they contribute nothing to the denominator $\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0]$. This is the first indication that the Wald ratio will be driven by the complier subgroup rather than by the full population.

7.7.2 The Monotonicity Assumption

The three core IV assumptions alone do not yield a clean causal interpretation for the Wald ratio when effects are heterogeneous. The difficulty is that the instrument may push some units toward treatment and others away from it: in the presence of defiers, the numerator and denominator of the Wald ratio conflate effects in opposite directions. A fourth assumption rules out this ambiguity.

Definition: Monotonicity [@imbens1994identification]

The treatment assignment is **monotone in** Z if $T_i(1) \geq T_i(0)$ for all i. Equivalently: there are no defiers in the population.

Interpretation. Monotonicity is not a generic law of causal inference. It is a design-specific claim about how this particular instrument changes treatment behavior. Switching the instrument from 0 to 1 may induce some units to take treatment (compliers) and leave others unaffected (always-takers or never-takers), but it should not reverse anyone's treatment decision. This is most plausible when the instrument is a randomized encouragement, access rule, or administrative assignment mechanism. In many observational IV settings the no-defiers assumption is substantively harder to defend and requires explicit justification.

7.7.3 The LATE Theorem

Under the three core IV assumptions, heterogeneous treatment effects, and monotonicity, the Wald ratio no longer identifies the ATE. Instead it recovers the average treatment effect for the units whose treatment status is actually changed by the instrument — the compliers. The effect is local not because it is estimated with nearby observations, but because it pertains to a local margin of behavioral response defined by the instrument itself.

Theorem 7.1 (LATE Theorem (Angrist and Imbens 1994))**.** *Suppose the following hold:*

1. ***Exogeneity (PO form):*** $Z \perp\!\!\!\perp (Y(0), Y(1), T(0), T(1))$*;*
2. ***Exclusion:*** $Y_i(t, z) = Y_i(t)$ *for all* z*;*
3. ***Relevance:*** $\mathbb{E}[T(1) - T(0)] \neq 0$*;*
4. ***Monotonicity:*** $T_i(1) \geq T_i(0)$ *for all* i*.*

Then the Wald estimand identifies the ***Local Average Treatment Effect*** *(LATE):*

$$\frac{\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0]}{\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0]} = \mathbb{E}\big[Y(1) - Y(0) \,\big|\, T_i(1) > T_i(0)\big] \equiv \tau_{\text{LATE}}. \tag{7.7}$$

That is, the Wald estimand identifies the average causal effect for compliers *only. When covariates are present, the same result holds with all assumptions and the Wald formula stated conditionally on* X*.*

Proof. We decompose the numerator and denominator by compliance type. Throughout, consistency gives $T_i = T_i(Z_i)$ and $Y_i = Y_i(T_i, Z_i)$, and exclusion reduces the latter to $Y_i = Y_i(T_i(Z_i))$.

Denominator. By consistency for T and exogeneity ($Z \perp\!\!\!\perp (T(0), T(1))$):

$$\begin{aligned}\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0] &= \mathbb{E}[T(1) \mid Z{=}1] - \mathbb{E}[T(0) \mid Z{=}0] \\ &= \mathbb{E}[T(1)] - \mathbb{E}[T(0)] = \mathbb{E}[T(1) - T(0)],\end{aligned}$$

and monotonicity gives $\mathbb{E}[T(1) - T(0)] = P(\text{complier})$, since always-takers contribute $1 - 1 = 0$ and never-takers contribute $0 - 0 = 0$.

Numerator. By consistency, exclusion, and exogeneity:

$$\begin{aligned}\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0] &= \mathbb{E}[Y(T(1))] - \mathbb{E}[Y(T(0))] \\ &= \sum_c P(c)\, \mathbb{E}\big[Y(T_c(1)) - Y(T_c(0)) \,\big|\, \text{type} = c\big].\end{aligned}$$

For always-takers, $T(1) = T(0) = 1$, so the contribution is 0. For never-takers, $T(1) = T(0) = 0$, contribution 0. For compliers, $T(1) = 1$ and $T(0) = 0$, so $Y(T(1)) - Y(T(0)) = Y(1) - Y(0)$. No defiers exist by monotonicity. Therefore

$$\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0] = P(\text{complier})\, \mathbb{E}\big[Y(1) - Y(0) \mid \text{complier}\big].$$

Ratio. Dividing numerator by denominator gives Equation 7.7. □

> Remark: The Two Senses of "Local"
>
> The LATE is local in two senses: it is local to the complier group defined by this instrument, and local to the particular instrument that defines that group. A different instrument, even for the same treatment, will generally select a different complier population and identify a different LATE. This is the subject of Section 7.8.3.

7.8 Interpreting IV Estimands

The two frameworks — linear homogeneous effects and the LATE framework — give the Wald estimand different interpretations. This section synthesizes those interpretations and clarifies when they agree, when they disagree, and what the difference means for applied work.

7.8.1 What the Two Frameworks Say

	Framework 1 (linear, homogeneous)	**Framework 2** (heterogeneous effects)
Key assumption	$\tau_i = \beta$ for all i	Monotonicity; no defiers
What IV identifies	$\beta = \text{ATE} = \text{ATT} = \text{LATE}$	$\tau_{\text{LATE}} = \mathbb{E}[\tau_i \mid \text{complier}]$
Estimand depends on instrument?	No (same β regardless of Z)	Yes (different $Z \implies$ different compliers $\implies$ different LATE)
Identifies the ATE?	Yes, automatically	Only if all units are compliers or effects homogeneous

Framework 1 is a special case of Framework 2: when $\tau_i = \beta$ for all i, the LATE equals the ATE equals β, and the instrument does not affect the estimand — only identification. Framework 2 is the more general and realistic setting. In applied work, the default interpretation of the Wald estimand is the LATE; the ATE interpretation requires the additional homogeneity argument of Framework 1.

> Remark: Same Formula, Different Estimand
>
> The two frameworks illustrate a broader principle in IV analysis. Across a range of fourth assumptions — constant effects, monotonicity, structural mean model restrictions, and others — the conditional Wald formula serves as the identifying expression in many (though not all) cases (Levis et al. 2024). Accordingly, two researchers who use the same instrument and compute the same Wald ratio may be consistently estimating different causal quantities if they maintain different structural assumptions: one estimates the ATE (under constant effects), another estimates the LATE (under monotonicity), and a third estimates the ATT (under a structural mean model restriction). The point estimates and confidence intervals can be numerically identical; what differs is the causal quantity they represent. This is not a deficiency of IV but a precise illustration of why causal modeling is not merely a formal preliminary to estimation. The data alone cannot resolve which estimand the Wald ratio identifies; that determination requires the researcher to commit to a structural assumption about treatment response. Stating that commitment explicitly — and defending it on substantive grounds — is as important as any statistical calculation.

7.8.2 When Does LATE Equal ATE?

LATE equals ATE only under additional structure, most notably treatment-effect homogeneity or special forms of heterogeneity that make complier effects representative of the full population. Neither condition should be assumed without argument.

In general, $\tau_{\text{LATE}} \neq \text{ATE}$ when treatment effects are heterogeneous and some units are not compliers. To see this, decompose the ATE by compliance type:

$$\text{ATE} = P(\text{co})\,\mathbb{E}[\tau_i \mid \text{co}] + P(\text{at})\,\mathbb{E}[\tau_i \mid \text{at}] + P(\text{nt})\,\mathbb{E}[\tau_i \mid \text{nt}],$$

where co, at, and nt abbreviate complier, always-taker, and never-taker. The LATE equals only the first term divided by its probability weight. The ATE and LATE coincide if and only if:

1. Mean treatment effects are equal across compliance types — a strict weakening of the constant-effect assumption of Section 7.3, but still a non-trivial restriction on heterogeneity; or
2. Everyone is a complier ($P(\mathrm{at}) = P(\mathrm{nt}) = 0$), which would require Z to perfectly determine T; or
3. The average effects for always-takers and never-takers happen to equal the LATE — an untestable coincidence.

In practice, none of these conditions is likely to hold exactly. The LATE is a well-defined, identifiable parameter, but it is not the ATE.

7.8.3 Different Instruments, Different Estimands

Because the LATE is specific to the complier population, and different instruments select different complier populations, two valid instruments for the same treatment can legitimately identify different LATEs. This is a precise, substantive fact — not a contradiction. Different instruments can legitimately identify different causal effects because they shift treatment for different margins of the population; the diversity of LATEs is informative about treatment effect heterogeneity, not a symptom of model failure.

Example: Compulsory Schooling Laws vs. Distance to College

Two classic instruments for years of schooling are (i) compulsory schooling laws (Angrist and Krueger 1991) and (ii) distance to the nearest college (Card 1995). The compliers for instrument (i) are individuals at the margin of dropping out before the compulsory leaving age — typically lower-income students. The compliers for instrument (ii) are students deterred from college by geographic distance — again, disproportionately lower-income. Both instruments identify the return to schooling for their respective complier populations, and the LATEs can legitimately differ even if both instruments are valid.

The dependence of the LATE on the instrument is sometimes described as a limitation of IV. It is better understood as a precise statement about what question is being answered. A researcher using compulsory schooling laws is estimating the return to schooling for students at the compulsory leaving margin; a researcher using distance to college is estimating the return for students deterred by geography. These are different causal questions, and it is informative — not troubling — that they can yield different answers.

7.8.4 The Policy Relevance of LATE

For many policy questions, the LATE is exactly the right estimand. If a policy is designed to encourage a subset of the population to take treatment — for example, an outreach program that reaches only some potential participants — then the effect on compliers is precisely what the policy-maker wants to know. LATE is most policy-relevant when the contemplated intervention resembles the instrument, because then the complier population under the study design is close to the policy-relevant margin.

When the ATE over the full population is required for policy analysis, IV alone is insufficient under heterogeneous effects. Additional assumptions — such as an explicit model of treatment effect heterogeneity, or a second instrument that identifies effects for a different subpopulation — are needed to extrapolate from the LATE to the ATE.

7.9 IV versus Back-Door Adjustment

Having developed both strategies in full, it is instructive to compare them directly. The two strategies differ in what they require, what they identify, and how they fail.

7.9.1 A Direct Comparison

Dimension	Back-door / propensity score	Instrumental variables
Core assumption	All confounders observed: $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$	Valid instrument: relevance, exogeneity, exclusion
Unobserved confounders	Fatal: back-door adjustment fails	Permitted: IV routes around U
Estimand	ATE, ATT, or ATC depending on design and overlap; all coincide under homogeneity	LATE (compliers only); reduces to a common β under homogeneous effects
Testability	Unconfoundedness untestable; overlap testable	Relevance testable; exogeneity and exclusion untestable (just-identified case)
Main threat	Unmeasured confounder	Exclusion restriction violation
Identifies the ATE?	Yes, under strong ignorability	Only under homogeneous effects
Typical setting	Rich administrative or survey data; RCT with imperfect compliance	Natural experiment; RCT with non-compliance; policy change

7.9.2 Complementary Failure Modes

Back-door adjustment fails when X does not capture all confounders — that is, when an unobserved variable U opens a back-door path. The estimator is then inconsistent even with infinite data, because the path $T \leftarrow U \rightarrow Y$ transmits spurious association that conditioning on X cannot close.

IV fails when the exclusion restriction is violated — that is, when Z has a direct effect on Y beyond its effect through T. As derived in Section 7.4, the bias in the Wald estimand is δ/π, amplified by weak instruments.

The two failures are in a sense orthogonal: back-door adjustment requires many observed covariates but tolerates no unobserved ones, while IV tolerates unobserved confounders but requires an instrument with no direct effect on the outcome. Back-door adjustment fails when confounding remains after conditioning; IV fails when the proposed source of exogenous variation is not truly exogenous or does not act solely through treatment. When designing a study, the choice of identification strategy should be guided by which assumption is more plausible in the specific empirical context.

7.9.3 When Both Strategies Are Available

When a valid instrument *and* a sufficient adjustment set X are both available, comparing the two estimates can be informative, but disagreement between them does not by itself tell us which method is wrong. The two strategies typically target different estimands — back-door adjustment identifies the ATE or ATT over the full population or the treated, while IV identifies the LATE for compliers — and they rely on different identifying assumptions. Agreement is therefore reassuring under treatment-effect homogeneity but is neither required nor sufficient otherwise.

The Hausman (1978) endogeneity test operationalizes this comparison: under the null hypothesis that T is exogenous given X, both the OLS and IV estimators are consistent, and a large discrepancy is evidence of endogeneity. Under the null and standard regularity conditions, an appropriately scaled quadratic form in the discrepancy $\widehat{\beta}_{\mathrm{IV}} - \widehat{\beta}_{\mathrm{OLS}}$ is asymptotically χ^2_p-distributed, where p is the number of components of T tested for endogeneity; the formal construction is given in Chapter 13. Rejection means at least one strategy is inconsistent; it does not identify which.

Remark: Interpreting a Hausman Rejection

A significant Hausman test can arise from two distinct sources: (a) back-door adjustment is biased because T is endogenous, or (b) IV is biased because the exclusion restriction is violated or the instrument is weak. Even when the test is non-significant, the two estimates may still differ for the legitimate reason that they identify different estimands — LATE versus ATE — rather than because either strategy has failed. External evidence — instrument strength diagnostics and institutional arguments for exclusion — is needed to interpret any discrepancy.

7.10 Practical Guidance on Defending an IV Design

There is no algorithm for finding a valid instrument; credible instruments arise from institutional knowledge and careful reasoning about the data-generating process. A strong IV design is defended primarily by institutional knowledge, design logic, and causal structure; statistical diagnostics are supportive but secondary.

A researcher proposing an instrument should be able to answer five questions explicitly:

1. **What exactly is the instrument?** Specify Z precisely: its source of variation, the level at which it varies, and the population to which it applies.

2. **Why does it shift treatment?** Articulate the causal mechanism by which Z moves T. Relevance can be verified empirically with the first-stage F-statistic, but the F-statistic is a diagnostic for instrument strength, not a substitute for a causal account of the $Z \to T$ link.

3. **Why is it as-if random relative to latent outcome determinants?** Argue why Z is unrelated to the unobserved causes of Y. The most credible sources are designed randomization (lotteries, randomized encouragement), natural experiments with institutional quasi-randomness (policy discontinuities, geographic boundaries, biological quirks), and shift-share designs (Bartik 1991; Goldsmith-Pinkham et al. 2020). Placebo regressions on pre-determined outcomes provide partial — but not definitive — evidence.

4. **Why can it affect the outcome only through treatment?** The exclusion restriction is untestable in just-identified models, so it must rest on a structural argument that no direct path $Z \to Y$ exists. A useful diagnostic is to ask: how large would the direct effect δ have to be, relative to the first-stage coefficient π, to overturn the estimated causal effect? When the first stage is weak, the answer is: not very large at all.

5. **What population margin does it shift?** Identify the complier population — the units whose treatment status changes with Z. This determines the LATE that is being identified and governs the external validity of the estimates for other populations or policy margins.

7.11 Applied Example: Charter School Lotteries and the KIPP Lynn Study

This example illustrates a canonical randomized-encouragement IV design. The key distinction is that the lottery randomizes *offer status*, not actual treatment. Winning the lottery does not mechanically force a student to attend KIPP, and losing the lottery does not make later attendance impossible in all cases. Thus the lottery offer is the instrument Z, while actual years of KIPP attendance is the endogenous treatment S. Here S plays the role of the generic treatment variable T used throughout the rest of this chapter; the paper's notation is retained to keep the empirical discussion close to the source.

This distinction is exactly what makes the design an IV design rather than a randomized controlled trial on treatment itself. The lottery generates exogenous variation in access to KIPP, and the IV analysis uses only the portion of attendance variation induced by that randomized offer to identify a causal effect.

7.11.1 Setting and Instrument

KIPP (Knowledge Is Power Program) schools follow a "No Excuses" model: extended school days, a longer academic year, selective teacher hiring, and strict behavioral norms. KIPP Academy Lynn was substantially oversubscribed beginning in 2005. Massachusetts law requires oversubscribed charter schools to select students by lottery, so the school conducted randomized admissions lotteries from 2005 through 2008. The treatment is years of KIPP attendance S; in the panel structure of the data this is denoted s_{igt}, the number of calendar years student i has spent at KIPP by the time of test (g, t) — a continuous, endogenous variable, because families self-select into applying and attending. The outcome Y_{igt} is the student's standardized score on the Massachusetts Comprehensive Assessment System (MCAS), normalized to mean zero and standard deviation one within each subject–grade–year cell statewide.

7.11.2 Mapping the Three Assumptions to the KIPP Context

The KIPP lottery design makes the logic of the three IV assumptions unusually transparent. One point deserves emphasis before proceeding. Relevance and exogeneity are especially natural in a lottery-based design: because lottery sequence numbers are randomly assigned within cohort, offer status is independent of the latent determinants of achievement by construction. Exclusion, however, still requires institutional argument. Random assignment of the instrument does not, by itself, imply that the instrument has no direct effect on the outcome — it only guarantees that the instrument is exogenous.

Relevance. The lottery offer must shift years of KIPP attendance. Lottery winners were offered a seat and about 80 percent accepted; losers rarely enrolled elsewhere at KIPP. The first-stage regression of s_{igt} on Z_i (with year and grade controls) yields a coefficient of approximately 1.2: at the time of each MCAS exam, lottery winners had spent about 1.2 more years at KIPP than lottery losers. The first-stage F-statistic is far above conventional thresholds. The first stage is less than the theoretical maximum because compliance is partial (some winners do not attend; some losers find entry through later cohorts or attrition slots) and the follow-up window captures students at different points in their KIPP tenure.

Exogeneity. The lottery offer must be independent of all potential outcomes and potential treatments — formally, $Z \perp\!\!\!\perp (Y(0), Y(1), T(0), T(1))$ within each application cohort. Because offer status was determined by randomly drawn lottery-sequence numbers, this independence holds by design — an especially strong basis for exogeneity compared to most observational IV applications. The authors verify it empirically: a joint test of covariate balance across lottery winners and losers yields a p-value of 0.615 for demographic characteristics and baseline test scores, consistent with the null of no pre-lottery differences. Pre-lottery variables are used for the balance check precisely because post-lottery variables such as LEP or SPED classification may themselves be affected by school attended.

Exclusion. The lottery offer must affect test scores only through KIPP attendance, not through any direct channel. Because the offer merely provides access to a school — it does not itself deliver instruction — this restriction is institutionally plausible. One potential violation is a discouragement effect: losing the lottery might demoralize students, suppressing their achievement regardless of where they enroll. The authors address this by noting that the scores of lottery losers are typical of demographically comparable students in Lynn, which is inconsistent with large discouragement effects.

7.11.3 First Stage, Reduced Form, and the 2SLS Estimand

The equations below illustrate the components of the Wald/2SLS logic — first stage, reduced form, and their ratio. Formal estimation theory for 2SLS, including asymptotic variance and cluster-robust inference, is deferred to Chapter 13.

The structural equation for test scores is

$$y_{igt} = \alpha_t + \beta_g + \sum_j \delta_j d_{ij} + \gamma' X_i + \theta\, s_{igt} + \varepsilon_{igt}, \tag{7.8}$$

where α_t and β_g are year-of-test and grade-of-test fixed effects, d_{ij} are application-cohort dummies (cohort membership determines the probability of winning, so cohort is an essential control), X_i is a vector of baseline demographics, and θ is the causal effect of interest per year at KIPP. We write θ rather than ρ (as in the original paper) to avoid clashing with the endogeneity correlation parameter of the same name used earlier in this chapter. The first-stage equation is

$$s_{igt} = \lambda_t + \kappa_g + \sum_j \mu_j d_{ij} + \Gamma' X_i + \pi Z_i + \eta_{igt}. \tag{7.9}$$

The model is just-identified (one excluded instrument per endogenous variable), so the 2SLS estimator of θ equals the ratio of the reduced-form coefficient on Z_i to the first-stage coefficient π.

Table 7.3: IV estimates of KIPP Lynn attendance on MCAS scores (per year at KIPP). Source: Angrist et al. (2012), Table 4. Standard errors clustered at the student level in parentheses. $N = 833$ student-by-test observations for both subjects. All regressions include year-of-test, grade-of-test, and application-cohort fixed effects, plus demographic and baseline-score controls.

Subject	First stage	Reduced form	2SLS
Math	1.221 (0.068)	0.430 (0.067)	0.352 (0.053)
ELA	1.228 (0.068)	0.164 (0.073)	0.133 (0.059)

Each year at KIPP raises math scores by approximately 0.35 standard deviations and ELA scores by approximately 0.13 standard deviations. The reduced-form estimate for math (0.43σ) is larger than the 2SLS estimate (0.35σ) because the first stage exceeds 1: lottery winners accumulated somewhat more than one additional year at KIPP per unit of follow-up time.

Remark: Treatment Heterogeneity and the Weighted-Average Interpretation

When the treatment is continuous, Angrist and Imbens (1995) show that the IV estimand is a weighted average of the marginal causal effects of each additional unit of treatment, with weights proportional to the first-stage effect of the instrument at each dose level. In the KIPP setting, θ is therefore a weighted average of the per-year effect across the different years a student might spend at KIPP, rather than the effect of a single fixed dose. The authors note this interpretation explicitly and adopt the conservative coding convention that partial years count as full years, which slightly attenuates the 2SLS estimates.

7.11.4 LATE Interpretation

The lottery design identifies the causal effect of KIPP attendance only for the students whose attendance behavior is changed by the lottery offer. These are the lottery compliers: students who attend KIPP if offered a seat and do not attend KIPP if not offered one. Compliance is partial in both directions: some lottery winners do not enroll and some lottery losers eventually find entry through later cohorts or attrition slots. Always-takers are largely ruled out at the extensive margin by the lottery's control over seat access; never-takers remain — winners who decline the offer — and both margins reappear in the intensive dimension (years of enrollment).

This is the key interpretive point of the example. The 2SLS estimand θ is not the effect of KIPP on all students in Lynn, nor on all applicants — it is the average per-year treatment effect for the *lottery compliers*, the specific margin of students whose enrollment decision is altered by randomized access to the school.

An important feature of this LATE is that it may differ from the effect the school would have on a randomly selected Lynn student who was not part of the applicant pool. KIPP applicants already had parents motivated enough to apply. The authors find that KIPP applicants have baseline test scores slightly *lower* than the district average, so the applicant pool is not positively selected on prior achievement, but it may still be selected on parental engagement in ways that affect how students respond to the KIPP environment.

The design is therefore strongest not because it identifies a universally generalizable effect, but because it identifies a clearly interpretable causal effect for a well-defined subpopulation under highly credible exogeneity.

7.11.5 Treatment Effect Heterogeneity

Subgroup analysis reveals that the LATE varies substantially across observable student characteristics. Reading gains ($\approx 0.13\sigma$ overall) are driven almost entirely by students classified as having limited English proficiency (LEP, $\approx 0.43\sigma$) and special education needs (SPED, $\approx 0.27\sigma$); non-LEP, non-SPED students show negligible ELA gains. Math effects are large and positive across all subgroups but are largest for LEP and lower-achieving students. An interaction model that adds the product of baseline score with years at KIPP — identified by including $Z_i \times$ baseline score as a second instrument — yields a significantly negative interaction term in both subjects: each additional standard deviation of baseline disadvantage is associated with an additional 0.08–0.17σ gain per year at KIPP.

These findings connect directly to Section 7.8.3: were we to define separate subgroup-specific instruments (LEP-lottery and non-LEP-lottery), each would identify a distinct subpopulation LATE. The overall 2SLS estimate is a weighted average of these subgroup LATEs, with weights proportional to each subgroup's share of the complier population. This heterogeneity is not a nuisance detail; it is exactly why IV estimates must be interpreted together with the margin of compliance they capture.

Remark: Lottery Design and Assumption Credibility

The KIPP study illustrates a general principle: instruments derived from *designed* randomization — lotteries, random assignment, randomized encouragement — provide the most transparent basis for the exogeneity assumption, because the distribution of Z is known by construction. The exclusion restriction still requires institutional argument (here, that the lottery offer affects only schooling, not morale or parental behavior in other dimensions), but exogeneity within the applicant sample, conditional on application cohort, is not merely plausible — it follows from the randomization protocol. This is why lottery-based IV designs occupy a privileged position in the program evaluation literature (Angrist and Pischke 2009).

The KIPP example ties together the main lessons of this chapter. All three IV assumptions are visible in the design: relevance is confirmed by a strong, precisely estimated first stage; exogeneity within the applicant sample follows from the randomization protocol; and exclusion rests on an institutional argument that the lottery offer affects achievement only through attendance. Because compliance is partial, the resulting estimate is a LATE for lottery compliers, not an ATE for all applicants or all students in Lynn. A lottery-based IV design is strongest not because it answers every causal question, but because it answers a clearly defined causal question for a clearly defined subpopulation.

7.12 Summary

1. **IV identifies effects from exogenous treatment variation.** When back-door adjustment fails because an unobserved U creates a path $T \leftarrow U \rightarrow Y$, a valid instrument Z identifies the causal effect by exploiting only the component of treatment variation that Z induces. IV does not block the confounding path — it avoids it by isolating exogenous variation and ignoring the rest.

2. **Three assumptions, ordered by testability.** Relevance can be assessed with the first-stage F-statistic, though the F-statistic is a sample diagnostic, not the assumption itself. Exogeneity and exclusion are primarily substantive and must be defended by institutional knowledge, design logic, and causal structure. Each violated assumption produces a distinct, quantifiable bias in the Wald estimand, amplified by weak instruments (Section 7.4).

3. **Framework 1: homogeneous-effect SEM $\Longrightarrow$ Wald identifies β.** Under constant treatment effects and the linear structural model, the Wald estimand identifies the single causal parameter β, which equals the ATE, ATT, and LATE simultaneously. The identification follows from a three-step argument using the reduced form and first stage (Section 7.3).

4. **Framework 2: heterogeneity + monotonicity $\Longrightarrow$ Wald identifies LATE.** Under heterogeneous treatment effects and monotonicity, the Wald estimand identifies the average treatment effect for *compliers* only — those whose treatment status changes with the instrument, a latent subgroup defined by potential treatment statuses $(T(0), T(1))$, not by observable characteristics (Section 7.7).

5. **Different instruments identify different effects.** The LATE depends on the instrument through the complier population it selects. Different instruments can legitimately identify different causal effects because they shift treatment for different margins of the population. LATE equals ATE only under additional structure, most notably treatment-effect homogeneity (Section 7.8).

6. **IV versus back-door adjustment.** The two strategies have complementary failure modes: back-door adjustment fails when confounders are unobserved; IV fails when the exclusion restriction is violated or the instrument is not truly exogenous. IV identifies the LATE (not the ATE) under heterogeneous effects; back-door adjustment identifies the ATE or ATT. They are tools for different identification problems, not competitors (Section 7.9).

7. **Applied example: the KIPP Lynn lottery.** Angrist et al. (2012) use a randomized admissions lottery as a canonical randomized-encouragement instrument for years of charter school attendance. Within the applicant sample, conditional on application cohort, random assignment of the offer supports exogeneity; exclusion is defended institutionally; relevance is confirmed by a first-stage coefficient of 1.2. The estimates identify the LATE for lottery compliers — 0.35σ per year in math, 0.13σ in ELA — with the largest gains for LEP, SPED, and low-baseline-score students (Section 7.11).

8. **Estimation deferred to Chapter 13.** This chapter establishes what IV identifies and under what assumptions. How the Wald ratio is estimated from finite data — the reduced form regression, two-stage least squares, asymptotic inference, and overidentification tests — is the subject of Chapter 13.

7.13 Problems

1. The three IV assumptions in three languages. Consider the DAG $\{Z \to T,\ T \to Y,\ U \to T,\ U \to Y,\ X \to T,\ X \to Y,\ X \to Z\}$ with U unobserved.

(a) List all back-door paths from T to Y. Does X alone satisfy the back-door criterion? Explain.
(b) Verify the three IV assumptions using d-separation: (i) relevance: show Z and T are not d-separated in $\mathcal{G}$; (ii) exogeneity: show $Z \perp\!\!\!\perp U \mid X$ in $\mathcal{G}$; (iii) exclusion: show $Y \perp\!\!\!\perp Z \mid T, X$ in $\mathcal{G}_{\overline{T}}$.
(c) Now add the arrow $Z \to Y$ to the DAG. Which IV assumption is violated? Show explicitly which step of the Wald derivation in Section 7.3 breaks down.
(d) Translate each of the three IV assumptions into the structural language: write the equations for T and Y and identify which coefficient restriction corresponds to each assumption.

2. Bias under assumption violations. Let $Y = \beta T + \varepsilon$ and $T = \pi Z + \eta$ with $\mathbb{E}[\varepsilon \mid Z] = 0$ and $\pi \neq 0$.

(a) Starting from $\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0]$, substitute the structural equation for Y and simplify. What role does exogeneity play?
(b) Show that $\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0] = \pi$ in the linear first-stage model. What role does relevance play?
(c) Derive the Wald estimand and confirm it equals β.
(d) Now suppose the exclusion restriction fails and $Y = \beta T + \delta Z + \varepsilon$ with $\delta \neq 0$. Derive the probability limit of the Wald estimator. Confirm the bias formula from Section 7.4.
(e) Suppose instead that exogeneity fails: $\mathbb{E}[\varepsilon \mid Z] = cZ$ for some constant $c \neq 0$. Derive the probability limit of the Wald estimator and express the bias in terms of c and π. Compare the structure of this bias with the exclusion violation bias.

3. Order, rank, and the limits of overidentification. Consider a model with one endogenous variable T and two instruments Z_1 and Z_2, both satisfying exogeneity and exclusion.

(a) State the order condition and verify it is satisfied.
(b) State the rank condition. What would it mean geometrically if the rank condition failed — i.e., if Z_1 and Z_2 were perfectly collinear in the first-stage regression?
(c) Explain intuitively why having two valid instruments rather than one should improve estimation precision.
(d) Now suppose Z_1 is valid but Z_2 violates the exclusion restriction. The Sargan–Hansen J-test is applied. Under what conditions does the J-test have power to detect Z_2's invalidity? Under what conditions does the test fail to detect it?
(e) Why does passing the J-test not confirm that both Z_1 and Z_2 are valid? Give a concrete example of a situation in which both instruments are invalid and the J-test has no power.

4. Compliance types and the LATE. In a binary instrument, binary treatment study, suppose the population has the following composition: 30% compliers with average treatment effect $\tau_c = 6$; 25% always-takers with $\tau_a = 3$; 45% never-takers with $\tau_n = 1$; no defiers.

(a) Compute $P(\text{complier}) = \mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0]$.
(b) Compute the ATE as a weighted average of τ_c, τ_a, τ_n with appropriate weights.
(c) The Wald estimand equals $\tau_c = 6$. By how much does this overstate the ATE, and why?

(d) A second study uses a different binary instrument Z' with a complier population of 50% and a LATE of 2. Is this contradictory? What can you infer about the relative treatment effect in the two complier populations?

(e) Explain, using compliance type language, why the denominator of the Wald estimand equals $P(\text{complier})$.

5. The exclusion restriction: plausibility and violations. Evaluate the exclusion restriction for each of the following proposed instruments. For each, (i) state whether the restriction is plausible and why; (ii) describe a specific mechanism by which it could be violated; and (iii) assess whether the violation would bias the IV estimate upward or downward.

(a) *Instrument*: rainfall in the home region of a politician, used as an instrument for government infrastructure spending. *Outcome*: local economic growth.

(b) *Instrument*: distance to the nearest hospital, used as an instrument for hospital admission. *Outcome*: 30-day mortality.

(c) *Instrument*: a randomly assigned financial incentive to enroll in a health screening program, used as an instrument for screening uptake. *Outcome*: health status two years later.

(d) *Instrument*: lottery number in the Vietnam-era draft lottery, used as an instrument for military service. *Outcome*: lifetime earnings. (This is the Angrist (1990) study; discuss why this instrument is widely regarded as satisfying the exclusion restriction.)

6. IV versus back-door adjustment. A researcher studies the effect of job training (T) on earnings (Y). Two strategies are available: (A) a rich set of pre-treatment covariates X and a propensity-score estimator; (B) a lottery that randomly selected units to be offered training (not required to attend), used as an instrument Z.

(a) Under what assumption does strategy (A) identify the ATE? What specific unobserved variable would most plausibly violate this assumption?

(b) Strategy (B) identifies a LATE. Describe the complier population in words. Is the LATE likely to be larger or smaller than the ATE in this setting? Explain.

(c) Both strategies are implemented and yield estimates of \$1,800 and \$2,400 per year, respectively. Describe a Hausman-type test that uses both estimates. Under what null hypothesis does the test have an approximate χ^2 distribution?

(d) If the two estimates differ significantly, which strategy would you trust more and why? What additional evidence would help distinguish the two explanations (endogeneity bias in (A) versus LATE $\neq$ ATE in (B))?

7. Nonparametric nonidentification of the ATE under valid IV assumptions *(advanced / second pass)*. Consider the IV DAG $Z \to T$, $T \to Y$, $U \to T$, $U \to Y$, with U unobserved, $Z \perp\!\!\!\perp U$, and no direct $Z \to Y$ edge, so that relevance, exogeneity, and exclusion all hold. This problem shows constructively that these assumptions do *not* nonparametrically identify the ATE, even with a genuinely relevant instrument, as the nonidentification remark opening Section 7.7 asserts.

(a) Construct two SEMs $\mathcal{M}_1$ and $\mathcal{M}_2$, each compatible with the IV graph, that are observationally indistinguishable but causally distinct:

$$P_{\mathcal{M}_1}(Z,T,Y) = P_{\mathcal{M}_2}(Z,T,Y), \qquad P_{\mathcal{M}_1}(Y{=}1 \mid \mathrm{do}(T{=}1)) \neq P_{\mathcal{M}_2}(Y{=}1 \mid \mathrm{do}(T{=}1)).$$

($Hint$: let $Z \sim \mathrm{Bern}(1/2)$ independently of a latent type $U \in \{\mathrm{C}, \mathrm{A}, \mathrm{N}\}$ (complier, always-taker, never-taker) with $P(\mathrm{C}) = 1/2$ and $P(\mathrm{A}) = P(\mathrm{N}) = 1/4$, and in both models set $T = Z$ for $U = \mathrm{C}$, $T = 1$ for $U = \mathrm{A}$, and $T = 0$ for $U = \mathrm{N}$, so that $P(T{=}1 \mid Z{=}1) - P(T{=}1 \mid Z{=}0) = 1/2$: the instrument is relevant. In both models set $Y = T$ for $U = \mathrm{C}$ and $Y = 1$ for $U = \mathrm{A}$; let the models differ only for $U = \mathrm{N}$, with $Y = 0$ in $\mathcal{M}_1$ and $Y = T$ in $\mathcal{M}_2$.)

(b) Verify that both models generate the same joint $P(Z,T,Y)$, then compute $P(Y{=}1 \mid \mathrm{do}(T{=}1))$ and $P(Y{=}1 \mid \mathrm{do}(T{=}0))$ in each model and show that the two ATEs differ.

(c) Verify that both models satisfy monotonicity (there are no defiers), compute the LATE $\mathbb{E}[Y(1)-Y(0) \mid U = \mathrm{C}]$ in each model, and confirm that it coincides across the two models and equals the Wald ratio computed from the common observed distribution. What does this comparison show about which causal quantity the instrument identifies without further assumptions?

Chapter 8

Mediation and Front-Door Identification

Learning Objectives

By the end of this chapter, students should be able to:

1. Explain the distinction between the total causal effect of T on Y and a pathway-specific effect that operates through a mediator M, and describe why this distinction matters scientifically.
2. Draw the prototype mediation DAG, write its structural equations, and identify the direct and indirect pathways.
3. Define the controlled direct effect (CDE) using the do-operator, identify it via the back-door formula for the joint intervention (T, M), and explain why it depends on the fixed level m.
4. Define the natural direct and indirect effects (NDE, NIE) using the potential outcomes notation $Y(t, M(t'))$, state the NDE + NIE = TE decomposition, and explain why these quantities involve cross-world counterfactuals.
5. State the sequential ignorability assumptions for identification of natural effects, write the mediation formula, and explain why a treatment-induced mediator–outcome confounder, whether measured or not, generally invalidates the cross-world independence.
6. Set up the Baron–Kenny three-equation system, derive the product and difference formulas for the indirect effect, and explain why the decomposition fails in nonlinear or interaction models.
7. State the three front-door conditions, derive the front-door formula using the do-calculus, and explain why the front-door graph enables identification despite unobserved T–Y confounding.
8. Contrast mediation analysis and instrumental variables on the dimensions of variable position, identification goal, and key assumption.

How to Read This Chapter

The chapter is long, and not all of it carries equal weight on a first pass.
Core first reading. Section 8.1–Section 8.5 (motivation, setup, TE, CDE, natural effects); Section 8.6 through the mediation formula (Theorem 8.2) and its proof; Section 8.8 through the failure examples; and the summary in Section 8.10.
Second pass. The plug-in estimation recipe in Section 8.6; the Baron–Kenny inference details (Sobel test and bootstrap) in Section 8.7; the front-door numerical example; and the mediation–IV comparison in Section 8.9.
Optional ($\star$). The starred remarks on alternative causal frameworks (NPSEM-IE vs. FFRCISTG, separable and interventional effects) and on principal stratification, and the problems marked *advanced* or *second pass*.

8.1 Motivation: Mechanisms

The identification results of Chapters 5–7 all answer the same question: *what is the total causal effect of T on Y?* Mediation analysis asks a finer question: *through what mechanism does that effect operate?*

More concretely, the total effect of T on Y may flow along multiple causal pathways. Some of this effect passes through an intermediate variable M — the *mediator* — along the path $T \to M \to Y$. The remainder flows directly along $T \to Y$, bypassing the mediator entirely. Mediation analysis aims to study mechanisms by defining direct and indirect effect concepts that target each pathway; as we will see, only some of these concepts yield an additive decomposition of the total effect.

This mechanism question matters for scientific and policy reasons. In a clinical trial of a behavioral intervention (T) on depression (Y), a researcher may want to know how much of the benefit operates through improved sleep quality (M) versus other pathways — because if sleep is the main channel, targeting sleep directly may be a more efficient intervention. In an economics study of education (T) on wages (Y), how much operates through occupation (M) versus cognitive skills? The answer determines whether a policy should target educational attainment or occupational access.

To keep the abstract formulas of this chapter tethered to a single concrete scenario, we adopt the depression intervention as a **running example**: T is a randomized behavioral intervention, M is self-reported sleep quality measured mid-trial, and Y is a depression score (e.g. on the PHQ-9 scale). We return to this example at each key estimand — TE, CDE, NDE/NIE, the Baron–Kenny decomposition — and note in Section 8.8 why the front-door criterion requires a different scenario.

The challenge is that mediators are *post-treatment variables*: they are affected by the treatment, and may themselves be confounded with the outcome. Conditioning on a post-treatment variable creates exactly the collider and selection-bias problems studied in Chapters 2 and 3. A naive approach — simply including M as a covariate in a regression of Y on T — conflates adjustment with mediation and can introduce bias even in a randomized experiment.

This chapter also develops the *front-door criterion*, a distinct identification strategy that uses the mediation structure of the DAG to identify causal effects even when treatment and outcome are confounded by an unobserved variable. This makes mediation analysis relevant not only to mechanism research but also to the core identification problem of earlier chapters.

8.2 The Mediation DAG

8.2.1 The Prototype Graph

Throughout this chapter we work with the following prototype graph.

The graph encodes two causal pathways: the **direct pathway** $T \to Y$, in which the treatment affects the outcome without passing through the mediator; and the **indirect pathway** $T \to M \to Y$, in which the treatment first shifts the mediator, which in turn shifts the outcome.

8.2.2 Structural Equations

The prototype graph corresponds to the following nonparametric structural equation model:

$$T = f_T(\mathbf{X}, U, \varepsilon_T), \tag{8.1}$$

$$M = f_M(T, \mathbf{X}, \varepsilon_M), \tag{8.2}$$

$$Y = f_Y(T, M, \mathbf{X}, U, \varepsilon_Y), \tag{8.3}$$

where each arrow corresponds to the presence of the parent in the child's structural equation. In particular, U enters both T's and Y's equations, making the confounding paths explicit; U is *absent* from M's equation, reflecting the absence of an arrow $U \to M$.

Disturbance structure. Writing a recursive system compatible with the graph is not yet a complete causal model: one must also say how the disturbances are jointly distributed. Unless stated otherwise, we adopt the NPSEM-IE reading of Equation 8.1–Equation 8.3: all shared causes represented in the model appear explicitly as variables (here $\mathbf{X}$ and U), and the equation-specific disturbances ε_T, ε_M, ε_Y are mutually independent given those explicitly represented background variables. This independent-errors assumption is innocuous for the total-effect results of Section 8.3, but it becomes load-bearing when cross-world counterfactuals appear in Section 8.5–Section 8.6.

8.2.3 What Makes Mediation Harder Than Total Effect Estimation

The key difficulty is that M is a *post-treatment* variable. This creates two interrelated problems.

Collider bias. Conditioning on M can open collider paths. Suppose $U \to T$ and $V \to M$ and $V \to Y$, with V unobserved. The path $T \to M \leftarrow V \to Y$ is blocked when M is not conditioned on, but opens as soon as M is included as a covariate, inducing spurious T–Y association through V. This is precisely why naively regressing Y on (T, M) does not isolate the direct effect.

Mediator–outcome confounding. Even when treatment is randomized — eliminating confounding on the T side — the mediator M is never randomized. An unobserved variable V with $V \to M$ and $V \to Y$ creates a back-door path from M to Y that randomization of T does not close. This is the central challenge of mediation analysis.

Post-Treatment Variables Are Dangerous to Condition On

The mediator M is caused by the treatment T. This single fact makes every operation involving M structurally different from operations involving pre-treatment covariates $\mathbf{X}$.
Conditioning on a post-treatment variable in a regression or matching procedure is *not* a neutral act. It can open collider paths that were previously blocked, introduce selection bias, and produce estimates of neither the total effect nor any well-defined direct effect. The appropriate response is not to avoid M entirely — mediation analysis requires reasoning about M — but to be precise about *which estimand* one is targeting and *which structural assumptions* justify the operation being performed.
This chapter studies three regimes in which conditioning on or marginalizing over M is justified:

1. **CDE** (Section 8.4): both T and M are intervened on simultaneously via $\mathrm{do}(T, M)$, so no conditioning on an observed M occurs.
2. **NDE/NIE** (Section 8.5–Section 8.6): M is marginalized over using a distribution from a different treatment arm, justified by sequential ignorability.
3. **Front-door** (Section 8.8): M is summed over in the front-door formula, justified by the graphical conditions of the front-door criterion.

Outside these three regimes, conditioning on a post-treatment variable should be treated as an error until proven otherwise.

8.2.4 A Working Graph for the Identification Sections

The prototype graph retains the unobserved confounder U on purpose: the chapter ultimately demonstrates, in Section 8.8, that the causal effect of T on Y can be identified even when U is hidden. In the intervening sections, however, we rely on back-door adjustment formulas that require every T–Y and T–M confounder to be captured by the observed covariate set $\mathbf{X}$.

Working Assumption (Section 8.3–Section 8.7)

Throughout Section 8.3–Section 8.7, we work with the **reduced prototype graph** shown below. Equivalently, the unobserved confounder U is assumed either (i) absent or (ii) observed and included in $\mathbf{X}$, so that $\mathbf{X}$ blocks every T–Y and T–M back-door path. The front-door section Section 8.8 restores the full prototype with U unobserved, and derives identification of the total effect without access to U.

Remark: Notation for Unobserved Confounders

Three latent variables recur in this chapter, each playing a distinct structural role: U is a *pre-treatment T–Y confounder* ($U \to T$, $U \to Y$); V is a *pre-treatment M–Y confounder* ($V \to M$, $V \to Y$); and L is a *treatment-induced M–Y confounder* ($T \to L$, $L \to M$, $L \to Y$). Keeping the three roles separate is essential: they break different identification arguments.

The Chapter's Estimands at a Glance

Estimand	**Intervention**	**Decomposes TE?**	**Main identification burden**
TE	do(T=t)	(reference)	T–Y confounding (Chs. 4–6)
CDE(m)	do(T=t), do(M=m)	No	Joint-intervention adjustment (Section 8.4)
NDE, NIE	Nested, cross-world	Yes	Sequential ignorability, incl. a cross-world independence (Section 8.5–Section 8.6)
TE via front-door	do(T=t)	Not a decomposition	Front-door graphical conditions (Section 8.8)

This matrix is the chapter's road map. A fuller version, the Framework Map, returns at the end of Section 8.7 as a synthesis.

8.3 Total Causal Effect

Before introducing direct and indirect effect concepts, we fix the reference quantity: the *total causal effect* of T on Y.

Definition: Total Effect

The **total effect** (TE) of T on Y is

$$\text{TE} = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y \mid \text{do}(T{=}1)] - \mathbb{E}[Y \mid \text{do}(T{=}0)].$$

The total effect captures the combined impact of all causal pathways from T to Y. It is identified by the back-door formula whenever $\mathbf{X}$ satisfies the back-door criterion:

$$\text{TE} = \sum_{\mathbf{x}} \big[\mathbb{E}[Y \mid T{=}1, \mathbf{X}{=}\mathbf{x}] - \mathbb{E}[Y \mid T{=}0, \mathbf{X}{=}\mathbf{x}]\big] P(\mathbf{X}{=}\mathbf{x}).$$

As throughout the book, the formula additionally presumes the intervention semantics and consistency established in Chapters 3–5, together with treatment positivity, so that both conditional means are well-defined wherever the outer average visits.

Running example. For the depression trial, the TE is the expected change in depression score if the entire study population were assigned to the intervention versus control. It combines the effect operating through sleep improvement with every other pathway by which the intervention acts on depression — cognitive restructuring, behavioral activation, therapeutic alliance, and so on. Mediation analysis asks: of this total, how much is due to sleep?

The goal of mediation analysis is to define direct and indirect effect estimands that target the separate pathways within this total. The next two sections introduce two such families, each formalizing a different notion of what "direct" and "indirect" mean; of the two, only the second — the natural direct and indirect effects — is constructed to yield an additive decomposition of the total effect.

8.4 Controlled Direct Effect

8.4.1 Definition

The do-operator provides a direct definition of the controlled direct effect: intervene on both T *and* M simultaneously, fixing M at a specified level m. Fixing $M = m$ shuts down causal pathways that pass through M; any remaining effect of T on Y operates through pathways that bypass M — in the prototype graph, precisely the direct edge $T \to Y$.

Definition: Controlled Direct Effect (CDE)

The **controlled direct effect** of changing T from 0 to 1 while fixing $M = m$ by intervention is

$$\mathrm{CDE}(m) = \mathbb{E}[Y \mid \mathrm{do}(T{=}1),\, \mathrm{do}(M{=}m)] - \mathbb{E}[Y \mid \mathrm{do}(T{=}0),\, \mathrm{do}(M{=}m)]\,. \tag{8.4}$$

Because Equation 8.4 involves two simultaneous interventions, it corresponds to the mutilated graph $\mathcal{G}_{\overline{T}\,\overline{M}}$ in which all edges into both T and M are deleted.

Running example. CDE(m) at $m =$ "poor sleep" is the expected change in depression score comparing intervention to control *if every participant's sleep quality were externally held at the poor-sleep level*, regardless of assignment. Any reduction in depression that survives this manipulation must come from non-sleep mechanisms. Whether sleep can be externally fixed in a clinical trial is a separate question — sleep is notoriously difficult to control — which is why the policy interpretation of the CDE in this scenario is strained even when the estimand is statistically identified.

Remark: The CDE Depends on m

The CDE depends on the level m at which M is fixed. In general, the direct effect may vary across values of m — this is *effect modification by the mediator*. In a linear additive model, $\mathrm{CDE}(m) = \tau'$ for all m, which is a special property of linearity. When T and M interact in their effect on Y, the CDEs at different values of m differ, and no single number summarizes the direct effect.

8.4.2 Identification of the CDE

Since the CDE involves a joint intervention $\mathrm{do}(T, M)$, its identification in this chapter's baseline-covariate setting reduces to a back-door-type adjustment applied to the pair (T, M) jointly.

Theorem 8.1 (Baseline Adjustment for the CDE)**.** *Suppose* $\mathbf{Z}$ *is a set of baseline (pre-treatment) covariates satisfying, for the joint intervention* $\mathrm{do}(T, M)$ *on* Y*: (i)* $\mathbf{Z}$ *contains no descendant of* T *or* M*, and (ii)* $\mathbf{Z}$ *d-separates the joint exposure set* $\{T, M\}$ *from* Y *in* $\mathcal{G}_{\underline{T}\,\underline{M}}$*, the graph obtained by deleting all arrows out of* T *and out of* M*. Suppose further that (iii)* ***joint positivity*** *holds at* (t, m)*:*

$$P(T{=}t,\ M{=}m \mid \mathbf{Z}{=}\mathbf{z}) > 0 \quad \textit{for } P\textit{-almost every } \mathbf{z} \textit{ in the support of } \mathbf{Z}.$$

Then

$$\mathbb{E}[Y \mid \mathrm{do}(T{=}t),\, \mathrm{do}(M{=}m)] = \sum_{\mathbf{z}} \mathbb{E}[Y \mid t,\, m,\, \mathbf{z}]\, P(\mathbf{z}). \tag{8.5}$$

Condition (iii) ensures that the conditional expectation is defined at every $\mathbf{z}$ that receives positive weight under $P(\mathbf{z})$. Joint positivity is the CDE counterpart of the overlap conditions (P1)–(P2) introduced for the mediation formula in Section 8.6. Even when T is randomized, (iii) is not automatic: it additionally requires that every level m of the mediator has positive conditional probability given the treatment arm and $\mathbf{Z}$. When some $(t, m, \mathbf{z})$ cells are never observed, CDE(m) is not empirically identified at that m.

Formula Equation 8.5 is the standard back-door adjustment formula applied to the pair (T, M) as a joint treatment. Under the working assumption of Section 8.2.4, $\mathbf{Z} = \mathbf{X}$ satisfies condition (ii): deleting the arrows out of T and M removes $T \to M$, $T \to Y$, and $M \to Y$, and the only remaining paths from $\{T, M\}$ to Y, namely $T \leftarrow \mathbf{X} \to Y$ and $M \leftarrow \mathbf{X} \to Y$, are blocked by $\mathbf{X}$.

Graph surgery and back-door adjustment are distinct steps. It is worth pausing on what the theorem does and does not say, because the two steps of identification can be conflated. Graph surgery *defines* the interventional target by deleting arrows into T and M; this specifies what we want to compute.

Expressing that target as a functional of the observed distribution is a separate step that requires a valid adjustment set $\mathbf{Z}$ in the *original* graph. The mutilated graph does not by itself justify the right-hand side of Equation 8.5. Consider what happens if we set aside the working assumption and return to the full prototype with U unobserved: the back-door path $T \leftarrow U \rightarrow Y$ is present in the original graph and is not blocked by any observed covariate, so $\mathbf{X}$ alone is *not* a valid adjustment set. Adjustment identifies the CDE in that setting only when $\mathbf{Z}$ actually blocks the U-path. When U remains unobserved, alternative strategies are required (e.g. the front-door formula of Section 8.8, or an instrumental variable as in Chapter 7).

Remark: Scope of the CDE Theorem — Sufficient, Not Necessary

Theorem 8.1 is a *baseline-adjustment* result: it treats (T, M) as a single joint exposure and standardizes over a covariate set that precedes both. In the point-treatment mediation graphs of this chapter, condition (ii) can equivalently be checked sequentially: $\mathbf{Z}$ blocks all back-door paths from T to Y, and $(T, \mathbf{Z})$ blocks all back-door paths from M to Y. The theorem's conditions are sufficient for this setting but stronger than necessary for time-ordered interventions in general. In particular, when a variable L with $T \rightarrow L$, $L \rightarrow M$, and $L \rightarrow Y$ is observed, L is a legitimate confounder for the later intervention on M even though it is a descendant of T; condition (i) excludes it, yet the CDE remains identified by the *sequential g-formula*, which adjusts for L at the second stage without treating it as a baseline covariate. Time-ordered joint interventions of this kind require the longitudinal g-formula and the dynamic treatment regime machinery treated later in the book.

8.4.3 Physical Manipulability and the Meaning of the CDE

The CDE answers the question: what is the effect of T on Y when the mediator is *prevented* from changing? This question has a clean causal answer precisely because $\mathrm{do}(M{=}m)$ is a genuine intervention: it severs all edges into M, placing M at m regardless of what T does.

Remark: Well-Defined Interventions on the Mediator

The CDE has its clearest interpretation when "set $M = m$" corresponds to a *sufficiently well-defined intervention* — or to a clearly specified class of interventions with equivalent effects on Y. Literal physical control is neither strictly necessary nor sufficient for this: hypothetical or stochastic interventions with clear consistency semantics can define a coherent CDE, while a physically performable manipulation may still admit multiple "versions" of $M = m$ with different effects on Y, leaving the estimand ambiguous. Settings where the intervention is well-defined include a drug trial where both the assigned treatment and a downstream biomarker can be externally controlled. In many substantive settings, however, no well-defined independent manipulation of the mediator is available:

- In an education study, one cannot intervene to hold occupation fixed while varying years of schooling.
- In a behavioral trial, one cannot set sleep quality to a predetermined level while varying the intervention.

When no sufficiently well-defined intervention on the mediator can be articulated, the symbols $Y(t, m)$ and $\mathrm{do}(M{=}m)$ can still be manipulated formally, but the causal estimand itself is ambiguous: different interventions producing the same recorded mediator value may have different effects on Y, so "the" CDE at level m is not a single well-posed quantity, and consistency has no unique referent. In those cases the natural direct effect (Section 8.5), which asks what would happen if M were allowed to take its natural value under the reference treatment, avoids specifying an external mediator level; note, however, that it does not resolve the underlying ambiguity, since the nested counterfactual $Y(t, M(0))$ presupposes the same well-defined mediator input — and it comes at the cost of stronger identification assumptions.

8.4.4 The CDE Does Not Have a Complementary Indirect Effect

A common misconception is that the CDE and some complementary "controlled indirect effect" sum to the total effect, analogous to the NDE + NIE = TE decomposition. This is not generally true.

Fixing $M = m$ by intervention removes treatment-induced variation in the mediator, but the residual quantity $\mathrm{TE} - \mathrm{CDE}(m)$ is not a controlled *indirect* effect. It is a well-defined causal contrast — every term below is an interventional mean — but it depends on the selected mediator level m, it does not correspond to any intervention that isolates only the indirect pathway, and it cannot be interpreted as the effect transmitted through M in a complementary pathway decomposition. Formally:

$$\begin{aligned} \mathrm{TE} - \mathrm{CDE}(m) = &\Big(\mathbb{E}[Y \mid \mathrm{do}(T{=}1)] - \mathbb{E}[Y \mid \mathrm{do}(T{=}1),\, \mathrm{do}(M{=}m)]\Big) \\ &- \Big(\mathbb{E}[Y \mid \mathrm{do}(T{=}0)] - \mathbb{E}[Y \mid \mathrm{do}(T{=}0),\, \mathrm{do}(M{=}m)]\Big), \end{aligned} \tag{8.6}$$

which mixes the effect of intervening on M under $T = 1$ and $T = 0$ in a way that has no simple pathway interpretation. The decomposition $\mathrm{TE} = \mathrm{NDE} + \mathrm{NIE}$ holds because the natural effects involve cross-world counterfactuals that are constructed precisely to be complementary; the CDE has no analogous cross-world partner.

The practical implication is that if the research goal is to decompose the total effect into direct and indirect components, the correct estimands are the NDE and NIE, not $\mathrm{CDE}(m)$ and $\mathrm{TE} - \mathrm{CDE}(m)$. The CDE is the right estimand when the question is specifically about the effect of T with M controlled at a given level — a different and more limited question.

Natural Effects Are a Strictly Harder Estimand Than the CDE

The step from the CDE to the NDE and NIE is not merely a notational upgrade — it is a conceptual escalation with serious identification consequences.
The CDE is a single-world estimand. The quantity $\mathbb{E}[Y \mid \mathrm{do}(T{=}t), \mathrm{do}(M{=}m)]$ involves a single joint intervention. Every individual lives in one hypothetical world where both T and M are externally fixed. The do-operator handles this directly, and identification reduces to the standard back-door criterion for the pair (T, M).
The NDE and NIE are cross-world estimands. The nested counterfactual $Y(t, M(t'))$ requires an individual to simultaneously inhabit two worlds: the world where $T = t$ (which determines Y) and the world where $T = t'$ (which determines what M would *naturally* be). When $t \neq t'$, these are logically distinct states of the world. No single experimental intervention can realize both at once for the same unit. The do-operator alone cannot express this quantity.
The identification price is steep. The CDE requires a back-door-type adjustment condition (plus positivity and consistency) — assumptions about observed confounding that can in principle be satisfied by design. The NDE and NIE require the sequential ignorability assumptions of Section 8.6, including a genuinely *cross-world* independence (Assumption 3 there), which cannot be satisfied by randomizing T and cannot be verified from data. Its structural sufficient condition — no treatment-induced mediator–outcome confounder — must be defended on subject-matter grounds.
Practical guidance. When the goal is a pathway-specific effect and the no-treatment-induced-confounder assumption is plausible, NDE and NIE are the right estimands. When that assumption is suspect, the CDE — or a sensitivity analysis around it — is the safer choice.

8.5 Natural Direct and Indirect Effects

8.5.1 Motivation: Cross-World Counterfactuals

The CDE fixes the mediator by external intervention. A more scientifically natural question is: what is the effect of T on Y that bypasses M when M is held at the value it *would naturally take* under the reference treatment $T = 0$?

Answering this requires a comparison across two intervention worlds: the world where $T = 1$ but M is held at the value it would naturally take under $T = 0$, versus the world where $T = 0$. The first of these cannot be realized by any single experiment on the same unit — T would have to be set to 1 (to determine Y) and to 0 (to fix M at its natural $T = 0$ value) simultaneously. It requires the *nested potential outcomes* notation $Y(t, M(t'))$, which denotes the outcome that would be observed if T were set to t and M were simultaneously set to the value it would naturally take if T were t'. These are called *cross-world counterfactuals.*

8.5.2 Definitions

Definition: Natural Direct and Indirect Effects [@pearl2001direct]

For binary $T \in \{0, 1\}$:

$$\text{NDE} = \mathbb{E}[Y(1, M(0)) - Y(0, M(0))]\,, \tag{8.7}$$

$$\text{NIE} = \mathbb{E}[Y(1, M(1)) - Y(1, M(0))]\,. \tag{8.8}$$

The **natural direct effect** (NDE) is the expected change in the outcome when T shifts from 0 to 1, holding the mediator at the value it would naturally take under $T = 0$. The **natural indirect effect** (NIE) is the expected change in the outcome due to the shift in the mediator from $M(0)$ to $M(1)$, holding $T = 1$ fixed.

Running example. The NDE is the expected reduction in depression score when the intervention is delivered but each participant's sleep is held at the value it *would have had* without the intervention — the part of the benefit that comes from cognitive, behavioral, or therapeutic-alliance pathways, not from sleep. The NIE is the complementary piece. By construction NDE + NIE = TE. One caveat carries over from the CDE discussion: the nested counterfactual $Y(t, m)$ presupposes that the mediator value is a well-defined causal input. If two different ways of arriving at the same sleep score would have different effects on depression, then "holding sleep at the value it would have had without the intervention" is ambiguous, and the NDE inherits that ambiguity. Throughout, we read $Y(t, m)$ structurally, as the output of the equation f_Y in Equation 8.3 evaluated at (t, m).

Decomposition. The total effect decomposes as

$$\text{TE} = \text{NDE} + \text{NIE}. \tag{8.9}$$

Proof.

$$\begin{aligned} \text{NDE} + \text{NIE} &= \mathbb{E}[Y(1, M(0)) - Y(0, M(0))] + \mathbb{E}[Y(1, M(1)) - Y(1, M(0))] \\ &= \mathbb{E}[Y(1, M(1)) - Y(0, M(0))] = \mathbb{E}[Y(1) - Y(0)] = \text{TE}. \qquad \square \end{aligned}$$

The final equality is not pure algebra: it uses *composition*, $Y(t, M(t)) = Y(t)$, which holds under the recursive structural interpretation adopted in Section 8.2 — setting $T = t$ and simultaneously setting M to the very value it would take under $T = t$ reproduces the single intervention do($T{=}t$).

Remark: Alternative Natural-Effects Decomposition

The decomposition Equation 8.9 is not the only one. The definition above fixes M at $M(0)$ in the NDE (the "pure" or "reference" direct effect) and contrasts $M(1)$ vs. $M(0)$ while holding $T = 1$ in the NIE (the "total" indirect effect). The mirror choice fixes M at $M(1)$ in the direct effect:

$$\text{NDE}^\star = \mathbb{E}[Y(1, M(1)) - Y(0, M(1))] \quad \text{("total direct")},$$

$$\text{NIE}^\star = \mathbb{E}[Y(0, M(1)) - Y(0, M(0))] \quad \text{("pure indirect")}.$$

This pair also sums to the total effect. The choice between the two decompositions is conventional, and the two agree when there is no $T \times M$ interaction in the structural model; when interaction is present, they generally differ, and the gap $\text{NDE} - \text{NDE}^\star = \text{NIE}^\star - \text{NIE}$ quantifies the discrepancy. In this textbook we use the Equation 8.9 decomposition throughout.

8.5.3 Interpreting the Cross-World Nature

The NDE involves the counterfactual $Y(1, M(0))$: the outcome when T is set to 1 but M is held at the value it would naturally take under $T = 0$. This is a *cross-world* quantity because the two coordinates of the argument refer to different intervention worlds. It cannot be observed for any individual, and in general it cannot be written as a do-expression.

This is not merely a philosophical subtlety. The cross-world nature has direct implications for identification: as shown in the next section, identifying the NDE and NIE requires assumptions that are strictly stronger than those needed for the CDE or the total effect.

Remark: CDE and NDE Coincide Only Without Interaction

In the linear additive Baron–Kenny SEM of Section 8.7 with no $T \times M$ interaction, the CDE and NDE coincide for all m: NDE $= \tau'$ and NIE $= ab$. The Baron–Kenny framework therefore implicitly satisfies the cross-world independence condition needed for this equality. When a $T \times M$ interaction δTM is added, the two concepts separate: $\text{CDE}(m) = \tau' + \delta m$ varies with m, while $\text{NDE} = \tau' + \delta\, \mathbb{E}[M(0)]$ is a single number. They then agree only at the one level $m = \mathbb{E}[M(0)]$, and in general answer different scientific questions (Problem 3 works this out).

Remark ($\star$ Advanced/Optional): NPSEM-IE vs. FFRCISTG and Alternatives

This remark compares causal frameworks and surveys alternatives to natural effects; it may be skipped on first reading. The main identification argument for the NDE and NIE does not depend on resolving the NPSEM-IE/FFRCISTG distinction.

The status of the cross-world counterfactual $Y(t', M(t))$ depends on the causal framework adopted, and this has direct consequences for the identification of the NDE and NIE.

Under Pearl's NPSEM-IE framework (Chapter 1), all relevant counterfactuals are jointly defined as functions of the same exogenous errors. With mutually independent errors and the graphical no-confounding restrictions, the cross-world independence used by the mediation formula (Assumption 3 of Section 8.6) follows. In particular, $Y(t', M(t))$ is a well-defined random variable for $t \neq t'$, and the NDE and NIE are identified under sequential ignorability.

Under the Finest Fully Randomized Causally Interpretable Structured Tree Graph (FFRCISTG) framework of Robins (1986) — developed graphically via single world intervention graphs by Richardson and Robins (2014) — the situation is different. An FFRCISTG is a *single-world* model: its assumptions concern counterfactuals within one intervention world at a time, and are, in principle, the assumptions that randomized experiments can enforce.

The key point is not primarily notational. It is that the single-world independences do *not* imply the cross-world independence $Y(t, m) \perp\!\!\!\perp M(t') \mid \mathbf{X}$: two models can agree on every single-world quantity and disagree on the joint law of $(Y(t, m), M(t'))$, and no experiment can distinguish them. Consequently, under FFRCISTG assumptions alone the NDE and NIE are not identified. Identification of natural effects requires additional cross-world structure, such as that supplied by an NPSEM-IE.

Two single-world alternatives deserve mention. First, *separable effects* (Stensrud et al. 2022, 2023) apply when the treatment can meaningfully be decomposed into distinct components — say T_M, acting on Y only through M, and T_Y, acting only directly — that could in principle be intervened on separately, together with dismissible-component (isolation) conditions on the graph. Separable effects answer a different scientific question, and they are not available merely by relabeling an ordinary treatment. In the canonical point-treatment setting, under full isolation, their identifying functional coincides with the mediation formula — another instance of the same-formula, different-interpretation principle encountered in the IV context (Chapter 7) — but in richer settings the formulas and assumptions diverge.

Second, *interventional* (also called *randomized*) direct and indirect effects replace the individual-level nested counterfactual $M(t')$ by a random draw from the conditional distribution of $M(t')$ given $\mathbf{X}$ (VanderWeele et al. 2014; Vansteelandt and Daniel 2017). Because no unit is required to inhabit two worlds at once, these estimands can be identified under single-world exchangeability assumptions — essentially Assumptions 1–2 of Section 8.6, without the cross-world Assumption 3 — and they remain well-defined even in the presence of a treatment-induced mediator–outcome confounder L. The price is that they decompose an "overall" effect that in general differs from the total effect, and they describe population-level mediator shifts rather than individual-level mechanisms.

For the purposes of this course, we work within the NPSEM-IE framework, where the NDE and NIE are well-defined and identified under sequential ignorability. Students who pursue the single-world literature will find that the estimation machinery of Part III transfers, since the observed-data functionals are the same or closely analogous.

8.6 Identification of Natural Effects

8.6.1 Sequential Ignorability

Identification of the NDE and NIE requires the conditions known collectively as *sequential ignorability* (Imai et al. 2010), together with consistency, composition, and positivity. We state sequential ignorability as three assumptions of two logically different kinds: Assumptions 1 and 2 below are *observed-world* statements — conditional independencies of the type met in Chapters 4–6 — while Assumption 3 is a *cross-world* statement relating counterfactuals from two different intervention worlds.

Despite the neutral-sounding name, these conditions are substantially stronger than the ignorability assumptions used for the total effect or the CDE. Assumptions 2 and 3 govern the mediator–outcome relationship, which is never directly controlled by the investigator; Assumption 3, moreover, is untestable *in principle*: no experiment, however elaborate, can place one unit in two intervention worlds at once. Students who have internalized the back-door criterion should resist the temptation to read sequential ignorability as a mild extension of it: the assumptions address different sub-problems, and the hardest one has nothing to do with how T was assigned.

Sequential Ignorability Assumptions

For all t, t', m:

1. **Treatment ignorability (joint form).**

$$\{Y(t', m),\, M(t)\} \;\perp\!\!\!\perp\; T \,\Big|\, \mathbf{X}.$$

Graphically: $\mathbf{X}$ blocks all back-door paths from T to Y and from T to M. Taking the $M(t)$ marginal gives the treatment–mediator component $M(t) \perp\!\!\!\perp T \mid \mathbf{X}$, which Step 5 of the mediation-formula proof uses directly; it is a consequence of the joint statement, not a separate assumption.

2. **Mediator–outcome ignorability given treatment.**

$$Y(t', m) \;\perp\!\!\!\perp\; M \,\Big|\, T, \mathbf{X}.$$

Graphically: $(T, \mathbf{X})$ blocks all back-door paths from M to Y.

3. **Cross-world independence.**

$$Y(t, m) \;\perp\!\!\!\perp\; M(t') \,\Big|\, \mathbf{X}, \quad \text{including } t \neq t'.$$

Unlike Assumptions 1–2, this statement pairs counterfactuals from two *different* intervention worlds. It has no observed-world analogue and cannot be enforced or tested by any experimental design.

In addition, the observed data are linked to the counterfactuals by **consistency** ($Y = Y(t, m)$ on $\{T{=}t, M{=}m\}$ and $M = M(t)$ on $\{T{=}t\}$) and **composition** ($Y(t, M(t)) = Y(t)$), both automatic under the recursive structural reading of Section 8.2.

Where Does the Cross-World Assumption Come From?

Assumption 3 does not come for free, but the NPSEM-IE structural model of Section 8.2 supplies a sufficient condition for it. Suppose the working graph and disturbance structure satisfy:

(i) $\mathbf{X}$ contains the pre-treatment common causes of M and Y represented in the model;

(ii) there is **no treatment-induced mediator–outcome confounder**: no variable L with $T \to L$, $L \to M$, and $L \to Y$;

(iii) the equation-specific disturbances are mutually independent given the explicitly represented background variables.

Then, conditional on $\mathbf{X}$, the counterfactuals $M(t') = f_M(t', \mathbf{X}, \varepsilon_M)$ and $Y(t, m) = f_Y(t, m, \mathbf{X}, U, \varepsilon_Y)$ are functions of *disjoint* disturbances (recall U does not enter f_M in the working graph, and any such U is absorbed into $\mathbf{X}$), so independent disturbances deliver $Y(t, m) \perp\!\!\!\perp M(t') \mid \mathbf{X}$.

If a treatment-induced confounder L exists, then $M(t')$ and $Y(t, m)$ both inherit dependence on L's disturbance, and the cross-world independence *generically* fails — whether or not L is observed. (As with d-connection throughout the book, a displayed path licenses dependence generically, not necessarily: a structural equation could ignore its L argument, or effects could cancel. What the presence of L destroys is the *sufficient condition*; the independence then holds only by such non-generic coincidence.) Measuring L does not rescue the mediation formula, because L is a descendant of T and cannot be adjusted for as a baseline covariate. In this sense the familiar "no treatment-induced mediator–outcome confounder" condition is the graphical face of Assumption 3, not a freestanding fourth assumption.

Positivity (Overlap) Conditions

The sequential ignorability assumptions above are conditional independence statements. A separate requirement — necessary for the mediation formula to be empirically identified — is overlap: every $(t, m, \mathbf{x})$ cell that enters the formula must occur with positive probability in the observed distribution.
(P1) Treatment overlap. $P(T = t \mid \mathbf{X} = \mathbf{x}) > 0$ for all $t \in \{0, 1\}$ and for P-almost every $\mathbf{x}$.
(P2) Mediator overlap across treatment arms. $P(M = m \mid T = t, \mathbf{X} = \mathbf{x}) > 0$ whenever $P(M = m \mid T = t', \mathbf{X} = \mathbf{x}) > 0$, for the pairs (t, t') appearing in the mediation formula and for P-almost every $\mathbf{x}$.
Condition (P1) ensures that $\mathbb{E}[Y \mid T = t, \mathbf{X}]$ and $P(M \mid T = t, \mathbf{X})$ are defined at every $\mathbf{x}$ that the outer average visits. Condition (P2) ensures that the outcome regression $\mathbb{E}[Y \mid T = t, M = m, \mathbf{X} = \mathbf{x}]$ is defined at every m that receives positive weight under the mediator distribution $P(M \mid T = t', \mathbf{X} = \mathbf{x})$. Conceptually, (P2) is a *support-inclusion* requirement,

$$\operatorname{supp}\{M \mid T{=}t', \mathbf{X}{=}\mathbf{x}\} \;\subseteq\; \operatorname{supp}\{M \mid T{=}t, \mathbf{X}{=}\mathbf{x}\},$$

where t is the outcome-regression arm and t' the mediator-distribution arm. For continuous M, positive point probabilities are the wrong currency; the requirement becomes conditional-support inclusion together with density regularity, and Theorem 8.2 below is accordingly stated for discrete M and $\mathbf{X}$. Randomization of T secures (P1) by design, but (P2) is a data-dependent requirement about the mediator's conditional distribution that no aspect of the experimental assignment guarantees.

Assumptions 1–2 are the natural extensions of the Baron–Kenny conditions of Section 8.7 to the potential outcomes setting: observed-world exchangeability for the treatment and for the mediator. Assumption 3 is the genuinely new requirement, and the second box explains its structural pedigree: it holds under the NPSEM-IE when, in addition to the pre-treatment confounding control already demanded by Assumptions 1–2, there is no treatment-induced mediator–outcome confounder L.

What randomization of T does and does not provide. Randomizing the treatment T satisfies Assumption 1 (both its T–Y and T–M components) by design. It does *not* satisfy Assumption 2 or Assumption 3. The mediator M is a post-treatment variable that is never randomized; any unobserved variable V with $V \to M$ and $V \to Y$ violates Assumption 2 regardless of how T was assigned. The structural condition behind Assumption 3 is even more demanding: it can be defeated by a variable L that is itself *caused* by the treatment, so randomizing T actually creates the conditions under which such violations can arise. The upshot is that a randomized experiment identifies the total effect and the first-stage $T \to M$ effect cleanly, but it does *not* by itself identify the second-stage $M \to Y$ effect, and therefore it does not by itself identify the NDE or NIE.

Randomization of T Does Not Secure the Cross-World Assumption

Unlike unmeasured T–Y confounders, which randomization of T can often eliminate, a treatment-induced mediator–outcome confounder L is itself caused by the treatment: the path $T \to L$ is set in motion by the intervention, so no aspect of how T is assigned closes the door on this violation of Assumption 3. The absence of such an L must instead be argued from design, timing, measurement, or subject-matter knowledge; when that argument is not credible, natural effects are fragile. This is why identification of natural effects is considered strictly harder than identification of the total effect or the CDE.

8.6.2 The Mediation Formula

Under the sequential ignorability assumptions, consistency and composition, and the positivity conditions (P1)–(P2), the cross-world expectation $\mathbb{E}[Y(t, M(t'))]$ is identified from observational data.

Theorem 8.2 (Mediation Formula (Pearl 2001))**.** *Under sequential ignorability (Assumptions 1–3), consistency and composition, positivity (P1)–(P2), and with discrete M and $\mathbf{X}$:*

$$\mathbb{E}[Y(t, M(t'))] = \sum_m \sum_{\mathbf{x}} \mathbb{E}[Y \mid T{=}t,\ M{=}m,\ \mathbf{X}{=}\mathbf{x}]\, P(M{=}m \mid T{=}t',\ \mathbf{X}{=}\mathbf{x})\, P(\mathbf{X}{=}\mathbf{x}). \tag{8.10}$$

Proof sketch. We derive the formula in five labeled steps, each invoking a specific identification assumption.

Step 1 (law of total expectation and composition). Condition on $\mathbf{X}$ and marginalize over $M(t')$:

$$\mathbb{E}[Y(t, M(t'))] = \sum_{m,\mathbf{x}} \mathbb{E}[Y(t,m) \mid M(t'){=}m,\ \mathbf{X}{=}\mathbf{x}]\, P(M(t'){=}m \mid \mathbf{X}{=}\mathbf{x})\, P(\mathbf{X}{=}\mathbf{x}),$$

using the composition axiom $Y(t, M(t')) = Y(t,m)$ on the event $\{M(t') = m\}$.

Step 2 (cross-world independence supplied by the causal model). Invoke Assumption 3: $Y(t,m) \perp\!\!\!\perp M(t') \mid \mathbf{X}$. This is the step where the structural model earns its keep. Under an NPSEM-IE compatible with the working graph, the counterfactuals $M(t')$ and $Y(t,m)$ are functions of disjoint disturbances conditional on $\mathbf{X}$. It is a cross-world restriction: strictly stronger than anything randomization of T provides, and not testable from the observed-data distribution. With the cross-world independence in hand, the conditioning on $M(t')$ in Step 1 can be dropped:

$$\mathbb{E}[Y(t,m) \mid M(t'){=}m,\ \mathbf{X}{=}\mathbf{x}] = \mathbb{E}[Y(t,m) \mid \mathbf{X}{=}\mathbf{x}].$$

This is the move that distinguishes the mediation formula from an ordinary back-door adjustment: the two counterfactual worlds must be decoupled before the observed-data substitutions that follow.

Step 3 (Assumptions 1 and 2). By $Y(t,m) \perp\!\!\!\perp T \mid \mathbf{X}$ (the $Y(t,m)$ marginal of Assumption 1) and $Y(t,m) \perp\!\!\!\perp M \mid T, \mathbf{X}$ (Assumption 2), conditioning on the observables T and M leaves the potential-outcome mean unchanged:

$$\mathbb{E}[Y(t,m) \mid \mathbf{X}{=}\mathbf{x}] = \mathbb{E}[Y(t,m) \mid T{=}t,\ M{=}m,\ \mathbf{X}{=}\mathbf{x}].$$

Step 4 (consistency for Y). On the event $\{T = t,\ M = m\}$, $Y(t,m) = Y$, and so the left side equals $\mathbb{E}[Y \mid T{=}t, M{=}m, \mathbf{X}{=}\mathbf{x}]$.

Step 5 (treatment–mediator component of Assumption 1, and consistency for M). Rewrite the mediator distribution using $M(t') \perp\!\!\!\perp T \mid \mathbf{X}$ and consistency ($M(t') = M$ when $T = t'$):

$$P(M(t'){=}m \mid \mathbf{X}{=}\mathbf{x}) = P(M{=}m \mid T{=}t',\ \mathbf{X}{=}\mathbf{x}).$$

Substituting Steps 2–5 into Step 1 yields Equation 8.10. The positivity conditions ensure that every conditional expectation and every conditional distribution on the right-hand side is well-defined at each cell that receives positive weight. □

Interpretation. The mediation formula "mixes" the outcome regression under $T = t$ with the mediator distribution under $T = t'$. To compute the NDE, set $t = 1$ and $t' = 0$: take the conditional mean of Y evaluated at treatment 1, but weight the mediator by its distribution under treatment 0. This counterfactual reweighting is what makes the formula non-trivial. Unlike an ordinary back-door adjustment, which is a functional of a single treatment arm, the mediation formula combines two arms, and the bridge between them is the cross-world independence used in Step 2. Whenever a student is tempted to read Equation 8.10 as a routine standardization formula, the presence of the second treatment index t' on the right-hand side is the visible trace of that cross-world step.

The NDE and NIE from the formula. Applying Equation 8.10 with $(t, t') = (1, 0)$ and $(0, 0)$:

$$\begin{aligned} \text{NDE} &= \sum_{m,\mathbf{x}} \left[\mathbb{E}[Y \mid T{=}1, M{=}m, \mathbf{X}{=}\mathbf{x}] - \mathbb{E}[Y \mid T{=}0, M{=}m, \mathbf{X}{=}\mathbf{x}]\right] \\ &\qquad \times P(M{=}m \mid T{=}0, \mathbf{X}{=}\mathbf{x})\, P(\mathbf{X}{=}\mathbf{x}), \\ \text{NIE} &= \sum_{m,\mathbf{x}} \mathbb{E}[Y \mid T{=}1, M{=}m, \mathbf{X}{=}\mathbf{x}] \\ &\qquad \times \left[P(M{=}m \mid T{=}1, \mathbf{X}{=}\mathbf{x}) - P(M{=}m \mid T{=}0, \mathbf{X}{=}\mathbf{x})\right] P(\mathbf{X}{=}\mathbf{x}). \end{aligned}$$

Estimation Recipe: Plug-In for the Mediation Formula

The mediation formula identifies $\mathbb{E}[Y(t, M(t'))]$ as a functional of the observed distribution, but it is not itself an estimator. Given data $\{(Y_i, T_i, M_i, \mathbf{X}_i)\}_{i=1}^n$, a plug-in estimator requires two working models and an averaging step:

1. **Fit an outcome model.** Regress Y on $(T, M, \mathbf{X})$ to obtain $\hat{\mu}(t, m, \mathbf{x})$.
2. **Fit a mediator model.** Regress M on $(T, \mathbf{X})$ to obtain $\hat{p}(m \mid t, \mathbf{x})$. For continuous M this may be a conditional density or a conditional sampler.
3. **Predict outcomes at the evaluation arm.** For each unit i and each mediator value m, compute $\hat{\mu}(t, m, \mathbf{X}_i)$ at the *evaluation* treatment level t (the first argument of $Y(t, M(t'))$).
4. **Reweight or simulate the mediator at the reference arm.** The cross-world ingredient pairs outcome predictions at t with the mediator distribution at the *reference* level t'. For discrete M, weight $\hat{\mu}(t, m, \mathbf{X}_i)$ by $\hat{p}(m \mid t', \mathbf{X}_i)$; for continuous M, draw Monte Carlo samples and average.
5. **Average over the empirical distribution of X.** The plug-in estimate is
$$\frac{1}{n}\sum_{i=1}^{n}\sum_{m}\hat{\mu}(t, m, \mathbf{X}_i)\,\hat{p}(m \mid t', \mathbf{X}_i), \tag{8.11}$$
with the Monte Carlo analogue for continuous M. The NDE and NIE are then differences of such terms at $(t, t') \in \{(1, 0),\ (0, 0),\ (1, 1)\}$.
6. **Quantify uncertainty.** Under correctly specified *regular* parametric mediator and outcome models and the usual smoothness and nonsingularity conditions, a nonparametric bootstrap over units provides asymptotically valid uncertainty quantification; boundary parameters and sparse cells can invalidate the ordinary bootstrap approximation. The influence-function methodology of Chapters 10–11 extends to mediation functionals, but the efficient influence function and the robustness structure are parameter-specific: the mediation functional has its own nuisance components and, depending on the parameterization, a *multiply* rather than doubly robust structure, so the ATE formulas of those chapters cannot simply be applied as-is.

Caveats. The plug-in estimator Equation 8.11 is consistent only when both $\hat{\mu}$ and $\hat{p}$ are correctly specified; misspecification in either produces bias. Identification is a separate matter from estimation: a correct mediation formula does not salvage a misspecified regression.

Remark: Three Cautions Before Applying the Mediation Formula

1. **Randomization handles one of the three assumptions, not all three.** Randomizing T secures Assumption 1 by design. It leaves Assumptions 2 and 3 entirely open: whether there is unmeasured M–Y confounding, and whether the treatment has induced a variable that confounds M and Y, are questions that the experimental design does not answer.
2. **The formula looks like routine adjustment but is not.** The expression $\sum_m \mathbb{E}[Y \mid t, m, \mathbf{x}]\, P(M{=}m \mid t', \mathbf{x})$ resembles a standardization formula, but it is computing a cross-world expectation $\mathbb{E}[Y(t, M(t'))]$, not a do-expression. Its validity rests on Assumptions 2 and 3; the latter cannot be read off the observed-data distribution or verified by any experiment.
3. **Sequential ignorability is an untestable assumption bundle.** Unlike treatment ignorability, which can sometimes be made credible by design or by a rich covariate set, the mediator–outcome ignorability in Assumption 2 and the cross-world independence in Assumption 3 must be defended on subject-matter grounds in every application. When that defense is weak, the CDE — which requires only a back-door-type adjustment condition — is the more credible estimand.

Remark (⋆ Advanced/Optional): Principal Stratification and SACE

This remark introduces an alternative decomposition framework used in clinical trials with truncation by death; it may be skipped on first reading.

The NDE and NIE decompose the total effect by causal pathway. An alternative approach — particularly relevant when the mediator is an *event* that can preclude observation of the outcome,

such as death — is *principal stratification* (Frangakis and Rubin 2002).
In this framework, units are classified by their potential mediator values under each treatment arm: the *principal stratum* $\{M(1) = m_1, M(0) = m_0\}$ is the subpopulation that would experience mediator level m_1 under $T = 1$ and m_0 under $T = 0$. This classification is latent — just as compliance types in the IV framework (Chapter 7) are unobservable — but it is well-defined within the potential outcomes language.
The leading application is **truncation by death**: when the mediator M is a survival indicator, units with $M(1) = M(0) = 0$ cannot be observed for any outcome Y. The estimand of interest is then the **Survival Average Causal Effect** (SACE):

$$\text{SACE} = \mathbb{E}\big[Y(1) - Y(0) \mid M(1) = M(0) = 1\big],$$

the average treatment effect among the stratum of units who would survive under both treatments. This estimand is analogous to the LATE in the IV framework: just as LATE conditions on the complier stratum, the SACE conditions on the always-survivor stratum.
Identification of the SACE is more delicate than identification of the total effect. Because the always-survivor stratum is a latent subpopulation whose membership is not observed for any unit, randomization of T alone does not point-identify the SACE. A monotonicity assumption — $M(1) \geq M(0)$ for all units — partitions the population into always-survivors, treatment-induced survivors, and never-survivors, and yields informative bounds on the SACE; it does not on its own deliver point identification. Further structure is needed, and several strategies are used: principal ignorability of the outcome given covariates, exclusion-style restrictions on stratum-specific effects, parametric models within strata, or sensitivity analyses. Unlike LATE, where a short standard list of IV assumptions point-identifies the stratum-specific effect, there is no single canonical bundle that does the same for the SACE. In every case, the SACE is not identified by the mediation formula: it requires separate identification arguments tailored to the principal stratification structure.
The principal stratification framework is conceptually distinct from the NDE/NIE framework. Where NDE and NIE decompose the *pathway* of the effect, the SACE asks about the effect within a *subpopulation* defined by potential mediator values.

Identification versus estimation. The last three sections have been entirely about causal estimands and their identification: what quantities are we trying to learn, and under what structural assumptions are they expressible as functionals of the observed data distribution? No specific model for $P(Y \mid T, M, \mathbf{X})$ or $P(M \mid T, \mathbf{X})$ was required. Section 8.7 turns to a parametric linear model that delivers familiar closed-form estimators under *additional modeling assumptions* — linearity, additivity, and no $T \times M$ interaction — that go beyond the identification conditions above. Fitting the regression system does not solve the identification problem; it assumes it has already been solved.

8.7 The Linear Mediation Model: A Historical Special Case

The framework developed above — potential outcomes, cross-world counterfactuals, sequential ignorability — is the modern conceptual foundation of mediation analysis. Before that framework existed, practitioners used a simpler regression-based approach that works cleanly under linearity and no interaction. That approach, due to Baron and Kenny (1986), was enormously influential and remains widely cited. We study it here for three reasons: it builds intuition for the two-stage structure of mediation, its coefficients a, b, τ' reappear as special cases of the NDE and NIE under linearity, and it is the dominant approach in many applied literatures that students will encounter.

It is not, however, the general framework. The algebraic decomposition $\tau = \tau' + ab$ is a consequence of linearity, not a causal identity. Outside linear, no-interaction models it fails, and the product and difference methods it produces are not estimates of the NDE or NIE.

8.7.1 The Baron–Kenny Three-Equation System

The regression-based approach restricts the reduced prototype graph to a linear SEM, conditioning throughout on $\mathbf{X}$:

$$Y = \alpha_1 + \tau T + \gamma_1^\top \mathbf{X} + \varepsilon_1, \tag{8.12}$$

$$M = \alpha_2 + aT + \gamma_2^\top \mathbf{X} + \varepsilon_2, \tag{8.13}$$

$$Y = \alpha_3 + \tau' T + bM + \gamma_3^\top \mathbf{X} + \varepsilon_3. \tag{8.14}$$

The four coefficients of interest are: τ, the total effect of T on Y; a, the first-stage effect of T on M; τ', the direct effect of T on Y controlling for M; and b, the second-stage effect of M on Y controlling for T.

8.7.2 The Component Pathways

Equations Equation 8.12–Equation 8.14 identify three distinct causal sub-problems, each by the back-door formula.

First stage ($T \to M$). Equation Equation 8.13 is the parametric implementation of the back-door formula for the effect of T on M: under linearity and Condition 2 below, conditioning on $\mathbf{X}$ blocks all back-door paths from T to M, and the OLS coefficient a identifies $\mathbb{E}[M(1)] - \mathbb{E}[M(0)]$. In a randomized experiment, a is identified without conditioning on anything.

Second stage ($M \to Y$ given T). Equation Equation 8.14 implements the back-door formula for the effect of M on Y given T. In the reduced prototype graph, M has parents T and $\mathbf{X}$, so every back-door path from M begins with $M \leftarrow T$ or $M \leftarrow \mathbf{X}$. The four such paths are $M \leftarrow T \to Y$ (blocked by T); $M \leftarrow T \leftarrow \mathbf{X} \to Y$ (blocked by T or $\mathbf{X}$); $M \leftarrow \mathbf{X} \to Y$ (blocked by $\mathbf{X}$); and $M \leftarrow \mathbf{X} \to T \to Y$ (blocked by T or $\mathbf{X}$). Conditioning on $(T, \mathbf{X})$ therefore blocks every back-door path from M to Y in this graph, provided Assumption 2 of Section 8.6 holds.

> The Second Stage Is Harder Than It Looks
>
> Even in a randomized experiment where T is randomized, the mediator M is never randomized. An unobserved variable V with $V \to M$ and $V \to Y$ creates a back-door path from M to Y that conditioning on $(T, \mathbf{X})$ cannot block. This is the central challenge of mediation analysis: identifying the second-stage effect b requires a no-unmeasured-confounding assumption for the mediator–outcome relationship, an assumption that randomization of the treatment does not provide.

8.7.3 The Product and Difference Formulas

Proposition 8.1 (Mediation Decomposition in the Linear Model)**.** *Under Equation 8.12–Equation 8.14,*

$$\tau \;=\; \tau' \;+\; ab. \tag{8.15}$$

*The indirect and direct effects are therefore identified by $\tau_{\text{ind}} = ab$ (*product method*) and $\tau_{\text{dir}} = \tau - ab = \tau'$ (*difference method*).*

Proof. Substitute Equation 8.13 into Equation 8.14:

$$\begin{aligned} Y &= \alpha_3 + \tau' T + b(\alpha_2 + aT + \gamma_2^\top \mathbf{X} + \varepsilon_2) + \gamma_3^\top \mathbf{X} + \varepsilon_3 \\ &= (\alpha_3 + b\alpha_2) + (\tau' + ab)\,T + (\gamma_3 + b\gamma_2)^\top \mathbf{X} + (b\varepsilon_2 + \varepsilon_3). \end{aligned}$$

Comparing with Equation 8.12 gives $\tau = \tau' + ab$. □

> The Decomposition $\tau = \tau' + ab$ Is an Algebraic Identity, Not a Causal Theorem
>
> The equality $\tau - \tau' = ab$ is a purely algebraic consequence of substituting one linear equation into another. It holds because linearity makes the indirect effect separable and additive. It does *not* hold in general:
>
> - In **nonlinear models** (binary outcomes, count outcomes, survival models), the product and difference methods yield numerically different estimates. Neither equals the NIE in general.
> - When T **and** M **interact** in their effect on Y, the CDE depends on m, the NDE and CDE diverge, and the indirect effect cannot be summarized by a single number ab.
> - For **non-continuous mediators**, the product ab has no simple causal interpretation outside the linear normal model.
>
> Students who learn the Baron–Kenny decomposition first often overgeneralize it. The correct generalization is the mediation formula (Theorem 8.2), which reduces to ab and τ' only in the linear, no-interaction special case.

Table 8.1: Mediation decomposition in the Baron–Kenny linear model.

Effect	**Formula**	**Path(s)**
Total	τ	$T \to Y$ and $T \to M \to Y$ combined
Direct	$\tau' = \tau - ab$	$T \to Y$ only
Indirect	ab	$T \to M \to Y$ only
Proportion mediated	ab/τ	Share of total effect via M

Running example: a numerical walk-through. Suppose a randomized trial of the behavioral intervention (T) on depression (Y), with sleep quality (M) as the mediator and baseline covariates $\mathbf{X}$, yields

$$\hat{\tau} = 0.50, \qquad \hat{a} = 0.40, \qquad \hat{b} = 0.60, \qquad \hat{\tau}' = 0.26.$$

For concreteness, orient the outcome so that larger Y means greater improvement (e.g. Y is the *reduction* in PHQ-9 score from baseline); a beneficial intervention then has positive coefficients throughout. Then:

- *Indirect effect via sleep* (product method): $\hat{a}\hat{b} = 0.40 \times 0.60 = 0.24$.
- *Direct effect* (difference method): $\hat{\tau} - \hat{a}\hat{b} = 0.50 - 0.24 = 0.26 = \hat{\tau}'$, which matches the fitted $\hat{\tau}'$, illustrating the algebraic identity.
- *Proportion mediated*: $0.24/0.50 = 0.48$, so roughly 48% of the total effect operates through sleep improvement in this sample.

These numbers serve the illustration only; Problem 4 asks students to recompute them and to construct a Sobel confidence interval.

8.7.4 Inference: The Sobel Test and Bootstrap

The delta method gives an approximate variance for the product $\hat{a}\hat{b}$:

$$\widehat{\mathrm{Var}}(\hat{a}\hat{b}) \approx \hat{b}^2\,\widehat{\mathrm{Var}}(\hat{a}) + \hat{a}^2\,\widehat{\mathrm{Var}}(\hat{b}) + 2\,\hat{a}\,\hat{b}\,\widehat{\mathrm{Cov}}(\hat{a}, \hat{b}). \tag{8.16}$$

The *Sobel test* (Sobel 1982) uses the simplified approximation that drops the cross-covariance:

$$\widehat{\mathrm{Var}}_{\mathrm{Sobel}}(\hat{a}\hat{b}) = \hat{b}^2\,\widehat{\mathrm{Var}}(\hat{a}) + \hat{a}^2\,\widehat{\mathrm{Var}}(\hat{b}), \tag{8.17}$$

yielding the z-statistic $z = \hat{a}\hat{b}/\sqrt{\widehat{\mathrm{Var}}_{\mathrm{Sobel}}(\hat{a}\hat{b})}$. Because $\hat{a}$ and $\hat{b}$ come from separate regressions, their cross-covariance is often treated as negligible, but it is not zero in general and dropping it is a convention rather than an exact simplification.

Statistical validity versus causal interpretation. The Sobel test is a statistical inference procedure for the product of two fitted regression coefficients. Under standard regression conditions, Equation 8.17 is an approximation to the sampling variance of $\hat{a}\hat{b}$, and the resulting z-statistic tests whether the regression product ab is zero. That is a property of the fitted model. Interpreting the tested product as an *indirect causal effect* is a separate claim that additionally requires the causal identification conditions and the structural restrictions below. When those assumptions fail, the Sobel approximation may still be a reasonable statistical test of whether the regression product is zero, but that product no longer corresponds to the indirect effect of T on Y through M.

In practice, bootstrap confidence intervals for ab are preferred over the Sobel test for inference, because the distribution of a product of estimates is skewed in finite samples — especially when either $\hat{a}$ or $\hat{b}$ is small.

The "Significance of Both Paths" Criterion Is Not a Test for Mediation

A common misuse of the Baron–Kenny framework, sometimes called the "causal steps" approach, declares mediation to be present when: (i) $T \to Y$ is significant in Equation 8.12; (ii) $T \to M$ is significant in Equation 8.13; (iii) $M \to Y$ is significant in Equation 8.14. This approach has three serious defects.

1. **Statistical significance $\neq$ mediation.** A significant $\hat{a}$ and $\hat{b}$ does not imply that the product

ab is meaningfully large, nor that it is identified. Low power for $\hat{a}$ or $\hat{b}$ individually does not imply low power for ab (and vice versa).

2. **The Sobel test tests a regression product, not a causal quantity.** Under standard regression conditions, Equation 8.17 is a valid statistical approximation regardless of whether the Baron–Kenny causal assumptions hold. What those assumptions control is the *interpretation* of the tested product as an indirect effect.
3. **Zero total effect does not preclude indirect effects.** It is possible for $\tau \approx 0$ while $ab \neq 0$ and $\tau' \neq 0$ (direct and indirect effects of opposite sign that cancel). Requiring significance of $\hat{\tau}$ as a precondition will miss these cases entirely.

The modern alternative is to estimate ab (or the NIE directly via the mediation formula), construct bootstrap or nonparametric confidence intervals, and interpret the result as a point estimate with uncertainty — not as a significance test.

8.7.5 The Baron–Kenny Assumptions: Two Distinct Categories

The conditions required for the Baron–Kenny decomposition fall into two fundamentally different categories that should not be conflated.

Baron–Kenny Causal Ignorability Conditions

These are causal identification assumptions. They concern unmeasured confounding and have graphical interpretations. Violating them introduces bias that no amount of additional data can remove given the observed variables.

1. **No unmeasured T–Y confounding.** $\varepsilon_1 \perp\!\!\!\perp T \mid \mathbf{X}$. Graphically: $\mathbf{X}$ blocks all back-door paths from T to Y.
2. **No unmeasured T–M confounding.** $\varepsilon_2 \perp\!\!\!\perp T \mid \mathbf{X}$. Graphically: $\mathbf{X}$ blocks all back-door paths from T to M.
3. **No unmeasured M–Y confounding given T.** $\varepsilon_3 \perp\!\!\!\perp M \mid T, \mathbf{X}$. Graphically: $(T, \mathbf{X})$ blocks all back-door paths from M to Y. This condition is *not* implied by randomization of T.

A caution on reading these displays: the ε_j are *structural* errors of the SEM, not fitted residuals. Population least-squares residuals are orthogonal to the regressors by construction, so the displayed independences are substantive restrictions only under the structural reading; they cannot be checked by inspecting regression output.

Baron–Kenny Structural Modeling Restrictions

These are parametric modeling assumptions. They concern the functional form of the structural equations, not unmeasured confounding. Violating them does not introduce identification bias in the causal sense, but it does mean that the product formula ab and the coefficient τ' no longer equal the NDE and NIE.

4. **Linearity and additivity.** The structural equations Equation 8.12–Equation 8.14 are correctly specified as linear and additive in their arguments.
5. **No $T \times M$ interaction.** The effect of M on Y does not depend on the level of T.

Restrictions 4 and 5 can be probed by residual diagnostics and interaction terms, but model adequacy cannot be conclusively established from observed data alone. Conditions 1–3 cannot be tested from the observed data at all.

In the reduced prototype graph, Conditions 1 and 2 hold because $\mathbf{X}$ blocks every T–Y and T–M back-door path by construction. Condition 3 is different: it requires the absence of any variable $V \to M$ and $V \to Y$ not captured by $(T, \mathbf{X})$, and it is never guaranteed by randomization of T because M is a post-treatment variable that is not itself randomized. Restrictions 4 and 5 have no graphical counterpart.

Baron–Kenny as a special case of the modern framework. Under Conditions 1–3 and Restrictions 4–5, the Baron–Kenny decomposition delivers valid *identification formulas* for the NDE and NIE: linearity and no interaction guarantee that the population coefficients satisfy NDE $= \tau'$ and NIE $= ab$. Translating these identification formulas into *consistent estimators* requires a separate set of conditions. Outside Conditions 1–3 and Restrictions 4–5, τ' and ab *need not* identify the CDE or the natural effects: the listed conditions are sufficient rather than necessary, and special structures can restore particular equalities, but in general the coefficients of a misspecified structural system carry no causal interpretation.

Framework Map

Estimand	**Framework needed**	**Section**
Total effect $P(y \mid \mathrm{do}(t))$	Do-calculus	Ch. 5 (back-door)
Controlled direct effect (CDE)	Do-calculus	Section 8.4
Natural direct effect (NDE)	Potential outcomes	Section 8.5
Natural indirect effect (NIE)	Potential outcomes	Section 8.5
Total effect via front-door	Do-calculus	Section 8.8

The first row and the last row name the same estimand — the total effect of T on Y — but identified by different routes: standard back-door adjustment in the first row, and the front-door formula in the last. The CDE, NDE, and NIE are distinct estimands.

Chapter pivot: from decomposing an identified total effect to identifying a confounded one. Everything up to this point has taken identification of the total effect for granted — the working assumption of Section 8.2.4 absorbed all T–Y confounding into observed covariates — and asked how that effect decomposes across pathways. The remainder of the chapter reverses the direction of travel. In Section 8.8, the mediator stops being the object of study and becomes an identification device: the total effect itself is unidentified by adjustment, and the mediation structure of the graph is what rescues it.

8.8 Front-Door Identification

8.8.1 The Front-Door DAG

Every identification strategy above assumed that an observed covariate set $\mathbf{X}$ exists that blocks the back-door paths from T to Y through U. What if U is wholly unobserved and no such adjustment set exists? In the prototype mediation DAG, the back-door criterion then fails for the total effect, and the methods of earlier sections are unavailable. Chapter 3 previewed the front-door strategy for exactly this situation; this section gives the full development — the identification theorem with its support conditions, the do-calculus proof, the failure modes, and a numerical illustration — and connects it to mediation.

The front-door criterion turns this obstacle into an opportunity: under two additional restrictions on the prototype mediation graph, the mediation structure itself provides identification of the total effect without conditioning on U. The two restrictions are: remove the direct $T \to Y$ edge, so that M fully mediates the effect of T on Y; and require that U has no arrow into M, so that the $T \to M$ sub-effect is unconfounded.

Why the running example does not apply here. The depression/sleep scenario fails Condition 1 (full mediation): a behavioral intervention on depression plausibly operates through several non-sleep channels — cognitive restructuring, behavioral activation, therapeutic alliance — so the direct edge $T \to Y$ is present. The front-door formula is therefore unavailable for that setting. The canonical example in which the three front-door conditions are all plausible is Pearl's smoking–tar–cancer graph: T denotes smoking, M the amount of tar deposited in the lungs, Y a lung-cancer outcome, and U an unobserved genetic susceptibility. If all of smoking's carcinogenic effect flows through tar, and the genetic susceptibility does not act on tar directly, the front-door formula identifies the causal effect of smoking on cancer without ever observing U. This is a *stylized* identification example, not a claim that measured tar burden satisfies the front-door conditions in actual epidemiologic data: a genetic susceptibility could plausibly affect inhalation behavior, tar metabolism, or the measurement of tar itself.

Two Different Questions About a Mediator-Like Variable

The front-door graph looks like the mediation graph with two edges removed, and M sits between T and Y in both. The resemblance is structural, but the questions being asked are fundamentally different.

	Ordinary mediation	**Front-door identification**
Question	How much of the total effect of T on Y flows through M?	Can M be used to identify the total effect despite unmeasured T–Y confounding?
Role of M	Pathway: carries part of the causal effect	Instrument-like relay: routes around unmeasured confounding
Target estimand	NDE, NIE (decomposition)	$P(y \mid \mathrm{do}(t))$ (the total effect itself)
What M **must satisfy**	Sequential ignorability for the M–Y sub-problem	Conditions 1–3 below (graphical, no potential outcomes required)
Direct $T \to Y$ **edge**	Present	Absent (required)

The front-door formula does not decompose the total effect into direct and indirect components. It identifies the total effect as a whole, using M as a relay that is unconfounded on the T-side; on the Y-side, the back-door path through T is closed by conditioning on T, after which the result is averaged over the observed marginal distribution of T. A researcher who wants to know *how much* of the effect goes through M needs ordinary mediation analysis.

“Instrument-Like” Does Not Mean Instrument

The phrase “instrument-like” above is a loose analogy for one feature: M helps route around unmeasured T–Y confounding, which is also what an instrumental variable does. M is *not* an instrument in the IV sense. An instrument Z satisfies the exclusion restriction — it affects Y only through T — so Z is excluded from the Y structural equation. By contrast, the front-door mediator M lies on the causal path from T to Y and enters the Y structural equation directly. The two strategies share the goal of identifying T's total effect under unobserved confounding, but they place the third variable in structurally opposite positions.

8.8.2 The Three Front-Door Conditions

Front-Door Conditions

A variable M satisfies the **front-door criterion** for the effect of T on Y if:

1. **Full mediation.** All directed paths from T to Y pass through M. (In the prototype graph this is equivalent to the absence of a direct $T \to Y$ edge; in richer graphs, deleting that edge is not sufficient if some other directed path bypasses M.)
2. **No unblocked back-door path from T to M.** (In the prototype graph, which carries no adjustment covariates, this means there is no latent common cause of T and M; in richer graphs a common cause may exist provided observed covariates block the resulting paths.)
3. **No unblocked back-door path from M to Y given T.** All back-door paths from M to Y are blocked by conditioning on T.

The criterion applies to the causal mediator represented in the graph: replacing M by a mismeasured proxy need not preserve any of the three conditions — a proxy may fail to intercept all directed T-to-Y paths, and measurement error can open both stages of the argument. Measurement error in M is thus an identification threat, not merely an estimation nuisance.

In the front-door graph: Condition 1 holds because there is no $T \to Y$ edge. Condition 2 holds, but for a subtler reason than “no back-door path exists”: there *is* a back-door path from T to M, namely $T \leftarrow U \to Y \leftarrow M$, but it is blocked by the unconditioned collider Y. Because U has no arrow into M, no *open* back-door path remains — and the criterion requires only the absence of open paths. Condition 3 holds because the only back-door path from M to Y is $M \leftarrow T \leftarrow U \to Y$, which is blocked by conditioning on T.

Remark: Single Mediator vs. a Set of Mediators

Pearl (1995) states the front-door criterion for a *set* of intermediate variables $\mathbf{M} = \{M_1, \ldots, M_k\}$: the set must intercept all directed paths from T to Y, have no unblocked back-door from T, and have all back-door paths to Y blocked by T. The formula generalizes accordingly, with the inner sum running over realizations of $\mathbf{M}$. For pedagogical clarity, this chapter treats the single-mediator case throughout; the generalization is straightforward conceptually but the multivariate $\mathbf{M}$ introduces additional positivity and estimation challenges.

Example 8.1 (When the Front-Door Criterion Fails)**.** Each of the three front-door conditions is load-bearing. We illustrate what breaks by modifying the front-door graph in two separate ways. As Case (a) shows, a single added edge can violate more than one condition at once.

Case (a): Conditions 2 and 3 fail — U also causes M. Start with the front-door graph and add the edge $U \to M$, so the unobserved confounder directly affects the mediator as well as the treatment and the outcome.

Condition 1 still holds: every directed path from T to Y passes through M. Condition 2, however, now **fails**: the path $T \leftarrow U \to M$ is an unblocked back-door path from T to M. Because U is unobserved, there is no observed variable that can block this path. Condition 3 **also fails**: the path $M \leftarrow U \to Y$ is an open back-door path from M to Y, and conditioning on T does not block it, since T does not lie on the path.

Why the formula breaks. Both stages collapse. Stage 1 uses $P(M{=}m \mid T{=}t)$ as if the $T \to M$ link were unconfounded. But with $U \to M$, the association between T and M is partly due to their shared cause U: the quantity $P(m \mid t)$ mixes cause and confounding and no longer identifies $P(m \mid \mathrm{do}(T{=}t))$. Likewise, because the open path $M \leftarrow U \to Y$ survives conditioning on T, the Stage 2 functional does not generally equal $P(y \mid \mathrm{do}(M{=}m))$. Failure of either condition alone would already suffice to invalidate the formula.

Case (b): Condition 3 fails — the $M \to Y$ link has an extra unobserved confounder. Return to the prototypical graph and add a second unobserved variable V that affects both M and Y.

Conditions 1 and 2 still hold. But Condition 3 now **fails**: the path $M \leftarrow V \to Y$ is a back-door path from M to Y that is *not* blocked by conditioning on T. The path does not pass through T, so fixing T does nothing to close it.

Why the formula breaks. Stage 2 uses $\sum_{t'} P(y \mid t', m)\, P(t')$ as the back-door adjustment formula for $P(y \mid \mathrm{do}(M{=}m))$, with T as the adjustment variable. But T only satisfies the back-door criterion for $M \to Y$ when it blocks *all* back-door paths from M to Y. The new path $M \leftarrow V \to Y$ is not blocked by T, so T is no longer a valid adjustment set for Stage 2.

Summary.

Modification	Condition(s) violated	Consequence
Add $T \to Y$	Cond. 1	A causal path bypasses M
Add $U \to M$	Conds. 2 and 3	Stages 1 and 2 both fail
Add $V \to M$ and $V \to Y$	Cond. 3	Stage 2 fails

In each case the graphical check detects the failure before any formula is written down. The first row's modification — adding a direct $T \to Y$ edge — shows that the absence of such an edge in the front-door graph is not an independent requirement: it is simply one way of satisfying Condition 1, which demands that *all* directed paths from T to Y pass through M.

8.8.3 Derivation of the Front-Door Formula

Theorem 8.3 (Front-Door Formula (Pearl 1995))**.** *Suppose M satisfies the front-door criterion for the effect of T on Y. Assume further the positivity conditions:*

(F1) Treatment positivity. *$P(T = t) > 0$ at the evaluation level t, so that $P(m \mid t)$ is well-defined; every t' entering the inner sum lies in the support of T.*

(F2) Mediator overlap across treatment arms. *For every mediator value m with $P(M{=}m \mid T{=}t) > 0$ and every t' with $P(T{=}t') > 0$, $P(M{=}m \mid T{=}t') > 0$.*

Then

$$P(y \mid \mathrm{do}(T{=}t)) = \sum_m P(m \mid t) \sum_{t'} P(y \mid m, t')\, P(t'). \tag{8.18}$$

Condition (F2) is the front-door analogue of the mediator overlap (P2): it ensures that $P(y \mid m, t')$ is empirically defined at every (m, t') cell that receives positive weight in Equation 8.18.

Proof. We apply the do-calculus in three steps, each exploiting one of the front-door conditions.

Step 1: Identify the effect of T on M. Apply Rule 2 with $X = \emptyset$, $Z = T$, $W = \emptyset$; the required graph is $\mathcal{G}_{\underline{T}}$, obtained by deleting T's outgoing arrow. The only remaining path between T and M is $T \leftarrow U \rightarrow Y \leftarrow M$, blocked at the unconditioned collider Y; Condition 2 is what guarantees that no open path survives (an edge $U \rightarrow M$ would leave $T \leftarrow U \rightarrow M$ open). Hence $(M \perp\!\!\!\perp T)_{\mathcal{G}_{\underline{T}}}$, and Rule 2 gives

$$P(m \mid \mathrm{do}(T{=}t)) = P(m \mid t).$$

Step 2: Identify the effect of M on Y. By Condition 3, conditioning on T blocks all back-door paths from M to Y, so $\{T\}$ is a valid back-door adjustment set:

$$P(y \mid \mathrm{do}(M{=}m)) = \sum_{t'} P(y \mid m, t')\, P(t').$$

Step 3: Combine via full mediation. The law of total probability applied under the intervention $\mathrm{do}(T{=}t)$ gives

$$P(y \mid \mathrm{do}(T{=}t)) = \sum_m P(m \mid \mathrm{do}(T{=}t))\; P(y \mid \mathrm{do}(T{=}t),\, m)\,, \tag{8.19}$$

where the second factor *conditions* on $M{=}m$. Converting that conditioning into the intervention $\mathrm{do}(M{=}m)$ is the delicate step, and it is here that Conditions 3 and 1 do their work, in that order.

First, apply Rule 2 with $X = T$, $Z = M$, $W = \emptyset$; the required graph is $\mathcal{G}_{\overline{T}\,\underline{M}}$. Deleting the arrows into T and out of M leaves only the edges $T \rightarrow M$ and $U \rightarrow Y$, so no path connects M to Y and $(Y \perp\!\!\!\perp M \mid T)_{\mathcal{G}_{\overline{T}\,\underline{M}}}$ holds — Condition 3 secures this in general, since any surviving path would be a back-door path from M to Y not blocked by T. Rule 2 gives

$$P(y \mid \mathrm{do}(T{=}t),\, m) = P(y \mid \mathrm{do}(T{=}t),\, \mathrm{do}(M{=}m))\,.$$

Second, apply Rule 3 with $X = M$, $Z = T$, $W = \emptyset$; the required graph is $\mathcal{G}_{\overline{M}\,\overline{T}}$. Deleting the arrows into both M and T isolates T, so the now-redundant $\mathrm{do}(T{=}t)$ can be dropped:

$$P(y \mid \mathrm{do}(T{=}t),\, \mathrm{do}(M{=}m)) = P(y \mid \mathrm{do}(M{=}m))\,.$$

Condition 1 secures this second step in general: every directed path from T to Y passes through M, whose incoming arrows have been deleted, so no route from T to Y survives. This is the formal content of "the total effect of T factors through M." Substituting Steps 1 and 2 into Equation 8.19 yields Equation 8.18.
□

> Remark: Two Routes to the Same Two-Stage Logic
>
> The two-stage structure of the proof — first identify the $T \rightarrow M$ effect, then the $M \rightarrow Y$ effect, and finally compose — is the do-calculus analogue of the two-stage reasoning in the mediation formula (Theorem 8.2). The mediation formula accomplishes this via sequential ignorability — conditional independence assumptions on potential outcomes. The front-door formula accomplishes it via the assumed causal graph and its intervention semantics: Condition 2 makes Step 1 unconfounded by construction, and Condition 3 makes Step 2 unconfounded because conditioning on T blocks the back-door path $M \leftarrow T \leftarrow U \rightarrow Y$. No covariate set $\mathbf{X}$ is needed, because the structural restrictions on the graph substitute for the ignorability assumptions — given, as always, the modularity of the structural equations, positivity (F1)–(F2), and a correctly measured mediator.

8.8.4 Intuition: Routing Around Confounding

The front-door formula achieves identification in two steps that each use only unconfounded variation.

1. **T to M:** There is no confounding on the $T \to M$ edge (U does not affect M), so $P(m \mid t)$ is the causal effect of T on M.
2. **M to Y:** There is back-door confounding on $M \to Y$ through the path $M \leftarrow T \leftarrow U \to Y$, but T is a non-collider on this path, so conditioning on T closes it. The resulting conditional $P(y \mid m, t')$ is then averaged over the observed marginal distribution of T.

The key insight is that U confounds T and Y but *not* the $T \to M$ edge. The front-door formula exploits this asymmetry to identify the total causal effect without ever observing or conditioning on U.

Example: Front-Door Formula — Binary Numerical Computation

Let $T, M, Y \in \{0, 1\}$ with U unobserved and the graph $T \to M \to Y$, $U \to T$, $U \to Y$. The observed joint distribution $P(T, M, Y)$ is given in the following population table.

T	M	Y	$P(T, M, Y)$
0	0	0	0.360
0	0	1	0.040
0	1	0	0.050
0	1	1	0.050
1	0	0	0.070
1	0	1	0.030
1	1	0	0.120
1	1	1	0.280

All quantities needed for the front-door formula can be read off by marginalization. For example, $P(T{=}1) = 0.070 + 0.030 + 0.120 + 0.280 = 0.5$, and $P(M{=}1 \mid T{=}1) = (0.120 + 0.280)/0.5 = 0.8$. The inner sum (Stage 2) does not depend on the intervention value t, so compute it once for each value of M:

$$S_0 = \sum_{t'} P(Y{=}1 \mid t', M{=}0)\, P(t') = 0.1 \times 0.5 + 0.3 \times 0.5 = 0.20,$$
$$S_1 = \sum_{t'} P(Y{=}1 \mid t', M{=}1)\, P(t') = 0.5 \times 0.5 + 0.7 \times 0.5 = 0.60.$$

Now apply the outer sum (Stage 1) for each intervention value:

$$P(Y{=}1 \mid \mathrm{do}(T{=}1)) = 0.2 \times 0.20 + 0.8 \times 0.60 = 0.52,$$
$$P(Y{=}1 \mid \mathrm{do}(T{=}0)) = 0.8 \times 0.20 + 0.2 \times 0.60 = 0.28.$$

Under the assumed front-door graph, the identified average causal effect is $0.52 - 0.28 = 0.24$. **Comparison with the naive approach.** A naive analyst ignoring the unobserved confounder U would estimate the effect by the observational contrast:

$$P(Y{=}1 \mid T{=}1) = \frac{0.030 + 0.280}{0.5} = 0.62, \qquad P(Y{=}1 \mid T{=}0) = \frac{0.040 + 0.050}{0.5} = 0.18.$$

The naive estimate is $0.62 - 0.18 = 0.44$, *almost twice* the effect of 0.24 identified under the assumed graph. The excess association of 0.20 is transmitted along the open back-door path $T \leftarrow U \rightarrow Y$: the graph determines that this non-causal route exists, and the observed table determines its net contribution, but the graph alone says nothing about the sign or magnitude of U's individual effects. The front-door formula removes this bias by routing the effect estimate through the unconfounded $T \rightarrow M$ link and the T-adjusted $M \rightarrow Y$ link, never directly comparing the $T{=}1$ and $T{=}0$ groups on Y.

Note how Stage 2 averages $P(Y \mid t', m)$ over the *marginal* distribution $P(t')$, not the treatment-conditional distribution $P(t' \mid m)$ — the re-weighting that removes the confounding introduced by U.

A final caveat: the observed table does not itself *validate* the assumed graph. Causal models on the front-door graph that reproduce this table do exist, so the computation is internally coherent, but the calculation is conditional on assuming that the data arose from *some* model satisfying the front-door graph. The graphical conditions are assumptions supplied from outside the data; the formula converts them into a number, it does not confirm them.

8.8.5 Front-Door vs. Prototype Mediation

Feature	**Prototype mediation DAG**	**Front-door DAG**
Direct $T \rightarrow Y$ edge	Present	Absent
U confounds T–Y	Yes	Yes
Goal	Mechanism analysis (decomposition under natural effects; level-specific direct effect under CDE)	Identify total effect despite T–Y confounding
Requires Assumption 2 (M–Y no confounding given T)	Yes	Satisfied by front-door Condition 3, not by randomization or by default
Identified by front-door formula	No (direct $T \rightarrow Y$ breaks Condition 1)	Yes

8.9 Mediation vs. Instrumental Variables

Mediation analysis and instrumental variables both involve a third variable connected to the treatment–outcome relationship, but the causal role of that variable is fundamentally different in the two frameworks.

8.9.1 Structural Comparison

Feature	**Instrumental Variables**	**Mediation Analysis**
Position of third variable	Pre-treatment (Z precedes T)	Post-treatment (M follows T)
Causal role	Exogenous source of variation in T	Pathway through which T affects Y
Primary goal	Identification of $T \to Y$ effect under confounding	Mechanism analysis of $T \to Y$ effect
Key assumption	Exclusion: Z affects Y only through T	Sequential ignorability: mediator–outcome ignorability plus the cross-world independence
Estimand	LATE (Wald, under monotonicity); under a homogeneous linear effect, the common structural coefficient (Ch. 7)	NDE, NIE, or CDE
Unobserved T–Y confounders	Permitted (IV routes around them)	Must be addressed separately (Assumption 1)
Testability	Relevance testable; exclusion untestable	Sequential ignorability untestable

8.9.2 The Conceptual Contrast

The contrast is sharpest in terms of what the third variable *does* in each framework.

In IV, Z is a *handle*: it generates exogenous variation in T that is free of the back-door confounding path $T \leftarrow U \rightarrow Y$. The instrument is valuable precisely because it is *not* on the causal path from T to Y — the exclusion restriction says that Z cannot directly affect Y.

In mediation analysis, M is a *pathway*: it transmits part of the causal effect of T to Y. The mediator is valuable precisely because it *is* on the causal path. The analysis goal depends on the estimand: under natural effects, the question is how much of the total effect flows through this pathway; under the CDE, the question is what part of T's effect on Y remains when M is held fixed at a chosen level.

The Key Conceptual Distinction

	IV	**Mediation**
What the third variable does	Generates clean variation in T (exogenous source)	Carries part of T's causal effect (pathway)
Exclusion vs. inclusion	Z excluded from Y's structural equation	M included in Y's structural equation
Question answered	Does T cause Y?	How does T cause Y?

8.9.3 Can the Same Variable Be Both?

It is worth asking whether the same variable M could serve as both a mediator and an instrument. The answer is: *not for the same treatment–outcome relation.* Within a single causal question of how T affects Y, the mediator role places M *on* the causal path (inclusion required), whereas the IV role demands the exclusion restriction — that M affects Y only through T. These are mutually incompatible structural assumptions about the same intermediate variable.

The qualifier *for the same treatment–outcome relation* matters. A variable that mediates one causal question may, in a differently oriented question, serve as an instrument-like exogenous source of variation. For example, occupation might mediate the effect of education on wages while functioning as an exogenous source of variation in some unrelated downstream analysis where neither education nor wages is the target.

The incompatibility is between the two roles *within one fixed* (T, Y) *pair*, not across different scientific questions.

The *front-door identification* formula is the closest bridge between the two within a single (T, Y) analysis: it uses the mediator M to identify the total effect of T on Y even when T is confounded — a goal that IV also pursues. But, as the warning box in Section 8.8 stresses, the front-door M is not an instrument: it lies on the causal path, satisfies a graphical front-door condition, and substitutes structural restrictions on the graph for the exclusion restriction that defines an IV.

8.10 Summary

1. **Mediation studies mechanisms.** Mediation analysis aims to study the mechanisms by which T affects Y by defining direct and indirect effect concepts that target the pathways $T \to Y$ and $T \to M \to Y$. Of the two direct-effect families developed in this chapter, only the natural effects are constructed to yield an additive decomposition of the total effect; the controlled direct effect is not. The challenge throughout is that the mediator is a post-treatment variable that may be confounded.
2. **The total effect is the baseline estimand.** The TE $= \mathbb{E}[Y(1) - Y(0)]$ captures all pathways combined. It may be identified by randomization, by back-door adjustment on a sufficient covariate set, by front-door identification, or by other identifying structures; which strategy applies depends on the available causal structure.
3. **The CDE uses do-calculus.** The controlled direct effect fixes $M = m$ by joint intervention $\mathrm{do}(T, M)$. Under consistency, a well-defined joint intervention, an appropriate adjustment condition for the pair (T, M), and joint positivity, it is identified by standardization (Theorem 8.1). The CDE depends on the fixed level m and does not have a natural "indirect" complement.
4. **Natural effects require potential outcomes.** The NDE and NIE involve cross-world counterfactuals $Y(t, M(t'))$ that cannot be expressed with the do-operator alone. They decompose the total effect as TE = NDE + NIE, and are identified by the mediation formula under sequential ignorability. The critical cross-world independence (Assumption 3) is not secured by randomization of T. Under the NPSEM-IE used in these notes, it follows from the independent-error structural model together with adequate control of pre-treatment common causes and the absence of a treatment-induced mediator–outcome confounder.
5. **The linear model simplifies but restricts.** The Baron–Kenny three-equation system gives the clean formulas $\tau_{\text{ind}} = ab$ and $\tau_{\text{dir}} = \tau'$, with $\tau = \tau' + ab$. This decomposition is purely algebraic and holds only under linearity and no interaction. In nonlinear or interaction settings, the product and difference methods disagree and NDE $\neq$ CDE.
6. **Front-door identification uses mediation structure.** When M fully mediates $T \to Y$, no $T \to M$ confounding exists, and T blocks the back-door paths from M to Y, the front-door formula identifies the total effect despite unobserved T–Y confounding. It achieves this by composing two unconfounded sub-effects.
7. **Mediation and IV are complementary, not equivalent.** IV uses a pre-treatment variable to generate exogenous variation in T; mediation uses a post-treatment variable to study how the causal effect operates. The same variable cannot simultaneously serve as a mediator and a valid IV for the same treatment–outcome relation.

8.11 Problems

1. Identifying the CDE. Consider the DAG: $T \to M$, $T \to Y$, $M \to Y$, $X \to T$, $X \to M$, $X \to Y$, with all variables observed.

(a) Write the identification formula for $\mathbb{E}[Y \mid \mathrm{do}(T{=}1), \mathrm{do}(M{=}m)]$ using baseline adjustment for the joint intervention (T, M) as in Theorem 8.1.
(b) Add an unobserved U with $U \to T$ and $U \to Y$. Is the CDE identified by *any* adjustment set consisting of observed variables? In particular, does standardization over X remain valid? Explain which condition of Theorem 8.1 fails.
(c) Instead add U with $U \to M$ and $U \to Y$. Answer the same questions.

2. CDE vs. total effect. In the reduced prototype graph, let $\mathbf{Z}$ satisfy the back-door criterion for both the total effect and the joint intervention (T, M).

(a) Write expressions for the total effect and the CDE(m) using the back-door formula.
(b) Give a sufficient *graphical* condition under which fixing M removes no causal pathway from T to Y — for example, that no directed path from T to Y passes through M. Interpret this condition.
(c) *(second pass)* Show by example that the graphical condition of part (b) alone does not make CDE(m) equal to the total effect for all m: exhibit a structural equation for Y with a $T \times M$ interaction in which the two quantities differ even though M is not affected by T. What additional *functional* condition closes the gap?

3. Natural direct and indirect effects. Verify the NDE + NIE = TE decomposition algebraically for the linear SEM $M = \alpha T + \eta$, $Y = \beta T + \gamma M + \varepsilon$ (no interaction).

(a) Compute $Y(t, M(t'))$ in the linear model, writing the noise terms explicitly: $Y(t, M(t')) = \beta t + \gamma\alpha t' + \gamma\eta + \varepsilon$. Note that the *same* η appears for every value of t' — the structural construction shares the mediator disturbance across intervention worlds, and this is exactly what makes the nested counterfactual meaningful.
(b) Derive NDE $= \beta$ and NIE $= \alpha\gamma$ from Equation 8.7–Equation 8.8.
(c) Confirm NDE + NIE $= \beta + \alpha\gamma =$ TE.
(d) Now suppose a $T \times M$ interaction is added: $Y = \beta T + \gamma M + \delta(T \cdot M) + \varepsilon$. Show that CDE$(m) = \beta + \delta m$ and NDE $= \beta + \delta\,\mathbb{E}[M(0)]$. Conclude that the two generally differ when $\delta \neq 0$, and identify the one mediator level at which they coincide.

4. The Baron–Kenny three-equation system. In the reduced prototype graph with the linear SEM Equation 8.12–Equation 8.14:

(a) State the three identification assumptions. For each, give the graphical condition in terms of back-door paths.
(b) Derive the equality $\tau = \tau' + ab$ algebraically.
(c) Suppose estimated coefficients are $\hat{\tau} = 0.50$, $\hat{a} = 0.40$, $\hat{b} = 0.60$, $\hat{\tau}' = 0.26$. Compute the indirect effect by both the product and difference methods. Do they agree? Compute the proportion mediated, and state the conditions — direct and indirect effects of the same sign, and a total effect bounded away from zero — under which calling this ratio a "proportion" is meaningful. What happens to ab/τ when $\tau' = -ab$?
(d) *(optional computational practice)* An analyst reports $\hat{a} = 0.40$ with $\widehat{\mathrm{SE}}(\hat{a}) = 0.08$, and $\hat{b} = 0.60$ with $\widehat{\mathrm{SE}}(\hat{b}) = 0.10$. Compute the Sobel standard error for $\hat{a}\hat{b}$ using Equation 8.17 and construct an approximate 95% confidence interval.

5. The critical role of Assumption 2 (mediator–outcome ignorability). Consider the graph in which an unobserved V has $V \to M$ and $V \to Y$, with T randomized.

(a) Identify all back-door paths from M to Y in this graph.
(b) Can any combination of observed variables $(T, \mathbf{X})$ block all of these paths? Explain using d-separation.
(c) Suppose an analyst fits Equation 8.14 ignoring V and obtains $\hat{b} = 0.80$. Under a *linear additive* structural model in which V has positive effects on both M and Y, in which direction is $\hat{b}$ biased? Explain why the DAG alone, without the linearity and sign restrictions, does not determine the direction.
(d) State *two* additional data structures or design features that would permit identification of the second-stage effect — for example: (i) measuring a covariate set sufficient for the M–Y back-door paths, or (ii) an experiment that directly assigns the mediator. For each, say which assumption of Section 8.6 it restores.

6. Front-door identification. *(Emphasis: the do-calculus derivation and diagnosing failure.)* Consider the front-door graph.

(a) Verify that the three front-door conditions hold.
(b) Walk through the three-step proof of Theorem 8.3 for this graph: identify which do-calculus rule justifies each step.
(c) Add a direct edge $T \to Y$ to the graph. Which front-door condition is violated? Does Equation 8.18 still hold?

(d) Explain how front-door Condition 3 corresponds to the Baron–Kenny requirement that the $M \to Y$ relation be unconfounded after conditioning on T (equivalently, Assumption 2 of Section 8.6). The front-door graph does not dispense with this requirement — it satisfies it *by construction of the graph*. Why does the condition hold in the front-door graph but fail in the graph with an M–Y confounder V?

7. Mediation vs. instrumental variables. A researcher studies the effect of a job training program (T) on wages (Y). She proposes two intermediate variables: (A) motivation (M_A), measured after the program starts; (B) a lottery that randomly selects applicants for admission (Z), measured before the program.

(a) For variable (A): draw the mediation DAG including M_A, T, Y, and an unobserved ability variable U. State the *complete* set of identification assumptions from Section 8.6 needed to identify the NIE through M_A, and explain why randomization of T secures only Assumption 1.
(b) For variable (B): draw the IV DAG with Z, T, Y, and U. State the three IV assumptions. Explain why the exclusion restriction and the "mediator inclusion" of mediation analysis are mutually incompatible conditions for the same intermediate variable.
(c) The researcher argues that M_A and Z are both "intermediate" variables and that the analyses are interchangeable. Write a one-paragraph critique of this argument.
(d) Can the front-door formula be applied if motivation M_A fully mediates the effect of T on Y and U does not directly affect M_A? State the three conditions and assess whether they hold.

8. Front-door identification with baseline covariates. *(Second pass; builds on Problem 6.)* Consider the DAG: $T \to M \to Y$, $\mathbf{X} \to T$, $\mathbf{X} \to M$, $\mathbf{X} \to Y$, $U \to T$, $U \to Y$, with $\mathbf{X}$ observed and pre-treatment, U unobserved, and no direct $T \to Y$ edge.

(a) Show that the three front-door conditions hold *conditionally on* $\mathbf{X}$: all directed paths from T to Y pass through M; every back-door path from T to M is blocked given $\mathbf{X}$ (enumerate the paths through $\mathbf{X}$ and through U, and say which node blocks each); and $(T, \mathbf{X})$ blocks every back-door path from M to Y.
(b) Derive the covariate-adjusted front-door formula

$$P(y \mid \mathrm{do}(T{=}t)) = \sum_{\mathbf{x}} P(\mathbf{x}) \sum_{m} P(m \mid t, \mathbf{x}) \sum_{t'} P(y \mid m, t', \mathbf{x})\, P(t' \mid \mathbf{x})$$

by repeating the three-step do-calculus argument of Theorem 8.3 with $W = \{\mathbf{X}\}$ carried through each rule application. (You will also need $P(\mathbf{x} \mid \mathrm{do}(t)) = P(\mathbf{x})$; which rule justifies it, and why does it require $\mathbf{X}$ to be pre-treatment?)
(c) Explain why the inner average uses the *covariate-conditional* distribution $P(t' \mid \mathbf{x})$ rather than the marginal $P(t')$ used in Equation 8.18.
(d) Now add the edge $U \to M$. Which conditional front-door condition fails, and which stage of the two-stage argument collapses?

9. Nonidentifiability by construction *(advanced / second pass)*. This problem practices the technique of the bow-graph example in Chapter 3: prove nonidentifiability by exhibiting two observationally equivalent models with different causal effects.

Consider the graph from Case (a) of Example 8.1: $T \to M \to Y$, $U \to T$, $U \to M$, $U \to Y$ (front-door Conditions 2 and 3 are violated). Let $T, M, Y, U \in \{0, 1\}$ with $U \sim \mathrm{Bern}(1/2)$.

(a) Construct two SEMs $\mathcal{M}_1$ and $\mathcal{M}_2$, each compatible with the graph, that induce the same observed distribution of (T, M, Y) but different interventional distributions: specifically, $P_{\mathcal{M}_1}(Y{=}1 \mid \mathrm{do}(T{=}1)) \neq P_{\mathcal{M}_2}(Y{=}1 \mid \mathrm{do}(T{=}1))$. (*Hint*: set $T = U$ in both models. In $\mathcal{M}_1$ set $M = T \wedge U$ and $Y = M \vee U$; in $\mathcal{M}_2$ set $M = T \vee U$ and $Y = M \vee U$, where $\wedge$ and $\vee$ denote Boolean AND and OR. Observationally $T = M = Y = U$ in both models, so both produce $P(T{=}0, M{=}0, Y{=}0) = P(T{=}1, M{=}1, Y{=}1) = \frac{1}{2}$. Then compute $P(Y{=}1 \mid \mathrm{do}(T{=}1))$ and $P(Y{=}1 \mid \mathrm{do}(T{=}0))$ in each model, and take their difference.)
(b) Explain precisely which identification strategy fails in this graph and why. Your explanation should reference (i) the back-door criterion (is there any observed variable that blocks $T \leftarrow U \to Y$?), (ii) the front-door criterion (which conditions fail and which stages collapse), and (iii) the completeness theorem (what does the existence of your two models imply about any identification method?).

Chapter 9

Sensitivity Analysis and Partial Identification

Learning Objectives

By the end of this chapter, students should be able to:

1. Distinguish sampling, model, and identification uncertainty, and explain why ordinary confidence intervals do not assess identifying assumptions.
2. Formulate and interpret a sensitivity curve, a sensitivity interval, and a tipping point.
3. Use a bias decomposition to analyze sensitivity to unmeasured confounding, and benchmark the sensitivity parameters against observed covariates on an estimator-matched, common scale.
4. Explain the assumptions and interpretations of the Rosenbaum Γ, E-value, and marginal sensitivity models.
5. Distinguish point from partial identification, derive Manski's bounded-outcome ATE bounds, and explain how positivity failures motivate retargeting.
6. Explain how sensitivity analysis applies to IV and mediation designs, and why robust estimation does not repair failed identification.

Proofs, derivations, extended lab material, and additional problems marked here as optional are collected in the separate chapter supplement; nothing needed for a first reading depends on it.

How to read this chapter. The core first reading comprises all of Section 9.1 (uncertainty and the general framework), the bias decomposition of Section 9.2, the E-value (Section 9.3.2), benchmarking and reporting (Section 9.4), partial identification and overlap (Section 9.5), and the boundary with modern estimation (Section 9.7). The Rosenbaum Γ-model (Section 9.3.1) and the marginal sensitivity model (Section 9.3.3) are specialized method capsules; the IV and mediation material (Section 9.6) gives design-specific applications; and the lab (Section 9.8) is optional computational material.

9.1 Why Sensitivity Analysis?

Chapters 5–8 developed a sequence of identification strategies: back-door adjustment, propensity-score re-weighting, instrumental variables, and front-door and mediation analysis. Each strategy identifies a causal parameter as a functional of the observed-data distribution under a corresponding set of assumptions. Each set of assumptions is, however, stated in terms of unobserved quantities — potential outcomes, hypothetical interventions, or unmeasured variables — and cannot be verified from the data alone.

The chapter ahead (Chapter 10) opens by noting that *identification does not by itself provide a statistically reliable estimator*: once a parameter is identified, a separate theory of estimation and inference is still needed. The present chapter makes the complementary point, which closes out Part II:

> *Statistical reliability does not by itself validate identification.*

A confidence interval reports sampling variation *conditional on* a maintained identification assumption. It is silent about whether that assumption is correct. A causal analysis is credible only if both components

— identification and estimation — are credible. Sensitivity analysis is the tool for reporting how much the conclusion depends on the first.

9.1.1 Sampling Uncertainty vs. Identification Uncertainty

Consider an analyst who reports $\hat{\tau} = 2.0$ with 95% CI $[1.2,\ 2.8]$. This interval answers one question and only one:

> *If the identifying assumptions are correct, what range of values of τ is consistent with the observed sampling variation?*

As $n \to \infty$ the interval will shrink around τ, provided the identifying assumptions hold. If the assumptions do not hold, the interval will instead shrink around a biased target. More data drawn from the same observational regime do not remove this bias: the identification gap can only be closed by adding new assumptions, new design information, or new measurements relevant to the source of nonidentification — validation data, randomized encouragement, negative controls, instrumental variables, repeated measurements, or richer confounder measurements. A real-data illustration of the negative-control idea appears in the college-selectivity case study of Chapter 5: the average SAT score of the colleges that *rejected* a student — a variable with no causal path to the student's earnings — strongly predicts earnings, directly exposing the confounding in the unadjusted comparison (Dale and Krueger 2002).

Formally, we may distinguish:

- **Sampling uncertainty:** $\hat{P}_n \neq P$. The empirical distribution differs from the population. This uncertainty vanishes as $n \to \infty$.
- **Model misspecification:** the working parametric or semiparametric model for a nuisance function (say the outcome regression $\mu_t(x)$ or the propensity score $\pi(x)$) is not correct. This uncertainty is partly addressable by flexible modeling, doubly robust construction, and cross-fitting (Chapters 10–12).
- **Identification uncertainty:** the causal parameter ψ is not in fact identified by the functional $\Psi(P)$ assumed by the analysis. This uncertainty does not vanish merely by increasing the sample size from the same observational regime.

The three sources of error compose. The usual confidence interval addresses only the first; the estimation chapters to come address primarily the second; this chapter addresses the third.

> Warning: Statistical Significance and Causal Credibility
>
> A narrow, highly significant confidence interval is not evidence that the identifying assumptions are correct. It is evidence of a conditional claim: *if* the assumptions are correct, *then* ψ is close to the estimate. Nothing in the interval speaks to whether the identifying assumption is true. A sensitivity analysis cannot verify that antecedent; it reports how strongly the conclusion depends on scientifically plausible violations of it.

9.1.2 The Role of Sensitivity Analysis

Sensitivity analysis is a structured way to vary the strength of violations of identifying assumptions and to report how the causal conclusion responds. It is not a test of the assumptions. Identifying assumptions are generally not *fully verifiable* from the observed outcome data, although some of their observed-data implications can be assessed — a proposed DAG may imply testable conditional independences, overidentified models imply moment restrictions, negative controls can falsify aspects of a design, known randomization protocols supply design information, and positivity can be partly diagnosed empirically. It is a statement of *conditional robustness*: how far can the assumptions be violated before the substantive conclusion changes?

Three interpretations of the same analysis are useful, and this chapter develops each:

- *Sensitivity curve.* A plot of the estimand against a scalar sensitivity parameter λ, with the neutral value $\lambda = \lambda_0$ recovering the baseline identifying assumption.
- *Sensitivity bounds.* The range of estimand values consistent with any λ in a specified plausibility set $\mathcal{L}$.

- *Tipping point.* The smallest violation magnitude sufficient to change the sign, magnitude, or statistical significance of the conclusion.

Remark: Sensitivity Analysis Disciplines the Discussion

Sensitivity analysis also has a second role, beyond robustness reporting: it disciplines the scientific discussion. Forcing the analyst to write down a specific sensitivity parameter and a plausibility set $\mathcal{L}$ forces explicit statements about what unmeasured confounders, instrument-validity violations, or cross-world dependencies could plausibly be present. A sensitivity analysis without a defensible $\mathcal{L}$ is a calculation, not an argument.

9.1.3 The General Framework

The three views of sensitivity analysis — curves, bounds, tipping points — all arise from a single construction. Let ψ be the causal estimand of interest, and let A_0 be a baseline identifying assumption such that, under A_0, $\psi = \Psi(P)$ for a known functional Ψ. A *sensitivity model* is an indexed relaxation $\{A_\lambda : \lambda \in \mathcal{L}\}$ of A_0, with A_0 recovered at a neutral value $\lambda_0 \in \mathcal{L}$. In the simplest, point-valued case, the estimand under A_λ is a function of both the observed-data distribution and the sensitivity parameter:

$$\psi(\lambda) = \Psi(P;\, \lambda), \qquad \Psi(P;\, \lambda_0) = \Psi(P); \tag{9.1}$$

the general case, in which a fixed λ pins ψ down only to a set, is developed in Section 9.5. The key reporting object is therefore not a single number but the attainable set

$$\{\psi(\lambda) : \lambda \in \mathcal{L}\}. \tag{9.2}$$

Across this chapter we will encounter several sensitivity parameters, each quantifying the magnitude of a specific violation:

- λ = strength of unmeasured confounding, parameterized either as an odds-ratio bound, as a pair of association strengths in the VanderWeele–Ding E-value framework, or as an odds-ratio bound in the marginal sensitivity model.
- δ = magnitude of a structural outcome-effect coefficient that the baseline identifying assumption treats as inert. Used in two contexts: the U-outcome effect in the linear and binary-confounder sensitivity models (Section 9.2.2–Section 9.2.3); and the direct effect of an instrument on the outcome violating the IV exclusion restriction (Section 9.6), where $\delta = 0$ is the baseline assumption. The two roles never appear together.
- α = propensity-score trimming threshold, addressing near-violations of positivity (Section 9.5.6).
- ρ = residual correlation between mediator and outcome disturbances, violating sequential ignorability (Section 9.6.4).

In each case the neutral value returns the baseline identifying assumption — $\Gamma = 1$ and $\Lambda = 1$ for the odds-ratio confounding bounds, and $\delta = 0$, $\alpha = 0$, $\rho = 0$ for the remaining parameters — and the distance from the neutral value quantifies the strength of the violation.

One entry in this list is of a different kind. The trimming threshold α does not relax an identifying assumption while holding the causal estimand fixed; it indexes the *retargeted* estimands $\tau_\alpha = \mathbb{E}\{Y(1) - Y(0) \mid \alpha \le \pi(X) \le 1-\alpha\}$. We therefore treat trimming as *target sensitivity* — sensitivity of the conclusion to the choice of population — rather than as an instance of the fixed-estimand sensitivity model.

A related notational point concerns the two capital lambdas. We write $\mathcal{L}$ for the *plausibility set* in which the sensitivity parameter is assumed to lie, and reserve the upright Λ for the scalar *level* of the marginal sensitivity model (Section 9.3.3), where it is standard in the literature (Tan 2006; Zhao et al. 2019). The two are different objects: $\mathcal{L}$ is a set of admissible parameter values, while Λ is one particular sensitivity parameter, whose own plausibility set is an interval $[1, \Lambda_{\max}] \subseteq \mathcal{L}$.

The symbol α appears elsewhere in this chapter in two unrelated roles: as a regression intercept in the linear outcome and IV models, and, in the lab, as the subscripted parameter α_U denoting the strength of U in the treatment mechanism. Similarly, in Section 9.6.3 the symbols π_c and π_d denote complier and defier population shares; they are unrelated to the propensity score $\pi(x)$.

Definition 9.1 (Sensitivity Model)**.** A *sensitivity model* for the causal estimand ψ and baseline identifying assumption A_0 is a triple $(\{A_\lambda\}_{\lambda\in\mathcal{L}}, \lambda_0, \Psi(\cdot;\cdot))$ such that (i) $\lambda_0 \in \mathcal{L}$ is a *neutral value* at which $A_{\lambda_0} = A_0$, recovering the original identifying assumption; and (ii) under A_λ and the observable restrictions imposed by the data, $\psi = \Psi(P;\lambda)$. The parameter λ may be scalar- or vector-valued.

Several of the models in this chapter pin ψ down only to a *set* at each λ — the MSM at fixed Λ delivers an interval for the estimand, a Rosenbaum analysis at fixed Γ bounds a randomization p-value, and the bias factor of Section 9.3.2 is two-dimensional before its scalar summary. The set-valued extension of this framework is developed in Section 9.5, where it merges naturally with partial identification.

Two further comments on the definition. First, the neutral value is not always zero. It is $\lambda_0 = 0$ for the additive parameters used in this chapter (δ, ρ, α_U) but $\lambda_0 = 1$ for the multiplicative ones (Γ, Λ), since those are odds-ratio bounds for which no violation means a ratio of one. Writing λ_0 rather than 0 keeps the framework honest about this.

Second, continuity of $\Psi(\cdot;\lambda)$ in λ is deliberately *not* part of the definition. It is a convenient regularity condition — it makes sensitivity curves drawable and tipping points well defined — and it holds for most models in this chapter. But bound endpoints obtained from an optimization can be nonsmooth, and a sensitivity region can change discretely as constraints become active, so continuity should be imposed where it is needed rather than assumed throughout.

9.1.3.1 Sensitivity Curves

When $\mathcal{L} \subseteq \mathbb{R}$ is one-dimensional, the natural reporting object is the *sensitivity curve* $\lambda \mapsto \hat{\psi}(\lambda)$, where $\hat{\psi}(\lambda)$ is a consistent estimator of $\psi(\lambda)$ under A_λ. A sensitivity curve is useful when the reader wishes to see the effect smoothly deform as the assumption is relaxed. Many sensitivity curves are monotone — the bias moves in one direction as λ increases — though this is not guaranteed.

9.1.3.2 Sensitivity Bounds

When multiple parameters index the violation, or when the sensitivity parameter is high-dimensional, reporting a full curve is infeasible. The natural alternative is to report the extremes over $\mathcal{L}$. One caveat should be recorded first: because continuity is not part of Definition 9.1, the image $\{\psi(\lambda) : \lambda \in \mathcal{L}\}$ need not be an interval. The bounds below are therefore an *enclosure* of that set; intermediate values inside $[\psi_L, \psi_U]$ need not be attainable by any admissible λ. With that caveat recorded, define

$$\psi_L = \inf_{\lambda\in\mathcal{L}} \psi(\lambda), \qquad \psi_U = \sup_{\lambda\in\mathcal{L}} \psi(\lambda). \tag{9.3}$$

The interval $[\psi_L, \psi_U]$ is the *sensitivity interval.* In practice, one estimates the lower and upper bounds and reports them with sampling-based uncertainty intervals; we caution that inference for the endpoints of a partially identified set is more delicate than inference for an ordinary scalar parameter, and the appropriate construction depends on whether the target is the identified set or the parameter itself.

9.1.3.3 Tipping-Point Analysis

A third summary is the smallest λ at which the substantive conclusion changes. If the conclusion of interest is "the effect is positive," the natural tipping point is

$$\lambda^\star = \inf\{\lambda \in \mathcal{L} : \psi(\lambda) \le 0\}, \tag{9.4}$$

or, if inference is at issue rather than the point estimate,

$$\lambda^\star_{\mathrm{CI}} = \inf\{\lambda \in \mathcal{L} : 0 \in \widehat{\mathrm{CI}}(\lambda)\}, \tag{9.5}$$

where $\widehat{\mathrm{CI}}(\lambda)$ is the confidence interval for $\psi(\lambda)$. Note that $\psi(\lambda)$ in Equation 9.4 is the pointwise value at λ, distinct from the global bound ψ_L of Equation 9.3.

Three cautions attach to Equation 9.4. First, the definition is deliberately *directional*: it is stated for the conclusion "the effect is positive," and the mirror image applies for a negative one. Second, writing $\psi(\lambda) \le 0$ rather than $\psi(\lambda) = 0$ avoids requiring an exact crossing, which need not exist once continuity is not assumed. Third, and most important, the infimum presupposes an *ordered, nested* scalar scale: $\lambda' > \lambda$ must mean that $A_{\lambda'}$ is a strictly weaker assumption, so that larger λ is unambiguously "more violation."

For a vector-valued λ, or a scalar one whose sensitivity sets are not nested, there is no unique smallest violation until one fixes a norm, a path through the parameter space, or a partial order — and that choice is a substantive modeling decision, not a technicality. If ψ is not monotone, $\lambda^\star$ is only the *first* crossing and should be reported as such.

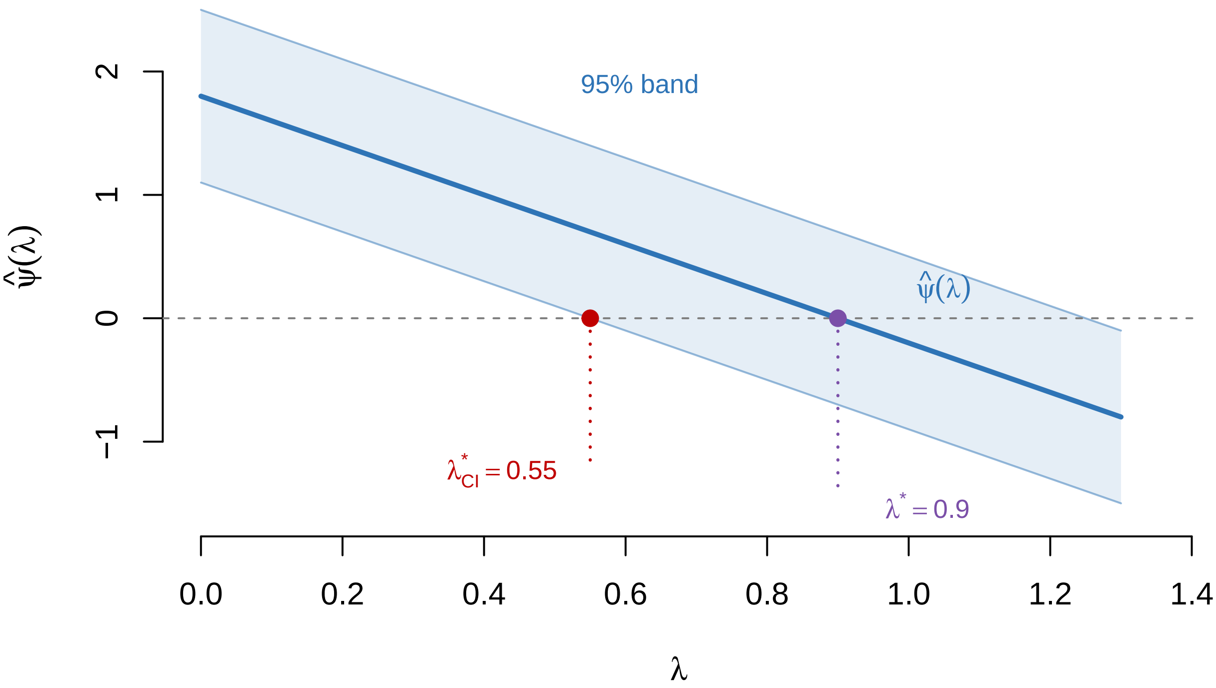

Figure 9.1: A sensitivity curve and its two tipping points, for the linear model of the example below: $\hat{\psi}(\lambda) = 1.8 - 2.0\lambda$ with a 95% band of half-width 0.7. The lower confidence limit reaches zero first, at $\lambda^\star_{\text{CI}} = 0.55$; the point estimate reaches zero at $\lambda^\star = 0.90$, so the significance tipping point is the stricter of the two — as it must be for any interval procedure that contains its point estimate at each λ.

Example 9.1 (Tipping-Point Reading)**.** Suppose $\hat{\psi}(0) = 1.8$ with 95% CI $[1.1, 2.5]$, and the sensitivity model produces $\hat{\psi}(\lambda) = 1.8 - 2.0\lambda$. Then:

- The sensitivity curve is linear in λ.
- The tipping point for the sign is $\lambda^\star = 0.9$.
- The tipping point for significance is $\lambda^\star_{\text{CI}}$, the value at which the lower endpoint of the CI — which in this simple linear model equals $1.1 - 2.0\lambda$ — hits zero, so $\lambda^\star_{\text{CI}} = 0.55$.

The significance tipping point is always at least as strict as the sign tipping point. An analysis is robust "at level $\lambda^\star$" for a conclusion that holds throughout $[0, \lambda^\star)$.

Remark: One Parameter Is a Design Choice, Not a Constraint

Most sensitivity models in this chapter have $\mathcal{L} \subseteq \mathbb{R}$ for the deliberate reason that a one-dimensional summary is reportable and interpretable. A multi-parameter model can be summarized along a chosen one-dimensional path through $\mathcal{L}$. This is a *conditional* sensitivity analysis, not a worst-case analysis over the full parameter region: because the path is only a subset of $\mathcal{L}$, its most adverse value is generally *less* extreme than the worst case over the entire region. The path therefore encodes a substantive tradeoff among the sensitivity components and must be reported explicitly. When a path summary does deliver the full-region worst case — as with the symmetric diagonal underlying the E-value — the worst-case reading is earned by additional structure, there the monotonicity of the bias factor in each argument, and not by the choice of path itself. The Ding–VanderWeele bias factor is genuinely a *two*-parameter model, and the E-value is a one-number summary of it, not the model itself.

9.2 Sensitivity to Unmeasured Confounding

The most common application of sensitivity analysis is to unmeasured confounding under the back-door framework of Chapter 5. The baseline identifying assumption is conditional exchangeability, $Y(t) \perp\!\!\!\perp T \mid X$, together with consistency and positivity. Under these,

$$\tau = \mathbb{E}\{\mu_1(X) - \mu_0(X)\}, \qquad \mu_t(x) = \mathbb{E}(Y \mid T = t, X = x). \tag{9.6}$$

Sensitivity analysis here contemplates a world in which there exists an unobserved U such that

$$Y(t) \perp\!\!\!\perp T \mid X, U, \qquad \text{but} \qquad Y(t) \not\perp\!\!\!\perp T \mid X. \tag{9.7}$$

In words: ignorability would hold if U were observed, but U is not observed, so the working identification formula Equation 9.6 is biased. The structure is the familiar back-door path $T \leftarrow U \rightarrow Y$ that adjustment for X alone cannot block.

9.2.1 The Bias Decomposition

Let

$$\Delta_{\text{obs}} = \mathbb{E}\{\mathbb{E}(Y \mid T = 1, X) - \mathbb{E}(Y \mid T = 0, X)\} \tag{9.8}$$

denote the *observed adjusted contrast* — the quantity that a back-door analysis actually computes when only X is adjusted for. Under ignorability conditional on X alone, $\Delta_{\text{obs}} = \tau$. Under the relaxed condition Equation 9.7, the observed adjusted contrast departs from τ by a quantity that depends on how U is distributed conditional on (T, X).

Theorem 9.1 (Bias Decomposition under Unmeasured Confounding)**.** *Suppose Equation 9.7 holds along with consistency and positivity* given (X, U), *that is* $P(T = t \mid X, U) > 0$ *almost surely for* $t \in \{0, 1\}$ *— the stronger requirement needed for* m_t *below to be well defined, and not implied by positivity given* X *alone. Let* $f(u \mid \cdot)$ *denote the conditional density (or mass function) of* U*, define*

$$m_t(x, u) = \mathbb{E}(Y \mid T = t, X = x, U = u),$$

and the distributional difference

$$g_t(x, u) = f(u \mid T = t, X = x) - f(u \mid X = x).$$

Then $\Delta_{\text{obs}} = \tau + B$*, where the* confounding bias *is*

$$B = \mathbb{E}\left\{\sum_t (-1)^{1-t} \int m_t(X, u)\, g_t(X, u)\, du\right\}, \tag{9.9}$$

with the integral replaced by a sum if U *is discrete.*

Proof sketch. Condition on (X, U); conditional exchangeability given (X, U) identifies $m_t(X, u)$ from the corresponding treatment arm, and the observed contrast differs from the causal one exactly through the distributional differences g_t, which yields Equation 9.9 upon rearranging. The full derivation is in the optional supplement. □

Equation Equation 9.9 is the master formula underlying most sensitivity models for unmeasured confounding. Each model amounts to a parameterization of the two quantities $m_t(x, u)$ and $g_t(x, u)$: the effect of U on Y, and the imbalance of U across treatment arms given X.

> Remark: Bias Decomposition in Words
>
> The observed adjusted contrast is the true effect plus a term controlled by *the U-outcome association* $\times$ *the U-treatment imbalance.* Every sensitivity model in Section 9.3 is a formal version of this slogan.

9.2.2 A Simple Linear Sensitivity Model

A closed-form version of Equation 9.9 is available in a fully linear model, which is the form most often encountered in econometrics and the one easiest to benchmark. Suppose the outcome follows

$$Y = \alpha + \tau T + \gamma^\top X + \delta U + \varepsilon, \qquad \mathbb{E}(\varepsilon \mid T, X, U) = 0, \tag{9.10}$$

and the treatment assignment depends on (X, U) through $T = h(X, U, \eta)$ with $\eta \perp\!\!\!\perp (U, \varepsilon)$.

Lemma 9.1 (Linear Sensitivity Bias)**.** *Under the model above and conditional exchangeability given* (X, U)*,*

$$B = \delta \cdot \mathbb{E}\{\mathbb{E}(U \mid T = 1, X) - \mathbb{E}(U \mid T = 0, X)\}\,. \tag{9.11}$$

Proof. Specializing Equation 9.9 to the linear outcome model, $m_t(x, u) = \alpha + \tau t + \gamma^\top x + \delta u$, so the inner integrals reduce to $\delta \int u\, g_t(X, u)\, du$, because the components of m_t that do not depend on u multiply integrals of $g_t(X, u)$, which is 0 by construction. Combining,

$$B = \delta\, \mathbb{E}\left\{\int u\, [g_1(X, u) - g_0(X, u)]\, du\right\} = \delta\, \mathbb{E}\{\mathbb{E}(U \mid T = 1, X) - \mathbb{E}(U \mid T = 0, X)\}. \qquad \square$$

Formula Equation 9.11 is the canonical teaching decomposition. Under the linear outcome model Equation 9.10 it is an exact identity, not an approximation; approximate language is appropriate only when Equation 9.10 is being used as a local working approximation to a nonlinear outcome model:

$$\text{bias} \;=\; \underbrace{\delta}_{U\text{-outcome effect}} \times \underbrace{\mathbb{E}\{\mathbb{E}(U \mid T = 1, X) - \mathbb{E}(U \mid T = 0, X)\}}_{U\text{-treatment imbalance}}. \tag{9.12}$$

Both factors are sensitivity parameters because neither is identified: U is unobserved, so its effect on Y and its imbalance across treatment arms can only be posited, not estimated.

9.2.3 Binary Unmeasured Confounder

A useful concrete case arises when $U \in \{0, 1\}$. Define $p_1(x) = P(U = 1 \mid T = 1, X = x)$, $p_0(x) = P(U = 1 \mid T = 0, X = x)$, and $\delta(t, x) = m_t(x, 1) - m_t(x, 0)$, the U-outcome contrast within the cell defined by (T, X).

Lemma 9.2 (Binary-Confounder Bias)**.** *If* $U \in \{0, 1\}$ *and the* U*-outcome contrast is constant across* (t, x) *at value* δ*, then*

$$B = \delta \cdot \mathbb{E}\{p_1(X) - p_0(X)\}. \tag{9.13}$$

Proof. With $U \in \{0, 1\}$, $m_t(x, u) = m_t(x, 0) + u\, \delta(t, x)$. The constant-contrast assumption gives $\delta(t, x) = \delta$. Applying Equation 9.9 and using that $g_t(x, u)$ integrates to 0 in u,

$$\int m_t(X, u)\, g_t(X, u)\, du = \delta\, [P(U{=}1 \mid T{=}t, X) - P(U{=}1 \mid X)],$$

so $B = \delta\, \mathbb{E}\{p_1(X) - p_0(X)\}$. $\square$

Formula Equation 9.13 is the workhorse of applied sensitivity analysis under the back-door design. It has two sensitivity parameters — the U-outcome effect δ and the mean U-treatment imbalance — each of which has a concrete interpretation. A sensitivity analysis then consists of computing $\hat{\tau}_{\text{sens}} = \hat{\Delta}_{\text{obs}} - \hat{B}$ over a grid of values and reporting the resulting surface, contours, tipping points, or bounds.

9.2.4 A Sensitivity Table

For reporting, a simple table tabulates the adjusted effect over a grid of sensitivity parameters, in the style of Rosenbaum (2002) and VanderWeele and Ding (2017), for an observed adjusted effect $\hat{\Delta}_{\text{obs}} = 2.0$.

Table 9.1: Sensitivity table for a binary unmeasured confounder. Observed adjusted effect $\hat{\Delta}_{\text{obs}} = 2.0$; cells show $\hat{\tau}_{\text{sens}} = \hat{\Delta}_{\text{obs}} - \delta \cdot (p_1 - p_0)$.

U-outcome effect δ	$p_1 - p_0 = 0.10$	0.30	0.50
$\delta = 2$	1.80	1.40	1.00
$\delta = 4$	1.60	0.80	0.00
$\delta = 8$	1.20	−0.40	−2.00

The table reads straightforwardly: the effect remains clearly positive when the posited U-outcome effect is small relative to the outcome scale, is substantially attenuated once δ and the imbalance are both moderate on that scale, and reverses when both are large. (Verbal labels such as "weak" or "strong" have no meaning without specifying the scale of Y, which is why the rows are indexed by δ itself.) The tipping point for the sign is not a single cell but the contour $\delta \cdot (p_1 - p_0) = 2.0$, a hyperbola in the $(\delta,\, p_1 - p_0)$ plane. The grid lands exactly on that contour at $(\delta = 4,\, p_1 - p_0 = 0.5)$; the cell $(\delta = 8,\, p_1 - p_0 = 0.3)$ lies just beyond it. The analyst then asks whether an unmeasured U anywhere near this strong is scientifically plausible — a question addressed through benchmarking in Section 9.4.

9.3 Three Canonical Sensitivity Models

The bias decomposition of Section 9.2.1 is a general identity. Three canonical *models* specialize it by imposing a structural restriction on either g_t (the U-treatment mechanism) or on the induced observed-data relationships. Each model is a named instance of the framework of Section 9.1.3 with a specific choice of sensitivity parameter.

We present the three models in the sequence of matched-design sensitivity analysis, risk-ratio-scale sensitivity summaries, and weighting-based sensitivity intervals. This is a sequence of settings, not an ordering by generality — the E-value is not tied to an estimator in the way the other two are.

9.3.1 Rosenbaum's Γ-Sensitivity Model

Rosenbaum (2002) formulated sensitivity analysis for matched observational studies in terms of an odds-ratio bound on the conditional treatment probabilities. Define the conditional treatment odds given (X, U) and the *sensitivity parameter*

$$\Gamma = \sup_{x,u,u'} \frac{\text{odds}(T = 1 \mid X = x, U = u)}{\text{odds}(T = 1 \mid X = x, U = u')}. \tag{9.14}$$

In words, Γ bounds the factor by which any two units with the same observed covariates can differ in their odds of treatment because of unobserved U.

Definition: Γ-Sensitivity Model

The Γ-sensitivity model at level $\Gamma \geq 1$ is the set of treatment-assignment mechanisms $P(T \mid X, U)$ that satisfy

$$\frac{1}{\Gamma} \leq \frac{P(T = 1 \mid X, U = u)/P(T = 0 \mid X, U = u)}{P(T = 1 \mid X, U = u')/P(T = 0 \mid X, U = u')} \leq \Gamma \tag{9.15}$$

for all x and all u, u'. $\Gamma = 1$ says that the treatment odds do not vary with the represented U within levels of X. That is equivalent to conditional exchangeability given X only when U stands for *all* hidden assignment variation relevant to the potential outcomes and the matched-design assignment model is correctly specified; read literally, it is the weaker statement about odds.

The parameter Γ is the sensitivity parameter in the sense of Definition 9.1, with $\lambda = \Gamma$, plausibility set $\mathcal{L} = [1, \infty)$, and neutral value $\lambda_0 = 1$. A value $\Gamma = 2$ states that units with the same X may differ by up to a factor of 2 in their treatment odds because of unmeasured U.

Under the Γ-model — testing Fisher's sharp null of no effect (or a specified constant additive effect), with assignments independent across matched pairs — worst-case null distributions for matched-pair test statistics are obtained by pushing every pair-specific treatment probability to the endpoints $1/(1+\Gamma)$ and $\Gamma/(1+\Gamma)$ permitted by Equation 9.15. For the sign test the resulting bounds are binomial; for Wilcoxon's signed-rank and other signed-score statistics they are distributions of rank-weighted sums of independent Bernoulli variables. The formal statement and its proof are in the optional supplement.

The practical outcome of a Γ-analysis is a tipping $\Gamma^\star$: the smallest Γ at which the test of no effect fails to reject. This value is not a property of the data alone. It is defined only relative to a stated test statistic, a stated worst-case direction, a one- or two-sided alternative, and a significance level; changing any of these changes $\Gamma^\star$, and all four should be reported alongside it. Small $\Gamma^\star$ (near 1) indicates a fragile conclusion; large $\Gamma^\star$ indicates a robust one.

Remark: Connection to Matched-Pair and Weighted Estimators

The Γ-model is most natural for matched-pair designs (Chapter 6) because the pair structure absorbs the X-conditioning exactly. Extensions to propensity-score stratification and IPW exist but involve additional bookkeeping; the marginal sensitivity model below is the cleanest framework for IPW and AIPW.

9.3.2 The E-Value and the VanderWeele–Ding Bound

Ding and VanderWeele (2016) and VanderWeele and Ding (2017) proposed a sensitivity summary that requires no matched design, takes the form of a single closed-form number, and can be read off a reported risk ratio, subject to the scope restrictions recorded at the end of this subsection. It has become the most widely reported sensitivity summary in the applied literature.

The setup uses two risk ratios as sensitivity parameters. Within a covariate stratum x, define

$$\mathrm{RR}_{TU}(x) = \sup_u \frac{dP(u \mid T=1, X=x)}{dP(u \mid T=0, X=x)}, \tag{9.16}$$

$$\mathrm{RR}_{UY}(x) = \max_{t\in\{0,1\}} \frac{\sup_u P(Y=1 \mid T=t, X=x, U=u)}{\inf_u P(Y=1 \mid T=t, X=x, U=u)}. \tag{9.17}$$

Write RR_{TU} and RR_{UY} for the maxima of these over x. These are, respectively, the maximum relative-risk association of U with T and the maximum relative-risk association of U with Y given (T, X); both are at least 1.

The force of the Ding–VanderWeele result is that with the parameters defined this generally, the bias-factor bound below requires *no* further structure: U need not be binary, need not be a single variable, and may interact with T in its effect on Y. That generality is why the E-value is usable in practice, where nothing is known about U. A transparent special-case derivation (binary U, no $T \times U$ interaction) appears in the optional supplement.

Let $\mathrm{RR}^{\mathrm{obs}}_{TY|X}(x)$ denote the observed risk ratio within stratum $X = x$, and $\mathrm{RR}^{\mathrm{true}}_{TY|X}(x)$ the corresponding causal risk ratio. Define the *bias factor*

$$B(\mathrm{RR}_{TU}, \mathrm{RR}_{UY}) = \frac{\mathrm{RR}_{TU}\,\mathrm{RR}_{UY}}{\mathrm{RR}_{TU} + \mathrm{RR}_{UY} - 1}. \tag{9.18}$$

Theorem 9.2 (VanderWeele–Ding Bound)**.** *Assume consistency; conditional exchangeability given the full confounder set, $\{Y(0), Y(1)\} \perp\!\!\!\perp T \mid X, U$; positivity of $P(T = t \mid X, U)$; and the binary-outcome risk-ratio setup above, with $P(U \mid T{=}1, X{=}x)$ absolutely continuous with respect to $P(U \mid T{=}0, X{=}x)$ so that Equation 9.16 is finite. Then for every x,*

$$\mathrm{RR}^{\mathrm{true}}_{TY|X}(x) \;\geq\; \frac{\mathrm{RR}^{\mathrm{obs}}_{TY|X}(x)}{B(\mathrm{RR}_{TU}, \mathrm{RR}_{UY})}. \tag{9.19}$$

Write $R = \mathrm{RR}^{\mathrm{obs}}_{TY|X}(x) \geq 1$ and define the E-value

$$\mathrm{EV} = R + \sqrt{R\,(R-1)}. \tag{9.20}$$

Then EV *is the smallest common value e such that the symmetric configuration* $\mathrm{RR}_{TU} = \mathrm{RR}_{UY} = e$ *drives the bound Equation 9.19 to the null. Equivalently,*

$$\mathrm{EV} = \min\{\max(\mathrm{RR}_{TU}, \mathrm{RR}_{UY}) : B(\mathrm{RR}_{TU}, \mathrm{RR}_{UY}) \geq R\}. \tag{9.21}$$

Consequently, for the observed risk ratio to be explained away completely, both *associations must be at least R and* at least one *must be at least* EV*:*

$$\min(\mathrm{RR}_{TU}, \mathrm{RR}_{UY}) \geq R, \qquad \max(\mathrm{RR}_{TU}, \mathrm{RR}_{UY}) \geq \mathrm{EV}. \tag{9.22}$$

A transparent closed-form derivation of Equation 9.19, the E-value formula, the min–max characterization, and the Cornfield pair — for the special case of a binary U with no $T \times U$ interaction — is given in the optional supplement.

The E-Value Bounds the Larger Association

It is a common and consequential misreading to say that an association can be explained away only if *both* RR_{TU} and RR_{UY} exceed the E-value. That condition is *sufficient* — if both exceed EV then $B \geq R$ — but it is not *necessary*. The correct pair of necessary conditions is Equation 9.22: both associations must be at least the observed risk ratio R, and at least one must be at least EV. Asymmetric pairs with one component well below the E-value can suffice: at $R = 2.5$, where $\mathrm{EV} = 4.44$, the pair $(\mathrm{RR}_{TU}, \mathrm{RR}_{UY}) = (3.00, 10.00)$ gives $B = 2.5$ and explains the effect away even though $3.00 < \mathrm{EV}$. Note that 3.00 still exceeds $R = 2.5$, as Equation 9.22 requires.

Remark: Standardizing over X

Theorem 9.2 is a *stratum-specific* statement: it bounds the ratio of observed to causal risk ratios within each level of X. It does not follow that the same bound applies to the standardized (marginal) risk ratios obtained by averaging $\mathbb{E}\{Y(t) \mid X\}$ over the distribution of X. A ratio of X-averaged risks is not obtained by simply averaging the X-specific risk ratios, and the relevant weights can differ between the observed and causal quantities, so a separate argument is required; numerical counterexamples satisfying the stratum-wise bound at every x while violating the standardized version are easy to construct. In applications the E-value is therefore most defensible when read at the level at which the risk ratio was estimated — typically a covariate-adjusted model-based risk ratio taken as approximately constant in X.

Example 9.2 (An E-Value Calculation)**.** An observational study reports an adjusted risk ratio of 2.5 with 95% CI $[1.7, 3.7]$. The E-value for the point estimate is

$$\mathrm{EV} = 2.5 + \sqrt{2.5 \cdot 1.5} = 2.5 + \sqrt{3.75} \approx 4.44.$$

The E-value for the lower confidence limit is $1.7 + \sqrt{1.7 \cdot 0.7} \approx 2.79$.

Interpretation: to reduce the point estimate to the null, an unmeasured confounder must have *both* risk-ratio associations — with treatment, and with the outcome given treatment — at least 2.5, and at least one of them at least 4.44. The symmetric configuration $\mathrm{RR}_{TU} = \mathrm{RR}_{UY} = 4.44$ is sufficient but not necessary: the asymmetric pair (3.00, 10.00) also explains the point estimate away. For the lower confidence limit the corresponding thresholds are 1.7 and 2.79.

Remark: Why a Single Number

Theorem 9.2 bounds the bias factor over the *product space* of $(\mathrm{RR}_{TU}, \mathrm{RR}_{UY})$. The E-value collapses the two parameters to one by asking about the symmetric diagonal, and by Equation 9.21 the resulting number is the smallest achievable value of the larger of the two associations. It is a genuine necessary threshold for that larger association. It does not, however, say nothing about the smaller one: by Equation 9.22 the smaller association must still be at least the observed risk ratio R. Asymmetric configurations with one parameter large and the other near R can also explain away the effect, and the full bivariate bound is more informative than either threshold alone. The E-value is reported because it is a single interpretable scalar, not because it is the tightest possible summary.

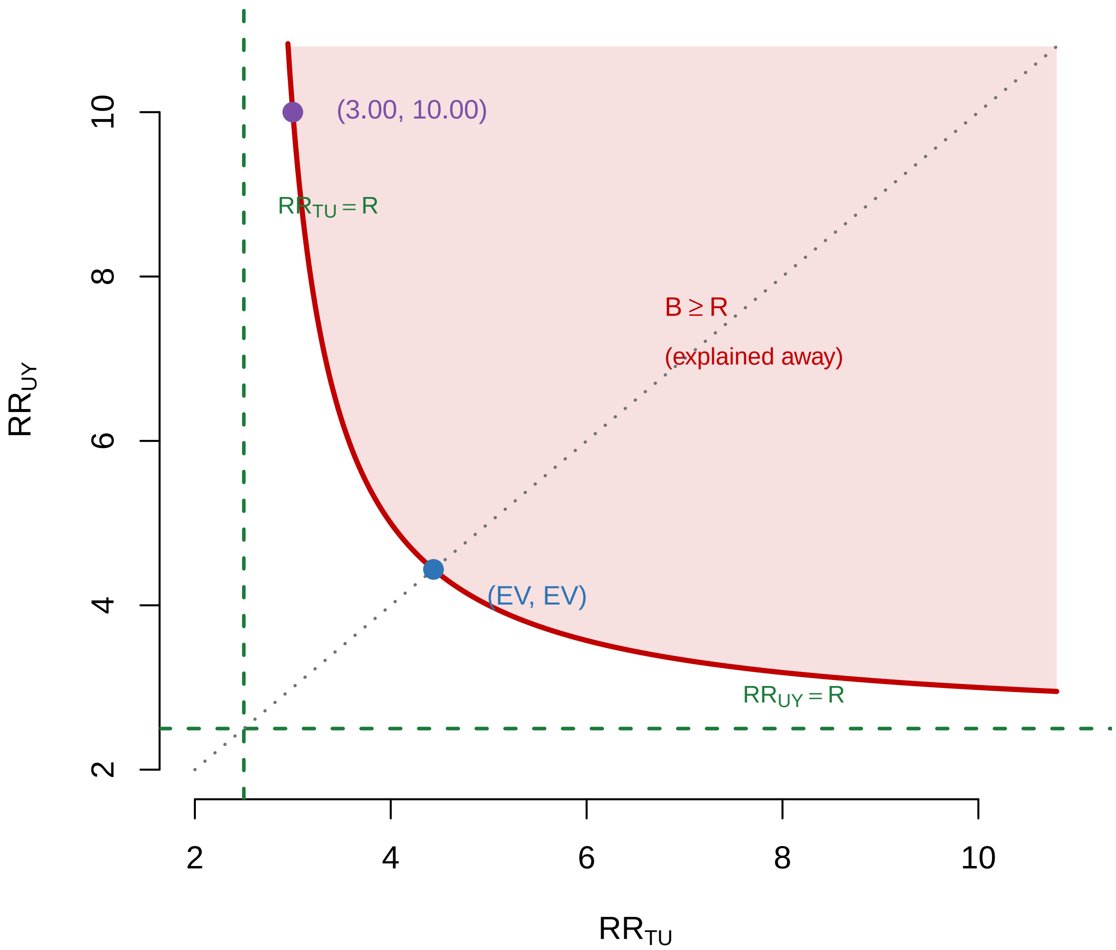

Figure 9.2: The two-parameter bias-factor region for an observed risk ratio $R = 2.5$, for which $\mathrm{EV} = 4.44$. The shaded set is the collection of pairs able to explain the association away; its boundary is the hyperbola $\mathrm{RR}_{UY} = R(\mathrm{RR}_{TU} - 1)/(\mathrm{RR}_{TU} - R)$. The dashed lines are the Cornfield thresholds. The E-value is where the region meets the symmetric diagonal (dotted), and is the smallest attainable value of the larger association. The point $(3.00, 10.00)$ lies inside the region with first coordinate well below EV — which is why 'both associations must reach the E-value' is false, while 'both must reach R' is true.

Scope of the E-value formula. Equation Equation 9.20 is stated for a risk ratio $R \geq 1$. For a protective effect, $R < 1$, the E-value is computed from $1/R$ and interpreted on that scale. For odds ratios and hazard ratios the formula is not automatically exact: an odds ratio approximates the corresponding risk ratio only under a rare-outcome condition, and a hazard ratio carries additional causal interpretive difficulties (selection over time among survivors) that the bias factor of Theorem 9.2 does not address. Applying the E-value to these measures requires the conversions and caveats given by VanderWeele and Ding (2017), not a direct substitution into Equation 9.20.

9.3.3 The Marginal Sensitivity Model

The Γ-model is formulated in terms of the unobservable conditional distribution $P(T \mid X, U)$; the E-value collapses the U-treatment and U-outcome associations to a symmetric scalar. Tan (2006) proposed an alternative formulation — the *marginal sensitivity model* (MSM) — that bounds the violation directly by an odds ratio comparing the nominal propensity $P(T = 1 \mid X)$ with the latent propensity $P(T = 1 \mid X, U)$, making it natural to combine with weighting estimators in Chapters 10–11.

Define the *nominal* and *true* propensity scores $\pi(x) = P(T = 1 \mid X = x)$ and $\pi^{\text{true}}(x, u) = P(T = 1 \mid X = x, U = u)$.

Definition: Marginal Sensitivity Model

The *marginal sensitivity model* at level $\Lambda \geq 1$ is the set of distributions $P(T \mid X, U)$ satisfying

$$\frac{1}{\Lambda} \leq \frac{\{1 - \pi^{\text{true}}(X, U)\}\, \pi(X)}{\pi^{\text{true}}(X, U)\, \{1 - \pi(X)\}} \leq \Lambda \qquad \text{almost surely.} \tag{9.23}$$

The constraint is symmetric: because both endpoints of the bound are imposed, the same constraint also bounds the inverse ratio by the same factor Λ.

The parameter Λ bounds, on the odds scale, how far the covariate-conditional propensity can depart from the true propensity once U is marginalized. Read literally, $\Lambda = 1$ says only that the represented U does not alter the treatment odds within levels of X; under the maintained latent exchangeability of Theorem 9.3 below, this eliminates the additional treatment-selection dependence on U and thereby recovers conditional exchangeability given X alone.

Theorem 9.3 (MSM Sensitivity Interval for the ATE)**.** *Assume consistency; latent conditional exchangeability $\{Y(0), Y(1)\} \perp\!\!\!\perp T \mid X, U$; positivity of the true propensity; suitable integrability conditions; and the marginal sensitivity model Equation 9.23 at level Λ. Then the average treatment effect satisfies $\tau_L(\Lambda) \leq \tau \leq \tau_U(\Lambda)$, where (τ_L, τ_U) are functionals of the observed-data distribution obtained by optimizing the weighting estimand over a valid outer envelope of weight perturbations implied by Equation 9.23. The interval is* valid *in the sense of Definition 9.2: it contains τ whenever the model holds. It is not claimed to be sharp.*

The interval is computed empirically by optimizing the self-normalized (Hájek) weighting estimator over a valid outer envelope of weight perturbations — a linear-fractional program whose solution is a threshold rule in the ordered outcomes, which is why it takes a percentile form — with percentile-bootstrap inference (Zhao et al. 2019). The derivation of the perturbation envelope and the optimization details are in the optional supplement.

Remark: MSM in Practice — Estimation-Oriented, Valid but Not Sharp

The MSM pairs naturally with weighting estimators (Chapters 10–12): it bounds the violation on the *weight* scale, so the analysis is a perturbation of the nominal weights that preserves the estimation architecture. Zhao et al. (2019) develop this for inverse-probability weighting with percentile-bootstrap inference; augmented and doubly robust extensions are separate developments. The resulting interval is *valid* but should not be called sharp: the constraint the MSM actually places on the perturbations is tighter and more structured than the envelope over which the optimization runs, so the interval is conservative in the sense of Definition 9.2, and later work characterizing the sharp identified set under the same odds-ratio restriction — the quantile-balancing analysis of Dorn and Guo (2023) — shows that the earlier intervals can be strictly wider.

9.3.4 Comparing the Three Models

The three models share the general logic of indexed assumption relaxation. The MSM and the E-value bias-factor model fit the causal identified-set formulation directly; Rosenbaum's usual p-value analysis is its inferential analogue and fits the estimand-set formulation after test inversion (Section 9.5). They differ in (i) what object the sensitivity parameter bounds, (ii) what estimator they naturally pair with, and (iii) how the sensitivity parameter is benchmarked.

Table 9.2: The three canonical sensitivity models for unmeasured confounding.

	Rosenbaum Γ	**E-value**	**MSM (Λ)**
Parameter bounds	Odds ratio of treatment given X, U	Pair of risk-ratio associations $\mathrm{RR}_{TU}, \mathrm{RR}_{UY}$	Odds ratio of nominal to true propensity
Natural design or estimator setting	Matched-pair and signed-score analyses	Estimator-agnostic risk-ratio-scale effect estimate	IPW (augmented extensions cited separately)
Typical sensitivity output	Tipping $\Gamma^\star$	EV (symmetric diagonal)	Sensitivity interval $[\tau_L, \tau_U]$ or tipping $\Lambda^\star$
Primary reference	Rosenbaum (2002)	VanderWeele and Ding (2017)	Tan (2006); Zhao et al. (2019)

Remark: Which Model to Report

In practice, reporting more than one summary is common and useful. The E-value is commonly included when the effect is naturally reported, or defensibly converted, to a risk-ratio scale, because it is a single number that is widely understood; the Γ-value is reported alongside it in matched or rank-based analyses; and the MSM bounds are reported alongside the others when the primary estimator is IPW or AIPW. The three are not substitutes: each bounds a different object — a treatment-odds ratio, a pair of risk-ratio associations, and an odds ratio of nominal to true propensity — and so each answers a distinct question. A rigorous applied analysis quotes at least one bound or tipping point together with benchmarking against observed covariates.

9.4 Benchmarking and Reporting

Every sensitivity model in the previous section has one or more parameters that are not identified by the data. A sensitivity curve of the form "at $\lambda = 0.2$ the effect is halved" is mathematically precise but scientifically empty until $\lambda = 0.2$ has been anchored to something concrete. *Benchmarking* is the discipline of giving sensitivity parameters an empirical referent by comparing them to the analogous quantities computed for the *observed* covariates (Cinelli and Hazlett 2020).

9.4.1 Why Sensitivity Parameters Need Interpretation

Consider a statement that would otherwise close a sensitivity analysis: "the conclusion is robust up to $\Gamma = 1.8$." To evaluate this statement scientifically, one must know whether an unmeasured confounder inducing a treatment-odds ratio of 1.8 is a plausible hypothesis in the particular study, or an implausible one. The same Γ-value can be a comfortable margin in one application and an alarming fragility in another.

Similarly, "the E-value equals 3.5" means that, to explain the effect away, an unmeasured confounder must have risk-ratio associations with T and with Y that are *both* at least the observed risk ratio R, with at least *one* of them at least 3.5. Whether a confounder of this magnitude is plausible depends on the context, and the benchmark must name its scale: in a study where the strongest observed covariate has an equivalent E-value of 2.0, an EV = 3.5 is reassuring; in a study where measured covariates routinely have equivalent E-values of 4 or more, it is not. Quoting a covariate's raw bias factor, or one of its two associations alone, in place of its equivalent E-value would compare different objects.

9.4.2 Benchmarking Against Observed Covariates

The core idea of benchmarking is to compute the sensitivity-parameter values implied by the observed covariates and to compare them to the proposed range for an unmeasured confounder.

For the binary-confounder setup, the sensitivity parameters are the U-outcome effect δ and the U-treatment imbalance $p_1(X) - p_0(X)$. For each observed covariate X_j, one can compute the analogous quantities. The treatment-side benchmark must match the estimator. If the primary estimator is OLS adjustment, the exact analogue is the Frisch–Waugh–Lovell auxiliary coefficient

$$d_j^{\mathrm{OLS}} = \text{coefficient on } T \text{ in the regression of } X_j \text{ on } (1, T, X_{-j}),$$

so the covariate's implied bias is $\hat{\gamma}_j\, d_j^{\mathrm{OLS}}$. If the primary estimator is a standardized conditional-mean contrast, the matching benchmark is instead $\mathbb{E}_{X_{-j}}\{\mathbb{E}(X_j \mid T{=}1, X_{-j}) - \mathbb{E}(X_j \mid T{=}0, X_{-j})\}$. The two need not coincide — the lab in Section 9.8 makes exactly this distinction for the unmeasured confounder itself, and the observed-covariate benchmark must respect it for the same reason.

For the E-value, benchmarking is particularly clean. For each observed covariate X_j, compute its risk-ratio association with T (call it RR_{TX_j}) and with Y (adjusted for T and the other covariates, call it RR_{X_jY}). The corresponding bias factor is

$$B_j = \frac{\mathrm{RR}_{TX_j}\,\mathrm{RR}_{X_jY}}{\mathrm{RR}_{TX_j} + \mathrm{RR}_{X_jY} - 1}.$$

By Theorem 9.2, a confounder with these associations can explain the observed effect away only if $B_j \geq \mathrm{RR}^{\mathrm{obs}}_{TY|X}$. Hence if $B_j < \mathrm{RR}^{\mathrm{obs}}_{TY|X}$ for every observed covariate, no unmeasured confounder as strong as *any* observed covariate can account for the effect. One precision matters here: the comparison is one *joint pair* at a time. "As strong as covariate X_j" means having the same pair as that single covariate. It does not rule out a hypothetical confounder combining the strongest treatment association seen for one covariate with the strongest outcome association seen for another: because $B(\cdot, \cdot)$ is increasing in each argument, such mixed maxima can yield a larger bias factor than any individual observed covariate exhibits.

Two scales are in play here, and mixing them is a common error. The bias factor B_j lives on the scale of the observed risk ratio, so the comparison just made is B_j against $\mathrm{RR}^{\mathrm{obs}}_{TY|X}$ — not B_j against EV, which is on the scale of an individual association. To compare on the E-value scale instead, convert each covariate to the symmetric pair of associations that would produce the same bias factor:

$$e_j = B_j + \sqrt{B_j\,(B_j - 1)}, \tag{9.24}$$

which solves $B(e_j, e_j) = B_j$ by the same algebra that produced Equation 9.20. Call e_j the *equivalent E-value* of covariate X_j. Because B is increasing in each argument, e_j and EV are directly comparable, and the conclusion survives a confounder as strong as X_j precisely when $\mathrm{EV} > e_j$.

Cinelli and Hazlett (2020) develop the idea systematically for the linear regression setting, introducing partial-R^2 benchmarks that quantify how much of the variation in T and Y an unmeasured confounder would need to explain — expressed as multiples of the partial-R^2 of the strongest observed covariate.

E-values are defined for ratio measures. When the outcome is continuous, an E-value can still be reported, but only after the effect has been placed on a ratio scale; the usual device is to convert a standardized mean difference d to an approximate risk ratio $\exp(0.91\,d)$ and compute the E-value from that (VanderWeele and Ding 2017). Quoting an E-value directly for a raw difference — a dollar amount, a test-score gap — is a category error, because the bias factor of Theorem 9.2 multiplies a ratio and has no interpretation as an additive displacement.

Remark: When Benchmarking Is Informative

Benchmarking is most informative when the observed covariate used as a benchmark is scientifically comparable to the hypothesized unmeasured confounder and is measured on a relevant scale. An unmeasured confounder that operates through a different causal pathway from any observed covariate — for example, a behavioral factor in a study whose adjustment set consists only of demographic controls — need not satisfy the same association bounds as the strongest observed covariate. In such settings a statement of the form "the unmeasured confounder would have to be 15% stronger than the strongest observed covariate" is a scale comparison, not a substantive ruling-out.

9.4.3 Reporting a Benchmarking Analysis

A minimal benchmarked sensitivity report contains three components: a point estimate with sampling-based confidence interval; a sensitivity summary (a tipping point, a bound, or the E-value); and a benchmarking comparison to observed covariates. The third component is essential. Without it, the sensitivity summary is a mathematical object without a scientific referent, and the report risks conveying a false precision about the robustness of the conclusion.

Warning: Sensitivity without Benchmarking

Reporting "the effect is robust up to $\Gamma = 2$" without a statement of whether $\Gamma = 2$ is plausible in the study at hand is the sensitivity-analysis analog of reporting a confidence interval without stating the confidence level: technically complete, but uninformative.

9.4.4 Reporting Guidelines

At minimum, an applied report should include the point estimate $\hat{\psi}$ and its sampling 95% confidence interval; a sensitivity summary — one of (a) a sensitivity curve, (b) a sensitivity bound over a specified $\mathcal{L}$, or (c) an E-value or tipping point; a benchmarking statement; and a plain-language interpretation.

Warning: Misleading Phrases

- "The result is causal because the estimate is statistically significant." Statistical significance addresses sampling uncertainty, not identification credibility.
- "The result is robust because we used machine learning." Flexible nuisance estimation improves the estimator; it does not validate the identifying assumptions.
- "Sensitivity analysis showed the result is not biased." Sensitivity analysis does not test for bias. It quantifies how much bias the identifying assumptions *could* produce if violated.
- "The E-value is X, therefore the effect is robust." An E-value is robust or not depending on whether X is plausible as an unmeasured-confounder strength. A large E-value without benchmarking is an unanchored statistic.

9.4.4.1 Recommended Template

A sample report paragraph, in the style recommended throughout this chapter:

> Under the back-door adjustment for X, the estimated risk ratio is 1.85 with 95% CI $[1.25, 2.70]$. The E-value is 3.10 for the point estimate and 1.81 for the lower confidence limit. Among observed covariates, the bias factors B_j range from 1.15 (age) to 1.60 (baseline health status); converting each to its equivalent E-value by Equation 9.24 gives 1.57 and 2.58 respectively. Comparing on that common scale, $3.10 > 2.58$: the point estimate is robust to an unmeasured confounder as strong as any single observed covariate, with roughly 20% to spare. The lower confidence limit is not, since $1.81 < 2.58$. Explaining away the point estimate would therefore require unmeasured confounding stronger, on the equivalent-E-value scale, than any single measured covariate. Whether confounding of that strength is scientifically implausible depends on whether baseline health is a credible comparator for the important pathways that may remain unmeasured; the numbers alone do not settle that question.

Two features of this template are deliberate. The effect is reported as a risk ratio rather than as an additive ATE, because the E-value formula is defined on the ratio scale; attaching an E-value to a mean difference would report two incompatible objects. And the comparison is e_j against EV throughout, never B_j against EV, which would mix scales.

9.5 Partial Identification and Overlap

9.5.1 From Sensitivity Models to Identified Sets

Definition 9.1 is stated for a *point-valued* relaxation, in which each λ pins ψ down to a single number. Several of the models in this chapter are not of that kind. The general object is therefore set-valued:

$$\mathcal{I}_\lambda(P) = \{\psi(Q) : Q \text{ compatible with } P \text{ and } A_\lambda\}, \tag{9.25}$$

the collection of estimand values consistent with the observed-data distribution P once the assumption has been relaxed to A_λ. The neutral value is characterized by $\mathcal{I}_{\lambda_0}(P) = \{\Psi(P)\}$. Write $\underline{\psi}(\lambda) = \inf \mathcal{I}_\lambda(P)$ and $\overline{\psi}(\lambda) = \sup \mathcal{I}_\lambda(P)$ for the *pointwise* lower and upper endpoints at a given λ. When $\mathcal{I}_\lambda(P)$ is a singleton these coincide and we recover the point-valued case; a sensitivity curve is the appropriate report exactly then. Otherwise the report is a band between $\underline{\psi}$ and $\overline{\psi}$.

One distinction within this formulation deserves emphasis. The MSM and the E-value bias-factor model fit the estimand-valued object directly: each restricts the possible values of, or discrepancy affecting, the causal estimand itself. Rosenbaum's Γ-analysis is ordinarily an *inferential* sensitivity analysis: at fixed Γ it bounds a randomization p-value under a sharp null, not the causal estimand. It enters the estimand-set framework only after *test inversion* — for example, by inverting tests over a constant additive treatment effect to obtain a Γ-sensitivity interval. The Γ-model thus shares the indexed-relaxation logic of Definition 9.1, but its usual p-value output is not literally an element of $\mathcal{I}_\Gamma(P)$ for ψ — a distinction worth marking in a chapter whose central theme is the separation of identification from inference.

For set-valued models, the tipping-point definition Equation 9.4 applies with ψ replaced by the pointwise lower endpoint $\underline{\psi}$.

The sensitivity models above each restrict the magnitude of a violation through a finite-dimensional sensitivity parameter or plausibility region. A more radical approach is to impose no parametric restriction on the violation at all — or, equivalently, to ask what the observed data can say about the causal parameter without any untestable identifying assumption. The answer is a set of values rather than a single point. This is the theory of *partial identification* (Manski 1990, 2003).

9.5.2 Point Identification vs. Partial Identification

We have so far worked under *point identification*: given the baseline assumption A_0, the estimand equals a specific functional of the observed-data distribution, $\psi = \Psi(P)$. Under *partial identification*, the assumptions pin down not a single value of ψ but a set. The *identified set* $\Psi_A(P)$ is the sharp answer to the question "given the assumption set A and the population distribution P, what values of ψ are possible?" Point identification is the limiting case $\Psi_A(P) = \{\psi\}$. Sensitivity bounds as defined in Equation 9.3 are a parametric restriction of the partial-identification idea.

> Definition: Identified Set
>
> Given an assumption set A (a statement restricting the joint distribution of observed and unobserved quantities), the *identified set* for ψ at distribution P is
>
> $$\Psi_A(P) = \{\psi(Q) : Q \in \mathcal{Q}_A,\ Q \text{ induces } P \text{ on the observables}\},$$
>
> where $\mathcal{Q}_A$ is the collection of joint distributions consistent with A.

The identified set is sharp in the sense that every value in it is consistent with some admissible Q, and every value outside is not. Refining A (adding assumptions) shrinks $\Psi_A(P)$; weakening A expands it.

Definition 9.2 (Sharp Endpoint Bounds and Conservative Bounds)**.** The bounds (ψ_L, ψ_U) are *sharp endpoint bounds* under assumption set A if $\psi_L = \inf \Psi_A(P)$ and $\psi_U = \sup \Psi_A(P)$. If $\Psi_A(P)$ is an interval, then $\Psi_A(P) = [\psi_L, \psi_U]$ and this interval is the *sharp identified set*; if $\Psi_A(P)$ is nonconvex, $[\psi_L, \psi_U]$ is only its convex hull, which contains values attainable under no admissible Q, and full sharpness requires reporting $\Psi_A(P)$ itself. A bound is *conservative* (or merely *valid*) if $\psi_L \le \inf \Psi_A(P)$ and $\psi_U \ge \sup \Psi_A(P)$: it contains the identified set but may be wider than A and P require.

Sharp endpoint bounds cannot be improved without adding assumptions; conservative bounds can sometimes be tightened by a more careful derivation under the same assumptions. Whenever bounds are reported, one should state which kind they are, because a sharpness claim has real content: it asserts that no further information can be extracted from (A, P). For the average treatment effect and the related scalar estimands of this chapter, $\Psi_A(P)$ is typically an interval, and sharp endpoint bounds coincide with the sharp identified set.

9.5.3 Manski's No-Assumption Bound

The widest identified set is obtained by imposing no identifying assumption at all beyond what the observed data mechanically reveal. The label "no-assumption" is conventional shorthand and slightly overstates the case: the result below still requires consistency, a binary treatment, and known bounds on the support of *both potential outcomes* — not merely of the observed outcome, since the proof bounds the unobserved counterfactual means. What it dispenses with is any assumption about the treatment-assignment mechanism.

Theorem 9.4 (Manski's No-Assumption Bound on the ATE)**.** *Assume consistency, $Y = T\,Y(1) + (1 - T)\,Y(0)$, and suppose both potential outcomes are bounded: $Y(t) \in [y_L, y_U]$ almost surely for $t \in \{0, 1\}$. Let $p = P(T = 1)$. Then $\tau = \mathbb{E}\{Y(1) - Y(0)\}$ satisfies $\tau \in [\tau_L, \tau_U]$ with*

$$\tau_L = \big[\mathbb{E}(Y \mid T = 1)\,p + y_L\,(1 - p)\big] - \big[\mathbb{E}(Y \mid T = 0)\,(1 - p) + y_U\,p\big], \tag{9.26}$$

$$\tau_U = \big[\mathbb{E}(Y \mid T = 1)\,p + y_U\,(1 - p)\big] - \big[\mathbb{E}(Y \mid T = 0)\,(1 - p) + y_L\,p\big]. \tag{9.27}$$

These bounds are sharp, and the width of the identified set is

$$\tau_U - \tau_L = y_U - y_L. \tag{9.28}$$

Proof. By the law of total probability,

$$\mathbb{E}\{Y(1)\} = \mathbb{E}\{Y(1) \mid T = 1\}\,p + \mathbb{E}\{Y(1) \mid T = 0\}\,(1 - p),$$

and symmetrically for $\mathbb{E}\{Y(0)\}$. Consistency identifies two of the four terms from the data: $\mathbb{E}\{Y(1) \mid T = 1\} = \mathbb{E}(Y \mid T = 1)$ and $\mathbb{E}\{Y(0) \mid T = 0\} = \mathbb{E}(Y \mid T = 0)$. The other two are unobserved counterfactual means, restricted only by the bounded-outcome condition to lie in $[y_L, y_U]$. The extremes of $\mathbb{E}\{Y(1)\}$ are therefore attained at $\mathbb{E}\{Y(1) \mid T = 0\} \in \{y_L, y_U\}$, giving

$$\mathbb{E}\{Y(1)\} \in \big[\mathbb{E}(Y \mid T = 1)\,p + y_L\,(1 - p),\ \mathbb{E}(Y \mid T = 1)\,p + y_U\,(1 - p)\big],$$

and similarly for $\mathbb{E}\{Y(0)\}$. The identified set for τ is the Minkowski difference of the two intervals, giving Equation 9.26–Equation 9.27. The width is $[y_U - y_L](1 - p) + [y_U - y_L]p = y_U - y_L$. Sharpness follows from the existence of potential-outcome distributions attaining each extreme, constructed explicitly as degenerate distributions concentrated at y_L or y_U in the unobserved arm. □

The width $y_U - y_L$ is the immediate and striking consequence: the no-assumption bound is as wide as the support of the outcome. If Y is a $[0, 1]$-valued indicator, the width is 1; if Y is a continuous health score on $[0, 100]$, the width is 100. In either case the bound is typically too wide to resolve the sign of τ.

Example 9.3 (Manski Bound in a Binary Outcome)**.** Suppose $Y \in \{0, 1\}$, $P(T = 1) = 0.4$, $\mathbb{E}(Y \mid T = 1) = 0.6$, and $\mathbb{E}(Y \mid T = 0) = 0.3$. The observed association is 0.3. Applying Theorem 9.4 with $y_L = 0$, $y_U = 1$:

$$\tau_L = (0.6)(0.4) + (0)(0.6) - [(0.3)(0.6) + (1)(0.4)] = 0.24 - 0.58 = -0.34,$$

$$\tau_U = (0.6)(0.4) + (1)(0.6) - [(0.3)(0.6) + (0)(0.4)] = 0.84 - 0.18 = 0.66.$$

The identified set is $\tau \in [-0.34, 0.66]$, which includes zero. With only the bounded-outcome restriction, the observed positive association of 0.3 is consistent with a causal effect anywhere from a meaningful harm to a substantial benefit.

This example illustrates both the discipline and the limitation of pure partial identification. The discipline: the width of the set transparently conveys how much the conclusion depends on untestable assumptions. The limitation: without assumptions, the set is usually too wide to be informative. Productive use of partial identification therefore typically proceeds by adding *shape restrictions* that are weaker than point identification but strong enough to narrow the set.

9.5.4 Shape Restrictions

Monotone treatment response. The assumption $Y(1) \geq Y(0)$ almost surely restricts the counterfactual outcomes to a half-plane: no unit is harmed by treatment. Under monotone treatment response, $\mathbb{E}\{Y(1) \mid T = 0\} \geq \mathbb{E}(Y \mid T = 0)$ and $\mathbb{E}\{Y(0) \mid T = 1\} \leq \mathbb{E}(Y \mid T = 1)$. Inserting these restrictions into the derivation of Theorem 9.4 tightens the bounds.

Monotone treatment selection. The assumption $\mathbb{E}\{Y(t) \mid T = 1\} \geq \mathbb{E}\{Y(t) \mid T = 0\}$ for each t states that treated units, on average, would have had better outcomes than untreated units *regardless of treatment level*. This is plausible in many observational studies where selection into treatment is favorable.

Combining monotone treatment response with monotone treatment selection can produce informative bounds even when neither assumption alone suffices. Manski (2003) develops these and other shape-restriction hierarchies.

> Remark: Sharpness of Shape-Restricted Bounds
>
> The bounds produced by inserting a single restriction into the proof of Theorem 9.4 are valid (the true τ lies in the reported set) but are not automatically sharp in the sense of Definition 9.2. Sharp bounds under a combination of restrictions require optimizing jointly over all joint distributions of $(Y(1), Y(0), T, X)$ consistent with both the observed-data distribution and the imposed restrictions, which can require linear or convex programming. When reporting bounds under shape restrictions, the analyst should state whether the bounds are sharp or merely conservative.

9.5.5 Bounds as Sensitivity Analysis

Partial identification and sensitivity analysis are two views of the same underlying object. A sensitivity model together with its bounds is a parametric partial-identification analysis; Manski's no-assumption bound is the limit when $\mathcal{L}$ is taken to permit arbitrary violations. Intermediate models — the Γ-model, the MSM — lie between these extremes, trading informativeness for robustness.

> Remark: When to Quote Bounds instead of a Sensitivity Curve
>
> A sensitivity curve reports a function; a bound reports an interval. Bounds are the appropriate summary when (i) the sensitivity parameter is high-dimensional so that no one-dimensional curve is natural; (ii) the analyst cannot defensibly bound λ by a specific plausibility set $\mathcal{L}$, and prefers to report what the data alone reveal under shape restrictions; or (iii) the identified set needs to be reported as a set, for example in regulatory contexts requiring robust lower bounds. Sensitivity curves are the appropriate summary when the analyst can defensibly order λ-values by plausibility.

9.5.6 Positivity, Overlap, and Retargeting

Chapter 6 introduced positivity — $0 < \pi(X) < 1$ almost surely — as a structural requirement for the identification formulas underlying propensity-score methods. When positivity fails, some counterfactual comparisons are not supported by the observed data, and the identification argument breaks down locally. When positivity holds formally but weakly, the estimators behave poorly even though the identification formula remains technically valid. Sensitivity analysis with respect to positivity addresses both failure modes.

9.5.6.1 Positivity as an Identification Assumption

The back-door identification formula involves the counterfactual outcome at both treatment levels for every value of X. If positivity fails on a set S_0 of positive P_X-probability — that is, $\pi(x) = 0$ throughout S_0 —

there are no treated units with $X \in S_0$ to inform μ_1 there, and the counterfactual mean $\mathbb{E}\{Y(1) \mid X \in S_0\}$ is not identified by the data. The qualification matters: an isolated zero at a P_X-null point does not obstruct identification of the population ATE. On such a region, $\mu_1(x_0)$ is not uniquely determined by the observed data, and any numerical value assigned to the missing arm-specific regression is supplied by a model-based extrapolation assumption rather than by observed support.

The statistical consequence is that the ATE itself may not be point-identified over the full covariate support even though it is identified over the subset where positivity holds. This motivates redefining the estimand to match the support where the data are informative.

9.5.6.2 Practical Positivity Violations

Even when $0 < \pi(x) < 1$ in a strict sense, estimation becomes unstable when $\pi(x)$ is near 0 or 1 over regions with non-negligible covariate mass. IPW weights are of order $1/\pi(X)$, so near-violations generate extreme weights that inflate variance and destabilize estimates. The instability is compounded by the fact that errors in $\hat{\pi}$ near the boundary translate into much larger errors in $1/\hat{\pi}$, so that low-overlap regions are also the regions where propensity misspecification is most damaging. Crump et al. (2009) address this directly by asking which subpopulation yields the most precisely estimable treatment effect; they characterize the variance-optimal overlap subsample and show that it takes the form of a threshold rule on $\pi(x)$, which in practice motivates discarding units with $\pi(x)$ outside roughly $[0.1, 0.9]$. That interval is a widely used rule of thumb motivated by the Crump criterion, not a variance-optimal threshold for every design and estimand.

9.5.6.3 Trimming as a Sensitivity Analysis

Define the *trimmed average treatment effect*

$$\tau_\alpha = \mathbb{E}\{Y(1) - Y(0) \mid \alpha \leq \pi(X) \leq 1 - \alpha\} \tag{9.29}$$

for $\alpha \in [0, 0.5)$. The choice $\alpha = 0$ recovers the full-population ATE; $\alpha > 0$ restricts attention to a *subpopulation with adequate overlap.*

Lemma 9.3 (Trimmed Estimand Identity)**.** *Let $\alpha \in (0, 1/2)$, let $S_\alpha = \{X : \alpha \leq \pi(X) \leq 1 - \alpha\}$, and suppose $q_\alpha = P(X \in S_\alpha) > 0$. Under consistency and conditional exchangeability,*

$$\tau_\alpha = \frac{1}{q_\alpha}\, \mathbb{E}\big\{[\mu_1(X) - \mu_0(X)]\, \mathbf{1}\{X \in S_\alpha\}\big\}\,. \tag{9.30}$$

Moreover, the IPW weights are bounded on the trimmed population: $w(T, X) \leq 1/\alpha$ on S_α.

The restriction to $\alpha \in (0, 1/2)$ is essential and not merely technical. At $\alpha = 0$ the set S_0 is the entire support, which in the setting motivating this section contains points with $\pi(x) \in \{0, 1\}$; at such points $\mu_1(x)$ or $\mu_0(x)$ is not nonparametrically identified, so Equation 9.30 does not hold as an observed-data identity. It holds at $\alpha = 0$ only if full-population positivity is separately assumed — which is precisely the assumption in question here. The causal estimand τ_0 still exists without positivity; what fails is its nonparametric identification by the displayed functional.

Proof. Conditional on $X \in S_\alpha$, exchangeability and positivity continue to hold, the latter with slack $\alpha > 0$, so the back-door formula applies pointwise. Integrating over the conditional distribution of X given $X \in S_\alpha$ gives Equation 9.30. The IPW weight bound follows directly from $\alpha \leq \pi(X) \leq 1 - \alpha$. □

The key conceptual point is that τ_α and τ_0 are *different estimands*: trimming is not a variance-reduction trick that leaves the target unchanged, but a restatement of the scientific question. A trimmed analysis answers "what is the causal effect on the subpopulation for which the treatment decision is not already essentially forced by covariates?" For some scientific questions this is the better target; for others, it is not.

Trimming should therefore be interpreted as changing the estimand to a region of covariate overlap. It is a robustness-and-retargeting strategy, not a sensitivity correction for the original population ATE: the bias produced by a positivity violation in the full population is not removed by reporting a trimmed estimand, it is sidestepped by redefining what is being estimated.

A further distinction: the estimand Equation 9.29 is defined through the *true* propensity π, but analysts trim on the *estimated* $\hat{\pi}$, which makes the retained population data-adaptive. A careful report separates the conceptual overlap population, the empirical rule, and the additional uncertainty introduced by estimating which units are retained.

> Warning: Trimming Changes the Estimand
>
> Reporting $\hat{\tau}_\alpha$ as a stabilized estimate of τ without acknowledging that the target population has changed is a common mistake. Whenever $\alpha > 0$, the estimand is Equation 9.29, not the full-population ATE. A proper sensitivity analysis reports $\hat{\tau}_\alpha$ and q_α together for a grid of positive thresholds $\alpha \in \{0.01, 0.05, 0.10\}$ — including $\alpha = 0$ only when full-population positivity is maintained as a separate assumption — and states explicitly how the target population shifts.

9.6 Sensitivity for Instrumental Variables and Mediation

Chapter 7 identified three assumptions that underwrite instrumental variables: relevance, exogeneity, and exclusion. Of the three, relevance is the most empirically diagnosable — weak-instrument F-tests and related diagnostics evaluate it directly from the first-stage regression. Exogeneity and exclusion, by contrast, involve unobserved relationships between the instrument and the outcome, and are not testable without additional structure. A sensitivity analysis for IV therefore concentrates on these two untestable assumptions.

9.6.1 What Weak-Instrument Diagnostics Do Not Address

A useful starting point is to be precise about what first-stage strength does and does not deliver. A strong first stage is not sufficient for instrument validity: it says nothing about exogeneity or exclusion, and a strong instrument may still directly affect the outcome. Conversely, an instrument may satisfy exogeneity and exclusion while having a weak first stage, in which case it is valid but provides practically uninformative — and possibly nonregular — inference. A *nonzero* first stage is of course necessary for identification by the Wald ratio; the point concerns strength, not relevance. The two concerns are otherwise orthogonal:

- *Relevance* is a statement about (Z, T) given X. It is observable: the first-stage coefficient is an identifiable functional of the observed distribution.
- *Exogeneity* restricts the relationship between Z and the latent determinants of treatment and outcome. *Exclusion* restricts the causal pathways from Z to Y. Neither condition is generally verifiable from the observed distribution alone.

A complete robustness argument therefore has two parts: (a) an evaluation of first-stage strength (Chapter 7) and (b) a sensitivity analysis for exogeneity and exclusion (this section). The two parts are complementary, not substitutes.

9.6.2 Direct-Effect Violation of Exclusion

Consider the canonical scalar-IV model of Chapter 7 with a direct effect δ of Z on Y:

$$Y = \alpha + \beta T + \delta Z + \varepsilon, \qquad \mathbb{E}(\varepsilon \mid Z) = 0. \tag{9.31}$$

The condition $\mathbb{E}(\varepsilon \mid Z) = 0$ *maintains* instrument exogeneity; the sensitivity parameter δ relaxes only the exclusion restriction, which is the condition $\delta = 0$. What follows is therefore an exclusion-only sensitivity analysis conditional on maintained exogeneity. A two-parameter analysis covering both assumptions would add a second term $\mathrm{Cov}(Z, \varepsilon)/\mathrm{Cov}(T, Z)$ to the bias formula; we do not pursue it here.

Theorem 9.5 (IV Bias under Exclusion Violation)**.** *Under Equation 9.31 and the standard IV regularity conditions (relevance* $\mathrm{Cov}(T, Z) \neq 0$ *and orthogonality of Z and ε),*

$$\frac{\mathrm{Cov}(Y, Z)}{\mathrm{Cov}(T, Z)} = \beta + \delta\, \frac{\mathrm{Var}(Z)}{\mathrm{Cov}(T, Z)}. \tag{9.32}$$

Equivalently, the Wald estimand at posited δ is

$$\beta(\delta) = \frac{\mathrm{Cov}(Y, Z) - \delta\, \mathrm{Var}(Z)}{\mathrm{Cov}(T, Z)}. \tag{9.33}$$

Proof. Take the covariance of Equation 9.31 with Z:

$$\mathrm{Cov}(Y, Z) = \beta\,\mathrm{Cov}(T, Z) + \delta\,\mathrm{Var}(Z) + \mathrm{Cov}(\varepsilon, Z),$$

and $\mathrm{Cov}(\varepsilon, Z) = 0$ by orthogonality. Dividing by $\mathrm{Cov}(T, Z)$ gives Equation 9.32. □

Formula Equation 9.32 has a striking pedagogical consequence: the bias term is proportional to $\mathrm{Var}(Z)/\mathrm{Cov}(T, Z)$, which is the inverse of the first-stage slope. A *weak* first stage therefore *amplifies* the bias from any exclusion violation: the same δ causes more damage in a weak-IV analysis than in a strong-IV analysis. This is the opposite of the intuition that weak instruments can at least be "rescued" by being exogenous.

Example 9.4 (IV Sensitivity Curve)**.** A study has $\mathrm{Cov}(Y, Z) = 3$, $\mathrm{Cov}(T, Z) = 1.5$, and $\mathrm{Var}(Z) = 1$. The Wald point estimate is $\hat{\beta}(0) = 3/1.5 = 2.0$. Applying Equation 9.33,

$$\hat{\beta}(\delta) = \frac{3 - \delta}{1.5}, \qquad \hat{\beta}(0.5) = 1.67,\ \hat{\beta}(1.0) = 1.33,\ \hat{\beta}(2.0) = 0.67,\ \hat{\beta}(3.0) = 0.$$

The tipping point for the sign is $\delta^\star = \mathrm{Cov}(Y, Z)/\mathrm{Var}(Z) = 3$. It coincides numerically with $\mathrm{Cov}(Y, Z)$ here only because $\mathrm{Var}(Z) = 1$; in general the two differ. In a weaker-instrument setting with $\mathrm{Cov}(T, Z) = 0.5$ but the same numerator, the tipping point would still be $\delta^\star = 3$ but the *rate* of bias accumulation would be tripled — each unit of δ moves the estimate by 2 rather than by 2/3.

9.6.3 Connection to LATE and Monotonicity

Under heterogeneous treatment effects, the Wald estimand identifies a local average treatment effect (LATE, Chapter 7) rather than an ATE, and the identification relies on a monotonicity assumption in addition to the three core IV assumptions. Sensitivity analysis can also address monotonicity violations, at the cost of additional modeling. Angrist et al. (1996) discuss a bound that includes "defiers" as an unobserved compliance type, and a natural first sensitivity parameter is the prevalence of defiers. Prevalence alone will not do, however. Writing π_c, π_d for the complier and defier proportions and τ_c, τ_d for their average treatment effects, the Wald estimand becomes

$$\frac{\Delta_Y}{\Delta_T} = \frac{\pi_c \tau_c - \pi_d \tau_d}{\pi_c - \pi_d},$$

so the distortion depends jointly on how many defiers there are, on how their average effect compares with the compliers', and on the net first stage $\pi_c - \pi_d$ in the denominator, which amplifies both. A sensitivity analysis must therefore posit at least two components, not one.

9.6.4 Sensitivity in Mediation Analysis

Sensitivity analysis is especially important in mediation analysis because the identifying assumptions are strictly stronger than those required for total effects. Chapter 8 derived the mediation formula under sequential ignorability (Imai et al. 2010): no unmeasured T–M confounding, no unmeasured M–Y confounding given T, and a cross-world independence condition. Even in a randomized trial, where randomization of T eliminates the first, the second and third cannot generally be guaranteed.

9.6.4.1 Why Mediation Needs Sensitivity Analysis

The controlled direct effect (CDE) is the effect of setting T and M jointly to specified values; the back-door formula for the joint intervention identifies it under the stated conditions, which require that all relevant M–Y confounders be observed and appropriately adjusted for. Natural direct and indirect effects (NDE, NIE) additionally require the relevant cross-world independence condition; the standard graphical sufficient condition is that no M–Y confounder — *observed or unobserved* — be affected by treatment. This is a strictly stronger condition, and observing such a confounder does not repair it: adjusting for the realized value of a treatment-induced confounder does not in general identify the natural effects.

Panel (a) already defeats M–Y ignorability. In panel (b), U is in addition affected by T; the cross-world independence condition then fails as well, and the NDE/NIE identification formula is biased even in a randomized trial. Strictly, the formal requirement for the natural effects is the cross-world independence; "no treatment-induced mediator–outcome confounder" is the standard *graphical sufficient condition* for it, which is why panel (b) is fatal. The residual-correlation parameter introduced below is aimed primarily at panel (a); panel (b) is the harder case, in which no scalar residual correlation fully captures the violation.

9.6.4.2 Residual-Correlation Sensitivity Parameter

A clean parameterization, following Imai et al. (2010), uses the correlation between the structural disturbances of the mediator and outcome equations. Consider the linear mediation system of Chapter 8:

$$M = a_0 + a\,T + a_X^\top X + \varepsilon_M, \tag{9.34}$$

$$Y = b_0 + \tau'\,T + b\,M + b_X^\top X + \varepsilon_Y, \tag{9.35}$$

and define the sensitivity parameter as the correlation between the structural disturbances:

$$\rho = \mathrm{Corr}(\varepsilon_M, \varepsilon_Y), \tag{9.36}$$

where $(\varepsilon_M, \varepsilon_Y)$ are assumed independent of (T, X) with a constant correlation. Within this maintained linear model, sequential ignorability implies $\rho = 0$: the disturbances are orthogonal because any shared source of variation has been conditioned out, and a nonzero ρ represents exactly the unobserved M–Y confounding that the assumption rules out. The converse direction is model-bound, and the equivalence should not be read as a general nonparametric statement: the entire Imai et al. (2010) sensitivity calculation is explicitly model-based. Under misspecification of the joint mediator–outcome system, $\rho = 0$ and sequential ignorability are not the same condition.

Under this parameterization, Imai et al. (2010) derive the bias in the estimated indirect effect as a smooth function of ρ and of the observed residual variances, giving a sensitivity curve $\rho \mapsto \widehat{\mathrm{NIE}}(\rho)$. The tipping point $\rho^\star$ at which the indirect effect reaches zero is the reportable summary.

9.6.4.3 Reporting Mediation Sensitivity

A well-reported mediation sensitivity analysis states three things: the point estimate of the indirect effect (with its sampling CI), the tipping value $\rho^\star$, and a benchmark for what magnitude of ρ is plausible in the given scientific setting. An example:

> Under sequential ignorability ($\rho = 0$), the estimated indirect effect of the behavioral intervention on depression through sleep quality is 0.38 with 95% CI $[0.18, 0.58]$. The sensitivity curve crosses zero at $\rho^\star = 0.31$, and the lower confidence limit crosses zero at $\rho^\star_{\mathrm{CI}} = 0.15$. A residual correlation of $\rho = 0.3$ corresponds to a squared residual correlation of 0.09.

It is tempting to describe $\rho^2 = 0.09$ as “9% of the joint variation explained,” and this is often done. It should be resisted. A squared correlation between two disturbances acquires a variance-explained reading only under an additional latent-factor model specifying how a common cause loads on M and on Y; without that model, ρ^2 is simply a squared correlation.

9.7 Sensitivity Analysis and Modern Estimators

Chapters 10–13 develop estimating-equation theory, doubly robust estimation, orthogonal scores, cross-fitting, and semiparametrically efficient IV estimation. These tools make estimators more robust — but robust to a specific class of perturbations, namely nuisance-model misspecification. They are silent about identification.

> **What Robust Estimation Does and Does Not Do**
>
> For the ATE procedures developed in Chapters 11–12, the AIPW/TMLE estimating equation is doubly robust, and its DML implementation uses Neyman orthogonality and cross-fitting. Under the product-rate, overlap, moment, consistency, and score-regularity conditions stated in those chapters, these properties protect the *identified observed-data functional* against first-order nuisance-estimation error and permit root-n inference with flexible learners. None of this establishes that the functional *equals the causal estimand*. Orthogonality is estimation robustness *conditional on* identification, not robustness of identification itself.

9.7.1 What Robust Estimation Does Not Do

Orthogonality and cross-fitting do not solve any of the following:

- *Unmeasured confounding.* The target functional $\mathbb{E}\{\mu_1(X) - \mu_0(X)\}$ does not equal τ if $Y(t) \not\perp\!\!\!\perp T \mid X$. No amount of accurate nuisance estimation corrects this; the nuisances are conditional on X, and X is the wrong conditioning set.
- *Positivity failure.* If $\pi(x) = 0$ on a region with positive covariate mass, the full-population target is not nonparametrically identified there and the influence-function theory underlying AIPW and DML breaks down. Note that an outcome-regression-based estimator may still return a finite number, because a fitted $\hat{\mu}_1(x)$ extrapolates into the region where no treated units were observed. Numerical finiteness is not identification: the reported value is then a property of the extrapolating model, not of the data.
- *IV invalidity.* If the exclusion restriction fails, 2SLS, efficient GMM (Chapter 13), and any other IV estimator converge to a biased target. The bias formula Equation 9.32 applies to any estimator that is consistent for the scalar Wald ratio in model Equation 9.31. Other IV estimands — overidentified GMM, nonlinear models, and heterogeneous-effect targets such as the LATE — require their own sensitivity derivations.
- *Mediation assumption failure.* The NIE and NDE are well-defined causal estimands whatever the data can support — they are fixed by the causal model, not by an assumption. What sequential ignorability governs is whether the mediation formula *identifies* them. When it fails, the estimand still exists, but the mediation functional no longer identifies it from the same observed data without additional assumptions.

A compact summary:

> *Neyman orthogonality and cross-fitting protect the estimator against nuisance-estimation error. They do not protect against violations of causal identification assumptions.*

This is the reason sensitivity analysis belongs in Part II (identification) rather than Part III (estimation). Every method in Chapters 10–13 assumes identification and refines the estimation. This chapter sits at the boundary: it refines the *assessment* of how much the identification can bend before the conclusion breaks.

Applied Workflow (Recommended)

1. State the estimand: the specific causal parameter of interest, in potential-outcome or do-notation.
2. State the identifying assumptions: which combination of back-door, IV, front-door, mediation, or other strategy is used.
3. Estimate the causal parameter using an appropriate method (outcome regression, IPW, AIPW, DML, 2SLS, etc.).
4. Quantify sampling uncertainty: standard errors, confidence intervals, or bootstrap distributions, under the maintained identifying assumptions.
5. Conduct sensitivity analysis for the least credible assumption: sensitivity curve, tipping point, or bounds, with benchmarking against observed covariates.
6. Report both the point estimate with sampling CI and the sensitivity summary, with plain-language interpretation of robustness.

Steps 1–4 are the customary content of a causal analysis. Step 5 is the contribution of this chapter, and step 6 brings the two outputs together. For observational analyses — or any analysis resting on substantively contestable, nonverifiable assumptions — omitting step 5 generally leaves the robustness argument incomplete.

9.8 Lab (Optional): A Tipping-Point Analysis for an Observational ATE

This lab implements a minimal sensitivity analysis end-to-end: a deliberately confounded observational ATE, a naive back-door adjustment that ignores the unmeasured confounder, and a tipping-point analysis that locates the strength of unmeasured confounding required to reverse the conclusion.

DGP for Lab 9

Let $X, U \overset{\text{i.i.d.}}{\sim} \text{Normal}(0,1)$ independently. The treatment is generated by

$$P(T = 1 \mid X, U) = \text{expit}(-0.3 + 0.6\,X + 1.0\,U),$$

and the outcome by

$$Y(t) = 1.5\,t + 0.8\,X + 2.0\,U + \varepsilon, \qquad \varepsilon \sim \text{Normal}(0,1).$$

The observed outcome is $Y = Y(T)$. The analyst observes (Y, T, X) but not U.

By construction, the true average treatment effect is $\tau = 1.5$. Ignorability given (X, U) holds, but ignorability given X alone is violated by the U arrows. The observed X enters the structural outcome equation with coefficient 0.8 — an oracle DGP coefficient, not the analyst-facing outcome-side benchmark, which is derived below as $\hat{\gamma}_X \approx 0.605$ from the analyst's observed regression.

9.8.1 Naive Adjusted Estimator

The analyst's first estimator adjusts only for X: $\hat{\tau}_X$ is the OLS coefficient on T in the regression of Y on $(1, T, X)$. Running 500 Monte Carlo replicates of sample size $n = 2{,}000$: the mean of $\hat{\tau}_X$ across replicates is 3.172; the Monte Carlo standard deviation is 0.099; the true ATE is 1.500.

The Monte Carlo standard deviation describes spread across simulation runs; the standard error an analyst computes from one realized data set is a different object, though here the usual OLS standard error tracks it closely, giving a typical single-replicate 95% interval of about [2.98, 3.37] — narrow, and not containing the truth. A consumer of this single report, relying on the CI for uncertainty, would conclude that the ATE is approximately 3.2 and confidently positive. This is the identification-uncertainty failure mode of Section 9.1.1 in concrete form.

9.8.2 Sensitivity Adjustment

Because U is Gaussian rather than binary, the relevant sensitivity model here is the *continuous-U linear* model of Section 9.2.2, not the binary-confounder formula. The bias quantity must also be matched to the estimator actually in use. Since $\hat{\tau}_X$ is the OLS coefficient on T in a regression of Y on $(1, T, X)$, the Frisch–Waugh–Lovell theorem gives the *exact* omitted-variable-bias identity

$$\text{plim}\,\hat{\tau}_X = \tau + \delta\,\theta_U(\alpha_U), \qquad \theta_U(\alpha_U) = \frac{\mathbb{E}(\tilde{T}U)}{\mathbb{E}(\tilde{T}^2)}, \tag{9.37}$$

where δ is the outcome coefficient on U and θ_U is the coefficient on T in the population regression of U on $(1, T, X)$, with $\tilde{T}$ the residual from projecting T on $(1, X)$.

Three quantities are easily confused here, and only the last is correct for this estimator: the marginal difference $\mathbb{E}(U \mid T{=}1) - \mathbb{E}(U \mid T{=}0)$, the standardized conditional contrast $\mathbb{E}\{\mathbb{E}(U \mid T{=}1, X) - \mathbb{E}(U \mid T{=}0, X)\}$, and the regression coefficient θ_U. In this particular numerical design the last two happen to agree to three decimals; that is a coincidence of the coefficients used here, not a general identity, and it is *not* implied by the independence or normality of X and U. Under other coefficient values the two differ appreciably — by more than 15% in some settings — because OLS weights strata by the conditional variance of T rather than by covariate mass. The marginal difference is smaller than either, because it ignores confounding by X. Note also that θ_U is built from conditional *means* of a standard normal variate, not from probabilities.

The analyst-facing analysis posits the pair (θ_U, δ) *directly*. This is the estimator-matched parameterization: by Equation 9.37 these two numbers are exactly what determine the bias of $\hat{\tau}_X$, and positing θ_U itself avoids routing the analysis through a treatment-model coefficient α_U that the analyst can neither observe nor estimate.

Table 9.3: Analyst-facing sensitivity surface for Lab 9, at $\hat{\tau}_X = 3.172$. Each cell is $\hat{\tau}_{\text{sens}} = \hat{\tau}_X - \delta\,\theta_U$ for posited (θ_U, δ).

Posited δ	$\theta_U = 0.25$	0.50	0.75	1.00
1.0	2.922	2.672	2.422	2.172
2.0	2.672	2.172	1.672	1.172
3.0	2.422	1.672	0.922	0.172

9.8.3 Tipping Contour and Benchmarking

The zero contour of the surface is the hyperbola

$$\delta \cdot \theta_U = \hat{\tau}_X = 3.172: \tag{9.38}$$

any posited pair whose product reaches 3.172 explains the entire adjusted estimate away, and no pair below the contour does.

The benchmark for the observed covariate must be estimator-matched, and *both* of its components must come from the analyst's observed regressions. The treatment-side component is the auxiliary coefficient $d_X^{\text{OLS}} = \text{coef}_T\{X \sim 1 + T\} = 0.472$. The outcome-side component is the coefficient on X in the regression the analyst actually ran:

$$\hat{\gamma}_X = \text{coef}_X\{Y \sim 1 + T + X\} \approx 0.605,$$

not the structural coefficient 0.8 of the data-generating process. The two differ because omitting U biases the coefficient on X as well: conditioning on the collider T induces a negative partial association between X and U ($\text{coef}_X\{U \sim 1 + T + X\} \approx -0.097$), and $0.8 + 2 \times (-0.097) \approx 0.605$. The observed covariate's implied bias is therefore

$$\hat{\gamma}_X\, d_X^{\text{OLS}} \approx 0.605 \times 0.472 \approx 0.286,$$

an order of magnitude below the contour value 3.172: no unmeasured confounder with the same joint pair of associations as X comes close to explaining the estimate away.

Oracle verification: the true configuration ($\alpha_U = 1.0$, $\delta = 2.0$) induces $\theta_U = 0.832$, and $3.172 - 2.0 \times 0.832 = 1.508 \approx \tau = 1.5$, confirming Equation 9.37.

9.8.4 Interpretation

The lab reinforces four points:

1. The naive single-replicate CI $[2.98, 3.37]$ is narrow but does not contain the true ATE $\tau = 1.5$. Sampling uncertainty is not identification uncertainty.
2. Sensitivity adjustment at the correct unmeasured-confounding magnitude recovers the truth to within Monte Carlo error. The method works when the sensitivity parameters are correctly posited.
3. The tipping point depends jointly on both sensitivity parameters, and two slices through the contour must not be conflated. At the *observed* imbalance of X, $d_X^{\text{OLS}} = 0.472$, the tipping value is $\delta^\star = 3.172/0.472 \approx 6.72$, about 11 times the observed partial outcome coefficient $\hat{\gamma}_X \approx 0.605$ — this is the observed-covariate benchmark. Along the oracle slice $\theta_U = 0.832$, $\delta^\star = 3.172/0.832 \approx 3.81$ — a hypothetical slice through the surface, not a "confounder-as-strong-as-X" comparison.
4. Benchmarking against the observed X anchors the analysis: X's implied bias ≈ 0.286 sits an order of magnitude below the tipping contour 3.172.

Remark: What the Lab Is and Is Not

The lab is a *demonstration* that sensitivity adjustment can recover the truth when the sensitivity parameters are correctly posited, and that the tipping-point framework gives a coherent way to report the dependence on those parameters. The lab is *not* a test of the assumptions: the analyst never learns α_U or δ from the data, because these quantities involve the unobserved U. The oracle computation of $\theta_U(\alpha_U)$ verifies that the bias formula Equation 9.37 is correct; it is not an implementable sensitivity analysis, because it uses the simulated U. What the analyst can actually do is posit a grid of (θ_U, δ) directly, compute the corresponding surface of sensitivity-adjusted estimates, locate its zero contour, and benchmark that contour against the pairs implied by observed covariates.

9.9 Chapter Summary

This chapter closed Part II by addressing the complement of what Chapters 5–8 provided. Those chapters gave conditions under which a causal parameter is identified as a functional of the observed-data distribution; this chapter provided tools for reporting how much the identification can be violated before the causal conclusion changes.

- Sampling uncertainty, model misspecification, and identification uncertainty are distinct sources of error. A confidence interval addresses only the first, and identification uncertainty does not vanish merely by collecting more data from the same observational regime.
- A sensitivity analysis introduces a parameter λ that quantifies the strength of a violation, with a neutral value λ_0 recovering the baseline assumption — zero for the additive parameters, one for the odds-ratio bounds Γ and Λ. The three reporting objects are the sensitivity curve, the sensitivity bounds, and the tipping point.
- The master bias decomposition $\Delta_{\text{obs}} = \tau + B$ underwrites sensitivity analysis for unmeasured confounding, and specializes cleanly in the linear and binary-U cases.
- Three canonical sensitivity models differ by the object that the sensitivity parameter bounds. Rosenbaum's Γ bounds a treatment-odds ratio. The VanderWeele–Ding E-value $\text{EV} = R + \sqrt{R(R-1)}$ is the smallest achievable value of the larger of RR_{TU} and RR_{UY} among confounders able to explain the effect away, so at least one of the two associations must reach it, while asymmetric pairs with one component below it can still suffice — though by the Cornfield inequality the smaller association must in every case be at least the observed risk ratio R. The marginal sensitivity model bounds an odds ratio of nominal to true propensity, yielding valid — not sharp — sensitivity intervals for weighting estimators.
- Sensitivity parameters need benchmarking against observed covariates to be interpretable, and the benchmark is informative only when the comparator is scientifically comparable to the hypothesized unmeasured confounder.
- Partial identification reports an identified set rather than a single value. Manski's no-assumption ATE bound has width $y_U - y_L$, the maintained common support range of the two potential outcomes; shape restrictions narrow it, and a careful report distinguishes sharp endpoint bounds from conservative ones.
- Exact positivity failure on a positive-mass region is an identification failure; near-positivity violations are primarily a source of estimation and inferential instability. Trimming changes the estimand to a region of covariate overlap; it is a retargeting strategy, not a correction for the original population ATE.
- An exclusion violation of size δ shifts the scalar Wald estimand by $\delta \cdot \text{Var}(Z)/\text{Cov}(T, Z)$ — an exclusion-only analysis under maintained exogeneity — and the bias is amplified by weak first stages. Weak-instrument diagnostics do not address exclusion or exogeneity violations.
- Mediation sensitivity analysis proceeds, within a maintained linear mediation model, by a residual-correlation parameter ρ that captures ordinary unmeasured M–Y confounding. A treatment-induced mediator–outcome confounder is the harder failure mode and is not captured by a scalar ρ.
- Modern estimation methods (AIPW, TMLE, DML, efficient GMM) address nuisance estimation, not identification. Neyman orthogonality protects the estimator; it does not protect the causal claim.

9.10 Problems

1. Sampling vs. identification uncertainty. In at most half a page, explain why identification uncertainty does not shrink as the sample size grows. Sketch on paper (no simulation required) a data-generating process with one observed covariate X and one unmeasured confounder U in which a back-door-adjusted estimator converges to a value different from the true ATE; state which identifying assumption is violated and give the limiting bias in terms of the model coefficients. (A full simulation version of this exercise is in the optional supplement.)

2. Binary unmeasured confounder. Suppose the observed adjusted effect is $\hat{\Delta}_{\text{obs}} = 2.0$. Assume a binary unmeasured confounder U with constant outcome contrast $\delta = 4$, and suppose the U-treatment imbalance is uniform across X with value $p_1 - p_0 = 0.3$.

(a) Use Lemma 9.2 to compute the confounding bias B and the sensitivity-adjusted estimate $\hat{\tau}_{\text{sens}}$.
(b) What outcome contrast δ would be required to zero out the estimate at the same imbalance level?
(c) What imbalance $p_1 - p_0$ would be required to zero out the estimate at $\delta = 4$?

3. Tipping point in a linear bias model. For $\hat{\Delta}_{\text{obs}} = 1.5$ and bias model $B(\lambda) = 0.4\,\lambda$:

(a) Find the tipping point $\lambda^\star$ for the sign of the adjusted estimate.
(b) Suppose the sampling standard error of the observed estimate is 0.3, independent of λ. Find the tipping point $\lambda^\star_{\text{CI}}$ for the lower confidence limit to reach zero (95% level). Compare to $\lambda^\star$.

4. Benchmarking. A researcher reports a sensitivity analysis with $\hat{\tau}_X = 2.4$ under the linear model Equation 9.10, in which the tipping point for the U-outcome effect is $\delta^\star = 3.0$ *at a posited treatment imbalance of* $d = 0.8$. The observed covariates have estimated outcome coefficients and treatment imbalances:

Covariate	$\hat{\gamma}_j$	$\hat{d}_j$
age	0.3	0.20
income	1.8	0.35
education	2.4	0.10

(a) Explain why the outcome coefficient $\hat{\gamma}_j$ alone is not a complete benchmark under the linear sensitivity model, and state what the second component contributes.
(b) Compute each covariate's implied bias $\hat{\gamma}_j\hat{d}_j$ and identify which covariate is the strongest benchmark on that scale. Note that the ranking differs from the ranking by $\hat{\gamma}_j$ alone.
(c) Compare $\delta^\star = 3.0$ with the largest observed outcome coefficient, while holding the imbalance fixed at $d = 0.8$. Is the result robust to an unmeasured confounder with the same outcome association as the strongest observed covariate?
(d) Write two sentences of interpretation suitable for an applied report, making clear which quantity is being held fixed.

5. E-value calculation. An observational study reports an adjusted risk ratio of 1.9 with 95% CI $[1.3, 2.8]$.

(a) Compute the E-value for the point estimate using Equation 9.20.
(b) Compute the E-value for the lower confidence limit.
(c) Write one sentence of interpretation for each, stating the symmetric reading correctly.
(d) Exhibit a pair $(\text{RR}_{TU}, \text{RR}_{UY})$ with RR_{TU} strictly below the E-value of part (a) that nonetheless explains the point estimate away, and verify it using Equation 9.18. Explain why this does not contradict Equation 9.21, and why one may not conclude that both associations must individually exceed the E-value.

6. Manski bound calculation. Suppose consistency holds and both potential outcomes satisfy $Y(t) \in [0, 10]$ almost surely. Also suppose $P(T = 1) = 0.3$, $\mathbb{E}(Y \mid T = 1) = 6.5$, and $\mathbb{E}(Y \mid T = 0) = 4.0$.

(a) Compute Manski's no-assumption bound for the ATE using Equation 9.26–Equation 9.27.
(b) Compute the width of the bound and verify Equation 9.28.
(c) The observed association is 2.5. Does the no-assumption identified set include zero? Discuss what this implies about the informational content of the data alone.

7. Modern estimators and identification. A colleague argues that because DML and AIPW are "doubly robust and orthogonalized," they "automatically correct for unmeasured confounding provided the machine-learning models are good enough." Explain why this is incorrect, referring to the bias decomposition of Theorem 9.1. Be precise about what is and is not estimable: machine learning can consistently estimate the observed-data functional Δ_{obs} and the observed conditional means $\mu_t(X)$, but it cannot recover the split of Δ_{obs} into τ and B, nor the U-indexed quantities $m_t(X, U)$ and $g_t(X, U)$, without information beyond the observed data. Explain why no amount of flexibility in the nuisance estimators changes this.

Chapter 9 Supplement: Optional Proofs, Derivations, and Extended Lab

This supplement collects material referenced as *optional* in Chapter 9: full proofs, the special-case E-value derivation, computational details for the marginal sensitivity model, the extended lab, a worked reporting example, and additional problems. References of the form Theorem 9.1 or Equation 9.20 point back into the main chapter.

S.1 Proof of the Bias Decomposition

Proof of Theorem 9.1. Under conditional exchangeability given (X, U) and consistency,

$$\mathbb{E}[Y(t) \mid X = x] = \int \mathbb{E}(Y \mid T = t, X = x, U = u)\, f(u \mid X = x)\, du = \int m_t(x, u)\, f(u \mid X = x)\, du.$$

Iterating expectation over X,

$$\mathbb{E}\{Y(t)\} = \mathbb{E}\left\{\int m_t(X, u)\, f(u \mid X)\, du\right\}, \qquad t \in \{0, 1\}.$$

In contrast, the conditional mean given (T, X) marginalizes U over its distribution *conditional on* T:

$$\mathbb{E}(Y \mid T = t, X) = \int m_t(X, u)\, f(u \mid T = t, X)\, du.$$

Taking expectations over X and subtracting,

$$\begin{aligned}\Delta_{\text{obs}} - \tau = \mathbb{E}&\left\{\int m_1(X, u)\, [f(u \mid T = 1, X) - f(u \mid X)]\, du\right\} \\ &- \mathbb{E}\left\{\int m_0(X, u)\, [f(u \mid T = 0, X) - f(u \mid X)]\, du\right\} = B. \qquad \square\end{aligned}$$

S.2 Rosenbaum Reference Distributions

Throughout, test Fisher's sharp null of no treatment effect, $Y_i(1) = Y_i(0)$ for every unit i (or, equivalently, outcomes adjusted by subtracting a specified constant additive effect); condition on the matched outcomes and covariates; and assume treatment assignments are independent across matched pairs, with exactly one treated unit per pair.

Theorem 9.6 (Γ-Bounds on the Rank Statistic)**.** *For a matched-pair design, the worst-case null distribution of a signed-score statistic under the* Γ*-model is obtained by letting each pair-specific treatment probability range over the interval* $[1/(1 + \Gamma),\ \Gamma/(1 + \Gamma)]$ *permitted by Equation 9.15. For the* sign test *the resulting bounding distributions are binomial with success probabilities* $\Gamma/(1 + \Gamma)$ *and* $1/(1 + \Gamma)$*. For* Wilcoxon's signed-rank statistic *they are not binomial: the bounds are distributions of weighted sums* $\sum_k c_k B_k$ *of independent Bernoulli variables* B_k*, where the* c_k *are the rank scores and each success probability is set to the corresponding endpoint.*

Proof sketch. Under matching on X, the two units in a pair have identical observed covariates, so which member of the pair receives treatment depends only on U, through Equation 9.15. That constraint confines each pair-specific treatment probability to $[1/(1+\Gamma),\, \Gamma/(1+\Gamma)]$. Because a signed-score statistic is a sum of independent pair contributions that is stochastically monotone in those probabilities, setting every probability to a common endpoint yields stochastic bounds on its null distribution. A full development is in Rosenbaum (2002). □

S.3 A Binary-Confounder Derivation of the VanderWeele–Ding Bound

Proof of Equation 9.19, binary-U no-interaction case. The restrictions imposed in this derivation — U binary, and a U–Y risk ratio $r(x)$ common to both treatment arms — are *not* required by the theorem. They are imposed only to make the optimization solvable in closed form.

Fix x, write $q_x = P(T{=}1 \mid X{=}x)$, $p_t = P(U{=}1 \mid T{=}t, X{=}x)$, and note that $p = P(U{=}1 \mid X{=}x) = q_x p_1 + (1-q_x)p_0$ necessarily lies between p_0 and p_1. Let r denote the common U–Y risk ratio. Then $p_1/p_0 \le \mathrm{RR}_{TU}$ and $r \le \mathrm{RR}_{UY}$. By the law of total probability applied to the U-marginal within each arm,

$$\mathrm{RR}^{\mathrm{obs}}_{TY|X}(x) = \frac{P(Y=1 \mid T=1, X=x, U=0)\,[1+p_1(r-1)]}{P(Y=1 \mid T=0, X=x, U=0)\,[1+p_0(r-1)]}.$$

The causal risk ratio at x is $\mathbb{E}[Y(1) \mid X=x]/\mathbb{E}[Y(0) \mid X=x]$, which, under conditional exchangeability given (X, U), marginalizes U over the *same* distribution $P(U \mid X=x)$ in both arms:

$$\mathrm{RR}^{\mathrm{true}}_{TY|X}(x) = \frac{P(Y=1 \mid T=1, X=x, U=0)\,[1+p(r-1)]}{P(Y=1 \mid T=0, X=x, U=0)\,[1+p(r-1)]}.$$

Because r is common across arms, the factors $[1+p(r-1)]$ cancel, and taking the ratio of the observed to the causal risk ratio gives

$$\frac{\mathrm{RR}^{\mathrm{obs}}_{TY|X}(x)}{\mathrm{RR}^{\mathrm{true}}_{TY|X}(x)} = \frac{1+p_1(r-1)}{1+p_0(r-1)}. \tag{9.39}$$

It remains to maximize Equation 9.39 over (p_0, p_1, r) subject to $p_1 \le 1$, $p_1/p_0 \le \mathrm{RR}_{TU}$, and $1 \le r \le \mathrm{RR}_{UY}$. Write $s = r - 1 \ge 0$. The ratio $(1+p_1 s)/(1+p_0 s)$ is nondecreasing in p_1 and nonincreasing in p_0, so the maximum is attained at $p_1 = 1$ and at the smallest p_0 the constraint permits, namely $p_0 = 1/\mathrm{RR}_{TU}$. Substituting,

$$\frac{1+s}{1+s/\mathrm{RR}_{TU}} = \frac{\mathrm{RR}_{TU}\, r}{\mathrm{RR}_{TU} + r - 1},$$

whose derivative in r has the sign of $\mathrm{RR}_{TU} - 1 \ge 0$, so it is maximized at $r = \mathrm{RR}_{UY}$. This yields the closed-form bound $B(\mathrm{RR}_{TU}, \mathrm{RR}_{UY})$.

Under strict treatment positivity, the boundary value $p_1 = 1$ is not admissible: with $p_0 < 1$ it would force $P(T{=}1 \mid X, U{=}0) = 0$. The displayed bound is nevertheless sharp as a *supremum*: it is approached arbitrarily closely by taking $p_1 \uparrow 1$, $p_0 = p_1/\mathrm{RR}_{TU}$, and $r \uparrow \mathrm{RR}_{UY}$, and equality is attained at $(p_1, p_0, r) = (1,\, 1/\mathrm{RR}_{TU},\, \mathrm{RR}_{UY})$ if boundary treatment probabilities are allowed. This proves Equation 9.19.

For the E-value, set $\mathrm{RR}_{TU} = \mathrm{RR}_{UY} = e$ and ask for the smallest e such that the bound on the causal risk ratio reaches 1. This requires $B(e,e) = R$, i.e. $e^2/(2e-1) = R$, so $e^2 - 2Re + R = 0$ and $e = R + \sqrt{R(R-1)}$, which is Equation 9.20.

For Equation 9.21, note that B is nondecreasing in each argument, so $B(\mathrm{RR}_{TU}, \mathrm{RR}_{UY}) \le B(m,m)$ for $m = \max(\mathrm{RR}_{TU}, \mathrm{RR}_{UY})$. Hence $B \ge R$ forces $B(m,m) \ge R = B(\mathrm{EV}, \mathrm{EV})$, and since $e \mapsto B(e,e)$ is strictly increasing this gives $m \ge \mathrm{EV}$. The symmetric pair attains the minimum. For the first half of Equation 9.22, note that $B(a,b)$ is increasing in b with $\lim_{b\to\infty} B(a,b) = a$, so $B(a,b) < a$ for every finite b; hence $B \ge R$ forces $a \ge R$, and symmetrically $b \ge R$. This is the classical Cornfield inequality. □

S.4 MSM Computation: Perturbation Envelope and Linear-Fractional Optimization

Computational sketch (estimating the interval). The population statement follows from the assumptions: under consistency, latent exchangeability, and true-propensity positivity, weighting by π^{true} identifies τ, and the MSM confines the true weights to an envelope around the nominal ones; optimizing the weighting estimand over this valid outer envelope yields $[\tau_L, \tau_U]$, which therefore contains τ.

What follows describes the empirical version of this optimization. Under the MSM the weights used in weighting estimators can be written as $w_i^{\text{true}} = w_i \cdot \phi_i$, where $w_i = T_i/\pi(X_i) + (1-T_i)/\{1-\pi(X_i)\}$ is the nominal weight and ϕ_i is a perturbation factor. Solving Equation 9.23 for the treated arm gives

$$\phi_i = \frac{\pi(X_i)}{\pi^{\text{true}}(X_i, U_i)} \in \left[\, \pi(X_i) + \{1-\pi(X_i)\}/\Lambda,\;\; \pi(X_i) + \Lambda\,\{1-\pi(X_i)\}\,\right],$$

and symmetrically for the control arm.

Two features of this interval matter. First, it depends on $\pi(X_i)$ and is strictly contained in $[\Lambda^{-1}, \Lambda]$; treating each ϕ_i as freely selectable in $[\Lambda^{-1}, \Lambda]$ therefore over-relaxes the constraint. Second, the perturbations across units are further linked by the normalization built into the estimator. The estimator in use is the self-normalized (Hájek) form, whose numerator and denominator are both linear in ϕ, so optimizing it is a *linear-fractional* program rather than a linear one. Such a program attains its extrema on the boundary, and the optimal assignment of units to endpoints is a threshold rule in the ordered outcome values, which is why the solution takes a percentile form. Zhao et al. (2019) derive the resulting percentile expressions and a percentile-bootstrap procedure for inference. □

S.5 The Basic Unmeasured-Confounding DAG

S.6 A Worked Benchmarked-Reporting Example

A well-benchmarked sensitivity analysis reads in the form:

> The estimated effect of the job-training program on employment at twelve months is a risk ratio of 1.60 with 95% CI $[1.20, 2.10]$. The E-value is 2.58 for the point estimate and 1.69 for the lower confidence limit. Among the observed covariates, the largest bias factor is that of prior-year earnings, $B = 1.35$, whose equivalent E-value is $e = 1.35 + \sqrt{1.35 \times 0.35} = 2.04$; baseline education has $B = 1.15$ and $e = 1.57$. Because $2.58 > 2.04$, the point estimate is robust to an unmeasured confounder as strong as any single observed covariate: explaining it away would require a confounder roughly 27% stronger than prior-year earnings on this scale. The lower confidence limit is not robust in the same sense, since $1.69 < 2.04$.

The last two sentences are the output of benchmarking: they anchor the E-value to the observed data in a way that a substantive reader can interpret. Note also that the comparison is made on a single scale throughout — equivalent E-values against E-values — as Equation 9.24 requires.

S.7 Double Robustness and Neyman Orthogonality

The AIPW score has two *distinct* robustness properties, and it is worth separating them, since they are often conflated.

Double robustness is a global property of the score: its population estimating equation remains unbiased if either the outcome model $\mu_t(x)$ or the propensity model $\pi(x)$ is correctly specified, even if the other is not.

Neyman orthogonality is a local property: the population score or moment map has zero Gateaux derivative with respect to the nuisance argument, evaluated at the truth. It says that first-order nuisance-estimation error does not propagate into the target.

The two properties are distinct but related, and the relationship is asymmetric. Under the differentiability conditions of Chapters 11–12, a doubly robust population moment is automatically Neyman orthogonal at the intersection model: double robustness makes its expectation identically zero along either nuisance axis, so both axis derivatives vanish, and by linearity of the Gateaux derivative the joint nuisance derivative vanishes as well. The converse fails — local orthogonality at the truth does not deliver global unbiasedness when one nuisance component is misspecified. The AIPW score has both properties.

Only orthogonality directly supplies the local product-rate argument used for root-n inference with flexible nuisance estimators. Together with cross-fitting, it means the requirement on the nuisance estimators is a *product*-rate condition,

$$\|\hat{\mu} - \mu_0\|_{L_2(P)} \, \|\hat{\pi} - \pi_0\|_{L_2(P)} = o_p(n^{-1/2}),$$

together with overlap, moment, consistency, and score-regularity conditions. The familiar symmetric sufficient condition is that each nuisance error be $o_p(n^{-1/4})$ — note o_p, not O_p — which is what delivers $\sqrt{n}$-asymptotics and a normal limit. Both properties address estimation under correctly maintained identifying assumptions.

S.8 Lab: Monte Carlo SD versus Single-Replicate Standard Error

The Monte Carlo standard deviation describes the repeated-sampling spread of the estimator across simulation runs. It is not itself the standard error an analyst would compute from a single realized data set; that requires a replicate-specific variance estimator, here the usual OLS standard error for the coefficient on T, which in this design tracks the Monte Carlo standard deviation closely. Using that estimator, a typical single-replicate 95% interval is approximately [2.98, 3.37]. The naive estimator is badly biased, and the interval is narrow and does not contain the truth.

S.9 Lab: Extended Sensitivity Grid, Tipping Table, and Oracle Benchmarking

Oracle parameter-scale comparison — not observed-covariate benchmarking. The following comparison is indexed by the structural coefficients of the simulated DGP. It is an oracle comparison, not an estimator-matched observed-covariate benchmark: matching X's structural treatment and outcome coefficients gives $(\alpha_U, \delta) = (0.6, 0.8)$, but these quantities are not available to the analyst and do not equal the benchmark pair required for the OLS estimator. The analyst-facing benchmark remains $(d_X^{\text{OLS}}, \hat{\gamma}_X) \approx (0.472, 0.605)$, as derived in the main chapter's lab.

The table below reports the sensitivity-adjusted estimate $\hat{\tau}_{\text{sens}} = \hat{\tau}_X - B$ over a grid of (α_U, δ) values. The coefficient θ_U is computed from a large oracle run under each posited α_U — that is, using the simulated U, which a real analyst would not have — and is displayed in the header row.

Table 9.4: Sensitivity-adjusted ATE estimate for Lab 9, at $\hat{\tau}_X = 3.172$. Each cell is $\hat{\tau}_{\text{sens}} = \hat{\tau}_X - \delta \cdot \theta_U(\alpha_U)$. The true configuration is $(\alpha_U = 1.00,\ \delta = 2.0)$, with sensitivity-adjusted value $1.508 \approx \tau = 1.5$ (checkmark).

Posited δ	$\alpha_U = 0.25$ ($\theta_U = 0.247$)	0.50 (0.473)	0.75 (0.670)	1.00 (0.832)	1.50 (1.068)
0.5	3.049	2.936	2.837	2.756	2.638
1.0	2.925	2.699	2.502	2.340	2.104
1.5	2.802	2.463	2.167	1.924	1.570
2.0	2.678	2.226	1.832	**1.508** ✓	1.036
3.0	2.431	1.753	1.162	0.676	−0.032

Tipping points and benchmarking. The tipping point along each column is the value of δ at which the sensitivity-adjusted estimate reaches zero. Analytically, $\delta^\star(\alpha_U) = \hat{\tau}_X / \theta_U(\alpha_U)$. Computed values:

α_U	θ_U	$\delta^\star$
0.25	0.247	12.84
0.50	0.473	6.71
0.75	0.670	4.73

α_U	θ_U	$\delta^\star$
1.00	0.832	3.81
1.50	1.068	2.97

On the structural-coefficient scale, X enters the treatment model with coefficient 0.6 and the outcome model with coefficient 0.8; the corresponding oracle auxiliary coefficient is $\theta_U(0.6) = 0.556$. All comparisons below are on this oracle scale.

- At $\alpha_U = 1.00$, $\delta^\star = 3.81$: at that imbalance an unmeasured confounder would need an outcome association (oracle structural-coefficient scale) about $3.81/0.8 \approx 4.8$ times X's structural coefficient to zero out the estimate. This holds the imbalance fixed at a value larger than X's and compares outcome coefficients only.
- A confounder matching X's *structural* coefficients on both margins ($\alpha_U = 0.6$, $\delta = 0.8$) adjusts the estimate to $\hat{\tau}_X - 0.8 \cdot 0.556 \approx 2.73$ — still positive and far from zero. At that imbalance the tipping value is $\delta^\star = 3.172/0.556 \approx 5.71$, roughly 7 times X's structural outcome coefficient.
- The true δ in the DGP is $2.0 = 2.5\gamma_X$ at $\alpha_U = 1.0$; the sensitivity adjustment at this configuration recovers $1.508 \approx \tau$, confirming Equation 9.37.

S.10 Optional Problems

S1. Sampling vs. identification uncertainty. Explain in your own words the difference between a narrow sampling confidence interval and a robust causal conclusion. Construct an example data-generating process, with numerical parameters and a stated sample size n (choose one and report it; $n = 5000$ is a convenient default), in which the confidence interval for a back-door-adjusted ATE is very narrow — take the estimator's standard error below 0.05 — yet the true ATE lies outside it. Specify the treatment and outcome models, the omitted confounder, and which estimator you are using. In reporting, distinguish carefully between the Monte Carlo standard deviation of the estimator across simulation replicates and the standard error estimated from a single realized data set; state which one your 0.05 refers to. Finally, identify the identifying assumption that is violated and the magnitude of the violation.

S2. IV sensitivity curve. In a scalar-IV model with $\mathrm{Cov}(Y, Z) = 3$, $\mathrm{Cov}(T, Z) = 1.5$, $\mathrm{Var}(Z) = 1$:

(a) Use Equation 9.33 to compute $\hat{\beta}(\delta)$ for $\delta \in \{0, 0.5, 1.0, 1.5, 2.0\}$.
(b) Find the tipping point $\delta^\star$.
(c) Repeat part (a) for the weaker-instrument case $\mathrm{Cov}(T, Z) = 0.5$ (keeping the other quantities fixed). Explain why the same δ produces a larger bias.

S3. Positivity sensitivity. Explain why the trimmed estimand τ_α of Equation 9.29 is generally not equal to the untrimmed ATE τ. In a data set with $\pi(X)$ distributed uniformly on $[0, 1]$, what fraction of the population is retained at trimming levels $\alpha \in \{0.01, 0.05, 0.10, 0.20\}$? Under what scientific questions is τ_α preferable to τ as a target?

S4. Mediation sensitivity. In a mediation study of a behavioral intervention (T) on depression (Y) through sleep quality (M), explain why randomization of T alone does not eliminate the need for a sensitivity analysis for the indirect effect. Then represent a nonzero residual correlation ρ in Equation 9.36 graphically, in two equivalent ways: as an unobserved common cause of M and Y, and as a bidirected edge between the two disturbances. Explain why no single *directed* arrow in the DAG is equivalent to a residual correlation. Using the two-panel figure in Section 9.6.4, state which panel your graph corresponds to and what would change if the unobserved common cause were itself affected by T. Finally, give a plausible scientific story in which the residual correlation could be large.

S5. Comparing the Γ and MSM models. Both models bound an odds ratio at a level (Γ or Λ). State precisely which conditional odds ratio each model bounds and which design each attaches to. Explain why equal numerical levels $\Gamma = \Lambda$ need not represent the same amount of unmeasured confounding, and identify which model's one-number summary is a statement about a p-value and which is an interval for the ATE.

Part III

Estimation

Chapter 10

Estimating Equations and Influence Functions

Learning Objectives

By the end of this chapter, students should be able to:

1. Explain why identification of a causal parameter does not automatically yield a statistically reliable estimator, and articulate the distinct questions that an estimation theory must address.
2. Define an estimating equation and verify the population moment condition for standard examples (sample mean, OLS, IPW).
3. State the definition of an asymptotically linear estimator and derive the influence function from a generic first-order expansion of an estimating equation.
4. Compute the influence function for simple causal estimators when nuisance functions are known, interpret it as the infinitesimal contribution of a single observation, and recognize that such oracle influence functions are specific to the restricted model in which the nuisance is known.
5. Distinguish the influence function of an *estimator* from a pathwise gradient of a *functional*, and state the one-directional relationship between them: a regular asymptotically linear estimator's influence function is necessarily a gradient, but not conversely.
6. Describe how estimation error in nuisance functions propagates into the target estimator, and explain why this is the central statistical challenge in causal inference.
7. Use the influence function to construct a consistent variance estimator and an approximate Wald confidence interval, and distinguish the conditions needed to establish asymptotic linearity from those needed to estimate the variance thereafter.
8. Define efficiency in the class of regular estimators, define the efficient influence function as the canonical gradient, and explain its role as the foundation for doubly robust estimation in Chapter 11.

How to Read This Chapter

This chapter repairs distinctions that Chapters 11 and 12 depend on, and it is correspondingly dense in places. Not everything is first-pass material.

Essential on a first reading. Section 10.1–Section 10.4; the two influence-function definitions (Definition 10.2 and Definition 10.3) together with the one-directional bridge and the two-track diagram that follows them; the oracle examples of Section 10.6; the stacked influence-function formula Equation 10.5 and the ratio-of-means Example 10.4; Section 10.8; and the definition of the EIF together with the ATE formula in Section 10.9.4.

Second reading or advanced. The full contamination derivation for OLS (Example 10.3); the variance calculation for IPW with an estimated propensity score (Example 10.5); and the formal statement of the convolution theorem with its vector-valued refinement (Theorem 10.2).

The formal semiparametric machinery — regular submodels, scores, tangent spaces, and the canonical gradient — is developed in Appendix C, which this chapter cites rather than reproduces.

10.1 Why Estimation Needs Its Own Theory

So far the focus has been on *identification*: under what assumptions can a causal parameter be written as a functional of the observed-data distribution? Identification does not by itself provide a statistically reliable estimator. Once a causal parameter is identified, several distinct questions remain:

- How should the estimator be constructed from the observed sample?
- How does estimation of nuisance functions affect the target estimator?
- What is the large-sample distribution of the estimator?
- How can we compute valid standard errors and confidence intervals?

These questions motivate a separate theory of *estimation and inference*. In causal inference this issue is especially important because identified parameters often depend on auxiliary, or *nuisance*, functions such as the outcome regression $\mu_t(x) = \mathbb{E}(Y \mid T = t, X = x)$ or the propensity score $\pi(x) = P(T = 1 \mid X = x)$. Even when the causal parameter is identified, different estimation strategies may behave quite differently in terms of robustness, efficiency, and sensitivity to model misspecification.

The goal of this chapter is to introduce a general framework for estimation based on *estimating equations* and *influence functions*. This framework will serve as the foundation for the doubly robust and semiparametric methods developed in Chapter 11; see also Imbens and Rubin (2015) and Hernán and Robins (2020) for complementary treatments.

10.2 A Running Example: The Average Treatment Effect

Throughout this chapter we use the average treatment effect (ATE) as a running example. Let $\tau = \mathbb{E}\{Y(1) - Y(0)\}$. Whenever the ATE is invoked, we assume the arm-specific potential outcomes are integrable, $\mathbb{E}|Y(t)| < \infty$ for $t = 0, 1$, so that τ is well defined; additional second-moment conditions are stated where asymptotic normality is required. Under consistency, conditional exchangeability, and positivity, the ATE is identified by the back-door formula (Chapter 5),

$$\tau = \mathbb{E}[\mu_1(X) - \mu_0(X)], \qquad \mu_t(x) = \mathbb{E}(Y \mid T = t, X = x), \quad t \in \{0, 1\}.$$

This identity tells us what the target parameter is, but it does not uniquely determine how to estimate it. For example, one may consider a *regression-based* estimator obtained by fitting models for $\mu_1(x)$ and $\mu_0(x)$; a *weighting* estimator based on the propensity score $\pi(x)$ (Chapter 6); or an *augmented* estimator combining both outcome and treatment models. All of these estimators may target the same causal parameter, yet they differ in their statistical properties. A main goal of this chapter is to develop a common language for describing and comparing such estimators.

10.3 Estimating Equations

A broad class of estimators can be defined as solutions to *estimating equations*. Let $O_1, \dots, O_n$ be i.i.d. observations from a distribution P, and let θ denote a finite-dimensional target parameter. Write $\mathbb{P}_n f = n^{-1} \sum_{i=1}^n f(O_i)$ for the empirical average.

Definition: Estimating Equation

An **estimating equation** for a parameter θ is an equation of the form

$$\mathbb{P}_n\{U(O; \theta)\} = 0,$$

where the **population moment condition** $\mathbb{E}\{U(O; \theta_0)\} = 0$ holds at the true parameter value θ_0. The function $U(O; \theta)$ is called the **estimating function**.

An estimating-equation estimator $\hat{\theta}$ is any solution to the sample moment condition above. Many familiar estimators take this form.

Example: Sample Mean

Let $\theta = \mathbb{E}(Y)$. The sample mean $\hat{\theta} = \bar{Y}$ solves $\mathbb{P}_n(Y - \theta) = 0$. The estimating function is $U(O;\theta) = Y - \theta$.

Example 10.1 (Ordinary Least Squares)**.** Let $O = (X, Y)$ with $\mathbb{E}\|X\|^2 < \infty$, $\mathbb{E}(Y^2) < \infty$, and $\mathbb{E}(XX^\top)$ nonsingular, and suppose β is defined by the linear moment condition $\mathbb{E}[X\{Y - X^\top\beta\}] = 0$; the nonsingularity makes β unique. If an intercept is intended, X is understood to include a constant component. Then the OLS estimator solves

$$\mathbb{P}_n[X\{Y - X^\top\beta\}] = 0,$$

which has a unique solution once the sample matrix $\mathbb{P}_n(XX^\top)$ is invertible, an event of probability tending to one. For the central limit theorem applied to this estimating function we additionally require $\mathbb{E}\|X(Y - X^\top\beta)\|^2 < \infty$.

Example 10.2 (A Simple IPW Estimating Equation)**.** Suppose the propensity score $\pi(X)$ is known. Under the identification assumptions of consistency, conditional exchangeability, and positivity (Chapter 5), the ATE $\tau = \mathbb{E}\{Y(1) - Y(0)\}$ admits the inverse-probability representation

$$\tau = \mathbb{E}\left\{\frac{TY}{\pi(X)} - \frac{(1-T)Y}{1-\pi(X)}\right\}.$$

Then τ solves $\mathbb{E}[TY/\pi(X) - (1-T)Y/\{1-\pi(X)\} - \tau] = 0$, and the corresponding estimator replaces the expectation by the empirical average. The right-hand side is an observed-data functional only after identification has been imposed; without those assumptions, knowing $\pi(X)$ alone does not turn the expression into a causal quantity.

A word about integrability, because weak positivity is treacherous here. Under consistency, conditional exchangeability, positivity, and the standing assumption $\mathbb{E}|Y(t)| < \infty$, the IPW moment is well defined and equals τ; equivalently, one may impose the observed-data condition

$$\mathbb{E}\left[\frac{T|Y|}{\pi(X)} + \frac{(1-T)|Y|}{1-\pi(X)}\right] < \infty.$$

The *observed* second moment $\mathbb{E}(Y^2) < \infty$ does *not* by itself deliver this: treatment may be rare precisely where the treated outcome is large, so that large values contribute little to the observed moment while being magnified by $1/\pi(X)$. A square-integrable influence function requires the stronger weighted-second-moment condition

$$\mathbb{E}\left[\frac{TY^2}{\pi(X)^2} + \frac{(1-T)Y^2}{\{1-\pi(X)\}^2}\right] < \infty,$$

which is what the asymptotic theory of Section 10.4 needs. Strong overlap together with $\mathbb{E}(Y^2) < \infty$ is a convenient sufficient condition for the latter; see Section 10.9.1 for the precise requirement.

The main advantage of the estimating-equation framework is that it provides a unified language for both estimator construction and asymptotic analysis.

10.4 From Estimating Equations to Asymptotic Linearity

Estimating equations are useful not only because they define estimators, but also because they often yield a convenient first-order expansion. Under suitable regularity conditions, an estimator solving an estimating equation can typically be approximated as

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\varphi(O_i) + o_p(1)$$

for some mean-zero function $\varphi(O)$.

Definition 10.1 (Asymptotic Linearity and Influence Function)**.** An estimator $\hat{\theta}$ is **asymptotically linear** with **influence function** $\varphi(O)$ if

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \varphi(O_i) + o_p(1), \qquad \mathbb{E}\{\varphi(O)\} = 0,$$

where $\varphi \in L_2^0(P)$, that is, $\mathbb{E}\|\varphi(O)\|^2 < \infty$.

The square-integrability requirement is what makes the representation useful: it is exactly the condition under which the central limit theorem applies to the leading sum.

This representation is fundamental because it immediately implies asymptotic normality by the multivariate central limit theorem. When $\theta \in \mathbb{R}^p$ the influence function $\varphi(O)$ is also $\mathbb{R}^p$-valued, and

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(\mathbf{0},\ \Sigma)\,, \qquad \Sigma = \mathbb{E}[\varphi(O)\varphi(O)^\top],$$

where Σ is the $p \times p$ asymptotic covariance matrix. In the scalar case $p = 1$ this reduces to $\Sigma = \mathbb{E}[\varphi(O)^2]$. Once the influence function is identified, we obtain both the asymptotic distribution and a principled basis for variance estimation and confidence intervals.

Proposition 10.1 (Generic First-Order Expansion)**.** *Under smoothness and regularity conditions, an estimator $\hat{\theta}$ solving $\mathbb{P}_n\{U(O;\theta)\} = 0$ admits a first-order expansion*

$$\sqrt{n}(\hat{\theta} - \theta_0) = -A^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} U(O_i;\theta_0) + o_p(1),$$

where $A = \mathbb{E}\{\partial_{\theta^\top} U(O;\theta_0)\}$. Hence the influence function is $\varphi(O) = -A^{-1}U(O;\theta_0)$.

Proof. A Taylor expansion of the sample moment condition around θ_0 gives

$$0 = \mathbb{P}_n\{U(O;\hat{\theta})\} \approx \mathbb{P}_n\{U(O;\theta_0)\} + \mathbb{P}_n\{\partial_{\theta^\top} U(O;\theta_0)\}\,(\hat{\theta} - \theta_0).$$

By the law of large numbers, $\mathbb{P}_n\{\partial_{\theta^\top} U(O;\theta_0)\} \to A$ in probability. Rearranging and multiplying by $\sqrt{n}$ yields the stated expansion. □

Proposition 10.1 is a standard Z-estimation result and shows why estimating equations naturally lead to asymptotic linearity.

10.5 Influence Functions: Intuition

10.5.1 Statistical Functionals

Many parameters of interest in statistics and causal inference can be written as *functionals* of the underlying distribution. We make this precise before defining the influence function.

> Definition: Statistical Functional
>
> Let $\mathcal{P}$ be a collection of probability distributions on the sample space of O. A **statistical functional** is a map $\Psi : \mathcal{P} \to \mathbb{R}^p$ that assigns a parameter value $\psi = \Psi(P)$ to each distribution $P \in \mathcal{P}$. A **plug-in estimator** of ψ has the form $\hat{\psi} = \Psi(\hat{P})$, where $\hat{P}$ is an estimate of the components of P on which Ψ depends. The **empirical plug-in** $\hat{\psi} = \Psi(\mathbb{P}_n)$ is the special case $\hat{P} = \mathbb{P}_n$, available when Ψ is well defined at the empirical distribution.

> The Empirical Plug-In Is Not Always Available
>
> For functionals depending only on unconditional moments — the mean, the variance, the OLS coefficient — evaluating Ψ at $\mathbb{P}_n$ is unproblematic. For functionals involving *conditional* distributions it generally is not. With continuous X, the expression $\Psi(\mathbb{P}_n) = \mathbb{E}_{\mathbb{P}_n}\{\mu_{1,\mathbb{P}_n}(X) - \mu_{0,\mathbb{P}_n}(X)\}$ is not meaningful as it stands: the empirical distribution places mass at finitely many covariate values, supplies no stable arm-specific conditional mean at each such value, and typically violates empirical overlap, since almost every X_i is observed under only one treatment arm. For causal functionals

of this kind, $\hat{P}$ must include *fitted or smoothed* nuisance functions rather than raw empirical conditionals.

Example: Common Statistical Functionals

(i) **Mean.** $\Psi(P) = \mathbb{E}_P(Y)$. The plug-in estimator is $\bar{Y}$.
(ii) **Variance.** $\Psi(P) = \mathbb{E}_P(Y^2) - [\mathbb{E}_P(Y)]^2$. The plug-in estimator is the sample variance (with n in the denominator).
(iii) **Quantile.** $\Psi(P) = F_P^{-1}(q) = \inf\{y : F_P(y) \geq q\}$ for $q \in (0,1)$. The plug-in estimator is the sample q-quantile. (We write q rather than τ for the quantile level, since τ denotes the ATE throughout this chapter.)
(iv) **OLS regression coefficient.** $\Psi(P) = [\mathbb{E}_P(XX^\top)]^{-1}\mathbb{E}_P(XY)$. The plug-in estimator is $(\sum_i X_i X_i^\top)^{-1} \sum_i X_i Y_i$.
(v) **Average treatment effect.** $\Psi(P) = \mathbb{E}_P[\mu_1(X) - \mu_0(X)]$ under the back-door identification formula. Here the empirical plug-in is unavailable for continuous X; the plug-in estimator is $\Psi(\hat{P}) = \mathbb{P}_n\{\hat{\mu}_1(X) - \hat{\mu}_0(X)\}$, which averages *fitted* regression functions over the empirical distribution of X.

Remark: Linearity vs. Nonlinearity

A functional Ψ is **linear** if $\Psi((1-\epsilon)P + \epsilon Q) = (1-\epsilon)\Psi(P) + \epsilon\Psi(Q)$ for all $P, Q \in \mathcal{P}$ and $\epsilon \in [0,1]$. The mean (i) is linear; the variance (ii), quantile (iii), OLS coefficient (iv), and ATE (v) are nonlinear. Linear functionals are straightforward to analyze because their plug-in estimators are sample averages. Nonlinear functionals require a local linearization, which is precisely what the influence function provides.

10.5.2 Influence Functions

Both objects called "influence function" formalize the same intuition: the first-order effect of a single observation.

> *An influence function records the infinitesimal contribution of one observation to a first-order expansion.*

But *whose* first-order expansion is left open by that sentence, and the two available answers are not interchangeable. One attaches to an *estimator*, describing how $\hat{\psi}$ responds to perturbing the sample; the other attaches to a *functional together with a model*, describing how $\Psi(P)$ responds to perturbing the distribution. Conflating them is the most common source of confusion in this material, so we define them separately and then state exactly how they are related.

Definition 10.2 (Influence Function of an Estimator (informal))**.** Let $\psi = \Psi(P)$ be a scalar parameter and let $\hat{\psi}$ be an estimator of ψ. A mean-zero function $\varphi \in L_2(P)$ is called the **influence function of the estimator** $\hat{\psi}$ if $\hat{\psi}$ is asymptotically linear at P, that is,

$$\sqrt{n}(\hat{\psi} - \psi) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \varphi(O_i) + o_p(1), \qquad \mathbb{E}\{\varphi(O)\} = 0.$$

This is a property of the *estimator* $\hat{\psi}$.

This is the same expansion as in Definition 10.1; the unscaled form $\hat{\psi} - \psi = n^{-1} \sum_i \varphi(O_i) + o_p(n^{-1/2})$ is equivalent and is also commonly used.

Definition 10.3 (Pathwise Gradient of a Functional (informal))**.** Let Ψ be a functional defined on a statistical model $\mathcal{P}$. A mean-zero function $\varphi \in L_2(P)$ is called a **gradient** of Ψ at P relative to $\mathcal{P}$ if, for every regular parametric submodel $\{P_\epsilon\} \subset \mathcal{P}$ passing through P at $\epsilon = 0$ with score S,

$$\frac{d}{d\epsilon}\Psi(P_\epsilon)\bigg|_{\epsilon=0} = \mathbb{E}_P\{\varphi(O)\, S(O)\}.$$

This is a property of the *parameter* Ψ and the *model* $\mathcal{P}$; it makes no reference to any estimator.

Two Notions, One Direction of Implication

Definition 10.2 and Definition 10.3 are *not* equivalent, and it is an error to present either as a restatement of the other. The correct relationship runs in one direction only:

> *If an estimator is regular and asymptotically linear, then its estimator influence function is also a gradient of the functional.*

The converse is not automatic: exhibiting a gradient of Ψ does not by itself establish that any particular estimator admits the corresponding asymptotically linear expansion. Furthermore, gradients need not be unique — in a restricted model a functional generally has many — and the distinguished one lying in the model's tangent space is the *canonical gradient* introduced in Section 10.9.3. Formal statements are given in Appendix C; see in particular Section C.3, Section C.4, and Section C.6.

The figure below collects this architecture in one place. The left column is the *estimator-side* theory developed in Section 10.4–Section 10.8: one starts from an estimating function, expands, and reads off a variance. The right column is the *functional-side* theory of Section 10.9 and Appendix C: one starts from a target functional and a model, differentiates along submodels, and projects. The two columns are developed independently and meet only through the bridge at the bottom. It is worth returning to this figure as each piece is introduced.

Remark: Point-Mass Contamination Is a Heuristic, Not the Definition

A convenient device for *guessing* a gradient is the *contamination path* $P_\epsilon = (1-\epsilon)P + \epsilon\delta_O$, the ϵ-mixture of P with a point mass at a fixed observation O, whose derivative at $\epsilon = 0$ is the classical Hampel influence curve. This calculation is often the fastest route to a candidate formula, and in a fully nonparametric model the candidate it produces is typically the right one.

It is not, however, the formal definition. When P is continuous, δ_O is not absolutely continuous with respect to P, so $d\delta_O/dP$ does not exist as an $L_2(P)$ function and the contamination path is not in general a regular parametric submodel with a well-defined score. Formal pathwise differentiation instead uses smooth dominated submodels and the score identity of Definition 10.3; see Section C.2 for the precise statement of why the contamination path fails to qualify. The heuristic should therefore be used to *find* a candidate gradient, which is then *verified* against the score identity.

The following calculation carries out the contamination heuristic in full for the OLS coefficient. On a first reading, the point to extract is the *conclusion* — the two routes agree here, and the paragraph headed "What this does and does not establish" explains why that agreement is special to the nonparametric model. The intervening matrix-derivative algebra can be deferred.

Example 10.3 (OLS Slope by Contamination Derivative)**.** *Second reading.*

Let $O = (X, Y) \in \mathbb{R}^{p+1}$, under the moment and nonsingularity conditions of Example 10.1, and consider the population linear regression coefficient

$$\beta = \Psi(P) = \left[\mathbb{E}(XX^\top)\right]^{-1}\mathbb{E}(XY),$$

a nonlinear functional of P. The OLS estimator $\hat\beta = (\sum_i X_i X_i^\top)^{-1}\sum_i X_i Y_i$ is the empirical plug-in $\Psi(\mathbb{P}_n)$.

Step 1: Construct the perturbed distribution. Let $P_\epsilon = (1-\epsilon)P + \epsilon\delta_O$ be the mixture of P with a point mass at a fixed observation $O = (X, Y)$. Then

$$\mathbb{E}_{P_\epsilon}(\tilde X\tilde X^\top) = (1-\epsilon)\mathbb{E}(XX^\top) + \epsilon XX^\top, \qquad \mathbb{E}_{P_\epsilon}(\tilde X\tilde Y) = (1-\epsilon)\mathbb{E}(XY) + \epsilon XY,$$

so the functional evaluated at P_ϵ is

$$\Psi(P_\epsilon) = \left[(1-\epsilon)\mathbb{E}(XX^\top) + \epsilon XX^\top\right]^{-1}\left[(1-\epsilon)\mathbb{E}(XY) + \epsilon XY\right].$$

Step 2: Differentiate at $\epsilon = 0$. Write $M(\epsilon) = (1-\epsilon)\mathbb{E}(XX^\top) + \epsilon XX^\top$ and $v(\epsilon) = (1-\epsilon)\mathbb{E}(XY) + \epsilon XY$, so $\Psi(P_\epsilon) = M(\epsilon)^{-1}v(\epsilon)$. Both paths are linear in ϵ, with derivatives

$$\dot M(0) = XX^\top - \mathbb{E}(XX^\top), \qquad \dot v(0) = XY - \mathbb{E}(XY).$$

Deriving the derivative of a matrix inverse. For any differentiable matrix-valued path $M(\epsilon)$ with $M(\epsilon)$ invertible, differentiating the identity $M(\epsilon)M(\epsilon)^{-1} = I$ gives $\dot{M}M^{-1} + M\frac{d}{d\epsilon}M^{-1} = 0$, so

$$\frac{d}{d\epsilon}M(\epsilon)^{-1} = -M(\epsilon)^{-1}\dot{M}(\epsilon)\, M(\epsilon)^{-1}.$$

Evaluating at $\epsilon = 0$, with $M(0) = \mathbb{E}(XX^\top)$,

$$\frac{d}{d\epsilon}M(\epsilon)^{-1}\Big|_{\epsilon=0} = -[\mathbb{E}(XX^\top)]^{-1}(XX^\top - \mathbb{E}(XX^\top))[\mathbb{E}(XX^\top)]^{-1}.$$

Applying the product rule. Using $\frac{d}{d\epsilon}[M^{-1}v] = (\dot{M^{-1}})v + M^{-1}\dot{v}$,

$$\begin{aligned}\varphi(O) &= \frac{d}{d\epsilon}\Psi(P_\epsilon)\Big|_{\epsilon=0} \\ &= -[\mathbb{E}(XX^\top)]^{-1}(XX^\top - \mathbb{E}(XX^\top))[\mathbb{E}(XX^\top)]^{-1}\mathbb{E}(XY) + [\mathbb{E}(XX^\top)]^{-1}(XY - \mathbb{E}(XY)).\end{aligned}$$

Substituting $\beta = [\mathbb{E}(XX^\top)]^{-1}\mathbb{E}(XY)$ and collecting terms,

$$\begin{aligned}\varphi(O) &= [\mathbb{E}(XX^\top)]^{-1}\{XY - \mathbb{E}(XY) - (XX^\top - \mathbb{E}(XX^\top))\beta\} \\ &= [\mathbb{E}(XX^\top)]^{-1}X(Y - X^\top\beta) - [\mathbb{E}(XX^\top)]^{-1}\{\mathbb{E}(XY) - \mathbb{E}(XX^\top)\beta\}.\end{aligned}$$

The second term is zero because $\beta = [\mathbb{E}(XX^\top)]^{-1}\mathbb{E}(XY)$ implies $\mathbb{E}(XY) - \mathbb{E}(XX^\top)\beta = 0$. Hence

$$\varphi(O) = [\mathbb{E}(XX^\top)]^{-1}X(Y - X^\top\beta).$$

Verification. One can confirm $\mathbb{E}\{\varphi(O)\} = 0$ directly: $\mathbb{E}[X(Y - X^\top\beta)] = \mathbb{E}(XY) - \mathbb{E}(XX^\top)\beta = 0$ by definition of β.

Connection to the estimating-equation approach. The OLS estimating function is $U(O;\beta) = X(Y - X^\top\beta)$, so $A = \mathbb{E}\{\partial_{\beta^\top}U\} = -\mathbb{E}(XX^\top)$ and Proposition 10.1 gives $\varphi(O) = -A^{-1}U(O;\beta_0) = [\mathbb{E}(XX^\top)]^{-1}X(Y - X^\top\beta)$, agreeing with the contamination calculation.

What this does and does not establish. The two routes answer different questions, and their agreement here is informative rather than automatic. The estimating-equation route delivers the influence function *of the OLS estimator* (Definition 10.2). The contamination route produces a candidate *gradient of the functional* Ψ (Definition 10.3); as noted above, it is a heuristic, and confirming that the candidate is a genuine gradient requires verifying the score identity against regular submodels. In the nonparametric model the tangent space is all of $L_2^0(P)$, the gradient is unique when it exists, and the two objects coincide — which is exactly why OLS is a clean illustration. In restricted models the two can differ, and the distinction of Definition 10.2 and Definition 10.3 becomes essential.

10.6 Influence Functions for Simple Causal Estimators

We now illustrate the idea in causal settings.

> These Are Oracle Estimators in Restricted Models
>
> The next two subsections concern *oracle* estimators, and the model in which each one lives must be kept in view. In Section 10.6.1 the outcome regressions μ_t are treated as *known*; in Section 10.6.2 the propensity score π is treated as *known*. Each influence function below is therefore both *estimator-specific* and *model-specific*.
>
> Two consequences matter for the rest of the chapter. First, the two influence functions are introduced under *different* oracle restrictions, so any efficiency comparison between them must first specify a common model in which both estimators are available. The natural choice is the intersection model in which μ_t *and* π are both known; there both estimators are regular, and the Rao–Blackwell comparison of Problems 3(a)–(b) is a legitimate within-model comparison. Second, because φ_{reg} corresponds to a model in which μ_t is known — a strictly smaller model than the nonparametric

one — its variance may fall *below* the nonparametric efficiency bound of Section 10.9 with no contradiction whatever. An efficiency bound constrains estimators within the model for which it was computed, and oracle knowledge shrinks that model. Neither φ_{reg} nor φ_{IPW} is in general the canonical gradient of the ATE in the unrestricted observed-data model.

10.6.1 Known Outcome Regression

Suppose $\tau = \mathbb{E}[\mu_1(X) - \mu_0(X)]$ and the functions $\mu_1(\cdot)$ and $\mu_0(\cdot)$ are known. The natural plug-in estimator is $\hat{\tau}_{\text{reg}} = \mathbb{P}_n\{\mu_1(X) - \mu_0(X)\}$. Its influence function, which we denote φ_{reg}, is

$$\varphi_{\text{reg}}(O) = \mu_1(X) - \mu_0(X) - \tau. \tag{10.1}$$

Note that φ_{reg} depends on O only through X.

10.6.2 Known Propensity Score

Suppose $\pi(X)$ is known and $\tau = \mathbb{E}\{TY/\pi(X) - (1-T)Y/[1-\pi(X)]\}$. The empirical analogue is

$$\hat{\tau}_{\text{IPW}} = \mathbb{P}_n\left\{\frac{TY}{\pi(X)} - \frac{(1-T)Y}{1-\pi(X)}\right\},$$

and its influence function, denoted φ_{IPW}, is

$$\varphi_{\text{IPW}}(O) = \frac{TY}{\pi(X)} - \frac{(1-T)Y}{1-\pi(X)} - \tau. \tag{10.2}$$

These two symbols are used throughout the remainder of the chapter and in the problems; both are defined only under the oracle conventions stated above.

Remark: Nuisance Functions and the Influence Function

Both examples above are *sample averages*: each has the form $\hat{\psi} = \mathbb{P}_n h(O)$ for a known function h, whose estimating function $U(O;\psi) = h(O) - \psi$ satisfies $A = -1$. For such estimators the influence function is exactly the centered summand, $\varphi(O) = h(O) - \mathbb{E}\{h(O)\}$.
This is a feature of the sample-average form, *not* of known nuisance in general. Even when every nuisance quantity is known, an estimator defined by $\mathbb{P}_n\{U(O;\theta)\} = 0$ has influence function $\varphi(O) = -A^{-1}U(O;\theta_0)$ by Proposition 10.1, and the normalization $-A^{-1}$ cannot be dropped. The OLS calculation of Example 10.3 is a counterexample within this chapter: there is no estimated nuisance parameter at all, yet the influence function is $[\mathbb{E}(XX^\top)]^{-1}X(Y - X^\top\beta)$ rather than the raw estimating function $X(Y - X^\top\beta)$.
The distinct issue taken up in the next section is what happens when nuisance functions are *unknown* and must be estimated from the same data: estimation error in $\hat{\mu}_t$ or $\hat{\pi}$ then perturbs the first-order expansion, adding a term that neither $-A^{-1}U$ with α held fixed nor the centered summand captures.

"Influence Function" and Its Neighbors

Several closely related but distinct objects appear in this chapter, and the same symbol φ is often overloaded across them. Keeping the following five concepts separate prevents confusion.

(i) **Estimating function.** A function $U(O;\theta)$ that defines an estimator through the sample moment equation $\mathbb{P}_n\{U(O;\theta)\} = 0$, with population condition $\mathbb{E}\{U(O;\theta_0)\} = 0$. Estimating functions are inputs to the construction of $\hat{\theta}$.

(ii) **Influence function of an estimator.** A function $\varphi(O)$ such that the estimator $\hat{\psi}$ admits the asymptotically linear expansion $\sqrt{n}(\hat{\psi} - \psi_0) = n^{-1/2}\sum_i \varphi(O_i) + o_p(1)$. This is a property of the estimator $\hat{\psi}$. In simple cases with known nuisance, φ is the centered estimating function; when nuisance is estimated, the two differ (Section 10.7).

(iii) **Gradient (pathwise influence function) of a functional.** A mean-zero $\varphi \in L_2(P)$ representing the pathwise derivative of Ψ at P through the score identity of Definition 10.3. This is a property of the parameter Ψ and the model $\mathcal{P}$, independent of any estimator. In a restricted model the gradient is generally *not unique*: any two gradients differ by an element orthogonal to the tangent space.

(iv) **Canonical gradient, or efficient influence function (EIF).** The unique gradient that lies in the model's tangent space $\mathcal{T}$, obtained by projecting any gradient onto $\mathcal{T}$. Among all gradients it has the smallest variance, and that variance is the semiparametric efficiency bound (Section 10.9.3; Section C.6). In a fully nonparametric model $\mathcal{T} = L_2^0(P)$, the gradient is unique, and the two notions collapse.

(v) **Estimated influence values.** The numerical quantities $\hat{\varphi}_i = \hat{\varphi}(O_i)$, $i = 1, \dots, n$, obtained by plugging estimated nuisance functions and parameters into the influence-function formula. These are the numbers used to compute the sandwich variance $\hat{V} = (n(n-1))^{-1} \sum_i (\hat{\varphi}_i - \bar{\varphi})^2$.

The connection. The two sides meet through a single bridge: if $\hat{\psi}$ is *regular and asymptotically linear*, then its estimator influence function (ii) must be a gradient (iii) of the functional. Being regular is not enough on its own — a regular estimator can be inefficient, and need not be asymptotically linear at all. The estimator is *efficient* precisely when its influence function equals the canonical gradient (iv), which for a well-designed estimator requires suitable conditions on the nuisance estimators, not merely their consistency. The estimated influence values (v) are then data-driven proxies used for variance estimation once the expansion has been established. Context disambiguates which sense of φ is intended.

10.7 Z-Estimation with Nuisance Parameters

The examples in Section 10.6 treated nuisance functions such as $\mu_t(x)$ and $\pi(x)$ as known. That simplification was pedagogically useful because it made the influence function transparent, but it is not realistic in practice. In real applications, nuisance functions must be estimated from the same data used to estimate the target parameter. Once this happens, the first-order expansion of $\hat{\psi}$ is no longer obtained by simply centering the estimating function with the nuisance held fixed: the randomness in the estimated nuisance introduces an additional term that the naive plug-in formula ignores. We therefore need a framework that accounts jointly for estimation of the target parameter and the nuisance parameter. Z-estimation provides that framework: it treats both as components of a single stacked estimating equation, so the corrected influence function — including the nuisance-estimation term — falls out of one unified first-order expansion.

10.7.1 The Stacked Estimating Equation Framework

The key idea is that once α is estimated rather than known, the target equation for ψ and the nuisance equation for α must be analyzed jointly rather than one after the other: only the joint analysis reveals the additional term that the estimated nuisance contributes to the first-order expansion of $\hat{\psi}$.

Suppose the target parameter $\psi \in \mathbb{R}^p$ depends on an unknown nuisance parameter $\alpha \in \mathbb{R}^q$ (e.g., regression coefficients in a parametric model for $\pi(x)$). Assume both are estimated by solving a *stacked* system of estimating equations:

$$\mathbb{P}_n\{U_1(O; \psi, \alpha)\} = 0, \tag{10.3}$$

$$\mathbb{P}_n\{U_2(O; \alpha)\} = 0. \tag{10.4}$$

Here $U_1(O; \psi, \alpha) \in \mathbb{R}^p$ defines the target estimator $\hat{\psi}$ given α, while $U_2(O; \alpha) \in \mathbb{R}^q$ defines the nuisance estimator $\hat{\alpha}$. These are solved *simultaneously* (or sequentially: first Equation 10.4 for $\hat{\alpha}$, then Equation 10.3 for $\hat{\psi}$). In the running ATE example, $p = 1$ and $\psi = \tau$, so U_1 is scalar.

Remark: Why Stack the Equations?

One might think it is enough to plug the estimated $\hat{\alpha}$ into the influence function derived under known α and proceed as in Section 10.6. This is incorrect in general: the randomness in $\hat{\alpha}$ contributes an additional term to the first-order expansion of $\hat{\psi}$ that is not captured by the plug-in influence function. The stacked framework keeps track of this extra term automatically.

10.7.2 The Z-Estimation Theorem

Write $\theta = (\psi^\top, \alpha^\top)^\top \in \mathbb{R}^{p+q}$ and let

$$U(O;\theta) = \begin{pmatrix} U_1(O;\psi,\alpha) \\ U_2(O;\alpha) \end{pmatrix}$$

be the stacked estimating function. The joint estimator $\hat{\theta} = (\hat{\psi}^\top, \hat{\alpha}^\top)^\top$ solves $\mathbb{P}_n\{U(O;\hat{\theta})\} = 0$.

Theorem 10.1 (Z-Estimation Theorem (Stacked Equations)). *Suppose the true parameter value is $\theta_0 = (\psi_0^\top, \alpha_0^\top)^\top$ and the following regularity conditions hold:*

(i) $\mathbb{E}\{U(O;\theta_0)\} = 0$ *(population moment condition);*
(ii) $\hat{\theta} \xrightarrow{p} \theta_0$ *(consistency of the joint estimator);*
(iii) the Jacobian $\mathcal{A} = \mathbb{E}\{\partial_{\theta^\top} U(O;\theta_0)\}$ *is invertible;*
(iv) $U(O;\theta)$ *is smooth enough in* θ *that a uniform law of large numbers and a CLT apply.*

Then

$$\sqrt{n}(\hat{\theta} - \theta_0) = -\mathcal{A}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} U(O_i;\theta_0) + o_p(1).$$

In particular, after partitioning $\mathcal{A}^{-1}$ *conformably with* (ψ, α) *as detailed below, the influence function of* $\hat{\psi}$ *is*

$$\varphi(O) = -A_{11}^{-1} U_1(O;\psi_0,\alpha_0) + A_{11}^{-1} A_{12} A_{22}^{-1} U_2(O;\alpha_0).$$

Remark: On the Consistency Hypothesis

Hypothesis (ii) is a separate prerequisite, not a consequence of (i), (iii), and (iv). Standard arguments establishing it require additional structure: identifiability of θ_0 via $\mathbb{E}\{U(O;\theta)\} = 0$ having a unique zero on the parameter space, together with uniform convergence of the sample moment $\mathbb{P}_n\{U(O;\theta)\}$ to its population counterpart on a compact neighborhood of θ_0. See Vaart (1998) (Theorem 5.9) for a standard set of sufficient conditions. In this chapter we treat consistency as given and focus on the asymptotic linearization that follows from it.

Proof. The argument mirrors Proposition 10.1 applied to the full stacked system. A Taylor expansion of $\mathbb{P}_n\{U(O;\hat{\theta})\} = 0$ around θ_0 gives

$$0 \approx \mathbb{P}_n\{U(O;\theta_0)\} + \mathbb{P}_n\{\partial_{\theta^\top} U(O;\theta_0)\} (\hat{\theta} - \theta_0).$$

By the law of large numbers, the matrix of derivatives converges to $\mathcal{A}$. Rearranging and multiplying by $\sqrt{n}$,

$$\sqrt{n}(\hat{\theta} - \theta_0) \approx -\mathcal{A}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} U(O_i;\theta_0).$$

Extracting the first component gives the influence function of $\hat{\psi}$. □

To see the structure explicitly, partition $\mathcal{A}$ conformably with (ψ, α):

$$\mathcal{A} = \begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix},$$

where $A_{11} = \mathbb{E}\{\partial_{\psi^\top} U_1\}$, $A_{12} = \mathbb{E}\{\partial_{\alpha^\top} U_1\}$, and $A_{22} = \mathbb{E}\{\partial_{\alpha^\top} U_2\}$. The lower-left block is zero because U_2 does not depend on ψ. By block-matrix inversion,

$$\mathcal{A}^{-1} = \begin{pmatrix} A_{11}^{-1} & -A_{11}^{-1} A_{12} A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{pmatrix}.$$

Reading off the first row, the influence function of $\hat{\psi}$ is

$$\varphi(O) = -A_{11}^{-1} U_1(O;\psi_0,\alpha_0) + A_{11}^{-1} A_{12} A_{22}^{-1} U_2(O;\alpha_0). \tag{10.5}$$

Remark: Just-Identified vs. Overidentified Moments

Formula Equation 10.5 presupposes that the system is **just identified**: U_1 has the same dimension as ψ and U_2 has the same dimension as α, so that A_{11} and A_{22} are square and invertible. This is the relevant representation for *finite-dimensional, just-identified parametric implementations* of the regression, IPW, and AIPW estimators studied in Chapter 11, in which the nuisance coefficients are estimated by a corresponding finite system of score or moment equations — one equation per unknown parameter.

When the nuisance functions are instead estimated nonparametrically or by machine learning, the nuisance is infinite-dimensional, there is no finite block A_{22} to invert, and the block-inverse representation does not apply; Chapters 11 and 12 describe the orthogonality and empirical-process or cross-fitting arguments used in that setting. When more moment conditions are available than parameters — the *overidentified* case familiar from generalized method of moments (GMM) — the simple block-inverse formula no longer applies. One then minimizes a quadratic form $\mathbb{P}_n U(O;\theta)^\top W \mathbb{P}_n U(O;\theta)$ for some weight matrix W, and the influence function takes the standard GMM sandwich form involving W and the Jacobian (Hansen 1982). The block formula above is the just-identified specialization.

Remark: Interpreting the Correction Term

Equation Equation 10.5 decomposes the influence function into two parts. The first term, $-A_{11}^{-1}U_1(O;\psi_0,\alpha_0)$, is exactly what we would obtain if α_0 were known. The second term, $A_{11}^{-1}A_{12}A_{22}^{-1}U_2(O;\alpha_0)$, is a *nuisance correction* that accounts for the additional variability introduced by estimating α. If $A_{12}=0$ the correction vanishes, and estimating α confers no first-order benefit or cost.

Two clarifications about what $A_{12}=0$ does and does not say. First, it is a statement about an *expectation*: $A_{12}=\mathbb{E}\{\partial_{\alpha^\top}U_1(O;\psi_0,\alpha_0)\}=0$ asserts that the *population moment* is insensitive to perturbations of α to first order at the truth. It does *not* assert that $U_1(O;\psi,\alpha)$ is free of α pointwise; the function may vary strongly in α observation by observation while these variations average to zero. This condition is the finite-dimensional analogue of *Neyman orthogonality*, taken up systematically in Chapter 12.

Second, orthogonality should not be identified with double robustness. They are related but logically distinct: orthogonality is a *local* derivative condition holding at the truth, whereas double robustness is a *global* property, namely that the moment condition retains mean zero when one of two nuisance components is misspecified anywhere in its model class. Neither implies the other without additional structure. Double robustness is developed in Chapter 11.

10.7.3 A Simple Stacked Example: A Ratio of Means

Before applying Equation 10.5 to the propensity-score problem, it is worth seeing the block correction operate in a setting with no causal structure and no matrix algebra, where every quantity can be computed in one line.

Example 10.4 (Ratio of Means)**.** Let $O=(Y,Z)$ with $\mathbb{E}(Z)\neq 0$ and finite second moments, and let the target be the ratio $\psi=\mathbb{E}(Y)/\mathbb{E}(Z)$. Treat $\alpha=\mathbb{E}(Z)$ as a nuisance parameter and estimate (ψ,α) jointly from the stacked system

$$U_1(O;\psi,\alpha)=Y-\psi\alpha, \qquad U_2(O;\alpha)=Z-\alpha.$$

Both population moment conditions hold at $(\psi_0,\alpha_0)=(\mathbb{E}Y/\mathbb{E}Z,\ \mathbb{E}Z)$.

Jacobian blocks. Differentiating,

$$A_{11}=-\alpha_0=-\mathbb{E}(Z), \qquad A_{12}=-\psi_0, \qquad A_{22}=-1.$$

Note $A_{12}=-\psi_0\neq 0$ whenever $\psi_0\neq 0$, so the nuisance correction genuinely matters here.

Influence function. Substituting into Equation 10.5,

$$\varphi(O)=-A_{11}^{-1}U_1+A_{11}^{-1}A_{12}A_{22}^{-1}U_2=\frac{Y-\psi_0\mathbb{E}(Z)}{\mathbb{E}(Z)}-\frac{\psi_0\{Z-\mathbb{E}(Z)\}}{\mathbb{E}(Z)},$$

and the terms in $\psi_0\mathbb{E}(Z)$ cancel, leaving the compact form

$$\varphi(O) = \frac{Y - \psi_0 Z}{\mathbb{E}(Z)}.$$

One checks directly that $\mathbb{E}\{\varphi(O)\} = \{\mathbb{E}(Y) - \psi_0\mathbb{E}(Z)\}/\mathbb{E}(Z) = 0$, as required.

What to notice. Had we ignored the nuisance and simply centered U_1 at the true α_0, we would have obtained $\{Y - \psi_0\mathbb{E}(Z)\}/\mathbb{E}(Z)$, which involves only the randomness in Y. Call that oracle influence function $\varphi_{\text{orc}}(O)$. The correction term supplies the $-\psi_0 Z$ piece, capturing the fact that $\hat{\alpha} = \bar{Z}$ is itself estimated.

Whether this raises or lowers the asymptotic variance is not settled by signs alone. Comparing the two,

$$\mathrm{Var}\{\varphi(O)\} - \mathrm{Var}\{\varphi_{\text{orc}}(O)\} = \frac{\psi_0\{\psi_0\mathrm{Var}(Z) - 2\,\mathrm{Cov}(Y,Z)\}}{\{\mathbb{E}(Z)\}^2},$$

so for $\psi_0 > 0$ the correction *reduces* the variance exactly when $\psi_0\mathrm{Var}(Z) < 2\,\mathrm{Cov}(Y,Z)$. Positive correlation between Y and Z is therefore not sufficient: if $\mathrm{Cov}(Y,Z)$ is positive but small relative to $\psi_0\mathrm{Var}(Z)$, estimating α *increases* the asymptotic variance.

This is worth dwelling on, because the propensity-score example of the next subsection behaves differently. There the correction can *only* reduce the variance, and that guarantee is not a general feature of the block formula: it comes from U_2 being a maximum likelihood score for a correctly specified model, which supplies the Bartlett and cross identities Equation 10.8 and Equation 10.9. Those identities fail here — $A_{12} = -\psi_0$ while $\mathbb{E}[U_1U_2] = \mathrm{Cov}(Y,Z)$, so $A_{12} = -\mathbb{E}[U_1U_2]$ only in the coincidental case $\psi_0 = \mathrm{Cov}(Y,Z)$. Absent that structure, the nuisance correction is simply a correction: it can move the variance in either direction.

10.7.4 Working Example: IPW with Estimated Propensity Score

Chapter 6 introduced IPW as an estimator motivated by the back-door identification formula and the propensity score. The present example revisits the same estimator from a different angle: not identification, but first-order asymptotic analysis when the propensity score is itself estimated. The generic target parameter ψ is now the ATE τ.

Setup. Assume a logistic regression model $\pi(X;\alpha) = \mathrm{expit}(X^\top\alpha)$. The IPW estimating equation for τ is

$$U_1(O;\tau,\alpha) = \frac{TY}{\pi(X;\alpha)} - \frac{(1-T)Y}{1-\pi(X;\alpha)} - \tau,$$

and the score equation for the logistic regression is $U_2(O;\alpha) = X\{T - \pi(X;\alpha)\}$. The stacked system $\mathbb{P}_nU_1 = 0$, $\mathbb{P}_nU_2 = 0$ yields $(\hat{\tau},\hat{\alpha})$.

Computing the Jacobian blocks. The blocks A_{11} and A_{22} are immediate:

$$A_{11} = -1, \qquad A_{22} = -\mathbb{E}\{\pi(X;\alpha_0)(1-\pi(X;\alpha_0))XX^\top\} =: -\Sigma_\pi.$$

For A_{12}, use the logistic score identity $\partial\pi/\partial\alpha^\top = \pi(1-\pi)X^\top$. Differentiating $1/\pi$ and $1/(1-\pi)$ then gives

$$\frac{\partial}{\partial\alpha^\top}\left[\frac{1}{\pi}\right] = -\frac{1-\pi}{\pi}X^\top, \qquad \frac{\partial}{\partial\alpha^\top}\left[\frac{1}{1-\pi}\right] = \frac{\pi}{1-\pi}X^\top,$$

so

$$\frac{\partial U_1}{\partial\alpha^\top} = -YX^\top\left\{\frac{T(1-\pi)}{\pi} + \frac{(1-T)\pi}{1-\pi}\right\}.$$

Using the identities $T(1-\pi)/\pi = T/\pi - T$ and $(1-T)\pi/(1-\pi) = (1-T)/(1-\pi) - (1-T)$, this simplifies to

$$\frac{\partial U_1}{\partial\alpha^\top} = -YX^\top\left\{\frac{T}{\pi} + \frac{1-T}{1-\pi} - 1\right\}.$$

Evaluating at α_0 and taking expectations,

$$A_{12} = -\mathbb{E}\left[YX^\top\left\{\frac{T}{\pi(X;\alpha_0)} + \frac{1-T}{1-\pi(X;\alpha_0)} - 1\right\}\right].$$

The corrected influence function. Substituting into Equation 10.5 with $A_{11} = -1$ (so $-A_{11}^{-1} = 1$) and $A_{22}^{-1} = -\Sigma_\pi^{-1}$, the two negative signs cancel in the second term, giving

$$\varphi(O) = U_1(O;\tau_0,\alpha_0) + A_{12}\,\Sigma_\pi^{-1}\,U_2(O;\alpha_0). \tag{10.6}$$

The first term is the naive IPW influence function from Section 10.6.2. The second term is the nuisance correction: it removes the first-order contribution of $\widehat{\alpha} - \alpha_0$ to the estimation error in $\widehat{\tau}$.

The next example works out the consequence of this correction for the asymptotic variance. It is the most technically demanding passage in the chapter, requiring the stacked expansion, the likelihood score, the information equality, and differentiation of an expectation whose law depends on α. On a first reading it is enough to take away the conclusion: estimating a correctly specified parametric propensity score by maximum likelihood can only lower the asymptotic variance of the IPW estimator.

Example 10.5 (Variance of IPW with Estimated Propensity Score)**.** *Advanced.*

We show that estimating α by maximum likelihood (ML) can only *reduce* the asymptotic variance of $\widehat{\tau}_{\mathrm{IPW}}$ relative to using the true α_0. Two Bartlett-type identities are the key.

Variance of the corrected influence function. Write $\sigma^2_{\mathrm{naive}} = \mathbb{E}[U_1(O;\tau_0,\alpha_0)^2]$ for the variance of the naive IPW estimating function when α_0 is treated as known. Expanding the square of Equation 10.6,

$$\mathbb{E}[\varphi(O)^2] = \sigma^2_{\mathrm{naive}} + 2\,A_{12}\Sigma_\pi^{-1}\mathbb{E}[U_1U_2] + A_{12}\Sigma_\pi^{-1}\mathbb{E}[U_2U_2^\top]\Sigma_\pi^{-1}A_{12}^\top. \tag{10.7}$$

Identity 1: Bartlett identity for the ML score. Because $U_2(O;\alpha) = X\{T - \pi(X;\alpha)\}$ is the ML score for α in a correctly specified logistic model, it satisfies the information equality

$$\mathbb{E}[\partial_{\alpha^\top}U_2(O;\alpha_0)] + \mathbb{E}[U_2U_2^\top] = 0.$$

Since $A_{22} = -\Sigma_\pi$, this gives

$$\mathbb{E}[U_2U_2^\top] = \Sigma_\pi. \tag{10.8}$$

Identity 2: a cross-identity linking A_{12} to $\mathbb{E}[U_1U_2^\top]$. A different identity constrains A_{12}. It comes not from U_1 being a score (it is not), but from the fact that $\mathbb{E}_\alpha[U_1]$ is identically zero as a function of α, where $\mathbb{E}_\alpha$ denotes expectation under a data-generating distribution in which the propensity score is $\pi(X;\alpha)$ and the distributions P_X and $P_{Y|T,X}$ are held fixed. Under this convention, and assuming consistency, conditional exchangeability, and correct specification of the propensity model,

$$\mathbb{E}_\alpha[U_1(O;\tau_0,\alpha)] = \mathbb{E}_X[\mu_1(X)] - \mathbb{E}_X[\mu_0(X)] - \tau_0 = 0 \quad \text{for all } \alpha.$$

Differentiating both sides with respect to $\alpha^\top$ and applying the Leibniz rule,

$$0 = \frac{\partial}{\partial\alpha^\top}\mathbb{E}_\alpha[U_1] = \mathbb{E}_\alpha\left[\frac{\partial U_1}{\partial\alpha^\top}\right] + \mathbb{E}_\alpha[U_1U_2^\top].$$

Evaluating at $\alpha = \alpha_0$ gives

$$A_{12} = -\mathbb{E}[U_1U_2^\top]. \tag{10.9}$$

Variance reduction. Identity Equation 10.9 gives $\mathbb{E}[U_1U_2] = -A_{12}^\top$, so the middle term of Equation 10.7 becomes $-2A_{12}\Sigma_\pi^{-1}A_{12}^\top$, while Equation 10.8 collapses the third term to $A_{12}\Sigma_\pi^{-1}A_{12}^\top$. Hence

$$\begin{aligned}\mathbb{E}[\varphi(O)^2] &= \sigma^2_{\mathrm{naive}} - 2\,A_{12}\Sigma_\pi^{-1}A_{12}^\top + A_{12}\Sigma_\pi^{-1}A_{12}^\top \\ &= \sigma^2_{\mathrm{naive}} - A_{12}\Sigma_\pi^{-1}A_{12}^\top = \sigma^2_{\mathrm{naive}} - \mathbb{E}[U_1U_2^\top]\Sigma_\pi^{-1}\mathbb{E}[U_2U_1].\end{aligned}$$

Since Σ_π is positive definite, the subtracted term is nonnegative, and therefore $\mathbb{E}[\varphi(O)^2] \le \sigma^2_{\mathrm{naive}}$, with equality only when $\mathbb{E}[U_1U_2^\top] = 0$. Thus *maximum likelihood estimation of a correctly specified parametric propensity-score model can reduce the asymptotic variance of the Horvitz–Thompson IPW estimator relative to using the true propensity score.*

We refer to this as *estimated-propensity-score variance reduction*; it is sometimes called the "propensity score paradox," though that phrase is also used for an unrelated phenomenon in matching, so the descriptive name is preferable. The phenomenon traces to Rosenbaum (1987), who observed in the context of model-based direct adjustment that weighting by an estimated propensity score can outperform weighting by the true one. It was placed on a formal semiparametric footing by Robins et al. (1992), and Hirano et al. (2003) later showed that weighting by their nonparametric series-logit propensity-score estimator attains the semiparametric efficiency bound for the ATE under the regularity conditions of that paper — a result that extends the present finite-dimensional calculation well beyond the parametric setting assumed here, though not automatically to every flexible propensity-score learner.

Scope of the result. The conclusion rests on three assumptions: (i) the propensity score model $\pi(X;\alpha)$ is a correctly specified *finite-dimensional* parametric model; (ii) $\hat{\alpha}$ is the maximum-likelihood estimator, so that U_2 is the score and the Bartlett identity holds; and (iii) the strong-overlap and moment conditions of Section 10.9.1 are met so that σ^2_{naive} is finite. When the parametric model is misspecified, when $\hat{\alpha}$ is computed by some other method, or when $\pi(X)$ is estimated nonparametrically by flexible machine learning, the identity $A_{12} = -\mathbb{E}[U_1 U_2^\top]$ need not hold and the variance-reduction conclusion fails in this simple form. The result should not be read as the blanket claim that *any* estimated propensity score improves IPW variance. It is worth noting, however, that a positive result *is* available in the nonparametric case, by a quite different argument: Hirano et al. (2003) show that weighting by their nonparametric series-logit propensity-score estimator attains the semiparametric efficiency bound for the ATE under suitable smoothness conditions. What fails here is the particular finite-dimensional Bartlett-identity argument, not the phenomenon itself.

Remark: Sandwich Variance Estimator

Equation Equation 10.5 also provides the basis for the *sandwich variance estimator*. Replacing θ_0 by $\hat{\theta}$ in all quantities, the asymptotic variance of $\hat{\tau}$ is consistently estimated by

$$\hat{V} = \frac{1}{n(n-1)} \sum_{i=1}^{n} (\hat{\varphi}_i - \bar{\varphi})^2, \qquad \bar{\varphi} = \frac{1}{n} \sum_{i=1}^{n} \hat{\varphi}_i,$$

where $\hat{\varphi}_i$ is $\varphi(O_i)$ evaluated at $(\hat{\tau}, \hat{\alpha})$. This estimator automatically accounts for nuisance estimation uncertainty and is valid without any re-sampling. A variance calculation for $\hat{\tau}$ that treats the fitted propensity-score weights as fixed — ignoring the link between $\hat{\alpha}$ and $\hat{\tau}$ — will in general be *wrong*, and by Example 10.5 typically too large. A logistic regression routine does not itself report a standard error for the resulting IPW estimate of the ATE.

The Central Challenge in Causal Estimation

The primary difficulty in practice is often not identifying the target parameter, but *correctly accounting for the effect of nuisance estimation* on the asymptotic distribution of the final estimator. Treating $\hat{\alpha}$ as if it were the true α_0 — that is, computing standard errors from the known-propensity influence function rather than from the corrected form Equation 10.5 — still yields a *consistent* point estimator under a correctly specified propensity-score model, but the variance it reports is generally wrong.
By the calculation just completed, the naive variance σ^2_{naive} exceeds the true asymptotic variance $\sigma^2_{\text{naive}} - A_{12}\Sigma_\pi^{-1}A_{12}^\top$, so, under the stated parametric model and the first-order normal approximation, treating the fitted weights as fixed is *asymptotically conservative*: the resulting intervals are wider than necessary and their limiting coverage exceeds the nominal level whenever $A_{12} \neq 0$. This is an asymptotic statement, not a finite-sample guarantee. The naive variance is exact only in the special case $A_{12} = 0$, equivalently $\mathbb{E}[U_1 U_2^\top] = 0$.
The stacked Z-estimation calculation supplies the appropriate first-order variance under the stated parametric assumptions. It does not, by itself, protect against model misspecification: if the parametric propensity-score model is wrong, $\hat{\tau}$ is inconsistent regardless of which variance formula is used.

Remark: Machine Learning for Nuisance Estimation

Theorem 10.1 is a *finite-dimensional, smooth* Z-estimation result with $\alpha \in \mathbb{R}^q$ for fixed q. Under its stated hypotheses it *derives* root-n convergence and joint asymptotic linearity of $(\hat{\psi}, \hat{\alpha})$; an $n^{-1/2}$ rate for $\hat{\alpha}$ is a conclusion, not a separate assumption.

The genuine limitation is dimensional. The theorem does not cover an *infinite-dimensional* nuisance function estimated nonparametrically or by machine learning, which may converge more slowly than $n^{-1/2}$ and for which the block-matrix correction in Equation 10.5 is no longer the appropriate general representation: there is no finite Jacobian block A_{12} to invert against.
In that setting asymptotic linearity is typically recovered by using an *orthogonal* score, whose first-order insensitivity to the nuisance tolerates slower rates, combined with *either* suitable empirical-process (for example, Donsker) conditions restricting the complexity of the estimated nuisance functions, *or* sample splitting and cross-fitting, which remove the need for such conditions altogether. Neither route is mandatory; they are alternatives. Chapter 11 develops the augmented and doubly robust estimators whose scores have the required orthogonality, and Chapter 12 develops sample splitting, cross-fitting, and debiased machine learning.

10.8 Variance Estimation and Confidence Intervals

Once an estimator admits the asymptotic linear representation

$$\sqrt{n}(\hat{\psi} - \psi) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \varphi(O_i) + o_p(1),$$

its asymptotic variance is $\mathrm{Var}(\hat{\psi}) \approx n^{-1}\mathrm{Var}\{\varphi(O)\}$. A natural estimator of this variance is obtained by replacing $\varphi(O_i)$ with estimated influence values $\hat{\varphi}_i$ and taking the sample variance:

$$\hat{V} = \frac{1}{n(n-1)} \sum_{i=1}^{n} (\hat{\varphi}_i - \bar{\varphi})^2, \qquad \bar{\varphi} = \frac{1}{n} \sum_{i=1}^{n} \hat{\varphi}_i.$$

The asymptotically equivalent uncentered form $\hat{V} = n^{-2} \sum_i \hat{\varphi}_i^2$ is also commonly used. Their absolute difference is $O_p(n^{-2})$; because $\hat{V} = O_p(n^{-1})$, their *relative* difference is $O_p(n^{-1})$, a consequence of $\bar{\varphi} = O_p(n^{-1/2})$ when $\mathbb{E}\varphi(O) = 0$. We adopt the centered form throughout this book, as it has slightly better finite-sample behavior and is the version employed in Chapter 11. This yields the approximate Wald confidence interval at level $1 - a$:

$$\hat{\psi} \pm z_{1-a/2} \sqrt{\hat{V}}.$$

We use a rather than the more conventional α for the significance level here and below to avoid collision with the nuisance parameter α of Section 10.7.

Proposition 10.2 (Variance from the Influence Function). *If $\sqrt{n}(\hat{\psi} - \psi) = n^{-1/2} \sum_{i=1}^{n} \varphi(O_i) + o_p(1)$ with $\mathbb{E}\{\varphi(O)\} = 0$ and $\mathbb{E}\{\varphi(O)^2\} < \infty$, then*

$$\sqrt{n}(\hat{\psi} - \psi) \xrightarrow{d} N(0,\, \mathbb{E}[\varphi(O)^2])\,.$$

If, in addition, the remainder in the expansion is negligible in L_2, or equivalently the sequence $n(\hat{\psi} - \psi)^2$ is uniformly integrable, then $\mathrm{Var}(\hat{\psi}) = n^{-1}\mathbb{E}[\varphi(O)^2] + o(n^{-1})$.

The distributional conclusion follows immediately from the central limit theorem applied to the i.i.d. sum. The second conclusion needs the extra hypothesis and does not follow from the first: an $o_p(1)$ remainder controls the remainder *in probability*, and convergence in probability or in distribution does not imply convergence of moments. A sequence can converge in distribution to a limit with finite variance while its own variances diverge, so an integrability condition of some kind is unavoidable. With that caveat noted, the influence function plays a dual role: it characterizes both the asymptotic distribution and — under the added condition — the asymptotic variance.

Remark: Estimating $\varphi(O_i)$ in Practice — Two Separate Steps

In practice $\varphi(O_i)$ depends on unknown parameters and nuisance functions, so it is replaced by an estimated influence value $\hat{\varphi}_i$. It is essential to separate two tasks that are often run together:

(i) establishing that $\hat{\psi}$ is asymptotically linear with influence function φ;
(ii) consistently estimating $\mathbb{E}\{\varphi(O)^2\}$ *once* (i) has been established.

These require different conditions, and the rate requirements are not the same.
For step (i), the nuisance-rate conditions depend on the estimator and, critically, on whether its score is orthogonal. A nonorthogonal score may require nuisance estimators converging fast enough that their contribution to the expansion is $o_p(n^{-1/2})$; an orthogonal score tolerates substantially slower rates, because the first-order nuisance sensitivity vanishes. This distinction is the subject of Chapter 12.
For step (ii), by contrast, $n^{-1/2}$ nuisance rates are generally *not* needed. Once the expansion in (i) holds, consistency of the influence-value variance estimator typically requires only L_2-consistency of the estimated influence function, $\|\hat{\varphi} - \varphi\|_{L_2(P)} = o_p(1)$, together with suitable moment conditions, plus either cross-fitting or empirical-process conditions restricting the complexity of the estimated nuisance functions. We return to these points in Chapter 11 for doubly robust estimators and in Chapter 12 for cross-fitting.

10.9 Toward Efficiency: Semiparametric Models and the Efficient Influence Function

Section 10.4 showed that the influence function of an asymptotically linear estimator determines both its asymptotic distribution and its asymptotic variance. Different regular estimators of the same parameter may therefore have different large-sample variances. This raises a natural question: among all regular estimators in a given statistical model, what is the smallest asymptotic variance that no regular estimator can improve upon? Answering that question leads to semiparametric efficiency theory. In this section we introduce only the basic objects — semiparametric models, regular estimators, and the efficient influence function — as preparation for Chapter 11. The formal semiparametric machinery underlying this section is developed in Appendix C, which supplies the proofs of the claims made informally here.

10.9.1 Semiparametric Models

In causal inference, the parameter of interest $\psi = \Psi(P)$ is a finite-dimensional quantity (e.g., the ATE), but the full distribution P of the observed data $O = (X, T, Y)$ is never fully specified. This structure motivates the following definition.

Definition: Semiparametric Model

A **semiparametric model** is a statistical model $\mathcal{P}$ for the distribution P of the observed data in which (i) the **parameter of interest** $\psi = \Psi(P) \in \mathbb{R}^p$ is finite-dimensional; and (ii) the remaining aspects of P — collectively called the **nuisance parameter** — are infinite-dimensional (i.e., $\mathcal{P}$ is not fully parametric).

The canonical example in this course is the ATE. Strictly speaking, the potential outcomes $Y(0)$ and $Y(1)$ are not part of the observed-data distribution P, so the causal parameter $\mathbb{E}\{Y(1) - Y(0)\}$ is not by itself a functional of P. Under the identification assumptions of consistency, conditional exchangeability, and positivity (Chapter 5), however, it equals the observed-data functional

$$\tau(P) = \mathbb{E}_P\{\mu_1(X) - \mu_0(X)\}, \qquad \mu_t(x) = \mathbb{E}_P(Y \mid T = t,\, X = x),$$

and we study estimation of this functional in the nonparametric observed-data model $\mathcal{P} = \{P : 0 < \pi(X) < 1 \text{ a.s.}\}$. Positivity defines the support restriction only; we restrict attention throughout to distributions $P \in \mathcal{P}$ for which $\tau(P)$ is finite, and at the reference distribution used for regular root-n inference we additionally require the efficient influence function to belong to $L_2(P)$.

Here the nuisance consists of the entire joint distribution of (X, T, Y) subject only to positivity; no parametric form is assumed for $\mu_t(x)$ or $\pi(x)$. The nuisance parameter accordingly includes the outcome regressions $\mu_t(x)$, the propensity score $\pi(x)$, and the marginal distribution of X. These are precisely the objects that must be estimated in practice. Finite-dimensional analogues of this nuisance-estimation problem were illustrated in Section 10.7, where the nuisance was a parametric coefficient vector; Chapters 11 and 12 treat the full nuisance functions. They are not themselves the target of inference.

Remark: Weak vs. Strong Overlap

The positivity condition $0 < \pi(X) < 1$ a.s. suffices for *identification* of τ, but not for stable regular root-n inference. The exact requirement for a finite efficiency bound is square integrability of the efficient influence function, that is,

$$\mathbb{E}\left[\frac{\sigma_1^2(X)}{\pi(X)} + \frac{\sigma_0^2(X)}{1-\pi(X)} + \{\mu_1(X) - \mu_0(X) - \tau\}^2\right] < \infty,$$

where $\sigma_t^2(x) = \mathrm{Var}(Y \mid T = t, X = x)$. A convenient *sufficient* condition is **strong overlap**, $\epsilon \leq \pi(X) \leq 1 - \epsilon$ a.s. for some $\epsilon > 0$, together with appropriate moment conditions on Y; it is not necessary, since the display above can hold when $\pi(X)$ approaches 0 or 1 provided the conditional variances shrink fast enough there. Under weak overlap the weight distribution need not be essentially bounded, and the largest observed weights may diverge with the sample size; the asymptotic variance of IPW-type estimators may then fail to be finite or fail to admit a normal limit at the $\sqrt{n}$ rate. The distinction between identification under weak overlap and regular inference under strong overlap is taken up systematically in Chapter 12.

Remark: Semiparametric vs. Parametric Efficiency

In regular finite-dimensional *parametric* models, the inverse Fisher information gives the asymptotic information bound for regular estimators of the full parameter θ. For a smooth scalar target $\psi = g(\theta)$ the corresponding bound is $\dot{g}(\theta_0)^\top I(\theta_0)^{-1}\dot{g}(\theta_0)$, reducing to $I(\theta_0)^{-1}$ when $g(\theta) = \theta$. (The classical Cramér–Rao inequality is the related finite-sample statement bounding the variance of unbiased estimators; the asymptotic version is the appropriate comparison here, since the semiparametric bound is likewise asymptotic and stated over regular estimators.) When the nuisance is infinite-dimensional, the inverse-information bound no longer applies directly: there are infinitely many directions in which the likelihood can vary, and the relevant lower bound must account for all of them. The semiparametric efficiency bound is the analogue of the parametric information bound for this setting; see Bickel et al. (1993) and Tsiatis (2006) for comprehensive treatments.

10.9.2 Regular Estimators

The efficiency bound is meaningful only within a restricted class of estimators. Without such a restriction, pathological estimators can behave well at a single distribution while behaving erratically under nearby perturbations, making variance comparisons meaningless from a local asymptotic point of view. The relevant class is that of *regular* estimators, which, informally, are estimators whose asymptotic distribution is stable under small perturbations of the data-generating distribution.

Definition: Regular Estimator (informal)

Let $\{P_{h/\sqrt{n}}\}$ denote a regular parametric submodel through P_0 perturbed in a fixed local direction h, and write $\psi_{h/\sqrt{n}} = \Psi(P_{h/\sqrt{n}})$. An estimator $\hat{\psi}_n$ of $\psi_0 = \Psi(P_0)$ is called **regular** at P_0 if there is a single limit law L such that

$$\sqrt{n}\{\hat{\psi}_n - \Psi(P_{h/\sqrt{n}})\} \rightsquigarrow L \qquad \text{under } P_{h/\sqrt{n}},$$

with the *same* L for every fixed direction h and every admissible regular submodel.

The point of indexing by a fixed h is that regularity is a statement about *local uniformity*: the estimator's limiting behavior, recentered at the moving target $\Psi(P_{h/\sqrt{n}})$, must not depend on which local direction the truth is perturbed in.

Regularity rules out pathological estimators that exploit specific features of the data-generating mechanism in a non-uniform way (so-called *superefficient* estimators). Under correct model specification and the regularity conditions stated for their respective statistical models, the finite-dimensional regression, IPW, and augmented estimators studied in this course are standard examples of regular estimators. The qualification matters: regularity is always relative to a specified model, and an estimator regular in one model need not be regular in a larger one. A parametric outcome-regression estimator, for instance, is not regular for the ATE in the unrestricted nonparametric model, since it fails to be consistent outside its working model. The formal theory of regular estimators and the convolution theorem underlying the efficiency bound are developed in Vaart (1998) (Chapters 8 and 25) and Tsiatis (2006) (Chapter 4).

10.9.3 The Semiparametric Efficiency Bound and the Efficient Influence Function

The central result of semiparametric efficiency theory is the convolution theorem of Bickel et al. (1993). Stating it requires the efficient influence function, so we define that object first: a function is not determined by its mean and covariance alone, and the EIF is characterized by *where it sits* — in the tangent space — not merely by how disperse it is.

Definition 10.4 and the ATE formula that follows from it are needed throughout Chapters 11 and 12 and should be read now. The theorem's regularity hypotheses and its vector-valued refinement are second-reading material: on a first pass it is enough to take from Theorem 10.2 that V^* is a floor no regular estimator can beat, and that a regular asymptotically linear estimator sits exactly on that floor precisely when its influence function is φ^*.

Definition 10.4 (Efficient Influence Function (canonical gradient))**.** Let Ψ be pathwise differentiable at P_0 relative to $\mathcal{P}$, with gradient set $\mathrm{IF}(\Psi, P_0)$ and tangent space $\mathcal{T}$. The **efficient influence function** (EIF), or **canonical gradient**, is the unique element

$$\varphi^* \in \mathrm{IF}(\Psi, P_0) \cap \mathcal{T},$$

obtained by projecting any gradient orthogonally onto $\mathcal{T}$. Its covariance $V^* = \mathbb{E}_{P_0}[\varphi^*(O)\varphi^*(O)^\top]$ is the **semiparametric efficiency bound**. Existence, uniqueness, and the variance-minimizing property are proved in Section C.6.

Theorem 10.2 (Semiparametric Convolution Theorem)**.** *Suppose Ψ is pathwise differentiable at P_0 relative to $\mathcal{P}$ with canonical gradient φ^* and bound $V^* = \mathbb{E}_{P_0}[\varphi^*\varphi^{*\top}]$, and suppose the model satisfies the local asymptotic normality and regularity conditions required by the Hájek–Le Cam convolution theorem. Then for any regular estimator $\hat\psi$ of $\psi_0 \in \mathbb{R}^p$,*

$$\sqrt{n}(\hat\psi - \psi_0) \xrightarrow{d} N(0, V^*) * M$$

for some distribution M, where $*$ denotes convolution. Moreover, a regular asymptotically linear *estimator is efficient — that is, it attains the limit $N(0, V^*)$ with degenerate M — if and only if its influence function equals φ^*.*

Remark (second reading): The Sense of "At Least as Dispersed"

The convolution theorem is fundamentally a statement about *distributions*; converting it into a covariance comparison requires the limiting convolution remainder M to have finite second moments. When it does, the limiting covariance of a regular estimator is $V = V^* + \mathrm{Cov}(M)$. For a scalar parameter ($p = 1$) this gives the familiar lower bound: any such regular estimator has asymptotic variance at least V^*. For vector-valued ψ, V^* is a $p \times p$ matrix and the comparison must be read in the **Loewner partial order**, $V \succeq V^*$, meaning $V - V^*$ is positive semidefinite. Equivalently, every linear combination $c^\top\hat\psi$ has asymptotic variance at least $c^\top V^* c$ for every $c \in \mathbb{R}^p$. If the remainder lacks finite second moments the convolution statement remains meaningful, but the covariance comparison does not. Throughout the rest of the chapter we phrase results in scalar terms, but the matrix generalization is automatic.

The efficient influence function is not merely any valid gradient; it is the unique one lying in the tangent space, and it minimizes variance over the gradient set. Two cautions about the scope of the theorem are worth stating explicitly. First, it is *not* true that every regular estimator has an influence function: a regular estimator may carry a nondegenerate convolution remainder M and need not be asymptotically linear at all. What is true is that every regular *asymptotically linear* estimator has an influence function, and that function must be a gradient of Ψ. Second, regularity alone does not imply efficiency; among regular asymptotically linear estimators, only the one whose influence function equals φ^* attains the bound V^* with no convolution remainder.

Remark: The EIF and the Parametric Score

It is tempting — and in some textbooks common — to describe the EIF as the "semiparametric score" of the model. The analogy is useful, since $\varphi^*(O)$ plays the role in the semiparametric efficiency bound that the score plays in the parametric information bound: its variance *is* the bound. But the two are different objects, and the difference is one of *role*, not of ambient space. Both are mean-zero elements of $L_2^0(P_0)$, and when the EIF exists both lie in the tangent space $\mathcal{T}$:

- a **score** S_{θ_0} is a tangent *direction*, generated by a regular parametric submodel;
- the **EIF** is the Riesz *representer* of the derivative of the target functional over all such directions.

The relation between them in the parametric case is not identity but a linear transformation. In a regular k-dimensional parametric model with score S_{θ_0}, nonsingular Fisher information $I(\theta_0)$, and smooth scalar target $\psi = g(\theta)$,

$$\varphi^*(O) = \dot{g}(\theta_0)^\top I(\theta_0)^{-1} S_{\theta_0}(O), \qquad \dot{g}(\theta_0) = \left.\frac{\partial g(\theta)}{\partial \theta}\right|_{\theta=\theta_0},$$

with the corresponding matrix form for a vector target. Only in the special case $g(\theta) = \theta$, where the target is the full parameter, does this reduce to $\varphi^*(O) = I(\theta_0)^{-1} S_{\theta_0}(O)$. We continue to call φ^* the canonical gradient when precision matters and the EIF when convenience matters.

Remark: How the EIF Is Computed

It is important to be precise about what is differentiated and what is projected. The pathwise derivative along a submodel with score S is a *scalar*, $\dot{\Psi}_P(S) = \partial_\epsilon \Psi(P_\epsilon)|_{\epsilon=0}$; it is not itself a function of O. The gradient is the *Riesz representer* of the linear map $S \mapsto \dot{\Psi}_P(S)$. Accordingly, the EIF is constructed in three steps:

(i) compute the directional derivative $S \mapsto \dot{\Psi}_P(S)$ for a *general* regular score S, not one fixed submodel;

(ii) find a gradient $\varphi \in L_2^0(P)$ representing it, so that $\dot{\Psi}_P(S) = \mathbb{E}_P\{\varphi(O)S(O)\}$ for *every* score S;

(iii) project φ orthogonally onto the tangent space $\mathcal{T}$ to obtain the canonical gradient φ^*.

One cannot in general obtain the EIF by differentiating along a single arbitrary submodel and projecting "the derivative," because a scalar derivative along one path does not determine the representer over the whole tangent space. A least favorable submodel is a useful device *after* the canonical gradient has been characterized, not a substitute for step (ii). In a fully nonparametric model $\mathcal{T} = L_2^0(P)$, the projection in step (iii) is the identity and the gradient is unique whenever it exists. The formal pathwise construction is carried out for the ATE in Section C.7; Chapter 11 gives a complementary estimator-side projection and augmentation argument that recovers the same AIPW form without a new semiparametric derivation.

Definition: Semiparametrically Efficient Estimator

A regular estimator $\hat{\psi}$ is called **semiparametrically efficient** at $P_0 \in \mathcal{P}$ if $\sqrt{n}(\hat{\psi} - \psi_0) \xrightarrow{d} N(0, V^*)$, i.e., it achieves the semiparametric efficiency bound V^* with no additional convolution component M. Equivalently, $\hat{\psi}$ is asymptotically linear with influence function $\varphi^*(O)$.

10.9.4 The Efficient Influence Function for the ATE

We record the EIF for the ATE as it will be derived and used extensively in Chapter 11.

Remark: EIF for the ATE

For the ATE $\tau = \mathbb{E}\{Y(1) - Y(0)\}$ in the nonparametric observed-data model under consistency, conditional exchangeability, and positivity, the efficient influence function is

$$\varphi^*(O) = \frac{T\{Y - \mu_1(X)\}}{\pi(X)} - \frac{(1-T)\{Y - \mu_0(X)\}}{1 - \pi(X)} + \mu_1(X) - \mu_0(X) - \tau. \tag{10.10}$$

This expression has a natural decomposition: the first two terms are an IPW-style residual correction, and the last two are the regression estimator centered at τ. Notably, the EIF depends on *both* nuisance functions $\pi(X)$ and $\mu_t(X)$.

Remark: From the EIF to an Estimator, and What It Costs

Because φ^* contains the unknown τ itself, one does not estimate τ by "averaging φ^*." The estimator is defined by solving $\mathbb{P}_n\{\varphi^*(O;\tau,\hat{\eta})\} = 0$ in τ for estimated nuisances $\hat{\eta} = (\hat{\pi}, \hat{\mu}_0, \hat{\mu}_1)$, which, since τ enters linearly with coefficient -1, has the closed form

$$\hat{\tau}_{\text{AIPW}} = \mathbb{P}_n\left[\hat{\mu}_1(X) - \hat{\mu}_0(X) + \frac{T\{Y - \hat{\mu}_1(X)\}}{\hat{\pi}(X)} - \frac{(1-T)\{Y - \hat{\mu}_0(X)\}}{1 - \hat{\pi}(X)}\right].$$

This is the augmented IPW estimator derived in Chapter 11.
What this estimator delivers depends on how much is assumed:

(i) **Doubly robust consistency.** Suppose the nuisance estimators converge to *some* limits, $\hat{\pi} \to \pi^\dagger$ and $\hat{\mu}_t \to \mu_t^\dagger$ in the relevant norms, with denominators bounded away from 0 and 1 and suitable integrability. If *either* $\pi^\dagger = \pi$ *or* $\mu_t^\dagger = \mu_t$ — that is, at least one limit is the truth, though not necessarily both — then $\hat{\tau}_{\text{AIPW}}$ is consistent for τ. Two qualifications. First, the *other* nuisance must still converge to a sufficiently regular limit; it need not converge to the truth, but it may not be allowed to drift arbitrarily. Second, the empirical mean of the estimated score must itself obey a law of large numbers: convergence of $\hat{\eta}$ in population norms does not automatically control the empirical-process term when the same data are used to fit and to average. This follows from finite-dimensional regularity, from suitable Glivenko–Cantelli or stability conditions on the nuisance estimators, or from cross-fitted evaluation (Chapter 12).

(ii) **Efficient root-n inference.** Attaining the bound $V^* = \mathbb{E}[\varphi^*(O)^2]$ requires more: *both* nuisances consistently estimated, at rates whose *product* satisfies

$$\|\hat{\pi} - \pi\|_{L_2(P)} \|\hat{\mu}_t - \mu_t\|_{L_2(P)} = o_p(n^{-1/2}),$$

together with either cross-fitting or empirical-process conditions controlling the complexity of $\hat{\eta}$. It is sufficient, for example, that each nuisance error be $o_p(n^{-1/4})$; note that $O_p(n^{-1/4})$ for each would give only $O_p(n^{-1/2})$ for the product, which is not enough. More generally one nuisance may converge more slowly provided the other converges correspondingly faster.

Consistency of the nuisance estimators alone does *not* imply the asymptotically linear expansion $\sqrt{n}(\hat{\tau} - \tau) = n^{-1/2}\sum_i \varphi^*(O_i) + o_p(1)$, and hence does not by itself deliver efficiency. The rate conditions in (ii) are established in Chapter 11 and the cross-fitting machinery in Chapter 12.

Remark: Bridge to Chapter 11

Two important properties of $\varphi^*(O)$ will be central in Chapter 11:

(i) **Double robustness.** Under the convergence, overlap, integrability, and empirical-law-of-large-numbers conditions above, the estimator $\hat{\tau}_{\text{AIPW}}$ — obtained by solving $\mathbb{P}_n\{\varphi^*(O;\tau,\hat{\eta})\} = 0$ rather than by averaging φ^* itself — is consistent for τ if *either* nuisance limit is correct, not necessarily both. This is the *doubly robust* property.

(ii) **Semiparametric efficiency.** When both nuisance estimators are consistent and converge at suitable rates, the resulting estimator is asymptotically linear with influence function $\varphi^*(O)$ and therefore achieves the efficiency bound V^*.

The *augmented IPW* (AIPW) estimator, sometimes called the one-step or debiased estimator, is the principal tool for exploiting these two properties simultaneously, and will be derived and analyzed in Chapter 11. The efficient influence function for the ATE will become central there, where we show that the same object generates both doubly robust estimating equations and the semiparametric efficiency bound.

10.10 Chapter Summary

This chapter introduced the statistical framework underlying estimation and inference for causal parameters.

1. **Identification is not estimation.** Identification provides a population formula, but not automatically a satisfactory estimator.

2. **Estimating equations.** Many estimators are defined as solutions to sample moment conditions $\mathbb{P}_n\{U(O;\theta)\} = 0$, with the population moment condition $\mathbb{E}\{U(O;\theta_0)\} = 0$ identifying θ_0 as a zero of the population analogue. Consistency of $\hat{\theta}$ requires additional structure (identifiability and uniform convergence).

3. **Asymptotic linearity.** Estimating equations often yield a first-order expansion $\sqrt{n}(\hat{\theta} - \theta_0) = n^{-1/2}\sum_i \varphi(O_i) + o_p(1)$, which immediately implies asymptotic normality via the CLT.

4. **Influence functions — two distinct notions.** The *estimator* influence function is the function $\varphi(O)$ appearing in the expansion above, describing the first-order sensitivity of that estimator to a single observation. A *gradient* of a functional is a mean-zero function representing the pathwise derivative of Ψ relative to a model. They are linked in one direction: a regular asymptotically linear estimator's influence function is necessarily a gradient.

5. **Nuisance estimation.** In causal inference, nuisance functions must be estimated, and this estimation error propagates into the target estimator. Controlling this propagation is the central statistical challenge.

6. **Variance from the influence function.** The asymptotic variance is $n^{-1}\mathbb{E}[\varphi(O)^2]$, estimated by $(n(n-1))^{-1}\sum_i(\hat{\varphi}_i - \bar{\varphi})^2$; this is consistent under the L_2, moment, and cross-fitting or empirical-process conditions discussed above, and only after asymptotic linearity has been established.

7. **Efficiency.** For regular asymptotically linear estimators in a *specified* model, the variance of the canonical gradient, or efficient influence function, is a universal *lower bound*. Whether the bound is *attained* is a separate question: pathwise differentiability supplies the bound but does not by itself construct an estimator achieving it. An estimator is efficient when it does attain the bound, equivalently when its influence function equals the canonical gradient; for the ATE, Chapters 11 and 12 give sufficient conditions. Because the bound depends on the model, oracle estimators in restricted models may have smaller variance without contradiction. This characterization guides estimator construction in Chapter 11.

Key Objects in This Chapter

Object	Role
Estimating function $U(O;\theta)$	Defines the estimator via $\mathbb{P}_n\{U(O;\hat{\theta})\} = 0$. A property of the construction, not of Ψ.
Estimator influence function $\varphi(O)$	First-order term in the expansion of $\hat{\psi} - \psi$ (Definition 10.2); a property of the *estimator*. For a smooth, just-identified Z-estimator, $\varphi = -A^{-1}U(O;\theta_0)$; with estimated nuisance parameters, apply this to the full stacked system.

Gradient of a functional	Mean-zero φ representing the pathwise derivative of Ψ via $\mathbb{E}_P\{\varphi S\}$ (Definition 10.3); a property of Ψ *and* the model $\mathcal{P}$. Generally non-unique.
Canonical gradient / EIF $\varphi^*(O)$	The unique gradient lying in the tangent space (Definition 10.4); its variance is the efficiency bound. Basis for doubly robust estimators (Chapter 11).
Asymptotic variance $\mathbb{E}[\varphi(O)^2]/n$	Governs precision; the target for efficiency comparisons *within a specified model.*
Estimated influence values $\hat{\varphi}_i$	Plug-in numerical values used for variance estimation *after* asymptotic linearity is established.

10.11 Problems

1. Estimating equations and moment conditions.

(a) Let $O = (X, Y)$ with X and Y scalar, $\mathbb{E}(X^2) < \infty$, $\mathbb{E}(Y^2) < \infty$, and $\mathrm{Var}(X) > 0$, and define $\theta = \mathrm{Cov}(X, Y)/\mathrm{Var}(X)$, the slope in the population simple regression of Y on X. Show first that the moment condition $\mathbb{E}[X(Y - \theta X)] = 0$ does *not* identify θ in general: its solution is instead the regression-through-the-origin coefficient $\theta_\circ = \mathbb{E}(XY)/\mathbb{E}(X^2)$. Writing $\mu_X = \mathbb{E}(X)$ and $\mu_Y = \mathbb{E}(Y)$, verify the identity

$$\mathbb{E}\{X(Y - \theta X)\} = \mu_X(\mu_Y - \theta\mu_X),$$

and conclude that $\theta_\circ = \theta$ if and only if $\mu_X(\mu_Y - \theta\mu_X) = 0$, that is, either $\mathbb{E}(X) = 0$ or the population intercept $\beta_0 = \mu_Y - \theta\mu_X$ vanishes. Show by example that $\mathbb{E}(Y) = 0$ alone is *not* sufficient. (*Hint*: take $\mathbb{E}(X) = 1$, $\mathrm{Var}(X) = 1$, and $Y = X - 1$.) Then give a correct estimating equation for θ by estimating the intercept jointly,

$$\mathbb{E}\begin{bmatrix} 1 \\ X \end{bmatrix}\{Y - \beta_0 - \theta X\} = 0,$$

and verify that the population moment condition holds at (β_0, θ_0).

(b) Let $\theta = F^{-1}(0.5)$ be the population median, and *assume F is continuous and strictly increasing in a neighborhood of its unique median.* Propose an estimating function $U(O;\theta)$ and verify the population moment condition. (*Hint*: consider $U(O;\theta) = \mathbf{1}(Y \le \theta) - 0.5$.) Explain why the continuity assumption is needed: if F has an atom at the median, $F(\theta)$ may strictly exceed 0.5 and the moment condition can fail at θ_0. Note also that U is discontinuous in θ, so the smooth Taylor argument of Proposition 10.1 does not apply directly; the asymptotic linear expansion for the sample median instead follows from a Bahadur representation and requires a positive density at θ_0. Finally, observe that at the sample level $\mathbb{P}_n U(Y;\theta) = 0$ need not have an exact solution — for instance when n is odd — so the sample median is properly defined through the empirical generalized inverse, or as an *approximate* Z-root satisfying $\mathbb{P}_n U(Y;\hat{\theta}) = o_p(n^{-1/2})$.

(c) Show that the OLS estimator and the IPW estimator of the ATE (Example 10.2) are both special cases of the general estimating-equation (Z-estimator) framework, in each case identifying U, A, and the resulting influence function.

2. Asymptotic linearity and the CLT.

(a) Let $\hat{\psi} = \bar{Y}$ be the sample mean of i.i.d. $Y_1, \dots, Y_n$ with $\mathbb{E}Y = \psi$ and $\mathrm{Var}(Y) = \sigma^2 < \infty$. Write down the influence function, state its asymptotic distribution, and give a consistent estimator of $\mathrm{Var}(\hat{\psi})$.

(b) Suppose $\hat{\psi}_1$ and $\hat{\psi}_2$ are two asymptotically linear estimators of the same parameter with influence functions φ_1 and φ_2. Show that $\hat{\psi}_\lambda = \lambda\hat{\psi}_1 + (1-\lambda)\hat{\psi}_2$ is also asymptotically linear for any fixed $\lambda \in \mathbb{R}$, and find its influence function.

(c) Use part (b) to derive the value of $\lambda \in \mathbb{R}$ that minimizes the asymptotic variance of $\hat{\psi}_\lambda$ (allow λ outside $[0, 1]$). Express the answer in terms of $\sigma_1^2 = \mathbb{E}[\varphi_1^2]$, $\sigma_2^2 = \mathbb{E}[\varphi_2^2]$, and $\rho = \mathbb{E}[\varphi_1\varphi_2]$. Your derivation should assume $\mathrm{Var}(\varphi_1 - \varphi_2) = \sigma_1^2 + \sigma_2^2 - 2\rho > 0$; explain what happens in the degenerate case $\mathrm{Var}(\varphi_1 - \varphi_2) = 0$, in which $\varphi_1 = \varphi_2$ almost surely and every λ yields the same first-order estimator. Under what condition on $(\sigma_1^2, \sigma_2^2, \rho)$ does the optimal λ equal $1/2$?

3. Influence functions for causal estimators. Consider the ATE $\tau = \mathbb{E}\{Y(1) - Y(0)\}$ identified under consistency, conditional exchangeability, and positivity. In parts (a)–(b) the nuisance functions are treated as *known*, so both estimators are oracle estimators in restricted models in the sense of the warning box in Section 10.6.

(a) Assume $\pi(X)$ and $\mu_t(X)$ are both known. Compute the influence functions of (i) the regression estimator $\hat{\tau}_{\mathrm{reg}} = \mathbb{P}_n\{\mu_1(X) - \mu_0(X)\}$; (ii) the IPW estimator $\hat{\tau}_{\mathrm{IPW}} = \mathbb{P}_n\{TY/\pi(X) - (1-T)Y/[1-\pi(X)]\}$.

(b) (*A Rao–Blackwell comparison.*) With φ_{reg} and φ_{IPW} as in Equation 10.1 and Equation 10.2, show that

$$\mathbb{E}\{\varphi_{\mathrm{IPW}}(O) \mid X\} = \mu_1(X) - \mu_0(X) - \tau,$$

using conditional exchangeability and consistency; since φ_{reg} is a function of X alone, this says $\mathbb{E}\{\varphi_{\mathrm{IPW}} \mid X\} = \varphi_{\mathrm{reg}}$. Deduce from the law of total variance that

$$\mathrm{Var}(\varphi_{\mathrm{IPW}}) = \mathrm{Var}(\varphi_{\mathrm{reg}}) + \mathbb{E}[\mathrm{Var}\{\varphi_{\mathrm{IPW}} \mid X\}] \geq \mathrm{Var}(\varphi_{\mathrm{reg}}),$$

so that in this oracle comparison the regression estimator is *never* asymptotically less efficient than the IPW estimator. State the condition on $\mathrm{Var}\{\varphi_{\mathrm{IPW}} \mid X\}$ under which the two variances are equal.

(c) The efficient influence function for the ATE is given in Equation 10.10. Verify that it has mean zero under P by computing $\mathbb{E}[\varphi^*(O)]$.

4. Variance estimation.

(a) Let $\hat{\varphi}_i = \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) - \hat{\tau}_{\mathrm{reg}}$ be the estimated influence values for the regression estimator. Write down the plug-in variance estimator $\hat{V}$ and simplify. Under what conditions is $\hat{V}$ consistent for $\mathrm{Var}(\hat{\tau}_{\mathrm{reg}})$? In answering, explain why the oracle restriction matters: this $\hat{\varphi}_i$ is the influence function of the estimator that treats μ_t as *known*, so $\hat{V}$ targets the correct variance only when the outcome regressions are effectively known, estimated on an independent and much larger sample, or otherwise contribute negligibly to the first-order expansion. Ordinary estimation of $\hat{\mu}_t$ from the same sample generally adds first-order terms that this $\hat{V}$ omits.

(b) A researcher reports $\hat{\tau} = 2.4$ and, from the estimated influence values, an empirical variance $\mathbb{P}_n\hat{\varphi}_i^2 = 36$ based on $n = 400$ observations, so that $\hat{V} = 36/400 = 0.09$ estimates $\mathrm{Var}(\hat{\tau})$ directly. Compute a 95% Wald confidence interval. Is the treatment effect statistically distinguishable from zero at the 5% level? (*Caution*: $\hat{V}$ is already an estimate of $\mathrm{Var}(\hat{\tau})$, not of $\mathrm{Var}\{\varphi(O)\}$; do not divide by n a second time.)

(c) Explain why plugging nuisance estimates into a candidate influence-function formula cannot repair an incorrect asymptotic linear expansion. In your answer, distinguish clearly between (i) the nuisance-rate conditions needed to *establish* that $\hat{\tau}$ is asymptotically linear with influence function φ — which depend on the estimator and on whether its score is orthogonal — and (ii) the L_2-consistency condition $\|\hat{\varphi} - \varphi\|_{L_2(P)} = o_p(1)$, together with moment and cross-fitting or empirical-process conditions, needed to *estimate the variance* once that expansion is already established. Explain why (ii) does not generally require $n^{-1/2}$ nuisance rates.

5. Efficiency comparisons.

(a) (*Efficiency bounds are model-specific.*) Problem 3(b) showed that, in the oracle setting where the nuisance functions are known, $\mathrm{Var}(\varphi_{\mathrm{reg}}) \leq \mathrm{Var}(\varphi_{\mathrm{IPW}})$ *always*. It is also possible for $\mathrm{Var}(\varphi_{\mathrm{reg}})$ to fall strictly *below* the nonparametric semiparametric efficiency bound $\mathbb{E}[\varphi^*(O)^2]$. Explain why this is not a contradiction. (*Hint*: an efficiency bound constrains regular estimators *within the model for which it was computed*. Identify the model to which each of the three influence functions belongs, and explain why oracle knowledge of μ_t corresponds to a strictly smaller model than the nonparametric one.) What, if anything, does this tell you about the relative merits of the two estimators in a practical setting where μ_t and π must both be estimated?

(b) The semiparametric efficiency bound for the ATE is $\mathbb{E}[\varphi^*(O)^2]$. Show that $\mathbb{E}[\varphi^*(O)^2] \leq \mathbb{E}[\varphi_{\mathrm{IPW}}(O)^2]$ by expanding the squared EIF. (*Hint*: use the law of iterated expectations to show that the cross-terms cancel.)

(c) Give one practical reason why an efficient estimator based on the EIF may not always be preferred over a simpler, less efficient estimator.

Chapter 11

Doubly Robust Estimation

Learning Objectives

By the end of this chapter, students should be able to:

Core (first reading).

1. Derive the AIPW estimator as a bias-corrected prediction estimator, and explain the role of each term.
2. Prove the double robustness property using the law of iterated expectations, and derive the exact double-robustness remainder identity that links double robustness to Neyman orthogonality.
3. Describe the class of augmented IPW estimators indexed by arbitrary control functions $b_t(x)$, verify unbiasedness for any b_t, and identify the optimal choice that minimizes total variance.
4. State the semiparametric efficiency bound for the ATE and explain why the AIPW estimating function is the efficient influence function.
5. Identify the product-rate condition as the key sufficient condition for nuisance estimation error to be asymptotically negligible in the AIPW estimator, explain its connection to Neyman orthogonality, and construct a consistent variance estimator and Wald confidence interval from the estimated influence values.

Second reading (starred sections).

6. Interpret the augmentation step as a Hilbert-space projection onto the orthogonal complement of the augmentation space, and derive the Pythagorean variance decomposition.
7. Carry out the projection argument in the causal inference setting, starting from the Horvitz–Thompson estimator.

Enrichment.

8. Derive the two approaches to building double robustness directly into the outcome model: IPW-weighted regression and the augmented model with clever covariate; explain why including $\hat{\pi}_i^{-1}$ acts as a debiasing correction and connect this to TMLE (Laan and Rubin 2006).
9. State the internal bias calibration (IBC) condition that forces the prediction estimator to coincide with AIPW, and explain how IPW-weighted regression and the clever-covariate augmented model each enforce IBC by construction.

How to read this chapter. Section 11.2–Section 11.4 (the AIPW construction, double robustness, the exact remainder, and the optimal-augmentation class with its perfect-square variance argument), Section 11.7 (the efficient influence function and the efficiency bound), Section 11.10 (asymptotic inference with estimated nuisances), and the simulation study of Section 11.9 form a complete first-reading path. The starred sections deepen the picture and may be deferred: Section 11.5–Section 11.6 recast the optimal augmentation as a Hilbert-space projection (the efficiency statement of Section 11.7 can be read without them and gains resonance on a second pass), and Section 11.8 is enrichment on building double robustness into the regression fit itself. The learning objectives above are tiered the same way.

11.1 Why Combine Outcome Regression and Weighting?

Under consistency, conditional exchangeability, and positivity (Chapter 6), the average treatment effect (ATE) $\tau = \mathbb{E}\{Y(1) - Y(0)\}$ is *identified* by either of the following equivalent expressions:

$$\tau = \mathbb{E}\{\mu_1(X) - \mu_0(X)\}, \qquad \mu_t(x) = \mathbb{E}(Y \mid T = t,\, X = x),$$

or

$$\tau = \mathbb{E}\left\{ \frac{TY}{\pi(X)} - \frac{(1-T)Y}{1-\pi(X)} \right\},$$

where $\pi(x) = P(T = 1 \mid X = x)$ is the propensity score. Without the causal assumptions, these expressions still define functionals of the observed-data distribution, but they do not in general equal τ.

These formulas suggest two basic estimation strategies.

- **Outcome regression (prediction estimator):** estimate $\mu_1(x)$ and $\mu_0(x)$, then average $\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)$ over the sample.
- **Inverse probability weighting (IPW):** estimate $\pi(x)$ and reweight observed outcomes to reconstruct the potential-outcome means.

Each strategy has strengths and weaknesses. Regression-based estimators can be statistically efficient when the outcome model is well specified, but may be biased under misspecification. IPW estimators are often conceptually appealing because they directly use the treatment model, but they can be unstable when estimated propensity scores are close to 0 or 1 (Rosenbaum and Rubin 1983).

This chapter develops a third strategy: combine both models to construct an estimator that is consistent when *either* model is correct, and achieves the semiparametric efficiency bound when both are correct (under regularity conditions developed in Section 11.10). We derive the same estimator in three complementary ways: first as a bias-corrected prediction estimator (Section 11.2), then as the optimal member of a class of augmented estimators (Section 11.3–Section 11.6), and finally as the efficient influence function in a semiparametric model (Section 11.7).

11.2 The Prediction Estimator and Its Bias

To understand the source of bias it is useful to work first with generic *working regression functions* $m_t(x)$, not necessarily equal to the true conditional means. The *prediction estimator* (also called the regression estimator) based on m_t is

$$\hat{\tau}_{\text{pred}}(m) = \frac{1}{n} \sum_{i=1}^{n} \{m_1(X_i) - m_0(X_i)\}.$$

Define the corresponding population prediction estimand $\tau_{\text{pred}}(m) = \mathbb{E}\{m_1(X) - m_0(X)\}$. Under ignorability, the target parameter is $\tau = \mathbb{E}\{\mu_1(X) - \mu_0(X)\}$, where $\mu_t(x) = \mathbb{E}\{Y(t) \mid X = x\} = \mathbb{E}(Y \mid T = t, X = x)$. Hence

$$\tau_{\text{pred}}(m) - \tau = \mathbb{E}\big[(m_1(X) - \mu_1(X)) - (m_0(X) - \mu_0(X))\big]. \tag{11.1}$$

Equation Equation 11.1 shows that $\tau_{\text{pred}}(m) = \tau$ if and only if $\mathbb{E}[\{m_1(X) - \mu_1(X)\} - \{m_0(X) - \mu_0(X)\}] = 0$. Correct specification of both arm-specific working regressions is a simple sufficient condition, but it is not necessary: the two integrated misspecification errors may cancel. In practice we substitute fitted models $m_t = \hat{\mu}_t$; if $\hat{\mu}_t$ converges to a deterministic limit $\mu_t^\dagger$, the same decomposition gives the probability-limit bias $\mathbb{E}[\{\mu_1^\dagger(X) - \mu_1(X)\} - \{\mu_0^\dagger(X) - \mu_0(X)\}]$ of the prediction estimator. (Exact finite-sample unbiasedness is not claimed when the regressions are estimated from the same data.)

If a propensity score estimator $\hat{\pi}$ is available, we can estimate the two bias terms in Equation 11.1 from the observed data. Define residuals $e_i(t) = Y_i(t) - \hat{\mu}_t(X_i)$. Since $Y_i(1)$ is observed only for treated units and $Y_i(0)$ only for control units,

$$\widehat{\text{Bias}}(\hat{\tau}_{\text{pred}}) = -\frac{1}{n} \sum_{i=1}^{n} \frac{T_i}{\hat{\pi}(X_i)}\, e_i(1) + \frac{1}{n} \sum_{i=1}^{n} \frac{1 - T_i}{1 - \hat{\pi}(X_i)}\, e_i(0).$$

The bias-corrected prediction estimator is therefore

$$\hat{\tau}_{\text{AIPW}} = \hat{\tau}_{\text{pred}} - \widehat{\text{Bias}}(\hat{\tau}_{\text{pred}}) = \hat{\mu}_{1,\text{dr}} - \hat{\mu}_{0,\text{dr}}, \tag{11.2}$$

where

$$\hat{\mu}_{1,\mathrm{dr}} = \frac{1}{n}\sum_{i=1}^{n}\frac{T_i}{\hat{\pi}(X_i)}\{Y_i - \hat{\mu}_1(X_i)\} + \frac{1}{n}\sum_{i=1}^{n}\hat{\mu}_1(X_i), \tag{11.3}$$

$$\hat{\mu}_{0,\mathrm{dr}} = \frac{1}{n}\sum_{i=1}^{n}\frac{1-T_i}{1-\hat{\pi}(X_i)}\{Y_i - \hat{\mu}_0(X_i)\} + \frac{1}{n}\sum_{i=1}^{n}\hat{\mu}_0(X_i). \tag{11.4}$$

Remark: Two Roles of the Augmentation Terms

The terms $T_i\{Y_i - \hat{\mu}_1(X_i)\}/\hat{\pi}(X_i)$ and $(1-T_i)\{Y_i - \hat{\mu}_0(X_i)\}/(1-\hat{\pi}(X_i))$ have two complementary roles. From the bias-correction perspective, they are IPW-weighted residuals that estimate the prediction error of the outcome regression. From an estimating-equation perspective, they are exactly the terms that make the estimating function orthogonal to nuisance perturbations *at the true nuisance functions* (Neyman orthogonality, Section 11.10). The estimated bias term has zero expectation when the outcome model is correct; the bias-corrected estimator is unbiased when the propensity model is correct. Hence the name *doubly robust*.

11.3 The Augmented IPW Estimator

Notation for the rest of the chapter. From this point on, $\mu_t(x)$ inside estimating functions and population identities denotes a *generic* (or *working*) outcome regression, not necessarily equal to the truth; the truth is written $\mu_t^*(x) = \mathbb{E}(Y \mid T = t, X = x)$, and the fitted version that replaces $\mu_t(x)$ in practice is $\hat{\mu}_t(x)$. Similarly, $\pi(x)$ inside estimating functions denotes a *working* propensity model; the true propensity score is written $\pi^*(x) = P(T = 1 \mid X = x)$, and the fitted version used in practice is $\hat{\pi}(x)$. Whether a particular theorem assumes $\pi = \pi^*$ or allows an arbitrary working π will be stated explicitly. We also use the unsubscripted scalar $\mu_t = \mathbb{E}\{Y(t)\}$ for the marginal counterfactual mean; the convention is that μ_t without an argument is the scalar mean and $\mu_t(x)$ with an argument is the regression function.

Collecting the terms in Equation 11.2 gives the estimating-equation form. The AIPW estimator solves $\mathbb{P}_n\{\phi(O;\tau,\hat{\eta})\} = 0$ where

$$\phi(O;\tau,\eta) = \left[\frac{T}{\pi(X)}\{Y - \mu_1(X)\} + \mu_1(X)\right] - \left[\frac{1-T}{1-\pi(X)}\{Y - \mu_0(X)\} + \mu_0(X)\right] - \tau, \tag{11.5}$$

and $O = (X, T, Y)$ denotes one observation. Solving explicitly,

$$\hat{\tau}_{\mathrm{AIPW}} = \frac{1}{n}\sum_{i=1}^{n}\left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{T_i}{\hat{\pi}(X_i)}\{Y_i - \hat{\mu}_1(X_i)\} - \frac{1-T_i}{1-\hat{\pi}(X_i)}\{Y_i - \hat{\mu}_0(X_i)\}\right]. \tag{11.6}$$

Definition: Augmented Inverse Probability Weighted Estimator

The estimator in Equation 11.6 is the *augmented inverse probability weighted (AIPW) estimator* of the ATE. It equals the prediction estimator plus a weighted-residual bias correction.

The AIPW estimator enjoys two important properties: double robustness and optimality within the augmented estimator class. We establish double robustness here; the optimality property is developed in Section 11.4.

Theorem 11.1 (Double Robustness). *Let $\phi(O;\tau,\eta)$ be the estimating function in Equation 11.5. Suppose consistency, conditional exchangeability, and positivity hold, so that $\tau = \mathbb{E}\{\mu_1^*(X) - \mu_0^*(X)\}$. Assume the working functions are admissible: $0 < \pi(x) < 1$ on the support of X, and the expectations displayed in Equation 11.5 exist. Then $\mathbb{E}\{\phi(O;\tau,\eta)\} = 0$ if either:*

1. *$\mu_t(x) = \mu_t^*(x)$ for $t = 0, 1$, regardless of whether $\pi(x)$ is correctly specified; or*
2. *$\pi(x) = \pi^*(x)$, regardless of whether μ_0 and μ_1 are correctly specified.*

Proof. **Case 1: outcome regression correct.** If $\mu_t(X) = \mathbb{E}(Y \mid T = t, X)$, then conditional on X,

$$\mathbb{E}\left[\frac{T}{\pi(X)}\{Y - \mu_1(X)\}\,\middle|\, X\right] = \frac{P(T=1 \mid X)}{\pi(X)}\,\mathbb{E}\{Y - \mu_1(X) \mid T = 1, X\} = 0,$$

and similarly the untreated augmentation term vanishes. Therefore $\mathbb{E}\{\phi\} = \mathbb{E}\{\mu_1(X) - \mu_0(X)\} - \tau = 0$.

Case 2: propensity score correct. Suppose $\pi(X) = \pi^*(X)$ but μ_0, μ_1 are arbitrary (possibly misspecified). We compute the conditional expectation of the first bracket of ϕ given X:

$$\begin{aligned}\mathbb{E}\left[\frac{T}{\pi(X)}\{Y - \mu_1(X)\} + \mu_1(X) \,\middle|\, X\right] &= \frac{\mathbb{E}[TY \mid X]}{\pi(X)} - \mu_1(X)\,\frac{\mathbb{E}[T \mid X]}{\pi(X)} + \mu_1(X)\\ &= \mu_1^*(X) - \mu_1(X) + \mu_1(X) = \mu_1^*(X),\end{aligned}$$

where the second line uses $\mathbb{E}[T \mid X] = \pi^*(X) = \pi(X)$ (so the μ_1 terms cancel) and $\mathbb{E}[TY \mid X] = \pi(X)\mu_1^*(X)$. By symmetry, the control bracket has conditional expectation $\mu_0^*(X)$. Hence $\mathbb{E}\{\phi \mid X\} = \mu_1^*(X) - \mu_0^*(X) - \tau$, and taking expectations over X and invoking the identification formula gives $\mathbb{E}\{\phi\} = 0$. □

Remark: The Algebra of Case 2

The key cancellation in Case 2 is that the working outcome function $\mu_1(X)$ enters the bracket twice — once multiplied by $T/\pi(X)$ and once as a standalone term — and the two instances have equal and opposite conditional expectations because $\mathbb{E}[T/\pi(X) \mid X] = 1$ when π is correct. The bracket thus "forgets" μ_1 entirely, even though it appeared there twice, and converges to $\mu_1^*(X)$. This is exactly what makes the augmentation doubly robust: a wrong μ_t is subtracted out by the correct π.

Theorem 11.2 (The Exact Double-Robustness Remainder)**.** *Under the assumptions of Theorem 11.1, for any admissible working pair* (π, μ_0, μ_1)*,*

$$\mathbb{E}\{\phi(O; \tau, \eta)\} = \mathbb{E}\left[\{\pi^*(X) - \pi(X)\}\left\{\frac{\mu_1^*(X) - \mu_1(X)}{\pi(X)} + \frac{\mu_0^*(X) - \mu_0(X)}{1 - \pi(X)}\right\}\right]. \tag{11.7}$$

Proof. Conditioning on X, $\mathbb{E}[\,T\{Y - \mu_1(X)\}/\pi(X) \mid X\,] = \{\pi^*(X)/\pi(X)\}\{\mu_1^*(X) - \mu_1(X)\}$, hence the treated bracket of Equation 11.5 satisfies

$$\mathbb{E}\left[\frac{T}{\pi(X)}\{Y - \mu_1(X)\} + \mu_1(X) \,\middle|\, X\right] - \mu_1^*(X) = \{\mu_1^*(X) - \mu_1(X)\}\,\frac{\pi^*(X) - \pi(X)}{\pi(X)},$$

and symmetrically the control bracket satisfies

$$\mathbb{E}\left[\frac{1-T}{1-\pi(X)}\{Y - \mu_0(X)\} + \mu_0(X) \,\middle|\, X\right] - \mu_0^*(X) = -\{\mu_0^*(X) - \mu_0(X)\}\,\frac{\pi^*(X) - \pi(X)}{1 - \pi(X)}.$$

Subtracting the two brackets, using $\tau = \mathbb{E}\{\mu_1^*(X) - \mu_0^*(X)\}$, and taking expectations over X gives Equation 11.7. □

Identity Equation 11.7 packages the main structural facts of this chapter into a single display.

Double robustness: the integrand is a product of a propensity-score error and an outcome-regression error, so it vanishes when either factor is identically zero — Cases 1 and 2 of Theorem 11.1 are both immediate.

Orthogonality at the truth: the remainder is bilinear in the two errors, so its Gateaux derivative with respect to either nuisance, evaluated at (π^*, μ^*), is zero — the property developed as Neyman orthogonality in Section 11.10.

The product rate: if the working functions are replaced by estimates $(\hat{\pi}, \hat{\mu}_t)$ fitted on an independent sample (so that Equation 11.7 applies conditionally on that sample) and the fitted propensity score satisfies $\varepsilon \le \hat{\pi} \le 1 - \varepsilon$, the Cauchy–Schwarz inequality bounds the conditional remainder by

$$|R_2| \;\le\; \varepsilon^{-1}\,\|\hat{\pi} - \pi^*\|(\|\hat{\mu}_1 - \mu_1^*\| + \|\hat{\mu}_0 - \mu_0^*\|), \tag{11.8}$$

with $\|\cdot\|$ the $L^2(P)$ norm — exactly the product structure behind the rate condition Equation 11.25 of Section 11.10.

Remark: Scope of Double Robustness

Theorem 11.1 is a *population moment identity*: it says that the estimating function has mean zero at the true τ when either working model is correct. It does not by itself assert anything about the fitted estimator, and it does not protect against misspecification of both models simultaneously. Consistency of $\hat{\tau}_{\mathrm{AIPW}}$ additionally requires convergence of the fitted nuisances, admissible denominators, and a law of large numbers for the estimated score, as the following result records.

Theorem 11.3 (Double-Robust Consistency)**.** *Write $\hat{\tau}_{\mathrm{AIPW}} = \mathbb{P}_n\,\psi(O;\hat{\eta})$ for the sample average in Equation 11.6, where $\psi(O;\eta) = \phi(O;\tau,\eta) + \tau$ does not involve τ. Suppose (i) $\hat{\mu}_t \to \mu_t^\dagger$ and $\hat{\pi} \to \pi^\dagger$ in $L^2(P_X)$-probability, with $\varepsilon \le \hat{\pi}(x), \pi^\dagger(x) \le 1-\varepsilon$ for some $\varepsilon > 0$ and $\mathbb{E}(Y^2) < \infty$; and (ii) $(\mathbb{P}_n - P)\,\psi(O;\hat{\eta}) = o_p(1)$. If either $\mu_t^\dagger = \mu_t^*$ for both t or $\pi^\dagger = \pi^*$, then $\hat{\tau}_{\mathrm{AIPW}} \xrightarrow{p} \tau$.*

Proof. Decompose

$$\hat{\tau}_{\mathrm{AIPW}} - \tau = (\mathbb{P}_n - P)\,\psi(O;\hat{\eta}) + P\{\psi(O;\hat{\eta}) - \psi(O;\eta^\dagger)\} + \mathbb{E}\{\phi(O;\tau,\eta^\dagger)\}.$$

The first term is $o_p(1)$ by (ii). The second tends to zero in probability by (i), the bounded denominators, and the Cauchy–Schwarz inequality. The third is zero by Theorem 11.1 — equivalently, by Equation 11.7 — under either correctness condition. □

The importance of the AIPW estimator goes beyond bias correction. In Section 11.4 we show that it is the variance-optimal member of a natural class of augmented estimators indexed by arbitrary control functions $b_t(x)$. Later, in Section 11.7, we will see that the same estimating function ϕ is exactly the efficient influence function for the ATE under the nonparametric model. Thus the same object arises from three perspectives: bias correction, optimal augmentation, and semiparametric efficiency.

11.4 A Class of Augmented Estimators

The AIPW estimator has a second important property beyond double robustness: it is the variance-minimizing estimator within a broad class of augmented IPW estimators. We now define this class and establish the optimality result.

Remark: Three Senses of "Optimal" in This Chapter

The word *optimal* will be used in three distinct senses before the chapter is done: (i) the *class-optimal* control function b_t^* that minimizes variance within the parametric family Equation 11.9–Equation 11.10 (this section); (ii) the *projection-optimal* estimator $\hat{\theta}_{\mathrm{opt}}$ that minimizes variance after an orthogonal augmentation correction (Section 11.5); and (iii) the *semiparametrically efficient* estimator, whose asymptotic variance attains the information lower bound over all regular asymptotically linear estimators under the nonparametric model (Section 11.7). The first two agree in the ATE setting, and Section 11.7 will show that they also agree with the third.

For any functions $b_1(x)$ and $b_0(x)$ that are square-integrable with respect to the distribution of X — we write $b_t \in \mathcal{L}^2 \equiv L^2(P_X)$ — define

$$\hat{\mu}_{1,b} = \frac{1}{n}\sum_{i=1}^{n} \frac{T_i}{\pi(X_i)}\,Y_i - \frac{1}{n}\sum_{i=1}^{n}\left\{\frac{T_i}{\pi(X_i)} - 1\right\} b_1(X_i), \tag{11.9}$$

$$\hat{\mu}_{0,b} = \frac{1}{n}\sum_{i=1}^{n} \frac{1-T_i}{1-\pi(X_i)}\,Y_i - \frac{1}{n}\sum_{i=1}^{n}\left\{\frac{1-T_i}{1-\pi(X_i)} - 1\right\} b_0(X_i), \tag{11.10}$$

and let $\hat{\tau}_b = \hat{\mu}_{1,b} - \hat{\mu}_{0,b}$. Think of $b_t(x)$ as any working approximation to $\mathbb{E}\{Y(t) \mid X = x\}$: the first term of $\hat{\mu}_{t,b}$ is an IPW estimator of μ_t, and the second subtracts a mean-zero correction indexed by b_t. The correction is mean-zero under the true π regardless of b_t, so every member of the family is unbiased; only the variance depends on the choice of b_t, which is what makes this a meaningful optimization problem.

Remark: Connection to the AIPW Form

The estimator $\hat{\mu}_{1,b}$ in Equation 11.9 and $\hat{\mu}_{1,\mathrm{dr}}$ in Equation 11.3 are identical when $b_1(x) = \mu_1(x)$: $\frac{T}{\pi}Y - (\frac{T}{\pi} - 1)\mu_1 = \frac{T}{\pi}\{Y - \mu_1\} + \mu_1$. So the class Equation 11.9–Equation 11.10 contains the standard AIPW estimator as the special case $b_t = \mu_t$.

The following theorem shows that every estimator in this class is unbiased for τ under a correctly specified propensity score, regardless of the choice of b_t, and provides an explicit formula for its total variance.

Theorem 11.4 (Unbiasedness and Variance of the AIPW Class)**.** *Under joint conditional exchangeability $\{Y(0), Y(1)\} \perp\!\!\!\perp T \mid X$ (the form used in the variance calculation below; Chapter 4 discusses its relation to the weaker per-arm form), SUTVA, and positivity strengthened to strict overlap ($\varepsilon \le \pi^*(x) \le 1 - \varepsilon$ for some $\varepsilon > 0$), with the true propensity score used as $\pi(x)$ in the estimator, for any $b_0, b_1 \in \mathcal{L}^2$ and $\mathbb{E}\{Y(t)^2\} < \infty$:*

1. ***Unbiasedness:*** $\mathbb{E}(\hat{\tau}_b) = \tau$.
2. ***Total variance:***

$$\begin{aligned}\mathrm{Var}(\hat{\tau}_b) = \frac{1}{n}\,\mathbb{E}\Bigg[\left(\frac{1}{\pi(X)} - 1\right)\{Y(1) - b_1(X)\}^2 + 2\{Y(1) - b_1(X)\}\{Y(0) - b_0(X)\} \\ + \left(\frac{1}{1 - \pi(X)} - 1\right)\{Y(0) - b_0(X)\}^2\Bigg] + \frac{1}{n}\,\mathrm{Var}\{Y(1) - Y(0)\}.\end{aligned} \tag{11.11}$$

Proof. Let $\xi = \frac{T}{\pi(X)}\{Y - b_1(X)\} + b_1(X) - \frac{1-T}{1-\pi(X)}\{Y - b_0(X)\} - b_0(X)$, so that $\hat{\tau}_b = n^{-1}\sum_{i=1}^n \xi_i$ and, for i.i.d. observations, $\mathrm{Var}(\hat{\tau}_b) = n^{-1}\mathrm{Var}(\xi)$. By SUTVA, $Y = TY(1) + (1 - T)Y(0)$, so

$$\xi = \frac{T}{\pi(X)}\{Y(1) - b_1(X)\} + b_1(X) - \frac{1 - T}{1 - \pi(X)}\{Y(0) - b_0(X)\} - b_0(X).$$

Part (i). Since $\pi(X) = \pi^*(X)$, we have $\mathbb{E}[T/\pi(X) \mid X] = 1$, so $\mathbb{E}[(T/\pi(X) - 1)b_1(X)] = 0$. Under conditional exchangeability, $T \perp\!\!\!\perp Y(1) \mid X$, so $\mathbb{E}[(T/\pi(X))Y(1) \mid X] = \mathbb{E}[Y(1) \mid X] = \mu_1^*(X)$. Hence $\mathbb{E}[\hat{\mu}_{1,b}] = \mu_1$ and, by symmetry, $\mathbb{E}[\hat{\mu}_{0,b}] = \mu_0$, so $\mathbb{E}(\hat{\tau}_b) = \tau$.

Part (ii). We apply the law of total variance, conditioning on $(X, Y(1), Y(0))$. By joint conditional exchangeability, the only remaining randomness is $T \mid \{X, Y(1), Y(0)\} \sim \mathrm{Bernoulli}(\pi(X))$.

Conditional mean. Since $\mathbb{E}[T/\pi(X) \mid X] = 1$ and $\mathbb{E}[(1 - T)/(1 - \pi(X)) \mid X] = 1$, we get $\mathbb{E}[\xi \mid X, Y(1), Y(0)] = Y(1) - Y(0)$.

Conditional variance. ξ is linear in T:

$$\xi = T\left[\frac{Y(1) - b_1(X)}{\pi(X)} + \frac{Y(0) - b_0(X)}{1 - \pi(X)}\right] - \frac{Y(0) - b_0(X)}{1 - \pi(X)} + b_1(X) - b_0(X).$$

Using $\mathrm{Var}(T \mid X) = \pi(X)(1 - \pi(X))$,

$$\begin{aligned}\mathrm{Var}(\xi \mid X, Y(1), Y(0)) &= \pi(X)(1 - \pi(X))\left[\frac{Y(1) - b_1(X)}{\pi(X)} + \frac{Y(0) - b_0(X)}{1 - \pi(X)}\right]^2 \\ &= \left(\frac{1}{\pi(X)} - 1\right)\{Y(1) - b_1(X)\}^2 + 2\{Y(1) - b_1(X)\}\{Y(0) - b_0(X)\} \\ &\quad + \left(\frac{1}{1 - \pi(X)} - 1\right)\{Y(0) - b_0(X)\}^2.\end{aligned}$$

Taking expectations over $(X, Y(1), Y(0))$ and adding the between-group term $\mathrm{Var}(\mathbb{E}[\xi \mid X, Y(1), Y(0)]) = \mathrm{Var}(Y(1) - Y(0))$ gives Equation 11.11. ☐

Every member of the class is unbiased for τ, for any choice of b_t. The first term of Equation 11.11 depends on b_t; the second, $n^{-1}\mathrm{Var}\{Y(1) - Y(0)\}$, does not.

Theorem 11.5 (Optimal Control Functions)**.** *The control functions $b_1^*(X) = \mathbb{E}\{Y(1) \mid X\}$ and $b_0^*(X) = \mathbb{E}\{Y(0) \mid X\}$ form a minimizing pair for the total variance in Equation 11.11; the full set of minimizing pairs, and the uniqueness of the resulting estimator, are described in the remark below. The resulting AIPW estimator coincides with the bias-corrected prediction estimator in Equation 11.2.*

Remark: Identification of b_t^* as the Observed-Data Regression

The optimal control function $b_t^*(X) = \mathbb{E}\{Y(t) \mid X\}$ is defined as a *causal regression* of the potential outcome on the covariates. Under conditional exchangeability and consistency (Chapter 6), it is identified by the observed-data regression μ_t^*:

$$b_t^*(X) = \mathbb{E}\{Y(t) \mid X\} = \mathbb{E}(Y \mid T = t,\, X) = \mu_t^*(X).$$

This identification is what makes the AIPW estimator implementable from observed data: the optimal control function is a regression we can fit directly from (Y, T, X).

Proof. Write $\mu_t^*(x) = \mathbb{E}\{Y(t) \mid X = x\}$ and set $u(X) = \mu_1^*(X) - b_1(X)$ and $v(X) = \mu_0^*(X) - b_0(X)$. Decompose

$$Y(t) - b_t(X) = (\mu_t^*(X) - b_t(X)) + (Y(t) - \mu_t^*(X)) = d_t(X) + \varepsilon_t,$$

where $d_1 = u$, $d_0 = v$, and $\mathbb{E}[\varepsilon_t \mid X] = 0$. Under strong ignorability, $T \perp\!\!\!\perp (Y(1), Y(0)) \mid X$, so ε_t is also independent of T given X.

Substituting into the b_t-dependent term of Equation 11.11 and using $\mathbb{E}[\varepsilon_t \mid X] = 0$, the law of iterated expectations eliminates every cross term between (u, v) and $(\varepsilon_1, \varepsilon_0)$, leaving

$$\underbrace{\mathbb{E}\left[\left(\frac{1}{\pi(X)} - 1\right) u(X)^2 + 2\,u(X)v(X) + \left(\frac{1}{1-\pi(X)} - 1\right) v(X)^2\right]}_{=:\,Q(b_0,b_1)} + C, \tag{11.12}$$

where $C = \mathbb{E}[(1/\pi - 1)\mathrm{Var}\{Y(1) \mid X\} + (1/(1-\pi) - 1)\mathrm{Var}\{Y(0) \mid X\}] + 2\mathbb{E}[\varepsilon_1\varepsilon_0]$ does not depend on b_0 or b_1.

It remains to show $Q(b_0, b_1) \geq 0$ with equality at $u \equiv 0$, $v \equiv 0$. Write

$$\left(\frac{1}{\pi} - 1\right) u^2 + 2uv + \left(\frac{1}{1-\pi} - 1\right) v^2 = \frac{1-\pi}{\pi}\, u^2 + 2uv + \frac{\pi}{1-\pi}\, v^2 = \left(\sqrt{\frac{1-\pi}{\pi}}\, u + \sqrt{\frac{\pi}{1-\pi}}\, v\right)^2,$$

a perfect square, so $Q(b_0, b_1) \geq 0$. The choice $b_t^*(X) = \mu_t^*(X)$, i.e. $u \equiv 0$ and $v \equiv 0$, achieves $Q = 0$ and hence $\mathrm{Var}(\hat\tau_b) \geq n^{-1}[C + \mathrm{Var}\{Y(1) - Y(0)\}] =: \mathrm{Var}(\hat\tau_{b^*})$. ☐

Remark: The Joint Minimizer Is Unique as an Estimator

The perfect square in the proof vanishes whenever $(1 - \pi(X))u(X) + \pi(X)v(X) = 0$ a.s., so $Q(b_0, b_1) = 0$ on a whole family of pairs (b_0, b_1), not only at (μ_0^*, μ_1^*). This is a reparameterization artifact, not a genuine tie. Collecting terms in Equation 11.9–Equation 11.10,

$$\hat\tau_b = \hat\tau_{\mathrm{HT}} - \frac{1}{n}\sum_{i=1}^{n}\{T_i - \pi(X_i)\}\, a_b(X_i), \qquad a_b(x) = \frac{b_1(x)}{\pi(x)} + \frac{b_0(x)}{1-\pi(x)}, \tag{11.13}$$

where $\hat\tau_{\mathrm{HT}}$ denotes the pure IPW (Horvitz–Thompson) estimator obtained at $b_0 = b_1 = 0$: the estimator depends on the pair (b_0, b_1) only through the single function a_b. Two pairs therefore yield the *identical estimator* — not merely the same variance — exactly when $(1-\pi)(b_1 - b_1') + \pi(b_0 - b_0') = 0$ a.s., and the $Q = 0$ family is precisely the set of pairs with $a_b = a_{b^*}$: one equivalence class of parameterizations of a single random variable. The variance-minimizing *estimator* is thus unique. The canonical representative $b_t^* = \mu_t^*$ of this class is distinguished by being the arm-wise optimum: it separately minimizes $\mathrm{Var}(\hat\mu_{t,b})$ for each t, as the projection argument of Section 11.6 makes explicit.

Remark: Which Parts of the Variance Formula Are Identified?

The decomposition Equation 11.11 is expressed in potential-outcome terms, and two of its ingredients — the cross-product term $\mathbb{E}[\{Y(1) - b_1(X)\}\{Y(0) - b_0(X)\}]$ and $\mathrm{Var}\{Y(1) - Y(0)\}$ — depend on the *joint* conditional distribution of $(Y(0), Y(1))$ given X, which is not identified in the nonparametric observed-data model without additional cross-world or structural assumptions: each unit reveals

only one potential outcome (the fundamental problem of causal inference, Chapter 4), so the two conditional marginals are identified but the association between them is not.
There is no contradiction, because $\mathrm{Var}(\hat{\tau}_b) = n^{-1}\mathrm{Var}(\xi)$ for the *observable* summand ξ defined in the proof, and the variance of an observable random variable is identified. Indeed, writing $W_t = Y(t) - b_t(X)$ and expanding $\mathrm{Var}\{Y(1) - Y(0)\} = \mathrm{Var}\{W_1 - W_0 + b_1(X) - b_0(X)\}$, the term $-2\,\mathbb{E}(W_1 W_0)$ that emerges cancels the $+2\,\mathbb{E}(W_1 W_0)$ contributed by the first bracket of Equation 11.11 exactly, for *every* choice of (b_0, b_1): the unidentified association drops out of the total, as it must. (Indeed, only the weak, per-arm form $Y(t) \perp\!\!\!\perp T \mid X$ is needed for the total and for Equation 11.14 below: conditioning on X alone and using $T(1-T) = 0$, every term of $\mathbb{E}(\xi^2 \mid X)$ involves one arm at a time.)
Carrying out the cancellation at the optimum $b_t^* = \mu_t^*$ yields a closed form. Write $\sigma_t^2(X) = \mathrm{Var}\{Y(t) \mid X\}$. At the optimum $u \equiv v \equiv 0$, so combining the constant C with

$$\mathrm{Var}\{Y(1) - Y(0)\} = \mathbb{E}\{\sigma_1^2(X) + \sigma_0^2(X)\} - 2\,\mathbb{E}[\mathrm{Cov}\{Y(1), Y(0) \mid X\}] + \mathrm{Var}\{\mu_1^*(X) - \mu_0^*(X)\}$$

and $\mathbb{E}(\varepsilon_1\varepsilon_0) = \mathbb{E}[\mathrm{Cov}\{Y(1), Y(0) \mid X\}]$ gives

$$\mathrm{Var}(\hat{\tau}_{b^*}) = \frac{1}{n}\left[\mathbb{E}\left\{\frac{\sigma_1^2(X)}{\pi(X)} + \frac{\sigma_0^2(X)}{1-\pi(X)}\right\} + \mathrm{Var}\{\mu_1^*(X) - \mu_0^*(X)\}\right], \tag{11.14}$$

in which every ingredient is a functional of the observed-data distribution. Anticipating Section 11.7, the bracketed quantity is exactly $\mathbb{E}\{\varphi_{\mathrm{eff}}(O)^2\}$ for the efficient influence function Equation 11.19: squaring Equation 11.19 conditionally on X — the cross terms vanish because $T(1-T) = 0$ and the residuals are conditionally mean zero — reproduces Equation 11.14 term by term. The optimal member of the augmented class therefore attains the semiparametric efficiency bound of Theorem 11.8, a fact verified here by elementary computation ahead of the general theory.

Remark: Interpretation of the Optimality Theorem

Theorem 11.5 shows that among the augmented estimators $\{\hat{\tau}_b : b_t \in \mathcal{L}^2\}$, the choice $b_t^*(X) = \mathbb{E}\{Y(t) \mid X\}$ minimizes asymptotic variance. Thus AIPW is optimal within this estimator class when both nuisance components are correctly specified. This is a class-specific optimality result; the broader semiparametric efficiency interpretation — showing that no regular asymptotically linear estimator, inside or outside this class, can improve upon AIPW — will be developed in Section 11.7. When either model is misspecified, the AIPW estimator retains consistency by double robustness but no longer necessarily achieves the minimum variance within the class.

11.5 The Projection Interpretation ⋆

The class-specific optimality of AIPW within the augmented IPW family (Theorem 11.5) has an elegant interpretation in terms of projections in a Hilbert space of estimators. This framing is a working analogy: the collection of square-integrable random variables, quotiented by a.s. equality, is a genuine Hilbert space $L^2(P)$ under the inner product $\langle X, Y\rangle = \mathbb{E}(XY)$, which coincides with $\mathrm{Cov}(X, Y)$ whenever either argument has mean zero. Theorem 11.6 is an instance of the L^2 projection theorem in that space, applied to the closed subspace Λ of mean-zero elements; see Tsiatis (2006) (Chapter 2) for the careful construction.

This projection view does not define a new estimator; rather, it explains why the augmentation used in Section 11.4 is variance reducing and why the optimal correction must be orthogonal to the augmentation space.

11.5.1 Optimal Estimation via Projection

Let $\hat{\theta}_0$ be an unbiased estimator of θ. Define the *augmentation space* Λ as a pre-specified closed linear subspace of mean-zero square-integrable random variables that are *computable from the observed data without knowledge of* θ. Elements of Λ are called *admissible augmentation terms*. The observability restriction is essential: if Λ were taken to be all mean-zero square-integrable random variables, the true parameter error $\hat{\theta}_0 - \theta$ itself would be in Λ, and the "projection" would yield the degenerate estimator θ

with zero variance — a construction that requires knowing the very quantity we are trying to estimate. Restricting Λ to observable corrections removes this degeneracy. For any $\hat{b} \in \Lambda$ the estimator $\hat{\theta}_b = \hat{\theta}_0 - \hat{b}$ remains unbiased for θ. We seek the $\hat{b}^* \in \Lambda$ that minimizes $\mathrm{Var}(\hat{\theta}_b)$.

Theorem 11.6 (Optimal Projection)**.** *The optimal correction is* $\hat{b}^* = \Pi(\hat{\theta}_0 \mid \Lambda)$*, the* L^2 *projection of* $\hat{\theta}_0$ *onto* Λ*. It is characterized by: (1)* $\hat{b}^* \in \Lambda$*; (2)* $\mathrm{Cov}(\hat{\theta}_0 - \hat{b}^*, \hat{b}) = 0$ *for all* $\hat{b} \in \Lambda$*. The optimal estimator is*

$$\hat{\theta}_{\mathrm{opt}} = \hat{\theta}_0 - \hat{b}^* = \Pi(\hat{\theta}_0 \mid \Lambda^{\perp}),$$

the projection of $\hat{\theta}_0$ *onto the orthogonal complement* $\Lambda^{\perp}$*. It satisfies the* Pythagorean identity

$$\mathrm{Var}(\hat{\theta}_0) = \mathrm{Var}(\hat{\theta}_{\mathrm{opt}}) + \mathrm{Var}(\hat{b}^*), \tag{11.15}$$

so in particular $\mathrm{Var}(\hat{\theta}_{\mathrm{opt}}) \leq \mathrm{Var}(\hat{\theta}_0)$.

Proof sketch. Λ is a closed linear subspace of the Hilbert space $L^2(P)$ of square-integrable random variables (its elements additionally have mean zero), so the L^2 projection theorem gives a unique $\hat{b}^* \in \Lambda$ satisfying the orthogonality condition $\mathrm{Cov}(\hat{\theta}_0 - \hat{b}^*, \hat{b}) = 0$ for all $\hat{b} \in \Lambda$. Any other choice $\hat{b} \in \Lambda$ yields

$$\mathrm{Var}(\hat{\theta}_0 - \hat{b}) = \mathrm{Var}(\hat{\theta}_{\mathrm{opt}} + (\hat{b}^* - \hat{b})) = \mathrm{Var}(\hat{\theta}_{\mathrm{opt}}) + \mathrm{Var}(\hat{b}^* - \hat{b}) \geq \mathrm{Var}(\hat{\theta}_{\mathrm{opt}}),$$

where the middle equality uses $\mathrm{Cov}(\hat{\theta}_{\mathrm{opt}}, \hat{b}^* - \hat{b}) = 0$. Specializing to $\hat{b} = 0$ gives Equation 11.15. □

The Pythagorean identity shows that the variance of the initial estimator decomposes orthogonally: the optimal estimator retains only the component in $\Lambda^{\perp}$, and the discarded component $\hat{b}^*$ accounts for the remainder. Because every element of Λ has mean zero, constants are orthogonal to Λ, so $\Pi(\hat{\theta}_0 \mid \Lambda) = \Pi(\hat{\theta}_0 - \theta \mid \Lambda)$: projecting the estimator and projecting its centered estimation error yield the same correction $\hat{b}^*$. The centered form is the one used in the semiparametric theory of Appendix C, where the projected objects — scores and influence functions — are mean zero by construction. Tsiatis (2006) calls Λ the *augmentation space* precisely because its elements are the corrections that reduce variance.

Remark: Finite-Sample Analogy for the Semiparametric Projection

Theorem 11.6 is a finite-sample $L^2(P)$ analogy for the projection arguments used in semiparametric efficiency theory. In the full theory the projection is performed at the level of *influence functions* and *tangent spaces*, but with an important difference of direction: every genuine influence function is already orthogonal to the nuisance tangent space, and the efficient influence function is the *canonical gradient* — the unique influence function lying in the model tangent space, obtained by projecting any influence function onto that space (Appendix C; Tsiatis (2006), Chapters 3–4). Projecting a score onto the orthogonal complement of the nuisance score space is the related but distinct *efficient-score* construction.
The Horvitz–Thompson expression, for its part, is an influence function only in the restricted model with π known; in the full model it is an unbiased estimating function whose orthogonalization against the treatment-mechanism score directions (Section 11.6) produces AIPW. Here we project a realized estimator $\hat{\theta}_0$ onto the orthogonal complement of an augmentation space Λ of observable corrections. The two constructions share a common Hilbert-space template, and — as Section 11.6 and Section 11.7 together verify — they yield the same correction in the ATE problem. Readers should not, however, take Theorem 11.6 as a substitute for the tangent-space derivation: it is a self-contained variance-reduction result for a class of unbiased estimators, not a derivation of the semiparametric efficiency bound itself.

11.6 Projection in the Causal Inference Setting ⋆

We now specialize the projection framework to the ATE, assuming $\pi(X)$ is known. To align with the arm-by-arm structure of Section 11.4, we decompose the Horvitz–Thompson estimator as $\hat{\tau}_{\mathrm{HT}} = \hat{\mu}_{1,\mathrm{HT}} - \hat{\mu}_{0,\mathrm{HT}}$, where

$$\hat{\mu}_{1,\mathrm{HT}} = \frac{1}{n}\sum_{i=1}^{n} \frac{T_i Y_i}{\pi_i}, \qquad \hat{\mu}_{0,\mathrm{HT}} = \frac{1}{n}\sum_{i=1}^{n} \frac{(1-T_i) Y_i}{1-\pi_i},$$

and $\pi_i = \pi(X_i)$.

11.6.1 Arm-wise Augmentation Spaces

Define separate augmentation spaces for the treated and control arms:

$$\Lambda_1 = \left\{ n^{-1} \sum_{i=1}^{n} \left(\frac{T_i}{\pi_i} - 1 \right) b_1(X_i) : b_1 \in \mathcal{L}^2 \right\}, \tag{11.16}$$

$$\Lambda_0 = \left\{ n^{-1} \sum_{i=1}^{n} \left(\frac{1 - T_i}{1 - \pi_i} - 1 \right) b_0(X_i) : b_0 \in \mathcal{L}^2 \right\}. \tag{11.17}$$

Because $\mathbb{E}[(T_i/\pi_i - 1) \mid X_i] = 0$ and $\mathbb{E}[(1 - T_i)/(1 - \pi_i) - 1 \mid X_i] = 0$, every element of Λ_1 (resp. Λ_0) has expectation zero, so $\hat{\mu}_{1,\mathrm{HT}} - \hat{b}_1$ (resp. $\hat{\mu}_{0,\mathrm{HT}} - \hat{b}_0$) remains unbiased for μ_1 (resp. μ_0) for any b_t. The combined augmentation space $\Lambda = \Lambda_1 + \Lambda_0$ is the arm-wise sum, whose elements are corrections that leave $\hat{\tau}_{\mathrm{HT}}$ unbiased for τ.

The sum adds nothing, however: Λ_1 and Λ_0 are the *same* subspace, parameterized in two different ways. Writing $T_i/\pi_i - 1 = (T_i - \pi_i)/\pi_i$ and $(1 - T_i)/(1 - \pi_i) - 1 = -(T_i - \pi_i)/(1 - \pi_i)$, and noting that when π is bounded away from 0 and 1 the maps $b_1 \mapsto b_1/\pi$ and $b_0 \mapsto -b_0/(1 - \pi)$ are bijections of $\mathcal{L}^2$,

$$\Lambda = \Lambda_1 = \Lambda_0 = \left\{ n^{-1} \sum_{i=1}^{n} (T_i - \pi_i)\, a(X_i) \ : \ a \in \mathcal{L}^2 \right\}. \tag{11.18}$$

This is the geometric face of the reparameterization identity Equation 11.13: the pair (b_0, b_1) supplies two functions to sweep out a space indexed by the single function a_b, which is why the joint indexing in Theorem 11.5 is redundant (the remark on uniqueness of the minimizing estimator). We nevertheless retain the arm-wise labels Λ_1 and Λ_0: they are the convenient coordinates for the arm-wise projections of Theorem 11.7 below.

> Remark: Connection to the Estimator Class
>
> The estimator $\hat{\mu}_{1,b}$ in Equation 11.9 is exactly $\hat{\mu}_{1,\mathrm{HT}}$ augmented by an element of Λ_1. Similarly, $\hat{\mu}_{0,b}$ augments $\hat{\mu}_{0,\mathrm{HT}}$ by an element of Λ_0. Subtracting gives $\hat{\tau}_b$, the full AIPW class of Section 11.4.

11.6.2 Computing the Projections

Theorem 11.7 (Arm-wise Projections onto the Augmentation Spaces)**.** *Assume the setting of Theorem 11.4: π known and bounded away from 0 and 1, and $\mathbb{E}\{Y(t)^2\} < \infty$. The optimal corrections are*

$$\Pi(\hat{\mu}_{1,\mathrm{HT}} \mid \Lambda_1) = n^{-1} \sum_{i=1}^{n} \left(\frac{T_i}{\pi_i} - 1 \right) b_1^*(X_i), \qquad b_1^*(x) = \mathbb{E}\{Y(1) \mid x\},$$

$$\Pi(\hat{\mu}_{0,\mathrm{HT}} \mid \Lambda_0) = n^{-1} \sum_{i=1}^{n} \left(\frac{1 - T_i}{1 - \pi_i} - 1 \right) b_0^*(X_i), \qquad b_0^*(x) = \mathbb{E}\{Y(0) \mid x\}.$$

Proof. We verify the treated-arm projection; the control arm follows by symmetry. Denote the residual $R_1 = \hat{\mu}_{1,\mathrm{HT}} - n^{-1} \sum_i (T_i/\pi_i - 1)\, b_1^*(X_i)$, which simplifies to

$$R_1 = \frac{1}{n} \sum_{i=1}^{n} \left[\frac{T_i}{\pi_i} \{Y_i - b_1^*(X_i)\} + b_1^*(X_i) \right] =: \frac{1}{n} \sum_{i=1}^{n} r_{1i}.$$

By Theorem 11.6, it suffices to show $\mathrm{Cov}(R_1, \hat{b}) = 0$ for every $\hat{b} = n^{-1} \sum_j (T_j/\pi_j - 1)\, b_1(X_j) \in \Lambda_1$. Writing $\hat{b}$ as $n^{-1} \sum_j b_j$ and using independence across units, cross terms with $i \neq j$ vanish, so $\mathrm{Cov}(R_1, \hat{b}) = n^{-1}\, \mathrm{Cov}(r_{1i}, b_i)$, and since $\mathbb{E}(b_i) = 0$ this reduces to showing $\mathbb{E}[r_{1i}\, b_i] = 0$.

Condition on $(X_i, Y_i(1), Y_i(0))$; the only remaining randomness is $T_i \mid X_i \sim \mathrm{Bernoulli}(\pi_i)$. Using $T_i Y_i = T_i Y_i(1)$ and $b_1^*(X_i) = \mathbb{E}\{Y_i(1) \mid X_i\}$,

$$\mathbb{E}\left[\frac{T_i}{\pi_i} \{Y_i(1) - b_1^*(X_i)\} \left(\frac{T_i}{\pi_i} - 1 \right) \middle| X_i, Y_i(1), Y_i(0) \right] = \{Y_i(1) - b_1^*(X_i)\}\, \mathbb{E}\left[\frac{T_i}{\pi_i} \left(\frac{T_i}{\pi_i} - 1 \right) \middle| X_i \right].$$

Since $T_i^2 = T_i$, we have $\mathbb{E}[T_i^2/\pi_i^2 \mid X_i] = 1/\pi_i$ and $\mathbb{E}[T_i/\pi_i \mid X_i] = 1$, so the inner expectation is $1/\pi_i - 1$. Taking the outer expectation and using $\mathbb{E}\{Y_i(1) - b_1^*(X_i) \mid X_i\} = 0$, this contribution to $\mathbb{E}[r_{1i}\, b_i]$ vanishes. The constant term $b_1^*(X_i)$ in r_{1i} also contributes zero because $\mathbb{E}[(T_i/\pi_i - 1) \mid X_i] = 0$. $\square$

11.6.3 Recovering the AIPW Estimator

By Theorem 11.6, the optimal estimators of μ_1 and μ_0 are the projections onto $\Lambda_1^{\perp}$ and $\Lambda_0^{\perp}$, respectively:

$$\hat{\mu}_{1,\text{opt}} = \hat{\mu}_{1,\text{HT}} - n^{-1}\sum_{i=1}^{n}\left(\frac{T_i}{\pi_i} - 1\right)\mu_1^*(X_i) = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{T_i}{\pi_i}\{Y_i - \mu_1^*(X_i)\} + \mu_1^*(X_i)\right],$$
$$\hat{\mu}_{0,\text{opt}} = \hat{\mu}_{0,\text{HT}} - n^{-1}\sum_{i=1}^{n}\left(\frac{1-T_i}{1-\pi_i} - 1\right)\mu_0^*(X_i) = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{1-T_i}{1-\pi_i}\{Y_i - \mu_0^*(X_i)\} + \mu_0^*(X_i)\right].$$

Subtracting, $\hat{\tau}_{\text{opt}} = \hat{\mu}_{1,\text{opt}} - \hat{\mu}_{0,\text{opt}}$, which is precisely the AIPW estimator Equation 11.6 with $b_t^*(X) = \mathbb{E}\{Y(t) \mid X\}$, confirming Theorem 11.5 by a different route.

Remark: What the Arm-wise Decomposition Adds

The remark on uniqueness of the minimizing estimator showed that the joint problem of Theorem 11.5 is over-parameterized: $\hat{\tau}_b$ depends on (b_0, b_1) only through the single function a_b, so many pairs describe the one optimal estimator. The arm-wise view resolves this redundancy. Within each arm the indexing is faithful — distinct b_1's produce distinct elements of Λ_1 in the parameterization Equation 11.16, and likewise for b_0 — so the projections of Theorem 11.7 pin down the unique pair $b_t^*(x) = \mathbb{E}\{Y(t) \mid X = x\}$ that minimizes each arm's variance separately. This pair is the canonical representative of the equivalence class in the remark on uniqueness of the minimizing estimator, and the resulting estimator coincides with AIPW.

11.7 The Efficient Influence Function and Semiparametric Efficiency

This section states the semiparametric efficiency result and interprets it in light of the estimator development above; no new derivation is given. A complete proof requires explicit characterization of the nuisance tangent space and is beyond the scope of these notes; see Tsiatis (2006) for a rigorous development.

We now return to the efficient influence function introduced in Chapter 10 and show that it is exactly the estimating function underlying AIPW. This identification is crucial: it explains why AIPW is not only doubly robust, but also semiparametrically efficient when both nuisance functions are estimated consistently at suitable rates.

Under the nonparametric model for $O = (X, T, Y)$, the efficient influence function for the ATE is

$$\varphi_{\text{eff}}(O) = \frac{T}{\pi^*(X)}\{Y - \mu_1^*(X)\} - \frac{1-T}{1-\pi^*(X)}\{Y - \mu_0^*(X)\} + \mu_1^*(X) - \mu_0^*(X) - \tau, \tag{11.19}$$

which by definition is a function of the *true* nuisance components π^* and μ_t^*. The AIPW estimating function $\phi(O; \tau, \eta)$ in Equation 11.5 has the same algebraic form as φ_{eff}, but with working nuisances substituted in place of the truths; the implementable AIPW estimator is obtained by further substituting the fitted versions. The identification that matters for efficiency is at the population level: when the working nuisances coincide with the truths, $\phi(O; \tau, \eta^*) = \varphi_{\text{eff}}(O)$, and this is the efficient influence function $\varphi^*(O)$ recorded in Chapter 10.

Chapter 10 also flagged two central properties of φ^* whose proofs were deferred to this chapter: double robustness and semiparametric efficiency. Both are now in hand: double robustness is Theorem 11.1, and semiparametric efficiency follows from the identification $\varphi_{\text{eff}} = \varphi^*$ together with the convolution theorem below. The remark on which parts of the variance formula are identified has already verified the variance side by direct computation: the minimized class variance Equation 11.14 equals $\mathbb{E}\{\varphi_{\text{eff}}(O)^2\}$.

The natural decomposition noted in Chapter 10 — an IPW-style residual correction plus a regression-centered term — is exactly the structure that emerged from both the bias-correction derivation of Section 11.2 and the optimal-augmentation derivation of Section 11.4. The augmentation space Λ corresponds to a specific component of the nuisance structure: it is the finite-sample analogue of the treatment-mechanism score subspace

$$\mathcal{H}_T = \{a(X)\{T - \pi(X)\} : a \in L^2(P_X)\},$$

as the common form Equation 11.18 makes explicit. This parametrization presumes strong overlap, $\epsilon \leq \pi(X) \leq 1-\epsilon$ almost surely. Appendix C defines the treatment-score space instead by the conditional mean-zero condition $\mathbb{E}\{h(T,X) \mid X\} = 0$, which under weak positivity is strictly larger; the two agree under strong overlap. Orthogonalizing the Horvitz–Thompson estimating function against $\mathcal{H}_T$ yields the AIPW form; this removes a nuisance-score component, but it is not the general definition of the efficient influence function, which is the canonical gradient (see also Appendix C).

To state the bound precisely, define the observed-data functional

$$\Psi(P) = \mathbb{E}_P\{\mu_{1,P}(X) - \mu_{0,P}(X)\}, \qquad \mu_{t,P}(x) = \mathbb{E}_P(Y \mid T = t, X = x),$$

on the nonparametric model for $O = (X, T, Y)$ satisfying positivity and $\mathbb{E}_P\{\varphi_{\text{eff}}(O)^2\} < \infty$. The functional $\Psi(P)$ is defined for every such P without causal assumptions; under consistency, conditional exchangeability, and positivity (Chapters 4–6), it equals the causal ATE τ. The efficiency theory below concerns estimation of $\Psi(P)$; the causal interpretation rides on the identification assumptions.

Theorem 11.8 (Semiparametric Efficiency Bound for the ATE)**.** *Under the nonparametric model for $O = (X, T, Y)$, every regular asymptotically linear estimator $\hat{\tau}$ of $\Psi(P)$ (equivalently, of the ATE under the identification assumptions) satisfies*

$$\operatorname{Avar}(\sqrt{n}(\hat{\tau} - \tau)) \;\geq\; \mathbb{E}\{\varphi_{\text{eff}}(O)^2\},$$

and the bound is attained by estimators whose influence function equals φ_{eff} in Equation 11.19.

Remark: Scope of the Efficiency Bound

The result is a specialization to the ATE of the Hájek–Le Cam convolution theorem in semiparametric theory; see Bickel et al. (1993) (Chapter 2) or Vaart (1998) (Chapter 25) for a rigorous development. Regularity rules out Hodges-type super-efficient estimators that achieve smaller variance on a measure-zero subset of the parameter space at the cost of instability elsewhere. Dropping regularity, or replacing the nonparametric model with a smaller semiparametric submodel, changes the bound.

Remark: EIF as the Semiparametric Analogue of the Score

The efficient influence function plays the same role in semiparametric theory that the score plays in parametric models: its variance gives the information lower bound, analogous to the Cramér–Rao bound. Strictly speaking, φ_{eff} is not itself a score (the score belongs to the tangent space of the model); it is the *canonical gradient* — the Riesz representer of the pathwise derivative of the parameter functional τ along regular submodels (Bickel et al. 1993).
The three derivations in this chapter — bias correction (Section 11.2), optimal augmentation (Section 11.4), and the present semiparametric identification — converge on the same estimating function, which is the deepest explanation of why AIPW occupies a central place in the causal inference toolkit.

11.8 Doubly Robust Regression: Weighted and Augmented Approaches ⋆

The AIPW estimator achieves double robustness by adding an explicit bias-correction term to the prediction estimator. An equally important question is how to build double robustness *directly into the outcome model fit*, so that the prediction estimator is itself doubly robust without any separate augmentation step. The unifying concept behind this section is the *internal bias calibration* (IBC) condition: a simple algebraic identity that, when enforced by the fitting procedure, forces the prediction estimator to coincide with AIPW. The two constructions below — IPW-weighted least squares and regression on a clever covariate — are distinct ways of ensuring IBC holds, and the second is the targeting step that underlies TMLE.

11.8.1 The Internal Bias Calibration Conditions

Let $\hat{\mu}_t(x)$ denote any fitted outcome model for arm t, and let $\hat{\pi}_i = \hat{\pi}(X_i)$. The prediction estimator $\hat{\tau}_{\text{pred}} = n^{-1} \sum_i \{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\}$ requires no augmentation if the IPW-weighted residuals vanish:

$$\sum_{i=1}^{n} \frac{T_i}{\hat{\pi}_i} \{Y_i - \hat{\mu}_1(X_i)\} = 0, \tag{11.20}$$

$$\sum_{i=1}^{n} \frac{1 - T_i}{1 - \hat{\pi}_i} \{Y_i - \hat{\mu}_0(X_i)\} = 0. \tag{11.21}$$

We call Equation 11.20–Equation 11.21 the *internal bias calibration* (IBC) conditions (Firth and Bennett 1998), following the terminology introduced in the context of prediction estimation in survey sampling. The word *internal* reflects the fact that these constraints are imposed within the outcome-model fitting step itself: once they hold, no separate bias-correction term is needed afterward. When both IBC conditions hold, the augmentation terms in Equation 11.6 are zero by construction, so $\hat{\tau}_{\text{pred}} = \hat{\tau}_{\text{AIPW}}$ and the prediction estimator is itself doubly robust. The two approaches below are two ways of fitting $\hat{\mu}_t$ so that the IBC conditions are satisfied by construction.

11.8.2 Weighted Regression Approach

Suppose the outcome model for arm t is parameterized as $\mu_t(X; \theta_t)$ for a finite-dimensional parameter θ_t, with the model including a constant term. Instead of fitting θ_t by ordinary least squares among the units with $T_i = t$, use *IPW-weighted* least squares: minimize

$$\sum_{i=1}^{n} \frac{T_i}{\hat{\pi}_i} \{Y_i - \mu_1(X_i; \theta_1)\}^2 \tag{11.22}$$

for $t = 1$, and analogously with $(1 - T_i)/(1 - \hat{\pi}_i)$ for $t = 0$. This is essentially IPW-weighted estimation of the regression coefficients: each unit's contribution to the least-squares criterion is upweighted by the inverse of its probability of being in that treatment arm. The normal equation of Equation 11.22 with respect to the intercept component of θ_1 is

$$\sum_{i=1}^{n} \frac{T_i}{\hat{\pi}_i} \{Y_i - \mu_1(X_i; \hat{\theta}_1)\} = 0,$$

which is exactly the IBC condition Equation 11.20. Hence the IPW-weighted fitted values automatically satisfy the IBC condition for any model containing a constant, and the resulting prediction estimator is doubly robust; see Robins et al. (1994) and Bang and Robins (2005) for a general treatment. The key point is not that weighted least squares creates a fundamentally different doubly robust estimator; rather, it constructs fitted values for which the prediction estimator algebraically equals the AIPW estimator.

11.8.3 Augmented Model Approach and the Clever Covariate

The augmented approach achieves the same calibration by modifying the covariate set rather than the fitting weights. Let $\hat{\mu}_1^{(0)}(X_i)$ be any initial fit of the treated outcome model (e.g., from OLS or a flexible machine-learning method). Augment it by including the *clever covariate* $\hat{\pi}_i^{-1}$ (Laan and Rubin 2006; Laan and Rose 2011) and run OLS of Y_i on $\hat{\mu}_1^{(0)}(X_i)$ and $\hat{\pi}_i^{-1}$ among treated units, fitting the model

$$Y_i = \alpha_1 + \beta_1 \hat{\mu}_1^{(0)}(X_i) + \gamma_1 \hat{\pi}_i^{-1} + e_i(1), \qquad T_i = 1. \tag{11.23}$$

Denote the OLS estimates by $\hat{\alpha}_1$, $\hat{\beta}_1$, $\hat{\gamma}_1$. The fitted outcome model evaluated at all units is $\hat{\mu}_1(X_i) = \hat{\alpha}_1 + \hat{\beta}_1 \hat{\mu}_1^{(0)}(X_i) + \hat{\gamma}_1 \hat{\pi}_i^{-1}$, and the prediction estimator of μ_1 is $\hat{\mu}_1^{\text{pred}} = n^{-1} \sum_{i=1}^{n} \hat{\mu}_1(X_i)$.

The normal equation of Equation 11.23 for $\hat{\gamma}_1$ (the coefficient on $\hat{\pi}_i^{-1}$) is

$$\sum_{i=1}^{n} \frac{T_i}{\hat{\pi}_i} \{Y_i - \hat{\mu}_1(X_i)\} = 0, \tag{11.24}$$

which is exactly the IBC condition Equation 11.20. Since Equation 11.24 equals zero, we can add it to $\hat{\mu}_1^{\text{pred}}$ without changing its value, giving the AIPW representation

$$\hat{\mu}_1^{\text{pred}} = \frac{1}{n}\sum_{i=1}^{n}\hat{\mu}_1(X_i) + \frac{1}{n}\sum_{i=1}^{n}\frac{T_i}{\hat{\pi}_i}\{Y_i - \hat{\mu}_1(X_i)\}.$$

The IPW-weighted residual sum is zero by the IBC condition, so the prediction estimator coincides with the AIPW estimator.

The model Equation 11.23 offers two additional degrees of freedom beyond the fixed-offset augmentation $\hat{\mu}_1^{(0)}(X_i) + \hat{\gamma}_1\hat{\pi}_i^{-1}$. First, the free coefficient $\hat{\beta}_1$ allows the scale of the initial fit to be recalibrated: if $\hat{\mu}_1^{(0)}$ systematically over- or under-predicts, $\hat{\beta}_1 \neq 1$ absorbs this scaling error. Second, the intercept $\hat{\alpha}_1$ absorbs any residual mean shift. The covariate $\hat{\pi}_i^{-1}$ is called *clever* because it is chosen so that the fitted regression satisfies the same score equation Equation 11.24 that appears in the AIPW bias correction. When the initial outcome model is misspecified, including this covariate lets the regression absorb the component of the model error that is correlated with $\hat{\pi}_i^{-1}$, so the fitted prediction estimator becomes algebraically equivalent to a debiased estimator. Whether the additional parameters $(\hat{\alpha}_1, \hat{\beta}_1)$ yield a measurable efficiency gain over the fixed-offset form depends on the data-generating process: the improvement is largest when the residual misspecification of $\hat{\mu}_1^{(0)}$ has substantial linear projection onto $\text{span}\{1, \hat{\mu}_1^{(0)}(X), \hat{\pi}^{-1}(X)\}$ and negligible otherwise. Section 11.9 provides a simulation in which the gain is substantial.

Remark: Behavior under Correct Outcome Model

If the initial outcome model $\hat{\mu}_1^{(0)}$ is correctly specified and the regressors $\{1, \mu_1^*(X), 1/\pi^\dagger(X)\}$ are linearly independent in the treated population (a full-rank condition), then the population least-squares coefficients are $(0,1,0)$: $\hat{\alpha}_1 \xrightarrow{p} 0$, $\hat{\beta}_1 \xrightarrow{p} 1$, and $\hat{\gamma}_1 \xrightarrow{p} 0$, so the augmented fit has the same probability limit as the initial outcome regression.
Coefficient convergence alone, however, does not make the two estimators first-order equivalent: the estimated coefficients are $O_p(n^{-1/2})$, and their effect on the asymptotic variance must be analyzed separately. When *both* nuisance models are correctly specified, that analysis is supplied by Neyman orthogonality (Section 11.10): the AIPW moment is first-order insensitive to $O_p(n^{-1/2})$ perturbations of the outcome fit at (π^*, μ^*), so including the additional covariates carries no asymptotic efficiency cost. When only the outcome model is correct and the propensity model is misspecified, the moment is not orthogonal in the outcome direction, and first-order equivalence is not automatic.

Remark: Connection to TMLE

The augmented model approach captures the same targeting idea that underlies *targeted minimum loss-based estimation* (TMLE), introduced by Laan and Rubin (2006) and developed comprehensively in Laan and Rose (2011). For a continuous outcome under squared-error loss, the standard TMLE targeting step uses exactly this fixed-offset, identity-link form (coefficient on $\hat{\mu}_1^{(0)}$ fixed at 1): a common one-dimensional least-favorable fluctuation sufficient to satisfy the IBC condition. TMLE is loss- and parameter-specific, however: for bounded or binary outcomes a link-appropriate fluctuation — typically logistic — is ordinarily used so that fitted conditional means remain in their admissible range. The model Equation 11.23 goes further by freely estimating $\hat{\beta}_1$, providing additional recalibration of the initial fit beyond the one-dimensional targeting step. A thorough treatment of TMLE, including its asymptotic theory and connections to cross-fitting, is beyond the scope of these notes; see Laan and Rose (2011) and Laan and Rose (2018) for details.

11.9 Lab: Simulation Study

This lab compares four estimators of the ATE: the IPW estimator $\hat{\tau}_{\text{HT}}$, the AIPW estimator $\hat{\tau}_{\text{AIPW}}$, the fixed-offset augmented-model estimator $\hat{\tau}_{\text{aug}}$, and the improved augmented-model estimator $\hat{\tau}_{\text{aug}^+}$ of Equation 11.23. A 2×2 design over nuisance-model correctness — outcome regression (OR) correct or misspecified, crossed with propensity score (PS) correct or misspecified — allows the lab to demonstrate double robustness directly: the augmented estimators remain consistent whenever at least one nuisance model is correct, and fail only when both are misspecified. The lab also reports the empirical coverage of the 95% Wald confidence interval based on the influence-function variance estimator of Section 11.10.2, illustrating the distinction between double-robust consistency and valid asymptotic inference. Full R code is provided in the supplementary file `chapter11_lab.R`.

11.9.1 Simulation Setup

Generate $n = 1000$ i.i.d. observations as follows. Draw $X_i \sim N(0,1)$, then $T_i \mid X_i \sim \text{Bernoulli}(\pi^*(X_i))$ with true propensity score

$$\pi^*(x) = \text{expit}\{0.2\,x + 0.2\,(x^2 - 1)\}.$$

The quadratic term is centered at $\mathbb{E}(X^2) = 1$ so that the marginal treatment probability is roughly balanced. The potential outcomes are

$$Y_i(1) = 1 + X_i + 0.5X_i^2 + \varepsilon_i(1), \qquad Y_i(0) = X_i + 0.5X_i^2 + \varepsilon_i(0),$$

with $\varepsilon_i(t) \overset{\text{i.i.d.}}{\sim} N(0,1)$ independent of (X_i, T_i), giving true ATE $\tau = 1$. The observed outcome is $Y_i = T_iY_i(1) + (1 - T_i)Y_i(0)$. The quadratic term enters both the true OR and the true PS; a linear-only fit omits it in either or both places, yielding the four scenarios below.

Table 11.1: The 2×2 design of nuisance-model specification.

Scenario	Outcome regression fit	Propensity score fit
S1	correct: $Y \sim (1, X, X^2)$	correct: $T \sim (1, X, X^2)$
S2	correct: $Y \sim (1, X, X^2)$	misspecified: $T \sim (1, X)$
S3	misspecified: $Y \sim (1, X)$	correct: $T \sim (1, X, X^2)$
S4	misspecified: $Y \sim (1, X)$	misspecified: $T \sim (1, X)$

In each scenario the OR is fitted arm-wise by linear regression and the PS by logistic regression, with the X^2 term either included or omitted as indicated. The fitted $\hat{\pi}$ is clipped to $[10^{-3}, 1 - 10^{-3}]$ for numerical stability. Clipping is a finite-sample regularization, not part of the theoretical estimator: in the runs reported below it is never active (zero of the four million fitted propensity scores per specification fall outside the clipping interval), so it leaves the probability limits of $\hat{\pi}$ — and hence the "PS correct" labels — untouched. Double robustness predicts that $\hat{\tau}_{\text{AIPW}}$ is consistent in Scenarios S1–S3 and loses its consistency guarantee in S4.

11.9.2 Estimator Construction

Given $\hat{\pi}_i = \hat{\pi}(X_i)$ and initial fitted values $\hat{\mu}_t(X_i) \equiv \hat{\mu}_t^{(0)}(X_i)$, let $c_{1i} = \hat{\pi}_i^{-1}$ and $c_{0i} = (1 - \hat{\pi}_i)^{-1}$.

IPW estimator. $\hat{\tau}_{\text{HT}} = n^{-1} \sum_i \{T_iY_i/\hat{\pi}_i - (1 - T_i)Y_i/(1 - \hat{\pi}_i)\}$.

AIPW estimator. $\hat{\tau}_{\text{AIPW}}$ as in Equation 11.6 with fitted nuisances.

Fixed-offset augmented-model estimator. Within each arm, regress $Y_i - \hat{\mu}_t(X_i)$ on c_{ti} without an intercept to obtain $\hat{\gamma}_t$, set $\hat{\mu}_t^{\text{aug}}(X_i) = \hat{\mu}_t(X_i) + \hat{\gamma}_t c_{ti}$, and compute $\hat{\tau}_{\text{aug}} = n^{-1} \sum_i \{\hat{\mu}_1^{\text{aug}}(X_i) - \hat{\mu}_0^{\text{aug}}(X_i)\}$.

Improved augmented-model estimator. Within each arm, regress Y_i on $(1, \hat{\mu}_t^{(0)}(X_i), c_{ti})$ by OLS to obtain $(\hat{\alpha}_t, \hat{\beta}_t, \hat{\gamma}_t)$, set $\hat{\mu}_t^{\text{aug}^+}(X_i) = \hat{\alpha}_t + \hat{\beta}_t\hat{\mu}_t(X_i) + \hat{\gamma}_t c_{ti}$, and compute $\hat{\tau}_{\text{aug}^+}$ analogously.

Variance estimators and Wald intervals. For each estimator $\hat{\tau}_\bullet$, the asymptotic variance is estimated by the empirical variance of the plug-in influence values, $\hat{V}_\bullet = \{n(n-1)\}^{-1} \sum_i (\hat{\varphi}_{\bullet,i} - \bar{\varphi}_\bullet)^2$, with $\hat{\varphi}_{\bullet,i}$ the influence-function contribution Equation 11.26 evaluated at the estimator-specific fitted values (for HT, the influence value is simply $T_iY_i/\hat{\pi}_i - (1-T_i)Y_i/(1-\hat{\pi}_i) - \hat{\tau}_{\text{HT}}$). The 95% Wald interval is $\hat{\tau}_\bullet \pm 1.96\sqrt{\hat{V}_\bullet}$.

11.9.3 Results

Repeat the simulation $B = 2000$ times and record bias, variance, MSE, and empirical 95% Wald coverage for each estimator.

Table 11.2: Simulation results ($n = 1000$, $B = 2000$ replications per scenario; scenario s uses `set.seed(2024 + s)`, i.e. seeds 2025–2028). Bias, variance, and MSE are in units of 10^{-3}; Cov is the empirical coverage of the 95% Wald interval.

Scenario	Metric	$\hat{\tau}_{\mathrm{HT}}$	$\hat{\tau}_{\mathrm{AIPW}}$	$\hat{\tau}_{\mathrm{aug}}$	$\hat{\tau}_{\mathrm{aug}^+}$
S1: OR ✓, PS ✓	Bias $\times 10^3$	0.55	−0.44	−0.45	−0.45
	Var $\times 10^3$	6.82	4.25	4.25	4.30
	MSE $\times 10^3$	6.82	4.25	4.25	4.30
	Cov (%)	99.9	95.0	94.9	94.5
S2: OR ✓, PS ×	Bias $\times 10^3$	188.56	0.62	0.62	0.62
	Var $\times 10^3$	6.32	4.33	4.33	4.33
	MSE $\times 10^3$	41.87	4.32	4.32	4.32
	Cov (%)	67.4	94.2	94.2	94.2
S3: OR ×, PS ✓	Bias $\times 10^3$	1.46	2.02	2.25	−25.26
	Var $\times 10^3$	5.74	4.71	4.41	8.02
	MSE $\times 10^3$	5.74	4.71	4.41	8.66
	Cov (%)	99.9	98.4	98.2	93.2
S4: OR ×, PS ×	Bias $\times 10^3$	188.42	188.77	188.62	−5.42
	Var $\times 10^3$	6.28	6.30	6.29	4.26
	MSE $\times 10^3$	41.78	41.93	41.87	4.29
	Cov (%)	67.5	33.1	33.1	94.7

Table 11.3: Coverage diagnostics from the same runs: average estimated standard error aveSE $= B^{-1}\sum_b (\hat{V}^{(b)})^{1/2}$ and empirical standard deviation empSD of the estimates across replications, both in units of 10^{-3}, and their ratio. Ratios above 1 indicate overstated standard errors (over-coverage); ratios below 1, understated standard errors.

Scenario	Metric	$\hat{\tau}_{\mathrm{HT}}$	$\hat{\tau}_{\mathrm{AIPW}}$	$\hat{\tau}_{\mathrm{aug}}$	$\hat{\tau}_{\mathrm{aug}^+}$
S1	aveSE $\times 10^3$	128.83	64.23	64.20	64.12
	empSD $\times 10^3$	82.58	65.20	65.20	65.56
	ratio	1.56	0.99	0.98	0.98
S2	aveSE $\times 10^3$	114.68	63.43	63.43	63.43
	empSD $\times 10^3$	79.50	65.77	65.77	65.77
	ratio	1.44	0.96	0.96	0.96
S3	aveSE $\times 10^3$	128.91	84.88	81.75	74.28
	empSD $\times 10^3$	75.74	68.60	66.39	89.57
	ratio	1.70	1.24	1.23	0.83
S4	aveSE $\times 10^3$	114.84	78.78	78.74	63.55
	empSD $\times 10^3$	79.27	79.37	79.34	65.27
	ratio	1.45	0.99	0.99	0.97

The results illustrate double robustness directly.

- **S1 (both correct).** All four estimators are approximately unbiased. $\hat{\tau}_{\mathrm{HT}}$ has about 60% higher MSE (6.82 vs. 4.25×10^{-3}) because it discards the outcome model. $\hat{\tau}_{\mathrm{AIPW}}$, $\hat{\tau}_{\mathrm{aug}}$, and $\hat{\tau}_{\mathrm{aug}^+}$ are essentially equivalent, confirming that the augmentation parameters converge to $(0, 1, 0)$ when the OR is correct; with both nuisance models correct, Neyman orthogonality implies that the recalibration carries no first-order efficiency cost (the remark on behavior under a correct outcome model). Wald coverage is close to nominal for the three PS-and-OR estimators. HT's coverage is far above nominal for a more subtle reason. The plug-in influence value treats the propensity score as known, so $\hat{V}_{\mathrm{HT}}$ targets the known-π asymptotic variance of the IPW estimator. The lab, however, *estimates* π by correctly specified logistic maximum likelihood, and Chapter 10 showed that ML estimation of the propensity parameters can only *reduce* the asymptotic variance of $\hat{\tau}_{\mathrm{HT}}$ relative to plugging in the true π^*. In this DGP the reduction is large: the average estimated standard error exceeds the empirical standard deviation by more than 50%, and the mismatch does not shrink as n grows, because it reflects the omitted first-order nuisance correction of Chapter 10, not a finite-sample effect. Re-running the lab with the true π^* plugged in restores HT coverage to the nominal level.

- **S2 (OR correct, PS misspecified).** HT is badly biased (bias ≈ 0.19, MSE $\approx 42 \times 10^{-3}$) and its Wald interval undercovers (67%). The three augmented estimators are essentially indistinguishable from one another: when the OR is correct, they converge asymptotically to the same limit regardless of PS specification. Their bias is under 1×10^{-3}, MSE is about 4.32×10^{-3}, and Wald coverage is $\approx 94.2\%$. This is the first direct demonstration of double robustness: PS misspecification is absorbed by the correctly specified OR model.
- **S3 (OR misspecified, PS correct).** HT, AIPW, and $\hat{\tau}_{\mathrm{aug}}$ remain consistent through the correct PS model, with biases under 3×10^{-3}. HT again over-covers for the reason given under S1; AIPW and $\hat{\tau}_{\mathrm{aug}}$, whose intervals were close to nominal in S1, now also over-cover (98.4% and 98.2%) — the Takeaways below explain why. $\hat{\tau}_{\mathrm{aug}}$ achieves the lowest MSE (4.41×10^{-3}): the targeted regression reduces residual variance without disturbing the IBC constraint. $\hat{\tau}_{\mathrm{aug}^+}$ shows a modest finite-sample bias (-25×10^{-3}) induced by extrapolating the three-term arm-wise regression through high-leverage clever-covariate values; its Wald coverage is 93.2%, still close to nominal. This is the second double-robustness demonstration: OR misspecification is absorbed by the correctly specified PS model.
- **S4 (both misspecified).** Double robustness provides no guarantee, and this is what the simulation shows. HT, AIPW, and $\hat{\tau}_{\mathrm{aug}}$ are all badly biased, and Wald coverage is catastrophic for AIPW and aug ($\approx 33\%$). The variance estimator still produces a number, but that number targets the spread of a sampling distribution that is not centered on τ: the Wald interval is concentrated around the wrong value. The diagnostics confirm the mechanism: the aveSE/empSD ratio is ≈ 0.99, so the interval has nearly the right width in the wrong place.

No Triple Robustness

The apparent "recovery" of $\hat{\tau}_{\mathrm{aug}^+}$ in S4 (bias -5×10^{-3}, coverage 94.7%) is a DGP-specific artifact and does *not* indicate a form of triple robustness. The regressors $(1, \hat{\mu}_t^{(0)}(X), \hat{\pi}^{-1}(X))$ form a three-dimensional function basis that, in this particular DGP, happens to capture enough of the omitted quadratic curvature through the nonlinearity of $\hat{\pi}^{-1}$ to render the prediction estimator nearly unbiased. Under a different DGP — for instance, one in which the X-dependence of the OR has a form not well approximated by any linear combination of the initial fit and $\hat{\pi}^{-1}$ — $\hat{\tau}_{\mathrm{aug}^+}$ would share the bias of the other estimators. Readers should treat S4 as a cautionary illustration of what goes wrong when double robustness fails, not as evidence of a distribution-free guarantee.

Takeaways. Comparing Scenarios S2–S3 against S4 is the crisp illustration of Theorem 11.1: AIPW is consistent whenever at least one nuisance model is correct, and only S4 breaks the guarantee. Wald coverage is more subtle, and the table repays a careful reading. AIPW coverage is close to nominal in S1 and S2 and collapses to 33% in S4 — the collapse is precisely the behavior anticipated by the remark on double robustness versus efficient inference, since the plug-in variance targets the spread around a misspecified limit, not around τ.

Scenario S3, however, shows persistent *over*-coverage (98.4% for AIPW), and this is not a finite-sample accident: S3 is exactly the $(\pi^*, \mu^\dagger)$ configuration analyzed in the remark on double robustness versus efficient inference. Because $\mu_t^\dagger \neq \mu_t^*$, the AIPW moment is not orthogonal in the π-direction, so estimating π affects the sampling distribution at first order; as in the discussion of HT under S1, ML estimation of a correctly specified propensity model *reduces* the true sampling variance below the quantity targeted by the plug-in influence values, so the Wald interval is conservative. The gap between average standard error and empirical standard deviation does not close as n grows — it is an asymptotic phenomenon — and re-running S3 with the true π^* plugged in restores nominal coverage.

The mirror-image configuration is S2, where the analogous first-order contribution comes from estimating the correct outcome model under a misspecified π; in this DGP that contribution happens to be negligible, which is why S2 coverage looks nominal. Neither the size nor the direction of these omitted contributions is guaranteed in general: under flexible machine-learning nuisances or in other designs, one-correct-model coverage can be conservative or anticonservative. The remedy for the omitted first-order term is a variance estimator that accounts for it — the stacked sandwich of the remark on double robustness versus efficient inference; cross-fitting, by contrast, addresses the same-sample empirical-process problem of flexible nuisance estimation (Chapter 12) and does not by itself restore orthogonality at a misspecified limit.

The lab also isolates a secondary phenomenon visible in S3: when the PS is correct but the OR is misspecified, the fixed-offset $\hat{\tau}_{\text{aug}}$ can reduce MSE relative to the plug-in AIPW estimator by shrinking residual variance through the IBC-enforced regression, without disturbing consistency. Efficiency gains of this type are the practical motivation for targeting methods such as TMLE (the remark on TMLE).

11.10 Asymptotic Inference with Estimated Nuisance Functions

A fundamental question in the practical use of the AIPW estimator is: under what conditions is the effect of estimating the nuisance functions π^* and μ_t^* asymptotically negligible, so that $\hat{\tau}_{\text{AIPW}}$ behaves as if the nuisance functions were known? Answering this question determines when plug-in nuisance estimators, including flexible machine-learning methods, can be used without invalidating large-sample inference on τ.

11.10.1 Asymptotic Normality

Under suitable regularity conditions,

$$\sqrt{n}(\hat{\tau}_{\text{AIPW}} - \tau) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \varphi_{\text{eff}}(O_i) + o_p(1) \xrightarrow{d} N(0,\ \mathbb{E}\{\varphi_{\text{eff}}(O)^2\}),$$

where φ_{eff} in Equation 11.19 is evaluated at the true nuisances. The $o_p(1)$ remainder captures the effect of nuisance estimation error. Neyman orthogonality implies that this error enters the AIPW estimator only through a second-order remainder: the Gateaux derivative of the estimating equation with respect to each nuisance function, evaluated at the truth, is zero. A sufficient condition for the remainder to be negligible is the *product-rate condition*

$$\|\hat{\pi} - \pi^*\| \cdot \|\hat{\mu}_t - \mu_t^*\| = o_p(n^{-1/2}), \qquad t = 0, 1, \tag{11.25}$$

where $\|\cdot\|$ denotes the $L_2(P)$ norm. The drift part of the remainder is exactly the identity Equation 11.7 evaluated at the fitted nuisances, and Equation 11.25 is the condition that the Cauchy–Schwarz bound Equation 11.8 converts into $o_p(n^{-1/2})$.

A symmetric sufficient condition for Equation 11.25 is that each nuisance estimator converge slightly faster than $n^{-1/4}$, namely $\|\hat{\pi} - \pi^*\| = o_p(n^{-1/4})$ and $\|\hat{\mu}_t - \mu_t^*\| = o_p(n^{-1/4})$. (Exact $n^{-1/4}$ rates would yield a product of order $O_p(n^{-1/2})$, not $o_p(n^{-1/2})$, and are therefore borderline.) This is much weaker than the parametric rate $n^{-1/2}$ required of each individually. In modern applications with flexible machine-learning methods, this statement is usually paired with *sample splitting* or *cross-fitting*; without such devices, additional empirical-process conditions are needed to justify the asymptotic expansion. Chapter 12 develops the cross-fitting approach in detail.

> Remark: Neyman Orthogonality
>
> The first-order insensitivity of the AIPW estimating function to nuisance perturbations is an instance of *Neyman orthogonality*, a property shared by a broad class of semiparametric estimators. Chapter 12 exploits this property systematically through cross-fitting, which avoids the same-sample Donsker or stochastic-equicontinuity conditions otherwise imposed on the nuisance classes, while retaining the nuisance-consistency, product-rate, overlap, and moment requirements.

11.10.2 Variance Estimation and Confidence Intervals

Because $\hat{\tau}_{\text{AIPW}}$ is asymptotically linear with influence function φ_{eff}, its asymptotic variance can be estimated consistently by the empirical variance of the plug-in influence values. Compute

$$\hat{\varphi}_i = \frac{T_i}{\hat{\pi}(X_i)}\{Y_i - \hat{\mu}_1(X_i)\} - \frac{1 - T_i}{1 - \hat{\pi}(X_i)}\{Y_i - \hat{\mu}_0(X_i)\} + \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) - \hat{\tau}_{\text{AIPW}}, \tag{11.26}$$

and set

$$\hat{V} = \frac{1}{n(n-1)} \sum_{i=1}^{n} (\hat{\varphi}_i - \bar{\varphi})^2, \qquad \bar{\varphi} = \frac{1}{n} \sum_{i=1}^{n} \hat{\varphi}_i. \tag{11.27}$$

The Wald confidence interval is $\hat{\tau}_{\text{AIPW}} \pm z_{1-a/2}\sqrt{\hat{V}}$.

Theorem 11.9 (Consistency of the Variance Estimator)**.** *Suppose* $\sqrt{n}(\hat{\tau}_{\text{AIPW}} - \tau) = n^{-1/2} \sum_i \varphi_{\text{eff}}(O_i) + o_p(1)$ *with* $\mathbb{E}\{\varphi_{\text{eff}}(O)^2\} < \infty$*, and that the estimated influence values Equation 11.26 satisfy*

$$\frac{1}{n} \sum_{i=1}^{n} \{\hat{\varphi}_i - \varphi_{\text{eff}}(O_i)\}^2 = o_p(1).$$

Then $n\hat{V} \xrightarrow{p} \mathbb{E}\{\varphi_{\text{eff}}(O)^2\}$*; that is,* $\hat{V}$ *is consistent for the asymptotic variance of* $\hat{\tau}_{\text{AIPW}}$*.*

The second condition is not implied by asymptotic linearity, which controls the average of the true influence values but not the empirical second moment of the estimation error $\hat{\varphi}_i - \varphi_{\text{eff}}(O_i)$. It holds, for example, when both nuisance estimators are L^2-consistent for the truth, the fitted propensity scores are bounded away from 0 and 1, and standard moment conditions hold.

Remark: Double Robustness versus Efficient Inference

It is important not to conflate two distinct results. *Double-robust consistency* (Theorem 11.1) requires that at least one of the two nuisance components be consistently estimated, together with overlap and standard law-of-large-numbers regularity conditions. What it does *not* require is the product-rate condition Equation 11.25 needed for root-n efficient inference.

Efficient asymptotic inference — the $\sqrt{n}$-normal expansion above and the Wald interval based on $\hat{V}$ — is a strictly stronger requirement. It requires *both* nuisance functions to be estimated consistently at rates satisfying Equation 11.25, together with overlap and moment conditions. Plugging estimated nuisance functions into Equation 11.27 estimates the asymptotic variance of the *efficient* influence function only when the asymptotic linear representation with φ_{eff} holds; that representation does *not* follow from double robustness alone.

In particular, suppose the propensity score is correct but the outcome regression is misspecified, so that $\hat{\pi} \to \pi^*$ but $\hat{\mu}_t \to \mu_t^\dagger \neq \mu_t^*$. AIPW remains consistent for τ (Theorem 11.1, Case 2): the AIPW moment evaluated at the limits is

$$\varphi^\dagger(O) = \frac{T}{\pi^*(X)}\{Y - \mu_1^\dagger(X)\} - \frac{1-T}{1-\pi^*(X)}\{Y - \mu_0^\dagger(X)\} + \mu_1^\dagger(X) - \mu_0^\dagger(X) - \tau,$$

with $\mathbb{E}\{\varphi^\dagger(O)\} = 0$.

It is tempting to conclude that $\varphi^\dagger$ is then the influence function of $\hat{\tau}_{\text{AIPW}}$ and that the plug-in variance consistently estimates $\mathbb{E}\{\varphi^\dagger(O)^2\}$. This conclusion is correct only when π^* is *known*, not estimated. When π is estimated and $\mu^\dagger \neq \mu^*$, the AIPW moment is no longer Neyman-orthogonal in the π-direction at $(\pi^*, \mu^\dagger)$. The Gateaux derivative of the population moment with respect to π at this point, along a perturbation $h(X)$, evaluates to

$$-\mathbb{E}\left[h(X)\left\{\frac{\mu_1^*(X) - \mu_1^\dagger(X)}{\pi^*(X)} + \frac{\mu_0^*(X) - \mu_0^\dagger(X)}{1-\pi^*(X)}\right\}\right],$$

which is generally nonzero whenever $\mu_t^\dagger \neq \mu_t^*$. The first-order error from estimating π therefore enters the asymptotic linear expansion of $\sqrt{n}(\hat{\tau}_{\text{AIPW}} - \tau)$, which acquires a contribution beyond $n^{-1/2}\sum_i \varphi^\dagger(O_i)$. By symmetry, the same phenomenon occurs in the mirror-image case where π is misspecified and μ is correct but estimated.

Consequently, the plug-in variance Equation 11.27 is not generally valid under one-correct-model misspecification, and the naive Wald interval can have incorrect coverage. Valid asymptotic inference in this regime requires one of the following: (i) treating $(\hat{\pi}, \hat{\mu}, \hat{\tau}_{\text{AIPW}})$ jointly as the solution of a stacked estimating-equation system and using the corresponding sandwich variance, which absorbs the nuisance-estimation contribution (see Tsiatis 2006, chap. 3); (ii) treating the correctly specified nuisance as known, or estimating it from a sufficiently large external sample, so that no first-order contribution arises; or (iii) verifying that the misspecified limit happens to coincide with the truth, which restores orthogonality. Sample splitting and cross-fitting do *not* belong on this list: they remove the same-sample empirical-process term, but the nonzero population derivative above — and hence the first-order contribution of estimating π — remains (Exercise 9). The plug-in variance is justified without qualification only when $\hat{\tau}_{\mathrm{AIPW}}$ admits an asymptotic linear representation with the stated influence function — a representation that holds at (π^*, μ^*) but generally fails at $(\pi^*, \mu^\dagger)$ once π^* is replaced by its estimator.

Extreme Propensity Scores

When $\hat{\pi}(X_i)$ is near 0 or 1, the IPW weights become large and can destabilize the estimator. Practical remedies include overlap diagnostics, truncation of extreme propensity scores, or restricting the target population to a subgroup with adequate overlap. The augmented model approach of Section 11.8 provides a complementary strategy by folding the propensity score into the outcome-model fit, which can improve finite-sample stability; it does not, however, eliminate the fundamental information loss that occurs when covariate support is lacking in one of the treatment arms.

11.10.3 Looking Ahead: When Plug-In Fails

For the fitted AIPW estimator, the root-n expansion and the plug-in influence-function variance estimator rest on the product-rate condition Equation 11.25 and an unstated empirical-process condition on the nuisance estimators; the finite-sample oracle projection identity of Section 11.5 involves no nuisance estimation and does not depend on them. For *correctly specified* parametric nuisance models of fixed dimension (e.g., logistic regression for π and linear regression for μ_t), under the usual identifiability, overlap, and moment conditions, both requirements are routinely verified: the parametric rate $n^{-1/2}$ is more than enough for the product-rate bound, and standard Donsker arguments dispose of the empirical-process remainder. (A misspecified parametric model converges to the wrong limit, and the product-rate condition to the truth fails entirely — the situation of the remark on double robustness versus efficient inference.)

The situation changes qualitatively when the nuisances are estimated with flexible machine-learning methods — random forests, gradient boosting, neural networks, regularized regressions in high dimensions, and the like. Two difficulties arise. First, the individual nuisance convergence rate may be slower than $n^{-1/4}$, or hard to characterize explicitly, so the product-rate condition becomes uncomfortably close to binding. Second, the machine-learning function classes are typically not Donsker, so the empirical-process remainder need not vanish when the same data are used for both nuisance estimation and score evaluation — plug-in inference can fail even when the nuisance estimates are predictively accurate.

Chapter 12 addresses both difficulties by (a) formalizing *Neyman orthogonality* as the first-order insensitivity property invoked above and exhibiting the AIPW score as an orthogonal score, and (b) decoupling nuisance estimation from score evaluation via *cross-fitting*. Under these devices, the same-data empirical-process problem is resolved: the product-rate condition suffices for the nuisance remainder to be $o_p(n^{-1/2})$, without a Donsker requirement on the nuisance function classes, provided that nuisance consistency, bounded moments, overlap, and score regularity conditions also hold. Each nuisance estimator is then allowed to converge slightly faster than $n^{-1/4}$. The variance estimator Equation 11.27 carries over essentially unchanged, using out-of-fold influence values in place of in-sample ones. The resulting *double/debiased machine learning* (DML) estimator (Chernozhukov et al. 2018) is thus a direct extension of the AIPW development here and the efficiency argument of Section 11.7.

11.11 Comparison of Regression, IPW, and AIPW

All consistency statements in the table below are conditional on the causal identification assumptions of Chapter 6: consistency, conditional exchangeability, and positivity. Without these, model correctness identifies a regression functional but not the causal estimand τ.

Table 11.4: Comparison of regression, IPW, and AIPW estimators of the ATE.

Estimator	Uses μ_t	Uses π	Consistent under regularity if	Main weakness
Regression (prediction)	Yes	No	outcome model correct	Sensitive to OR misspecification
IPW (Horvitz–Thompson)	No	Yes	propensity model correct	Unstable under weak overlap
AIPW	Yes	Yes	either model correct	Requires estimating both nuisances and careful inference (cross-fitting or regularity conditions)

When overlap is weak, the IBC-based approaches of Section 11.8 may improve numerical stability and finite-sample behavior by folding the propensity score into the outcome-model fit rather than relying on a separate multiplicative correction. They do not, however, eliminate the underlying weak-overlap problem: where covariate support is lacking for one of the treatment arms, no nonparametric estimator under the observed-data model can recover the unsupported portion of the ATE without additional extrapolation assumptions. In such cases the practical recommendation is to change the target estimand — through trimming, overlap weighting, or restriction to a subgroup with adequate support — rather than to expect any algebraic refinement of AIPW to fully repair it.

11.12 Chapter Summary

1. The prediction estimator is biased when the outcome model is misspecified; the bias can be estimated with the propensity score and subtracted to yield the AIPW estimator (Section 11.2).

2. The AIPW estimator has two key properties. First, double robustness (Theorem 11.1): it is consistent whenever either the outcome model or the propensity score model is correctly specified. Second, within the class of augmented IPW estimators indexed by $b_t(x)$, the choice $b_t^*(x) = \mathbb{E}\{Y(t) \mid X\}$ minimizes variance when both nuisance components are correctly specified (Theorem 11.4–Theorem 11.5). This is a class-specific optimality result.

3. The same class-specific optimum can be derived from a Hilbert-space projection: the optimal estimator is the projection of the HT estimator onto $\Lambda^\perp$, and the Pythagorean identity governs the variance reduction (Theorem 11.6).

4. Under the nonparametric observed-data model and regularity conditions, the AIPW estimating function is the efficient influence function for the ATE, and its variance gives the semiparametric efficiency bound (Section 11.7). The class-specific optimum therefore coincides with the *global* semiparametric efficiency bound: no regular asymptotically linear estimator, inside or outside the augmented IPW class, can improve upon AIPW's asymptotic variance. The bias-correction, optimal-augmentation, and semiparametric-efficiency derivations all pick out the same object.

5. Double robustness can also be enforced within the outcome-regression fitting step, either by IPW-weighted regression or by an augmented regression model that includes the inverse propensity score as a clever covariate. The latter is closely related to the targeting idea used in TMLE (Laan and Rubin 2006) (Section 11.8).

6. The simulation study (Section 11.9) shows that when the propensity score model is correct, IPW, AIPW, and the augmented-model estimators are all consistent. When the outcome regression is misspecified but the propensity score is correct, the fixed-offset augmented-model estimator can outperform the plug-in AIPW estimator in MSE because the targeted regression fit reduces residual variance while preserving the IBC condition. The improved augmented-model estimator adds additional recalibration but can introduce finite-sample bias (S3) and offers no general guarantee when both nuisance models are misspecified (S4).

7. For asymptotic inference, the central issue is whether the effect of estimating nuisance functions is asymptotically negligible. Neyman orthogonality yields a second-order remainder, and the product-rate condition Equation 11.25 is sufficient for root-n inference. In modern machine-learning settings, cross-fitting is often used to make these conditions more plausible; Chapter 12 develops this approach.

Key Objects in This Chapter

Symbol	**Meaning**
τ	ATE $= \mathbb{E}\{Y(1) - Y(0)\}$
$\mu_t^*(x)$, $\pi^*(x)$	True outcome regression $\mathbb{E}(Y \mid T = t, X = x)$ and true propensity score $P(T = 1 \mid X = x)$
$\mu_t(x)$, $\pi(x)$	Generic *working* nuisance functions (convention of Section 11.3); fitted versions $\hat{\mu}_t$, $\hat{\pi}$
$b_t(x)$	Control function; optimal $b_t^*(x) = \mathbb{E}\{Y(t) \mid X = x\} = \mu_t^*(x)$ under identification
$\hat{\tau}_{\text{AIPW}}$	AIPW estimator Equation 11.6
Λ, $\Lambda^\perp$	Augmentation space and its orthogonal complement
$\varphi_{\text{eff}}(O)$	Efficient influence function Equation 11.19
$\hat{\pi}_i^{-1}$, $(1 - \hat{\pi}_i)^{-1}$	Clever covariates (treated and control arms); their normal equations enforce the IBC conditions
$\hat{\gamma}_1$	Targeting coefficient; $\hat{\gamma}_1 \xrightarrow{p} 0$ when outcome model correct
IBC	Internal bias calibration conditions Equation 11.20–Equation 11.21

11.13 Exercises

1. Bias of the prediction estimator.

(a) Verify the bias formula Equation 11.1 by writing $\tau_{\text{pred}}(m) - \tau$ as a difference of working-model errors and simplifying.
(b) Assume the working propensity score equals the true propensity score, and treat the working regressions m_t as fixed functions. Show, using iterated expectations, that $\mathbb{E}\{\widehat{\text{Bias}}(\hat{\tau}_{\text{pred}})\} = \tau_{\text{pred}}(m) - \tau$, i.e., the weighted-residual expression is unbiased for the *actual* prediction bias, which is generally nonzero. Under what condition on the working outcome regressions is this expectation zero?
(c) Construct a simple example (binary X, binary T, binary Y) in which both the estimated bias is nonzero and the outcome model is misspecified, but the AIPW estimator is still consistent.

2. The AIPW class and optimal control functions.

(a) Verify that $\hat{\tau}_b$ with $b_t = \mu_t$ equals the AIPW estimator Equation 11.6.
(b) Using Equation 11.11, show that $b_t^*(x) = \mathbb{E}\{Y(t) \mid x\}$ minimizes conditional variance by completing the square in b_t. Then use the reparameterization Equation 11.13 to characterize the full set of minimizing pairs (the remark on uniqueness of the minimizing estimator).
(c) Suppose $\pi(x) = 1/2$ for all x. Simplify Equation 11.11 and interpret the result.

3. Double robustness.

(a) Verify Case 1 of Theorem 11.1: with $\mu_t = \mu_t^*$ and arbitrary π, show $\mathbb{E}\{\phi\} = 0$.
(b) Verify Case 2: with $\pi = \pi^*$ and arbitrary μ_t, show $\mathbb{E}\{\phi\} = 0$.
(c) Provide a counterexample showing $\mathbb{E}\{\phi\} \neq 0$ when both models are misspecified.

4. Projection and the Pythagorean identity.

(a) Verify $\text{Cov}(\hat{\theta}_{\text{opt}}, \hat{b}^*) = 0$ from the definition of $\hat{b}^*$.

(b) Deduce Equation 11.15 from the decomposition $\hat{\theta}_0 = \hat{\theta}_{\text{opt}} + \hat{b}^*$.
(c) Explain why $\hat{\theta}_{\text{opt}}$ is still unbiased for θ even though $\hat{b}^* \in \Lambda$ is subtracted.

5. Semiparametric efficiency. Let $\varphi_{\text{IPW}}(O) = TY/\pi^*(X) - (1-T)Y/\{1-\pi^*(X)\} - \tau$ denote the influence function of the IPW estimator with known propensity score.

(a) Show that

$$\varphi_{\text{IPW}}(O) - \varphi_{\text{eff}}(O) = \{T - \pi^*(X)\}\, a^*(X), \qquad a^*(x) = \frac{\mu_1^*(x)}{\pi^*(x)} + \frac{\mu_0^*(x)}{1-\pi^*(x)},$$

and conclude $\mathbb{E}\{\varphi_{\text{eff}}(O)^2\} \le \mathbb{E}\{\varphi_{\text{IPW}}(O)^2\}$ by verifying that the cross term $\mathbb{E}(\varphi_{\text{eff}}\,\Delta)$ with $\Delta = \varphi_{\text{IPW}} - \varphi_{\text{eff}}$ vanishes. (*Hint*: condition on X and use $T^2 = T$.) Observe that the difference has exactly the form of the augmentation elements in Equation 11.18.
(b) Conclude from part (a) that the efficiency gain is

$$\mathbb{E}\{\varphi^2_{\text{IPW}}\} - \mathbb{E}\{\varphi^2_{\text{eff}}\} = \mathbb{E}[\pi^*(X)\{1-\pi^*(X)\}\, a^*(X)^2],$$

which depends on the *levels* of the regression functions μ_t^*, not only on the conditional variances of the potential outcomes. In particular, show that adding the same constant c to both potential outcomes leaves τ, φ_{eff}, and every conditional variance unchanged, yet the gain depends on c. What does this suggest about centering outcomes before IPW estimation? (Relate your answer to the augmented class Equation 11.9–Equation 11.10 with constant b_t.)
(c) Give one reason why an efficient estimator based on φ_{eff} may still be disfavored in practice relative to a simpler, less efficient alternative.

6. Augmented model and the clever covariate.

(a) Let $\hat{\mu}_1^{(0)}(X_i)$ be any initial outcome model fit. Show that the normal equation for $\hat{\gamma}_1$ in the augmented model Equation 11.23 is exactly the IBC condition Equation 11.20, and conclude that the prediction estimator can always be written in AIPW form.
(b) Explain why $\hat{\gamma}_1 \xrightarrow{p} 0$ when the initial model $\hat{\mu}_1^{(0)}$ is correctly specified, stating the full-rank condition you need. Does coefficient convergence alone imply that including the clever covariate carries no first-order efficiency cost? Explain what additional property delivers that conclusion when both nuisance models are correct (the remark on behavior under a correct outcome model).
(c) Explain in words why $\hat{\pi}_i^{-1}$ is called the clever covariate: what bias does it absorb, and why does this make the prediction estimator a debiased estimator even when $\hat{\mu}_1^{(0)}$ is misspecified?

7. Coverage of the Wald confidence interval (computational). Use the data-generating process of Section 11.9 with the propensity score generalized to $\pi^*(x) = \operatorname{expit}\{0.2\,x + \gamma(x^2-1)\}$, so that the baseline ($\gamma = 0.2$) coincides with the lab and larger γ degrades overlap. Implement the AIPW estimator and the influence-function variance estimator Equation 11.27. Following Scenario S3 of the lab, use logistic regression of T on $(1, X, X^2)$ for $\hat{\pi}$ (correct PS) and OLS of Y on $(1, X)$ separately in each arm for $\hat{\mu}_t$ (misspecified OR); truncate $\hat{\pi}$ to $[10^{-3}, 1-10^{-3}]$.

(a) With $\gamma = 0.2$, $n = 500$, $B = 2000$ replications, compute the empirical coverage of the 95% Wald interval. Compare the average standard error with the empirical standard deviation of $\hat{\tau}_{\text{AIPW}}$ across replications.
(b) Repeat part (a) with $\gamma \in \{0.35, 0.45\}$, which produces increasingly weak overlap at large $|X|$. Report coverage, average SE, and empirical SD. What do you observe? (The grid stops below 1/2 deliberately: for $X \sim N(0,1)$, $\mathbb{E}\{1/(1-\pi^*(X))\} = 1 + e^{-\gamma}\,\mathbb{E}\{e^{\gamma X^2 + 0.2X}\}$ is finite only for $\gamma < 1/2$, so at $\gamma \ge 1/2$ the efficient influence function has infinite variance and the root-n Wald theory under study no longer applies. Verify the threshold with the Gaussian moment-generating function.)
(c) Which assumption of the asymptotic theory (Section 11.10) is stressed as γ grows, and which of the remedies in the *Extreme Propensity Scores* warning box would you try first? (No simulation required; a paragraph suffices.)

8. The exact double-robustness remainder. Let $R(\pi, \mu_0, \mu_1)$ denote the right-hand side of Equation 11.7.

(a) Evaluate R at the probability limits of the four lab scenarios: argue that $R = 0$ in S1–S3 while $R \ne 0$ in general in S4, and reconcile this with the bias pattern in the results table.

(b) Prove the bound Equation 11.8 from Equation 11.7 under $\varepsilon \leq \hat{\pi} \leq 1-\varepsilon$, and explain why the *product* structure — rather than a sum of the two nuisance errors — is what makes the rate condition Equation 11.25 attainable with flexible machine-learning nuisances.

(c) Show directly from Equation 11.7 that the Gateaux derivatives of the population moment with respect to each nuisance vanish at (π^*, μ^*), and relate this to Neyman orthogonality.

9. Cross-fitting does not restore orthogonality. Consider the setting of the remark on double robustness versus efficient inference: the propensity model is correctly specified and fitted by maximum likelihood at the parametric rate, so $\hat{\pi} \to \pi^*$, while the outcome model is misspecified, so $\hat{\mu}_t \to \mu_t^\dagger \neq \mu_t^*$. Suppose the nuisances are fitted on one half of the sample and the AIPW score is averaged over the other half (a single sample split).

(a) Expand $\sqrt{n}(\hat{\tau}_{\text{AIPW}} - \tau)$ around $(\pi^*, \mu^\dagger)$ and identify the term $\sqrt{n}\, D_\pi[\hat{\pi} - \pi^*]$, where D_π is the Gateaux derivative displayed in the remark on double robustness versus efficient inference.

(b) Explain why independence of the two half-samples removes the empirical-process remainder but leaves $\sqrt{n}\, D_\pi[\hat{\pi} - \pi^*] = O_p(1)$ intact, so the plug-in variance Equation 11.27 remains invalid.

(c) Which of the remedies in the remark on double robustness versus efficient inference addresses this term, and what does it require in practice?

Chapter 12

Cross-Fitting and Double Machine Learning

Learning Objectives

By the end of this chapter, students should be able to:

1. Explain why flexible nuisance estimation can produce misleading inference for a causal parameter even when predictive accuracy is high.
2. Define Neyman orthogonality, verify it for a given estimating function, and explain how it reduces sensitivity to first-order nuisance estimation error.
3. Identify the orthogonal score for the ATE, recognize it as the efficient influence function from Chapter 11, and state the three roles it simultaneously fulfills.
4. Describe sample splitting, explain why independence between nuisance estimation and score evaluation simplifies asymptotic arguments, and articulate its efficiency cost.
5. Describe K-fold cross-fitting, write the cross-fitted moment equation explicitly, and explain how cross-fitting avoids the first-order information loss of a single evaluation split, permitting efficiency under the conditions of the main theorem.
6. State the product-rate condition for the cross-fitted AIPW estimator, connect it to Neyman orthogonality, and explain why a sufficient symmetric condition is that both nuisance blocks converge slightly faster than $n^{-1/4}$, rather than requiring each nuisance estimator to achieve the parametric $O_p(n^{-1/2})$ rate.
7. Construct a consistent variance estimator from out-of-fold estimated influence values and form a Wald confidence interval.
8. Articulate what machine learning can and cannot contribute to causal inference, and follow the practical workflow for cross-fitted orthogonal-score estimation.

12.1 Why Flexible Nuisance Estimation Is Both Attractive and Dangerous

In Chapters 10 and 11 we developed estimation and inference using estimating equations, influence functions, doubly robust scores, and semiparametric efficiency. A central feature of these methods is that the causal parameter of interest depends on nuisance functions such as the outcome regressions

$$\mu_t(x) = \mathbb{E}(Y \mid T = t, X = x), \qquad t \in \{0, 1\},$$

and the propensity score

$$\pi(x) = P(T = 1 \mid X = x).$$

To reduce notation, this chapter drops the stars used for the true nuisance functions in Chapter 11: here μ_t and π denote the truth, while hats denote estimated functions.

In many real applications, these nuisance functions may be too complex to model accurately using low-dimensional parametric forms. This makes flexible methods attractive, including penalized regression, splines and sieves, random forests, boosting, neural networks, and ensemble learning.

These methods can substantially improve predictive accuracy, but they also create new statistical difficulties: they may converge more slowly than parametric estimators; they may overfit the observed sample; their asymptotic behavior may be difficult to characterize; and naive plug-in inference can fail even when prediction quality is good.

Better nuisance prediction therefore does not automatically imply valid inference for the target causal parameter. The goal of this chapter is to explain how orthogonal scores, sample splitting, and cross-fitting make it possible to combine flexible nuisance estimation with valid large-sample inference. The solution developed here combines an orthogonal score, which removes first-order sensitivity to nuisance estimation error, with cross-fitting, which separates nuisance estimation from score evaluation to control the remaining empirical process remainder.

The entire chapter is organized by a single decomposition. Writing $\varphi_\eta(O) = \varphi_{\text{eff}}(O;\, \tau_0, \eta)$ for the ATE score at nuisance η (the score φ_{eff} is introduced formally in Section 12.4), $\mathbb{P}_n$ for the empirical average, and P for the population expectation (notation reviewed in Section 12.5.1), the error of the estimator splits into three terms:

$$\hat{\tau} - \tau_0 = \underbrace{(\mathbb{P}_n - P)\,\varphi_{\eta_0}}_{\text{ordinary CLT}} + \underbrace{P\,(\varphi_{\hat{\eta}} - \varphi_{\eta_0})}_{\substack{\text{orthogonality}\\ +\text{ product rate}}} + \underbrace{(\mathbb{P}_n - P)(\varphi_{\hat{\eta}} - \varphi_{\eta_0})}_{\substack{\text{Donsker control}\\ \text{or cross-fitting}}}. \tag{12.1}$$

The first term is a sample average of a fixed function and obeys the central limit theorem. The second is the *population drift*, tamed by Neyman orthogonality together with the product-rate condition. The third is the *empirical-process remainder*, tamed either by a Donsker condition or — the route taken by DML — by cross-fitting. The figure below maps each term to the sections that control it; the rest of the chapter fills in this map.

12.2 Why Naive Plug-In Estimation Can Fail

Suppose the parameter of interest is written as a functional $\psi = \Psi(\eta)$, where η denotes a nuisance object such as (μ_0, μ_1, π). A natural estimator is the *plug-in estimator* $\hat{\psi}_{\text{plug}} = \Psi(\hat{\eta})$, obtained by replacing the nuisance functions with estimators.

12.2.1 The First-Order Taylor Expansion

To see precisely how nuisance error propagates, assume that Ψ is twice Fréchet differentiable in a neighborhood of η_0, with locally bounded second derivative in the norm $\|\cdot\|$, and apply a first-order Taylor expansion of Ψ around the true nuisance η_0:

$$\hat{\psi}_{\text{plug}} - \psi = \underbrace{D_\eta \Psi(\eta_0)[\hat{\eta} - \eta_0]}_{\text{linear term}} + \underbrace{R(\hat{\eta},\, \eta_0)}_{\text{second-order remainder}}, \tag{12.2}$$

where $D_\eta \Psi(\eta_0)[h]$ denotes the derivative of Ψ at η_0 in direction h; under the smoothness assumption above, the remainder satisfies $|R(\hat{\eta}, \eta_0)| \lesssim \|\hat{\eta} - \eta_0\|^2$. (Mere existence of a Gateaux derivative would not deliver a quadratic remainder.) The expansion is schematic: a causal plug-in estimator typically depends on the empirical distribution of X as well as on estimated conditional nuisance functions, but the schematic form suffices to display the roles of the two terms.

For root-n inference we need $\sqrt{n}(\hat{\psi}_{\text{plug}} - \psi)$ to have a centered limiting distribution. The second-order remainder is typically manageable: if $\|\hat{\eta} - \eta_0\| = o_p(n^{-1/4})$ then $R = o_p(n^{-1/2})$, which is negligible at the root-n scale. The obstacle is the *linear term*, which is proportional to the nuisance estimation error $\hat{\eta} - \eta_0$.

12.2.2 Finite-Dimensional Nuisance: Orthogonality Is Sufficient

Suppose first that $\eta \in \mathbb{R}^d$ for a fixed finite d, and that $\hat{\eta}$ is a standard parametric estimator satisfying $\hat{\eta} - \eta_0 = O_p(n^{-1/2})$. Then the linear term in Equation 12.2 is also $O_p(n^{-1/2})$, the same order as the leading term in the asymptotic expansion of $\hat{\psi}_{\text{plug}} - \psi$, so nuisance estimation contributes at first order to the limiting distribution of $\sqrt{n}(\hat{\psi}_{\text{plug}} - \psi)$. This does not by itself invalidate inference: in a finite-dimensional regular model, the contribution can be incorporated through the joint influence function of the pair $(\hat{\psi}_{\text{plug}}, \hat{\eta})$, the delta method, or a stacked sandwich variance (Chapter 10). The nuisance-estimation term is part of the estimator's influence function, not an extraneous contamination.

If the functional Ψ satisfies $D_\eta \Psi(\eta_0) = 0$ — the *Neyman orthogonality* condition introduced formally in Section 12.3 — then the linear term vanishes identically. The expansion Equation 12.2 reduces to the second-order remainder alone, which under parametric rates is $O_p(n^{-1})$ and hence asymptotically negligible. In the finite-dimensional setting, Neyman orthogonality is therefore **sufficient rather than necessary** for valid inference: it makes the nuisance-estimation contribution asymptotically negligible, so no joint variance correction is required.

One distinction is worth flagging. For an estimator that also contains an oracle empirical-average term — such as the AIPW estimator studied from Section 12.4 onward — orthogonality removes the first-order nuisance contribution while leaving the oracle sampling term unchanged, so the estimator behaves to first order as though the nuisance were known. For a *pure* direct plug-in $\Psi(\hat{\eta})$ with no such term, a zero first derivative can instead produce a degenerate root-n limit or a nonstandard second-order limit; that case requires a separate analysis.

12.2.3 Infinite-Dimensional Nuisance: Orthogonality Is Not Enough

When η belongs to a function space — as is the case for the nuisance triple (μ_0, μ_1, π) — orthogonality is no longer sufficient on its own. Orthogonality removes the first derivative of the target with respect to nuisance perturbations, but it does not by itself control the stochastic fluctuation of the score when the nuisance estimate is a random function chosen from a large class. A second source of first-order error arises from evaluating the estimating equation on the same sample used to estimate $\hat{\eta}$. Write the plug-in estimating equation as $\mathbb{P}_n\{\phi(\,\cdot\,;\psi,\hat{\eta})\} = 0$, and decompose the deviation from the population equation as

$$\begin{aligned}
&\mathbb{P}_n\{\phi(\,\cdot\,;\psi,\hat{\eta})\} - \mathbb{P}_n\{\phi(\,\cdot\,;\psi,\eta_0)\} \\
&\quad = \underbrace{P\{\phi(\,\cdot\,;\psi,\hat{\eta}) - \phi(\,\cdot\,;\psi,\eta_0)\}}_{\text{population drift}} + \underbrace{(\mathbb{P}_n - P)\{\phi(\,\cdot\,;\psi,\hat{\eta}) - \phi(\,\cdot\,;\psi,\eta_0)\}}_{\text{empirical process term}}.
\end{aligned} \tag{12.3}$$

Neyman orthogonality controls the *population drift*: because the Gateaux derivative of $P\phi$ with respect to η vanishes at the truth, the population drift is second order in $\|\hat{\eta} - \eta_0\|$ under the smoothness conditions of Section 12.2.1.

Neyman orthogonality, however, says nothing about the *empirical process term*, whose behavior depends on the complexity of the function class $\{\phi(\,\cdot\,;\psi,\eta) : \eta \in \mathcal{H}\}$ traced out as $\hat{\eta}$ varies. When the class is small — as for parametric estimators — classical empirical-process arguments control this term at order $o_p(n^{-1/2})$. For flexible machine-learning estimators the class is typically too rich for such arguments, and the empirical process term can remain non-negligible even as $\hat{\eta}$ converges consistently. Section 12.5 makes this term precise for the ATE score and shows why *sample splitting* or *cross-fitting* (Section 12.6–Section 12.7) are needed alongside orthogonality for valid root-n inference.

> Remark: Not a Machine-Learning-Specific Problem
>
> This problem is not specific to machine learning. It arises whenever nuisance parameters are estimated in an infinite-dimensional model. Machine learning makes the issue more visible because it encourages highly adaptive nuisance estimation; see Chernozhukov et al. (2018) for a systematic treatment.

12.3 Orthogonal Scores

The key device for reducing sensitivity to nuisance estimation error is an *orthogonal score*. Let $\mathbb{E}\{\phi(O;\,\psi,\eta)\} = 0$ be an estimating equation for the target parameter ψ, where η denotes nuisance parameters.

Remark: From Functionals to Moment Equations

Section 12.2 formulated orthogonality at the level of a functional Ψ as $D_\eta\Psi(\eta_0) = 0$. Most practical estimators are not defined as direct plug-ins of a closed-form functional but as solutions to a moment equation $\mathbb{P}_n\{\phi(\,\cdot\,;\widehat{\psi},\widehat{\eta})\} = 0$, so it is more useful to phrase orthogonality in terms of the score ϕ. Assume the population moment $P\phi(\,\cdot\,;\psi,\eta)$ is differentiable in (ψ,η) in a neighborhood of (ψ_0,η_0) and that the derivative $\partial_\psi P\phi(\,\cdot\,;\psi_0,\eta_0)$ is nonsingular. Then the implicit-function theorem applies and the parameter $\psi(\eta) = \operatorname{argzero}_\psi\, P\{\phi(\,\cdot\,;\psi,\eta)\}$ defined by the moment equation satisfies

$$D_\eta\psi(\eta_0)[h] = -\left[\partial_\psi P\phi(\,\cdot\,;\psi_0,\eta_0)\right]^{-1}\partial_r\, P\{\phi(\,\cdot\,;\psi_0,\eta_0+rh)\}\big|_{r=0},$$

so vanishing of the nuisance Gateaux derivative of $P\phi$ is equivalent to vanishing of $D_\eta\psi$. The two formulations are therefore the same orthogonality condition expressed in two languages, and we use whichever is more convenient for the argument at hand.

Definition 12.1 (Neyman Orthogonality)**.** The score function $\phi(O;\,\psi,\eta)$ is called *orthogonal* (or *Neyman orthogonal*) at the truth (ψ_0,η_0) if the Gateaux derivative of the estimating equation with respect to the nuisance, evaluated at the truth, vanishes:

$$\frac{\partial}{\partial r}\mathbb{E}\{\phi(O;\,\psi_0,\,\eta_0+rh)\}\bigg|_{r=0} = 0$$

for all perturbation directions h in a suitable class.

This condition means that small local perturbations of the nuisance parameter have no first-order effect on the estimating equation at the truth. Under suitable second-order smoothness — or, for the ATE score, by the exact drift calculation in the proof of Theorem 12.1 — the remaining population drift is second order in the nuisance error; the zero derivative alone gives local first-order insensitivity, not automatically a quadratic bound. Orthogonality does not mean nuisance estimation no longer matters: it controls only the population-level drift, and a separate argument is needed to rule out the additional bias that can arise when the same sample is used for both nuisance estimation and score evaluation. That argument is the subject of Section 12.5–Section 12.7.

Orthogonality is the principal reason that semiparametric estimators can remain valid under flexible nuisance estimation. In particular, the doubly robust score introduced in Chapter 11 is orthogonal, which is exactly why the product-rate condition Equation 12.13 below (identical to the condition stated in Chapter 11) is sufficient for asymptotic negligibility of nuisance estimation error.

Remark: Neyman's Original Insight

The terminology honors Jerzy Neyman, whose work on hypothesis testing introduced the idea of constructing test statistics that are insensitive to nuisance parameters. In the modern semiparametric literature the concept was formalized by Robins et al. (1994) and others, and later made systematic in the double machine learning framework of Chernozhukov et al. (2018).

12.4 The Orthogonal Score for the Average Treatment Effect

We continue to use the average treatment effect $\tau = \mathbb{E}\{Y(1) - Y(0)\}$ as the running example. Under consistency, conditional exchangeability, and positivity, the efficient influence function derived in Chapter 11 is

$$\varphi_{\text{eff}}(O;\,\tau,\eta) = \frac{T}{\pi(X)}\{Y - \mu_1(X)\} - \frac{1-T}{1-\pi(X)}\{Y - \mu_0(X)\} + \mu_1(X) - \mu_0(X) - \tau, \tag{12.4}$$

where $\eta = (\mu_0,\mu_1,\pi)$. Chapter 11 identified this as the right score for the ATE; the role of Chapter 12 is to explain how to estimate its nuisance components safely when flexible learners are used. DML does not introduce a new causal estimand or a new ATE score; it is the cross-fitted implementation of the AIPW score, together with rate and regularity conditions that permit flexible nuisance learning. (The cross-fitted estimator is, of course, a different finite-sample object from the same-sample AIPW estimator; what is unchanged is the estimand and the orthogonal score.) We now prove that this score is Neyman orthogonal. The proof verifies Definition 12.1 separately for perturbations in each of the three nuisance components.

Lemma 12.1 (Neyman Orthogonality of the ATE Score)**.** *The score $\varphi_{\text{eff}}(O;\tau,\eta)$ in Equation 12.4 is Neyman orthogonal at every truth $(\tau_0,\eta_0)=(\tau_0,\mu_0,\mu_1,\pi)$ whose nuisance components are the observed-data functions defined below, provided strong overlap holds ($c \le \pi(X) \le 1-c$ for some $c>0$) and $\mathbb{E}|Y|<\infty$. That is, for every bounded perturbation direction $h(x)$, $g(x)$, or $\ell(x)$,*

$$\frac{\partial}{\partial r}\mathbb{E}\{\varphi_{\text{eff}}(O;\,\tau_0,\,\mu_0,\,\mu_1,\,\pi+rh)\}\Big|_{r=0}=0,$$
$$\frac{\partial}{\partial r}\mathbb{E}\{\varphi_{\text{eff}}(O;\,\tau_0,\,\mu_0,\,\mu_1+rg,\,\pi)\}\Big|_{r=0}=0,$$
$$\frac{\partial}{\partial r}\mathbb{E}\{\varphi_{\text{eff}}(O;\,\tau_0,\,\mu_0+r\ell,\,\mu_1,\,\pi)\}\Big|_{r=0}=0.$$

Proof. Throughout, the perturbation paths are assumed to remain inside the nuisance parameter space — in particular, $\pi+rh$ stays in $(0,1)$ for all sufficiently small r, which holds for bounded h under the strong-overlap hypothesis — and differentiation and expectation are interchanged under standard domination arguments, which use the bounds on $1/\pi$ and $1/(1-\pi)$ supplied by strong overlap together with $\mathbb{E}|Y|<\infty$; the law of iterated expectations is applied by conditioning on X. By definition of the observed-data nuisance functions,

$$\mathbb{E}(Y\mid T=1,X)=\mu_1(X),\qquad \mathbb{E}(Y\mid T=0,X)=\mu_0(X),\qquad \mathbb{E}(T\mid X)=\pi(X).$$

These equalities hold by definition and require no causal assumptions. The identification assumptions — consistency, conditional exchangeability, and positivity — serve a different purpose: they identify $\tau_0=\mathbb{E}\{Y(1)-Y(0)\}=\mathbb{E}\{\mu_1(X)-\mu_0(X)\}$ as a causal quantity. They play no role in the algebraic verification of Neyman orthogonality once the observed-data score has been specified; the calculation below is an observed-data probability calculation.

The calculations below rest on two complementary families of conditional-mean identities. The defining identity $\mathbb{E}(T\mid X)=\pi(X)$ implies the two balancing identities

$$\mathbb{E}\left(\frac{T}{\pi(X)}\,\middle|\,X\right)=1,\qquad \mathbb{E}\left(\frac{1-T}{1-\pi(X)}\,\middle|\,X\right)=1,\tag{12.5}$$

which eliminate the perturbations in μ_1 and μ_0. The defining identities $\mathbb{E}(Y\mid T=t,X)=\mu_t(X)$ imply the two residual identities $\mathbb{E}\{T(Y-\mu_1(X))\mid X\}=0$ and $\mathbb{E}\{(1-T)(Y-\mu_0(X))\mid X\}=0$ (for the first, condition on X and use $\mathbb{E}\{TY\mid X\}=\pi(X)\,\mathbb{E}(Y\mid T{=}1,X)=\pi(X)\mu_1(X)$), which eliminate the perturbation in π.

Perturbation in π. Replace π by $\pi+rh$ and differentiate. The only terms in Equation 12.4 that depend on π are the two IPW residual terms, so

$$\frac{\partial}{\partial r}\mathbb{E}\{\varphi_{\text{eff}}(O;\,\tau_0,\mu_0,\mu_1,\pi+rh)\}\Big|_{r=0}=\mathbb{E}\left[-\frac{T\,h(X)}{\pi(X)^2}\{Y-\mu_1(X)\}-\frac{(1-T)\,h(X)}{(1-\pi(X))^2}\{Y-\mu_0(X)\}\right]=0$$

by the two residual identities above (factor out $h(X)/\pi(X)^2$ and $h(X)/(1-\pi(X))^2$ respectively, then condition on X).

Perturbation in μ_1. Replace μ_1 by μ_1+rg. Only $-(T/\pi(X))\mu_1(X)$ and $\mu_1(X)$ in Equation 12.4 depend on μ_1, so

$$\frac{\partial}{\partial r}\mathbb{E}\{\varphi_{\text{eff}}(O;\,\tau_0,\mu_0,\mu_1+rg,\pi)\}\Big|_{r=0}=\mathbb{E}\left[g(X)\left(1-\frac{T}{\pi(X)}\right)\right]=0$$

by the first identity in Equation 12.5.

Perturbation in μ_0. Replace μ_0 by $\mu_0+r\ell$. Symmetrically,

$$\frac{\partial}{\partial r}\mathbb{E}\{\varphi_{\text{eff}}(O;\,\tau_0,\mu_0+r\ell,\mu_1,\pi)\}\Big|_{r=0}=\mathbb{E}\left[\ell(X)\left(\frac{1-T}{1-\pi(X)}-1\right)\right]=0$$

by the second identity in Equation 12.5. This completes the proof. □

Remark: What the Proof Reveals

The two types of nuisance direction are eliminated by two complementary families of conditional-mean identities. For perturbations of μ_1 and μ_0, the derivative vanishes by the balancing identities Equation 12.5, which follow from the defining identity $\mathbb{E}(T \mid X) = \pi(X)$ — the same identity behind the balancing theorem and the IPW identification formula of Chapter 6. For perturbations of π, the derivative vanishes by the residual identities $\mathbb{E}\{Y - \mu_t(X) \mid T = t, X\} = 0$, which follow from the defining property of the outcome regressions. Neyman orthogonality of the ATE score is therefore a joint consequence of how the propensity score *and* the outcome regressions are defined, rather than an additional requirement imposed on either. This dual structure is also what produces the exact product form of the population drift in the proof of Theorem 12.1.

The score in Equation 12.4 simultaneously fulfills three purposes:

1. its zero-moment equation $\mathbb{E}\{\varphi_{\text{eff}}(O;\, \tau, \eta)\} = 0$ represents, under the causal identification assumptions of Chapter 5, the identified observed-data functional for the ATE (the score does not create causal identification by itself);
2. it is doubly robust, yielding a consistent estimator whenever either the outcome model or the propensity score model is correctly specified; and
3. it is the efficient influence function, so an estimator that is regular, asymptotically linear with this influence function, and whose nuisance remainders are asymptotically negligible achieves the semiparametric efficiency bound (Theorem 11.8).

This is why the score in Equation 12.4 is central in modern causal inference.

Remark: Double Robustness versus Semiparametric Efficiency

Properties (ii) and (iii) above are distinct claims and require distinct conditions, which it is worth stating separately.
Double robustness (ii) is a consistency claim: under standard overlap and law-of-large-numbers regularity, the AIPW estimator is consistent for τ if either $\hat{\mu}_t$ converges to the true μ_t *or* $\hat{\pi}$ converges to the true π, but not necessarily both. *Semiparametric efficiency* (iii) is a distributional claim about $\sqrt{n}(\hat{\tau} - \tau_0)$: it requires that the nuisance estimators converge to the truth at rates satisfying the product-rate condition Equation 12.13 introduced below. When only one nuisance block is correctly specified the estimator can remain consistent by double robustness, but the asymptotic-linearity expansion with influence function φ_{eff} and the efficient-bound variance need not apply, and inference based on them can be misleading without further analysis of the actual limiting estimating function.

12.5 Why Reusing the Same Data Can Be Problematic

Section 12.2 identified two obstacles to valid root-n inference: the population drift and the empirical process term in decomposition Equation 12.3. We showed there that Neyman orthogonality reduces the population drift to a second-order quantity in $\|\hat{\eta} - \eta_0\|$ but leaves the empirical process term uncontrolled. This section makes the empirical process term precise in the context of the ATE estimating equation and introduces the Donsker condition — the classical tool for controlling it when the same data are reused for both nuisance estimation and score evaluation.

12.5.1 The Plug-In Remainder for the AIPW Estimator

Recall from Chapter 10 that we write $\mathbb{P}_n f = n^{-1} \sum_{i=1}^n f(O_i)$ for the empirical average of a function f, and define the *centered empirical process*

$$\mathbb{G}_n f = \sqrt{n}\,(\mathbb{P}_n - P)f = \frac{1}{\sqrt{n}} \sum_{i=1}^n \{f(O_i) - \mathbb{E}f(O)\}.$$

Consider the plug-in AIPW estimator $\hat{\tau}_{\text{plug}}$ defined by the moment equation $\mathbb{P}_n\{\varphi_{\text{eff}}(O;\, \hat{\tau}_{\text{plug}}, \hat{\eta})\} = 0$, where the same sample is used for both nuisance estimation and score evaluation. Because φ_{eff} is linear in its τ-argument with $\partial_\tau \varphi_{\text{eff}} = -1$, we may rearrange and add and subtract $\varphi_{\text{eff}}(O;\,\tau_0,\eta_0)$ to obtain

$$\hat{\tau}_{\text{plug}} - \tau_0 = \mathbb{P}_n\{\varphi_{\text{eff}}(O;\,\tau_0,\eta_0)\} + \underbrace{\mathbb{P}_n\{\varphi_{\text{eff}}(O;\,\tau_0,\hat{\eta}) - \varphi_{\text{eff}}(O;\,\tau_0,\eta_0)\}}_{=:\,R_n}. \tag{12.6}$$

The first term is a sample average of a fixed function, satisfies the CLT, and is $O_p(n^{-1/2})$. The remainder R_n is exactly the left-hand side of decomposition Equation 12.3 specialized to $\phi = \varphi_{\text{eff}}$ at (τ_0,η_0) — together with the leading CLT term, this is precisely the three-term roadmap Equation 12.1 of Section 12.1:

$$R_n = \underbrace{P\{\varphi_{\text{eff}}(\,\cdot\,;\tau_0,\hat{\eta}) - \varphi_{\text{eff}}(\,\cdot\,;\tau_0,\eta_0)\}}_{\text{population drift}} + \underbrace{(\mathbb{P}_n - P)\{\varphi_{\text{eff}}(\,\cdot\,;\tau_0,\hat{\eta}) - \varphi_{\text{eff}}(\,\cdot\,;\tau_0,\eta_0)\}}_{\text{empirical process term}}. \tag{12.7}$$

By Lemma 12.1, Neyman orthogonality eliminates the first-order term in the expansion of the population drift around η_0, leaving a remainder bilinear in $(\hat{\mu}_t - \mu_t,\, \hat{\pi} - \pi)$. Bounding this bilinear remainder by Cauchy–Schwarz under strong overlap and the product-rate condition Equation 12.13 then yields population drift $= o_p(n^{-1/2})$ (the explicit calculation appears as Step 1 in the proof of Theorem 12.1). Writing $f_\eta(\cdot) = \varphi_{\text{eff}}(\,\cdot\,;\tau_0,\eta)$ (the object written φ_η in the roadmap Equation 12.1), the empirical process term takes the form

$$(\mathbb{P}_n - P)\{f_{\hat{\eta}} - f_{\eta_0}\} = \frac{1}{\sqrt{n}}\,\mathbb{G}_n\{f_{\hat{\eta}} - f_{\eta_0}\}. \tag{12.8}$$

This term depends on $\hat{\eta}$, which is estimated from the same sample. Whether it is $o_p(n^{-1/2})$ depends on how complex the class $\{f_\eta : \eta \in \mathcal{H}\}$ is — a question the Donsker condition answers.

12.5.2 The Donsker Condition ⋆

This subsection provides empirical-process background and may be treated as second-pass material. For a first reading it suffices to know: (i) what same-sample stochastic equicontinuity is supposed to accomplish; (ii) that highly adaptive learners may fall outside classical fixed-complexity regimes; and (iii) that cross-fitting supplies a different route (Section 12.6–Section 12.7).

The Donsker condition is a technical way of asking whether the function class used by the nuisance learner is tame enough that same-sample score evaluation remains asymptotically well behaved.

Definition 12.2 (Donsker Class)**.** A class of measurable functions $\mathcal{F}$ is a *Donsker class* (with respect to P) if the empirical process $\{\mathbb{G}_n f : f \in \mathcal{F}\}$ converges weakly in $\ell^\infty(\mathcal{F})$ to a tight Gaussian process indexed by $\mathcal{F}$. In particular, every Donsker class satisfies the uniform bound

$$\sup_{f\in\mathcal{F}} \big|(\mathbb{P}_n - P)f\big| = O_p(n^{-1/2})$$

and the empirical process $\mathbb{G}_n = n^{1/2}(\mathbb{P}_n - P)$ is asymptotically equicontinuous over $\mathcal{F}$ with respect to an appropriate semimetric. Conversely, weak convergence in $\ell^\infty(\mathcal{F})$ follows from convergence of the finite-dimensional distributions — which holds by the multivariate CLT whenever every $f \in \mathcal{F}$ is square-integrable — together with asymptotic equicontinuity under a totally bounded semimetric and the relevant measurability conditions. The displayed uniform bound is a consequence of tightness; it does not by itself replace the finite-dimensional convergence requirement. See Vaart (1998), Chapter 19, for a complete treatment.

A key sufficient condition for $\mathcal{F}$ to be Donsker involves its *bracketing entropy.* The bracketing number $N_{[]}(\epsilon, \mathcal{F}, L_2(P))$ counts the minimum number of ϵ-brackets $[\ell_j, u_j]$ needed to cover $\mathcal{F}$, where a bracket is a pair of functions with $P(u_j - \ell_j)^2 \le \epsilon^2$; its logarithm

$$H_{[]}(\epsilon, \mathcal{F}, L_2(P)) = \log N_{[]}(\epsilon, \mathcal{F}, L_2(P))$$

is the bracketing entropy. Subject to measurability and a square-integrable envelope, the class $\mathcal{F}$ is Donsker whenever

$$\int_0^{\delta_0} \sqrt{H_{[]}(\epsilon, \mathcal{F}, L_2(P))}\, d\epsilon < \infty$$

for some $\delta_0 > 0$ (the integral is over the relevant range of bracket sizes; classes are typically bounded in $L_2(P)$ so the upper limit can be taken as the L_2-diameter of $\mathcal{F}$). Intuitively, the integral is finite when the class is not too large: a sufficient condition is that the bracketing entropy grow at most polynomially with exponent below two, $H_{[]}(\epsilon, \mathcal{F}, L_2(P)) \lesssim \epsilon^{-v}$ for some $v < 2$. Note carefully that it is the *entropy*, not the bracketing number itself, that may grow polynomially in $1/\epsilon$; the bracketing number then grows roughly exponentially in ϵ^{-v}.

Example 12.1 (Donsker and Non-Donsker Classes)**.**

- **Parametric classes** (e.g., logistic regression models indexed by a finite-dimensional parameter) are Donsker under mild moment conditions, since the bracketing entropy grows only logarithmically.
- **Hölder-smooth function classes** on $[0,1]^d$ with smoothness index $s > d/2$ are Donsker: a bounded Hölder ball has bracketing entropy $H_{[]}(\epsilon, \mathcal{F}, L_2(P)) \lesssim \epsilon^{-d/s}$ for any P (via the sup-norm entropy bound), so $s > d/2$ suffices for the entropy integral to converge. Under regular full-dimensional support conditions on P, a matching lower bound holds, making $s > d/2$ the sharp threshold for this class; for P concentrated on a lower-dimensional set, the $L_2(P)$ entropy can be much smaller.
- **Indicator classes** $\mathcal{F} = \{x \mapsto \mathbf{1}(x \leq t) : t \in \mathbb{R}\}$ are Donsker by the classical Donsker theorem for empirical distribution functions.
- **Random forests and neural networks.** Common modern implementations typically fall outside classical fixed Donsker regimes, especially when their complexity grows with n, because the function classes they implicitly explore can approximate arbitrarily complex functions. Restricted or fixed-complexity variants can nevertheless be Donsker.
- **High-dimensional lasso.** With a growing number of selected variables the effective model dimension also grows with n, so the class typically falls outside fixed Donsker regimes; such procedures are often analyzed using sparsity-based empirical-process arguments rather than Donsker theorems.

12.5.3 Connecting the Two Terms

Returning to the remainder Equation 12.7, the two terms correspond exactly to the two challenges identified in Section 12.2.3:

- **Population drift.** Its first-order nuisance derivative is eliminated by Neyman orthogonality. For the ATE score, the resulting exact bilinear remainder is $o_p(n^{-1/2})$ under strong overlap and the product-rate condition, regardless of how $\hat{\eta}$ is estimated.
- **Empirical process term.** Requires a separate argument. If $\{f_\eta : \eta \in \mathcal{H}\}$ is Donsker, $f_{\hat{\eta}}$ belongs to this class with probability approaching one, and the *induced score* converges in the sense $P\{f_{\hat{\eta}} - f_{\eta_0}\}^2 = o_p(1)$ — convergence of $\hat{\eta}$ in an unspecified norm does not by itself imply this — then by stochastic equicontinuity,
$$\mathbb{G}_n\{f_{\hat{\eta}} - f_{\eta_0}\} = o_p(1), \tag{12.9}$$
making the empirical process term $o_p(n^{-1/2})$. Combined with the controlled population drift, the full remainder $R_n = o_p(n^{-1/2})$, and root-n asymptotic linearity follows from Equation 12.6.

When the Donsker condition fails, Equation 12.9 need not hold: the empirical process term can remain non-negligible even as $\hat{\eta} \to \eta_0$. Root-n asymptotic linearity of the plug-in estimator may then fail unless some other empirical-process or stability argument is available — for example, algorithm-specific bounds for sparse high-dimensional learners, or the sample-splitting arguments of Section 12.6–Section 12.7. The Donsker condition is sufficient for same-sample empirical-process control, not necessary, but absent some such argument the limiting distribution can degrade. This is why orthogonality and cross-fitting are complements rather than substitutes: orthogonality controls the population drift and cross-fitting controls the empirical process term, and the two together yield valid root-n inference with flexible nuisance learners under the broadest set of conditions.

> Remark: Donsker Conditions and Machine Learning
>
> Common modern implementations of random forests, neural networks, and high-dimensional regularized learners often fall outside classical fixed Donsker regimes, especially when their complexity grows with n. This is a genuine obstruction to same-sample empirical-process arguments for plug-in estimators using such learners. Cross-fitting addresses it by changing the dependence structure: because $\hat{\eta}^{(-k)}$ is independent of the observations in $\mathcal{I}_k$, the empirical process term Equation 12.8 can be controlled by a conditional argument instead of a Donsker assumption, under L_2 consistency, overlap, and moment conditions.

The practical solution is therefore to separate nuisance estimation from score evaluation, which leads to sample splitting and, in its more efficient form, cross-fitting.

12.6 Sample Splitting

Section 12.5 showed that the plug-in remainder R_n in Equation 12.7 has two parts: the population drift, handled by Neyman orthogonality, and the empirical process term Equation 12.8, which is not controlled by orthogonality alone when the Donsker condition fails. Sample splitting resolves the second problem by construction.

Partition the indices $\{1, \dots, n\}$ into a training sample $\mathcal{I}_{\text{train}}$ and an evaluation sample $\mathcal{I}_{\text{eval}}$. Fit the nuisance estimators $\widehat{\eta}^{(\text{train})} = (\widehat{\mu}_0^{(\text{train})}, \widehat{\mu}_1^{(\text{train})}, \widehat{\pi}^{(\text{train})})$ on $\mathcal{I}_{\text{train}}$, then define $\widehat{\tau}$ by solving the estimating equation on the held-out sample:

$$\frac{1}{|\mathcal{I}_{\text{eval}}|} \sum_{i \in \mathcal{I}_{\text{eval}}} \varphi_{\text{eff}}(O_i;\, \tau,\, \widehat{\eta}^{(\text{train})}) = 0.$$

To see why this removes the same-sample empirical-process dependence, write the analog of the remainder decomposition Equation 12.7 on the evaluation sample. Let $n_{\text{e}} = |\mathcal{I}_{\text{eval}}|$ and $\mathbb{P}_{n_{\text{e}}}$ denote the empirical measure over $\mathcal{I}_{\text{eval}}$. The remainder on the evaluation fold is

$$\begin{aligned} R_n^{(\text{e})} &= \underbrace{P\{\varphi_{\text{eff}}(\,\cdot\,; \tau_0, \widehat{\eta}^{(\text{train})}) - \varphi_{\text{eff}}(\,\cdot\,; \tau_0, \eta_0)\}}_{\text{population drift: } o_p(n^{-1/2}) \text{ by orthogonality}} \\ &\quad + \underbrace{(\mathbb{P}_{n_{\text{e}}} - P)\{\varphi_{\text{eff}}(\,\cdot\,; \tau_0, \widehat{\eta}^{(\text{train})}) - \varphi_{\text{eff}}(\,\cdot\,; \tau_0, \eta_0)\}}_{\text{empirical process term}}. \end{aligned} \tag{12.10}$$

The population drift is $o_p(n^{-1/2})$ by the same orthogonality plus product-rate argument used in Section 12.5.1 and made explicit as Step 1 of the proof of Theorem 12.1. The empirical process term is now different in character: because $\widehat{\eta}^{(\text{train})}$ is estimated on $\mathcal{I}_{\text{train}}$, it is *independent* of the observations $\{O_i : i \in \mathcal{I}_{\text{eval}}\}$. Conditioning on $\widehat{\eta}^{(\text{train})}$, the summands in the empirical process term are independent and mean zero, so under L_2 consistency of $\widehat{\eta}^{(\text{train})}$, strong overlap, and finite-variance moment conditions, a conditional variance bound (made precise in the proof of Theorem 12.1) gives

$$(\mathbb{P}_{n_{\text{e}}} - P)\{\varphi_{\text{eff}}(\,\cdot\,; \tau_0, \widehat{\eta}^{(\text{train})}) - \varphi_{\text{eff}}(\,\cdot\,; \tau_0, \eta_0)\} = o_p(n^{-1/2}).$$

No Donsker condition on the function class is needed: independence allows a conditional argument in place of the stochastic equicontinuity that Donsker theory would otherwise supply. The full remainder $R_n^{(\text{e})} = o_p(n^{-1/2})$, and root-$n$ asymptotic linearity of the split estimator follows.

The main disadvantage of sample splitting is inefficiency: only $|\mathcal{I}_{\text{eval}}|$ observations contribute to the estimation of τ, and the estimate depends on the particular random partition chosen. In other words, sample splitting buys cleaner asymptotic theory by sacrificing training sample size for each nuisance learner, which can slow the convergence of $\widehat{\eta}$ and inflate the variance of the final estimator.

> Remark: A Single Split Is Noisy
>
> A single split can be noisy. In practice, sensitivity to the split is often assessed by repeating the procedure with several random partitions and aggregating results. Cross-fitting provides a more systematic solution by rotating the training and evaluation roles across all observations.

12.7 Cross-Fitting

Cross-fitting extends sample splitting to avoid the first-order information loss of a single evaluation split — permitting full-sample efficiency under the conditions of Theorem 12.1 — while preserving the independence argument that controls the empirical-process term. The key observation from Section 12.6 was that conditioning on $\widehat{\eta}^{(\text{train})}$ makes the evaluation-fold summands independent and mean zero, so the empirical-process term can be made $o_p(n^{-1/2})$ by a conditional variance argument under L_2 consistency, overlap, and moment conditions. Cross-fitting applies this argument simultaneously to every observation by rotating the training and evaluation roles across K folds.

Partition the indices $\{1, \dots, n\}$ into K approximately equal folds $\mathcal{I}_1, \dots, \mathcal{I}_K$. For each fold k:

1. Estimate the nuisance functions using all observations *not* in fold k, obtaining $\hat{\eta}^{(-k)} = (\hat{\mu}_0^{(-k)}, \hat{\mu}_1^{(-k)}, \hat{\pi}^{(-k)})$.
2. Evaluate the orthogonal score on the held-out fold $\mathcal{I}_k$ using $\hat{\eta}^{(-k)}$.
3. Aggregate the score contributions across all K folds.

The cross-fitted estimator $\hat{\tau}_{\mathrm{DML}}$ solves

$$\frac{1}{n} \sum_{k=1}^{K} \sum_{i \in \mathcal{I}_k} \varphi_{\mathrm{eff}}(O_i;\, \tau,\, \hat{\eta}^{(-k)}) = 0. \tag{12.11}$$

Throughout this section we write $n_k = |\mathcal{I}_k|$, so that $\sum_k n_k = n$ and the fold proportions n_k/n sum to one. When folds are exactly balanced, $n_k/n = 1/K$ and the weighted averages below collapse to the unweighted form $K^{-1} \sum_k$. The figure below depicts the procedure for one representative fold.

To see why the same-sample empirical-process dependence is no longer an obstacle, write the remainder Equation 12.7 as a sum over folds,

$$R_n = \sum_{k=1}^{K} \frac{n_k}{n} R_n^{(k)},$$

where $R_n^{(k)}$ is the fold-k remainder evaluated on $\mathcal{I}_k$, decomposed exactly as in Equation 12.10 with $\mathcal{I}_{\mathrm{eval}} = \mathcal{I}_k$ and $\hat{\eta}^{(\mathrm{train})} = \hat{\eta}^{(-k)}$. For each k, the nuisance estimate $\hat{\eta}^{(-k)}$ is trained on $\{1, \dots, n\} \setminus \mathcal{I}_k$, which is independent of the observations in $\mathcal{I}_k$.

Care is needed about what is conditionally mean zero. Writing $f_k = \varphi_{\mathrm{eff}}(\,\cdot\,; \tau_0, \hat{\eta}^{(-k)}) - \varphi_{\mathrm{eff}}(\,\cdot\,; \tau_0, \eta_0)$, the *centered* variables $f_k(O_i) - P f_k$, $i \in \mathcal{I}_k$, are conditionally independent and mean zero given the training sigma-field; the uncentered score differences have conditional mean $P f_k = B_n^{(k)}$, the population drift, which is controlled separately by orthogonality and the product-rate condition (Step 1 of the proof of Theorem 12.1 gives $\sqrt{n}\, B_n^{(k)} = o_p(1)$). For the centered part, a conditional variance argument under L_2 consistency of $\hat{\eta}^{(-k)}$, strong overlap, and finite-variance moment conditions (Step 2 of the proof) gives $\sqrt{n}\, E_n^{(k)} = o_p(1)$ without any Donsker assumption on the nuisance class. Together, $\sqrt{n}\, R_n^{(k)} = \sqrt{n}\,(B_n^{(k)} + E_n^{(k)}) = o_p(1)$ for each k. Since R_n is a convex combination of the $R_n^{(k)}$ with weights $n_k/n \le 1$, the fold-wise bound transfers directly: $\sqrt{n}\, R_n = o_p(1)$.

Remark: Cross-Fold Independence Is Not Required

The argument above is fold-by-fold: it bounds each $R_n^{(k)}$ separately, conditioning on the corresponding training set for the empirical-process part. The nuisance estimates $\hat{\eta}^{(-1)}, \dots, \hat{\eta}^{(-K)}$ are *not* mutually independent — any two of them share most of their training observations — but cross-fold independence is never invoked. What is required is only that each $\hat{\eta}^{(-k)}$ be independent of its own evaluation fold $\mathcal{I}_k$, which holds by construction. Because the $o_p(1)$ bound on $R_n^{(k)}$ is uniform across k for fixed K, averaging over folds preserves the bound.

Cross-fitting therefore achieves two goals simultaneously: it preserves the held-out independence structure that removes the need for Donsker conditions, and it ensures every observation serves once as an evaluation point, so the full sample contributes to the estimation of τ. For this reason, cross-fitting has become the standard device for combining orthogonal scores with flexible nuisance estimation (Zheng and Laan 2011; Chernozhukov et al. 2018; Laan and Rose 2011). Cross-fitting removes the same-sample empirical-process obstruction, but it does not by itself guarantee good finite-sample performance if the nuisance estimators are inaccurate or overlap is poor.

Algorithm: Cross-Fitted Orthogonal-Score Estimator for the ATE

1. Partition the sample into K folds $\mathcal{I}_1, \dots, \mathcal{I}_K$, generating the fold assignment independently of the observed covariates, treatments, and outcomes.
2. For each $k = 1, \dots, K$, fit nuisance estimators $\hat{\eta}^{(-k)} = (\hat{\mu}_0^{(-k)}, \hat{\mu}_1^{(-k)}, \hat{\pi}^{(-k)})$ on the complement $\{1, \dots, n\} \setminus \mathcal{I}_k$.
3. Repeat *all* data-dependent preprocessing, variable selection, hyperparameter tuning, and ensemble construction within each training complement. No information from $\mathcal{I}_k$ — including

its covariates, outcomes, or treatment indicators — may be used to construct $\hat{\eta}^{(-k)}$; a learner tuned or standardized on the full sample and then refit within folds is not independent of its evaluation fold.

4. For each observation $i \in \mathcal{I}_k$, compute the out-of-fold score contribution as a function of τ,

$$\hat{s}_i(\tau) = \varphi_{\text{eff}}(O_i;\, \tau,\, \hat{\eta}^{(-k)}).$$

5. Solve the cross-fitted moment equation Equation 12.11, $n^{-1}\sum_i \hat{s}_i(\tau) = 0$: because the score is linear in τ, the solution is

$$\hat{\tau}_{\text{DML}} = \hat{\mu}_{1,\text{DML}} - \hat{\mu}_{0,\text{DML}}, \tag{12.12}$$

where

$$\hat{\mu}_{1,\text{DML}} = \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{I}_k}\Big[\hat{\mu}_1^{(-k)}(X_i) + \frac{T_i}{\hat{\pi}^{(-k)}(X_i)}\{Y_i - \hat{\mu}_1^{(-k)}(X_i)\}\Big],$$
$$\hat{\mu}_{0,\text{DML}} = \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{I}_k}\Big[\hat{\mu}_0^{(-k)}(X_i) + \frac{1-T_i}{1-\hat{\pi}^{(-k)}(X_i)}\{Y_i - \hat{\mu}_0^{(-k)}(X_i)\}\Big].$$

12.8 Double Machine Learning for the Average Treatment Effect

The estimator in Equation 12.12 is simply the AIPW estimator from Chapter 11 with out-of-fold nuisance estimates substituted in place of estimates fit on the full sample. The term *double machine learning* (DML), introduced by Chernozhukov et al. (2018), refers to the combination of three ingredients:

- *Neyman orthogonality*, which reduces the population drift term in the remainder decomposition Equation 12.7 to a second-order product in nuisance estimation errors;
- *cross-fitting*, which removes the same-sample dependence in the empirical-process term of Equation 12.7 by separating training and evaluation folds, so the term can be controlled by a conditional variance argument in place of a Donsker assumption; and
- *flexible machine-learning estimation* of the nuisance functions μ_0, μ_1, and π, which is made valid by the first two ingredients.

The word "double" should not be read as merely meaning "two models"; it refers to the combination of nuisance learning with orthogonalization and debiasing. The crucial feature is the orthogonal moment structure combined with cross-fitting, which together make flexible nuisance estimation compatible with root-n inference. In this sense, DML is not a new causal estimand and not a new identification strategy; it is a modern estimation protocol built around an orthogonal score.

Remark: Cross-Fitted AIPW

The estimator in Equation 12.12 is often called the *cross-fitted AIPW estimator* in the biostatistics literature. The two names refer to the same object and may be used interchangeably when the context is clear.

Remark: DML1 versus DML2

The pooled moment equation Equation 12.11 corresponds to the aggregation scheme called *DML2* in Chernozhukov et al. (2018). An alternative, *DML1*, solves a target equation separately within each fold and averages the K fold-specific estimates. For the linear ATE score, DML1 and DML2 are *exactly equal* when all folds have the same size, as a finite-sample identity; with unequal fold sizes, DML2 weights the fold-specific estimates by n_k/n whereas DML1 gives each fold equal weight, and under standard conditions the difference is asymptotically negligible. The pooled version is generally preferred in practice.

"Double" Does Not Mean "Two Models Are Enough"

It is tempting to think that estimating any two of μ_0, μ_1, and π flexibly is sufficient. What matters is that all nuisance components entering the orthogonal score are estimated out-of-fold, and that the product-rate condition (Section 12.9) is satisfied. Using flexible methods for only one nuisance function while misspecifying another can still produce invalid inference.

12.9 Rate Conditions and Asymptotic Normality

The main theoretical advantage of orthogonality and cross-fitting is that valid root-n inference can hold even when nuisance estimators converge more slowly than $n^{-1/2}$. The following result makes this precise.

Theorem 12.1 (Asymptotic Linearity Under Cross-Fitting)**.** *Let $O_1, \dots, O_n$ be i.i.d. from P. Let the number of folds K be fixed, with fold sizes satisfying $\min_k n_k/n$ bounded away from zero, and assume the fold partition is either deterministic or generated by randomization independent of $O_1, \dots, O_n$. Suppose that for each fold k all nuisance fitting, feature selection, preprocessing, and tuning use only observations outside $\mathcal{I}_k$. Assume $\mathbb{E}|Y|^{2+\delta} < \infty$ for some $\delta > 0$ and $0 < \sigma^2 = \mathbb{E}\{\varphi_{\text{eff}}(O;\, \tau_0, \eta_0)^2\} < \infty$. (The causal identification assumptions of Chapter 5 are required if τ_0 is to be interpreted as the ATE; the distributional conclusion below concerns the observed-data functional.) Suppose further:*

(i) the score $\varphi_{\text{eff}}(O;\, \tau, \eta)$ is orthogonal at the truth (τ_0, η_0);

(ii) each component of $\widehat{\eta}^{(-k)}$ is $L_2(P)$-consistent for the corresponding component of η_0, that is, $\|\widehat{\eta}^{(-k)} - \eta_0\|_{L_2(P)} = o_p(1)$ for each fold k;

(iii) strong overlap and bounded fitted propensities *hold: there exists a constant $c > 0$ such that $c \le \pi(X) \le 1 - c$ almost surely (a property of the data-generating law), and, as an assumption on (or enforced constraint of) the nuisance estimator,*

$$P\big(c \;\le\; \widehat{\pi}^{(-k)}(X) \;\le\; 1 - c \text{ for almost every } X\big) \to 1$$

as $n \to \infty$, for each fold k;

(iv) the nuisance estimators satisfy the product-rate condition

$$\|\widehat{\pi}^{(-k)} - \pi\| \cdot \|\widehat{\mu}_t^{(-k)} - \mu_t\| = o_p(n^{-1/2}), \qquad t = 0, 1, \tag{12.13}$$

where $\|\cdot\|$ denotes the $L_2(P)$ norm.

Then the cross-fitted estimator satisfies

$$\sqrt{n}(\widehat{\tau}_{\text{DML}} - \tau_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \varphi_{\text{eff}}(O_i;\, \tau_0, \eta_0) + o_p(1),$$

and consequently

$$\sqrt{n}(\widehat{\tau}_{\text{DML}} - \tau_0) \xrightarrow{d} N\Big(0,\ \mathbb{E}[\varphi_{\text{eff}}(O;\, \tau_0, \eta_0)^2]\Big).$$

The asymptotic variance equals the semiparametric efficiency bound (Theorem 11.8). Under the standard regularity conditions ensuring that the estimator is regular over the maintained semiparametric model, the influence-function expansion above therefore implies attainment of the semiparametric efficiency bound; pointwise asymptotic linearity alone does not constitute efficiency in the formal sense.

Remark: Weak Overlap, Strong Overlap, and Bounded Fitted Propensities

Condition (iii) in the theorem combines two distinct requirements. *Strong overlap* is a property of the data-generating law: $c \le \pi(X) \le 1 - c$. *Bounded fitted propensities* are an additional property imposed on — or enforced for — the nuisance estimator: $c \le \widehat{\pi}^{(-k)}(X) \le 1 - c$ with probability approaching one. Under condition (iii), both the true and the estimated propensity scores are bounded uniformly away from 0 and 1 by a positive constant c. It is important to distinguish this from the weaker positivity condition $0 < \pi(X) < 1$ a.s., which serves a different role.

Weak overlap ($0 < \pi(X) < 1$ a.s.) is the condition needed for *identification*: it ensures that every covariate stratum contains units from both treatment arms, so that the counterfactual means $\mathbb{E}[Y(t) \mid X]$ are identified from observed data. Without weak overlap, certain strata have no counterfactual information and the ATE is not nonparametrically identified.
Strong overlap ($c \le \pi(X) \le 1 - c$ and $c \le \hat{\pi}^{(-k)}(X) \le 1 - c$ for some $c > 0$) is what the proof below actually uses: it bounds the inverse-probability weights $1/\hat{\pi}$ and $1/(1-\hat{\pi})$ away from infinity, ensuring that the Cauchy–Schwarz step in Step 1 and the variance bound in Step 2 go through. Without strong overlap, the influence function values can have infinite variance, and $\sqrt{n}$-consistent, asymptotically normal inference for the standard ATE can break down even if the ATE is identified (Khan and Tamer 2010); recovery is sometimes possible under stronger moment conditions or by changing the target estimand (for example through trimming or overlap weighting).
The practical consequence is that weak overlap is necessary but not sufficient for regular inference. When the propensity score approaches 0 or 1 in parts of the covariate space, the inverse weights blow up, influence function values become extreme, and the variance estimator from Theorem 12.2 can be severely unstable. Trimming extreme weights or restricting the target population to a region of stronger overlap are common practical responses. The requirement on $\hat{\pi}^{(-k)}$ is equally important: even if the true π is well-behaved, an estimated propensity score that strays near 0 or 1 in a given fold will cause the same instability through the finite-sample weights.
Note also that numerically clipping $\hat{\pi}$ to an interval such as $[0.01, 0.99]$, as in the lab of Section 12.10, bounds the fitted weights but neither demonstrates that the true treatment mechanism has adequate overlap nor formally redefines the target population; if clipping remains active asymptotically, it introduces its own approximation bias. Weight truncation and support-based retargeting (changing the estimand to a subpopulation with adequate overlap) are different operations and should be reported as such.

Proof sketch. In addition to the conditions stated in the theorem, we assume measurability of the nuisance estimators wherever it is needed; such conditions are standard in the double machine learning literature (Chernozhukov et al. 2018).

Write $n_k = |\mathcal{I}_k|$ and $\mathbb{P}_{n_k}$ for the empirical measure over fold k. The cross-fitted moment equation Equation 12.11 can be written as

$$0 = \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{I}_k} \varphi_{\text{eff}}(O_i;\, \hat{\tau}_{\text{DML}}, \hat{\eta}^{(-k)}) = \sum_{k=1}^{K}\frac{n_k}{n}\,\mathbb{P}_{n_k}\{\varphi_{\text{eff}}(O;\, \hat{\tau}_{\text{DML}}, \hat{\eta}^{(-k)})\}.$$

Adding and subtracting $\varphi_{\text{eff}}(O;\, \tau_0, \eta_0)$ inside each fold and rearranging, using the fact that the score is linear in τ with $\partial_\tau \varphi_{\text{eff}} = -1$,

$$\hat{\tau}_{\text{DML}} - \tau_0 = \frac{1}{n}\sum_{i=1}^{n}\varphi_{\text{eff}}(O_i;\, \tau_0, \eta_0) + \sum_{k=1}^{K}\frac{n_k}{n}\,R_n^{(k)},$$

where each fold remainder $R_n^{(k)}$ decomposes as in Equation 12.10:

$$R_n^{(k)} = \underbrace{P\{\varphi_{\text{eff}}(\,\cdot\,;\tau_0,\hat{\eta}^{(-k)}) - \varphi_{\text{eff}}(\,\cdot\,;\tau_0,\eta_0)\}}_{\text{population drift } B_n^{(k)}} + \underbrace{(\mathbb{P}_{n_k} - P)\{\varphi_{\text{eff}}(\,\cdot\,;\tau_0,\hat{\eta}^{(-k)}) - \varphi_{\text{eff}}(\,\cdot\,;\tau_0,\eta_0)\}}_{\text{empirical process term } E_n^{(k)}}.$$

Since $n_k/n \le 1$ and $\sum_k n_k/n = 1$, it suffices to show $\sqrt{n}\,R_n^{(k)} = o_p(1)$ for each k.

Step 1: population drift. By iterated expectations conditioning on X, and using $\mathbb{E}(T \mid X) = \pi(X)$ and $\mathbb{E}(Y \mid T{=}t, X) = \mu_t(X)$,

$$\begin{aligned} B_n^{(k)} &= \mathbb{E}\Bigg[\frac{\pi(X)}{\hat{\pi}^{(-k)}(X)}\{\mu_1(X) - \hat{\mu}_1^{(-k)}(X)\} - \frac{1-\pi(X)}{1-\hat{\pi}^{(-k)}(X)}\{\mu_0(X) - \hat{\mu}_0^{(-k)}(X)\} \\ &\qquad + \hat{\mu}_1^{(-k)}(X) - \mu_1(X) - \hat{\mu}_0^{(-k)}(X) + \mu_0(X)\Bigg] \\ &= \mathbb{E}\Bigg[(\hat{\mu}_1^{(-k)} - \mu_1)(X)\cdot\frac{\hat{\pi}^{(-k)}(X) - \pi(X)}{\hat{\pi}^{(-k)}(X)} + (\hat{\mu}_0^{(-k)} - \mu_0)(X)\cdot\frac{\hat{\pi}^{(-k)}(X) - \pi(X)}{1-\hat{\pi}^{(-k)}(X)}\Bigg]. \end{aligned}$$

Under strong overlap (condition (iii)), $\hat{\pi}^{(-k)}$ is bounded away from 0 and 1 with probability approaching one, so the denominators are bounded below. Applying the Cauchy–Schwarz inequality,

$$|B_n^{(k)}| \le C\Big(\|\hat{\mu}_1^{(-k)} - \mu_1\|\cdot\|\hat{\pi}^{(-k)} - \pi\| + \|\hat{\mu}_0^{(-k)} - \mu_0\|\cdot\|\hat{\pi}^{(-k)} - \pi\|\Big) = o_p(n^{-1/2})$$

by the product-rate condition (iv). Hence $\sqrt{n}\, B_n^{(k)} = o_p(1)$.

Step 2: empirical process term. (The detailed calculation below may be skimmed on first reading; the two essential facts are that $E_n^{(k)}$ has conditional mean zero given $\widehat{\eta}^{(-k)}$ and conditional variance bounded by $\|f_k\|_{L_2}^2/n_k = o_p(1/n_k)$.) Condition on $\widehat{\eta}^{(-k)}$. Since $\widehat{\eta}^{(-k)}$ is trained on $\{1,\dots,n\} \setminus \mathcal{I}_k$, the summands $\{\varphi_{\text{eff}}(O_i;\tau_0,\widehat{\eta}^{(-k)}) - \varphi_{\text{eff}}(O_i;\tau_0,\eta_0)\}$ for $i \in \mathcal{I}_k$ are conditionally i.i.d. with conditional mean $B_n^{(k)}$, which is removed by writing $E_n^{(k)} = (\mathbb{P}_{n_k} - P)\{f_k\}$ as a centered empirical average of $f_k = \varphi_{\text{eff}}(\cdot;\tau_0,\widehat{\eta}^{(-k)}) - \varphi_{\text{eff}}(\cdot;\tau_0,\eta_0)$. The conditional variance of $E_n^{(k)}$ satisfies

$$\operatorname{Var}(E_n^{(k)} \mid \widehat{\eta}^{(-k)}) = \frac{1}{n_k}\operatorname{Var}(f_k(O) \mid \widehat{\eta}^{(-k)}) \le \frac{1}{n_k}\|f_k\|_{L_2}^2.$$

Under strong overlap and consistency (conditions (ii)–(iii)), the L_2 norm satisfies $\|f_k\|_{L_2}^2 = o_p(1)$, so

$$\operatorname{Var}(\sqrt{n}\, E_n^{(k)} \mid \widehat{\eta}^{(-k)}) = \frac{n}{n_k}\|f_k\|_{L_2}^2 = O(1)\cdot o_p(1) = o_p(1),$$

using $n/n_k = O(1)$ uniformly over the fixed number of folds, which holds because $\min_k n_k/n$ is bounded away from zero by assumption. By conditional Chebyshev, $\sqrt{n}\, E_n^{(k)} = o_p(1)$.

Conclusion. Combining Steps 1 and 2, $\sqrt{n}\, R_n^{(k)} = o_p(1)$ for each k, and hence $\sqrt{n}\sum_{k=1}^{K}(n_k/n)\, R_n^{(k)} = o_p(1)$ since $n_k/n \le 1$. The first term $n^{-1/2}\sum_i \varphi_{\text{eff}}(O_i;\tau_0,\eta_0)$ converges in distribution to $N(0,\mathbb{E}[\varphi_{\text{eff}}^2])$ by the ordinary CLT. Asymptotic linearity and asymptotic normality follow by Slutsky's theorem. □

Remark: Interpreting the Product-Rate Condition

Condition Equation 12.13 requires the *product* of the two nuisance estimation errors to be $o_p(n^{-1/2})$. This is substantially weaker than requiring each error individually to be $o_p(n^{-1/2})$. A sufficient symmetric condition is that each nuisance estimator converges at rate $o_p(n^{-1/4})$ in $L_2(P)$ norm, equivalently that its squared $L_2(P)$ error is $o_p(n^{-1/2})$. This rate is achievable by many nonparametric and machine-learning methods under regularity conditions, and is what makes flexible nuisance estimation feasible in semiparametric causal inference.

Remark: Connection to Chapter 11

Theorem 12.1 is the cross-fitting version of the asymptotic normality result stated in Chapter 11. The product-rate condition Equation 12.13 is identical to the condition stated there; the difference is that here the nuisance estimates are out-of-fold, which removes the need for Donsker conditions. Cross-fitting thus provides the same asymptotic conclusion while accommodating a substantially broader class of nuisance learners.

Remark: Dependence and the i.i.d. Assumption

The theory in this chapter assumes i.i.d. observational units. With cluster dependence — patients within hospital wards, students within schools — folds should generally be formed at the level of the independent sampling unit (for example, by cluster), and inference should aggregate influence contributions at the cluster level. Interference, in which one unit's treatment affects another unit's outcome, is a different problem: it changes the estimand and the identification analysis rather than merely the fold construction (see Exercise 6).

12.10 Lab: Simulation Study of the DML Estimator

This lab uses simulation to illustrate the finite-sample behavior that motivates the DML estimator of Theorem 12.1. The central message of this chapter is that flexible nuisance estimation alone is not sufficient for valid inference: even when the nuisance estimators are consistent, same-sample score evaluation can leave a non-negligible empirical-process remainder at the root-n scale, so the resulting AIPW estimator may fail to be asymptotically linear with influence function φ_{eff} — invalidating ordinary influence-function inference even where the point estimate remains consistent. Cross-fitting addresses this by removing the same-sample dependence in the empirical-process term of the remainder decomposition Equation 12.7, so that the term can be controlled by a conditional argument.

To avoid deliberate parametric misspecification, the three *feasible* estimators below use identical random-forest learners for the nuisance functions; the oracle estimator instead plugs in the true nuisance functions and serves only as an infeasible benchmark. The primary controlled comparison is between same-sample RF AIPW and cross-fitted RF AIPW — the only difference between them is whether the nuisance estimates are produced on the full sample (same data reused) or on held-out folds (cross-fitted) — with the single-split estimator added to separate held-out evaluation from fold rotation.

12.10.1 Data-Generating Process

Let $n = 2000$ and generate covariates $X_1, X_2, X_3 \overset{\text{iid}}{\sim} \text{Uniform}(-1, 1)$, collected as $X = (X_1, X_2, X_3)^\top$. Define the true nuisance functions by

$$\begin{aligned}\pi(X) &= \frac{\exp\bigl(\sin(\pi X_1) + X_2^2 - 0.5\bigr)}{1 + \exp\bigl(\sin(\pi X_1) + X_2^2 - 0.5\bigr)},\\ \mu_1(X) &= 2\sin(\pi X_1) + X_2^2 + X_3,\\ \mu_0(X) &= \sin(\pi X_1) + X_2^2 - X_3.\end{aligned}$$

The true ATE is

$$\tau = \mathbb{E}\{\mu_1(X) - \mu_0(X)\} = \mathbb{E}\{\sin(\pi X_1) + 2X_3\} = 0,$$

since X_1 and X_3 are symmetric about zero. Treatment and outcome are drawn as

$$T_i \mid X_i \sim \text{Bernoulli}\{\pi(X_i)\}, \qquad Y_i = \mu_{T_i}(X_i) + \varepsilon_i, \qquad \varepsilon_i \sim N(0, 1).$$

The nonlinearity of π and μ_t means that random forests are flexible enough to approximate them well in this simulation; formal L_2 consistency would require additional conditions on the forest construction and the tuning sequence, which we do not verify here. The random-forest classes implemented in the lab are not covered by the classical Donsker sufficient conditions of Section 12.5 (we do not formally establish that the induced score classes are non-Donsker) — precisely the setting in which same-sample plug-in estimation lacks a supporting empirical-process argument and can become problematic.

12.10.2 The Four Estimators

1. **Oracle AIPW.** Plug in the true nuisance functions $\pi(X_i)$, $\mu_1(X_i)$, $\mu_0(X_i)$ directly. This estimator is infeasible but provides the semiparametric efficiency benchmark.
2. **Same-sample AIPW** (often called “naive” AIPW). Estimate π, μ_1, μ_0 using random forests on the *full sample*, then evaluate the AIPW score on the same sample. Data reuse means the empirical process term in Equation 12.7 is not controlled by any argument available to us: the fixed random-forest implementation used here is not covered by the classical Donsker sufficient conditions, and we make no formal claim that the fitted forests are L_2-consistent.
3. **DML (cross-fitted AIPW).** Estimate π, μ_1, μ_0 using the same random forest learners, but via $K = 5$ fold cross-fitting. Training and evaluation folds are separated, so the empirical-process term is controlled by the conditional argument of Section 12.7 without any Donsker assumption on the nuisance class — under the L_2-consistency, overlap, and moment conditions of Theorem 12.1, which we do not verify for this implementation.
4. **Single-split AIPW.** Randomly split the sample in half; fit the same random-forest learners on one half and evaluate the AIPW score on the other half only. This estimator enjoys the same independence argument as cross-fitting (Section 12.6) but uses only $n/2$ evaluation points, so it isolates the role of held-out evaluation from the efficiency gain of rotating folds.

Estimators (2)–(4) use identical learners; the differences are data reuse versus held-out evaluation, and — between (3) and (4) — whether evaluation roles rotate over folds.

12.10.3 Implementation and Reproducibility

The simulation reported below was run in Python using scikit-learn random forests. The implementation uses fixed hyperparameters throughout; *no nested cross-validation is used to tune the learners inside the* *cross-fitting loop*. This keeps the comparison between the same-sample and DML estimators clean: any difference between them is attributable to cross-fitting alone, not to different tuning paths.

Simulation Settings

Software	Python 3.12, `scikit-learn` 1.8.0, `numpy` 2.4.4
Sample size	$n = 2000$
Replications	$n_{\text{sim}} = 200$
Folds	$K = 5$, shuffled, fixed per replication
Nuisance learners	random forest regressor for μ_t; random forest classifier for π
Hyperparameters	`n_estimators=100`, `min_samples_leaf=5`; all other settings at library defaults
Propensity trimming	$\hat{\pi}$ clipped to [0.01, 0.99] after fitting
Tuning	none (fixed hyperparameters)
Seeds	replication seed $1000 + r$ for $r = 0, \dots, 199$; derived seeds for each learner, fold, and the single-split permutation

The pseudocode below describes one replication; the full procedure applies it n_{sim} times and collects the four estimators per replication.

Pseudocode: One Replication

For convenience, write the uncentered AIPW pseudo-outcome

$$\text{aipw}_i(\eta) = \mu_1(X_i) - \mu_0(X_i) + \frac{T_i}{\pi(X_i)}\{Y_i - \mu_1(X_i)\} - \frac{1 - T_i}{1 - \pi(X_i)}\{Y_i - \mu_0(X_i)\},$$

evaluated with whatever nuisance triple $\eta = (\mu_0, \mu_1, \pi)$ is supplied; we write $\text{aipw}_i(\eta_0)$ when η is the truth and $\text{aipw}_i(\hat{\eta})$ when it is an estimate. The four estimators all average this pseudo-outcome (over the full sample, or over the evaluation half for the single-split estimator) but differ in which nuisances they use.

1. **Generate data.** Draw (X_i, T_i, Y_i) for $i = 1, \dots, n$ from the DGP of Section 12.10.1.
2. **Oracle AIPW.** Compute $\hat{\tau}_{\text{oracle}} = n^{-1} \sum_i \text{aipw}_i(\eta_0)$ using the true $\eta_0 = (\mu_0, \mu_1, \pi)$.
3. **Same-sample RF AIPW.** Fit $\hat{\pi}$, $\hat{\mu}_0$, $\hat{\mu}_1$ as random forests on the full sample; clip $\hat{\pi}$ to $[0.01, 0.99]$; compute $\hat{\tau}_{\text{ss}} = n^{-1} \sum_i \text{aipw}_i(\hat{\eta})$ with the same-sample fits, where $\hat{\eta} = (\hat{\mu}_0, \hat{\mu}_1, \hat{\pi})$.
4. **DML (cross-fitted AIPW).** Partition $\{1, \dots, n\}$ into K shuffled folds $\mathcal{I}_1, \dots, \mathcal{I}_K$. For each fold k: (a) fit $\hat{\pi}^{(-k)}$, $\hat{\mu}_0^{(-k)}$, $\hat{\mu}_1^{(-k)}$ on $\{1, \dots, n\} \setminus \mathcal{I}_k$; (b) clip $\hat{\pi}^{(-k)}$ to $[0.01, 0.99]$; (c) predict on $\mathcal{I}_k$. Compute $\hat{\tau}_{\text{DML}} = n^{-1} \sum_{k=1}^{K} \sum_{i \in \mathcal{I}_k} \text{aipw}_i(\hat{\eta}^{(-k)})$, equivalently formula Equation 12.12.
5. **Single-split AIPW.** Randomly permute $\{1, \dots, n\}$; fit $\hat{\eta}^{(\text{train})}$ on the first half (with clipping as above); compute $\hat{\tau}_{\text{split}}$ as the average of $\text{aipw}_i(\hat{\eta}^{(\text{train})})$ over the second half only.
6. **Inference.** For each estimator, compute the plug-in variance estimator Equation 12.14 from the corresponding plug-in score values (in-sample, out-of-fold, or evaluation-half) and the nominal 95% Wald interval. The number of contributing observations is the denominator: n for the full-sample estimators and $n_e = n/2$ for the single-split estimator, whose variance estimator is $\hat{V}_{\text{split}} = n_e^{-2} \sum_{i \in \mathcal{I}_{\text{eval}}} (\hat{\varphi}_i - \bar{\hat{\varphi}}_{\text{eval}})^2$.
7. **Return.** $(\hat{\tau}_{\text{oracle}}, \hat{\tau}_{\text{ss}}, \hat{\tau}_{\text{DML}}, \hat{\tau}_{\text{split}})$ with standard errors and intervals.

The reference implementation used to produce the numbers below, `chapter12_lab.py`, is available in the companion repository. For exact reproducibility, the repository records the Python and `scikit-learn` versions and the seed-derivation rule; "library defaults" are not stable across software versions, so version pinning is part of the specification. Readers can adapt the implementation to R (for example, using `ranger` for random forests and `origami` or a hand-coded `KFold` for the cross-fitting loop).

12.10.4 Simulation Results and Interpretation

Running the simulation with $n = 2000$ and $n_{\text{sim}} = 200$ replications produces the point-estimation results below.

Table 12.1: Point-estimation performance over $n_{\text{sim}} = 200$ replications, with RMSE $= \sqrt{\text{Monte Carlo variance} + \text{bias}^2}$. The Monte Carlo standard errors of the reported biases, $\sqrt{\text{variance}/n_{\text{sim}}}$, are approximately 0.0038 (oracle), 0.0039 (same-sample), 0.0042 (DML), and 0.0065 (single-split).

Estimator	Bias	Monte Carlo variance	RMSE
Oracle AIPW	0.009	0.0029	0.055
Same-sample RF AIPW	0.039	0.0031	0.068
DML ($K = 5$)	0.004	0.0036	0.060
Single-split AIPW	0.004	0.0084	0.092

The simulation illustrates the asymptotic logic of cross-fitting: it reduces the bias created by same-sample nuisance fitting, but finite-sample performance still depends on the quality of the nuisance learners, the degree of overlap, and the sample size.

- **Oracle AIPW** is nearly unbiased with small variance, serving as the semiparametric efficiency benchmark.
- **Same-sample RF AIPW** exhibits a clear positive bias (0.039, roughly ten Monte Carlo standard errors from zero). This behavior is consistent with the mechanism analyzed in Section 12.5: reusing the same sample for nuisance estimation and score evaluation allows overfitting in the random forests to propagate into the AIPW score through the empirical process term Equation 12.8.
- **DML** reduces the bias to 0.004 — within Monte Carlo error of zero and indistinguishable from the oracle — with a slightly larger Monte Carlo variance (0.0036 versus 0.0031). The bias reduction is the intended effect of cross-fitting: by separating training and evaluation folds (Section 12.7), the empirical-process term is controlled by a conditional variance argument rather than by a Donsker assumption on the random-forest class, under the L_2-consistency, overlap, and moment conditions of Theorem 12.1. The smaller nuisance-training sample within each fold ($(K-1)n/K = 1600$ observations rather than $n = 2000$) and the additional randomness from fold assignment are plausible contributors to the variance difference, but the experiment does not identify a variance decomposition; in particular, the lower Monte Carlo variance of the same-sample estimator should not be attributed to overfitting without a separate analysis. Under the assumptions of Theorem 12.1, the first-order efficiency comparison between the two estimators is asymptotic rather than a general finite-sample variance ordering. The net effect here is nevertheless a lower RMSE for DML (0.060) than for same-sample RF AIPW (0.068): the bias reduction outweighs the variance difference.
- **Single-split AIPW** is also essentially unbiased (0.004): held-out evaluation alone removes the same-sample problem. But its Monte Carlo variance (0.0084) is more than twice DML's, because only $n/2$ observations contribute to the target average. The comparison between the single-split and cross-fitted estimators isolates the second role of cross-fitting: rotating the folds restores the full sample as evaluation points without giving up the independence argument.

Remark: The Key Comparison

The key comparison is between same-sample RF AIPW and DML, not between either and the oracle. Both use identical random forest learners; the difference is the cross-fitting protocol as a whole. Note that the two estimators also differ in effective training sample size (n versus $(K-1)n/K$ per fold), so the bias reduction from 0.039 to 0.004 demonstrates the effect of the protocol, not an exact empirical decomposition of the bias into population-drift and empirical-process components. The accompanying small increase in variance is a finite-sample phenomenon, not an asymptotic penalty: under the conditions of Theorem 12.1, cross-fitting attains first-order efficiency, and the gap should narrow as n grows. The fact that DML already attains a lower RMSE than same-sample RF AIPW at $n = 2000$, while also being essentially unbiased, is consistent with the behavior predicted by Theorem 12.1.

12.10.5 Inference Performance

The chapter promises valid *inference*, not merely accurate point estimation, so the lab also evaluates the plug-in influence-function standard errors of Section 12.11. For each estimator and each replication we compute $\widehat{V}$ from Equation 12.14 (using in-sample plug-in score values for the same-sample estimator, out-of-fold values for DML, and evaluation-half values with denominator $n_e = n/2$ for the single-split estimator), the nominal 95% Wald interval $\hat{\tau} \pm 1.96\sqrt{\widehat{V}}$, and its length.

Table 12.2: Inference performance at nominal level 0.95 over $n_{\text{sim}} = 200$ replications. The Monte Carlo standard error of a coverage estimate near 0.95 is $\sqrt{0.95 \times 0.05/200} \approx 0.015$, so coverages within roughly 0.92–0.98 are compatible with the nominal level; 0.790 is not.

Estimator	aveSE	empSD	aveSE/empSD	Coverage	CI length
Oracle AIPW	0.057	0.054	1.06	0.960	0.223
Same-sample RF AIPW	0.045	0.056	0.81	0.790	0.178
DML ($K = 5$)	0.066	0.060	1.10	0.950	0.257
Single-split AIPW	0.093	0.092	1.02	0.955	0.365

Three features stand out. First, the same-sample estimator fails at the inferential level even more clearly than at the point-estimation level: its Wald intervals cover the truth in only 79% of replications. The failure has two reinforcing sources: the interval is centered at a biased estimate, and the estimated standard error is too small (aveSE/empSD = 0.81). The understatement is consistent with in-sample overfitting: fitting and evaluating the nuisance functions on the same observations can compress the residual and inverse-weight contributions appearing in the plug-in score, and the lower mean maximum absolute plug-in score value (8.4 for the same-sample estimator versus 13.6 for the oracle) is a diagnostic consistent with this explanation, not a proof or a variance decomposition. Because the same-sample estimator has not been shown to be asymptotically linear with influence function φ_{eff}, its plugged-in quantities $\varphi_{\text{eff}}(O_i; \hat{\tau}, \hat{\eta})$ should be read as *plug-in score values* rather than automatically as valid estimated influence values.

Second, DML delivers coverage at the nominal level (0.950, ratio 1.10), consistent with the Wald theory of Section 12.11. Third, in this experiment the single-split estimator also attains nominal coverage (0.955, ratio 1.02). Held-out evaluation controls the same-sample empirical-process remainder, but valid Wald inference additionally relies on the population-drift, nuisance-consistency, overlap, and moment conditions of Theorem 12.1; the simulation shows the procedure works here, not that held-out independence is sufficient in general. The wider single-split intervals (0.365 versus 0.257, 42% longer than DML's) reflect the fact that only the evaluation half contributes directly to the target average, together with any finite-sample effect of fitting the nuisances on the smaller training half. Held-out evaluation addresses the same-sample dependence problem; fold rotation recovers evaluation-sample efficiency; valid inference requires the full collection of conditions in Theorem 12.1.

12.10.6 Nuisance and Influence Diagnostics

Because the true nuisance functions are known in a simulation, the lab can also report the nuisance diagnostics that practice cannot observe. The reported nuisance errors are *empirical* root-mean-square errors against the true functions, computed at the observed covariate values: all n points with in-sample predictions for the same-sample estimator, the same n points with out-of-fold predictions for DML, and the $n_e = n/2$ evaluation points for the single-split estimator. Each is a Monte Carlo approximation to the $L_2(P_X)$ norm; the same-sample and DML errors are evaluated at identical design points, so their comparison is design-comparable, while the single-split errors use half the points.

Table 12.3: Nuisance and influence diagnostics, averaged over the 200 replications. Columns 2–4: empirical RMSE of each fitted nuisance against the truth (see text for the evaluation design); columns 5–6: mean products of the nuisance errors entering the drift bound; "Clip": number of fitted propensities falling outside $[0.01, 0.99]$ before clipping; last column: mean over replications of the largest absolute plug-in score value.

Estimator	$\hat{\pi}$	$\hat{\mu}_0$	$\hat{\mu}_1$	$\hat{\mu}_1 \cdot \hat{\pi}$	$\hat{\mu}_0 \cdot \hat{\pi}$	Clip	$\max_i \lvert\hat{\varphi}_i\rvert$
Oracle AIPW	–	–	–	–	–	–	13.6
Same-sample RF AIPW	0.139	0.363	0.374	0.052	0.050	0	8.4
DML ($K = 5$)	0.107	0.347	0.372	0.040	0.037	0	26.6
Single-split AIPW	0.111	0.369	0.404	0.045	0.041	0	22.5

Three lessons emerge. First, both error products for DML (0.040 and 0.037) exceed $n^{-1/2} = 0.022$ at this sample size, so the Cauchy–Schwarz bound on the population drift is *not* small relative to the target root-n scale — and yet the realized bias of DML is negligible. This is a useful reminder that the bound is conservative: the exact drift is a signed bilinear integral in which cancellation can occur. It is equally a reminder that the product-rate condition is not verified here, consistent with the caveats below. Second, the same-sample in-sample fits show a *larger* empirical error against the true nuisance functions than DML's out-of-fold predictions (0.139 versus 0.107 for $\hat{\pi}$), which is consistent with overfitting toward the realized labels and away from the smooth truth, although the experiment does not isolate that mechanism. Third, the propensity clipping to $[0.01, 0.99]$ never binds — zero fitted propensities fall outside the interval across all estimators and all 200 replications — confirming that clipping plays no role in these results.

This finite-sample experiment illustrates the behavior motivating cross-fitting; it does not by itself verify the assumptions of Theorem 12.1. With the fixed random-forest implementation used here, we do not establish L_2-consistency, the product-rate condition, or membership outside every possible Donsker regime. The contrast between the same-sample and cross-fitted estimators is consistent with the removal of same-sample overfitting effects, but the simulation does not separately identify the population-drift term and the empirical-process term of the remainder decomposition Equation 12.7. In this particular DGP, the true logit lies in $[-1.5, 1.5]$, so the true propensity score is bounded in approximately $[0.18, 0.82]$; consistent with this, the clipping of $\hat{\pi}$ to $[0.01, 0.99]$ never binds in the reported runs.

12.11 Variance Estimation and Confidence Intervals

Inference for $\hat{\tau}_{\mathrm{DML}}$ proceeds exactly as in Chapter 11, except that the nuisance estimates are now out-of-fold. Cross-fitting changes how the nuisance functions are estimated, not the asymptotic variance formula itself; the variance is still the variance of the efficient score, estimated empirically using the out-of-fold score contributions. Let $k(i)$ denote the fold containing observation i, and define the estimated influence value as the score contribution $\hat{s}_i$ from the algorithm box, evaluated at the solution $\hat{\tau}_{\mathrm{DML}}$:

$$\hat{\varphi}_i = \hat{s}_i(\hat{\tau}_{\mathrm{DML}}) = \varphi_{\mathrm{eff}}(O_i;\, \hat{\tau}_{\mathrm{DML}},\, \hat{\eta}^{(-k(i))}).$$

A natural variance estimator is

$$\hat{V} = \frac{1}{n(n-1)} \sum_{i=1}^{n} (\hat{\varphi}_i - \bar{\varphi})^2, \qquad \bar{\varphi} = \frac{1}{n} \sum_{i=1}^{n} \hat{\varphi}_i. \tag{12.14}$$

The Wald confidence interval at level $1 - a$ is $\hat{\tau}_{\mathrm{DML}} \pm z_{1-a/2} \sqrt{\hat{V}}$. (We write a for the significance level to avoid collision with other uses of α.)

Theorem 12.2 (Consistency of the Variance Estimator)**.** *Suppose the conditions of Theorem 12.1 hold* *and, in addition, the estimated influence values converge in the sense*

$$\frac{1}{n}\sum_{i=1}^{n}\{\hat{\varphi}_i - \varphi_{\text{eff}}(O_i;\, \tau_0, \eta_0)\}^2 = o_p(1).$$

Then the variance estimator $\hat{V}$ *in Equation 12.14 is consistent for the asymptotic variance of* $\hat{\tau}_{\text{DML}}$*; equivalently,* $n\hat{V} \xrightarrow{p} \mathbb{E}[\varphi_{\text{eff}}(O;\, \tau_0, \eta_0)^2]$*. The Wald confidence interval* $\hat{\tau}_{\text{DML}} \pm z_{1-a/2}\sqrt{\hat{V}}$ *is therefore asymptotically valid.*

Theorem 12.2 is deliberately stated in a general form: the estimated-influence-value convergence condition, not the particular estimator, is what drives variance consistency. For the cross-fitted AIPW estimator specifically, the condition is implied by the assumptions of Theorem 12.1 — strong overlap bounds the fitted inverse weights, cross-fitted L_2 consistency gives $\|f_k\|_{L_2} = o_p(1)$ fold by fold exactly as in Step 2 of the proof, and consistency of $\hat{\tau}_{\text{DML}}$ handles the τ-argument, which enters the score linearly — so variance consistency for $\hat{\tau}_{\text{DML}}$ holds as a corollary under those assumptions alone.

From a practical standpoint, Equation 12.14 is the sample variance of the out-of-fold estimated influence values divided by n, i.e., the usual plug-in estimate of the variance of a sample mean. The formula is identical to the one in Chapter 11; the only difference is that $\hat{\varphi}_i$ now uses the out-of-fold nuisance estimate $\hat{\eta}^{(-k(i))}$ rather than a full-sample nuisance estimate.

Remark: Why the Centering Is Numerically Inconsequential

Because φ_{eff} is linear in τ with $\partial_\tau \varphi_{\text{eff}} = -1$, evaluating the score at the solution $\hat{\tau}_{\text{DML}}$ of the cross-fitted moment equation Equation 12.11 forces $\bar{\varphi} = n^{-1}\sum_{i=1}^{n}\hat{\varphi}_i = 0$ exactly (see Exercise 5). The centering by $\bar{\varphi}$ in Equation 12.14 is therefore numerically inconsequential; it is included only because the formula then matches the conventional sample-variance expression for an i.i.d. mean and remains stable under finite-precision arithmetic.

12.12 A Practical Workflow

The theory above leads to the following workflow for applied cross-fitted orthogonal-score estimation. The steps are grouped into three phases: identification and setup, estimation, and diagnostics and interpretation.

Workflow: Cross-Fitted Causal Estimation

Phase 1: Identification and setup

1. **Specify the estimand and defend the identification strategy.** State the target causal parameter and the assumptions under which it is identified — conditional exchangeability, positivity, and consistency for the ATE (Chapter 5) — and explain the substantive or design-based basis for each assumption.
2. **Choose the orthogonal score.** For the ATE under conditional exchangeability, this is the efficient influence function Equation 12.4. The score fixes what nuisance components must be estimated.
3. **Select nuisance learners.** Choose learners for $\mu_0(x)$, $\mu_1(x)$, and $\pi(x)$ with theoretical or empirical support for adequate L_2 prediction in the scientific setting. The product-rate condition Equation 12.13 cannot be verified directly from a single observed data set: use learner sensitivity analyses, out-of-fold predictive diagnostics, and available smoothness or sparsity theory as indirect evidence, while recognizing that none proves the required asymptotic rate. Remember also that the condition constrains the *product* of the two errors; the symmetric $o_p(n^{-1/4})$ requirement is sufficient, not necessary.
4. **Assess covariate overlap.** Before fitting, examine arm-specific supports and distributions using density or empirical-CDF plots, ranges and tail quantiles, or nearest-neighbor distances across arms. Standardized mean differences may be reported as balance summaries, but they are not by themselves positivity diagnostics: equal means can coexist with poor tail overlap. Near-violation of positivity at the design level will inflate influence values and destabilize inference regardless of the estimator used. A complementary post-fit check on the estimated

propensity scores appears in Phase 3.

Phase 2: Estimation

5. **Partition and cross-fit.** Partition the sample into K folds ($K = 5$ is a common default). For each fold k, fit the nuisance learners on the complement $\{1, \ldots, n\} \setminus \mathcal{I}_k$ to obtain out-of-fold estimates $\hat{\eta}^{(-k)}$.
6. **Solve the moment equation.** Apply the explicit AIPW formula Equation 12.12 with out-of-fold nuisance estimates to obtain $\hat{\tau}_{\mathrm{DML}}$.
7. **Compute estimated influence values.** For each $i \in \mathcal{I}_k$, compute $\hat{\varphi}_i = \varphi_{\mathrm{eff}}(O_i; \hat{\tau}_{\mathrm{DML}}, \hat{\eta}^{(-k)})$. These values are used for variance estimation and can also be inspected diagnostically for extreme influence contributions and weak-overlap symptoms.
8. **Estimate the variance.** Compute $\hat{V}$ as the sample variance of the $\hat{\varphi}_i$ divided by n, as in Equation 12.14, and form the Wald confidence interval.

Phase 3: Diagnostics and interpretation

9. **Inspect estimated propensity scores.** Examine the distribution of $\hat{\pi}^{(-k)}(X_i)$ across folds; extreme values near 0 or 1 inflate influence-value variance and can destabilize coverage.
10. **Assess sensitivity.** Repeat the analysis with alternative learners for the nuisance functions. Substantial sensitivity to learner choice signals that the product-rate condition may not be satisfied at this sample size.
11. **Interpret relative to identification assumptions.** The credibility of the causal estimate rests on the identification assumptions, not on the predictive accuracy of the nuisance learners. Conditional exchangeability cannot be verified from the observed-data distribution alone and must be defended substantively. Consistency is a claim about how the recorded treatment maps to the intervention of interest, a matter of design and measurement. Positivity has observable implications and should be investigated empirically, although finite samples cannot conclusively establish support throughout a high-dimensional covariate space.

Step 11 is the most important step in the workflow. The core of identification — conditional exchangeability — sits outside what any algorithm can verify, and a careful discussion of identification often does more to make a causal estimate credible than any choice of nuisance learner. We return to this point in Section 12.13.

12.13 What Machine Learning Does Not Solve

Flexible nuisance estimation is a genuine improvement over parametric methods, but it does not resolve the fundamental difficulties of causal inference. In particular, machine learning does not solve:

- **Unmeasured confounding.** If important confounders are absent from X, conditional exchangeability $Y(t) \perp\!\!\!\perp T \mid X$ fails. No amount of flexibility in estimating $\pi(x)$ or $\mu_t(x)$ can correct this.
- **Ill-defined interventions, hidden versions, or interference.** These problems invalidate the simple unit-level representation $Y_i(t)$ and the corresponding ATE used in this chapter. They do not invalidate the potential-outcomes framework itself; rather, the intervention and estimand must be reformulated, for example by indexing Y_i by the treatment vector $\mathbf{T}$ or by a specified exposure mapping.
- **Violations of consistency.** Even after the intervention is well defined, one must justify that the observed outcome equals the potential outcome corresponding to the treatment version actually received. No amount of flexibility in nuisance estimation bears on this claim.
- **Exact positivity failure.** If $\pi(x) = 0$ or $\pi(x) = 1$ on a set of positive probability relevant to the target population, the standard nonparametric ATE is not identified without extrapolation assumptions.
- **Weak overlap or near-positivity.** If $\pi(x)$ is positive but approaches 0 or 1 in part of the covariate space, the ATE may remain identified, but influence values become unstable, the efficient variance can be very large or infinite, and coverage of Wald confidence intervals can degrade severely.
- **Ambiguity about the estimand.** Flexible estimation cannot resolve disagreement about what causal quantity is scientifically meaningful.
- **Invalid instrumental variable assumptions.** The exclusion restriction and exogeneity conditions (Chapter 7) are not testable from the data; machine learning cannot substitute for subject-matter justification.

- **Fragile mediation assumptions.** Identification of natural direct and indirect effects requires the sequential ignorability assumptions of Chapter 8 (Section 8.6) — treatment ignorability, mediator–outcome ignorability, and a cross-world independence — together with consistency, composition, and suitable positivity. A treatment-induced mediator–outcome confounder, whether measured or not, generically invalidates the cross-world independence and with it the standard mediation formula. Flexible nuisance learning does not supply any of these assumptions.

Machine learning improves the estimation of nuisance functions *within* a given identification strategy. It does not create identification where none exists. Machine learning can improve estimation after the causal problem has been formulated and identified, but it cannot repair a failure of identification assumptions.

Remark: What Modern Causal Estimation Is

Modern causal estimation is not merely machine learning plus a causal estimand: it is machine learning embedded inside a carefully constructed semiparametric procedure, whose validity rests on identification assumptions that no algorithm can verify.

12.14 Chapter Summary

1. Flexible methods are attractive for nuisance estimation when the true nuisance functions are complex, but naive plug-in estimators can fail for two reasons: orthogonality may be absent, and even when orthogonality holds, same-sample nuisance fitting can leave an additional empirical-process remainder (Section 12.2).
2. A score is Neyman orthogonal if the Gateaux derivative of its population moment with respect to the nuisance vanishes at the truth (Definition 12.1). Under suitable local smoothness, this removes the first-order nuisance term from the population moment drift; it does not by itself control the same-sample empirical-process remainder.
3. The ATE score Equation 12.4 is orthogonal and doubly robust and is the efficient influence function in the maintained nonparametric model; a regular estimator that is asymptotically linear with this influence function attains the semiparametric efficiency bound. It serves as the orthogonal score throughout this chapter (Section 12.4).
4. Even with an orthogonal score, reusing the same data for nuisance estimation and score evaluation can leave a non-negligible empirical-process remainder, especially when the induced score class falls outside classical fixed Donsker regimes (Section 12.5).
5. Sample splitting separates nuisance fitting from score evaluation, allowing the empirical-process remainder to be controlled conditionally without a Donsker assumption, at the cost of using only the evaluation subsample for the target estimate (Section 12.6).
6. Cross-fitting rotates training and evaluation roles across K folds, so each observation is evaluated only with nuisance estimates trained on other observations; this avoids the first-order information loss of a single split and permits efficiency under the conditions of Theorem 12.1 (Section 12.7).
7. The cross-fitted AIPW estimator $\hat{\tau}_{\mathrm{DML}}$ is the standard double machine learning estimator for the ATE (Section 12.8).
8. Under the full conditions of Theorem 12.1 — i.i.d. sampling, fixed data-independent folds, moment conditions, $L_2(P)$-consistency, strong overlap with bounded fitted propensities, and the product-rate condition Equation 12.13 — orthogonality and cross-fitting together yield asymptotic linearity, root-n consistency, and asymptotic normality of $\hat{\tau}_{\mathrm{DML}}$, with asymptotic variance equal to the semiparametric efficiency bound. A sufficient symmetric condition is that each nuisance estimator converges at rate $o_p(n^{-1/4})$ in $L_2(P)$ norm, equivalently that its squared $L_2(P)$ error is $o_p(n^{-1/2})$.
9. The asymptotic variance is estimated consistently from the empirical variance of the out-of-fold estimated influence values (Theorem 12.2).
10. Machine learning improves nuisance estimation, but it does not resolve identification problems: unmeasured confounding, positivity failures, and untestable structural assumptions remain the researcher's responsibility (Section 12.13).

Key Objects in This Chapter

Symbol	Meaning
τ	ATE $= \mathbb{E}\{Y(1) - Y(0)\}$
$\mu_t(x)$	Outcome regression $\mathbb{E}(Y \mid T = t, X = x)$
$\pi(x)$	Propensity score $P(T = 1 \mid X = x)$
η	Nuisance tuple (μ_0, μ_1, π)
$\phi(O;\, \psi, \eta)$	Generic orthogonal score; zero mean at truth
$\varphi_{\text{eff}}(O;\, \tau, \eta)$	Efficient influence function for the ATE Equation 12.4
$\mathcal{I}_k$	k-th fold; $k = 1, \ldots, K$
$\hat{\eta}^{(-k)}$	Nuisance estimates trained without fold k
$\hat{\tau}_{\text{DML}}$	Cross-fitted AIPW (DML) estimator Equation 12.12
$\hat{\varphi}_i$	Out-of-fold estimated influence value for observation i
$\hat{V}$	Variance estimator Equation 12.14

12.15 Exercises

1. Verifying orthogonality.

(a) Write out the efficient influence function $\varphi_{\text{eff}}(O;\, \tau, \eta)$ for the ATE and compute its expectation. Confirm it equals zero at the truth.
(b) Consider a perturbation $\pi \mapsto \pi + r \cdot h$ for a bounded function $h(x)$. Differentiate $\mathbb{E}\{\varphi_{\text{eff}}(O;\, \tau, \mu_0, \mu_1, \pi + rh)\}$ with respect to r and evaluate at $r = 0$. Show the derivative is zero, confirming Neyman orthogonality in the propensity score direction.
(c) Repeat the calculation for a perturbation in μ_1.

2. First-order bias of the plug-in estimator. Suppose $\hat{\mu}_1 = \mu_1 + \delta$ for a deterministic function $\delta(x)$, and $\hat{\mu}_0 = \mu_0$, $\hat{\pi} = \pi$.

(a) Show that the bias of the resulting prediction estimator $\hat{\tau}_{\text{pred}} = n^{-1} \sum_i \{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\}$ is $\mathbb{E}\{\delta(X)\}$.
(b) Show that the population drift of the plug-in AIPW estimator (using $\hat{\mu}_1 = \mu_1 + \delta$ with the true π and μ_0) is zero. Explain which property of the AIPW score is responsible. Does this zero population drift guarantee that the same-sample AIPW estimator is root-n asymptotically linear with influence function φ_{eff} when $\hat{\mu}_1$ is estimated flexibly from the same sample used to evaluate the score? Explain why the empirical-process term must still be controlled.

3. Cross-fitting with $K = 2$. Partition a sample of $n = 200$ observations into two halves $\mathcal{I}_1$ and $\mathcal{I}_2$.

(a) Write down the cross-fitted moment equation Equation 12.11 explicitly for $K = 2$.
(b) Explain why the contribution from $\mathcal{I}_1$ uses nuisance estimates trained on $\mathcal{I}_2$ and vice versa.
(c) Compare this procedure to single sample splitting with $\mathcal{I}_1$ as the training fold. What is gained and what is lost?

4. The product-rate condition. Suppose $\|\hat{\pi} - \pi\|_{L_2} = o_p(n^{-\alpha})$ and $\|\hat{\mu}_t - \mu_t\|_{L_2} = o_p(n^{-\beta})$ for $\alpha, \beta > 0$.

(a) State the condition on α and β under which the product-rate condition Equation 12.13 holds.
(b) Show that $\alpha = \beta = 1/4$ satisfies your condition.
(c) Now suppose that the propensity score is estimated at the parametric rate, $\|\hat{\pi} - \pi\|_{L_2} = O_p(n^{-1/2})$, while the outcome model is only known to be consistent, $\|\hat{\mu}_t - \mu_t\|_{L_2} = o_p(1)$ (no rate is specified). Show that the product-rate condition Equation 12.13 still holds. What does this say about the standard situation in which the propensity score is estimated by a parametric logistic regression while the outcome model is fit by a flexible learner?

5. Variance estimation. Using the explicit formula Equation 12.12, write out the estimated influence value $\hat{\varphi}_i$ for a generic observation $i \in \mathcal{I}_k$. Show that $n^{-1} \sum_i \hat{\varphi}_i = 0$ exactly when $\hat{\tau}_{\text{DML}}$ solves the cross-fitted moment equation, and explain why the centering in Equation 12.14 is therefore numerically inconsequential.

6. What machine learning cannot do. For each of the following scenarios, identify the identification assumption that is violated and explain why flexible nuisance estimation cannot remedy the problem.

(a) A study of job-training effects on earnings omits pre-program earnings, which is a pretreatment common cause of program participation and subsequent earnings and is not otherwise captured by the measured covariates X.
(b) In a study of a medical treatment assigned within hospital wards, a patient's outcome may depend on both that patient's own treatment and the proportion of other patients treated in the same ward. Which component of SUTVA fails? How does this differ from ordinary within-ward correlation of outcomes that does not represent treatment interference?
(c) An instrument is used to estimate the effect of education on wages, but the instrument also directly affects wages through a channel not related to education.

7. Cross-fitting leakage. A researcher creates five folds but selects variables, tunes hyperparameters, and standardizes covariates using the full sample before fitting the fold-specific nuisance models. Does the independence argument that justifies cross-fitting (Section 12.7) remain valid? Identify which operations must be repeated within each training complement, and explain what goes wrong in the conditional variance argument of Step 2 of the proof of Theorem 12.1 when they are not.

8. Coverage experiment (optional computational project). Using the reference implementation `chapter12_lab.py` (record your software versions; results vary slightly across library releases). A full run of parts (a)–(c) involves thousands of random-forest fits — on the order of 10–15 minutes per configuration at $n = 2000$ on a single core, and several times longer at $n = 8000$ — so reducing n_{sim} to 100 and reporting the correspondingly larger Monte Carlo standard errors is acceptable, as is completing a single part.

(a) Rerun the simulation with $n = 500$ and $n = 8000$, keeping $K = 5$. Report the point-estimation and inference tables for each n, including the Monte Carlo standard error of each coverage estimate. How do the same-sample coverage and the DML aveSE/empSD ratio change with n, and why?
(b) At $n = 2000$, rerun DML with $K \in \{2, 10\}$. Relate any changes in bias and interval length to the effective nuisance-training sample size $(K-1)n/K$.
(c) The out-of-fold nuisance error products reported in the diagnostics table exceed $n^{-1/2}$, yet DML's realized bias is negligible. Reconcile these two facts using the exact drift identity in Step 1 of the proof of Theorem 12.1 and the role of Cauchy–Schwarz.

Chapter 13

Estimation for Instrumental Variables

Learning Objectives

By the end of this chapter, students should be able to:

1. Construct the Wald estimator as the sample analog of the Wald estimand and interpret its numerator and denominator as the reduced form and first stage. Extend this to the IV regression estimator with covariates as the direct sample analog of the residualized-covariance ratio $\mathrm{Cov}(\tilde{Z}, Y)/\mathrm{Cov}(\tilde{Z}, T)$, where $\tilde{Z}$ is the linear-projection residual of Z on the intercept and X, and recognize the Wald estimator as a special case.
2. Distinguish the structural form of a linear SEM from its reduced form; derive the reduced form by solving out endogenous variables; interpret the reduced-form coefficient $\phi = \beta\pi$ as the product of the structural parameter and the first-stage strength; define the reduced form regression estimator as OLS applied to the reduced form equation; and interpret $\hat{\phi}_{\mathrm{RF}}$ as the intent-to-treat (ITT) effect when the instrument is binary and randomly assigned.
3. Derive the scalar IV regression estimator from the valid sample moment equations, derive 2SLS through projection, and prove their equivalence in the single-instrument case.
4. Interpret 2SLS as a method-of-moments estimator based on the IV orthogonality condition, connect it to the estimating-equation framework of Chapter 10, and extend to GMM for overidentified models.
5. State the asymptotic distribution of the GMM estimator, specialize the sandwich variance formula to the exactly identified and efficient GMM cases, and construct cluster-robust standard errors.
6. Explain the weak-instrument problem, state precisely in what sense 2SLS is "biased toward OLS," distinguish the conventional homoskedastic first-stage F-statistic from its robust and effective counterparts, and construct and interpret an Anderson–Rubin confidence set obtained by test inversion.

Learning Objectives — Advanced Enrichment (Section 13.8)

Students who complete the advanced section should additionally be able to describe generalized empirical likelihood as a one-step alternative to efficient GMM, identify empirical likelihood, exponential tilting, and the continuous updating estimator as representative members of the GEL family, and state their first-order equivalence with efficient GMM under standard regularity and strong-identification conditions.

13.1 From Identification to Estimation

Chapter 7 established what IV identifies and under what assumptions. This chapter shows how to estimate that target from finite data using sample analogs of the same orthogonality restrictions.

Chapter 7 showed that when an instrument Z satisfies relevance, exogeneity, and exclusion, the residualized-covariance ratio $\mathrm{Cov}(\tilde{Z}, Y)/\mathrm{Cov}(\tilde{Z}, T)$ — where $\tilde{Z} = Z - L(Z \mid 1, X)$ is the linear-projection residual of Z

on the intercept and X — is a well-defined function of observable quantities, but it does not identify any specific causal parameter without a fourth structural assumption. This chapter works within the linear constant-effect model

$$Y = \alpha + \beta T + \gamma^\top X + \varepsilon, \tag{13.1}$$

$$T = a_T + \pi Z + \delta^\top X + \eta, \tag{13.2}$$

where a_T is the first-stage intercept and $X \in \mathbb{R}^p$ is the unit-level vector of non-constant covariates (the corresponding stacked sample object $\mathbf{X} \in \mathbb{R}^{n\times p}$ is the covariate matrix).

The fourth structural assumption is the constant-effect restriction: for binary T, $Y_i(1) - Y_i(0) = \beta$ for all i; for continuous T, $Y_i(t) - Y_i(t') = \beta(t - t')$ for all i and all t, t' in the treatment domain. The difference form is the primitive assumption; when $Y_i(\cdot)$ is differentiable on a connected domain it is equivalent to $\partial Y_i(t)/\partial t = \beta$. Under the three core assumptions plus this restriction, the structural coefficient β is identified by the Wald formula:

$$\beta = \frac{\mathrm{Cov}(\tilde{Z}, Y)}{\mathrm{Cov}(\tilde{Z}, T)}.$$

Under heterogeneous treatment effects, the interpretation depends on the nature of the treatment. In the binary-instrument, binary-treatment setting, the same ratio identifies the LATE for the complier population, provided the additional monotonicity assumption ($T_i(1) \geq T_i(0)$ for all i) serves as the fourth structural assumption instead. With a continuous or multi-valued treatment the corresponding IV estimand is generally a weighted average of marginal or incremental causal effects rather than a single unqualified LATE; that interpretive point is developed in Chapter 7. In the simplest binary-instrument, no-covariate case, this reduces to the Wald estimand:

$$\beta = \frac{\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0]}{\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0]}.$$

Identification establishes that β is a well-defined functional of the observable distribution $P(Y, T, Z, X)$. This chapter addresses the estimation problem: how to recover β from a finite sample. The answer is not a single formula but a family of estimators — the Wald estimator, two-stage least squares (2SLS), and the generalized method of moments (GMM) — each of which is a sample analog of the same underlying orthogonality restriction, viewed from a different angle. This chapter develops estimators for the IV estimands identified in Chapter 7; it does not weaken or replace the substantive assumptions needed for identification.

This chapter assumes that a valid IV design has already been justified. Our goal here is not to discover instruments, but to estimate an already-identified IV estimand and quantify its uncertainty.

The chapter has a core sequence and one optional extension.

Core material (Section 13.2–Section 13.7). These sections develop the Wald estimator, IV regression, two-stage least squares, moment-condition estimation, GMM, and robust variance estimation. Section 13.2 introduces the Wald estimator and derives the scalar IV regression estimator with covariates from the valid sample IV moment equations. Section 13.3 extends this to two-stage least squares for multi-valued and multiple instruments. Section 13.4 proves the equivalence of the IV regression estimator and 2SLS. Section 13.5 interprets both as method-of-moments estimators and extends to GMM for overidentified models. Section 13.6 develops the asymptotic distribution of the GMM estimator, specializes the sandwich variance formula, and covers cluster-robust inference. Section 13.7 studies weak instruments, first-stage diagnostics, and weak-instrument-robust inference.

Advanced enrichment (Section 13.8). This section introduces generalized empirical likelihood as an alternative implementation of the same IV moment restrictions. The detailed convex-duality derivation, the minimum-discrepancy representation, the computational algorithm, and the extended table of special cases are collected in the Chapter 13 Supplement. The section presupposes the core material but can be omitted without loss of continuity.

The figure below displays this architecture. Each arrow is a step taken in this chapter; the identification assumptions at the top are the subject of Chapter 7 and are maintained, not re-examined, below that point.

13.2 The Wald Estimator and the IV Regression Estimator

The Wald estimator is the canonical just-identified special case. It is the cleanest entry point for understanding IV estimation, but most empirical applications with covariates or multiple instruments are implemented through regression or GMM formulations. We begin with the Wald estimator to establish the sample-analog logic before generalizing.

13.2.1 The Wald Estimator

We begin with the simplest setting: a binary instrument $Z \in \{0, 1\}$, a scalar treatment T, a scalar outcome Y, and no additional covariates X. Chapter 7 established that under relevance, exogeneity, exclusion, and the constant-effect assumption ($\tau_i = \beta$ for all i), the parameter β is identified by the Wald estimand

$$\beta = \frac{\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0]}{\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0]}.$$

Each expectation in this ratio is a population mean conditional on instrument status. Replacing each with its sample analog yields the Wald estimator.

Definition 13.1 (Wald Estimator)**.** Let $n_z = \sum_{i=1}^{n} \mathbf{1}(Z_i = z)$ for $z \in \{0, 1\}$, and define the within-group sample means

$$\bar{Y}_z = \frac{1}{n_z} \sum_{i:\, Z_i = z} Y_i, \qquad \bar{T}_z = \frac{1}{n_z} \sum_{i:\, Z_i = z} T_i.$$

The *Wald estimator* is

$$\hat{\beta}_{\text{Wald}} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{T}_1 - \bar{T}_0}. \tag{13.3}$$

Under heterogeneous treatment effects, the Wald estimator remains consistent for the IV estimand defined by the same ratio formula, but the causal interpretation of that estimand depends on the structure of T: for binary T it is the LATE, and for continuous T it is generally a weighted average of marginal causal effects rather than a single treatment-control contrast.

The Wald estimator has a transparent structural interpretation. The numerator $\bar{Y}_1 - \bar{Y}_0$ estimates the *reduced form*: the total effect of the instrument on the outcome. The denominator $\bar{T}_1 - \bar{T}_0$ estimates the *first stage*: the effect of the instrument on treatment uptake. Their ratio recovers the effect of treatment on the outcome by attributing all of the instrument's effect on Y to the path $Z \to T \to Y$. Three assumptions do distinct work here. Exogeneity rules out any noncausal Z–Y association; exclusion rules out a direct path from Z to Y, so that the reduced-form effect operates through T; and relevance guarantees a nonzero denominator. The fourth structural restriction then determines what the ratio means — a homogeneous effect, a LATE, or a weighted average causal response.

The three regression objects serve distinct purposes: the first stage answers how strongly the instrument moves treatment; the reduced form answers how the instrument changes the outcome; and the Wald (or 2SLS) ratio converts that intent-to-treat effect into a per-unit-of-treatment causal effect under the IV assumptions. All three should be reported in applied work — not because they are redundant, but because each is informative about a different aspect of the design.

> Remark: Consistency by the Continuous Mapping Theorem
>
> The within-group means $\bar{Y}_z$ and $\bar{T}_z$ are consistent estimators of $\mathbb{E}[Y \mid Z{=}z]$ and $\mathbb{E}[T \mid Z{=}z]$ by the law of large numbers. Consistency of $\hat{\beta}_{\text{Wald}}$ for β then follows from the continuous mapping theorem, provided the denominator $\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0] \neq 0$ (the relevance condition).

> Remark: What the Wald Estimator Targets under Heterogeneous Effects
>
> When treatment effects are heterogeneous, the causal estimand depends on the structure of T.
> **Binary treatment.** For binary T, Chapter 7 showed that under the additional monotonicity assumption ($T_i(1) \geq T_i(0)$ for all i, ruling out defiers) — the fourth structural assumption in Framework 2 — the Wald estimand identifies the Local Average Treatment Effect (LATE) for the

subpopulation of *compliers*: units whose treatment status is changed by the instrument. The Wald estimator therefore consistently estimates the LATE, not the population average treatment effect. **Continuous treatment.** For continuous T with binary Z, no single per-unit-of-treatment causal contrast exists for the estimator to target. Under monotonicity in the treatment response function, the IV estimand identifies a weighted average of marginal causal effects $\partial Y_i(t)/\partial t$ across the treatment distribution, with weights determined by the instrument-induced shift in the distribution of T (Angrist, Graddy, and Imbens, 2000). The numerator-denominator algebra of the Wald estimator carries through, but "the LATE" is no longer the right name for what it estimates.
The estimator's consistency for its IV estimand is statistical and holds in both cases; the interpretation of that estimand as a causal parameter depends on the structure of T and the substantive identifying assumptions.

13.2.2 The IV Regression Estimator with Covariates

When the IV identifying restrictions are conditional on observed covariates $X \in \mathbb{R}^p$, or when covariate adjustment is desired for precision, the unadjusted group-mean Wald ratio is generally not the appropriate sample analog: the binary group-mean difference does not partial out X. (The presence of covariates does not by itself invalidate the unadjusted ratio — under unconditional randomization of Z it can remain consistent, with adjustment serving only precision — but whenever the exogeneity conditions hold only given X, partialling out is required.) We derive the general IV estimator directly from the estimating-equation principle: an exogenous variable is valid as an instrument precisely because it satisfies an orthogonality condition with the structural error that the endogenous treatment does not.

Notation. Stack observations into vectors $\mathbf{Y}, \mathbf{T} \in \mathbb{R}^n$, $\mathbf{Z} \in \mathbb{R}^n$ (scalar instrument), and the covariate matrix $\mathbf{X} \in \mathbb{R}^{n\times p}$. Let $\bar{\mathbf{X}} = [\mathbf{1}_n, \mathbf{X}] \in \mathbb{R}^{n\times(p+1)}$ denote the design matrix obtained by adjoining a column of ones to $\mathbf{X}$, and define the *annihilator matrix*

$$M_X = I_n - \bar{\mathbf{X}}(\bar{\mathbf{X}}^\top \bar{\mathbf{X}})^{-1}\bar{\mathbf{X}}^\top, \tag{13.4}$$

the orthogonal projection onto the orthogonal complement of $\mathrm{col}(\bar{\mathbf{X}})$. For any vector $\mathbf{W} \in \mathbb{R}^n$, $M_X\mathbf{W}$ is the vector of OLS residuals from regressing $\mathbf{W}$ on the intercept and the columns of $\mathbf{X}$; the i-th entry of $M_X\mathbf{W}$ is the within-X residual $\tilde{W}_i$. The matrix M_X is symmetric and idempotent. Throughout this chapter, "partialling out X" always means partialling out the intercept together with the columns of X.

Derivation from the sample IV equations. The structural model imposes

$$Y_i = \alpha + \beta T_i + \gamma^\top X_i + \varepsilon_i, \qquad \mathbb{E}\left[\begin{pmatrix}1\\ X_i\\ Z_i\end{pmatrix}\varepsilon_i\right] = 0, \qquad \mathbb{E}[T_i\,\varepsilon_i] \neq 0. \tag{13.5}$$

The components of the moment vector encode exogeneity of the intercept (trivially), exogeneity of X, and the joint exogeneity–exclusion restriction on Z; the inequality $\mathbb{E}[T\varepsilon] \neq 0$ is the endogeneity that motivates IV in the first place and is *not* a mean-independence statement that one is willing to assume.

To translate this moment condition into estimation, define the structural residual as a function of the parameters,

$$\mathbf{e}(\beta, \bar{\gamma}) = \mathbf{Y} - \mathbf{T}\beta - \bar{\mathbf{X}}\bar{\gamma}, \qquad \bar{\gamma} = (\alpha, \gamma^\top)^\top,$$

and consider the three families of sample equations that could be imposed on it:

$$\mathbf{T}^\top \mathbf{e}(\beta, \bar{\gamma}) = \mathbf{0}, \tag{S-T}$$

$$\bar{\mathbf{X}}^\top \mathbf{e}(\beta, \bar{\gamma}) = \mathbf{0}, \tag{S-X}$$

$$\mathbf{Z}^\top \mathbf{e}(\beta, \bar{\gamma}) = \mathbf{0}. \tag{S-Z}$$

Equations (S-X) and (S-Z) are the sample analogs of the population-valid moment conditions in Equation 13.5: they are exactly the exogeneity restrictions on the intercept, on X, and on Z. Equation (S-T), by contrast, is the equation OLS would impose on the endogenous regressor, and its population analog $\mathbb{E}[T_i\varepsilon_i] = 0$ is false by assumption.

Note that these are *not* the normal equations of any single least-squares problem. In particular, they are not obtained by regressing $\mathbf{Y}$ on $(\mathbf{T}, \bar{\mathbf{X}}, \mathbf{Z})$: that regression would give the residual a coefficient on $\mathbf{Z}$, and constraining that coefficient to zero does not produce (S-Z). The argument here is a moment-equation argument, not a normal-equation argument. IV estimation retains the valid exogenous-regressor equations and *replaces* the invalid treatment equation (S-T) by the valid instrument equation (S-Z).

We therefore solve the $(p+2)$-dimensional system (S-X), (S-Z). From (S-X):

$$\hat{\bar{\gamma}} = (\bar{\mathbf{X}}^\top \bar{\mathbf{X}})^{-1}(\bar{\mathbf{X}}^\top \mathbf{Y} - \bar{\mathbf{X}}^\top \mathbf{T}\beta).$$

Substituting into (S-Z) and simplifying using M_X gives $\mathbf{Z}^\top M_X \mathbf{T}\,\beta = \mathbf{Z}^\top M_X \mathbf{Y}$. Provided $\mathbf{Z}^\top M_X \mathbf{T} \neq 0$ (the relevance condition after partialling out the intercept and X), this has a unique solution.

Definition 13.2 (IV Regression Estimator)**.** The *IV regression estimator* in the linear IV model with scalar instrument Z and covariates X is

$$\hat{\beta}_{\text{IV}} = \frac{\mathbf{Z}^\top M_X \mathbf{Y}}{\mathbf{Z}^\top M_X \mathbf{T}} = \frac{\widehat{\text{Cov}}(\tilde{Z}, Y)}{\widehat{\text{Cov}}(\tilde{Z}, T)}, \tag{13.6}$$

where $\tilde{Z}_i$, $\tilde{Y}_i$, $\tilde{T}_i$ are the within-X residuals (entries of $M_X\mathbf{Z}$, $M_X\mathbf{Y}$, $M_X\mathbf{T}$).

The estimator forces the residual $\hat{\mathbf{e}} = \mathbf{Y} - \mathbf{T}\hat{\beta}_{\text{IV}} - \bar{\mathbf{X}}\hat{\bar{\gamma}}$ to satisfy both (S-X) and (S-Z); it is the unique solution obtained by substituting the instrument equation for the corrupted treatment equation.

Remark: Wald Estimator as a Special Case

When $Z \in \{0,1\}$ is binary and there are no covariates beyond the intercept (so $\bar{\mathbf{X}} = \mathbf{1}_n$), $M_X = I_n - n^{-1}\mathbf{1}\mathbf{1}^\top$ is the centering matrix and $M_X\mathbf{W}$ is the mean-centered $\mathbf{W}$. One can verify that $\mathbf{Z}^\top M_X \mathbf{Y} \propto (\bar{Y}_1 - \bar{Y}_0)$ and $\mathbf{Z}^\top M_X \mathbf{T} \propto (\bar{T}_1 - \bar{T}_0)$, so $\hat{\beta}_{\text{IV}} = \hat{\beta}_{\text{Wald}}$.

13.2.3 Structural Form, Reduced Form, and the Reduced Form Regression

The terminology *reduced form* only makes sense in contrast to something that has not been reduced. This subsection introduces that contrast, derives the reduced form of the linear SEM, and then presents the reduced form regression as the procedure for estimating the relevant coefficient.

The structural form. The system Equation 13.1–Equation 13.2 is called the *structural form* of the model. It represents the causal mechanism directly: each equation describes how one variable is determined by others, including endogenous variables on the right-hand side. In particular, the outcome equation $Y = \alpha + \beta T + \gamma^\top X + \varepsilon$ contains the endogenous regressor T on its right-hand side. The structural coefficient β has a causal interpretation (it is the effect of a unit change in T on Y, holding X and ε fixed), but T is correlated with ε, so OLS applied to the structural form is inconsistent.

The reduced form. The *reduced form* is obtained by solving the structural system so that each endogenous variable is expressed purely as a function of the exogenous variables Z and X, with no endogenous regressors on the right-hand side. Substituting the first-stage equation Equation 13.2 into the outcome equation Equation 13.1 gives

$$Y = \underbrace{(\alpha + \beta a_T)}_{\alpha_{\text{rf}}} + \underbrace{\beta\pi}_{\phi}\, Z + \underbrace{(\beta\delta + \gamma)^\top}_{\gamma_{\text{rf}}^\top}\, X + \underbrace{(\beta\eta + \varepsilon)}_{\nu}. \tag{13.7}$$

Writing this compactly,

$$Y = \alpha_{\text{rf}} + \phi\, Z + \gamma_{\text{rf}}^\top X + \nu, \tag{13.8}$$

where $\alpha_{\text{rf}} = \alpha + \beta a_T$, $\phi = \beta\pi$, $\gamma_{\text{rf}} = \beta\delta + \gamma$, and $\nu = \beta\eta + \varepsilon$. The reduced-form equation for T is simply the first-stage equation Equation 13.2 itself, which is already in reduced form because Z and X are exogenous.

Two features of the reduced form are immediate from Equation 13.7. First, both right-hand-side variables (Z and X) are exogenous, so OLS applied to Equation 13.8 is consistent for ϕ. Second, $\phi = \beta\pi$ is a product of the structural parameter of interest β and the first-stage strength π. Neither β nor π is separately identified from the reduced form alone; identifying β requires also using the first-stage equation — which is the essence of IV.

Because the exclusion restriction rules out any direct path $Z \to Y$, the full effect of the instrument on the outcome flows through the treatment channel $Z \to T \to Y$, so $\phi = \beta\pi$ exactly — and not $\phi = \beta\pi + \kappa$ for some additional direct-effect coefficient $\kappa \neq 0$ that would be present if Z entered the outcome equation directly.

The reduced form regression. The reduced form regression is simply OLS applied to the reduced form equation Equation 13.8. Its purpose is to estimate the reduced-form coefficient ϕ.

Definition 13.3 (Reduced Form Regression Estimator)**.** The **reduced form regression estimator** is the OLS estimator of ϕ in the reduced form equation Equation 13.8:

$$\hat{\phi}_{\mathrm{RF}} = \frac{\widehat{\mathrm{Cov}}(\tilde{Z}, Y)}{\widehat{\mathrm{Var}}(\tilde{Z})}, \tag{13.9}$$

where $\tilde{Z}$ is the residual from projecting Z on $(1, X^\top)^\top$. It is a consistent estimator of $\phi = \beta\pi$ under the reduced-form exogeneity condition $\mathbb{E}[\nu \mid Z, X] = 0$. Since $\nu = \beta\eta + \varepsilon$, this condition follows from two conditions: $\mathbb{E}[\varepsilon \mid Z, X] = 0$, instrument exogeneity for the structural outcome disturbance; and $\mathbb{E}[\eta \mid Z, X] = 0$, the first-stage conditional-mean restriction under which π and δ are identified by OLS. Exclusion is a distinct structural restriction and is already encoded by the absence of Z from the outcome equation Equation 13.1; it entered when the reduced form Equation 13.7 was derived, not here. All three conditions are assumed throughout this chapter.

Intent-to-treat interpretation. When $Z \in \{0, 1\}$ is binary and randomly assigned (as in a lottery or randomized encouragement design), $\hat{\phi}_{\mathrm{RF}}$ estimates the *intent-to-treat* (ITT) effect: the average effect on Y of being assigned $Z = 1$ rather than $Z = 0$, regardless of whether treatment is actually taken up. For a continuous or multi-valued instrument, ϕ is instead the coefficient in the specified linear reduced-form projection, and carries an ITT reading only when a particular intervention contrast on Z is separately defined. The ITT is a valid causal estimand in its own right — it requires only exogeneity of Z, not the exclusion restriction — and it is often of direct policy interest when the instrument corresponds to an implementable intervention.

Relationship to the first stage and the IV estimator. The first-stage regression estimator is

$$\hat{\pi}_{\mathrm{FS}} = \frac{\widehat{\mathrm{Cov}}(\tilde{Z}, T)}{\widehat{\mathrm{Var}}(\tilde{Z})}.$$

Since $\phi = \beta\pi$, we have $\beta = \phi/\pi$, and the sample analog gives

$$\hat{\beta}_{\mathrm{IV}} = \frac{\hat{\phi}_{\mathrm{RF}}}{\hat{\pi}_{\mathrm{FS}}}, \tag{13.10}$$

which reproduces the IV regression estimator Equation 13.6. The reduced form delivers the instrument's total effect on the outcome; the first stage scales it by the instrument's effect on treatment; the ratio removes the scaling and recovers the structural parameter. Equation Equation 13.10 also shows that the IV estimator is undefined when $\hat{\pi}_{\mathrm{FS}} = 0$ — the sample-level manifestation of the relevance requirement.

The ratio form Equation 13.10 is useful because it exposes the logic of IV, not because all IV estimation should be carried out by separately estimating and then dividing two coefficients. In practice, 2SLS (Section 13.3) solves the estimating equations jointly and produces valid standard errors automatically; forming $\hat{\phi}_{\mathrm{RF}}/\hat{\pi}_{\mathrm{FS}}$ by hand and dividing their separate standard errors is incorrect.

> Remark: Reporting All Three Quantities
>
> In applied work it is standard practice to report the first stage, reduced form, and IV (or 2SLS) estimates side by side, as in the KIPP table of Chapter 7. The reduced form and first stage each have a transparent OLS interpretation and can be assessed independently before the ratio is formed. Reporting all three allows the reader to verify instrument strength (first stage), the ITT effect (reduced form), and the per-unit-of-treatment causal effect (IV/2SLS) simultaneously. The IV estimate carries no identifying content beyond what the reduced form and first stage together contain.

13.3 Two-Stage Least Squares

The Wald estimator is limited to binary instruments and no covariates. Two-stage least squares (2SLS) extends the identification-to-estimation step to settings with continuous or multi-valued instruments, multiple instruments, and observed covariates. The key idea is that 2SLS replaces the endogenous regressor by its projection onto the instrument space, thereby retaining only the variation in treatment that is explained by the exogenous instruments and covariates. Although it is defined through a two-stage procedure below, 2SLS is a single estimator with a closed-form expression; the stages are a transparent derivation device, not a description of two separate estimation problems.

13.3.1 Setup

Return to the structural model Equation 13.1–Equation 13.2, where T is a scalar endogenous variable, $X \in \mathbb{R}^p$ is a vector of exogenous covariates, and $Z \in \mathbb{R}^q$ is a vector of instruments with $q \geq 1$. The IV assumptions require:

1. **Relevance**: the first-stage coefficient $\pi \neq 0$. Provided the residualized second-moment matrix $\mathbb{E}[\tilde{Z}\tilde{Z}^\top]$ is nonsingular, this is equivalent to $\mathrm{Cov}(\tilde{Z}, T) \neq 0$, where $\tilde{Z} = Z - L(Z \mid 1, X)$ is the linear-projection residual of Z on the intercept and X.
2. **Exclusion**: Z does not enter the structural outcome equation Equation 13.1 directly.
3. **Instrument exogeneity**: $\mathbb{E}[\varepsilon \mid Z, X] = 0$.
4. **First-stage conditional-mean specification**: $\mathbb{E}[\eta \mid Z, X] = 0$.

Exclusion is a restriction on which variables appear in the structural equation; exogeneity is a conditional-mean restriction on the structural disturbance. They are logically separate and are listed separately here. The last two conditions are stronger than what GMM requires: the estimation theory of Section 13.5–Section 13.6 uses only the unconditional moments $\mathbb{E}[W\varepsilon] = 0$. The conditional-mean form is maintained here because it is what justifies interpreting the first-stage and reduced-form OLS fits as conditional expectations rather than as mere linear projections. Endogeneity means $\mathrm{Cov}(T, \varepsilon) \neq 0$, arising because ε and η share an unobserved common driver.

13.3.2 Stage 1: The First-Stage Regression

Regress T on Z and X by OLS using the first-stage model $T = a_T + \pi^\top Z + \delta^\top X + \eta$, and denote the OLS fitted values $\hat{T}_i = \hat{a}_T + \hat{\pi}^\top Z_i + \hat{\delta}^\top X_i$. In matrix form, $\hat{\mathbf{T}} = P_W \mathbf{T}$ where P_W is the projection onto the column space of the instrument design matrix $\mathbf{W}$ with rows $W_i = (1, X_i^\top, Z_i^\top)^\top$. The fitted value $\hat{T}_i$ is the linear projection of T_i onto the space spanned by the intercept, Z_i, and X_i. Because OLS residuals are orthogonal to the regressors, $\hat{T}_i$ is by construction uncorrelated (in the sample) with the first-stage residual $\hat{\eta}_i = T_i - \hat{T}_i$. The fitted values $\hat{T}_i$ are not used because they predict treatment well in a generic regression sense; they are used because they isolate the component of treatment variation spanned by $(1, Z, X)$ — the variation that is exogenous under IV validity.

13.3.3 Stage 2: The Second-Stage Regression

Regress Y on $\hat{T}$ and X by OLS: $Y = \alpha + \beta\hat{T} + \gamma^\top X + \text{error}$.

Definition 13.4 (2SLS Estimator)**.** The *two-stage least squares estimator* $\hat{\beta}_{\text{2SLS}}$ is the OLS coefficient on $\hat{T}$ in the second-stage regression.

Equivalently, 2SLS projects the endogenous regressors onto the linear span of the instruments and exogenous covariates, and then uses only that projected component in the structural regression. The component of T orthogonal to (Z, X) — the residual $\hat{\eta}_i = T_i - \hat{T}_i$, which is correlated with ε_i when T is endogenous — is dropped before the causal coefficient is estimated. The fitted values $\hat{T}_i$ are not literally randomized; they are the variation in T that is linearly explained by (Z, X), and their exogeneity with respect to ε is inherited from the maintained IV assumptions (instrument exogeneity and exclusion), not guaranteed by the projection algebra alone.

To obtain a closed form, let $\tilde{Y}_i$, $\tilde{T}_i$, and $\tilde{Z}_i$ denote the residuals from projecting Y_i, T_i, and Z_i, respectively, onto X_i by OLS (i.e., the within-X-residuals). In the single-instrument case ($q = 1$), the 2SLS estimator simplifies to

$$\hat{\beta}_{2SLS} = \frac{\sum_{i=1}^{n} \tilde{Z}_i Y_i}{\sum_{i=1}^{n} \tilde{Z}_i T_i} = \frac{\widehat{\mathrm{Cov}}(\tilde{Z}, Y)}{\widehat{\mathrm{Cov}}(\tilde{Z}, T)}. \tag{13.11}$$

This is structurally parallel to the Wald ratio: a ratio of the instrument-by-outcome sample covariance to the instrument-by-treatment sample covariance, after partialling out the observed covariates X.

Standard Errors from the Second-Stage OLS Are Incorrect

A common error is to report the standard errors from the second-stage OLS regression directly. Those standard errors use $\hat{T}$ rather than T and residuals that do not equal the structural errors ε_i, so the resulting standard errors are invalid. Correct inference requires the sandwich variance formula derived in Section 13.6, or software that implements 2SLS natively.

13.4 Equivalence of the IV Regression Estimator and 2SLS

The IV regression estimator Equation 13.6 and the 2SLS estimator look like different procedures: one computes a ratio of partialled-out covariances directly, while the other constructs first-stage fitted values and runs a second regression. In the single-instrument case they are in fact numerically identical for any covariate vector X. This theorem is not just algebraic housekeeping: it shows that the ratio-of-covariances view and the staged-regression view are two computational faces of the same IV estimator. Knowing both is useful — the ratio form exposes the logic of IV estimation while the staged-regression form generalizes to multiple instruments and connects naturally to GMM.

Theorem 13.1 (IV Regression–2SLS Equivalence)**.** *In the linear IV model with a scalar instrument Z and covariates $X \in \mathbb{R}^p$, the IV regression estimator and the 2SLS estimator coincide: $\hat{\beta}_{2SLS} = \hat{\beta}_{IV}$.*

The intuition is compact. In Section 13.2 we derived $\hat{\beta}_{IV}$ by replacing the invalid (S-T) with the valid instrument equation and solving the system (S-X), (S-Z), obtaining $\hat{\beta}_{IV} = (\mathbf{Z}^\top M_X \mathbf{T})^{-1} \mathbf{Z}^\top M_X \mathbf{Y}$. The 2SLS procedure reaches the same formula by a projection argument.

Proof. Retain the matrix notation of Section 13.2: $\bar{\mathbf{X}} = [\mathbf{1}_n, \mathbf{X}]$, $M_X = I_n - \bar{\mathbf{X}}(\bar{\mathbf{X}}^\top \bar{\mathbf{X}})^{-1} \bar{\mathbf{X}}^\top$, scalar instrument $\mathbf{Z} \in \mathbb{R}^n$.

First-stage projection. The first-stage fitted value is $\hat{\mathbf{T}} = P_{[\bar{\mathbf{X}}, \mathbf{Z}]} \mathbf{T}$, where $P_{[\bar{\mathbf{X}}, \mathbf{Z}]}$ projects onto $\mathrm{col}(\bar{\mathbf{X}}, \mathbf{Z})$. By the Frisch–Waugh–Lovell theorem applied to Stage 1:

$$M_X \hat{\mathbf{T}} = M_X \mathbf{Z} (\mathbf{Z}^\top M_X \mathbf{Z})^{-1} \mathbf{Z}^\top M_X \mathbf{T}.$$

2SLS second-stage closed form. Applying FWL to the second-stage regression of $\mathbf{Y}$ on $(\hat{\mathbf{T}}, \bar{\mathbf{X}})$:

$$\hat{\beta}_{2SLS} = \frac{\hat{\mathbf{T}}^\top M_X \mathbf{Y}}{\hat{\mathbf{T}}^\top M_X \hat{\mathbf{T}}} = \frac{\mathbf{T}^\top M_X \mathbf{Z} (\mathbf{Z}^\top M_X \mathbf{Z})^{-1} \mathbf{Z}^\top M_X \mathbf{Y}}{\mathbf{T}^\top M_X \mathbf{Z} (\mathbf{Z}^\top M_X \mathbf{Z})^{-1} \mathbf{Z}^\top M_X \mathbf{T}}.$$

Under exact identification ($q = 1$), the scalars $\mathbf{T}^\top M_X \mathbf{Z}$ and $(\mathbf{Z}^\top M_X \mathbf{Z})^{-1}$ cancel from numerator and denominator, leaving

$$\hat{\beta}_{2SLS} = \frac{\mathbf{Z}^\top M_X \mathbf{Y}}{\mathbf{Z}^\top M_X \mathbf{T}} = \hat{\beta}_{IV}. \qquad \square$$

In the scalar-instrument linear model, Wald, IV regression, and 2SLS are therefore not competing methods; they are different representations of the same sample analog of the identification formula.

Remark: What the Proof Reveals

The proof makes the logic of IV transparent at the level of estimating equations. OLS imposes residual orthogonality with every regressor. When T is endogenous, the treatment equation $\mathbf{T}^\top \mathbf{e}(\theta) = 0$ is invalid at the structural parameter, since $\mathbb{E}(T_i \varepsilon_i) \neq 0$. IV retains the valid equations for the exogenous regressors and replaces the invalid treatment equation by the instrument equation $\mathbf{Z}^\top \mathbf{e}(\theta) = 0$, which is valid because Z is exogenous. The result is a method-of-moments construction, not an OLS normal-equation system: the retained equations happen to coincide with OLS equations for the exogenous block, but the system as a whole is not the first-order condition of any least-squares problem. The 2SLS derivation reaches the same estimator from a different direction, by projection.

Remark: Wald Estimator as a Special Case of the Equivalence

When $Z \in \{0,1\}$ is binary and X is absent, the IV regression estimator reduces to the Wald estimator (Definition 13.1), as shown in Section 13.2. Theorem 13.1 therefore implies $\hat{\beta}_{\text{2SLS}} = \hat{\beta}_{\text{Wald}}$ in this case as well.

Remark: Generalization to Multiple Instruments

With $q > 1$ instruments, the IV regression formula no longer applies directly because the system $\sum_i \tilde{Z}_i Y_i / \sum_i \tilde{Z}_i T_i$ is not well defined for vector $\tilde{Z}_i$. 2SLS resolves this by using the first-stage projection of T onto the full instrument vector, producing a single scalar $\hat{T}_i$. The resulting estimator is efficient in the class of linear IV estimators under homoskedasticity and reduces to the IV regression estimator when $q = 1$.

13.5 The Moment-Condition View and GMM

The proof of Theorem 13.1 already contains the key idea: IV estimation amounts to solving the normal equations of the structural model with the invalid T-block replaced by the valid instrument equation $\mathbf{Z}^\top \hat{\mathbf{e}} = 0$. This replacement is not an algebraic trick but a direct expression of the *moment condition* implied by IV validity: $\mathbb{E}[(1, X^\top, Z^\top)^\top \varepsilon] = 0$, which follows from the IV exogeneity statement $\mathbb{E}[\varepsilon \mid Z, X] = 0$ together with the (trivial) exogeneity of the intercept and of X. The advantage of the moment-condition view is that it separates the identifying restrictions from the specific algebra of 2SLS, making clear which part of the problem is about causal assumptions and which part is about estimation efficiency. Recognizing 2SLS as a method-of-moments estimator for this condition connects it to the general estimating-equation framework of Chapter 10 and motivates the generalized method of moments for overidentified models.

13.5.1 2SLS as a Method-of-Moments Estimator

Stack the constant, the covariates, and the instrument into the $(p+1+q)$-dimensional vector $W = (1,\, X^\top,\, Z^\top)^\top$, which contains every variable that should be exogenous under IV validity. The IV moment condition is

$$\mathbb{E}[W\,(Y - \alpha - \beta T - \gamma^\top X)] = 0.$$

Setting $\theta = (\alpha, \beta, \gamma^\top)^\top \in \mathbb{R}^{p+2}$ and defining the moment function $U(O;\, \theta) = W\,(Y - \alpha - \beta T - \gamma^\top X)$, the identifying condition is $\mathbb{E}\{U(O;\, \theta_0)\} = 0$, which is exactly an estimating equation in the sense of Chapter 10. The number of moment conditions is $p + 1 + q$ and the number of parameters is $p + 2$; the model is exactly identified when $q = 1$ and overidentified when $q > 1$.

Theorem 13.2 (2SLS as a Method-of-Moments Estimator)**.** *In the exactly identified linear IV model ($q = 1$), suppose the sample rank condition holds: $n^{-1} \sum_i W_i D_i^\top$ is nonsingular. (Equality of the number of equations and the number of parameters is only the order condition; it does not by itself guarantee a unique solution.) Then the 2SLS estimator $\hat{\theta}_{\text{2SLS}} = (\hat{\alpha},\, \hat{\beta}_{\text{2SLS}},\, \hat{\gamma}^\top)^\top$ is the unique solution to the sample moment system*

$$\frac{1}{n}\sum_{i=1}^{n} W_i\,(Y_i - \hat{\alpha} - \hat{\beta}_{\text{2SLS}}\, T_i - \hat{\gamma}^\top X_i) = 0, \qquad W_i = (1,\, X_i^\top,\, Z_i)^\top \in \mathbb{R}^{p+2}.$$

The first row of this $(p+2)$-dimensional system is the intercept-orthogonality condition (residuals sum to zero); the middle p rows are the X-orthogonality conditions; the last row is the instrument-orthogonality condition.

This moment-condition view clarifies why the first-stage fitted values $\hat{T}_i$ are not merely a computational device. They encode the orthogonality structure implied by instrument validity: replacing T_i with $\hat{T}_i$ ensures the second-stage regression uses only the exogenous variation in treatment. IV is therefore not a separate estimation philosophy from the rest of Part III; it is a special case of the general estimating-equation framework of Chapter 10, with moment functions induced by instrument validity.

13.5.2 Overidentification and GMM

When $q > 1$ instruments are available, the model is *overidentified*: there are more moment conditions than parameters. The system $\mathbb{P}_n U(O;\, \theta) = 0$ is generically overdetermined and has no exact solution.

When the model is overidentified, different linear combinations of the sample moments contain the same target parameter but with different noise levels. GMM chooses the combination that solves the moment equations as well as possible, with a weighting scheme that can exploit the heteroskedastic structure of the moment scores for greater efficiency. The weighting matrix $\hat{\Omega}_n$ determines which combination is used: identity weighting treats all moment conditions equally, while efficient weighting downweights noisy *directions* in the moment vector and accounts for correlation among the moment conditions through the full inverse covariance matrix Σ^{-1}. Only when Σ is diagonal does this reduce to weighting individual moments by their inverse variances.

The *generalized method of moments* (GMM) formalizes this by minimizing a weighted quadratic form in the sample moments:

$$\hat{\theta}_{\mathrm{GMM}} = \arg\min_{\theta} \, [\mathbb{P}_n U(O;\, \theta)]^\top \, \hat{\Omega}_n \, [\mathbb{P}_n U(O;\, \theta)],$$

where $\hat{\Omega}_n$ is a positive-definite weighting matrix. Different choices of $\hat{\Omega}_n$ yield different estimators within the IV class:

- The choice $\hat{\Omega}_n = (n^{-1} \sum_i W_i W_i^\top)^{-1}$ — the inverse of the second-moment matrix of the instrument vector $W_i = (1, X_i^\top, Z_i^\top)^\top$ — yields the 2SLS estimator.
- The choice $\hat{\Omega}_n = \left[\mathbb{P}_n U(O;\, \hat{\theta})\, U(O;\, \hat{\theta})^\top\right]^{-1}$ — the inverse of the estimated moment covariance — yields the *efficient GMM estimator*, which achieves the semiparametric efficiency bound for the IV moment-condition model.

These comparisons have content only when $m > k$. In an exactly identified model every positive-definite weighting matrix yields the same estimator and the same asymptotic variance, so 2SLS and efficient GMM coincide whatever the conditional variance structure. In an overidentified model, homoskedasticity is sufficient for 2SLS to be efficient; under general heteroskedasticity, overidentified 2SLS need not be efficient, and efficient GMM is weakly (often strictly) more efficient.

Remark: Overidentification as a Testable Restriction

An overidentified model imposes more moment conditions than are needed for point identification. If all instruments are valid, the sample moments should be jointly near zero. The Sargan–Hansen J-statistic (Sargan 1958; Hansen 1982) tests this restriction: under the null that all moment conditions hold, $J \xrightarrow{d} \chi^2_{q-1}$. A rejection indicates that the full set of maintained moment conditions is incompatible with the data. This may reflect an exclusion failure, an exogeneity failure, or other model misspecification; the test does not identify which assumption failed or which instrument is responsible. Overidentification therefore allows us to check whether the full set of moment restrictions is jointly compatible with the data, but a failure to reject is consistent with all instruments being valid, or with violations that happen to cancel in the joint test.

Example 13.1 (GMM with Two Instruments)**.** We illustrate the GMM procedure in the simplest overidentified setting: one endogenous regressor, two instruments, no additional covariates.

Setup. Consider the structural model $Y = \beta T + \varepsilon$ (intercept suppressed after demeaning). Two instruments Z_1 and Z_2 are available, giving the moment conditions $\mathbb{E}[Z_1 \varepsilon] = 0$ and $\mathbb{E}[Z_2 \varepsilon] = 0$. The model is overidentified ($q = 2$, one free parameter), so the two moment conditions cannot generally be satisfied simultaneously in finite samples.

Sample moments. Suppose the following sample covariances are observed:

$$\widehat{\mathrm{Cov}}(Z_1, Y) = 0.40, \quad \widehat{\mathrm{Cov}}(Z_1, T) = 0.50, \quad \widehat{\mathrm{Cov}}(Z_2, Y) = 0.60, \quad \widehat{\mathrm{Cov}}(Z_2, T) = 0.80.$$

The just-identified IV estimate using each instrument alone is

$$\hat{\beta}^{(1)} = \frac{0.40}{0.50} = 0.80, \qquad \hat{\beta}^{(2)} = \frac{0.60}{0.80} = 0.75.$$

The two instruments give different point estimates. Under the linear constant-effect model maintained in this example, both moment conditions target the same parameter β, so the discrepancy between $\hat{\beta}^{(1)}$ and $\hat{\beta}^{(2)}$ reflects sampling variation alone. (Under treatment-effect heterogeneity, instruments that move different complier subpopulations can additionally target different local estimands, in which case overidentified combinations weight several distinct LATEs, and the resulting combination need not be a convex average of the instrument-specific local estimands: depending on the first-stage structure and the estimator, some weights can be negative.) The GMM objective combines both moment conditions.

GMM moment function. With $a = (\widehat{\mathrm{Cov}}(Z_1, T),\ \widehat{\mathrm{Cov}}(Z_2, T))^\top = (0.50,\ 0.80)^\top$ and $c = (\widehat{\mathrm{Cov}}(Z_1, Y),\ \widehat{\mathrm{Cov}}(Z_2, Y))^\top = (0.40,\ 0.60)^\top$, the sample moment vector is

$$\hat{U}(\beta) = c - a\beta = \begin{pmatrix} 0.40 - 0.50\beta \\ 0.60 - 0.80\beta \end{pmatrix}.$$

GMM with identity weighting ($\Omega = I_2$). The GMM criterion is $Q(\beta) = \hat{U}(\beta)^\top \hat{U}(\beta)$. The first-order condition $\partial Q/\partial\beta = 0$ gives

$$a^\top(c - a\beta) = 0 \implies \hat{\beta}_{\mathrm{GMM}} = \frac{a^\top c}{a^\top a} = \frac{(0.50)(0.40) + (0.80)(0.60)}{(0.50)^2 + (0.80)^2} = \frac{0.68}{0.89} \approx 0.764.$$

Under identity weighting this is a *first-stage-sensitivity-weighted* average of the two just-identified estimates, with weights proportional to a_j^2: it lies between 0.75 and 0.80, with more weight on Z_2 because it has the larger first-stage covariance ($0.80 > 0.50$). These weights are not in general inverse-variance (precision) weights; identity weighting ignores Σ entirely.

Effect of the weighting matrix. With a diagonal weight $\Omega = \mathrm{diag}(w_1, w_2)$, the GMM estimator becomes

$$\hat{\beta}_{\mathrm{GMM}}(\Omega) = \frac{w_1(0.50)(0.40) + w_2(0.80)(0.60)}{w_1(0.50)^2 + w_2(0.80)^2}.$$

Setting $w_1 = 4$ and $w_2 = 1$ (upweighting Z_1) gives

$$\hat{\beta}_{\mathrm{GMM}} = \frac{4(0.20) + 1(0.48)}{4(0.25) + 1(0.64)} = \frac{1.28}{1.64} \approx 0.780,$$

closer to $\hat{\beta}^{(1)} = 0.80$ as expected. As $w_1/w_2 \to \infty$, the estimate converges to $\hat{\beta}^{(1)}$; as $w_1/w_2 \to 0$, it converges to $\hat{\beta}^{(2)}$. The efficient weighting $\Omega^* = \hat{\Sigma}^{-1}$ (Section 13.6) minimizes the asymptotic variance over all weightings. If $\hat{\Sigma}$ is diagonal, this amounts to choosing $w_j = \hat{\Sigma}_{jj}^{-1}$, so noisier moments receive less weight. In general $\hat{\Sigma}^{-1}$ is *not* diagonal: efficient GMM also uses the off-diagonal covariances among the moment conditions, and so weights directions in moment space rather than each instrument separately — the diagonal family considered in this example does not contain the optimum unless the moments are uncorrelated.

Sargan–Hansen J-test. At $\hat{\beta} \approx 0.764$ under identity weighting, the residual sample moments are

$$\hat{U}_1 = 0.40 - (0.50)(0.764) = 0.018, \qquad \hat{U}_2 = 0.60 - (0.80)(0.764) = -0.011.$$

Neither is exactly zero, reflecting the overidentification. The J-statistic is $J = n\,\hat{U}(\hat{\beta})^\top \hat{\Sigma}^{-1} \hat{U}(\hat{\beta})$, and has a χ_1^2 null distribution (one overidentifying restriction) when evaluated at the efficiently weighted estimator using the corresponding $\hat{\Sigma}$. For this example we adopt the convention $\hat{\Sigma} = I_2$ with $n = 200$, so that identity weighting is the assumed optimal weighting; then

$$J = 200\{(0.018)^2 + (-0.011)^2\} \approx 0.09,$$

far below the 5% critical value $\chi^2_{1,0.95} = 3.84$.

The maintained moment restrictions are therefore not rejected, and that is the whole of the conclusion. Failure to reject does *not* establish that both instruments satisfy exclusion. A rejection could arise from exogeneity failure, exclusion failure, failure of the common constant-effect restriction, functional-form misspecification, or any other failure of the maintained moments, and the test cannot say which.

13.6 Asymptotic Theory of the GMM Estimator

Section 13.5 placed IV estimation within the GMM framework. This section derives the asymptotic distribution of the GMM estimator for the linear IV model and then specializes to 2SLS and to the efficient GMM estimator.

13.6.1 Setup and Notation

Stack the structural regressors and instruments into vectors:

$$D_i = \begin{pmatrix} 1 \\ T_i \\ X_i \end{pmatrix} \in \mathbb{R}^k, \qquad W_i = \begin{pmatrix} 1 \\ X_i \\ Z_i \end{pmatrix} \in \mathbb{R}^m,$$

where $k = p + 2$ and $m = p + 1 + q$. The structural model is $Y_i = D_i^\top \theta_0 + \varepsilon_i$ with $\theta_0 = (\alpha_0, \beta_0, \gamma_0^\top)^\top$, and the IV moment condition is

$$\mathbb{E}[U(O_i;\, \theta_0)] = 0, \qquad U(O_i;\, \theta) = W_i\,(Y_i - D_i^\top \theta).$$

The order condition requires $m \geq k$ (at least as many instruments as parameters), i.e., $q \geq 1$. For identification, the rank condition requires $A = \mathbb{E}[W_i D_i^\top] \in \mathbb{R}^{m \times k}$ to have full column rank k.

Define the moment variance matrix

$$\Sigma = \mathbb{E}[U(O_i;\, \theta_0)\, U(O_i;\, \theta_0)^\top] = \mathbb{E}[\varepsilon_i^2\, W_i W_i^\top] \ \in \mathbb{R}^{m \times m}.$$

Under homoskedasticity $\mathbb{E}[\varepsilon_i^2 \mid W_i] = \sigma^2$, this simplifies to $\Sigma = \sigma^2\, \mathbb{E}[W_i W_i^\top]$.

13.6.2 Asymptotic Distribution of the GMM Estimator

The large-sample logic is the same as for any method-of-moments estimator: sample orthogonality converges to population orthogonality by the law of large numbers, and the estimator fluctuates around the truth according to how sensitive the moment equations are to the parameter and how noisy the sample moments are. The sensitivity is captured by the matrix $A = -\mathbb{E}[\partial U / \partial \theta^\top]$, which measures how sharply the moments respond to a change in θ; the noise is captured by $\Sigma = \mathbb{E}[U(O;\theta_0) U(O;\theta_0)^\top]$, the variance of the moment scores at the truth. The asymptotic variance is a sandwich that combines both.

Recall that the GMM estimator minimizes $[\mathbb{P}_n U(O;\theta)]^\top \widehat{\Omega}_n [\mathbb{P}_n U(O;\theta)]$ for some positive-definite weighting matrix $\widehat{\Omega}_n \xrightarrow{p} \Omega$.

Theorem 13.3 (Asymptotic Distribution of the GMM Estimator)**.** *Suppose the standard GMM regularity conditions hold: the IV moment condition is satisfied at a unique population minimizer θ_0 lying in the interior of a compact parameter space of fixed finite dimension; $O_i = (Y_i, T_i, X_i, Z_i)$ are i.i.d. with finite fourth moments; the number of moment conditions m is fixed; $\widehat{\Omega}_n \xrightarrow{p} \Omega$ positive definite; $\hat{\theta}_{\text{GMM}} \xrightarrow{p} \theta_0$; A has full column rank; and Σ is positive definite. Then*

$$\sqrt{n}\,(\hat{\theta}_{\text{GMM}} - \theta_0) \ \xrightarrow{d}\ N(0,\ V_{\text{GMM}}(\Omega)),$$

where

$$V_{\text{GMM}}(\Omega) = (A^\top \Omega A)^{-1}\, A^\top \Omega\, \Sigma\, \Omega A\, (A^\top \Omega A)^{-1}. \tag{13.12}$$

> Intuition: the Sandwich Structure
>
> The asymptotic variance has the same logic as every sandwich formula in semiparametric estimation: one part measures how responsive the moments are to the parameter (A), and another measures how variable the moments themselves are (Σ). Estimation is more precise when the moments react strongly to the parameter (large A) and fluctuate little across samples (small Σ). The weighting matrix Ω scales the two layers but does not alter their fundamental roles.

The formula Equation 13.12 is the *sandwich variance* for GMM, and is an instance of the general estimating-equation theory from Chapter 10: the sensitivity matrix A plays the role of $-\mathbb{E}[\partial U / \partial \theta^\top]$ and the moment variance Σ plays the role of $\mathbb{E}[UU^\top]$.

13.6.3 Two Important Special Cases

Exactly identified case ($q = 1$, $m = k$). When there is a single instrument, A is square and invertible by the rank condition. The GMM objective has an exact solution $\mathbb{P}_n U(O; \hat{\theta}) = 0$ regardless of Ω, so all weighting matrices yield the same estimator. The sandwich variance simplifies to

$$V_{\mathrm{IV}} = A^{-1}\,\Sigma\,(A^{\top})^{-1}, \tag{13.13}$$

matching the general moment-estimator formula from Chapter 10. In the scalar no-covariate case ($p = 0$), $W_i = (1, Z_i)^{\top}$ and $D_i = (1, T_i)^{\top}$, so $A = \mathbb{E}[W_i D_i^{\top}]$ is a 2×2 matrix. Partitioning the inverse and reading off the block corresponding to β yields the scalar variance

$$V_\beta = \frac{\mathbb{E}[\varepsilon^2 \tilde{Z}^2]}{\left(\mathbb{E}[\tilde{Z}\,T]\right)^2}, \qquad \tilde{Z} = Z - \mathbb{E}[Z].$$

Centering the *instrument* is essential when an intercept is included. The uncentered expression $\mathbb{E}[\varepsilon^2 Z^2]/(\mathbb{E}[ZT])^2$ is valid under the convention $\mathbb{E}[Z] = 0$; if in addition $\mathbb{E}[T] = 0$, then $\mathbb{E}[ZT] = \mathrm{Cov}(Z, T)$. Mean-centering Y and T alone does *not* remove the need to center Z: the numerator $\mathbb{E}[\varepsilon^2 Z^2]$ still differs from $\mathbb{E}[\varepsilon^2 \tilde{Z}^2]$ whenever $\mathbb{E}[Z] \neq 0$. Under homoskedasticity $\mathbb{E}[\varepsilon^2 \mid Z] = \sigma^2$ and the first-stage relation $T = a_T + \pi Z + \eta$ with $\mathbb{E}[\eta \mid Z] = 0$, the variance reduces to

$$V_\beta = \frac{\sigma^2}{\pi^2\,\mathrm{Var}(Z)}.$$

Efficient GMM. Among all choices of Ω, the asymptotic variance $V_{\mathrm{GMM}}(\Omega)$ is minimized (in the positive semidefinite sense) by $\Omega^* = \Sigma^{-1}$. Substituting into Equation 13.12:

$$V_{\mathrm{eff}} = \left(A^{\top}\Sigma^{-1}A\right)^{-1}. \tag{13.14}$$

This is the *efficient GMM variance* and is the semiparametric efficiency lower bound for IV estimation within the linear IV moment-restriction model defined by $\mathbb{E}[W\varepsilon] = 0$; it is the best achievable variance among all estimators that use only these moment conditions, but it does not in general coincide with efficiency bounds from richer semiparametric models that exploit additional structure. For any other weighting matrix Ω, $V_{\mathrm{GMM}}(\Omega) - V_{\mathrm{eff}} \succeq 0$.

> Remark: When 2SLS Is Efficient
>
> Begin with the exactly identified case $m = k$. There A is square and invertible, every positive-definite Ω yields the same estimator, and
>
> $$A^{-1}\Sigma(A^{\top})^{-1} = (A^{\top}\Sigma^{-1}A)^{-1},$$
>
> so 2SLS *is* efficient GMM regardless of the conditional variance structure — heteroskedasticity included. The efficiency question only has content when $m > k$.
> In the overidentified case, the 2SLS estimator corresponds to the weighting matrix $\Omega_{\mathrm{2SLS}} = \mathbb{E}[W_i W_i^{\top}]^{-1}$. Under homoskedasticity, $\Sigma = \sigma^2 \mathbb{E}[WW^{\top}]$, so $\Sigma^{-1} = \sigma^{-2}\mathbb{E}[WW^{\top}]^{-1} \propto \Omega_{\mathrm{2SLS}}$. Since scalar multiples of Ω do not change the minimizer, 2SLS achieves V_{eff} under homoskedasticity. Under heteroskedasticity, Σ is not in general a scalar multiple of $\mathbb{E}[WW^{\top}]$, and the optimal weighting $\hat{\Omega}_n = (\mathbb{P}_n \hat{\varepsilon}_i^2 W_i W_i^{\top})^{-1}$ is weakly more efficient than Ω_{2SLS}, with strict improvement except in the special heteroskedastic designs where Σ happens to remain proportional to $\mathbb{E}[WW^{\top}]$ — one simple sufficient condition for no gain, though equality can also occur in special designs where the 2SLS and efficient-GMM influence matrices coincide on the identifying directions. Whenever homoskedasticity is plausible, efficient GMM brings no gain over 2SLS; under heteroskedasticity, efficient GMM is generically strictly more efficient, and the gain can be substantial when the conditional variance of ε varies sharply with W. This tells students when they can stop at 2SLS without sacrificing asymptotic efficiency.

13.6.4 Consistent Variance Estimation

Let $\hat{\varepsilon}_i = Y_i - D_i^\top \hat{\theta}$ denote the structural residuals evaluated at the GMM estimates. Consistent estimators of A and Σ are

$$\hat{A} = \frac{1}{n}\sum_{i=1}^{n} W_i D_i^\top, \qquad \hat{\Sigma} = \frac{1}{n}\sum_{i=1}^{n} \hat{\varepsilon}_i^2 \, W_i W_i^\top.$$

Substituting into Equation 13.12 gives the heteroskedasticity-robust sandwich variance estimator:

$$\hat{V}_{\mathrm{GMM}} = (\hat{A}^\top \hat{\Omega}_n \hat{A})^{-1}\, \hat{A}^\top \hat{\Omega}_n \,\hat{\Sigma}\, \hat{\Omega}_n \hat{A}\, (\hat{A}^\top \hat{\Omega}_n \hat{A})^{-1}. \tag{13.15}$$

In the exactly identified case, $\hat{A}$ is square and Equation 13.15 reduces to $\hat{A}^{-1}\hat{\Sigma}(\hat{A}^\top)^{-1}$, independent of $\hat{\Omega}_n$. For the efficient GMM estimator, the optimal weighting is $\hat{\Omega}_n = \hat{\Sigma}^{-1}$, and the variance estimator simplifies to $(\hat{A}^\top \hat{\Sigma}^{-1} \hat{A})^{-1}$.

Do Not Forget the Factor $1/n$

Theorem 13.3 is stated for $\sqrt{n}(\hat{\theta} - \theta_0)$, so $\hat{V}_{\mathrm{GMM}}$ in Equation 13.15 estimates the covariance matrix of the *root-n scaled* estimator, not of $\hat{\theta}$ itself. Standard errors for the coefficients are therefore

$$\widehat{\mathrm{Var}}(\hat{\theta}) = \frac{\hat{V}_{\mathrm{GMM}}}{n}, \qquad \widehat{\mathrm{se}}(\hat{\theta}_j) = \sqrt{\frac{\hat{V}_{\mathrm{GMM},jj}}{n}}.$$

Reporting $\sqrt{\hat{V}_{\mathrm{GMM},jj}}$ directly inflates every standard error by a factor of $\sqrt{n}$.

In applications, the default variance estimator should rarely be the i.i.d. formula. Heteroskedasticity-robust, and when appropriate cluster-robust, standard errors are the practical baseline for IV work. For independent clusters $c = 1, \ldots, G$, define the *cluster score* $\hat{S}_c = \sum_{i \in c} \hat{\varepsilon}_i W_i$, and estimate the moment covariance by

$$\hat{\Sigma}_{\mathrm{cl}} = \frac{1}{n}\sum_{c=1}^{G} \hat{S}_c \hat{S}_c^\top. \tag{13.16}$$

Substituting $\hat{\Sigma}_{\mathrm{cl}}$ for $\hat{\Sigma}$ in Equation 13.15 and dividing the result by n gives $\widehat{\mathrm{Var}}(\hat{\theta})$ as above. Note that $\hat{\Sigma}_{\mathrm{cl}}$ reduces to the heteroskedasticity-robust $\hat{\Sigma}$ when every cluster is a singleton. The asymptotics here are in the number of clusters, not the number of observations: they require independence across clusters, $G \to \infty$, no single cluster dominating the sample, and finite cluster-level second moments.

Cluster-robust standard errors are required whenever observations within a group share unmodeled common shocks — as in the KIPP application of Chapter 7, revisited in Section 13.7.4, where each student contributes several test observations and the errors are clustered at the student level.

Remark: Two-Step Efficient GMM

Because the optimal weighting $\hat{\Omega}_n = \hat{\Sigma}^{-1}$ requires preliminary residuals $\hat{\varepsilon}_i$, efficient GMM is typically implemented in two steps: obtain a consistent first-step estimator (e.g., 2SLS) to form $\hat{\varepsilon}_i$, construct $\hat{\Sigma}$, and then re-minimize the GMM criterion with $\hat{\Omega}_n = \hat{\Sigma}^{-1}$. The resulting estimator achieves V_{eff} asymptotically.

13.7 Weak Instruments and Inferential Fragility

Every result in Section 13.6 was obtained under *strong identification*.

Where the Standard Asymptotic Theory Stops

Theorem 13.3 assumes that the population Jacobian $A = \mathbb{E}(W D^\top)$ has full rank and stays bounded away from singularity along the sequence of models being approximated. The weak-instrument sequences studied in this section violate exactly that regularity condition. Consequently the normal approximation of Theorem 13.3 and the Wald confidence interval built from the sandwich variance estimator Equation 13.15 are *not uniformly valid*: their coverage can be arbitrarily poor over the region of the parameter space in which the first stage is close to zero, no matter how large n is.

The GMM and 2SLS estimators are well behaved when the instrument strongly predicts treatment. When the first-stage relationship is weak, IV estimation becomes fragile.

It is worth being precise about what kind of fragility this is. A small fixed first-stage coefficient is not an asymptotic impossibility: if $\pi \neq 0$ is held fixed, increasing n eventually restores consistency and asymptotic normality. Weak-instrument asymptotics instead study *sequences* such as $\pi_n = c/\sqrt{n}$, along which the concentration parameter stays bounded and the conventional normal approximation does not improve with sample size. That device is a modeling choice designed to approximate the finite-sample behavior of a design in which π is small relative to its sampling variability, and it is the right approximation to have in mind whenever the first stage is weak relative to n. A weak instrument is not merely an efficiency problem: it distorts the sampling distribution of IV estimators, pulls their center toward OLS, and undermines conventional confidence intervals. These are not asymptotic curiosities but practical hazards that arise routinely in empirical work, and they are among the most important practical concerns in IV design.

13.7.1 The Weak-Instrument Problem

The 2SLS closed form Equation 13.11 divides by the sample covariance $\widehat{\mathrm{Cov}}(\tilde{Z}, T)$. When the population first-stage coefficient π is near zero, this denominator is near zero in the population and highly variable in finite samples. Dividing a noisy numerator by a noisy, near-zero denominator is the source of every symptom listed below (Bound et al. 1995).

- **Center pulled toward OLS.** Under weak-instrument asymptotics the center of the 2SLS sampling distribution is pulled toward the OLS probability limit.
- **Non-Gaussian sampling distribution.** The distribution of $\hat{\beta}_{2SLS}$ can be markedly skewed, bimodal, or heavy-tailed, so the $N(0, V_{2SLS})$ approximation is unreliable.
- **Size distortion.** Wald-type confidence intervals derived from the asymptotic normal approximation can severely undercover the true parameter, and the corresponding t-tests can reject far more often than their nominal level.

The first of these three statements requires care, because the phrase "finite-sample bias toward OLS" is often used more loosely than the underlying mathematics permits.

In What Sense Is 2SLS "Biased Toward OLS"?

Under standard weak-instrument asymptotics, the center of the 2SLS distribution is pulled toward the OLS probability limit in a local-to-zero, Nagar-expansion, or median-bias sense. In just-identified ratio models, however, the ordinary finite-sample moments of $\hat{\beta}_{2SLS}$ may fail to exist, so "bias toward OLS" should *not* automatically be read as a statement about $\mathbb{E}(\hat{\beta}_{2SLS} - \beta)$. A fixed nonzero population first stage still yields consistency as $n \to \infty$; weak instruments instead make the finite-sample approximation and conventional Wald inference unreliable.

The distinction matters because it identifies the correct remedy. If the problem were a bias in the usual sense, one would look for a bias correction. The problem is instead that the *approximating distribution* is wrong, which is why the remedy in Section 13.7.3 is a different *test*, not a different point estimator. The figure below conveys the contrast schematically.

13.7.2 First-Stage Diagnostics

The most widely used diagnostic for weak instruments is an F-statistic from the first-stage regression, testing the joint significance of the excluded instruments Z after partialling out X. There is, however, no single "first-stage F": at least three distinct statistics circulate under that name, each with its own critical values.

1. **The conventional (homoskedastic) first-stage F.** This is the classical F-statistic for $H_0: \pi = 0$ computed under homoskedastic, serially uncorrelated errors. Staiger and Stock (1997) argued informally for the rule of thumb $F \geq 10$, and Stock and Yogo (2005) supplied formal critical values,

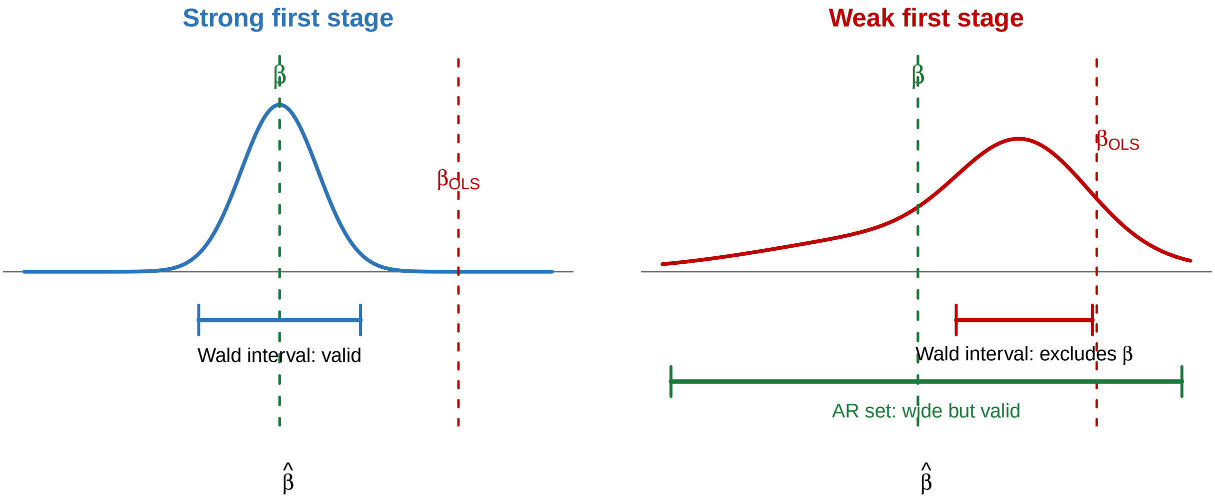

Figure 13.1: Schematic comparison of the sampling behavior of $\hat{\beta}_{2SLS}$ under strong and weak first stages. With a strong first stage the distribution is approximately normal and concentrated at β, and the conventional Wald interval is reliable. With a weak first stage the distribution is displaced toward the OLS probability limit β_{OLS}, skewed, and heavy-tailed; the conventional Wald interval is too short and centered in the wrong place, whereas the Anderson–Rubin set remains valid at the cost of being wide. The curves are conceptual and are not the exact density of any particular weak-instrument model.

tabulating thresholds at which either the 2SLS bias relative to OLS bias or the size of the nominal 5% Wald test exceeds a stated tolerance. These critical values depend on the number of instruments and on which of the two criteria is used; 10 is a rounded stand-in for one cell of that table, not a universal constant.

2. **Robust and cluster-robust first-stage diagnostics.** When the errors are heteroskedastic or correlated within clusters, the conventional F no longer has the distribution assumed by the Stock–Yogo tabulation. The robust analog is a Wald statistic for $\pi = 0$ built from a robust or cluster-robust variance estimator; with multiple endogenous regressors, the Kleibergen–Paap rk Wald F-statistic plays this role.
3. **The effective F-statistic.** Montiel Olea and Pflueger (2013) proposed an *effective* F-statistic designed specifically for the case of heteroskedasticity, serial correlation, or clustering, with its own critical values indexed by a chosen tolerance for relative bias. It is formulated for a *single* endogenous regressor and permits multiple instruments; with several endogenous regressors, further diagnostics are required.

> Do Not Compare a Robust F to the Threshold 10
>
> The number 10 belongs to the conventional homoskedastic first-stage F. A heteroskedasticity-robust, cluster-robust, or effective F-statistic should be compared with the critical value derived for *that* statistic, not with the conventional rule of thumb. Two further cautions apply to all three statistics. First, 10 is a coarse threshold even in its home setting: it may be far too lenient with multiple endogenous regressors or many instruments, and Lee et al. (2022) show that in just-identified models a much larger first-stage F is needed before a conventional t-test attains its nominal size. Second, and more fundamentally, every first-stage diagnostic is a *relevance* diagnostic. A large F says nothing about exogeneity or exclusion: a strong but invalid instrument passes the first-stage test with no difficulty.

Because the diagnostics are noisy and the thresholds are statistic-specific, a natural response is to stop conditioning inference on a preliminary strength test and instead use a procedure whose validity does not depend on first-stage strength at all. That is the subject of the next subsection.

13.7.3 Weak-Instrument-Robust Inference

The **Anderson–Rubin (AR) test** (Anderson and Rubin 1949) inverts the question. Rather than asking whether the data are consistent with a particular estimate, it asks whether a hypothesized value β_0 is consistent with the IV moment condition. Substituting β_0 into the structural model gives

$$Y - \beta_0 T = \alpha + \gamma^\top X + \{\varepsilon + (\beta - \beta_0)T\}.$$

If $\beta_0 = \beta$, the composite error reduces to ε and is uncorrelated with Z by instrument validity. The AR test therefore regresses $Y - \beta_0 T$ on Z and X and tests the null that the coefficient on Z is zero.

The decisive feature of this construction is that the tested restriction is linear in the observables once β_0 is fixed: the null hypothesis has been converted into a statement about a reduced-form coefficient, and no division by an estimated first stage occurs anywhere. Under the classical homoskedastic Gaussian linear model the resulting F-statistic has an *exact* finite-sample F distribution regardless of instrument strength; without normality the same statistic has an asymptotic χ^2_q/q distribution under homoskedasticity, and heteroskedasticity- and cluster-robust versions remain asymptotically valid more generally. In all cases the size of the test does not depend on π.

Definition 13.5 (Anderson–Rubin Confidence Set)**.** For a nominal level $1-a$, the Anderson–Rubin confidence set is obtained by *test inversion*:

$$\mathcal{C}_{1-a} = \{\beta_0 : \text{the AR test of } H_0\colon \beta = \beta_0 \text{ is not rejected at level } a\}.$$

In practice $\mathcal{C}_{1-a}$ is computed by evaluating the AR statistic on a grid of candidate values β_0 and retaining those that are not rejected; in the single-instrument case it can also be obtained in closed form by solving a quadratic inequality in β_0.

Because the AR test has correct size at every β_0 whatever the first-stage strength, $\mathcal{C}_{1-a}$ has correct coverage whether or not the instrument is weak. What varies with instrument strength is not the validity of the set but its *shape*.

Reading an Unusual Anderson–Rubin Set

An inverted AR confidence set may be

- **wide** or even **unbounded** (possibly the whole real line), when the instrument is too weak to rule out large values of β;
- **disconnected**, a union of intervals, when the quadratic inequality defining the set opens downward;
- **empty** — but only in an *overidentified* model, when the maintained moment restrictions are rejected at every candidate value β_0.

The last case cannot arise when the model is just identified with $\hat{\pi} \neq 0$: at $\beta_0 = \hat{\phi}/\hat{\pi}$ the estimated reduced-form coefficient on Z is exactly zero, so the AR statistic is zero and that value is never rejected. A just-identified AR set therefore always contains the point estimate and is never empty. None of these outcomes is a numerical failure. They are substantive information. An unbounded set reports honestly that the data identify β weakly or not at all — precisely the information a spuriously short Wald interval conceals. An empty set is an overidentification rejection: no single value of β reconciles all the instruments, so the maintained model, not just the candidate parameter value, is in doubt.

What Weak-Instrument Robustness Does *Not* Deliver

The AR test is robust to weak instruments *under maintained instrument validity*. Its derivation uses $\mathbb{E}[Z\varepsilon] = 0$ and the exclusion restriction at the very first step. If exogeneity or exclusion fails, the AR test is testing the wrong null, and no amount of identification-robustness repairs it. Identification strength and identification validity are separate questions, and only the first is addressed here.
Robustness to weak identification is also not automatic robustness to arbitrary heteroskedasticity or clustering. Those require the appropriate robust or cluster-robust covariance estimator for the reduced-form coefficient, together with the matching asymptotic reference distribution; the exact F result above holds only in the classical homoskedastic Gaussian model.

The **conditional likelihood ratio (CLR) test** of Moreira (2003) extends the AR idea more efficiently to the multiple-instrument case. With a single endogenous regressor and a single instrument the practical distinction between AR and CLR is small; with several instruments, CLR can be substantially more powerful, because AR spends degrees of freedom testing all q reduced-form restrictions jointly rather than concentrating power on the direction of interest.

Remark: Two Distinct IV Concerns

The weak-instrument problem is a *statistical* concern: it questions the quality of estimation and inference given a valid identification strategy. It is distinct from the *estimand* concern raised in Chapter 7. With a binary instrument and binary treatment under monotonicity, a valid instrument generally identifies a LATE rather than the population ATE; with a continuous or multi-valued treatment, it generally identifies an instrument-specific weighted average causal response. Instrument strength does not convert either local or instrument-specific estimand into a population ATE. Any complete IV analysis must address both concerns.

13.7.4 Applied Example: End-to-End Estimation for the KIPP Lottery

Section 7.11 used the KIPP Lynn admission lottery of Angrist et al. (2012) to illustrate what the IV assumptions identify. We now carry the same design through the estimation machinery of this chapter. Recall the structure: Z_i is the lottery offer, s_{igt} is years of KIPP attendance, y_{igt} is the MCAS score, and the model is just identified with one excluded instrument. The KIPP table of Chapter 7 reports, for math scores, a first stage of $\hat{\pi} = 1.221$ (0.068), a reduced form of $\hat{\phi} = 0.430$ (0.067), and a 2SLS estimate of $\hat{\beta} = 0.352$ (0.053), all with standard errors clustered at the student level and $N = 833$ student-by-test observations.

Step 1: the point estimate. The model is just identified, so 2SLS is the Wald ratio of Section 13.2:

$$\hat{\beta}_{\text{2SLS}} = \frac{\hat{\phi}}{\hat{\pi}} = \frac{0.430}{1.221} = 0.352.$$

Each additional year at KIPP raises math scores by about 0.35 standard deviations. Because the endogenous treatment is *years* of attendance rather than a binary attendance indicator, this is not a binary-complier LATE. Under the maintained monotone-dose interpretation it is a lottery-specific weighted average causal response per additional year of KIPP attendance, over the attendance margins the offer actually shifted — the interpretation given in Chapter 7.

Step 2: the robust standard error. Because students contribute several test observations, the moment function is correlated within student, and the cluster-robust form of the sandwich estimator Equation 13.15 described in Section 13.6 is the appropriate variance estimator; the reported 0.053 is of this form. Equivalently, by the delta method applied to the ratio $\hat{\phi}/\hat{\pi}$,

$$\widehat{\text{se}}(\hat{\beta}) = \frac{1}{|\hat{\pi}|}\Big\{\widehat{\text{se}}(\hat{\phi})^2 - 2\hat{\beta}\,\widehat{\text{Cov}}(\hat{\phi},\hat{\pi}) + \hat{\beta}^2\,\widehat{\text{se}}(\hat{\pi})^2\Big\}^{1/2}.$$

Matching this expression to the reported standard errors implies a correlation of about 0.27 between $\hat{\phi}$ and $\hat{\pi}$. That covariance is not itself reported; it is backed out from rounded published estimates and standard errors, so every quantity below that uses it is a reconstruction from summary statistics rather than a direct analysis of the microdata. The conventional 95% Wald interval is $0.352 \pm 1.96 \times 0.053 = (0.248, 0.456)$.

Step 3: the strength diagnostic. The reported first-stage standard error is cluster-robust, so the statistic this computes is the cluster-robust first-stage Wald F: the t-ratio is $1.221/0.068 = 17.96$, giving $F = 17.96^2 \approx 322$. Because the model is just identified with a single endogenous regressor, this statistic also coincides with the effective F of Section 13.7.2, and it is far above the relevant effective-F critical values, so weak identification is not a practical concern in this application. Note what has and has not been computed: the conventional homoskedastic first-stage F is a different statistic and has not been calculated here. This is the expected outcome for a well-run admission lottery: the offer is a direct administrative determinant of enrollment.

Step 4: the identification-robust interval. With a single instrument, the AR statistic for $H_0 : \beta = \beta_0$ is the squared t-ratio of the coefficient on Z in the regression of $y_{igt} - \beta_0 s_{igt}$ on the covariates and Z. That coefficient is estimated by $\hat{\phi} - \beta_0\hat{\pi}$, so

$$\text{AR}(\beta_0) = \frac{(\hat{\phi} - \beta_0\hat{\pi})^2}{\widehat{\text{se}}(\hat{\phi})^2 - 2\beta_0\widehat{\text{Cov}}(\hat{\phi},\hat{\pi}) + \beta_0^2\,\widehat{\text{se}}(\hat{\pi})^2},$$

and the 95% AR set collects the β_0 with $\mathrm{AR}(\beta_0) \le 1.96^2$. This inequality is quadratic in β_0; solving it with the covariance implied by the rounded reported estimates gives the approximate reconstructed set

$$\mathcal{C}_{0.95} \approx (0.249,\ 0.458),$$

essentially indistinguishable from the Wald interval (0.248, 0.456). That agreement is the point: when the first stage is strong, the identification-robust set and the conventional interval nearly coincide, and the extra work buys reassurance rather than a different answer.

Step 5: what a weak first stage would have done. The coefficient on β_0^2 in the quadratic inequality is $\hat{\pi}^2 - 1.96^2\,\widehat{\mathrm{se}}(\hat{\pi})^2$, which is positive precisely when the first-stage F exceeds $1.96^2 = 3.84$. Suppose counterfactually that the lottery had moved attendance by only $\hat{\pi} = 0.10$ years with the same standard error 0.068 (so $F \approx 2.2$), with a reduced form of $\hat{\phi} = 0.035$ (0.067) chosen to leave the point estimate at 0.35. The conventional Wald interval would then be $(-0.918, 1.618)$: wide, but bounded and apparently informative. The AR set, by contrast, would be the *entire real line*. Both procedures use the same data; only one of them reports honestly that those data exclude nothing.

Step 6: interpretation. Four distinct claims have been kept separate throughout, and a complete report should keep them separate too.

- *Instrument validity* — exogeneity holds by randomization of the lottery; exclusion is an assumption, defended institutionally in Section 7.11 and not testable here, since the model is just identified.
- *Instrument strength* — established empirically by $F \approx 322$, which licenses the conventional normal approximation.
- *The estimand* — with heterogeneous effects, 0.352 is a lottery-specific weighted average causal response per additional year of attendance, over the dose margins shifted by the offer (Chapter 7). It is neither a binary-complier LATE nor the population ATE.
- *Sampling uncertainty* — quantified by the cluster-robust interval (0.248, 0.456), confirmed by the AR set (0.249, 0.458).

Chapter 7 supplied the first and third of these; this chapter supplies the second and fourth.

Advanced Enrichment: Section 13.8

The core development of the chapter is complete. The remaining section presents one extension beyond it. It presupposes familiarity with Section 13.2–Section 13.7 but can be omitted without loss of continuity, and nothing later in the book depends on it. Section 13.8 introduces generalized empirical likelihood (GEL) as a one-step alternative to efficient GMM for the same moment restrictions. Its convex-duality derivation, minimum-discrepancy representation, computational algorithm, extended special-cases table, and additional problems are collected in the Chapter 13 Supplement.

13.8 Generalized Empirical Likelihood

13.8.1 GEL as a One-Step Moment Estimator

Section 13.6 showed that efficient GMM attains the bound $(A^\top \Sigma^{-1} A)^{-1}$ but requires two steps: a preliminary consistent estimator is used to form $\hat{\Sigma}$, which is then plugged in as the weighting matrix for a second optimization. That structure raises a practical question of how many iterations to run, and the estimated weighting matrix contributes a source of finite-sample bias (Newey and Smith 2004).

Generalized empirical likelihood (GEL) is a one-step alternative that uses exactly the same moment restrictions. The contrast with GMM is structural rather than substantive:

> GMM minimizes a quadratic form in the sample moments, with a separately estimated weighting matrix; GEL profiles a concave criterion over a multiplier attached to the same moments, and the optimal weighting emerges from that inner problem rather than being supplied in advance.

Retain the notation of Section 13.6: let $U_i(\theta) = U(O_i;\theta) = W_i(Y_i - D_i^\top \theta)$ denote the m-dimensional moment function, and write $\bar{U}(\theta) = n^{-1}\sum_{i=1}^{n} U_i(\theta)$.

GEL introduces a strictly convex function $G\colon \mathcal{V} \to \mathbb{R}$, where $\mathcal{V}$ is an open interval containing zero. The GEL estimator solves the saddle-point problem

$$\hat{\theta}_{\text{GEL}} = \arg\min_{\theta} \sup_{\lambda\in\Lambda_n(\theta)} \frac{1}{n}\sum_{i=1}^{n}[-G(\lambda^\top U_i(\theta))], \tag{13.17}$$

where $\lambda \in \mathbb{R}^m$ is an auxiliary dual variable and $\Lambda_n(\theta) = \{\lambda : \lambda^\top U_i(\theta) \in \mathcal{V},\ i = 1, \dots, n\}$ is the feasible set for λ given θ. No preliminary weighting matrix appears anywhere in Equation 13.17: the inner supremum over λ performs the role of weighting implicitly. The sign and normalization convention of Equation 13.17 is used throughout the chapter and the supplement.

We normalize G to satisfy

$$G(0) = 0, \qquad g(0) = 1, \qquad g'(0) = 1, \tag{13.18}$$

writing $g = G'$. For the regular generators considered here — those with nonzero first derivative and positive finite second derivative at zero, as all named members have — the conditions $g(0) = g'(0) = 1$ can be imposed by an affine rescaling of the criterion together with a corresponding rescaling of λ, without changing the estimator of θ; they ensure that the GEL first-order conditions reduce to the efficient GMM conditions at the population level. The condition $G(0) = 0$ is a normalization of the additive constant of G; it likewise leaves the estimator unchanged, but it pins down the correspondence between G and its convex conjugate in the duality developed in the supplement.

Remark: The Dual Variable as Observation Weights

At the solution, the quantities $\hat{\omega}_i = g(\hat{\lambda}^\top U_i(\hat{\theta}))$ act as weights attached to individual observations, and normalizing them to sum to one produces a re-weighted *weight vector* — an empirical distribution proper when the weights are positive, as they are for EL and ET — under which the sample moment condition holds exactly. This is the sense in which GEL re-weights *observations* where GMM re-weights *moments*. The precise statement — including the distinction between the raw weights $\hat{\omega}_i$ and the normalized probabilities $\hat{\pi}_i$, which is easy to get wrong — is developed in the Chapter 13 Supplement: the exact profile identities in its section on profile weights, and the normalization question alongside them.

13.8.2 Representative Members: EL, ET, and CUE

Different choices of G give different members of the family. Three are standard.

Definition 13.6 (Representative Members of the GEL Family)**.**

1. **Empirical likelihood (EL):** $G(v) = -\log(1 - v)$, $\mathcal{V} = (-\infty, 1)$.
2. **Exponential tilting (ET):** $G(v) = e^v - 1$, $\mathcal{V} = \mathbb{R}$.
3. **Continuous updating estimator (CUE):** $G(v) = v + v^2/2$, $\mathcal{V} = \mathbb{R}$.

Each satisfies the normalization Equation 13.18.

The third case connects the GEL family directly back to GMM.

Theorem 13.4 (CUE as a GEL Estimator (Newey and Smith 2004))**.** *If $G(v)$ is quadratic (i.e. $G(v) = v + v^2/2$ up to rescaling), then $\hat{\theta}_{\text{GEL}} = \hat{\theta}_{\text{CUE}}$, where the continuous updating estimator solves*

$$\hat{\theta}_{\text{CUE}} = \arg\min_{\theta} \bar{U}(\theta)^\top \left[n^{-1}\textstyle\sum_i U_i(\theta)U_i(\theta)^\top\right]^{-1} \bar{U}(\theta).$$

Unlike two-step efficient GMM, CUE uses a weighting matrix that is continuously updated as θ changes, so no preliminary estimator is needed. It has the same first-order asymptotics as efficient GMM but in practice requires a global nonlinear optimizer, because the criterion is no longer quadratic in θ. Among the three named members, CUE is the most transparent computationally and EL is often preferred for its bias properties; ET, whose weights $e^{\lambda^\top U_i}$ are positive and defined for every real argument, avoids the domain restriction that constrains EL. Additional members, the full table of generating functions and their conjugates, and the two-loop algorithm used in practice are given in the Chapter 13 Supplement.

13.8.3 First-Order Equivalence with Efficient GMM

The central asymptotic result is that the choice within the family, and the choice between the family and efficient GMM, is asymptotically immaterial.

Theorem 13.5 (First-Order Asymptotic Equivalence (Newey and Smith 2004))**.** *Suppose the moment condition $\mathbb{E}[U(O;\theta_0)] = 0$ holds with θ_0 in the interior of a compact parameter space of fixed finite dimension, O_i are i.i.d. with sufficient moments, the rank condition holds so that identification is strong in the sense required at the start of Section 13.7, and G satisfies the regularity conditions of Newey and Smith (2004). Then for any GEL estimator,*

$$\sqrt{n}\,(\hat{\theta}_{\mathrm{GEL}} - \theta_0) \xrightarrow{d} N(0,\ (A^\top \Sigma^{-1} A)^{-1})\,,$$

the same limiting distribution as the efficient GMM estimator.

The theorem is stated under the same strong-identification premise flagged in Section 13.7; it is a statement about which estimators agree in large samples, not a statement about which approximations are trustworthy in a given data set. Each GEL member also supplies an overidentification statistic — a profile-divergence analog of the Sargan–Hansen J-statistic of Section 13.5 — with the same χ^2_{m-k} null limit; its exact form and its relation to J are derived in the Chapter 13 Supplement.

13.8.4 What GEL Does and Does Not Improve

Differences among asymptotically equivalent estimators appear at higher order.

Remark: A Qualified Higher-Order Statement

Newey and Smith (2004) show that, to higher asymptotic order, GEL estimators remove the bias component that arises in two-step GMM from estimating the Jacobian $\mathbb{E}[\partial U_i(\theta_0)/\partial\theta^\top]$, and that EL additionally removes the component arising from estimating the weighting matrix Σ. These are genuine improvements in models with several overidentifying restrictions, the number of moment conditions being held fixed, where such bias components are a real concern. They are also *particular* higher-order components, established under additional smoothness and moment assumptions, and they are not statements about many-instrument asymptotics in the modern sense of m growing with n. No universal finite-sample ranking of GEL over two-step GMM follows, and none should be asserted: which estimator performs better in a given design remains an empirical question.

The Boundary of What GEL Solves

GEL changes how a given moment model is *estimated*. It does not change the model. In particular, GEL does not solve

- invalid instruments — if $\mathbb{E}[W\varepsilon] \neq 0$, every member of the family is inconsistent, exactly as GMM is;
- weak identification — Theorem 13.5 assumes strong identification, and the fragility described in Section 13.7 applies to GEL as well;
- incorrect structural moment conditions — a misspecified $U_i(\theta)$ is misspecified under any estimation criterion;
- poor support or finite-sample feasibility problems. For the positive-weight members EL, ET, and Hellinger, feasibility requires the origin to lie in the appropriate convex hull of the sample moments $\{U_i(\theta)\}$, which can fail for small n or many moments. CUE permits signed weights and imposes no convex-hull requirement; it instead requires nonsingularity of the sample second-moment matrix entering its quadratic inner problem.

GEL belongs to the same maintained model as GMM, and inherits every identification assumption of that model.

Remark: Where the Rest of the GEL Theory Lives

The Chapter 13 Supplement develops the material omitted here: the Fenchel-conjugate derivation and the minimum-discrepancy primal problem, the raw-versus-normalized weight distinction and the exact profile identities, the full table of generators, conjugates, and implied divergences, the two-loop computational algorithm, higher-order comparisons, and additional problems.

13.9 Chapter Summary

This chapter developed the main estimation tools for the linear instrumental-variable model. The organizing principle is the same as in Chapter 10: identification provides a target, and the estimator is a sample analog of the identification formula. Chapter 7 established what the IV assumptions identify; this chapter showed how the resulting moment restrictions are estimated, how the associated uncertainty is quantified, and why weak identification requires different inferential tools. The chapter's architecture is the single chain

$$\text{identification} \longrightarrow \text{sample moments} \longrightarrow \text{2SLS/GMM} \longrightarrow \text{robust inference},$$

and the key results along it are the following.

1. **Wald estimator, reduced form, and first stage.** The **Wald estimator** is the direct sample analog of the Wald estimand for a binary instrument with no covariates: the ratio of the reduced-form difference-in-means to the first-stage difference-in-means. The first stage answers how strongly the instrument moves treatment; the reduced form answers how the instrument changes the outcome; the ratio converts the intent-to-treat effect into a per-unit-of-treatment causal effect under the IV assumptions.

2. **IV regression estimator and 2SLS.** The **IV regression estimator** extends the Wald estimator to the general single-instrument setting with covariates X, as the ratio of within-X instrument-outcome to instrument-treatment sample covariances, derived from the valid sample moment equations. **Two-stage least squares** extends this further to multiple instruments: the first stage projects T onto (Z, X) and the second stage regresses Y on the fitted values and X.

3. **Equivalence and moment conditions.** The IV regression estimator and 2SLS are **numerically identical** in the single-instrument case for any X (Theorem 13.1), with the Wald estimator as the further special case $Z \in \{0, 1\}$, X absent. Both are **method-of-moments estimators** based on the IV orthogonality condition $\mathbb{E}[W\varepsilon] = 0$, an instance of the general estimating-equation framework of Chapter 10.

4. **GMM and efficient weighting.** In an exactly identified model all positive-definite weighting matrices give the same estimator and the same asymptotic variance, so 2SLS and efficient GMM coincide whatever the conditional variance structure. In *overidentified* models, **efficient GMM** generalizes 2SLS and achieves the semiparametric efficiency bound within the maintained IV moment-restriction model; homoskedasticity is sufficient for 2SLS to attain that bound, while under general heteroskedasticity overidentified 2SLS need not, and optimally weighted GMM is weakly (often strictly) more efficient.

5. **Robust variance estimation under strong identification.** Provided the population Jacobian $A = \mathbb{E}(WD^\top)$ has full rank and is bounded away from singularity, the **asymptotic distribution** of the GMM estimator is normal with sandwich variance $V_{\text{GMM}} = (A^\top \Omega A)^{-1} A^\top \Omega \Sigma \Omega A (A^\top \Omega A)^{-1}$, where A is the sensitivity matrix and Σ is the moment variance. In applications, the default should be heteroskedasticity-robust standard errors; cluster-robust standard errors are required when observations within groups share common shocks.

6. **Weak instruments and identification-robust inference.** Strong identification is a premise, not a guarantee. When the first stage is weak, the sampling distribution of $\widehat{\beta}_{\text{2SLS}}$ is displaced toward the OLS probability limit and is neither normal nor well approximated by one, and conventional Wald intervals undercover. "Bias toward OLS" is a statement about the center of a limiting approximation, not necessarily about $\mathbb{E}(\widehat{\beta}_{\text{2SLS}} - \beta)$, whose finite-sample moments may not exist. First-stage diagnostics come in conventional, robust, and effective versions with different critical values, and all of them speak only to relevance, never to validity. **Anderson–Rubin** confidence sets, obtained by inverting a test of $H_0 : \beta = \beta_0$, have correct coverage regardless of first-stage strength; they may be wide, unbounded, or disconnected, and — in an overidentified model — empty. Each of those shapes is informative rather than a malfunction.

7. **GEL as an optional extension. Generalized empirical likelihood** provides alternative one-step implementations of the same moment restrictions — empirical likelihood, exponential tilting, and the continuous updating estimator among them — and shares the first-order limiting distribution of efficient GMM under standard regularity and strong-identification conditions (Theorem 13.5). Higher-order bias improvements are available for particular components under additional conditions; no universal finite-sample ranking follows, and GEL changes only how a moment model is estimated, never whether the model is correct.

Remark: Omitted Topics and Further Reading

Two topics adjacent to this chapter are deliberately left out. The *control function* approach rewrites the endogeneity problem by augmenting the outcome regression with the first-stage residual; in the linear model it reproduces the 2SLS coefficient exactly, so it adds a representation rather than an estimator, and its useful extensions require a triangular structural model with a scalar monotone first-stage disturbance — a different set of assumptions from those maintained here (Imbens and Newey 2009). *Many-instrument* and *many-weak-instrument* asymptotics, in which the number of moment conditions grows with n, likewise change the approximating sequence rather than the estimator. Both belong to a separate treatment of nonlinear and high-dimensional IV models.
More generally, this chapter stays within a low-dimensional linear model with a fixed number of moment conditions. The general principles of orthogonal scores, sample splitting, and cross-fitting — which allow nuisance components to be estimated by machine learning at slower than $\sqrt{n}$ rates — are developed in Chapter 12, with the ATE as the principal example. Their IV-specific implementations, including orthogonalized first-stage and covariate-adjustment scores, lie outside the scope of these notes.

13.10 Exercises

1. The Wald estimator as a ratio-of-moments estimator. Consider the binary-instrument, no-covariate setting of Section 13.2. Let $\mu_Y(z) = \mathbb{E}[Y \mid Z = z]$ and $\mu_T(z) = \mathbb{E}[T \mid Z = z]$ for $z \in \{0, 1\}$, and write $\Delta_Y = \mu_Y(1) - \mu_Y(0)$, $\Delta_T = \mu_T(1) - \mu_T(0)$.

(a) Augment β with an intercept parameter α and express the structural model $Y = \alpha + \beta T + \varepsilon$ as the solution to the two-dimensional moment condition

$$\mathbb{E}\left[\begin{pmatrix} 1 \\ Z \end{pmatrix} (Y - \alpha - \beta T)\right] = 0,$$

so that $U(O; \alpha, \beta) = (1, Z)^\top (Y - \alpha - \beta T) \in \mathbb{R}^2$. Verify that the system has two moment conditions and two parameters (α, β), so the order condition for exact identification is satisfied. State the accompanying rank condition — here $\Delta_T \neq 0$ — and explain why counting equations alone does not guarantee a unique solution. Then solve the system and recover $\beta = \Delta_Y / \Delta_T$ together with an expression for α.

(b) Using the fact that $\bar{Y}_z \xrightarrow{p} \mu_Y(z)$ and $\bar{T}_z \xrightarrow{p} \mu_T(z)$, prove that $\hat{\beta}_{\text{Wald}} \xrightarrow{p} \beta$ via the continuous mapping theorem. State the condition on Δ_T that is required.

(c) Apply the delta method to the vector $(\hat{\Delta}_Y, \hat{\Delta}_T)^\top$ to show that

$$\sqrt{n}\,(\hat{\beta}_{\text{Wald}} - \beta) \;\xrightarrow{d}\; N\left(0,\; \frac{1}{\Delta_T^2} \sum_{z \in \{0,1\}} \frac{\text{Var}(Y_i - \beta T_i \mid Z_i = z)}{p_z}\right),$$

where $p_z = \Pr(Z = z)$. Confirm that this matches the IV variance formula Equation 13.13 in the scalar no-covariate case ($p = 0$).

2. Matrix form of 2SLS and why the second-stage standard errors are wrong. Stack observations into matrices: let $\mathbf{Y} \in \mathbb{R}^n$, $\mathbf{D} \in \mathbb{R}^{n \times k}$ be the full regressor matrix (including the intercept, T, and X), and $\mathbf{W} \in \mathbb{R}^{n \times m}$ be the full instrument matrix (including the intercept, X, and Z). Let $P_{\mathbf{W}} = \mathbf{W}(\mathbf{W}^\top\mathbf{W})^{-1}\mathbf{W}^\top$ be the orthogonal projection onto the column space of $\mathbf{W}$.

(a) Show that the 2SLS estimator can be written as $\hat{\theta}_{\text{2SLS}} = (\mathbf{D}^\top P_{\mathbf{W}} \mathbf{D})^{-1} \mathbf{D}^\top P_{\mathbf{W}} \mathbf{Y}$. (*Hint*: the first stage replaces $\mathbf{D}$ by its projection $\hat{\mathbf{D}} = P_{\mathbf{W}}\mathbf{D}$; then apply OLS to the second-stage regression of $\mathbf{Y}$ on $\hat{\mathbf{D}}$, using idempotency of $P_{\mathbf{W}}$.)

(b) In the single-instrument, no-covariate case ($k = m = 2$, intercept plus T or Z), verify from the matrix formula above that $\hat{\beta}_{2SLS} = \hat{\beta}_{IV}$, reproducing Theorem 13.1 without the Frisch–Waugh–Lovell argument.

(c) The second-stage OLS regression uses the fitted regressor matrix $\hat{\mathbf{D}}$ in place of $\mathbf{D}$. Let $\hat{\varepsilon}_{2nd} = \mathbf{Y} - \hat{\mathbf{D}}\hat{\theta}_{2SLS}$ denote the second-stage residuals actually used by OLS. Show that $\hat{\varepsilon}_{2nd} \neq \mathbf{Y} - \mathbf{D}\hat{\theta}_{2SLS}$ in general. Explain why this discrepancy makes the second-stage OLS standard errors invalid, and identify the correct residuals that should enter the sandwich variance estimator Equation 13.15.

3. Efficiency of GMM and the Sargan–Hansen J-statistic.

(a) Prove that $V_{GMM}(\Omega) \succeq V_{eff}$ for every positive-definite Ω, where $V_{eff} = (A^\top \Sigma^{-1} A)^{-1}$ is the efficient GMM variance Equation 13.14. (*Hint*: factor $V_{GMM}(\Omega) - V_{eff}$ as a product of the form $C^\top \Sigma^{-1} C$ for a suitable matrix C by completing the square; recall that $B^\top \Sigma^{-1} B \succeq 0$ for any matrix B and positive-definite Σ.)

(b) Under homoskedasticity, show that 2SLS is the efficient GMM estimator by verifying that the 2SLS weighting matrix $\Omega_{2SLS} = \mathbb{E}[W_i W_i^\top]^{-1}$ is a scalar multiple of Σ^{-1}. Conclude that no efficiency gain over 2SLS is possible in the homoskedastic linear IV model.

(c) Return to Example 13.1 with two instruments and no covariates. The GMM estimator under identity weighting gives $\hat{\beta} \approx 0.764$ and residual sample moments $\hat{U}_1 = 0.018$, $\hat{U}_2 = -0.011$. Treating $\hat{\Sigma} = I_2$ and $n = 200$, compute the Sargan–Hansen J-statistic $J = n\,\hat{U}(\hat{\beta})^\top \hat{\Sigma}^{-1} \hat{U}(\hat{\beta})$ and determine, using the χ^2_1 critical value at the 5% level, whether the overidentifying restriction is rejected. Interpret the result in terms of the joint maintained moment restrictions, and explain why failure to reject does not establish that either instrument is valid.

4. Constructing a cluster-robust sandwich by hand. Consider a just-identified IV model with $W_i = (1, Z_i)^\top$ and $D_i = (1, T_i)^\top$, so $m = k = 2$. Suppose $n = 6$ observations fall into $G = 3$ clusters, $c_1 = \{1, 2\}$, $c_2 = \{3, 4\}$, $c_3 = \{5, 6\}$, and that the estimated moment contributions $\hat{\varepsilon}_i W_i \in \mathbb{R}^2$ are

$\hat{\varepsilon}_1 W_1 = (0.6,\ 1.2)^\top$	$\hat{\varepsilon}_2 W_2 = (0.4,\ 0.8)^\top$	$\hat{\varepsilon}_3 W_3 = (-0.5,\ 0.3)^\top$
$\hat{\varepsilon}_4 W_4 = (-0.3,\ -0.1)^\top$	$\hat{\varepsilon}_5 W_5 = (0.2,\ -0.7)^\top$	$\hat{\varepsilon}_6 W_6 = (-0.4,\ -1.5)^\top$

(a) Compute the three cluster scores $\hat{S}_c = \sum_{i \in c} \hat{\varepsilon}_i W_i$, and then form $\hat{\Sigma}_{cl}$ using Equation 13.16.

(b) Compute the heteroskedasticity-robust $\hat{\Sigma} = n^{-1} \sum_i \hat{\varepsilon}_i^2 W_i W_i^\top$ for the same data, and compare the two. Which entries grow, and what feature of the within-cluster scores explains the difference? Construct a rearrangement of the same six vectors into three clusters for which *both diagonal elements* of $\hat{\Sigma}_{cl}$ are smaller than the corresponding diagonal elements of $\hat{\Sigma}$, and explain why clustering is not automatically conservative.

(c) Suppose $\hat{A} = n^{-1} \sum_i W_i D_i^\top$ equals $\begin{pmatrix} 1 & 2 \\ 0.5 & 3 \end{pmatrix}$. Form the exactly identified sandwich $\hat{V} = \hat{A}^{-1} \hat{\Sigma}_{cl} (\hat{A}^\top)^{-1}$ and report the cluster-robust standard error of $\hat{\beta}$. Do not forget the factor $1/n$.

(d) State the conditions under which the cluster-robust approximation is justified, and explain why $G = 3$ would be inadequate in practice even though the arithmetic goes through.

5. The continuous updating estimator as a GEL member. For the continuous updating estimator (CUE), the GEL generating function is $G(v) = v + v^2/2$ on $\mathcal{V} = \mathbb{R}$.

(a) Verify that G satisfies the GEL normalization Equation 13.18.

(b) Solve the inner supremum over λ in Equation 13.17 explicitly at fixed θ: show that

$$\hat{\lambda} = -\left[n^{-1} \sum_i U_i(\theta) U_i(\theta)^\top\right]^{-1} \bar{U}(\theta),$$

and substitute back to confirm that the profile objective equals $\frac{1}{2} \bar{U}(\theta)^\top [n^{-1} \sum_i U_i(\theta) U_i(\theta)^\top]^{-1} \bar{U}(\theta)$, establishing Theorem 13.4.

(c) Explain why the factor $\frac{1}{2}$ in part (b) has no effect on $\hat{\theta}_{CUE}$, and why the objective in Theorem 13.4, unlike the two-step efficient GMM criterion, is not quadratic in θ even in the linear IV model.

6. Weak-instrument-robust inference by Anderson–Rubin. Consider the linear IV model $Y_i = \alpha + \beta T_i + \varepsilon_i$ with a scalar instrument Z_i. Maintain the joint moment restriction

$$\mathbb{E}\left[\begin{pmatrix} 1 \\ Z_i \end{pmatrix} \varepsilon_i\right] = 0,$$

that is, $\mathbb{E}[\varepsilon_i] = 0$ together with $\mathbb{E}[(Z_i - \mathbb{E}Z_i)\varepsilon_i] = 0$, but do *not* assume that the first-stage coefficient is bounded away from zero.

(a) For a proposed value β_0, define $R_i(\beta_0) = Y_i - \beta_0 T_i$. Show that under $H_0\colon \beta = \beta_0$, $\mathbb{E}[Z_i\{R_i(\beta_0) - \alpha\}] = 0$.

(b) Explain why testing H_0 can therefore be implemented by testing the coefficient on Z_i in a regression of $R_i(\beta_0)$ on an intercept and Z_i, and identify where the intercept moment $\mathbb{E}[\varepsilon_i] = 0$ is used. (With only $\mathbb{E}[Z_i\varepsilon_i] = 0$ and no intercept restriction, the population slope in this regression is $\mathrm{Cov}(Z_i, \varepsilon_i)/\mathrm{Var}(Z_i)$, which need not vanish.) State how the procedure should be modified when additional exogenous covariates X_i are included, using the residualized instrument $\tilde{Z}_i = Z_i - L(Z_i \mid 1, X_i)$.

(c) Explain why the null-size validity of this test does not require the first-stage coefficient to be bounded away from zero. Contrast this with the Wald interval built from Theorem 13.3, and identify exactly which step of that construction fails when $\pi \approx 0$.

(d) Describe how inversion of the test over candidate values β_0 produces an Anderson–Rubin confidence set (Definition 13.5).

(e) Fix a significance level $a \in (0, 1)$ (the symbol α is in use as the structural intercept). Write the level-a AR set as the solution of a quadratic inequality $c_2\beta_0^2 + c_1\beta_0 + c_0 \leq 0$ in the just-identified case, using the reduced-form and first-stage coefficient estimates $(\hat{\phi}, \hat{\pi})$ and their joint covariance matrix. Show that the sign of c_2 is determined by whether the first-stage Wald statistic $\hat{\pi}^2/\widehat{\mathrm{Var}}(\hat{\pi})$ exceeds the critical value, and use this to enumerate the possible shapes of the set: a bounded interval, an unbounded or disconnected set, a half-line in the degenerate linear case, or the whole real line.

(f) Show that in this just-identified setting the AR set can *never* be empty when $\hat{\pi} \neq 0$, by evaluating the AR statistic at $\beta_0 = \hat{\phi}/\hat{\pi}$. Then explain why an AR set *can* be empty in an overidentified model, and what an empty set says about the maintained moment restrictions.

(g) Reproduce the two numerical calculations of Section 13.7.4: the AR set for the reported KIPP first stage $\hat{\pi} = 1.221$, and the AR set for the counterfactual weak first stage $\hat{\pi} = 0.10$. Confirm that the first is essentially the Wald interval and the second is the whole real line, and explain in one or two sentences what the contrast shows about relying on a Wald interval when the first stage is uncertain.

(h) Explain why weak-instrument robustness does not protect the procedure against violation of instrument exogeneity or exclusion. In particular, explain what the AR test is testing when the exclusion restriction fails, and why an empty AR set is evidence against the maintained model rather than against any particular value of β.

Chapter 13 Supplement: Convex Duality, Minimum Discrepancy, and Computation

This supplement collects the material referenced as optional in Section 13.8: the convex-conjugate duality between the GEL saddle-point problem and a minimum-discrepancy primal problem, the distinction between raw dual weights and normalized probabilities, the full table of generating functions and their conjugates, the two-loop computational algorithm, higher-order comparisons with two-step GMM, and additional problems. References of the form Equation 13.17 or Section 13.8 point back into the main chapter, and the notation is that of Section 13.8.1: $U_i(\theta) = W_i(Y_i - D_i^\top \theta)$, $\bar{U}(\theta) = n^{-1}\sum_i U_i(\theta)$, G is the strictly convex generating function with $g = G'$, and $\Lambda_n(\theta) = \{\lambda : \lambda^\top U_i(\theta) \in \mathcal{V} \text{ for all } i\}$.

S13.1 Convex-Conjugate Duality

The GEL saddle-point problem Equation 13.17 is the Lagrangian dual of a natural primal problem that re-weights observations rather than moments. The bridge between the two is the Legendre–Fenchel conjugate of G.

The Primal: Minimum-Discrepancy Estimation

The *minimum-discrepancy* (MD) estimator minimizes a convex divergence between a vector of observation weights $\omega = (\omega_1, \dots, \omega_n)$ and the reference weight $\omega_i = 1$ (one unit of mass per observation), subject to the sample moment condition being exactly satisfied:

$$\hat{\theta}_{\mathrm{MD}} = \arg\min_{\theta,\, \omega_i \in \mathcal{W}} \sum_{i=1}^{n} F(\omega_i) \quad \text{subject to} \quad \sum_{i=1}^{n} \omega_i U_i(\theta) = 0, \tag{13.19}$$

where F is a strictly convex function with minimum at $\omega = 1$. The domain of F is member-specific and must be carried along explicitly. Write

$$\mathcal{W} = \operatorname{int}\operatorname{dom} F,$$

so that the primal optimization is taken over $\omega_i \in \mathcal{W}$. For EL, ET, and Hellinger, $\mathcal{W} = (0, \infty)$; for CUE, $F(\omega) = \frac{1}{2}(\omega - 1)^2$ is finite on the whole line and $\mathcal{W} = \mathbb{R}$, so CUE weights may be negative. This distinction propagates through Equation 13.19, Theorem 13.6, and the feasibility discussion below, and it is the reason the convex-hull condition stated later cannot be applied uniformly across members.

The problem asks: what is the re-weighting of observations, closest to the uniform reference $\omega_i = 1$ in the F-divergence sense, that exactly satisfies the moment restriction?

Note that Equation 13.19 imposes *no* constraint $\sum_i \omega_i = n$. This is a modeling choice with consequences that are developed in Section S13.3; it is made here because it produces a duality in the single dual variable λ, whereas imposing the normalization explicitly introduces a second Lagrange multiplier.

The Conjugate Relationship

The key structural fact is that the MD divergence F and the GEL generating function G are related through the Legendre–Fenchel conjugate. Define

$$F(\omega) = \sup_{v \in \mathcal{V}} \left[\omega v - G(v)\right], \tag{13.20}$$

the Legendre–Fenchel conjugate of G. Under the Legendre-type assumptions of Theorem 13.6 below (properness, closedness, strict convexity, differentiability, and essential smoothness of G), the conjugate $F = G^*$ is essentially strictly convex and differentiable on $\operatorname{int}\operatorname{dom} F$, with the inverse-gradient relation $F' = (G')^{-1}$; strict convexity of G alone would not deliver these differentiability properties. The normalization Equation 13.18 determines the behavior of F at the reference point $\omega = 1$: the supremum in Equation 13.20 is attained at $v^* = 0$ (since the first-order condition gives $\omega = g(v^*)$, so $\omega = 1 \Rightarrow v^* = 0$), yielding

$$F(1) = 1 \cdot 0 - G(0) = 0, \qquad f(1) = v^*(1) = 0, \tag{13.21}$$

where $f = F'$ and the second equality uses the envelope theorem. The location of the minimizer follows entirely from $g(0) = 1$: the first-order condition $\omega = g(v^*)$ gives $v^* = 0$ at $\omega = 1$, and hence $f(1) = 0$, regardless of the value of $G(0)$. The condition $G(0) = 0$ has no bearing on which ω minimizes F; it normalizes the *level* of F so that $F(1) = 0$, making the divergence interpretable as a non-negative cost with zero cost at the reference weight $\omega_i = 1$.

The Duality Theorem

Theorem 13.6 (Fixed-θ Fenchel Duality)**.** *Let G be proper, closed, strictly convex, differentiable, and essentially smooth on the open interval $\mathcal{V} \ni 0$, and satisfy the normalization Equation 13.18. Let $F = G^*$ be its Legendre–Fenchel conjugate Equation 13.20, with $\mathcal{W} = \operatorname{int}\operatorname{dom} F$. Fix θ, and suppose the primal problem*

$$\inf_{\omega_i \in \mathcal{W}} \left\{ \sum_{i=1}^{n} F(\omega_i) : \sum_{i=1}^{n} \omega_i U_i(\theta) = 0 \right\}$$

has a relative-interior feasible point and that the primal and dual optima are attained. Then strong duality holds:

$$\inf_{\omega} \sum_{i=1}^{n} F(\omega_i) = \sup_{\lambda \in \Lambda_n(\theta)} \left\{ -\sum_{i=1}^{n} G\big(\lambda^\top U_i(\theta)\big) \right\},$$

and the optimal weights satisfy

$$\hat{\omega}_i(\theta) = g\big(\hat{\lambda}(\theta)^\top U_i(\theta)\big), \qquad f(\hat{\omega}_i(\theta)) = \hat{\lambda}(\theta)^\top U_i(\theta).$$

If in addition the outer optima over θ are attained, then minimizing the common profiled value over θ yields the same set of estimators, so $\hat{\theta}_{\mathrm{GEL}} = \hat{\theta}_{\mathrm{MD}}$.

Two points about the hypotheses deserve emphasis, because weaker statements circulate.

First, the normalization Equation 13.18 is nowhere near sufficient on its own. Properness, closedness, essential smoothness, differentiability with invertible gradient on the relevant interior, primal feasibility with a relative-interior point, and attainment of the optima all do real work. Nor does strict convexity of G by itself deliver every property of F used later: it gives strict convexity of the conjugate on the appropriate domain, but the differentiability and inverse-gradient identity $f^{-1} = g$ require the essential-smoothness and Legendre-type conditions.

Second, the estimator-level conclusion is obtained from equality of the *globally profiled objective values* under strong duality, not from the observation that first-order conditions coincide. The outer problem in θ is generally nonconvex, so matching stationarity conditions would establish only that the two procedures share critical points.

Feasibility Is Member-Specific

The Slater-type condition is often summarized as "the origin must lie in the relative interior of the convex hull of $\{U_i(\theta)\}$" (the interior proper if the sample moments span $\mathbb{R}^m$). That is the correct condition for the *positive-weight* members EL, ET, and Hellinger, for which $\mathcal{W} = (0, \infty)$, and it can fail when n is small relative to the number of moment conditions. It is *not* the condition for CUE.

Because $\mathcal{W} = \mathbb{R}$ there, CUE admits signed weights and its inner problem is a concave quadratic in λ; it is well defined whenever the sample second-moment matrix $n^{-1}\sum_i U_i(\theta)U_i(\theta)^\top$ is nonsingular, with no convex-hull requirement at all.

A note on the relation of Theorem 13.6 to the literature. Newey and Smith (2004) develop the minimum-discrepancy representation for the *Cressie–Read family* (Cressie and Read 1984), with explicitly normalized probability weights; in that family $G(v) = (1+\gamma v)^{(\gamma+1)/\gamma}/(\gamma+1) - 1/(\gamma+1)$ and $F(\omega) = [(\omega)^{\gamma+1} - 1 - (\gamma+1)(\omega-1)]/[\gamma(\gamma+1)]$, with EL, ET, and CUE at $\gamma=-1$, $\gamma=0$, and $\gamma=1$. Ragusa (2011) studies the broader relationship between minimum-divergence and GEL formulations, and shows that the standard tractable probability-constrained MD problem corresponds to GEL only for an appropriate subclass of divergences — the correspondence is not free of charge outside that subclass. The *unnormalized* primal Equation 13.19 used here is neither of these: it is introduced directly through Legendre–Fenchel duality, and under the regularity, relative-interior feasibility, attainment, and strong-duality conditions of Theorem 13.6 it is dual to the GEL inner problem for the specified conjugate pair (G,F).

S13.2 Minimum-Discrepancy Formulations and the Profile Identity

Although the MD problem Equation 13.19 jointly optimizes over $(\theta,\omega_1,\dots,\omega_n)$, it can be solved by a two-loop profile procedure that exploits the convexity of F in the weights.

Inner loop: profile weights at fixed θ. For a given θ, the minimization over $(\omega_1,\dots,\omega_n)$ subject to $\sum_i \omega_i U_i(\theta) = 0$ is a convex program. Its Lagrangian is

$$L(\omega,\lambda) = \sum_{i=1}^n F(\omega_i) - \lambda^\top \sum_{i=1}^n \omega_i U_i(\theta),$$

and the first-order condition $f(\omega_i) = \lambda^\top U_i(\theta)$ gives the profile weight

$$\hat\omega_i(\theta) = f^{-1}(\lambda^\top U_i(\theta)) = g(\lambda^\top U_i(\theta)),$$

where the last equality uses the conjugate identity $f^{-1} = g$. Substituting into the moment constraint $\sum_i \hat\omega_i(\theta)U_i(\theta) = 0$ determines the Lagrange multiplier $\hat\lambda(\theta)$ as the solution to

$$\hat\lambda(\theta) = \arg\min_{\lambda\in\Lambda_n(\theta)} \frac{1}{n}\sum_{i=1}^n G(\lambda^\top U_i(\theta)). \tag{13.22}$$

This inner minimization is a convex optimization in λ alone and a damped or line-search Newton method often converges rapidly for each fixed θ when initialized in the feasible interior.

Profile divergence. Once $\hat\omega_i(\theta) = g(\hat\lambda(\theta)^\top U_i(\theta))$ is substituted back into the MD objective, the Fenchel–Young equality $F(\omega) = \omega f(\omega) - G(f(\omega))$ and the moment constraint $\sum_i \hat\omega_i(\theta)U_i(\theta) = 0$ together give

$$Q_{\mathrm{MD}}(\theta) \;\equiv\; \sum_{i=1}^n F(\hat\omega_i(\theta)) = -\sum_{i=1}^n G(\hat\lambda(\theta)^\top U_i(\theta)). \tag{13.23}$$

The profiled divergence therefore reduces to a function of θ alone, evaluated at the inner-loop solution.

Outer loop: minimize over θ. The MD estimator is

$$\hat\theta_{\mathrm{MD}} = \arg\min_\theta \, Q_{\mathrm{MD}}(\theta) = \arg\max_\theta \min_{\lambda\in\Lambda_n(\theta)} \frac{1}{n}\sum_{i=1}^n G(\lambda^\top U_i(\theta)),$$

where the second equality follows from Equation 13.23. This is exactly the GEL saddle-point problem Equation 13.17 written as a max–min, confirming at the algorithmic level that the two procedures find the same estimator.

The Profile Identity Holds for the Raw Weights

The identity Equation 13.23 is stated for the *unnormalized* weights $\hat\omega_i(\theta) = g(\hat\lambda(\theta)^\top U_i(\theta))$ produced by the unconstrained primal Equation 13.19. It is not, without modification, an identity for the normalized probabilities of Section S13.3. See the warning there before reusing it.

S13.3 Raw Dual Weights, Normalized Probabilities, and the Overidentification Statistic

Three objects circulate in this literature under similar names, and conflating them is the most common technical error in accounts of GEL. They are:

1. the **raw dual weights**
$$\hat{\omega}_i = g(\hat{\lambda}^\top U_i(\hat{\theta})), \tag{13.24}$$
which solve the unconstrained primal Equation 13.19 and satisfy $\sum_i \hat{\omega}_i U_i(\hat{\theta}) = 0$ but in general satisfy $\sum_i \hat{\omega}_i \neq n$;
2. the **normalized weights**
$$\hat{\pi}_i = \frac{\hat{\omega}_i}{\sum_{j=1}^n \hat{\omega}_j} = \frac{g\big(\hat{\lambda}^\top U_i(\hat{\theta})\big)}{\sum_{j=1}^n g\big(\hat{\lambda}^\top U_j(\hat{\theta})\big)}, \qquad i = 1, \dots, n, \tag{13.25}$$
which are defined only when the denominator $\sum_j \hat{\omega}_j$ is nonzero — automatic for the positive-weight members, an assumption for CUE, whose signed raw weights could in principle sum to zero. When defined they satisfy $\sum_i \hat{\pi}_i = 1$ and, because normalization is by a common nonzero scalar, also $\sum_i \hat{\pi}_i U_i(\hat{\theta}) = 0$;
3. the **probability-constrained MD weights**, obtained from a primal problem that imposes $\sum_i \pi_i = 1$ explicitly as a second constraint, with its own Lagrange multiplier.

Normalization Preserves the Moment Equation but Changes the Objective

Rescaling by a common positive constant leaves the weighted moment equation $\sum_i \hat{\pi}_i U_i(\hat{\theta}) = 0$ intact, because the right-hand side is zero. It does *not* leave the discrepancy objective intact: $\sum_i F(\hat{\pi}_i \cdot c)$ is not $\sum_i F(\hat{\omega}_i)$ up to an additive constant for general F. Consequently the exact profile identity Equation 13.23, which is derived for the raw weights, should not be reused with normalized weights unless the corresponding normalization multiplier and the resulting objective adjustment are carried along explicitly. Objects (2) and (3) above coincide at the solution for the Cressie–Read members under the standard normalization, but the reasoning that establishes this must be supplied, not assumed.

For EL, ET, and Hellinger, the function g is strictly positive on the relevant domain $\mathcal{V}$, so the raw weights are all strictly positive and the normalized $\hat{\pi}_i$ in Equation 13.25 are genuine probability weights.

Empirical Probabilities and CUE

For CUE, $g(v) = 1 + v$ can be negative when $\hat{\lambda}^\top U_i(\hat{\theta}) < -1$, in which case some $\hat{\omega}_i$ are negative and the normalized $\hat{\pi}_i$ form a signed *weight vector*, not a probability distribution on $\{1, \dots, n\}$. Whenever raw or CUE weights are under discussion, "weight vector" is the accurate term and "empirical distribution" should be avoided, unless an explicit positivity constraint $\hat{\omega}_i > 0$ is imposed. This is one reason CUE is sometimes treated as a GMM-type estimator with a one-step weight-update interpretation rather than as a pure empirical-likelihood method.

The GEL Overidentification Statistic

Theorem 13.7 (Wilks' Theorem for GEL (Newey and Smith 2004))**.** *Under the first-order conditions of Theorem 13.5, together with the additional smoothness, moment, interiority, and profile regularity* *conditions required for the GEL profile statistic (first-order estimator equivalence alone is not sufficient), the GEL profile divergence evaluated at the estimator satisfies*

$$T_{\mathrm{GEL}} \;\equiv\; 2\sum_{i=1}^{n} F(\hat\omega_i) \;=\; -2\sum_{i=1}^{n} G(\hat\lambda^\top U_i(\hat\theta_{\mathrm{GEL}})) \;\xrightarrow{d}\; \chi^2_{m-k}, \tag{13.26}$$

where m is the number of moment conditions, k is the number of parameters, and $m-k$ is the number of overidentifying restrictions.

The two expressions in Equation 13.26 are equal by the profile identity Equation 13.23: the total primal divergence from the reference weights equals minus the profiled GEL objective at the solution. The χ^2_{m-k} limit parallels Wilks' theorem for parametric likelihood-ratio tests: the degrees of freedom count the overidentifying restrictions, and the statistic is zero under exact identification ($m = k$). This is the GEL counterpart of the Sargan–Hansen J-statistic of Section 13.5.

For EL specifically, $F(\omega) = \omega - 1 - \log\omega$. Setting $\hat\omega_i = n\hat\pi_i$ — which for EL is the normalization delivered by the probability-constrained primal, so that $\sum_i \hat\omega_i = n$ — and substituting,

$$T_{\mathrm{EL}} = 2\sum_{i=1}^{n} F(\hat\omega_i) = 2\sum_{i=1}^{n}(n\hat\pi_i - 1 - \log(n\hat\pi_i)) = -2\sum_{i=1}^{n}\log(n\hat\pi_i),$$

the empirical likelihood ratio statistic of Qin and Lawless (1994). This is -2 times the log empirical likelihood ratio of the constrained EL distribution $\{\hat\pi_i\}$ relative to the unconstrained nonparametric maximum at the uniform empirical distribution $\{1/n\}$.

Remark S13.1: Comparison with the Sargan–Hansen J-Statistic

The J-statistic of Section 13.5 equals $n\,\bar U(\hat\theta)^\top\hat\Sigma^{-1}\bar U(\hat\theta)$ and also converges to χ^2_{m-k} under the null. The two statistics test the same null hypothesis — all m moment conditions hold simultaneously — but differ in construction. The J-statistic uses the sample mean $\bar U(\hat\theta)$ and a separate estimate $\hat\Sigma$, while T_{GEL} uses the profile divergence at the re-weighted distribution. For CUE, $F(\omega) = (\omega-1)^2/2$, so $T_{\mathrm{CUE}} = \sum_i(\hat\omega_i - 1)^2$, a quadratic form closely related to the J-statistic evaluated at the continuously updated weighting matrix. Under the null all GEL statistics share the same χ^2_{m-k} limit; they differ in local power and higher-order properties.

Remark S13.2: CUE Objective Conventions and the Factor of $\frac{1}{2}$

The CUE objective in Theorem 13.4 is written without a leading $\frac{1}{2}$, while the divergence $F(\omega) = \frac{1}{2}(\omega-1)^2$ in the table of Section S13.4 carries one. These are two different quantities — the CUE estimator objective minimized over θ, and the per-observation MD divergence F defined as the conjugate of G — so neither is "the right" one. The factor is harmless for the minimizer $\hat\theta_{\mathrm{CUE}}$ but matters when comparing $T_{\mathrm{CUE}} = 2\sum_i F(\hat\omega_i)$ to the Sargan–Hansen J-statistic, which is what motivates the factor of 2 in T_{CUE}. The convention is fixed throughout: the CUE estimator objective omits the $\frac{1}{2}$, the MD divergence F includes it, and the Wilks statistic accounts for the factor explicitly.

S13.4 Detailed Special Cases

The conjugate relationship Equation 13.20 determines the MD divergence F from G. The table below summarizes the four leading special cases; the derivations follow by solving the first-order condition $\omega - g(v^*) = 0$ for v^* and substituting back into Equation 13.20.

Table 13.1: The four main GEL special cases. G is the strictly convex function defining the GEL estimator Equation 13.17; $g = G'$ generates the raw weights Equation 13.24; $F(\omega) = \sup_v [\omega v - G(v)]$ is the Legendre–Fenchel conjugate of G and defines the MD primal Equation 13.19. All four satisfy $G(0) = 0$, $g(0) = 1$, $F(1) = 0$, $f(1) = 0$. The last column evaluates the discrepancy at the *normalized* weights $\bar{\omega}_i = n\hat{\pi}_i$, so that $\sum_i \bar{\omega}_i = n$, with $\hat{\pi}_i$ as in Equation 13.25.

Estimator	$G(v)$	$g(v) = G'(v)$	$F(\omega)$	$\sum_i F(\bar{\omega}_i)$
EL	$-\log(1-v)$	$\dfrac{1}{1-v}$	$\omega - 1 - \log\omega$	$-\sum_i \log(n\hat{\pi}_i)$
Hellinger	$\dfrac{2v}{2-v}$	$\dfrac{4}{(2-v)^2}$	$2(\sqrt{\omega}-1)^2$	$2n\sum_i\left(\sqrt{\hat{\pi}_i} - \frac{1}{\sqrt{n}}\right)^2$
ET	$e^v - 1$	e^v	$\omega\log\omega - \omega + 1$	$n\sum_i \hat{\pi}_i \log(n\hat{\pi}_i)$
CUE	$v + \frac{1}{2}v^2$	$1 + v$	$\frac{1}{2}(\omega-1)^2$	$\frac{n^2}{2}\sum_i(\hat{\pi}_i - \frac{1}{n})^2$

Two features of the table require comment. First, the symbol $\bar{\omega}_i$ is deliberately distinct from the raw weight $\hat{\omega}_i$ of Equation 13.24, which need not sum to n; the warning in Section S13.3 applies to reusing these expressions outside this normalization. Second, each entry in the last column is, up to the factor n, a familiar discrepancy between $\hat{\pi}$ and the uniform reference P_n, the uniform distribution on $\{1, \dots, n\}$: EL gives $n\,D_{\mathrm{KL}}(P_n \| \hat{\pi})$ (reverse KL), ET gives $n\,D_{\mathrm{KL}}(\hat{\pi} \| P_n)$ (forward KL), Hellinger gives $4n\,H^2(\hat{\pi}, P_n)$ under the standard convention $H^2(P, Q) = \frac{1}{2}\sum_i(\sqrt{p_i} - \sqrt{q_i})^2$, and CUE gives $\frac{n}{2}\,\chi^2(\hat{\pi} \| P_n)$ under the standard Pearson convention $\chi^2(\hat{\pi} \| P_n) = \sum_i(\hat{\pi}_i - 1/n)^2/(1/n)$. EL, Hellinger, ET, and CUE correspond to $\gamma = -1$, $\gamma = -\frac{1}{2}$, $\gamma = 0$, and $\gamma = 1$ in the Cressie–Read family, respectively. The domain of G is $\mathcal{V} = (-\infty, 1)$ for EL, $\mathcal{V} = (-\infty, 2)$ for Hellinger, and $\mathcal{V} = \mathbb{R}$ for ET and CUE.

These are discrepancies evaluated at the normalized weights $\bar{\omega}_i = n\hat{\pi}_i$, not at the general raw GEL weights $\hat{\omega}_i$. For EL, $\sum_i F(\bar{\omega}_i) = -\sum_i \log(n\hat{\pi}_i)$ is n times the Kullback–Leibler divergence *from* the uniform reference $1/n$ *to* the EL probabilities $\hat{\pi}$ (the reverse KL); equivalently, EL maximizes $\sum_i \log\hat{\pi}_i$ subject to $\sum_i \hat{\pi}_i = 1$ and the moment restriction, which is the nonparametric maximum likelihood estimator of the observation probabilities. (For EL the distinction is immaterial at the solution: the EL first-order conditions themselves imply $\sum_i \hat{\omega}_i = n$, so raw and normalized weights coincide there. That automatic identity does not hold uniformly across GEL members, which is why the general expressions are stated in $\bar{\omega}_i$.) For ET, $\sum_i F(\bar{\omega}_i) = n\sum_i \hat{\pi}_i \log(n\hat{\pi}_i)$ is n times the KL divergence in the opposite direction (forward KL). For CUE, the divergence is a quadratic distance from the uniform reference $1/n$.

Remark S13.3: Re-weighting Observations vs. Re-weighting Moments

The convex-conjugate duality highlights a structural contrast between GMM and GEL. GMM re-weights the *moment conditions* via the matrix $\hat{\Omega}$, assigning more importance to moment conditions that are precisely estimated. MD — and its GEL dual — instead re-weights the *observations* relative to the reference vector of ones. For the positive-weight members, after normalization the weights define an empirical distribution, and the procedure may be read as selecting the distribution closest to the uniform empirical distribution in the corresponding F-divergence, subject to the moment restrictions. For CUE the raw weights may be signed, so the geometric reading is instead a quadratic perturbation of the reference weight vector, not a probability distribution. The two approaches exploit overidentification through structurally different mechanisms, yet they are asymptotically equivalent to first order.

S13.5 Computation

The GEL saddle-point problem Equation 13.17 is a joint nonlinear optimization in (θ, λ). In practice it is solved by the two-loop scheme of Section S13.2: the outer loop iterates over θ, calling the inner loop at each step to evaluate $\hat{\lambda}(\theta)$ from Equation 13.22 and the gradient of $Q_{\mathrm{MD}}(\theta)$.

Remark S13.4: Connection to the GEL Saddle-Point

The two-loop MD procedure and the GEL saddle-point are two computational routes to the same estimator. The GEL formulation jointly optimizes (θ, λ) in Equation 13.17, while the MD two-loop approach decouples the problem: the inner loop solves a convex minimization in λ for fixed θ, and the outer loop minimizes the profiled divergence in θ. The inner-loop solution $\widehat{\lambda}(\theta)$ from Equation 13.22 is precisely the dual variable appearing in the GEL first-order conditions, and the profile divergence Equation 13.23 is the GEL profile objective.

Three practical points deserve emphasis.

1. **The inner problem is convex; the outer problem is not.** For fixed θ, Equation 13.22 is a smooth convex minimization in $\lambda \in \mathbb{R}^m$, and a damped or line-search Newton method typically converges rapidly when started from a feasible interior point. The outer objective $Q_{\mathrm{MD}}(\theta)$ is generally non-convex in θ even in the linear IV model, so a global optimizer or multiple starting values is advisable. This is most visible for CUE, whose criterion in Theorem 13.4 is a ratio of quadratics.
2. **Feasibility can fail, and how it fails depends on the member.** For the positive-weight members EL, ET, and Hellinger, the inner supremum is finite only when the origin lies in the relative interior of the convex hull of $\{U_i(\theta)\}$ (the interior proper under full-dimensionality of the sample moments). With small n or many moment conditions this fails for some θ, and implementations must either restrict the search region or use a penalized or "adjusted" version of the criterion. CUE is not subject to this restriction: its signed weights make the inner problem a concave quadratic, well defined whenever $n^{-1}\sum_i U_i(\theta)U_i(\theta)^\top$ is nonsingular.
3. **Starting values.** Two-stage least squares provides a consistent and cheap starting value for the outer loop; the corresponding starting value for λ is 0, which is always feasible since $0 \in \mathcal{V}$.

Among the three named members of Definition 13.6, CUE is the most transparent computationally — the inner supremum has a closed form at fixed θ — but its objective is non-convex in θ and requires a global optimizer. EL is often preferred in practice for its bias properties. It also supports likelihood-ratio-type confidence regions with Wilks calibration (Theorem 13.7); under additional smoothness and moment conditions, Bartlett correction and related refinements can improve their coverage accuracy.

S13.6 Higher-Order Comparisons

Newey and Smith (2004) show that, to higher asymptotic order, GEL estimators have smaller bias than standard two-step GMM in two specific respects.

First, GEL eliminates the bias component that arises from estimating the Jacobian

$$\mathbb{E}[\partial U_i(\theta_0)/\partial\theta^\top].$$

This is an important source of finite-sample bias in overidentified IV models. It appears in GMM because the gradient term in the first-order conditions is estimated by a simple sample average rather than by an efficiently weighted one. The inner saddle-point over λ in GEL implicitly uses an efficient estimator of the Jacobian, so this component vanishes.

Second, EL additionally eliminates the bias from estimating the weighting matrix Σ, because EL uses the re-weighted probabilities $\widehat{\pi}_i$ as an efficient estimate of the second moment. Under symmetric disturbances — a condition satisfied in many IV settings — these bias reductions extend to the full GEL family.

What the Higher-Order Results Do and Do Not Say

These are statements about *particular components* of a higher-order expansion, obtained under smoothness and moment conditions beyond those needed for Theorem 13.5, and with the number of moment conditions held fixed. They do not establish a uniform finite-sample ranking of GEL over two-step GMM, and they say nothing about behavior under weak identification, where the expansions themselves are not valid. In particular, EL's bias advantage in overidentified models is often accompanied by higher variance and by the feasibility failures described in Section S13.5, so the mean-squared error comparison is design-specific. The practical conclusion is that GEL estimators can be attractive in overidentified IV models with moderate sample sizes and several overidentifying restrictions, where the Jacobian bias component is a real concern — the number of moments being held fixed in the asymptotic approximation. This conclusion should not be extrapolated to many-instrument sequences in which the number of moments grows with n, and it does not say that GEL dominates GMM.

S13.7 Optional Problems

S1. GEL first-order conditions and the empirical probabilities. For empirical likelihood (EL), $G(v) = -\log(1-v)$ on $\mathcal{V} = (-\infty, 1)$, so that $g(v) = G'(v) = (1-v)^{-1}$ satisfies the GEL normalization $g(0) = 1$.

(a) At fixed θ, the inner problem in Equation 13.17 maximizes $-n^{-1}\sum_i G(\lambda^\top U_i(\theta))$ over λ; its first-order condition is equivalently obtained by minimizing $n^{-1}\sum_i G(\lambda^\top U_i(\theta))$. Write this condition, $\partial/\partial\lambda\,[\,n^{-1}\sum_i G(\lambda^\top U_i(\theta))] = 0$, and show that it implies $\sum_{i=1}^n \hat{\pi}_i U_i(\theta) = 0$, where $\hat{\pi}_i \propto (1 - \hat{\lambda}^\top U_i(\theta))^{-1}$.

(b) Verify that the normalization in Equation 13.25 is consistent with $g(v) = (1-v)^{-1}$, and explain why the conclusion of part (a) is unaffected by whether one uses the raw weights $\hat{\omega}_i$ or the normalized $\hat{\pi}_i$, even though the value of the discrepancy objective is affected.

S2. The exactly identified case. In the exactly identified case ($m = k$), assume in addition that $n^{-1}\sum_i U_i(\hat{\theta})U_i(\hat{\theta})^\top \succ 0$, so that the inner problem is strictly concave in λ near the solution. Show that $\hat{\lambda} = 0$ is then the unique solution of the inner problem at any GEL solution $\hat{\theta}$. Conclude that every GEL estimator coincides with the just-identified GMM estimator, and that the empirical probabilities Equation 13.25 all equal $1/n$. (*Hint*: in the exactly identified case $\bar{U}(\hat{\theta}) = 0$ exactly; $\lambda = 0$ solves the inner first-order condition, and the rank condition rules out other solutions. Verify that the corresponding primal weights from the conjugate relationship Equation 13.20 all equal the reference value $\hat{\omega}_i = g(0) = 1$.) Relate the conclusion to Theorem 13.7: what is T_{GEL} when $m = k$?

S3. Deriving a conjugate pair. For exponential tilting, $G(v) = e^v - 1$ on $\mathcal{V} = \mathbb{R}$.

(a) Verify the normalization Equation 13.18.

(b) Compute the Legendre–Fenchel conjugate Equation 13.20 directly and confirm the entry $F(\omega) = \omega\log\omega - \omega + 1$ of the table in Section S13.4.

(c) Confirm $F(1) = 0$ and $f(1) = 0$, and identify which of the three normalizations in Equation 13.18 each of these two facts depends on.

S4. Why normalization is not innocuous. Let $\hat{\omega}_i > 0$ be raw EL weights with $c = n^{-1}\sum_i \hat{\omega}_i \neq 1$, and let $\hat{\pi}_i = \hat{\omega}_i/(nc)$ be the normalized weights, so that $n\hat{\pi}_i = \hat{\omega}_i/c$.

(a) Show that $\sum_i \hat{\pi}_i U_i(\hat{\theta}) = 0$ follows from $\sum_i \hat{\omega}_i U_i(\hat{\theta}) = 0$.

(b) With $F(\omega) = \omega - 1 - \log\omega$, compute $\sum_i F(n\hat{\pi}_i) - \sum_i F(\hat{\omega}_i)$ explicitly and show that it is not zero in general. Explain, in one sentence, what goes wrong if the profile identity Equation 13.23 is applied to the normalized weights without adjustment.

Appendix A

Graphical Intuition for Conditional Independence and d-Separation

This appendix provides a gentle introduction to the probabilistic and graphical ideas that underlie Chapters 2 and 3. Our goal is not to give a complete treatment of graphical models, but rather to develop enough intuition so that the formal machinery of d-separation, back-door adjustment, and intervention graphs does not appear abruptly.

A directed acyclic graph (DAG) is more than a picture. It is a compact language for expressing assumptions about how variables are related. Once those assumptions are represented graphically, the graph tells us which variables may be associated, which paths transmit dependence, and which variables should or should not be conditioned on. These ideas become central in Chapter 2, where we study d-separation, and in Chapter 3, where we use graph surgery to derive identification results.

The key pedagogical idea of this appendix is simple: before learning the full d-separation criterion, it is helpful to understand three elementary three-node patterns. Those local patterns explain most of what happens later in larger graphs. Chapter 2 follows the same route — the three motifs with examples, then the blocking rules, then the d-separation criterion — so this appendix can be read as a gentler first pass over the same arc.

A.1 Conditional Independence: The Probabilistic Language Behind Graphs

Before introducing graphs, we first recall the probabilistic notion that graphs are designed to encode.

Definition A.1 (Conditional Independence)**.** Let X, Y, and Z be random variables. We say that X and Y are *conditionally independent given* Z, written $X \perp\!\!\!\perp Y \mid Z$, if

$$p(x, y \mid z) = p(x \mid z)\, p(y \mid z) \qquad \text{for } P_Z\text{-almost every } z.$$

Equivalently, once Z is known, learning X gives no additional information about Y, and learning Y gives no additional information about X.

In terms of densities (with respect to a dominating measure on (X, Y, Z)), conditional independence admits the following equivalent characterizations, each interpreted almost everywhere:

$$X \perp\!\!\!\perp Y \mid Z \iff f(x, y, z)\, f(z) = f(x, z)\, f(y, z) \iff \exists\, a, b\colon f(x, y, z) = a(x, z)\, b(y, z).$$

The last form is especially useful: it says that the joint density factors into one piece depending on (x, z) and another depending on (y, z), with no cross-term in x and y.

Proposition A.1 (Fundamental Properties of Conditional Independence)**.** *For random variables X, Y, Z, and W, the following hold:*

(C1) ***Symmetry.*** $X \perp\!\!\!\perp Y \mid Z \Rightarrow Y \perp\!\!\!\perp X \mid Z$. *(C2)* ***Decomposition.*** $X \perp\!\!\!\perp (Y, W) \mid Z \Rightarrow X \perp\!\!\!\perp Y \mid Z$. *(C3)* ***Weak Union.*** $X \perp\!\!\!\perp (Y, W) \mid Z \Rightarrow X \perp\!\!\!\perp Y \mid (Z, W)$. *(C4)* ***Contraction.*** $X \perp\!\!\!\perp Y \mid Z$ *and*

$X \perp\!\!\!\perp W \mid (Y, Z) \;\Rightarrow\; X \perp\!\!\!\perp (Y, W) \mid Z$. *(C5)* ***Intersection.*** *If* $f(x, y, z, w) > 0$ *for all* (x, y, z, w)*, then* $X \perp\!\!\!\perp Y \mid (Z, W)$ *and* $X \perp\!\!\!\perp Z \mid (Y, W) \;\Rightarrow\; X \perp\!\!\!\perp (Y, Z) \mid W$*.*

Source*: Dawid (1979); Dawid (1980).*

Properties (C1)–(C4) hold for any probability distribution and are known as the *semigraphoid axioms*. Property (C5) additionally requires the joint density to be strictly positive; together with (C1)–(C4) it forms the *graphoid axioms*. These properties are used implicitly throughout the course whenever conditional independence statements are combined or simplified.

Conditional independence is not the same as marginal independence. Two variables may be dependent marginally but independent after conditioning on a third variable. Conversely, two variables may be independent marginally but become dependent after conditioning. Both phenomena occur repeatedly in causal inference.

Example A.1 (Ice Cream, Drowning, and Season)**.** Let X = ice cream sales, Y = drowning incidents, and Z = season. Marginally, X and Y are positively associated because both tend to be higher in summer. This does not mean that ice cream sales cause drowning. A more plausible explanation is that season is a common cause of both variables. Once season is fixed, the association largely disappears: $X \perp\!\!\!\perp Y \mid Z$.

This example illustrates a recurring theme in causal inference: an observed association may be induced by a third variable, and conditioning on that variable can remove the spurious dependence.

Remark: The Central Question

At this stage, the main question for the reader is:

Does conditioning on a variable remove association, preserve it, or create it?

Graphs provide a systematic answer to exactly this question.

A.2 Three Basic Motifs: Chain, Fork, and Collider

Every path in a DAG is built from local three-node configurations. There are three fundamental types: a *chain*, a *fork*, and a *collider*. Their behavior under conditioning is the foundation of d-separation.

A.2.1 Chain

Consider the pattern $X_1 \to X_2 \to X_3$, where the middle node X_2 lies on a directed pathway from X_1 to X_3.

Example A.2 (Exercise, Body Weight, and Blood Pressure)**.** Let X_1 = exercise, X_2 = body weight, and X_3 = blood pressure. Exercise may affect blood pressure partly through its effect on body weight, so the path $X_1 \to X_2 \to X_3$ transmits association from X_1 to X_3. Marginally, exercise and blood pressure are associated; conditioning on body weight blocks this particular pathway. In the isolated three-node DAG as drawn, this is the only pathway, so $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

Once the middle variable is fixed, the chain no longer transmits additional information from X_1 to X_3.

A.2.2 Fork

Consider the pattern $X_1 \leftarrow X_2 \to X_3$, where the middle node X_2 is a common cause of X_1 and X_3.

Example A.3 (Ice Cream, Season, and Drowning)**.** With X_1 = ice cream sales, X_2 = season, and X_3 = drowning incidents, season affects both variables, so X_1 and X_3 are associated even though neither causes the other. Conditioning on season blocks this path: $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

A fork is the simplest graphical form of confounding: the middle node creates association, and conditioning on it blocks the path.

A.2.3 Collider

Consider the pattern $X_1 \to X_2 \leftarrow X_3$, where the middle node X_2 is a common effect of X_1 and X_3.

Example A.4 (Talent, Legacy Status, and College Admission)**.** Let X_1 = academic talent, X_3 = legacy status, and X_2 = admission to an elite university. Talent and legacy status may be unrelated in the general applicant pool. But among admitted students they can become statistically associated: low legacy status makes unusually high talent more likely among admitted applicants, and vice versa. Symbolically, $X_1 \perp\!\!\!\perp X_3$, but conditioning on X_2 opens the path between them, so X_1 and X_3 are typically dependent given X_2.

Unlike chains and forks, a collider *blocks* the path by default. Conditioning on the collider opens the path and may induce association that was not present marginally.

A.2.4 Summary of the Three Motifs

The three motifs behave as follows. A chain ($X_1 \to X_2 \to X_3$) and a fork ($X_1 \leftarrow X_2 \to X_3$) are each open by default and blocked by conditioning on the middle node X_2. A collider ($X_1 \to X_2 \leftarrow X_3$) is blocked by default and opened by conditioning on the middle node or any of its descendants. The stories here deliberately differ from those in Chapter 2 (smoking–tar–cancer, poverty–diet–exercise, accident–hospitalization–cancer), so that each motif is anchored in two contexts; the admission example reappears as the talent–wealth example in the IV-DAG section of Chapter 2.

Remark: Path-Blocking versus Conditional Independence

The independence statements in the chain, fork, and collider examples above are read in the isolated three-node DAGs as drawn. In a larger DAG, conditioning on the middle node X_2 of a chain or fork blocks *that particular path*, but X_1 and X_3 are conditionally independent only if every other path between them is also blocked. We return to this point in Section A.3.

Conditioning Is Not Always Beneficial

Many adjustment mistakes arise from forgetting this asymmetry. Conditioning on a collider can create bias rather than remove it. The informal rule “control for more variables” is unsafe in causal inference; d-separation teaches a more precise lesson: condition on the *right* variables, not simply on many variables.

None of these behaviors needs to be taken on faith: each is provable from the DAG factorization of Section A.4. The chain-proof example of Chapter 2 carries out the chain computation, and the blocking-derivation remark there obtains the remaining rules from the same factorization. Chapter 2 develops these three motifs into the full d-separation criterion; see in particular its sections on the blocking rules, on the criterion itself, and on the extended treatment of collider bias.

A.3 d-Separation: When Does Conditioning Block a Path?

The three-node motifs explain what happens locally on a path. The next step is to extend this logic to a general DAG, where two variables may be connected by many paths.

Definition A.2 (Path)**.** A *path* between two nodes is a sequence of distinct nodes such that each consecutive pair is connected by an edge, regardless of edge direction.

Definition A.3 (Blocked Path)**.** A path is *blocked* by a conditioning set S if at least one of the following holds:

1. the path contains a chain or fork node that belongs to S, or
2. the path contains a collider such that neither the collider nor any of its descendants belongs to S.

A path is *open* given S if it is not blocked.

Definition A.4 (d-Separation)**.** Two nodes X and Y are *d-separated* by a set S if every path between X and Y is blocked by S. More generally, two disjoint sets of nodes $\mathbf{X}$ and $\mathbf{Y}$, each disjoint from S, are d-separated by S if every $X \in \mathbf{X}$ is d-separated from every $Y \in \mathbf{Y}$ by S.

Remark: d-Separation versus Conditional Independence

d-Separation is a graphical condition; conditional independence is a probabilistic one. The Markov property guarantees only one direction: d-separation in $\mathcal{G}$ implies the corresponding conditional independence in every distribution that factorizes according to $\mathcal{G}$. The reverse implication — that d-connection forces conditional dependence — requires a faithfulness or no-cancellation assumption, since path coefficients in a parametric model can cancel exactly (for instance, in a linear Gaussian DAG). We therefore read an open path as "the graph does not force independence," not as "dependence is guaranteed." This distinction is taken up in detail in Chapter 2.

A.3.1 A Confounding Example

Consider the DAG with edges $X \to T$, $T \to Y$, and $X \to Y$. Here T is the treatment, Y is the outcome, and X is a pre-treatment covariate that affects both. There are two paths from T to Y: the directed causal path $T \to Y$, and the back-door path $T \leftarrow X \to Y$.

Without conditioning, the back-door path is open, so the observed association between T and Y mixes the causal effect with confounding. Conditioning on X blocks the fork $T \leftarrow X \to Y$, thereby isolating the causal path.

This confounding graph anticipates the back-door criterion of Chapter 3: the graphical condition that makes adjustment valid is precisely that all back-door paths are blocked.

A.3.2 A Collider Warning

Now consider the DAG with edges $T \to C$, $U \to C$, and $U \to Y$. Here C is a collider on the path $T \to C \leftarrow U \to Y$. Without conditioning on C, the path is blocked at the collider. Conditioning on C opens the path, creating a spurious association between T and Y through U.

Even more subtly, conditioning on a *descendant* of C can also open the path. If additionally $C \to D$, then conditioning on D may also induce association between T and Y through the collider at C: observing D amounts to observing a noisy proxy of C. The blocking-derivation remark of Chapter 2 derives this descendant rule from the factorization.

A.3.3 A Practical Checklist

To decide whether X and Y are d-separated by S, proceed as follows. First, list all paths between X and Y. On each path, classify each interior node as part of a chain, fork, or collider. Then check whether each path is blocked by S. Finally, conclude that X and Y are d-separated if and only if every path is blocked.

For beginners, this path-by-path procedure is often more useful than starting with the most compact formal definition.

A.4 DAG Factorization and the Markov Property

So far the graph has served purely qualitative purposes: classifying paths as open or blocked. The same parent structure carries quantitative information as well, because it prescribes how a joint distribution can be broken into simple pieces.

Any joint distribution can be written by conditioning each variable on all of its predecessors, but that generic decomposition involves high-dimensional conditional distributions and reveals little. The graphical model earns its keep by replacing the generic decomposition with a sparse one, in which each variable is conditioned only on its parents.

Definition A.5 (DAG Factorization)**.** Let $\mathcal{G}$ be a DAG with nodes $V_1, \dots, V_p$. A joint distribution $p(v_1, \dots, v_p)$ is said to *factorize according to $\mathcal{G}$* if

$$p(v_1, \dots, v_p) = \prod_{j=1}^{p} p\big(v_j \mid \mathrm{Pa}(V_j)\big)\,,$$

where $\mathrm{Pa}(V_j)$ denotes the set of parents of V_j in $\mathcal{G}$.

This factorization implies the local Markov property, defined below: once its parents are known, a node is conditionally independent of all variables that are neither its descendants nor its parents. In a causal DAG, drawing a parent arrow is an additional substantive claim that the parent is a direct causal input; this informally rules out exact cancellations or redundant arrows, and is connected to the notions of minimality and faithfulness discussed in Chapter 2.

Example A.5 (A Simple Confounding Graph)**.** For the DAG with edges $X \to T$, $T \to Y$, and $X \to Y$, the joint density factorizes as

$$p(x, t, y) = p(x)\, p(t \mid x)\, p(y \mid t, x).$$

Thus the graph determines which local conditional distributions appear in the factorization.

Definition A.6 (Local Markov Property)**.** Let $\mathrm{Nd}(V_i) = V \setminus (\{V_i\} \cup \mathrm{De}(V_i))$ denote the set of *non-descendants* of V_i, where $\mathrm{De}(V_i)$ is the set of descendants of V_i in $\mathcal{G}$. A distribution P satisfies the *local Markov property* with respect to $\mathcal{G}$ if, for every node $V_i \in V$,

$$V_i \perp\!\!\!\perp (\mathrm{Nd}(V_i) \setminus \mathrm{Pa}(V_i)) \mid \mathrm{Pa}(V_i).$$

That is, once we condition on the direct causes of a node, that node is independent of all variables that are neither its descendants nor its parents.

> Remark: Global Markov Property
>
> The global Markov property is the statement that every d-separation in $\mathcal{G}$ implies a conditional independence in P. Thus, if two sets of variables are d-separated by a set S in the graph, then they are conditionally independent given S in the distribution. This is studied in detail in Chapter 2.

Theorem A.1 (Equivalence of the Three Markov Properties (Lauritzen 1996, Theorem 3.27))**.** *Let $\mathcal{G}$ be a DAG and let P admit a joint density with respect to a product dominating measure, with regular conditional distributions. Then the following are equivalent:*

(i) P admits the recursive factorization $p(v_1, \ldots, v_k) = \prod_i p(v_i \mid \mathrm{Pa}(v_i))$;
(ii) P satisfies the local Markov property;
(iii) P satisfies the global Markov property.

Proof. We sketch the cycle (iii) $\Rightarrow$ (ii) $\Rightarrow$ (i) $\Rightarrow$ (iii); full details are given by Lauritzen (1996).

Global $\Rightarrow$ local (instantiation). Each local statement is itself a d-separation, hence one entry of the global dictionary. Take any path from V_i to a non-descendant W. If the path leaves V_i through a parent, it is blocked there: the parent is a conditioned non-collider. If it leaves through a child, then reaching a *non*-descendant forces the path to reverse direction somewhere, and the reversal point is a collider that is a descendant of V_i; by acyclicity no descendant of V_i can lie in $\mathrm{Pa}(V_i)$, so the collider is unactivated and the path is blocked.

Local $\Rightarrow$ factorization (chain rule). List the nodes in a topological ordering, so that every predecessor of V_i is a non-descendant. The chain rule gives $p(v_1, \ldots, v_k) = \prod_i p(v_i \mid v_1, \ldots, v_{i-1})$, and the local statement for V_i licenses shrinking each conditioning set to the parents: $p(v_i \mid v_1, \ldots, v_{i-1}) = p(v_i \mid \mathrm{Pa}(v_i))$. The k local statements are thus exactly calibrated to the k factors of the chain rule: they pin the functional form of the joint density to a product.

Factorization $\Rightarrow$ global (the substantive arrow). This is the soundness theorem of Chapter 2, and the standard proof runs through the moral-graph criterion of Section A.5: restrict to the ancestral set of $A \cup B \cup \mathbf{S}$ (marginalizing over non-ancestors preserves the product form), moralize (each factor $p(v_i \mid \mathrm{pa}_i)$ becomes a function on one clique of the moral graph, so the density is a product of clique functions), and observe that separation of A from B by $\mathbf{S}$ in the moralized ancestral graph splits that product into $g(a, \mathbf{s})\, h(b, \mathbf{s})$ — which is conditional independence. □

> Remark: Reading the Equivalence
>
> Once the density has product form, which marginalizations and conditionings preserve it becomes a purely combinatorial question, and d-separation is exactly the graphical language of that combinatorics; the equivalence amounts to one theorem plus two lines of bookkeeping. Notably, the implication from factorization to the global Markov property does not require strict positivity of

the joint density; strict positivity is needed only for axioms involving the intersection property among general conditional-independence relations. The local property is the version most commonly verified in practice, since the factorization makes it immediate.

Example A.6 (Education–Earnings Graph)**.** Consider the DAG above, with edges $N \to E$, $B \to E$, $E \to Y$, and $B \to Y$. The corresponding factorization is

$$p(n, b, e, y) = p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b).$$

The parent set of Y is $\mathrm{Pa}(Y) = \{E, B\}$, so the local Markov property gives $Y \perp\!\!\!\perp N \mid \{E, B\}$.

The same conclusion follows by d-separation. There are two paths from N to Y:

$$N \to E \to Y \qquad \text{and} \qquad N \to E \leftarrow B \to Y.$$

Given $\{E, B\}$, the first path is blocked at the non-collider E. The second path is opened at the collider E, but blocked at the non-collider B. Thus every path from N to Y is blocked by $\{E, B\}$.

The factorization above also implies that $N \perp\!\!\!\perp B$, since neither node has a parent or a common ancestor in the graph. This is a substantive modeling assumption: if neighborhood and family background are believed to be associated, the graph should include an edge between them, a common cause, or a different ordering.

Remark: Edges as Scientific Claims

In a causal DAG, an arrow is typically drawn only when a direct dependence-generating relation is believed to be present. This explains why edges are not included gratuitously: every arrow represents a substantive scientific claim. We do not formalize minimality or faithfulness here; both are discussed in Chapter 2.

Chapter 2 states the local and global Markov properties formally and uses them to connect d-separation to conditional independence; it then applies the same factorization to the education–earnings DAG introduced above.

A.5 Optional: Moralization as an Alternative Criterion

Note to Reader

This section may be skipped on a first reading.

There is an alternative graph-theoretic way to check d-separation based on constructing an undirected graph called the *moral graph*. The method is elegant and useful in more advanced graphical-model theory, but it is not necessary for understanding the main causal ideas of Chapters 2 and 3. For most students, the path-by-path method using chains, forks, and colliders is more intuitive on first exposure.

To check whether $X \perp\!\!\!\perp Y \mid S$, the procedure is as follows. First, take the induced subgraph on $\mathrm{An}(X \cup Y \cup S)$, the ancestors of all variables under consideration, including those in the conditioning set S. Second, connect any two parents of a common child by an undirected edge. Third, drop all arrow directions. Finally, check whether S separates X and Y in the resulting undirected graph; equivalently, whether X and Y become disconnected once the nodes in S and their incident edges are removed. This procedure yields a criterion equivalent to d-separation.

Example A.7 (Moral Graph of a Collider)**.** Consider the DAG $A \to C \leftarrow B$. For the conditional query $A \perp\!\!\!\perp B \mid C$, the ancestral set is $\mathrm{An}(\{A, B, C\}) = \{A, B, C\}$, so C is retained. Moralization connects the two parents A and B, and after deleting the conditioned node C the edge $A - B$ remains; therefore A and B are not separated, matching the fact that conditioning on a collider opens the path. By contrast, for the marginal query $A \perp\!\!\!\perp B$, the ancestral set is $\mathrm{An}(\{A, B\}) = \{A, B\}$, so C is discarded before moralization and no moral edge is added; A and B are separated, as expected.

A.6 The ID Algorithm and the Hedge: A Proof Sketch of Completeness

This section collects the graph terminology behind the completeness theorem of Chapter 3 and sketches its proof. It is reference material for independent reading: the machinery below — ADMGs, districts, kernels, fixing, and the hedge — is not required elsewhere in the course.

A.6.1 ADMGs and Semi-Markovian Models

Chapters 2 and 3 represent latent common causes as explicit unobserved nodes (e.g. $U \to T$ and $U \to Y$). The completeness theorem uses a more general graph class that encodes latent common causes compactly as *bidirected edges*: a bidirected edge $X \leftrightarrow Y$ records latent common-cause dependence between X and Y; in the canonical semi-Markovian representation, it is realized by an exogenous latent common cause U with $U \to X$ and $U \to Y$. A graph containing both directed and bidirected edges, with the directed part acyclic, is called an *acyclic directed mixed graph* (ADMG): directed edges represent causal relations among the observed variables, while bidirected edges represent latent common-cause relations. In the *canonical semi-Markovian representation*, every latent variable is exogenous, the exogenous variables are mutually independent, and each latent common cause has at most two observed children, so that it enters the ADMG as a bidirected edge between the pair of observed variables it affects. Causal models of this form are called *semi-Markovian*. (Conditional interventional distributions $P(y \mid \mathrm{do}(t), c)$ require an extension of the ID algorithm known as IDC, for which an analogous completeness result holds; Chapter 3 treats only the unconditional case.)

> Remark: A Convention for Nonidentification Witnesses
>
> Chapter 2 adopted an informal minimality convention: arrows are not drawn gratuitously. For *nonidentification* arguments the opposite convention is the appropriate one. A structural causal model is *compatible* with $\mathcal{G}$ if each structural equation depends on some subset of the displayed parents — possibly a strict subset, so that a displayed arrow may carry a null effect; $\mathcal{G}$ acts as an allowed-parent supergraph. This matches the definition of identifiability, which quantifies over *every* model compatible with the graph in this permissive sense. The two-model witness constructions used to demonstrate nonidentification are to be read this way.

A.6.2 How ID Succeeds: Districts and Kernels

Shpitser and Pearl (2006) introduced the *identification (ID) algorithm*: a procedure that takes as input an ADMG $\mathcal{G}$, a target intervention distribution $P(y \mid \mathrm{do}(t))$, and the observed joint distribution P, and either returns an explicit observational formula for $P(y \mid \mathrm{do}(t))$ or reports that the quantity is not identifiable from P given $\mathcal{G}$.

Sufficiency (do-calculus identifies whenever identification is possible): proved constructively via the ID algorithm. The key structural concept is the *district* (called a *c-component*, for confounded component, in the original ID literature): a maximal set of observed nodes connected by paths of bidirected edges. Its members are linked through a network of latent common-cause relations, though they need not all share a single latent parent. For a single node V with no bidirected edges, its district is $\{V\}$ itself.

The ID algorithm first restricts attention to the relevant ancestral set Y^* — the vertices with a directed path to Y that avoids T, together with Y itself — and decomposes the induced subgraph $\mathcal{G}[Y^*]$, denoted below by $\mathcal{G}_{Y^*}$ (not to be confused with the edge-deletion operations $\mathcal{G}_{\overline{X}}$ and $\mathcal{G}_{\underline{X}}$), into districts. It then attempts to recover one interventional kernel for each district by a recursive reduction of the observed distribution (Tian's *Q-factorization* (Tian and Pearl 2002); in modern terminology, a sequence of *fixing* operations). If every required district kernel is recoverable — every district is *intrinsic* — the kernels combine into an identifying functional; the output is best viewed as a modified *nested* Markov factorization of the ADMG, in the same way that the g-formula is a modified Markov factorization of a DAG. If some required district is not intrinsic, the algorithm fails and exposes a hedge (defined below). When the algorithm succeeds, it has produced an explicit identification formula, demonstrating that the do-calculus is sufficient.

A.6.3 How ID Fails: The Hedge and the Two-Model Witness

Necessity (do-calculus cannot identify nonidentifiable quantities): proved by showing that whenever the ID algorithm fails, the causal effect is genuinely nonidentifiable — not merely hard to compute. The key graphical obstruction is the *hedge*.

> Informally, a *hedge* for $P(y \mid \mathrm{do}(t))$ in $\mathcal{G}$ is a pair of nested *C-forests* $\langle F, F' \rangle$ with $F' \subseteq F$, both rooted at the same set $R \subseteq F'$, such that F intersects the treatment set T while F' does not, and every node of R has a directed path to Y that avoids T. (A C-forest is a subgraph whose nodes form a single district — all connected through bidirected edges — and whose directed edges form a forest with root set R.) Intuitively, the hedge encodes a "loop" of confounding surrounding the treatment that the do-calculus cannot break.

Shpitser and Pearl (2006) prove that whenever a hedge for $P(y \mid \mathrm{do}(t))$ exists in $\mathcal{G}$, the target is not identifiable. Conversely, whenever the ID algorithm fails, it exposes a hedge for the target query: the algorithm fails if and only if $\mathcal{G}$ contains a hedge *for* $P(y \mid \mathrm{do}(t))$ *itself* — equivalently, some district of $\mathcal{G}_{Y^*}$ is not intrinsic (Shpitser 2023). (An older formulation characterized identifiability by the absence of hedges for every reduced subquery $P(y' \mid \mathrm{do}(t'))$ with $T' \subseteq T$ and $Y' \subseteq Y$. That subset-subquery formulation is incorrect as stated: a target such as $P(y \mid \mathrm{do}(t_1, t_2))$ can be identifiable even though the reduced query $P(y \mid \mathrm{do}(t_2))$ has a hedge (Shpitser 2023). The corrected criterion concerns the target query itself.) When a hedge for the target exists, one can explicitly construct two semi-Markovian models $\mathcal{M}_1$ and $\mathcal{M}_2$ that:

- induce the same observed joint distribution P, so no data can distinguish them; but
- assign different values to $P(y \mid \mathrm{do}(t))$, so the interventional distribution is not a function of P alone.

The construction builds $\mathcal{M}_1$ and $\mathcal{M}_2$ by altering the causal mechanisms of the *observed* variables in the relevant district (and, if needed, the distribution of the exogenous variables — which, being exogenous, have no structural equations of their own), keeping the observed joint distribution identical. Since the two models are observationally indistinguishable yet causally distinct, no method — including the do-calculus — can identify $P(y \mid \mathrm{do}(t))$ from P alone. This establishes necessity.

Combining sufficiency and necessity: the ID algorithm (and hence the do-calculus) succeeds if and only if no hedge exists for the target query, which is if and only if $P(y \mid \mathrm{do}(t))$ is identifiable. □

A.7 Summary

This appendix introduced the graphical ideas that support Chapters 2 and 3. Conditional independence is the probabilistic language that graphs are designed to encode. The three local motifs — chain, fork, and collider — determine how conditioning affects association along a path. d-Separation extends these local rules to arbitrary graphs: two variables are d-separated by S if every path between them is blocked by S. The most important application in causal inference is to distinguish confounding paths from causal paths and to identify valid adjustment sets. The Markov property gives a probabilistic interpretation to the graph by linking graphical structure to a factorization of the joint distribution. Finally, an optional last section sketches the ID algorithm and the hedge obstruction — the machinery behind the completeness theorem of Chapter 3.

Appendix B

Single World Intervention Graphs

This appendix develops single world intervention graphs (SWIGs), a graphical device introduced by Richardson and Robins (2014) and given an accessible practical treatment by Bezuidenhout et al. (2025). A SWIG provides a formal bridge between the potential outcomes framework and the do-calculus by encoding both in a single diagram; see Wang et al. (2026) for a recent overview situating SWIGs within the broader landscape of causal inference frameworks. A reader familiar with Chapters 2–4 will find that SWIGs require no new conceptual ingredients: they are ordinary DAGs in which the treatment node is split into two pieces, making potential outcomes and identifiability assumptions graphically explicit rather than leaving them implicit.

Chapter 4 used the back-door criterion and the adjustment formula to identify causal effects. SWIGs make that identification argument graphically explicit: potential outcomes appear as labeled nodes inside the graph, so ignorability becomes a routine d-separation statement rather than a free-standing probabilistic assumption. Section B.5 restates the back-door identification result of Chapter 4 as a graph-theoretic theorem inside the SWIG, and examines a partial converse: under faithfulness of the SWIG and a restriction of the adjustment set to nondescendants of the treatment, the single-world ignorability statement $Y(t) \perp\!\!\!\perp T \mid X$ entails the back-door criterion.

B.1 The SWIG Construction

In the original DAG $\mathcal{G}$, the node T represents the treatment as it naturally occurs. Under do(T=t), however, the treatment is externally fixed. The SWIG separates these two roles by *splitting* T into two pieces displayed side by side in the graph:

- a **random half** (left side, capital letter T): represents the treatment value that would naturally occur, and retains all incoming edges from T's causal parents;
- a **fixed half** (right side, lowercase letter t): represents the externally imposed intervention value, carries all outgoing edges that formerly left T, and has *no* edge to or from the random half.

Definition: Single World Intervention Graph [@richardson2014ace]

The *SWIG* $\mathcal{G}(t)$ is constructed from $\mathcal{G}$ by:

1. Replacing T with two nodes: random half T (all incoming edges retained) and fixed half t (all outgoing edges retained).
2. Severing the direct connection between the two halves.
3. Redrawing causal arrows at the split node: *all arrows into the split node land on the random half*; *all arrows departing the split node leave from the fixed half.*
4. Relabeling every descendant V of T in $\mathcal{G}$ as $V(t)$, indicating it is a potential outcome under do(T=t).

```
\usetikzlibrary{arrows.meta,positioning}
\definecolor{isublue}{RGB}{46,117,182}
\definecolor{defbg}{RGB}{238,244,251}
```

```
\tikzset{rnd/.style={circle,draw=isublue,fill=defbg,thick,minimum size=9mm,font=\small\bfseries},
         fix/.style={rectangle,draw=isublue,fill=gray!15,thick,minimum size=9mm,font=\small\bfseries
\begin{tikzpicture}[>=stealth]
  \begin{scope}
    \node[rnd](L) at (1.4,1.5){$L$};
    \node[rnd](A) at (0,0){$A$};
    \node[rnd](Y) at (2.8,0){$Y$};
    \draw[->,thick,color=isublue](L)--(A); \draw[->,thick,color=isublue](L)--(Y); \draw[->,thick,col
    \node[font=\small\itshape,below=5mm of A,xshift=14mm]{DAG};
  \end{scope}
  \node[font=\large] at (3.9,0.4){$\Rightarrow$};
  \begin{scope}[xshift=4.9cm]
    \node[rnd](Lr) at (1.9,1.5){$L$};
    \node[rnd](Ar) at (0.4,0){$A$};
    \node[fix](af) at (1.5,0){$a$};
    \node[rnd](Ya) at (3.5,0){$Y(a)$};
    \draw[->,thick,color=isublue](Lr)--(Ar); \draw[->,thick,color=isublue](Lr)--(Ya); \draw[->,thick
    \node[font=\small\itshape,below=5mm of Ar,xshift=16mm]{SWIG $\mathcal{G}(a)$};
  \end{scope}
\end{tikzpicture}
```

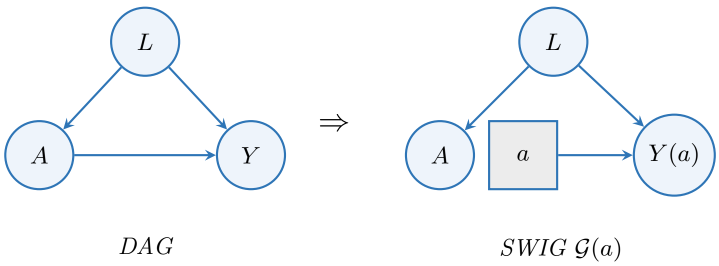

Figure B.1: Converting the confounded-treatment DAG into the SWIG for do(A=a).

B.2 The Fixed Half as a Source Node

After the SWIG $\mathcal{G}(t)$ is constructed, d-separation among its nodes is evaluated using the standard rules of Chapter 2. The key structural fact is:

The Fixed Half Is a Source Node

In $\mathcal{G}(t)$, **no directed edge points into** the fixed half t. Two structural consequences follow.

1. **The route $T \to t \to Y(t)$ does not exist.** Because no edge connects the random half T to the fixed half t, the directed concatenation $T \to t \to Y(t)$ is *not a path in the graph* — there is nothing for d-separation to block.
2. **Every path from T to a potential outcome $V(t)$ leaves T through one of T's natural causes.** The only edges incident to the random half T in $\mathcal{G}(t)$ are the incoming edges from T's parents in $\mathcal{G}$. Any path from T to $V(t)$ must therefore start with an arrow pointing *into* T — that is, every such path is a back-door path.

"Source Node" Means No Parents, Not "On No Path"

Calling t a source node refers to the absence of *directed edges into* t, nothing more. An undirected path can still pass through a source node. What the source-node property delivers is the asymmetry that matters for the back-door argument: at the random half T, *all* edges point inward, so any path leaving T must do so through one of T's causes.

Reading the back-door criterion off the SWIG. By the source-node analysis above, every path from the random half T to a potential outcome $V(t)$ in $\mathcal{G}(t)$ is a back-door path in $\mathcal{G}$, so blocking all back-door paths is exactly what d-separation of T and $V(t)$ in the SWIG asks for.

Recovering ignorability from the SWIG. In the SWIG of the confounded-treatment example, the only path from the random half A to $Y(a)$ is $A \leftarrow L \rightarrow Y(a)$: open, but blocked by conditioning on L. Conditioning on L achieves d-separation, and the standard rules give $A \perp\!\!\!\perp Y(a) \mid L$ in $\mathcal{G}(a)$, which is exactly the conditional ignorability of Chapter 4, now *derived* as a graph-structural consequence rather than asserted as an assumption.

B.3 When Ignorability Fails: Hidden Confounding

The SWIG is equally useful for diagnosing *failures* of ignorability. Suppose the DAG is augmented by a hidden common cause U that affects both A and Y but is not recorded in the data.

```
\usetikzlibrary{arrows.meta,positioning}
\definecolor{isublue}{RGB}{46,117,182}
\definecolor{defbg}{RGB}{238,244,251}
\tikzset{rnd/.style={circle,draw=isublue,fill=defbg,thick,minimum size=9mm,font=\small\bfseries},
         fix/.style={rectangle,draw=isublue,fill=gray!15,thick,minimum size=9mm,font=\small\bfseries
         hid/.style={circle,draw=isublue,fill=white,thick,dashed,minimum size=9mm,font=\small\bfseri
\begin{tikzpicture}[>=stealth]
  \begin{scope}
    \node[rnd](L) at (1.4,1.5){$L$}; \node[rnd](A) at (0,0){$A$}; \node[rnd](Y) at (2.8,0){$Y$}; \no
    \draw[->,thick,color=isublue](L)--(A); \draw[->,thick,color=isublue](L)--(Y); \draw[->,thick,col
    \draw[->,thick,dashed,color=isublue](U)--(A); \draw[->,thick,dashed,color=isublue](U)--(Y);
    \node[font=\small\itshape,below=3mm of U,xshift=0]{DAG (with $U$)};
  \end{scope}
  \node[font=\large] at (3.9,0.15){$\Rightarrow$};
  \begin{scope}[xshift=4.9cm]
    \node[rnd](Lr) at (1.9,1.5){$L$}; \node[rnd](Ar) at (0.4,0){$A$}; \node[fix](af) at (1.5,0){$a$}
    \draw[->,thick,color=isublue](Lr)--(Ar); \draw[->,thick,color=isublue](Lr)--(Ya); \draw[->,thick
    \draw[->,thick,dashed,color=isublue](Ur)--(Ar); \draw[->,thick,dashed,color=isublue](Ur)--(Ya);
    \node[font=\small\itshape,below=3mm of Ur,xshift=0]{SWIG (with $U$)};
  \end{scope}
\end{tikzpicture}
```

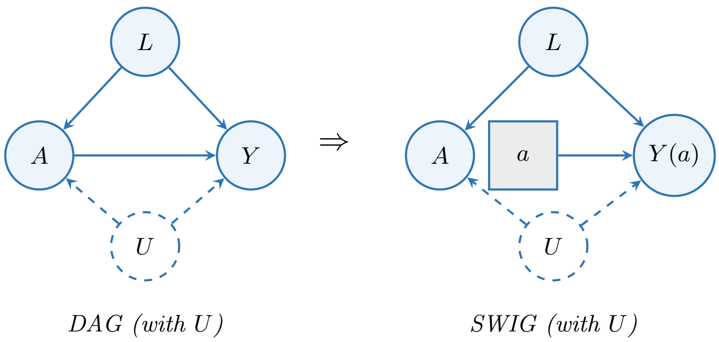

Figure B.2: Adding a hidden confounder U (dashed circle). Ignorability fails.

In the SWIG $\mathcal{G}(a)$ with hidden U, the paths from A to $Y(a)$ are: (1) $A \leftarrow L \rightarrow Y(a)$: blocked by conditioning on L; and (2) $A \leftarrow U \rightarrow Y(a)$: open regardless of L, since U is unobserved and cannot be conditioned on. Even after conditioning on L, $A \not\perp\!\!\!\perp Y(a) \mid L$, so ignorability fails and no adjustment for observed covariates alone can identify the ATE. The failure is *immediately visible* from the SWIG: an unblocked dashed arrow entering the random half A signals that selection into treatment carries information about the potential outcome that no set of observed covariates can remove.

Remark

This failure motivates the instrumental variable designs of Chapter 7. When a valid instrument Z is available — one that affects A but has no direct effect on Y and is independent of U — the IV strategy sidesteps the unblocked U-path rather than attempting to block it by covariate adjustment.

B.4 What SWIGs Achieve

1. Potential outcomes as random variables in a DAG. In $\mathcal{G}(t)$, the variable $Y(t)$ is an ordinary node with well-defined parents: the fixed half t and any other causes of Y retained from $\mathcal{G}$. Its marginal distribution equals the interventional distribution $f(y \mid \mathrm{do}(T{=}t))$.

2. The mutilated graph as a special case. Removing the random half T from $\mathcal{G}(t)$, and suppressing the fixed-half labeling on its descendants, recovers the usual mutilated graph $\mathcal{G}_{\overline{T}}$ of Chapter 1 representing $\mathrm{do}(T{=}t)$. The SWIG is more informative: it retains the random half alongside the fixed half, making it possible to ask questions about the relationship between the natural treatment T and the potential outcomes $V(t)$ via d-separation.

3. Unconfoundedness as a d-separation statement. The ignorability condition $Y(t) \perp\!\!\!\perp T \mid X$ corresponds, under the Markov property of the SWIG, to a d-separation statement *in* $\mathcal{G}(t)$ verifiable by path-tracing. Ignorability is no longer an assertion encoded in potential-outcome notation alone; it is a structural claim about the graph.

4. A single graph for both frameworks. Before SWIGs, combining the potential outcomes framework and the do-calculus required switching notation. The SWIG eliminates this translation step: one graph simultaneously hosts the natural treatment T (random half), the intervention value t (fixed half), the potential outcomes $V(t)$, and all d-separation relationships among them.

Cross-World Independence and Its Limits

Consider $Y(1) \perp\!\!\!\perp Y(0)$: $Y(1)$ lives in SWIG $\mathcal{G}(1)$ and $Y(0)$ lives in SWIG $\mathcal{G}(0)$. No unit is ever observed in both worlds simultaneously, so no data can directly test this independence. A more substantive cross-world example arises in mediation analysis (Chapter 8): identification of natural direct and indirect effects relies on assumptions of the form $Y(t, m) \perp\!\!\!\perp M(t') \mid X$ $(t \neq t')$, which combine potential outcomes from *different SWIGs*. SWIGs make such assumptions explicit: the analyst must specify a joint distribution across multiple SWIGs. The single-world results of Section B.5 therefore cannot themselves supply cross-world assumptions.

Remark

More than one node may be split in a single SWIG. In longitudinal settings with time-varying treatments (A_0, A_1), splitting both nodes yields the sequential-randomization SWIG, from which the two *sequential exchangeability* conditions needed to identify the g-formula follow directly by d-separation.

B.5 Single-World Ignorability and the Back-Door Criterion

This section restates the back-door identification result of Chapter 4 graphically as a theorem inside the SWIG, and then examines a partial converse. The target conditional independence is the *single-world* (or *weak*) ignorability statement $Y(t) \perp\!\!\!\perp T \mid X$ $(t = 0, 1)$. The cross-world joint statement $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ involves $Y(0)$ and $Y(1)$, which live in two distinct SWIGs; a single SWIG cannot encode it as a d-separation.

Definition: Faithfulness of the SWIG

Let $\mathcal{G}(t)$ be the SWIG induced by an intervention do($T{=}t$), and let P_t denote the joint distribution it induces over the random nodes. The distribution P_t is **faithful** to $\mathcal{G}(t)$ if every conditional independence that holds in P_t corresponds to a d-separation in $\mathcal{G}(t)$:

$$A \perp\!\!\!\perp B \mid C \text{ in } P_t \implies A \text{ and } B \text{ are d-separated by } C \text{ in } \mathcal{G}(t).$$

Faithfulness rules out accidental cancellations of causal paths in the structural model. In smooth parametric families with a fixed graph, the parameter configurations that produce unfaithfulness typically form a lower-dimensional subset, so faithfulness is, in that sense, generic. The qualifier matters: deterministic relationships, context-specific independencies, and graph misspecification can all produce substantive failures of faithfulness.

Faithfulness is invoked only for the converse direction below.

Theorem: Back-Door Criterion Implies Single-World Ignorability

Fix an intervention value t. Suppose the structural causal model underlying $\mathcal{G}$ induces a distribution P_t over the random nodes of the SWIG $\mathcal{G}(t)$ that is Markov with respect to $\mathcal{G}(t)$. If X contains no descendants of T in $\mathcal{G}$ and blocks every back-door path from T to Y, then $Y(t) \perp\!\!\!\perp T \mid X$.

Proof. By the source-node analysis of Section B.2, every path from the random half T to the potential outcome $Y(t)$ in $\mathcal{G}(t)$ leaves T through one of T's parents in $\mathcal{G}$ and is therefore a back-door path. By hypothesis, X blocks every such path; the no-descendants clause ensures that conditioning on members of X does not open a previously blocked collider path. Hence T and $Y(t)$ are d-separated by X in $\mathcal{G}(t)$, and the Markov property of P_t delivers $Y(t) \perp\!\!\!\perp T \mid X$. □

For binary treatment, applying the theorem separately at $t = 0$ and $t = 1$ yields the weak ignorability conditions $Y(0) \perp\!\!\!\perp T \mid X$ and $Y(1) \perp\!\!\!\perp T \mid X$ that justify the adjustment formula of Chapter 4.

Remark: A Partial Converse

If X is restricted *a priori* to pre-treatment (nondescendant) variables in $\mathcal{G}$ and P_t is faithful to $\mathcal{G}(t)$, the implication above can be partially reversed:

$$Y(t) \perp\!\!\!\perp T \mid X \text{ in } P_t \implies X \text{ blocks every back-door path from } T \text{ to } Y.$$

By faithfulness, the assumed independence corresponds to d-separation of the random half T and $Y(t)$ by X in $\mathcal{G}(t)$; since every path from T to $Y(t)$ is a back-door path, X must block all of them. Without the nondescendant restriction, however, conditional exchangeability alone does not imply the full back-door criterion.

Why "No Descendants of T" Is a Hypothesis, Not a Conclusion

The partial converse imposes "no descendants of T" as a hypothesis. This is essential: the conditional independence $Y(t) \perp\!\!\!\perp T \mid X$ does *not* by itself imply that X contains no descendant of T. For instance, take the DAG with edges $T \to Y$ and $T \to D$ and no confounding, and let $X = \{D\}$. D is a descendant of T, so X violates the back-door criterion. Nonetheless, $Y(t) \perp\!\!\!\perp T \mid D$ may continue to hold because there is no back-door path from T to Y at all. Conditioning on a descendant of T *may* open a previously blocked collider path, but it need not. The no-descendants clause belongs to the *definition* of an admissible graphical adjustment set and cannot be inferred from observed conditional independencies alone.

From single-world to cross-world. The strong ignorability condition $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ is a statement about the joint distribution of $Y(0)$ and $Y(1)$ together with T. Because $Y(0)$ and $Y(1)$ are nodes in two distinct SWIGs, no single SWIG represents this joint distribution as a d-separation, and the theorem above correspondingly speaks only about the single-world form. For ATE identification this is no obstacle: the single-world form $Y(t) \perp\!\!\!\perp T \mid X$ for each $t \in \{0, 1\}$ is enough. Identification of joint or distributional functionals of $(Y(0), Y(1))$ — for instance, the variance of the unit-level treatment effect, or the proportion of units harmed by treatment — requires the cross-world joint form and assumptions across multiple SWIGs that single-world graphical reasoning cannot supply.

> Remark: How Much Extra Structure Does NPSEM-IE Impose?
>
> A binary example, following Wang et al. (2026), makes the comparison quantitative. Take X, T, and Y all binary, so the full joint distribution of $(X, T(x), Y(t, x))$ for $t, x \in \{0, 1\}$ has $2^{1+2+4} - 1 = 127$ free parameters. Three nested assumptions shrink this model:
>
> - Weak (single-world) ignorability $T \perp\!\!\!\perp Y(t) \mid X$ for each t separately: 123 parameters.
> - Strong cross-world ignorability $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$: 121 parameters.
> - Full NPSEM-IE (all exogenous errors $(\varepsilon_X, \varepsilon_T, \varepsilon_Y)$ mutually independent): 19 parameters.
>
> NPSEM-IE therefore imposes $123 - 19 = 104$ additional constraints beyond what is needed for ATE identification. Most of these are cross-world statements such as $T(X{=}0) \perp\!\!\!\perp Y(t, X{=}1)$ that remain untestable even under randomized treatment assignment. This makes concrete the point above: working under NPSEM-IE is a substantive modeling choice, not a free consequence of using structural equations.

Conceptual takeaway. The back-door criterion is a graphical sufficient condition for the single-world conditional exchangeability assumption needed for ATE identification. Under the additional restriction that the candidate adjustment set is composed of nondescendants of T, and under faithfulness of the relevant counterfactual distribution to the SWIG, the absence of unblocked back-door paths corresponds to the single-world independence $Y(t) \perp\!\!\!\perp T \mid X$. In this sense the graphical and counterfactual languages describe the same identifying structure. Strict logical equivalence in full generality is more delicate: ignorability can hold in particular models for parametric reasons that no graphical criterion will detect.

Problems

1. SWIG construction. Consider the DAG: $X \to T$, $X \to Y$, $T \to Y$, with X fully observed.

(a) Construct the SWIG $\mathcal{G}(t)$ by splitting T into its random and fixed halves. Draw the result, labeling the random half, fixed half, and the potential outcome $Y(t)$.

(b) In $\mathcal{G}(t)$, identify all paths between the random half T and $Y(t)$. Apply d-separation to determine which are blocked and which are open before conditioning.

(c) Verify that X satisfies the back-door criterion in $\mathcal{G}$: show that X blocks the unique back-door path $T \leftarrow X \to Y$ and that X contains no descendant of T. Apply the theorem above to conclude $Y(t) \perp\!\!\!\perp T \mid X$, and confirm that the same conclusion can be read directly off $\mathcal{G}(t)$ via d-separation.

(d) Now add a hidden common cause $U \to T$, $U \to Y$. Draw the revised SWIG. Does the ignorability argument still hold? State the correct conclusion and identify what additional structure (if any) would be needed for identification.

2. Sequential exchangeability in a longitudinal SWIG. Suppose the treatment is time-varying: T_0 is assigned at baseline and T_1 at a second time point, and a time-varying covariate L is measured between the two assignments. The DAG has edges $X \to T_0$, $X \to L$, $X \to Y$, $T_0 \to L$, $T_0 \to T_1$, $T_0 \to Y$, $L \to T_1$, $L \to Y$, and $T_1 \to Y$.

(a) Construct the SWIG $\mathcal{G}(t_0, t_1)$ by splitting both T_0 and T_1. Label the random halves, the fixed halves t_0 and t_1, and the potential outcomes. In particular, the random half of T_1 should be labeled $T_1(t_0)$ and the time-varying covariate should be labeled $L(t_0)$.

(b) List the back-door paths in $\mathcal{G}(t_0, t_1)$ from the random half T_0 to $Y(t_0, t_1)$, and from $T_1(t_0)$ to $Y(t_0, t_1)$. For each, identify the smallest subset of $\{X, T_0, L(t_0)\}$ whose conditioning achieves d-separation.

(c) State the two sequential exchangeability conditions that the SWIG makes graphically transparent: $Y(t_0, t_1) \perp\!\!\!\perp T_0 \mid X$ and $Y(t_0, t_1) \perp\!\!\!\perp T_1(t_0) \mid X, T_0 = t_0, L(t_0)$. Explain in plain language what each says about treatment assignment at the corresponding time point.

(d) In observed-data shorthand, the second condition is often written $Y(\bar{t}) \perp\!\!\!\perp T_1 \mid X, T_0 = t_0, L$. State the implicit consistency step that licenses replacing $L(t_0)$ with the observed L on the event $\{T_0 = t_0\}$.

(e) What goes wrong if the analyst omits L from the conditioning set at the second stage? What goes wrong if the analyst conditions on L but treats it as a baseline covariate? Connect your answers to the role of $L(t_0)$ as a treatment-induced confounder.

3. Why the no-descendants clause is a hypothesis. Consider the DAG with edges $T \to Y$ and $T \to D$, no confounders, and let $X = \{D\}$.

(a) Construct the SWIG $\mathcal{G}(t)$. Identify all paths between the random half T and $Y(t)$, and determine whether any back-door paths exist.
(b) Argue that, in this graph, $Y(t) \perp\!\!\!\perp T \mid D$ holds in P_t even though $X = \{D\}$ violates the back-door criterion.
(c) Explain in your own words why this example shows that the no-descendants clause must be imposed as a hypothesis on the adjustment set rather than derived from a conditional independence statement.

Appendix C

Pathwise Differentiability, Tangent Spaces, and the Efficient Influence Function

Chapter 10 introduced the efficient influence function of the average treatment effect functional and stated, without proof, that this object is the *canonical gradient* of Ψ relative to the nonparametric tangent space. This appendix supplies the geometric machinery behind that statement. The treatment follows Bickel et al. (1993) and Tsiatis (2006), with notation aligned to Chapter 10.

The reader should think of this appendix as the formal counterpart of the asymptotic-linearity and efficiency sections of Chapter 10: those sections showed how an influence function determines the asymptotic distribution of an estimator; here we develop the parallel notion of an influence function as a *derivative of a functional*, and characterize the unique influence function that achieves the semiparametric efficiency bound. The notation is standard in semiparametric theory and is introduced as needed.

The applied consequences — the AIPW estimator, double robustness, Neyman orthogonality, cross-fitting, and rate conditions — are developed in Chapters 11 and 12. This appendix supplies only the foundational geometry on which those chapters rest. Section C.1 collects the Hilbert-space background used throughout, for readers who have not previously encountered the projection theorem and Riesz representation. Section C.7 closes the appendix by carrying out the EIF derivation explicitly for the average treatment effect, recovering the AIPW influence function from the projection construction of Theorem C.1.

Learning Objectives

After studying this appendix, students should be able to:

1. Define a regular parametric submodel through differentiability in quadratic mean, identify its score, and construct the tangent space of a model at P.
2. Distinguish an estimating function, the influence function of an estimator, a pathwise gradient of a functional, and the efficient influence function.
3. Explain why pathwise gradients may be non-unique in a restricted model, and why the canonical gradient is nonetheless unique.
4. Derive the efficient influence function either as the Riesz representer of the pathwise derivative on $\mathcal{T}$ or as a projection, and state the normalization that a projection construction requires.
5. Verify the AIPW efficient influence function for the ATE, read off the semiparametric efficiency bound, and state the moment condition under which that bound is finite.

How to read this appendix. The material divides into a core path and a set of refinements, and it is worth separating them on a first reading. The *core* is Section C.1 through the projection theorem and Riesz representation; Definition C.1 and the meaning of a score; Definition C.2 with Example C.1; Definition C.3 and Definition C.4; Theorem C.1; the ATE derivation of Section C.7 through Equation C.13 and the bound Equation C.14; and the synthesis table of Section C.8. That path is self-contained and suffices for reading Chapters 11 and 12.

The remaining material is *second-pass*: the density counterexample (Example C.2); the attainability discussion after Definition C.4; the normal-location example (Example C.3); the convolution discussion following Theorem C.2; the weak-positivity subtleties in the definition of $\mathcal{H}_T$; and the IPW-versus-AIPW remark at the end of Section C.7. Exercises marked *[Advanced]* belong to the same tier. None of it is optional in the long run — each item exists because a natural first guess about the theory is wrong — but none of it is needed to follow the main line.

C.1 Hilbert-Space Background

The geometry of pathwise differentiability lives in the Hilbert space $L_2(P)$ of square-integrable real-valued functions on $\mathcal{O}$, equipped with the inner product

$$\langle f, g\rangle_P \;=\; \mathbb{E}_P[f(O)\,g(O)], \qquad \|f\|_P \;=\; \langle f, f\rangle_P^{1/2}.$$

Two functions are **orthogonal**, written $f \perp g$, when $\langle f, g\rangle_P = 0$. All scores and influence functions introduced below live in the closed subspace of mean-zero functions,

$$L_2^0(P) \;=\; \{f \in L_2(P) : \mathbb{E}_P[f(O)] = 0\}.$$

This section records the four Hilbert-space facts used repeatedly in the rest of the appendix. Readers familiar with these may skip ahead; for a full treatment, see Vaart (1998) (§25.7) or the appendices of Tsiatis (2006).

Closed linear span. For a subset $A \subset L_2^0(P)$, the **closed linear span** $\overline{\operatorname{span}}(A)$ is the smallest closed subspace of $L_2^0(P)$ containing A; concretely, it consists of $L_2(P)$-limits of finite linear combinations of elements of A. Tangent spaces below are defined as closed linear spans, rather than as raw sets of scores, because the set of scores of regular parametric submodels need not itself be closed.

Projection theorem. If $V \subset L_2^0(P)$ is a closed subspace, every $f \in L_2^0(P)$ admits a unique decomposition

$$f \;=\; f_V \;+\; f_{V^\perp}, \qquad f_V \in V, \quad f_{V^\perp} \in V^\perp,$$

where $V^\perp = \{g \in L_2^0(P) : \langle g, h\rangle_P = 0 \text{ for all } h \in V\}$ is the orthogonal complement. The map $\Pi[\,\cdot \mid V\,] : f \mapsto f_V$ is the **orthogonal projection** onto V and is characterized equivalently by either of the following:

(i) $f - \Pi[f \mid V] \perp V$ (orthogonality of the residual);
(ii) $\Pi[f \mid V]$ is the unique element of V minimizing $\|f - g\|_P$ over $g \in V$ (best approximation).

The decomposition $L_2^0(P) = V \oplus V^\perp$ is the engine behind the canonical-gradient construction in Section C.6.

Riesz representation. Every continuous linear functional $\Lambda : L_2^0(P) \to \mathbb{R}$ admits a unique representer $r_\Lambda \in L_2^0(P)$ such that

$$\Lambda(f) \;=\; \langle r_\Lambda, f\rangle_P \qquad \text{for all } f \in L_2^0(P),$$

and conversely every $r \in L_2^0(P)$ defines such a continuous functional via $f \mapsto \langle r, f\rangle_P$. The same statement holds verbatim with $L_2^0(P)$ replaced by any closed subspace, in particular by the tangent space $\mathcal{T}$.

Pathwise differentiability of a functional Ψ (Definition C.2 below) is precisely the statement that the score-to-derivative map $S \mapsto \partial_\varepsilon \Psi(P_\varepsilon)|_0$ extends to a continuous linear functional

$$\dot{\Psi}_P : \mathcal{T} \to \mathbb{R}$$

on the tangent space. Riesz representation then supplies a *unique* $\varphi^* \in \mathcal{T}$ with

$$\dot{\Psi}_P(g) \;=\; \mathbb{E}_P[\varphi^*(O)\,g(O)], \qquad g \in \mathcal{T},$$

and this unique representer in $\mathcal{T}$ is the canonical gradient. It is important to keep two statements apart. Uniqueness holds *within* $\mathcal{T}$; a function $\varphi \in L_2^0(P)$ represents the same derivative on every score if and only if

$$\varphi \;=\; \varphi^* + h \qquad \text{for some } h \in \mathcal{T}^\perp,$$

so representers in the ambient space $L_2^0(P)$ are generally non-unique whenever $\mathcal{T} \neq L_2^0(P)$. Their projections onto $\mathcal{T}$ all coincide. Section C.4 takes up this non-uniqueness, and Theorem C.1 states the resulting dichotomy formally.

Codimension. A non-zero continuous linear functional Λ on a Hilbert space H has closed kernel $\ker(\Lambda) := \{f \in H : \Lambda(f) = 0\}$ of codimension one, and the orthogonal complement of that kernel *within* H is the line spanned by the Riesz representer:

$$H \cap \ker(\Lambda)^{\perp} \;=\; \operatorname{span}\{r_\Lambda\}.$$

Applied with $H = \mathcal{T}$ and $\Lambda = \dot{\Psi}_P$, this describes the geometry of a scalar target: the tangent space splits into the nuisance directions, along which ψ does not move to first order, and the single remaining direction along which it does. The codimension-one statement is a geometric consequence, not the source of uniqueness — uniqueness of φ^* comes from Riesz representation on $\mathcal{T}$ and requires no rank condition. In the degenerate case $\dot{\Psi}_P = 0$ one has $\varphi^* = 0$ and $\ker(\dot{\Psi}_P) = \mathcal{T}$, which has codimension zero.

C.2 Regular Parametric Submodels and Scores

Let $\mathcal{P}$ be a statistical model for the distribution of the observed data O on a sample space $\mathcal{O}$, with true distribution $P \in \mathcal{P}$. For simplicity this appendix works throughout in a *dominated* model: all densities below are taken with respect to one common σ-finite reference measure μ. This is an assumption on $\mathcal{P}$, not a triviality. The general theory of differentiable paths does not require a single measure dominating the whole model — domination local to each path is enough — but the global convention keeps the notation light and costs nothing in the examples treated here. The parameter of interest is a smooth real-valued functional

$$\Psi : \mathcal{P} \to \mathbb{R}, \qquad \psi = \Psi(P).$$

For vector-valued $\Psi : \mathcal{P} \to \mathbb{R}^p$, each component is pathwise differentiable in the sense of Definition C.2, and we write $\varphi_j^* \in \mathcal{T}$ for the *canonical* gradient of the jth component (Definition C.5). Stacking the componentwise canonical gradients gives $\varphi^* = (\varphi_1^*, \dots, \varphi_p^*)^\top$, and the semiparametric efficiency bound is the covariance matrix

$$V^* \;=\; \mathbb{E}_P[\varphi^*(O)\,\varphi^*(O)^\top] \;\in\; \mathbb{R}^{p\times p}.$$

For any regular asymptotically linear estimator with influence function φ, the matrix difference $\mathbb{E}_P[\varphi(O)\,\varphi(O)^\top] - V^*$ is positive semidefinite. The word *canonical* is doing real work here: stacking arbitrary componentwise gradients would admit terms in $\mathcal{T}^\perp$ and inflate the covariance matrix, so the resulting object would not be the bound.

Asymptotic-variance comparisons are then made in the Loewner partial order, or equivalently through all scalar linear contrasts $c^\top\Psi$, $c \in \mathbb{R}^p$. No new Hilbert-space principle is needed. Two caveats are worth recording. The bound matrix may be singular: there is then a non-zero c with $c^\top V^* c = 0$, i.e. $c^\top\varphi^* = 0$ almost surely, so the contrast $c^\top\Psi$ has zero canonical gradient and zero first-order efficiency bound. This usually signals a redundant target vector or a locally flat contrast, and it should not be read as an impossibility theorem ruling out every non-degenerate estimator of that contrast. Second, individual contrasts may differ in their regularity properties, so pathwise differentiability of every component is a genuine requirement rather than a formality. For clarity we restrict attention to $p = 1$ throughout.

To define what it means for Ψ to be "smooth," we need a notion of a path through $\mathcal{P}$ that begins at P. The standard device is the regular parametric submodel.

Definition C.1 (Regular Parametric Submodel)**.** A **regular parametric submodel** of $\mathcal{P}$ passing through P is a one-parameter family

$$\{P_\varepsilon : \varepsilon \in (-\delta, \delta)\} \subset \mathcal{P}, \qquad P_0 = P,$$

that is **differentiable in quadratic mean** (DQM) at $\varepsilon = 0$: there exists $S \in L_2(P)$ such that

$$\int \Big\{ \sqrt{p_\varepsilon(o)} \;-\; \sqrt{p(o)} \;-\; \tfrac{1}{2}\,\varepsilon\, S(o)\,\sqrt{p(o)} \Big\}^2 \, d\mu(o) \;=\; o(\varepsilon^2) \qquad (\varepsilon \to 0). \tag{C.1}$$

The function S is the **score** of the submodel at P.

The definition differentiates the *square-root density* $\varepsilon \mapsto p_\varepsilon^{1/2}$ in $L_2(\mu)$ rather than the log density in $L_2(P)$. This is the standard construction, and the choice is not cosmetic: $L_2(P)$-differentiability of $\log p_\varepsilon$ is neither the usual definition nor, without further integrability conditions, sufficient to license the likelihood and integration manipulations used below. Differentiability in quadratic mean supplies a well-defined square-integrable score and the likelihood expansion underlying local asymptotic normality. It does not by itself license differentiation of an *unbounded* functional such as $\mathbb{E}_{P_\varepsilon}[Y]$; that step may additionally require local moment or uniform-integrability conditions, as Example C.1 illustrates. In particular Equation C.1 implies

$$\mathbb{E}_P[S(O)] = 0, \qquad \mathbb{E}_P[S(O)^2] < \infty, \tag{C.2}$$

so every score lies in $L_2^0(P)$, and the score S is determined by the path up to P-null sets. See Vaart (1998) (§7.2) for the proof and for the local asymptotic normality that DQM buys.

The familiar log-derivative formula survives as a *consequence* rather than a definition. When $\varepsilon \mapsto p_\varepsilon(o)$ is differentiable pointwise and the usual domination conditions permit interchanging differentiation and integration, the DQM score satisfies

$$S(o) \;=\; \frac{\partial}{\partial\varepsilon} \log p_\varepsilon(o)\Big|_{\varepsilon=0} \qquad \text{for } P\text{-almost every } o,$$

and this is the representation used in every concrete calculation below. A regular submodel is, intuitively, a smooth one-dimensional curve through $\mathcal{P}$ that can be probed by ordinary parametric methods, and its score is the directional derivative of the log-likelihood at P along that curve.

Remark: Why Submodels Rather Than Point-Mass Contamination

A heuristic device sometimes used for "probing" a functional is the contamination path $P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_o$. The corresponding Hampel influence function is the derivative

$$\mathrm{IF}(o;\Psi,P) \;=\; \frac{\partial}{\partial\varepsilon}\Psi\{(1-\varepsilon)P + \varepsilon\delta_o\}\Big|_{\varepsilon=0},$$

which reduces to $\Psi(\delta_o) - \Psi(P)$ only for linear functionals (Vaart 1998, sec. 2.1). Contamination is intuitive but does not yield a regular submodel in the sense of Definition C.1: when P is continuous, δ_o is not absolutely continuous with respect to P, so the Radon–Nikodym derivative $d\delta_o/dP$ does not exist as an $L_2(P)$ function, and the contamination path lacks a well-defined score in the sense used here. All formal results below use Definition C.1, while the contamination heuristic remains useful for guessing the form of an influence function in concrete examples.

C.3 Pathwise Differentiability

The functional Ψ is differentiable along a submodel if the map $\varepsilon \mapsto \Psi(P_\varepsilon)$ is differentiable at $\varepsilon = 0$ in the ordinary sense. Pathwise differentiability asserts that this derivative can be represented as an inner product between a fixed function and the score, with the *same* representer for every regular submodel.

Definition C.2 (Pathwise Differentiability and Influence Function)**.** The functional $\Psi : \mathcal{P} \to \mathbb{R}$ is **pathwise differentiable** at P if there exists a function $\varphi \in L_2^0(P)$ such that, for every regular parametric submodel $\{P_\varepsilon\}$ with score S,

$$\frac{\partial}{\partial\varepsilon}\Psi(P_\varepsilon)\Big|_{\varepsilon=0} \;=\; \mathbb{E}_P[\varphi(O)\,S(O)]. \tag{C.3}$$

Any such φ is called an **influence function** (or **gradient**) of Ψ at P. The set of influence functions is denoted $\mathrm{IF}(\Psi, P)$.

Equation Equation C.3 is the defining identity of semiparametric theory. It says the perturbation of Ψ along a submodel is fully encoded by the $L_2(P)$ inner product of a fixed function φ with the score S. Geometrically, φ is a representer for the linear functional $S \mapsto \partial_\varepsilon\Psi(P_\varepsilon)|_0$ restricted to the space of scores. Because the right-hand side of Equation C.3 is continuous in S with respect to the $L_2(P)$ norm, this map extends uniquely by linearity and continuity to the whole tangent space; we write

$$\dot\Psi_P : \mathcal{T} \to \mathbb{R}, \qquad \dot\Psi_P(g) \;=\; \mathbb{E}_P[\varphi(O)\,g(O)],$$

and call $\dot\Psi_P$ the **pathwise derivative** of Ψ at P. Note that $\dot\Psi_P$ does not depend on which gradient φ is used to define it, since any two agree on $\mathcal{T}$.

Example C.1 (Mean Functional)**.** Let $O = Y$ with $\mathbb{E}_P[Y^2] < \infty$ and $\Psi(P) = \mathbb{E}_P[Y]$. Let $\{P_\varepsilon\}$ be a regular submodel with score S, and assume the local integrability condition that permits differentiating $\varepsilon \mapsto \mathbb{E}_{P_\varepsilon}[Y]$ under the integral sign at $\varepsilon = 0$. Then

$$\frac{\partial}{\partial \varepsilon} \int y\, p_\varepsilon(y)\, d\mu(y)\bigg|_0 = \int y\, S(y)\, p(y)\, d\mu(y) = \mathbb{E}_P[Y \cdot S(O)].$$

Subtracting $\mathbb{E}_P[Y] \cdot \mathbb{E}_P[S] = 0$ gives $\partial_\varepsilon \Psi(P_\varepsilon)|_0 = \mathbb{E}_P[(Y - \mathbb{E}_P Y)\, S(O)]$, so

$$\varphi(O) = Y - \Psi(P)$$

is an influence function of the mean functional. Under the nonparametric model (Section C.5), this is in fact the *unique* influence function, and hence the efficient influence function.

The example illustrates two features common to all gradient calculations. First, φ is identified only up to functions orthogonal to the model's tangent space (defined in Section C.5); the constant $\Psi(P)$ above was subtracted to make φ mean-zero. Second, the calculation is an exchange of differentiation and integration. It is worth being precise about what licenses that exchange. Differentiability in quadratic mean supplies the score structure — that $\partial_\varepsilon p_\varepsilon$ behaves like Sp in the appropriate L_2 sense — but it does not by itself justify differentiating the integral of an *unbounded* integrand such as y against p_ε. For bounded Y the interchange is immediate. For unbounded Y it follows from a local second-moment condition, for instance $\sup_{|\varepsilon|<\delta} \mathbb{E}_{P_\varepsilon}[Y^2] < \infty$, which gives uniform integrability along the path. Conditions of this kind are assumed silently in most treatments; they are recorded here because the appendix is meant to be the formal counterpart of Chapter 10 rather than a summary of it.

Not every functional of interest is pathwise differentiable, and the failure is not an edge case. The next example is the reason the hypothesis in Definition C.2 has content.

Example C.2 (A Functional with No Gradient)**.** *Second reading.*

Let $O = Y$ and let $\mathcal{P}$ consist of the distributions on $\mathbb{R}$ whose density admits a continuous version in a neighborhood of a fixed point y_0, with $p(y_0) > 0$. Evaluating that continuous version at y_0 defines $\Psi(P) = p(y_0)$.

Fixing a version matters: an element of $L_2(P)$ is an equivalence class, so pointwise evaluation is not otherwise well defined. For the same reason the perturbations must be restricted to *bounded continuous* $h \in L_2^0(P)$, so that $p_\varepsilon = (1 + \varepsilon h)\, p$ remains a continuous density near y_0 and stays inside $\mathcal{P}$. Each such path is regular with score h, by the computation in Section C.5, and

$$\frac{\partial}{\partial \varepsilon} \Psi(P_\varepsilon)\bigg|_0 = p(y_0)\, h(y_0).$$

Now let h_k be continuous mean-zero bumps supported on $[y_0 - 1/k,\, y_0 + 1/k]$ with peak height $\sqrt{k}$. Then $\|h_k\|_P = O(1)$, because the squared peak k is offset by the support width $2/k$, while $\partial_\varepsilon \Psi(P_{\varepsilon,k})|_0 \asymp \sqrt{k}\, p(y_0) \to \infty$. The score-to-derivative map is therefore unbounded on the unit ball of $\mathcal{T}$: it is not continuous, and it has no Riesz representer in $L_2(P)$. Informally, the only object that would represent point evaluation is a Dirac delta, which is not a function in $L_2(P)$.

Two conclusions follow, and they are not the same conclusion. The functional is not pathwise differentiable, so by the remark on the connection to estimation below, no regular *asymptotically linear* estimator of $p(y_0)$ exists in this model. Ruling out every regular $\sqrt{n}$-consistent estimator is a strictly stronger statement and needs the local asymptotic minimax or convolution machinery of Vaart (1998) (Chapter 25); under that framework it does indeed follow. This is the analytic content of the familiar fact that density estimation converges more slowly than $\sqrt{n}$.

The same phenomenon affects causal targets. A regression function $\mathbb{E}_P[Y \mid X = x_0]$ at a fixed value of a continuously distributed covariate, or a dose-response curve $\mathbb{E}_P[Y(t_0)]$ at a single level of a continuous treatment, is not pathwise differentiable in the unrestricted model. Students sometimes assume that any causal estimand that is *identified* must possess an efficient influence function; identification and pathwise differentiability are independent properties. It is equally tempting, and equally wrong, to suppose that adding smoothness fixes the problem. Restrictions do change the tangent space and can improve rates, but ordinary finite-order smoothness or shape constraints — Hölder classes, monotonicity — generally leave pointwise targets slower than $\sqrt{n}$. What typically restores pathwise differentiability is smoothing or averaging the *target* itself, as the ATE averages a regression contrast over P_X in Section C.7, or imposing a sufficiently strong structural, often finite-dimensional, restriction on the model.

> Remark: Connection to Estimation
>
> If an estimator $\hat{\psi}$ is asymptotically linear at P with influence function φ in the sense of Chapter 10, and if $\hat{\psi}$ is regular along every submodel, then φ satisfies Equation C.3 and is therefore a gradient of Ψ. Pathwise differentiability is thus a necessary condition for the existence of a regular asymptotically linear estimator of Ψ (Tsiatis 2006, Theorem 3.1).

C.4 Non-Uniqueness of Influence Functions

Definition C.2 does not produce a unique influence function. If φ satisfies Equation C.3 and $h \in L_2^0(P)$ is orthogonal to every score of the model in $L_2(P)$, then $\varphi + h$ also satisfies Equation C.3, since $\mathbb{E}_P[(\varphi + h)S] = \mathbb{E}_P[\varphi S] + \mathbb{E}_P[hS] = \mathbb{E}_P[\varphi S]$. Whether φ is unique depends on how rich the collection of scores is — a point made precise in Section C.5 through the notion of the tangent space.

The terminological distinction Chapter 10 alluded to can now be made precise:

- An **estimating function** is any function $U(O;\theta)$ whose root defines an estimator.
- An **asymptotic influence function** of an estimator is the function φ in its asymptotic expansion.
- A **(pathwise) influence function** of a functional is a $\varphi \in L_2^0(P)$ satisfying Equation C.3.

For a regular asymptotically linear estimator of a pathwise-differentiable functional, the estimator's asymptotic influence function is also a pathwise influence function of the functional. The relationship to estimating functions is looser: for an M-estimator solving $n^{-1}\sum_{i=1}^n U(O_i;\theta) = 0$, the standard expansion gives

$$\varphi(O) \;=\; -A^{-1}\,U(O;\theta_0), \qquad A \;=\; \mathbb{E}_P\big[\partial U(O;\theta_0)/\partial\theta^\top\big]\,,$$

so a generic estimating function U equals the influence function φ only after this normalization (or when U is written directly in influence-function form). The first two notions can be defined without choosing a statistical model class $\mathcal{P}$ — an estimator influence function is still defined at a particular distribution P — while the third depends crucially on $\mathcal{P}$ through the collection of admissible submodels. This dependence is what makes $\mathrm{IF}(\Psi, P)$ generally non-unique in semiparametric models.

C.5 The Tangent Space

The space of admissible perturbations of P within the model $\mathcal{P}$ is called the tangent space.

Definition C.3 (Tangent Space)**.** The **tangent space** of $\mathcal{P}$ at P, denoted $\mathcal{T}_P(\mathcal{P})$ or simply $\mathcal{T}$, is the closed linear span in $L_2^0(P)$ of all scores of regular parametric submodels of $\mathcal{P}$ passing through P.

The tangent space is a closed subspace of the Hilbert space $L_2^0(P)$. Two extreme cases are illustrative.

Unrestricted nonparametric model. Suppose $\mathcal{P}$ is the unrestricted dominated model, containing every distribution on $\mathcal{O}$ that is absolutely continuous with respect to μ. For *bounded* $h \in L_2^0(P)$ the tilted family

$$p_\varepsilon(o) \;=\; (1 + \varepsilon\, h(o))p(o)$$

is a genuine density for $|\varepsilon| < 1/\|h\|_\infty$: it is non-negative there, and it integrates to $1 + \varepsilon\,\mathbb{E}_P[h] = 1$ exactly, so no normalizing constant is required. The path is regular with score $S = h$. Since bounded mean-zero functions are dense in $L_2^0(P)$, taking the closed linear span gives $\mathcal{T} = L_2^0(P)$.

It is worth resisting the habit of treating "nonparametric" and "maximal tangent space" as synonyms. The conclusion $\mathcal{T} = L_2^0(P)$ holds because the model admits every bounded tilt. A model can be infinite-dimensional and still impose support, shape, monotonicity, smoothness, or moment restrictions, any of which excludes some tilts and yields a proper closed subspace $\mathcal{T} \subsetneq L_2^0(P)$.

Fully parametric model. If $\mathcal{P} = \{P_\theta : \theta \in \Theta \subset \mathbb{R}^k\}$ is parametric with score components $S_{\theta_0,1}, \dots, S_{\theta_0,k}$ at θ_0, then $\mathcal{T} = \operatorname{span}\{S_{\theta_0,1}, \dots, S_{\theta_0,k}\}$ is finite-dimensional, of dimension $\operatorname{rank}\{I(\theta_0)\} \le k$, where $I(\theta_0)$ is the Fisher information matrix. The dimension is exactly k precisely when the score components are linearly independent in $L_2(P)$, that is, when $I(\theta_0)$ is non-singular. A singular information matrix means that some parameter direction produces no first-order perturbation of P. That may reflect local non-identification or a redundant parameterization, but it may also arise from an irregular yet perfectly injective parameterization: the model $P_\theta = N(\theta^3, 1)$ has zero score at $\theta_0 = 0$ while $\theta \mapsto P_\theta$ remains one-to-one. Singular information is therefore best described as first-order degeneracy rather than as failure of identification.

Many semiparametric causal problems have a finite-dimensional target $\psi = \Psi(P)$ and infinite-dimensional nuisance. In fully nonparametric observed-data models, the tangent space is often the whole space $L_2^0(P)$; in restricted semiparametric models — for instance, those imposing a known treatment mechanism, a known missingness mechanism, or a structural restriction on P — the tangent space $\mathcal{T}$ is a proper infinite-dimensional subspace of $L_2^0(P)$.

Definition C.4 (First-Order Nuisance Tangent Space)**.** Suppose Ψ is pathwise differentiable at P with pathwise derivative $\dot{\Psi}_P : \mathcal{T} \to \mathbb{R}$. The **nuisance tangent space** of Ψ at P is the kernel of that derivative,

$$\mathcal{T}_\eta \;:=\; \ker(\dot{\Psi}_P) \;=\; \{\, g \in \mathcal{T} \;:\; \dot{\Psi}_P(g) = 0 \,\}.$$

Because $\dot{\Psi}_P$ is continuous, $\mathcal{T}_\eta$ is a closed linear subspace of $\mathcal{T}$.

A regular submodel whose score lies in $\mathcal{T}_\eta$ is called a **nuisance submodel**: it moves P without moving ψ to first order. Defining $\mathcal{T}_\eta$ as a kernel rather than as the closed span of attainable nuisance scores is a deliberate choice, and the two are not automatically the same object. Every score of a nuisance submodel does lie in the kernel, so the closed span of such scores is contained in $\mathcal{T}_\eta$; the reverse inclusion requires that every tangent direction annihilated by $\dot{\Psi}_P$ be attainable — or at least approximable in $L_2(P)$ — by scores of actual nuisance submodels. In sufficiently rich models, including the unrestricted nonparametric models used throughout Chapters 10–12, that local richness holds and the two descriptions agree. It is not automatic in a general model, and taking the kernel as the definition avoids leaving an attainability assumption unstated.

The nuisance tangent space is a closed subspace of $\mathcal{T}$, and hence of $L_2^0(P)$. Throughout the appendix, its orthogonal complement is taken in $L_2^0(P)$:

$$\mathcal{T}_\eta^\perp \;:=\; \{f \in L_2^0(P) : \mathbb{E}_P[fg] = 0 \text{ for all } g \in \mathcal{T}_\eta\}. \tag{C.4}$$

With this convention, the Hilbert-space decomposition

$$L_2^0(P) \;=\; \mathcal{T}_\eta \,\oplus\, \mathcal{T}_\eta^\perp \tag{C.5}$$

holds automatically by the projection theorem, and provides the geometric structure underlying semiparametric efficiency.

Proposition C.1 (Every Influence Function Lies in $\mathcal{T}_\eta^\perp$)**.** *Let Ψ be pathwise differentiable at P. Then every influence function $\varphi \in \mathrm{IF}(\Psi, P)$ satisfies $\varphi \in \mathcal{T}_\eta^\perp$.*

Proof. Let $\varphi \in \mathrm{IF}(\Psi, P)$. By Definition C.2 and the continuous extension described in Section C.3, $\mathbb{E}_P[\varphi\, g] = \dot{\Psi}_P(g)$ for every $g \in \mathcal{T}$. If in addition $g \in \mathcal{T}_\eta = \ker(\dot{\Psi}_P)$, the right-hand side vanishes, so $\mathbb{E}_P[\varphi\, g] = 0$. Hence $\varphi \perp \mathcal{T}_\eta$, i.e. $\varphi \in \mathcal{T}_\eta^\perp$. □

Proposition C.1 is the key structural fact: being orthogonal to $\mathcal{T}_\eta$ is automatic for influence functions, not a restriction. The distinguishing property of the efficient influence function, introduced in the next section, is instead that it is the unique influence function lying in the full tangent space $\mathcal{T}$.

$\mathcal{T}_\eta^\perp$ Is Not the Same as $\mathcal{T} \cap \mathcal{T}_\eta^\perp$

Because Equation C.4 takes the complement in all of $L_2^0(P)$, the nuisance orthocomplement decomposes as

$$\mathcal{T}_\eta^\perp \;=\; (\mathcal{T} \cap \mathcal{T}_\eta^\perp) \,\oplus\, \mathcal{T}^\perp.$$

For a scalar target with $\dot{\Psi}_P \neq 0$, the first summand is the one-dimensional line $\mathrm{span}\{\varphi^*\}$, but the second can be large. In a restricted model $\mathcal{T}_\eta^\perp$ is therefore generally *not* one-dimensional; it becomes so only when $\mathcal{T} = L_2^0(P)$, as in the nonparametric ATE model of Section C.7. Keeping the two spaces apart is what makes Theorem C.1 say something: orthogonality to $\mathcal{T}_\eta$ is free, whereas membership in $\mathcal{T}$ is the selection criterion.

Remark: Causal Example

For the ATE under consistency, conditional exchangeability, and positivity, the causal estimand is identified with the observed-data functional

$$\tau(P) \;=\; \mathbb{E}_P[\mu_1(X) - \mu_0(X)]\,, \qquad \mu_t(X) \;=\; \mathbb{E}_P[Y \mid T = t,\, X].$$

In the nonparametric observed-data model for $O = (X, T, Y)$, $\mathcal{T} = L_2^0(P)$, and $\mathcal{T}_\eta$ is the closed linear span of scores of submodels along which the pathwise derivative of τ vanishes. Section C.7 carries out the construction in detail and shows that, provided $\dot{\tau}_P \neq 0$, the space $\mathcal{T}_\eta^\perp$ is one-dimensional and spanned by the AIPW influence function of Chapter 10; this is the formal content of the claim made there that φ^* is uniquely determined. Here the complement happens to be one-dimensional because the model is unrestricted, so $\mathcal{T}^\perp = \{0\}$.

The abstract claim that gradients may be non-unique deserves a model in which one can watch it happen. The nonparametric examples above are the wrong place to look: there $\mathcal{T} = L_2^0(P)$, so $\mathcal{T}^\perp = \{0\}$ and the gradient is unique. Non-uniqueness requires a *restricted* model, and the smallest interesting one suffices.

Example C.3 (Non-Uniqueness in a Normal Location Model)**.** *Second reading.*

Let $Y \sim N(\mu, 1)$ with $\mathcal{P} = \{N(\mu, 1) : \mu \in \mathbb{R}\}$ and target $\Psi(P_\mu) = \mu$. The model is one-dimensional, with score $S(Y) = Y - \mu$ and

$$\mathcal{T} \;=\; \operatorname{span}\{Y - \mu\} \;\subsetneq\; L_2^0(P).$$

Write $Z = Y - \mu \sim N(0, 1)$. Since $\partial_\mu \Psi(P_\mu) = 1$ and $\mathbb{E}_P[Z^2] = 1$, the function $\varphi_1(Y) = Y - \mu$ is a gradient. So, however, is

$$\varphi_2(Y) \;=\; Y - \mu + c\left\{(Y - \mu)^2 - 1\right\} \qquad \text{for every } c \in \mathbb{R},$$

because $\mathbb{E}_P[(Z^2 - 1)Z] = \mathbb{E}_P[Z^3] - \mathbb{E}_P[Z] = 0$, so the added term is orthogonal to the only score in the model and contributes nothing to Equation C.3. Indeed $\mathbb{E}_P[\varphi_2 S] = 1$ for all c. The added term lies in $\mathcal{T}^\perp$, exactly as Theorem C.1 (i) will require.

Projecting onto $\mathcal{T}$ recovers the canonical gradient:

$$\Pi[\varphi_2 \mid \mathcal{T}] \;=\; \frac{\langle \varphi_2,\, Z\rangle_P}{\|Z\|_P^2}\, Z \;=\; Z \;=\; \varphi_1,$$

independently of c. Comparing variances,

$$\mathbb{E}_P[\varphi_2^2] \;=\; 1 + 2c^2 \;\geq\; 1 \;=\; \mathbb{E}_P[\varphi_1^2],$$

with equality only at $c = 0$ — the Pythagorean inequality of Theorem C.1 (iii) in the simplest possible setting. The efficiency bound is $V^* = 1$, attained by the sample mean, and the extra term $c\{(Y - \mu)^2 - 1\}$ is precisely the kind of variance inflation that projection removes.

The non-canonical gradients are not merely formal. Each is realized by an actual estimator: for c fixed, put

$$\hat{\mu}_c \;=\; \bar{Y} \;+\; c\left\{\frac{1}{n}\sum_{i=1}^n (Y_i - \bar{Y})^2 - 1\right\}.$$

Because $n^{-1}\sum_i (Y_i - \bar{Y})^2 = n^{-1}\sum_i (Y_i - \mu)^2 - (\bar{Y} - \mu)^2$ and $\sqrt{n}(\bar{Y} - \mu)^2 = O_P(n^{-1/2})$,

$$\sqrt{n}(\hat{\mu}_c - \mu) \;=\; \frac{1}{\sqrt{n}}\sum_{i=1}^n \Big[(Y_i - \mu) + c\{(Y_i - \mu)^2 - 1\}\Big] \;+\; o_P(1),$$

so $\hat{\mu}_c$ is regular and asymptotically linear with influence function φ_2, and its asymptotic variance is $1 + 2c^2$. Only $c = 0$, the sample mean, attains the bound. This matters because the connection between asymptotic linearity and gradients runs in one direction only — exhibiting a gradient does not by itself produce an estimator — and here the estimator can be written down.

Two lessons generalize. First, non-uniqueness is a statement about the *model*, not the functional: enlarging $\mathcal{P}$ to the unrestricted dominated finite-variance model on $\mathbb{R}$ shrinks $\mathcal{T}^\perp$ to $\{0\}$ and makes the gradient unique. Second, the non-canonical gradients are not wrong — each is a legitimate influence function of a regular asymptotically linear estimator — they are merely inefficient.

C.6 The Canonical Gradient and the Efficiency Bound

Proposition C.1 showed that every influence function lies in $\mathcal{T}_\eta^\perp$. What distinguishes a single influence function within this set? The answer is orthogonality to $\mathcal{T}^\perp$, equivalently membership in the full tangent space $\mathcal{T}$: among all influence functions, there is a unique one lying in $\mathcal{T}$, and it is the one with smallest variance.

Theorem C.1 (Canonical Gradient)**.** *Let Ψ be pathwise differentiable at P, so that the set $\mathrm{IF}(\Psi, P)$ of gradients is non-empty. Then:*

(i) Any two influence functions differ by an element of $\mathcal{T}^\perp$: for $\varphi_1, \varphi_2 \in \mathrm{IF}(\Psi, P)$, $\varphi_1 - \varphi_2 \in \mathcal{T}^\perp$.
(ii) There exists a unique $\varphi^* \in \mathrm{IF}(\Psi, P)$ with $\varphi^* \in \mathcal{T}$, namely

$$\varphi^* \;=\; \Pi[\varphi \mid \mathcal{T}] \qquad \textit{for any } \varphi \in \mathrm{IF}(\Psi, P),$$

where $\Pi[\,\cdot \mid \mathcal{T}\,]$ denotes orthogonal projection in $L_2^0(P)$ onto $\mathcal{T}$.
(iii) For every $\varphi \in \mathrm{IF}(\Psi, P)$, $\mathbb{E}_P[\varphi(O)^2] \ge \mathbb{E}_P[\varphi^*(O)^2]$, with equality iff $\varphi = \varphi^*$ in $L_2(P)$.

Proof. (i) For $\varphi_1, \varphi_2 \in \mathrm{IF}(\Psi, P)$ and every score S of a regular submodel, $\mathbb{E}_P[(\varphi_1 - \varphi_2)S] = \partial_\varepsilon \Psi|_0 - \partial_\varepsilon \Psi|_0 = 0$. By linearity and L_2-continuity, $\mathbb{E}_P[(\varphi_1 - \varphi_2)g] = 0$ for every $g \in \mathcal{T}$, i.e., $\varphi_1 - \varphi_2 \in \mathcal{T}^\perp$.

(ii) Fix any $\varphi \in \mathrm{IF}(\Psi, P)$ and set $\varphi^* := \Pi[\varphi \mid \mathcal{T}]$, so $\varphi - \varphi^* \in \mathcal{T}^\perp$. For any score $S \in \mathcal{T}$,

$$\mathbb{E}_P[\varphi^* S] \;=\; \mathbb{E}_P[\varphi S] - \mathbb{E}_P[(\varphi - \varphi^*)S] \;=\; \mathbb{E}_P[\varphi S] \;=\; \frac{\partial}{\partial \varepsilon}\Psi(P_\varepsilon)\bigg|_0,$$

so $\varphi^* \in \mathrm{IF}(\Psi, P)$. Uniqueness: if $\tilde{\varphi}^* \in \mathrm{IF}(\Psi, P) \cap \mathcal{T}$, then by (i) $\varphi^* - \tilde{\varphi}^* \in \mathcal{T}^\perp \cap \mathcal{T} = \{0\}$.

(iii) Write $\varphi = \varphi^* + (\varphi - \varphi^*)$ with the two summands orthogonal in $L_2(P)$ (since $\varphi^* \in \mathcal{T}$ and $\varphi - \varphi^* \in \mathcal{T}^\perp$). The Pythagorean identity gives

$$\mathbb{E}_P[\varphi^2] \;=\; \mathbb{E}_P[(\varphi^*)^2] + \mathbb{E}_P[(\varphi - \varphi^*)^2] \;\ge\; \mathbb{E}_P[(\varphi^*)^2],$$

with equality iff $\varphi = \varphi^*$. □

The figure below summarizes the geometry of the theorem in a single picture, and is worth consulting alongside Example C.3.

Definition C.5 (Efficient Influence Function and Efficiency Bound)**.** The unique element $\varphi^* \in \mathrm{IF}(\Psi, P) \cap \mathcal{T}$ is called the **efficient influence function** (EIF), or **canonical gradient**, of Ψ at P. Its variance, $V^*(\Psi, P) = \mathbb{E}_P[\varphi^*(O)^2]$, is the **semiparametric efficiency bound** for estimating $\Psi(P)$ in the model $\mathcal{P}$.

As stated, $V^*(\Psi, P)$ is finite whenever it is defined at all, since $\varphi^* \in L_2^0(P)$ by construction. If the pathwise derivative has no continuous $L_2(P)$ representer, then Ψ is not pathwise differentiable at P and no bound exists in the sense of Definition C.5. It is convenient to record that situation through the extended-value convention

$$V^*(\Psi, P) \;=\; +\infty \qquad \text{when no } L_2(P) \text{ canonical gradient exists,} \tag{C.6}$$

and we use Equation C.6 below when discussing what weak overlap does to the ATE bound.

Remark: Role of the Nuisance Tangent Space

Since every influence function already lies in $\mathcal{T}_\eta^\perp$ (Proposition C.1), the EIF can equivalently be described as the unique influence function in $\mathcal{T} \cap \mathcal{T}_\eta^\perp$. This intersection contains exactly the directions in $\mathcal{T}$ that are orthogonal to nuisance perturbations — i.e., directions along which ψ genuinely moves. Two distinct projection constructions are used in practice, and it is worth stating both precisely, because only the first is automatic.

(a) Projection of a known gradient. If φ is already a pathwise gradient of Ψ, then $\varphi^* = \Pi[\varphi \mid \mathcal{T}]$ by

Theorem C.1 (ii). Nothing further is needed: projection preserves the derivative identity, because the discarded part lies in $\mathcal{T}^{\perp}$.
(b) Projection of a target score. In a model carrying a regular parametrization with target score S_ψ, the **efficient score** is the residual of S_ψ after removing nuisance directions,

$$S_{\text{eff}} \;=\; S_\psi \;-\; \Pi[S_\psi \mid \mathcal{T}_\eta],$$

and the efficient influence function is the *normalized* version

$$\varphi^* \;=\; I_{\text{eff}}^{-1}\, S_{\text{eff}}, \qquad I_{\text{eff}} \;=\; \mathbb{E}_P[S_{\text{eff}}(O)^2],$$

provided $I_{\text{eff}} > 0$. In the degenerate case $I_{\text{eff}} = 0$ one has $S_{\text{eff}} = 0$: the target is locally flat, the canonical gradient is $\varphi^* = 0$, and the normalization is neither available nor needed.

Projection Alone Does Not Produce the EIF

It is tempting to summarize both constructions as "take an unbiased estimating function and subtract its projection onto $\mathcal{T}_\eta$." That recipe is too broad. Subtracting the nuisance projection from an arbitrary function of $\mathcal{T}$ produces *some* element of $\mathcal{T} \cap \mathcal{T}_\eta^{\perp}$, and for a scalar target every multiple $c\,\varphi^*$ lies in that one-dimensional space. Only $c = 1$ represents the pathwise derivative. The missing ingredient is the normalization in (b), or equivalently a direct verification that the candidate satisfies Equation C.3. Removing nuisance directions fixes the *direction* of the answer; it does not fix its *scale*.

The efficiency bound has the following interpretation, which is the formal version of the heuristic statement in Chapter 10.

Theorem C.2 (Asymptotic Efficiency Bound)**.** *Let $O_1, \ldots, O_n$ be i.i.d. from P, and let $\hat{\psi}_n$ be a regular asymptotically linear estimator of $\Psi(P)$ in the model $\mathcal{P}$. Then its asymptotic variance satisfies*

$$\mathrm{AVar}(\sqrt{n}(\hat{\psi}_n - \psi)) \;\geq\; V^*(\Psi, P),$$

and the bound is achieved if and only if the influence function of $\hat{\psi}_n$ is φ^* in $L_2(P)$.

For regular asymptotically linear estimators, the variance bound in Theorem C.2 follows directly from Theorem C.1 (iii) applied to the estimator's influence function, which is itself a gradient of Ψ. This is the statement the appendix actually needs, and it is elementary given the projection theorem.

The Hájek–Le Cam convolution theorem extends the lower-bound interpretation beyond the asymptotically linear class, but its content should be stated carefully. Under the local asymptotic normality and regularity conditions the theorem requires, every weak limit law of a regular estimator has the convolution form $N(0, V^*) * M$ for some probability measure M; equivalently $\sqrt{n}(\hat{\psi}_n - \psi) \rightsquigarrow Z + W$ with $Z \sim N(0, V^*)$ independent of $W \sim M$. If that limit law has a finite second moment, its variance is at least V^*, and one recovers a variance comparison. Without a finite second moment the comparison by "taking variances" is unavailable — a regular estimator need not have a limit distribution with any moments at all — and the correct conclusion is the convolution statement itself, or the equivalent local asymptotic minimax bound. The limits are also in general subsequential. A precise treatment requires the local asymptotic normality framework of Vaart (1998) (Chapter 25). For our purposes the content is that no regular estimator improves on V^* in the convolution ordering, and that a regular asymptotically linear estimator achieves the bound exactly when its influence function equals the EIF.

Remark: Why the EIF Appears in Concrete Formulas

Theorem C.1 explains why specific objects such as the AIPW influence function of Chapter 10 look "constructed" rather than guessed: they are canonical gradients, and the construction is projection. The lesson to carry forward is that an unbiased estimating function need not be a pathwise gradient at all, so unbiasedness is evidence neither of efficiency nor of gradient status; which functions qualify depends on the tangent space, and hence on the model. Section C.7 settles the case of IPW for the ATE.

The relationship to Chapter 11 deserves care, since the two projections are not the same operation. The finite-sample projection carried out there is an estimator-side variance-reduction argument over an augmentation space, and it recovers the same AIPW algebra. Theorem C.1 (ii) instead projects a genuine pathwise gradient onto a model tangent space. The two agree literally for U_{IPW} in the restricted model with known π, where that function really is a gradient; in the unrestricted model it is merely an unbiased estimating function, and Section C.7 supplies the functional-side canonical-gradient derivation instead.

C.7 Worked Example: The ATE Functional

This section makes the geometric machinery of the appendix concrete by carrying out the EIF derivation for the average treatment effect under the nonparametric observed-data model. The end product is the AIPW influence function of Chapter 10. The value of the derivation lies in showing how it arises directly from Theorem C.1 as the Riesz representer in $L_2^0(P)$ of the pathwise derivative of τ — recovering the AIPW formula from first principles rather than guessing it.

Setup. The observed data are $O = (X, T, Y)$ with X pre-treatment covariates, $T \in \{0, 1\}$ a treatment indicator, and Y the outcome. Under the identification assumptions of Chapters 4–5 — consistency, conditional exchangeability, and positivity — the ATE is identified with the observed-data functional

$$\tau(P) \;=\; \mathbb{E}_P[\mu_1(X) - \mu_0(X)]\,, \qquad \mu_t(X) \;=\; \mathbb{E}_P[Y \mid T = t,\, X].$$

Write

$$\pi_t(x) \;=\; P(T = t \mid X = x), \qquad \pi(x) \;:=\; \pi_1(x), \quad \pi_0(x) = 1 - \pi(x),$$

for the propensity score, and $\sigma_t^2(X) = \mathrm{Var}_P(Y \mid T = t, X)$ for the conditional outcome variances. The model $\mathcal{P}$ is the unrestricted dominated model for the joint law of (X, T, Y) subject to positivity, $0 < \pi(X) < 1$ almost surely, so by the nonparametric example in Section C.5 the tangent space is $\mathcal{T} = L_2^0(P)$.

Positivity Alone Is Not Enough

Pointwise positivity does not guarantee that the object we are about to construct lies in $L_2(P)$, and if it does not, then τ is not pathwise differentiable in the sense of Definition C.2 and the efficiency bound is infinite. Propensity scores approaching 0 or 1 are an important way this can fail, since they amplify conditional outcome noise, but they are not the only way: heavy-tailed outcomes, large conditional variances, or a treatment-effect contrast $\mu_1 - \mu_0$ that is not square integrable will do it too. We therefore assume throughout this section that $\tau(P)$ is finite and that

$$\mathbb{E}_P\left[\frac{\sigma_1^2(X)}{\pi(X)} + \frac{\sigma_0^2(X)}{1 - \pi(X)} + \{\mu_1(X) - \mu_0(X) - \tau(P)\}^2\right] \;<\; \infty. \tag{C.7}$$

Condition Equation C.7 is exactly square integrability of the candidate efficient influence function Equation C.13 derived below, as the variance calculation at the end of this section makes plain. A convenient sufficient condition is **strong overlap**,

$$\epsilon \;\le\; \pi(X) \;\le\; 1 - \epsilon \quad \text{almost surely for some } \epsilon > 0,$$

together with $\mathbb{E}_P[Y^2] < \infty$. When Equation C.7 fails we write $V^*(\tau, P) = +\infty$ under the convention Equation C.6, with the understanding that τ is then not pathwise differentiable at P in the sense of Definition C.2. This is not a technical footnote: overlap is one principal component of Equation C.7, which is the precise sense in which the practical overlap diagnostics of Chapters 6 and 9 bear on whether root-n efficient estimation of the ATE is possible.

Score factorization. The joint density factors as

$$p(x, t, y) \;=\; p_X(x) \cdot p_{T|X}(t \mid x) \cdot p_{Y|T,X}(y \mid t, x).$$

Differentiating $\log p_\varepsilon$ along any regular submodel $\{P_\varepsilon\}$ yields the additive score decomposition

$$S(O) \;=\; S_X(X) \,+\, S_T(T \mid X) \,+\, S_Y(Y \mid T, X),$$

where $S_X = \partial_\varepsilon \log p_{X,\varepsilon}|_0$ and analogously for S_T and S_Y. These satisfy the conditional mean-zero relations

$$\mathbb{E}_P[S_X(X)] = 0, \qquad \mathbb{E}_P[S_T(T \mid X) \mid X] = 0, \qquad \mathbb{E}_P[S_Y(Y \mid T, X) \mid T, X] = 0.$$

Define the closed subspaces of $L_2^0(P)$:

$$\begin{aligned}
\mathcal{H}_X &= \{a(X) : a \in L_2(P_X),\ \mathbb{E}_P[a(X)] = 0\},\\
\mathcal{H}_T &= \{h(T, X) \in L_2^0(P)\ :\ \mathbb{E}_P[h(T, X) \mid X] = 0\},\\
\mathcal{H}_Y &= \{c(O) \in L_2^0(P)\ :\ \mathbb{E}_P[c(O) \mid T, X] = 0\}.
\end{aligned}$$

The treatment space $\mathcal{H}_T$ is specified by a conditional mean-zero condition rather than by an explicit parametrization, because the two are not equivalent under weak positivity. For binary T, every $h \in \mathcal{H}_T$ can indeed be written as

$$h(T, X) \ = \ b(X)\,\{T - \pi(X)\}, \qquad b(X) \ = \ \frac{h(1, X)}{1 - \pi(X)}, \tag{C.8}$$

but the square-integrability requirement on b is weighted by the conditional variance of T:

$$\mathbb{E}_P[h(T, X)^2] \ = \ \mathbb{E}_P\big[b(X)^2\,\pi(X)\{1 - \pi(X)\}\big] \ < \ \infty.$$

A treatment score can therefore be square-integrable while $b \notin L_2(P_X)$, precisely in the regions where $\pi(X)$ approaches 0 or 1. Defining $\mathcal{H}_T$ as $\{b(X)(T - \pi(X)) : b \in L_2(P_X)\}$ would exclude such directions and give a space strictly smaller than the treatment tangent space. Under strong overlap the weighted condition and $b \in L_2(P_X)$ are equivalent, and the distinction is immaterial.

A short conditioning calculation shows that the three subspaces are pairwise orthogonal in $L_2^0(P)$. For $a(X) \in \mathcal{H}_X$ and $h(T, X) \in \mathcal{H}_T$,

$$\mathbb{E}_P[a(X)\,h(T, X)] \ = \ \mathbb{E}_P[a(X)\,\mathbb{E}_P\{h(T, X) \mid X\}] \ = \ 0,$$

and the other two pairs follow analogously by conditioning on (T, X). Joint spanning is explicit rather than merely assertible. Given any $g \in L_2^0(P)$, set

$$\begin{aligned}
g_X &\ = \ \mathbb{E}_P[g \mid X],\\
g_T &\ = \ \mathbb{E}_P[g \mid T, X] \ - \ \mathbb{E}_P[g \mid X],\\
g_Y &\ = \ g \ - \ \mathbb{E}_P[g \mid T, X].
\end{aligned} \tag{C.9}$$

These three terms sum to g by cancellation, and each lies in the advertised subspace: $\mathbb{E}_P[g_X] = \mathbb{E}_P[g] = 0$ puts g_X in $\mathcal{H}_X$; $\mathbb{E}_P[g_T \mid X] = 0$ by the tower property puts g_T in $\mathcal{H}_T$; and $\mathbb{E}_P[g_Y \mid T, X] = 0$ puts g_Y in $\mathcal{H}_Y$. Since Equation C.9 is a nested sequence of conditional expectations, it is precisely the orthogonal projection of g onto the three subspaces. Hence

$$L_2^0(P) \ = \ \mathcal{H}_X \oplus \mathcal{H}_T \oplus \mathcal{H}_Y. \tag{C.10}$$

The score decomposition $S = S_X + S_T + S_Y$ above is precisely the projection decomposition of $S \in L_2^0(P)$ relative to Equation C.10.

The pathwise derivative. The calculation below differentiates the conditional outcome means $\mu_{t,\varepsilon}(x) = \mathbb{E}_{P_\varepsilon}[Y \mid T = t, X = x]$ under the conditional integral, which as in Example C.1 requires more than differentiability in quadratic mean. We therefore restrict attention to regular submodels satisfying the local conditional-moment conditions needed for that interchange; equivalently, one may establish the identity first for bounded-score, locally square-integrable paths and then extend it to the tangent-space closure by continuity. With that understood, differentiating $\tau(P_\varepsilon) = \int (\mu_{1,\varepsilon}(x) - \mu_{0,\varepsilon}(x))\,p_{X,\varepsilon}(x)\,d\mu(x)$ at $\varepsilon = 0$ and applying the product rule gives

$$\begin{aligned}
\frac{\partial}{\partial\varepsilon}\tau(P_\varepsilon)\Big|_0 \ = \ & \underbrace{\int \{\mu_1(x) - \mu_0(x)\}\,S_X(x)\,p_X(x)\,d\mu(x)}_{(I)}\\
& + \underbrace{\int \Big(\partial_\varepsilon \mu_{1,\varepsilon}(x) - \partial_\varepsilon \mu_{0,\varepsilon}(x)\Big)\Big|_{\varepsilon=0} p_X(x)\,d\mu(x)}_{(II)}.
\end{aligned} \tag{C.11}$$

No S_T term appears: $\tau(P)$ is a functional of p_X and $p_{Y|T,X}$ only, so perturbations of the treatment law do not affect τ to first order. This already shows that $\mathcal{H}_T \subset \mathcal{T}_\eta$.

For **Term (I)**, since $\mathbb{E}_P[S_X] = 0$ we may subtract any constant from $\mu_1(X) - \mu_0(X)$ without changing the inner product; the choice $\tau = \mathbb{E}_P[\mu_1(X) - \mu_0(X)]$ is the unique one that places the result in $\mathcal{H}_X$, giving

$$(I) \;=\; \mathbb{E}_P[\{\mu_1(X) - \mu_0(X) - \tau\}\, S_X(X)] \;=\; \langle \varphi_X,\, S_X \rangle_P,$$

with $\varphi_X(X) := \mu_1(X) - \mu_0(X) - \tau \in \mathcal{H}_X$.

For **Term (II)**, fix $t \in \{0, 1\}$ and compute

$$\partial_\varepsilon \mu_{t,\varepsilon}(x)\Big|_0 \;=\; \int y\, S_Y(y \mid t, x)\, p(y \mid t, x)\, d\mu(y) \;=\; \mathbb{E}_P\big[(Y - \mu_t(x))\, S_Y(Y \mid t, X)\,\big|\, T = t,\, X = x\big]\,,$$

where the second equality uses $\mathbb{E}_P[S_Y(Y \mid t, x) \mid T = t, X = x] = 0$ to subtract $\mu_t(x)$. Multiplying by $p_X(x)$ and integrating gives a conditional expectation that can be re-expressed as an unconditional one through the identity

$$p_X(x)\, p(y \mid t, x) \;=\; \frac{p(x, T = t, y)}{\pi_t(x)},$$

which follows from the factorization $p(x, T = t, y) = p_X(x)\, \pi_t(x)\, p(y \mid t, x)$, with π_t as defined in the setup above. Applying this identity for $t = 1$ and $t = 0$ and combining,

$$(II) \;=\; \mathbb{E}_P\left[\left\{\frac{T}{\pi(X)}(Y - \mu_1(X)) \;-\; \frac{1 - T}{1 - \pi(X)}(Y - \mu_0(X))\right\} S_Y(Y \mid T, X)\right] \;=\; \langle \varphi_Y,\, S_Y \rangle_P,$$

with

$$\varphi_Y(O) \;:=\; \frac{T\,(Y - \mu_1(X))}{\pi(X)} \;-\; \frac{(1 - T)\,(Y - \mu_0(X))}{1 - \pi(X)}.$$

A short conditioning check confirms $\mathbb{E}_P[\varphi_Y \mid T, X] = 0$, so $\varphi_Y \in \mathcal{H}_Y$.

The canonical gradient. Combining the two terms and using the orthogonality of the three subspaces in Equation C.10,

$$\frac{\partial}{\partial \varepsilon} \tau(P_\varepsilon)\Big|_0 \;=\; \langle \varphi^*,\, S \rangle_P, \qquad \varphi^*(O) \;:=\; \varphi_X(X) + \varphi_Y(O), \tag{C.12}$$

for every score $S = S_X + S_T + S_Y \in \mathcal{T} = L_2^0(P)$. By Definition C.2, φ^* is an influence function of τ. By construction $\varphi^* \in \mathcal{H}_X \oplus \mathcal{H}_Y \subset L_2^0(P) = \mathcal{T}$, so φ^* is the canonical gradient of Theorem C.1 (ii). Writing it out explicitly,

$$\varphi^*(O) \;=\; \mu_1(X) - \mu_0(X) - \tau \;+\; \frac{T\,(Y - \mu_1(X))}{\pi(X)} \;-\; \frac{(1 - T)\,(Y - \mu_0(X))}{1 - \pi(X)}, \tag{C.13}$$

which is the AIPW influence function of Chapter 10. The figure below records the bookkeeping: each of the three score directions contributes its own piece of the gradient, and the treatment direction contributes nothing.

Dimension of $\mathcal{T}_\eta^\perp$. The pathwise derivative $\dot\tau_P : \mathcal{T} \to \mathbb{R}$, $\dot\tau_P(S) = \partial_\varepsilon \tau(P_\varepsilon)|_0$, is by Equation C.12 a continuous linear functional on $\mathcal{T} = L_2^0(P)$ with Riesz representer φ^*, and its kernel is $\mathcal{T}_\eta$ by Definition C.4. Suppose $\dot\tau_P \neq 0$, equivalently $\varphi^* \neq 0$ in $L_2(P)$; by the efficiency-bound formula below this holds unless Y is almost surely a deterministic function of (T, X) *and* the conditional effect $\mu_1(X) - \mu_0(X)$ is almost surely constant. Then the codimension fact of Section C.1 gives $\ker(\dot\tau_P)$ codimension one in $L_2^0(P)$, whence

$$\mathcal{T}_\eta^\perp \;=\; \mathcal{T} \cap \mathcal{T}_\eta^\perp \;=\; \mathrm{span}\{\varphi^*\},$$

the one-dimensional subspace promised above. Two features of the unrestricted model are doing work here and should be named: the tangent space is all of $L_2^0(P)$, so $\mathcal{T}^\perp = \{0\}$ and the warning following Proposition C.1 is vacuous; and the derivative is non-zero. In the degenerate case $\dot\tau_P = 0$ one has $\varphi^* = 0$, $\mathcal{T}_\eta = L_2^0(P)$, and $\mathcal{T}_\eta^\perp = \{0\}$ rather than a line.

Reading off the efficiency bound. By Definition C.5, the semiparametric efficiency bound for estimating $\tau(P)$ in the nonparametric model is $V^*(\tau, P) = \mathbb{E}_P[\varphi^*(O)^2]$. Since $\varphi_X \in \mathcal{H}_X$ and $\varphi_Y \in \mathcal{H}_Y$ are orthogonal, the Pythagorean identity splits the bound into an outcome-noise term and a treatment-effect-heterogeneity term:

$$V^*(\tau, P) \;=\; \mathbb{E}_P[\varphi_Y(O)^2] \;+\; \mathbb{E}_P[\varphi_X(X)^2].$$

The second term is $\mathbb{E}_P[\{\mu_1(X) - \mu_0(X) - \tau\}^2]$ by definition. For the first, note that $T(1-T) = 0$ kills the cross product, so

$$\varphi_Y(O)^2 \;=\; \frac{T\{Y - \mu_1(X)\}^2}{\pi(X)^2} \;+\; \frac{(1-T)\{Y - \mu_0(X)\}^2}{\{1 - \pi(X)\}^2},$$

and conditioning on (T, X) and then on X gives, for $t = 1$,

$$\begin{aligned}\mathbb{E}_P\left[\frac{T\{Y - \mu_1(X)\}^2}{\pi(X)^2}\right] \;&=\; \mathbb{E}_P\left[\frac{\mathbb{E}_P[T\{Y - \mu_1(X)\}^2 \mid X]}{\pi(X)^2}\right] \\ &=\; \mathbb{E}_P\left[\frac{\pi(X)\,\sigma_1^2(X)}{\pi(X)^2}\right] \;=\; \mathbb{E}_P\left[\frac{\sigma_1^2(X)}{\pi(X)}\right],\end{aligned}$$

with the analogous computation for $t = 0$. Collecting terms,

$$V^*(\tau, P) \;=\; \mathbb{E}_P\left[\frac{\sigma_1^2(X)}{\pi(X)} + \frac{\sigma_0^2(X)}{1 - \pi(X)} + \{\mu_1(X) - \mu_0(X) - \tau\}^2\right], \tag{C.14}$$

the classical semiparametric variance bound for the ATE (Hahn 1998; Robins et al. 1994). Comparing Equation C.14 with Equation C.7 confirms the claim made at the outset: the assumed moment condition is exactly the statement $V^*(\tau, P) < \infty$, equivalently $\varphi^* \in L_2^0(P)$. The $1/\pi$ and $1/(1-\pi)$ weights show directly how weak overlap inflates the bound and, in the limit, destroys pathwise differentiability altogether.

It remains to say what this does and does not establish. Equation Equation C.14 is a **parameter-side** result: it is a property of τ and P, computed before any estimator is proposed, and by Theorem C.2 no regular asymptotically linear estimator of τ can have smaller asymptotic variance. It does not follow that every procedure labeled AIPW attains it. A feasible AIPW estimator attains the bound when the estimator-side conditions developed in Chapters 11 and 12 hold: consistent nuisance estimation, a product-rate condition controlling the second-order remainder, empirical-process control or cross-fitting to license the linearization, and a consistent variance estimator for inference. The EIF calculation supplies the target; those chapters supply the argument that a particular estimator hits it.

C.7.1 IPW versus AIPW: The Same Function in Two Models

Second reading.

Now that $\mathcal{H}_X$, $\mathcal{H}_T$, $\mathcal{H}_Y$ and φ^* are all in hand, the status of the IPW estimating function

$$U_{\text{IPW}}(O) \;=\; \frac{T\,Y}{\pi(X)} \;-\; \frac{(1-T)\,Y}{1 - \pi(X)} \;-\; \tau$$

can be settled exactly. A first point to clear away: evaluated at the *true* propensity score, and provided the displayed terms are integrable, U_{IPW} has mean zero whether or not the analyst happens to know π. Unbiasedness is a property of P, not of the analyst's information. What knowledge of π changes is the *model*, and therefore the tangent space.

Throughout this remark we assume in addition that

$$U_{\text{IPW}} \in L_2^0(P), \qquad \text{equivalently} \qquad \mathbb{E}_P\left[\frac{\mathbb{E}_P[Y^2 \mid T = 1, X]}{\pi(X)} + \frac{\mathbb{E}_P[Y^2 \mid T = 0, X]}{1 - \pi(X)}\right] < \infty. \tag{C.15}$$

This is a genuinely stronger requirement than Equation C.7: the canonical gradient involves only the residual variances σ_t^2 and the contrast $\mu_1 - \mu_0$, whereas U_{IPW} also carries the common level of μ_1 and μ_0, inverse-weighted. A function must lie in $L_2^0(P)$ before it can be a gradient in any model, so Equation C.15 is needed before the question below is even well posed. Strong overlap together with $\mathbb{E}_P[Y^2] < \infty$ suffices.

Direct algebra gives the exact decomposition

$$U_{\text{IPW}}(O) \;=\; \varphi^*(O) \;+\; \{T - \pi(X)\}\left\{\frac{\mu_1(X)}{\pi(X)} \;+\; \frac{\mu_0(X)}{1 - \pi(X)}\right\}, \tag{C.16}$$

in which the second term, written

$$q(O) \;=\; \{T-\pi(X)\}\left\{\frac{\mu_1(X)}{\pi(X)} + \frac{\mu_0(X)}{1-\pi(X)}\right\},$$

has the form $b(X)\{T-\pi(X)\}$ and hence lies in the treatment-score space $\mathcal{H}_T$ (see Equation C.8). Everything follows from where q sits.

Unrestricted treatment mechanism. Here $\mathcal{H}_T \subset \mathcal{T}$, while $\dot{\tau}_P$ vanishes on $\mathcal{H}_T$, so $\mathcal{H}_T \subset \mathcal{T}_\eta$. By Proposition C.1 every gradient of τ is orthogonal to $\mathcal{H}_T$. Since $\varphi^* \perp \mathcal{H}_T$ and $q \in \mathcal{H}_T$,

$$\mathbb{E}_P[U_{\mathrm{IPW}}\, q] \;=\; \mathbb{E}_P[\varphi^* q] + \mathbb{E}_P[q^2] \;=\; \mathbb{E}_P[q^2].$$

So U_{IPW} is orthogonal to $\mathcal{H}_T$ if and only if $q = 0$ in $L_2(P)$. Generically $q \neq 0$, and then U_{IPW} is not a gradient of τ at all — efficient or otherwise. The exceptional case is worth stating because it is a genuine equivalence rather than a one-way implication: if $q = 0$ almost surely, then $U_{\mathrm{IPW}} = \varphi^*$ and the IPW function *is* the canonical gradient, even in the unrestricted model.

Known treatment mechanism. If π is known, no submodel may perturb $p_{T|X}$, so treatment-score directions drop out of the tangent space: $\mathcal{T} = \mathcal{H}_X \oplus \mathcal{H}_Y$ and $\mathcal{H}_T \subset \mathcal{T}^\perp$. The second term of Equation C.16 now lies in $\mathcal{T}^\perp$, so by Theorem C.1 (i) U_{IPW} is a perfectly valid gradient of τ — just not the canonical one. Projecting it onto $\mathcal{T}$ deletes the $\mathcal{H}_T$ component and returns φ^*.

The efficiency bound is the same in both models, since φ^* is unchanged; knowing π does not help one estimate τ more precisely. What changes is which functions count as gradients, and hence which estimators are regular. This is the cleanest illustration in the appendix of the fact that "efficient" is always efficient *relative to a model*, and it explains the otherwise puzzling result of Chapter 10 that estimating π can reduce the variance of the unaugmented Horvitz–Thompson IPW estimator relative to using the true π.

C.8 Synthesis

The appendix has introduced several objects that are easy to conflate, partly because the literature calls more than one of them an "influence function." The table below collects them.

Table C.1: The objects of semiparametric efficiency theory. A score is a property of a particular regular submodel. Pathwise gradients, canonical gradients, and efficiency bounds are properties of a functional relative to a statistical model. An estimator influence function is a property of a particular estimator at P. The canonical gradient and the efficiency bound are unique once the functional, the distribution, and the model have been fixed, which is why efficiency statements must always name the model.

Object	**Lives in**	**Defining property**	**Unique?**
Score S	$\mathcal{T} \subset L_2^0(P)$	DQM derivative along a regular submodel (Definition C.1)	No: one per submodel
Pathwise gradient φ	$L_2^0(P)$	$\dot{\Psi}_P(S) = \mathbb{E}_P[\varphi S]$ for every score (Definition C.2)	No, unless $\mathcal{T} = L_2^0(P)$
Canonical gradient φ^*	$\mathcal{T} \cap \mathcal{T}_\eta^\perp$	Riesz representer of $\dot{\Psi}_P$ on $\mathcal{T}$; equivalently $\Pi[\varphi \mid \mathcal{T}]$ (Theorem C.1)	Yes
Estimator influence function	$L_2^0(P)$	$\sqrt{n}(\widehat{\psi}_n - \psi) = n^{-1/2}\sum_i \varphi(O_i) + o_P(1)$	Yes, for a fixed estimator
Efficiency bound V^*	$[0,\infty)$, or $+\infty$ by convention	$\mathbb{E}_P[(\varphi^*)^2]$ (Definition C.5); finite whenever Ψ is pathwise differentiable, and $+\infty$ under Equation C.6 when no L_2 representer exists	Yes

Three points are worth carrying away. First, an estimating function is not a gradient until it is normalized and shown to satisfy Equation C.3; unbiasedness is not enough. Second, non-uniqueness of gradients is a symptom of a restricted model, and projection onto $\mathcal{T}$ is the cure. Third, every efficiency claim is relative to a model: the same function can be a valid gradient in one model and no gradient at all in a larger one, as Section C.7.1 shows for IPW.

C.9 Exercises

1. A regular tilted submodel. *[(b)–(d) advanced]* Let $h \in L_2^0(P)$ be bounded and set $p_\varepsilon(o) = \{1 + \varepsilon h(o)\}\, p(o)$ for $|\varepsilon| < 1/\|h\|_\infty$.

(a) Verify that p_ε is non-negative and integrates to one, with no normalizing constant.
(b) Verify differentiability in quadratic mean Equation C.1 directly, and show that the score is h. (*Hint*: expand $\sqrt{1+\varepsilon h}$ and use boundedness of h to control the remainder.)
(c) Conclude that the tangent space of the unrestricted dominated model is $L_2^0(P)$, explaining where the density of bounded mean-zero functions in $L_2^0(P)$ is used.
(d) Where does this *linear-tilt* construction fail when h is unbounded? Explain why this does not show that an unbounded $h \in L_2^0(P)$ can never be the score of some other regular submodel.

2. Non-uniqueness in a normal location model. Let $Y \sim N(\mu, 1)$ with $\Psi(P_\mu) = \mu$, as in Example C.3.

(a) Identify $\mathcal{T}$ and $\mathcal{T}^\perp$.
(b) Show that $\varphi_1(Y) = Y - \mu$ is a gradient.
(c) Show that $\varphi_1(Y) + c\{(Y-\mu)^2 - 1\}$ is a gradient for every $c \in \mathbb{R}$, and explain which property of the normal distribution makes this work.
(d) Project this gradient onto $\mathcal{T}$ and compare variances, identifying the EIF.
(e) Now enlarge $\mathcal{P}$ to the unrestricted dominated model of distributions on $\mathbb{R}$ with finite variance, keeping $\Psi(P) = \mathbb{E}_P[Y]$. What happens to the family of gradients, and why?
(f) Verify that $\hat{\mu}_c$ of Example C.3 is asymptotically linear with influence function φ_2, and compute its asymptotic variance directly.

3. Kernel and nuisance tangent space. Let $\dot{\Psi}_P : \mathcal{T} \to \mathbb{R}$ be a non-zero continuous linear functional with representer $\varphi^* \in \mathcal{T}$. Prove that

$$\ker(\dot{\Psi}_P) \;=\; \{g \in \mathcal{T} : \mathbb{E}_P[\varphi^* g] = 0\}, \qquad \mathcal{T} \;=\; \ker(\dot{\Psi}_P) \oplus \operatorname{span}\{\varphi^*\}.$$

Then state what each assertion becomes when $\dot{\Psi}_P = 0$, and reconcile your answer with the codimension discussion of Section C.1.

4. Orthogonal decomposition for (X, T, Y). For arbitrary $g \in L_2^0(P)$ define g_X, g_T, g_Y as in Equation C.9.

(a) Show $g = g_X + g_T + g_Y$ and that the three terms are pairwise orthogonal.
(b) For binary T, show that any g_T takes the form $b(X)\{T - \pi(X)\}$ and identify b.
(c) *[Advanced]* Show that the correct integrability condition on b is $\mathbb{E}_P[b(X)^2 \pi(X)\{1-\pi(X)\}] < \infty$, and construct a P satisfying positivity for which some $g_T \in \mathcal{H}_T$ has $b \notin L_2(P_X)$.
(d) Show that under strong overlap the two conditions coincide.

5. IPW versus AIPW geometry. With U_{IPW}, q, and φ^* as in Section C.7.1, and assuming $U_{\text{IPW}} \in L_2^0(P)$ as in Equation C.15:

(a) Verify Equation C.16 by direct algebra.
(b) Explain why q is a treatment-score direction.
(c) Show that U_{IPW} is a gradient of τ in the unrestricted model if and only if $q = 0$ almost surely, and that otherwise it is not a gradient. Then explain why it is a valid non-canonical gradient in the known-propensity model.
(d) Verify that projecting U_{IPW} onto the known-propensity tangent space returns φ^*.
(e) Show that $\mathbb{E}_P[U_{\text{IPW}}^2] - V^*(\tau, P) = \mathbb{E}_P[q^2]$, and interpret the excess as a variance penalty.
(f) Exhibit a P satisfying Equation C.7 but violating Equation C.15, and explain what goes wrong.

6. Deriving the ATE efficiency bound. Starting from Equation C.13, prove Equation C.14. Then:

(a) Give sufficient conditions on π and the conditional distribution of Y for the bound to be finite.

(b) Construct a P satisfying $0 < \pi(X) < 1$ almost surely for which the bound is infinite. What does Definition C.2 say about τ at such a P? What qualitative instability should one expect from IPW and AIPW estimators there, and why is this an asymptotic warning rather than a universal finite-sample theorem?
(c) Explain why the bound does not depend on whether π is known.

7. Pointwise versus integrated functionals. *[Advanced]* Let X have a density that is continuous and positive near x_0, and suppose the regression $m_P(x) = \mathbb{E}_P[Y \mid X = x]$ has a specified continuous version near x_0. Consider the pointwise functional $\Psi_{\text{pt}}(P) = m_P(x_0)$.

(a) Using bounded *continuous* bump-function tilts as in Example C.2, show that the score-to-derivative map for Ψ_{pt} is not continuous in the $L_2(P)$ norm. Explain why the tilts must be continuous and why a version of m_P must be fixed in advance.
(b) Conclude that no regular asymptotically linear estimator of Ψ_{pt} exists in the unrestricted model, and state the additional local asymptotic minimax conclusion needed to rule out regular root-n estimation more generally.
(c) Contrast this with the integrated functional $\tau(P) = \mathbb{E}_P[\mu_1(X) - \mu_0(X)]$. Explain why averaging over P_X smooths the target, and state the positivity and moment condition Equation C.7 under which its canonical gradient lies in $L_2^0(P)$.

C.10 Bibliographic Notes

The modern formulation of pathwise differentiability and tangent spaces is developed in the monographs of Bickel et al. (1993) and Vaart (1998) (Chapter 25); the latter is the standard reference for the convolution theorem and local asymptotic normality underlying Theorem C.2. Tsiatis (2006) gives a treatment oriented specifically toward missing data and causal inference, and is a natural companion to the material developed here.

For the ATE functional specifically, Hahn (1998) derives the efficiency bound Equation C.14 and analyzes the role of the propensity score, including the point developed in Section C.7.1 that knowledge of π does not lower the bound; Robins et al. (1994) obtain the same influence function from the augmented estimating-equation perspective that Chapter 11 follows. The failure of pathwise differentiability for pointwise functionals (Example C.2) and the resulting slower-than-$\sqrt{n}$ rates are treated in Vaart (1998) (§25.3).

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